THE RIFLE COMPANY, PLATOON AND SECTION IN BATTLE

CHAPTER 1
INTRODUCTION

SECTION 1 – GENERAL

Purpose

1. This pamphlet provides doctrinal and training guidance for the employment of the rifle company, platoons, sections, organic to the infantry battalion.

Scope

2. This pamphlet is designed as a reference for the rifle company commander and his platoon and section commanders. It covers:-

a. The organization of the company and its organic elements.

b. The organization for battle.

c. The tactical employment in all operations of war.

3. The pamphlet is to be read in conjunction with "GSP-1549 Training for War 1995 and GSP-1808 Infantry Battalion in Battle 2000.

Mission

4. The basic combat mission of the Rifle Company and its elements is to close with the enemy, by means of fire and manoeuvre, in order to destroy or capture him or to repel his assault by fire, close combat and counter-attack.

Characteristics and Capabilities

5. The Rifle Company is the basic tactical element of an infantry battalion and the rifle platoons with their sections are the basic tactical elements of the Rifle Company. Every platoon and section is organized, equipped and trained to use a variety of techniques to accomplish a mission based on certain capabilities which are as following:-

a. Close with the enemy by means of fire and manoeuvre in order to destroy or capture him.

b. Repulse enemy attack by fire, close combat and (usually as part of a large force) counter-attack.

c. Provide a base-of-fire element and a manoeuvre element.

d. Seize and hold ground.

e. Conduct independent operations on a limited scale.

f. Participate in air transport operations when provided with sufficient air effort.

g. Conduct operations in all types of terrain and climatic conditions.

h. Participate in amphibious operations.

j. Take part in different type of operations by day or night.

k. Move by any method i.e. cross-country, road, rail, sea and air.

l. Provide limited anti-tank protection.

m. Conduct heliborne/anti-heliborne operations.

SECTION 2 – **ORGANIZATION – THE INFANTRY BATTALION**

General

1. Annex "A" shows the outline organization of an infantry battalion. It is important that every man in the company should understand this organization and know what assistance can be given to the rifle companies from the combat support company.

Battalion Headquarters

2. The battalion headquarters is made up of:-

a. A command element which includes the commanding officer, Second-in-Command (2IC), adjutant and intelligence officer. The battalion office works under the adjutant and comprises of six clerks and two dispatch riders. When the battalion headquarters is split into main headquarters and a command post, the battalion office will form part of the main headquarters.

b. *Intelligence/Education Section.* It consists of two havildars, four naiks and two lance naiks. In peace, they act as education

instructors carrying out intelligence duties, while in operations they work under the adjutant/intelligence officer. The section is responsible for sifting and recording information. It will often man Observation Posts in forward platoon areas and will need protection.

c. *Regimental Police Section.* It consists of a havildar and four sepoys trained in provost duties. They are responsible for traffic control duties within the battalion and for implementing certain security measures.

d. *Medical Detachment.* It consists of a havildar and a naik stretcher-bearer. In operations a medical officer and a havildar nursing orderly are attached from the army medical corps, and they establishes the Regimental Aid Post (RAP)

Combat Support Company

2. The combat support company is organized into a company headquarters, and four platoons:-

a. *Company Headquarters.* It consists of a company commander (Major), senior junior commissioned officer (Subedar), company havildar major, company quartermaster havildar, one pay/kote naik, one sepoy clerk, one General Duty (GD) sepoy, two drivers, one radio/wireless operator and five cooks (unit).

b. *The Signal Platoon.* This platoon provides communications within the battalion in all operations of war. In static situations, telephone lines are laid while in mobile operations wireless sets are normally used. The platoon consists of a signal officer, a naib subedar, a havildar, a naik storeman, two naiks of wire and line party, two naiks as key board operators, two general duty sepoys, three drivers, eight sepoys as wireless and radio operators and two sepoys as key board operators. Total strength of signal platoon is 23 all ranks. The other rank lay line and man the telephone exchange. The wireless sets used for inter-communication between the battalion headquarters and rifle companies are AN/PRC-77 or AN/PRC-4620 while those used within rifle companies are AN/PRC-77 and PK/PRC-2505 (walkie-talkie). The means of communication within the battalion are sufficient to provide flexibility. No rigid distribution of wireless sets/telephones can be laid down, as it will depend on the prevailing tactical situation.

c. *Mortar Platoon.* It has a total strength of 36 all ranks and possesses six 81mm mortars divided into two sections of a three mortars each. These mortars can fire high explosive and smoke ammunition. It constitutes the principal means of providing fire support under the direct control of the battalion commander. Owing to the high trajectory of the shell, mortar fire is particularly suitable for engaging targets in dead ground or behind high cover, which cannot be engaged by other means. In attack, a Mobile Fire Controller (MFC) often accompanies a rifle company, while in defence a mortar Observation Post (OP) may be located in a platoon locality. A mortar platoon can provide either two MFCs or two OPs at a time. The mortar platoon can perform the following tasks: -

(1) Assist in preparatory bombardment.

(2) Provide close support fire.

(3) Help in reorganization of the battalion after an attack.

(4) Create smoke screens.

(5) Support protective detachments and patrols.

(6) Provide defensive and harassing fire.

(7) Support counter-attacks.

(8) Perform counter mortar bombardment tasks.

d. *Anti-Tank Guided Missile (ATGM) Platoon*. This platoon has a total of 20 all ranks. The platoon is organized into two sections. The ATGM platoon forms an integral part of the infantry battalion. Its primary role is to provide anti-tank support to the battalion; it can, however, be given other roles when not required for its primary role such as engaging enemy bunkers, Observation Posts, non armoured vehicles, crew served weapons et cetera. The ATGM platoon can perform following tasks:-

(1) In advance and attack:-

(a) To cover the movement of reconnaissance vehicles and tanks.

(b) Flank protection.

(c) To provide covering fire for the assault against AFVs, bunkers and buildings.

(d) On reorganization, provide the framework of the anti-armour defence.

(2) In defence and withdrawal:-

(a) To provide the framework of anti-armour defence.

(b) Covering gaps between battalion/company boundaries in conjunction with integral resources/ other arms.

(c) Acting as part of protective detachments in conjunction with other battalion resources/other arms.

(d) Destruction of enemy troop lift helicopters if hovering or landing.

e. *Assault Pioneer Platoon.* This platoon has a total strength of 23 all ranks consisting of a subedar platoon commander and twenty-two other rank. The platoon is organized into three sections. The platoon havildar and one of the section commanders (Naik) are attached from the engineers, whereas, two remaining section commanders are from infantry. Besides carrying out following tasks, individuals of this platoon are also trained to perform NBC related duties:-

(1) Laying lifting anti-tank and anti-personnel mines removal of booby traps laid by the enemy.

(2) Carrying out minor demolitions.

(3) Construction of obstacles.

(4) Assist rifle companies in the handling of river crossing equipment.

(5) Close in defence of Battalion Headquarters.

Administrative Platoon

2. This platoon is responsible for the administrative echelons of the battalion and the control and maintenance of transport other than that with rifle companies. It is organized into two sections:-

a. *Administrative Section.* This section has strength of 39 all ranks and functions under the Battalion Quartermaster and contains his

staff i.e. Naib Subedar Quarter Master (JQ) and his store men et cetera.

b. *Transport Section.* Consists of a total of 13 personnel. The battalion Mechanical Transport Officer (MTO) who is of the rank of a Captain commands this section. He is assisted by a Subedar commonly known as Mechanical Transport Junior Commissioned Officer (MT JCO). A naik, two craftsmen vehicle mechanics and two craftsmen electricians are attached from the Electrical and Mechanical Engineers (EME).

Rifle Company

3. Total strength of a rifle coy is 146 all ranks. It consists of a company headquarters, three rifle platoons and a weapon section:-

a. *Company Headquarters.* It consists of a company commander (Major), company senior Junior Commissioned Officer (Subedar), company havildar major, company quartermaster havildar, one pay/kote naik, a company clerk and fifteen lance naiks/sepoys. The latter include three drivers, three signalers, four stretcher-bearers, four cooks and one Sepoy GD.

b. *Rifle Platoon.* It consists of a platoon headquarters and three rifle sections. The platoon headquarters consists of a platoon commander (Captain/Subaltern), one Junior Commissioned Officer as platoon Second-in-Command, two rocket launcher sepoys, one wireless operator and one sepoy GD. A rifle section is commanded by a havildar. In addition there is a naik and eight sepoys in each section. The section has two light machine-guns and can be split up into two groups. One tripod per platoon is provided for use of MG1A3 in MG role.

c. *Weapon Section.* Consists of seventeen other ranks. It has two sections, one of Recoilless Rifles and the other of 60 mm mortars and a GLHMG detachment. Each section has two 106 mm Recoilless Rifles and two 60 mm mortars respectively where as GLHMG detachment holds one 40 mm GLHMG. Both the sections and GLHMG detachment are commanded by a havildar each.

SECTION 3 – **TRAINING**

Role of the Infantry

1. The primary role of the infantry is to close with and destroy the enemy. This is achieved by fire and movement and the infantry company is designed to carry-out fire and movement in all operations of war, in all types of terrain and under all types of climatic conditions. It is also designed to hold ground. The demands thus made on infantrymen in battle are so heavy that they have to be extremely well trained in peace to meet these demands fully in war.

Need For Planning

2. The time available for training is always too short; the company commander will rarely have time to do all he wants to do. He must, therefore plan all his training with utmost care to make the best use of available time. The first essential in any planning is a clear aim. Before he begins to plan any training the company commander must be quite clear about the aim in his own mind.

3. Instructors are often young and inexperienced. They have a hard and responsible task and will look to their company commander for help and guidance. He must always ensure that they get required assistance.

4. For planning and conduct of advance infantry training see Chapter 17.

SECTION 4 – **TACTICS**

1. Minor tactics is the application of weapons and formations to ground. Every company, platoon and section commander must master:-

   a. Weapon handling.

   b. Fire control.

   c. Fieldcraft and appreciation of ground.

   d. Selection and construction of fire positions/weapon trenches.

   e. Concealment and the use and construction of cover.

2. Once a battalion has been committed to battle, success or failure depends almost entirely on the initiative of the junior leaders and the efficiency with which they handle their commands.

3. Junior Leaders must always try to improve their military knowledge. They must also use their common-sense, as tactics is essentially the common-sense. To help them master tactics junior leaders should remember these four constant factors: -

a. Have a clear aim.

b. Make a simple plan.

c. Try to achieve surprise.

d. Be bold and aggressive.

CHAPTER 2
MAN MANAGEMENT AND ADMINISTRATION

SECTION 5 – **MORALE**

General

1. Morale is founded on discipline, leadership, ideological motivation and comradeship. High morale can turn a defeat into victory, and, similarly low morale will hamper success, no matter how well trained is a unit and how perfect is its equipment.

2. Morale is the degree of confidence in the mind of a soldier when he identifies himself with a group, accepts its aims and works hard to achieve them. In a company, therefore, a soldier's morale means how he feels about himself, his job and the rest of his company. If he feels that he is accepted as a member of the company, the chances are that he will be a happy and enthusiastic soldier. If all the members of a company feel like this and are well trained and well led, they will pull together as an efficient team. Each and every man will be determined to avoid letting down either his own friends or the reputation of his section, platoon, company or battalion.

Discipline

3. Discipline helps the soldier to overcome fear and fatigue. It is founded on self-respect, self-control and a sense of duty. It enables a soldier to carry-out orders without any regard to his own safety in the interest of the other members of his company, it calls for a high standard of leadership. Good discipline results as much from cheerful cooperation as from plain obedience. If morale is high, a man

will impose a much more exacting discipline on himself than anyone else can impose on him.

Leadership

4. Leadership amounts to winning the confidence and cooperation of men to such an extent that they will readily and cheerfully undertake any task given to them. Men expect their leaders to set a good example in every way i.e to be efficient, just and genuinely interested in their personal welfare. Junior leaders will not gain the confidence of their men unless and until they measure up to these standards.

SECTION 6 – **CARE OF MEN**

General

1. The first responsibility of all commanders is the care of their men. A company commander must begin by getting to know the men in his company. They will have different personalities, back grounds and aptitudes. Company, platoon and section commanders must learn what these individual characteristics are if they are to handle their men to the best advantage of the subunit as a whole.

2. A sense of comradeship between all ranks should be encouraged. Men should not be transferred between subunits without good reason; should this prove necessary, the reasons should be explained. When reinforcements arrive, friends or individuals coming from the same village should be posted to the same section whenever this is possible.

3. As far as the essential needs of security allow, men should always be kept informed of the latest situation. They will work better for knowing what they are doing and why they are doing it.

4. The company commander must continually look after his men’s interest and welfare. He should see that they are comfortable, well cared for as the situation allows and that their duties are fairly divided. Some men have a knack of dodging duties whereas others undertake cheerfully any task given to them. He must always be ready to listen to complaints and to remedy the cause if they prove justifiable. If men have problems, he should help them if he can, or failing that seek his battalion commander’s help.

5. The company and platoon commanders will find it useful to keep an up to date roll of their men. Details that should be included are:-

a. Number, rank and name.

b. Home address.

c. Blood group.

d. Name and address of next of kin, details of family.

e. Date of enrolment.

f. Date of posting to the platoon.

g. Trade and specialist qualifications.

h. Courses attended.

j. Category of firer (Class of shot i.e. Standard Shot, Marks Man or Weak).

k. Rates of pay and allowances.

l. Details of leave.

6. The company commander should study each man's personal documents kept in the company office. This study, coupled with his personal contact with the men, will soon familiarise him with their service, family background, trade and individual interests.

7. There is a mistaken idea that the question of fitness of the men is the sole responsibility of the medical officer. The medical officer is the commanding officer's adviser on medical matters. The care of men, however, is the responsibility of the company commander advised by the medical officer. He must always watch for signs of illness or strain in his men. He must insist on the highest possible standard of personal cleanliness. Men must shave, clean their teeth daily and take bath as often as circumstances allow.

Sanitation

8. All men must be made to realize the importance of sanitation and the dangers of not maintaining the desired standards in this respect. All refuse, urine, empty tins and waste food must be buried or disposed of in order to prevent the breeding of flies, which is one of the leading dangers to the health of the troops. One man having dirty habits with no regard for sanitation may be responsible for the illness of many of his comrades from diseases like enteric fever, dysentery, cholera et cetera. Lice, which may be found in blankets or picked up in old

dugouts or dirty houses, are often carriers of disease. Blankets should be aired regularly and men who become infested with lice must immediately report the fact.

9. Only water from sources passed as pure should be drunk. Sterilizing tablets or boiling should be used in all doubtful circumstances.

Health of Troops

10. Maintenance and preservation of health of troops is a command responsibility. One of the major factors affecting the health of troops in the field is communicable diseases. These diseases, unless detected and controlled immediately on their onset, may spread and cause undue casualties. Casualties can also be caused by certain other factors e.g. heat, cold and neglect of feet et cetera. Measures to prevent and control the communicable diseases and for minimizing casualties due to other factors in the field are discussed in succeeding paragraphs. Medical officers are responsible for recommending any additional measures. It is the duty of all company commanders to know and implement these measures and to ensure compliance. Common communicable diseases are:-

a. *Malaria.* Of all the communicable diseases, malaria requires special mention as it continues to be one of the prime factors in causing wastage of manpower in our region. The following measures will help in controlling malaria in the field:-

(1) *Personal Protective Measures.* These are observed during malaria season. These measures are designed to exclude the possibility of mosquitoes biting the men. They include use of liquid repellent on the exposed parts of the body, wearing of protective clothing between dusk and dawn and using mosquito nets wherever possible.

(2) *Drug Treatment.* Its aim is to administer suitable anti-malaria drugs so as to suppress or prevent the disease. All fresh recruits, personnel returning from leave, personnel having enlarged spleen diagnosed through medical check-ups or who have contacted malaria within the last twelve months, will be given anti-malaria drug treatment, under the supervision of an officer, and as advised by the medical officer. The formation commander under extra-ordinary circumstances orders suppressive treatment. This point is mentioned here only to emphasize the fact that the

suppressive treatment should not be resorted to by the company commander on his own.

b. *Infectious Hepatitis.* It is an infectious disease caused by a virus and can be recognized by loss of appetite, vomiting, headache and diarrhea followed by fever and jaundice and finally skin turning yellow. The virus is spread through breathing and also excreted through the feces. It enters another human body either through food and water, which has been contaminated by fecal matter or soiled hands of a carrier or even by inhalation of contaminated air. The disease attacks the liver and in severe cases may lead to death. As a rule an affected individual takes two to three months to recover and should be given an easy time after rejoining from sick leave till he recovers his health. The disease can be prevented by early detection through proper medical inspections, observance of anti-fly measures in cook houses, covering food and water from flies, and food handlers observing elementary rules of personal hygiene. All ranks must be stopped from eating from unauthorized sources. All waste products must be properly disposed of; proper latrines should be constructed and used. Earth should not be used for washing utensils and ice should not be used in cold drinks.

c. *Enteric Group of Fevers.* This group includes typhoid and para typhoid fevers caused by bacteria. These diseases are characterised by continuous fever, severe headache, diarrhea and sometimes-rose spots on the trunk. Bacteria are spread in the feces and urine of the infected persons who may be patients, or carriers. Human beings are infected when they come into contact directly or indirectly with such feces or urine, because of flies, careless passage of urine directly into water sources, improper sitting or construction of latrines or due to the dirty hands of patient. Preventive measures include education of all ranks regarding personal cleanliness, protection of water supplies from pollution, sterilizing water by chlorinating or boiling, use of properly constructed latrines and burial of all excreta, protection of food from flies and dust, sterilisation of milk by boiling, proper supervision of processing, preparation and serving of all food including proper washing of fruit and vegetables, prohibition regarding eating from un-authorized places, immediate evacuation of suspected cases to nearest medical facility and once yearly TAB inoculation of all ranks and their families.

d. *Cholera.* It is an intestinal infection characterised by a sudden onset of vomiting, diarrhea, rapid loss of fluids from the body, leading to muscular cramps, suppression of urine and collapse. It is spread through feces and vomit of patients and carriers. It is also transmitted by contamination of water and food, dirty hands, utensils or flies. Preventive measures, therefore, are the same as for the enteric group of fevers. In addition early detection of infected persons, isolation of suspects and patients and immediate inoculation of exposed persons with cholera vaccine is necessary.

e. *Dysentery.* An intestinal infection characterised by frequent passage of loose stool containing blood, pus and mucus accompanied by abdominal cramps, fever and diarrhea. Feces of infected persons are the source of infection. Preventive measures are the same as for the enteric group of fevers. Group cases of acute diarrhea disorder must be reported to the medical officer at once.

SECTION 7 - **ADMINISTRATION**

Care of Arms, Ammunition and Equipment

1. It is useless to have a weapon, which does not fire at the right time. Unflagging attention must, therefore, be given at all times to the care and maintenance of arms. All the automatic weapons must not be stripped at the same time. Company commander must lay down timings for cleaning of automatic weapons in such a way so that enough automatic weapons are available in the platoon at all times to deal with any unforeseen situation. The magazines of rifles, light machine-guns and sub machine-guns/machine pistols (MP5) should be emptied of ammunition when out of the line to prevent magazine spring from becoming weak. In action, the platoon Second-in-Command (2IC) is responsible for maintaining the ammunition supply to the platoon, while the company Havildar Major is responsible for the company.

2. The platoon Second-in-Command in conjunction with section commanders must maintain a list of equipment in charge of the men and platoon. Frequent inspections carried out by the company and platoon commanders to ensure that it is correct and action taken to replace deficiencies.

Rations

3. In the field, the company commander is responsible for ensuring that men's food is palatable and that it is provided regularly. In this he is assisted by the company Second-in-Command (2IC) and quartermaster havildar. The company commander should personally investigate complaints regarding the quality and quantity of food and take action to rectify those, which are justified.

4. Platoon and section commanders must see that rations are evenly distributed and due portion is kept for any man who is late for his meals being on some duty.

5. If the company is doing its own cooking in a forward area, care must be taken to guard against smoke or fires disclosing the position to the enemy.

Routine

6. Whether in or out of the line a routine should be adhered to. The maintenance of a routine is important for the following reasons:-

a. It helps to ensure that no points of administration are overlooked.

b. Strictly applied, it avoids unfairness and ensures that high standards are maintained.

c. It helps to relieve boredom, particularly during quiet periods in a defensive position.

7. Timings should be laid down for the following:-

a. Washing and shaving.

b. Men to report sick.

c. Inspection of arms and ammunition.

d. Inspection of equipment, tools and clothing.

e. Repair of clothing (socks et cetera) in need of minor repairs.

f. Foot inspection.

g. Meals.

h. Rest periods.

8. In forward areas inspections may have to be staggered since it will probably not be possible to release all men of the company at the same time. In

addition, since men in forward localities will get little sleep by night, they must rest by day. Rest periods must be properly organized and every effort made to prevent interference with them.

Administration on the March

9. In spite of the fact that infantry may often be moved in Mechanical Transport (MT), no unit can be considered fit for war unless it can march long distances and fight when it reaches its destination. The importance of march discipline and the necessity of maintaining a high standard in all the conditions must be realized by all ranks.

10. To assist in maintaining a high standard of march discipline, the following points should be observed:-

a. Periodic inspection of:-

(1) *Socks* – To see that they are of the right size and free from holes or lumpy darns, both of which cause blisters.

(2) *Boots* – To see that they are fit, soft and in good condition. New boots should be "broken in" by wear before being used for a long march.

(3) *Feet* – To ensure that they are clean and hard and that toe nails are kept short. Genuinely ill men should not be made to march.

b. Immediately before a march, the company will be inspected to ensure that:-

(1) Equipment is in order.

(2) Water bottles are filled.

(3) Loads are fairly distributed.

c. On the line of march:-

(1) Men should keep in step. A steady, even pace must be maintained. Doubling to regain lost distance should be avoided.

(2) Men must be correctly covered off, in the ranks if marching in threes, or at the correct distance in single file.

(3) At the hourly halt, troops will fall out on the same side of the road on which they are marching. All ranks, except

those detailed as sentries, will take off their web equipment and lie down. If however, march is taking place close to the forward area or through a hostile country where there may be a possibility of hostile attacks or attacks by enemy long-range patrols penetrating through forward defended localities, equipment will not be taken off; instead the buckle will be undone and the troops will lie down taking support of their haversacks.

(4) Smoking must not be permitted on the line of march, but only at the hourly halts. By night, smoking should be forbidden if there is any doubt as to the line of march being exposed to enemy observation.

d. After a march:-

(1) If possible, all men should be made to wash their feet and foot inspection should be held. Blistered feet must receive immediate attention.

(2) Socks and boots should be inspected. If fresh socks are not available, dirty ones should be changed to the opposite feet.

Casualties

11. If the wounded can walk, he will do so to the company aid post/ nearest medical post. Whatever the type of operation, ammunition and automatic weapon or Rocket Launcher (RPG-7) will be removed from the casualty and instead a rifle will be left with him. Further instructions/actions are tabulated below:-

a. *In Advance*

(1) Administer first aid and morphia injection if necessary.

(2) Leave that casualty near the axis of advance and give its position to the higher headquarters for subsequent collection.

(3) Mark the position of casualty by sticking his rifle into the ground by bayonet.

b. *In Attack*

(1) Men will not fall out to assist the casualty.

(2) Giving first aid and marking position of a casualty will be done after completion of that phase and higher headquarters

informed accordingly. The company stretcher-bearers would evacuate the casualty to a point notified before.

c. *In Defence*

(1) Administer first aid and morphia injection, if necessary.

(2) Company stretcher-bearers would take the casualty to the company aid post for evacuation back to the regimental/ battalion aid post.

d. *Withdrawal.* Make every effort to evacuate casualty before company finally withdraws. The details of actions regarding first aid and evacuation remain the same as discussed above.

e. *Medical Kit.* Consists of:-

(1) Sulpha Diazine Cream (for application on skin).

(2) Mydriacyl Eye Drops (for dilation of pupil).

(3) Atropine Injection – Auto Injector (for treatment against nerve agents).

(4) Diazepam Tablets (10 tablets of 2 mg each).

f. *Training.* Every man must:-

(1) Have knowledge of first aid.

(2) Be able to give morphia injection. It is not given to a wounde man if he has a head/chest injury or is unconscious.

(3) Know application of, tourniquets and when to be used, splints, treatment of shock and anti-snake-bite measures.

Burial of The Dead

12. Dead should be marked by sticking the rifle into the ground by bayonet and putting steel helmet on top of the rifle butt. Stretcher-bearers should remove the dead from the company area. If stretcher-bearers are not available, men from the platoons must be detailed to do it.

13. Dead must be collected as soon as possible, if necessary under cover of darkness. Personal effects and one portion of the identity disc will be removed, leaving the other portion on the body. The platoon commander will make a duplicate list of personal effects, retaining one copy and sending the other,

together with the effects, in a sandbag, or the individual's haversack to company headquarters.

14. Normally the burial of the dead will be carried out under company or battalion arrangements and conducted by the religious teacher, if available. On occasions, it may be necessary for the company commander to perform this duty. It is most important that all graves are registered and marked with temporary nameplates. The registration, burial and respect shown to enemy dead is an obligation that must be carried out. Failure to meet this obligation is the hallmark of a bad unit.

CHAPTER 3
INFORMATION AND INTERCOMMUNICATION

SECTION 8 – INFORMATION

General

1. A commander in the field has to base his plan on the information available to him. The more accurate and up to date the information is; more likely his plan is to succeed. Forward troops act as eyes and ears of their commander and they must send frequent reports of anything they have seen which will help to build-up an overall picture of the situation.

2. It is often difficult for junior leaders in close contact with the enemy to find time to send back reports, but they must do so as their commanders cannot act or plan without upto date information.

Information Required

3. Information required by a commander in battle falls under three main headings:-

a. Enemy forces.

b. Own forces.

c. Ground.

Enemy Forces

4. a. Where are they?

b. In what strength? (Infantry, Artillery, Transport)

c. What are they doing?

d. What have been their losses? (Men, Tanks, Guns, Vehicles)

5. It is important that hostile shelling and mortaring should be reported on the correct shelrep/morrep form.

Own Forces

6. a. Where are they?

b. What are they doing and what do they intend to do?

c. What state are they in? Have they lost any men or equipment?

d. What troops are on the flanks and what are they doing?

Ground

7. a. What is the country like? Is hilly or flat, close or open? Does it offer cover, concealment and observation?

b. Are there any obstacles, natural or artificial?

c. Can they be crossed and how?

d. What is the state of roads and tracks?

Reporting Information

8. It will not always be possible to answer every question, especially about the enemy. Negative information may be most helpful; for example, that the enemy is not holding a particular place on the front.

9. Once an event has been reported, the information must be followed up. Higher headquarters will always wish to know what has happened to the enemy once reported, even if they have stayed still and done nothing at all.

10. Reports should always include:-

a. The time of the event; there is always some delay between an event occurring and the time news of it reaches various headquarters.

b. The place where the event happened, including the grid reference; a report from a company/platoon commander referring to his front without grid reference may mean nothing at all to the battalion/brigade headquarters.

c. The source of the information; some reports are consistently more reliable than others. Reports from civilians and sometimes from

other troops may be unreliable and have to be judged on their merits.

d. The action, if any, which the sender is taking as a result of the information obtained.

11. In addition to sending information back, it should also be passed to the subunits on both flanks. The men must also be kept informed of the progress of the battle on their own and, if possible, other fronts.

SECTION 9 – **PRISONERS OF WAR**

Enemy Prisoners

1. Prisoners of war are a valuable source of information. Their examination and interrogation is the responsibility of specially trained intelligence staffs. Company commanders are responsible for sending back prisoners under escort to battalion headquarters immediately after capture. The earlier they are interrogated, the more valuable their information is likely to be.

2. When enemy prisoners are taken, the procedure to be followed is:-

a. *Disarming.* Prisoners must be fully disarmed. They must not be allowed to destroy any part of their equipment.

b. *Segregation.* Officers, Junior Commissioned Officers and men must be separated from each other. Talking must be forbidden; prisoners must never be left unguarded.

c. *Searching.* Prisoners must be thoroughly searched for maps and papers. These will be taken from them to prevent their destruction and will be sent back with the prisoner's escort as they may contain valuable information. It is forbidden to remove prisoner's identity tokens, badges of rank, decorations and personal valuable. Money will not be taken from prisoners except on the orders of an officer. In such cases, the sum taken must be recorded and a receipt must be given to the prisoner.

3. If possible, our walking wounded should be used as escort. The strength of the escort will vary according to the state of the prisoner's morale and the type of country. One man can escort one prisoner, but for more than one, a minimum of two guards are required. At night, more guards are needed than by day. This

requirement may cause a serious drain on manpower, so prisoners are escorted as many as thirty or more at a time back to battalion headquarters; the size of the escort depending entirely on the state of the prisoner's morale.

4. The escort should take with them:-

a. All maps and papers which have been taken from the prisoners.

b. A report on when, where and how the prisoners were captured.

Conduct of Pakistani Prisoners of War

5. It is the duty of all ranks to avoid capture by the enemy. Information obtained from prisoners could be of great value to the enemy. According to the Geneva Convention of 1949, a prisoner must give his number, rank and date of birth when called upon to do so. He must refuse to give other information. It is the duty of every prisoner of war to:-

a. Destroy anything, which may give information to the enemy and to ensure that no weapon falls intact into enemy hands.

b. Try to escape the earlier the escape is attempted the better the chances of success, since the guarding of prisoners of war in the field is always difficult.

c. Give only his number, rank, name and date of birth. He must refuse to give other information. A prisoner who refuses to answer questions will defeat any methods of interrogation. Ill-treatment of our prisoners by the enemy may lead to reprisals against his men in our hands. Finally a prisoner who is steadfast in his refusal to answer questions not permitted under the Geneva Convention of 1949 will gain the respect of his captors.

6. The enemy will try various methods to obtain information from prisoners including:-

a. Impersonation of Pakistani or Allied prisoners and mixing with genuine prisoners.

b. Offering special privileges or by friendly reception on capture.

c. The use of concealed microphones in rooms.

d. Suggesting that another officer, Junior Commissioned Officer or man has talked freely, giving the impression that silence is no longer necessary.

e. Renewed interrogation long after capture is made in a casual and friendly way.

7. There are many other ways, which the enemy will use to obtain information.

SECTION 10 - **REPORTS**

Verbal Reports

1. In the battle the company/platoon commander will, whenever possible, pass information by a verbal report. The advantages of a verbal report are that it cannot be intercepted by the enemy and cannot be misunderstood by the recipient.

2. A verbal report must be short and clear. Therefore, the company/platoon commander must:-

a. Think out beforehand what must be reported.

b. If there is time, write down a few short notes so that no important details are forgotten.

3. A verbal report should be made clearly and without hesitation. No one can make a report when confused or out of breath, and the officer or Non Commissioned Officer to whom it is to be made will always allow the bearer of the report or message, time to recover his breath and think out what he has to say.

Report by Radio and Telephone

4. Whenever practicable, a platoon commander should make verbal reports in person to his company commander. There are times, however, when it is neither possible nor desirable for him to do so and he must report on the radio or the telephone, if he has one. In either case, the same rules of brevity and clarity apply as for verbal reports. Section commanders may also be required to report by these means, especially when in command of patrols.

5. Provided the rules of security are observed, a normal conversation can be carried on by telephone but it must be remembered that enemy patrols may tap the lines and intercept conversations.

6. Talking on a radio set is quite different. Before anyone speak on the radio, he must be taught voice procedure. Constant practice is needed to ensure good and secure communications. Practice is necessary to:-

a. Distinguish and understand words during bad working conditions caused by atmospheric interference and interference from other stations.

b. Become accustomed to one-way conversations.

c. Acquire the particular manner of speaking (pronunciation, enunciation and special stress on words or syllables) which is necessary on the radio.

SECTION 11 - **MESSAGES**

Written Messages

1. It may be neither possible nor desirable to report by any of the means so far considered and it may be necessary to send a written message. This should be:-

a. *Clear.* There must be no doubt about the meaning of a message. Handwriting must be legible.

b. *Brief.* The shorter the message the better it is, provided that it's meaning is clear. Unnecessary words are left out and the standard list of abbreviations should be used in message writing.

c. *Accurate.* Such details as grid references, figures, dates and time must be carefully checked before the message is sent.

2. If possible messages should be written on the standard message from (PAFZ-3051) but if these are not available, they may be written on any paper. The following layout should be used in writing a message:-

From 2 Sec Date . . . ….. …. Time of Origin 091530

{Note (a) } {Note (b) }

To 5 Pl

{Note (a) }

have reached TEKRI (.) no enemy seen

Muhammad Khan Naik

{Note (c)}

Notes: (a) After the word “From”, the unit (in this case No.2 Section) will be written and not the name of person. The message is likewise addressed to unit and noted a person.

(b) After “Date . . . . . Time of Origin” should be inserted six figures, the first two for the date of the month and the last four for the time. If the date of the month is the 9th or earlier, a “0” is inserted before the figure, as shown above.

(c) The message should be signed by the sender with name and rank only.

Instructions for Message Writing

3. Block capitals will be used for all place names and grid references will be given if possible. The four cardinal points of the compass and their derived adjectives will be written in full, in block capitals, for example, SOUTH, SOUTHERN. For intermediate points the capital letters N,S,E and W will be used, for example, SE of RAWALPINDI. Proper nouns will be written in block capitals. In addition words requiring emphasis will be written in capital letters.

4. The 24 hours clock system will be used when giving the time examples:-

Noon	.. 1200 hours
9 am	.. 0900 hours
9.30 pm	.. 2130 hours
1.30 am	.. 0130 hours

5. These rules have been framed to make message brief and easier to understand. However, a Junior Commissioned Officer or a Non Commissioned Officer should not hesitate to send a message merely because he is doubtful about these rules or the abbreviations. It is the contents of the message, which matters most.

Verbal Messages by Runner

6. This is the least satisfactory way of passing information since misunderstandings can easily occur. Even a simple message may be distorted by changing or leaving out a single word. A verbal message should take the same form as a written message, except that the time only and not the date need be given.

7. Verbal messages should always be short. The sender of the message should:-

a. Pick a sensible man to take it.

b. Think it out carefully so that he can give to the messenger without hesitation.

c. Dictate it slowly and clearly.

d. After a pause, make the messenger repeat twice to ensure that he has memorized it.

e. Ensure that the messenger is quite clear where and to whom to take the message and what route he is to follow.

SECTION 12 – **SECURITY OF INFORMATION**

General

1. No information may be included in a private letter, which may give away information to the enemy. It is forbidden to use codes, cipher or shorthand in letters or to include any maps or documents. Photographs of a purely personal nature may be sent if no prohibited subject is included, either in the foreground or in the background. Photographs may not be included in privileged mail or in correspondence to neutral or enemy countries. Soldier's letters are censored in wartime by their officers.

2. The location of a unit must not be disclosed to the enemy. Only the army post office, will be used. The correct form of address, which will be made known to all rank in unit orders, must be used. The name and address of the writer will not be shown on the outside of envelopes nor will they be shown on postcards.

3. Troops in forward areas should never carry anything, which may give information to the enemy. In this category are:-

a. Marked maps showing positions of our troops.

b. Any document, including private letters, which shows the name of the unit or formation.

4. All waste paper in camps and billets must be destroyed. None must be left behind when troops leave.

Communication and Telephone Security

5. Every transmission by radio is liable to interception by the enemy. The information gained in this way will supplement that received from other sources, such as patrols, prisoners of war and air reconnaissance and may enable the enemy intelligence to build-up an accurate picture of our future operations and order of battle. The rules of voice procedure must, therefore, always be obeyed.

6. The rules for speaking on the radio are:-

a. Before speaking, think out what is to be said and which portion of the message is to be encoded.

b. Use correct voice procedure.

c. Use only the authorized call signs, appointment codes, code (SLIDEX) and grid reference code (GRIDDLE).

d. Refer to units by their correct address groups or call signs if they are on the same radio set and never by their names or nicknames.

e. Never use jargon. For example, do not refer to aircraft as “Our winged friends”.

f. Do not encode facts about the enemy or his locations.

g. Give the position of our own troops in code but do not qualify the code. For example:-

(1) Do not give the code for the location with a fact known to the enemy as in “Have just captured enemy gun at PLUM (code word) or JONDPR (griddle code)”.

(2) Do not connect descriptions of places with code words or encoded grid references as “I am at village PLUM” or “I am at village JONDPR”. Say “I am at JONDPR”.

7. It is also possible for the enemy to tap telephone lines, especially when the battle is static. The security rules for speaking on the radio therefore apply to telephone conversations. It is particularly dangerous to talk about future events either on the radio or on the telephone and this should not be done unless it is certain that the enemy will not have time to make use of any information he may intercept.

The Phonetic Alphabet

8.

Letter	Word	Letter	Word
A	ALFA	N	NOVEMBER
B	BRAVO(BRAHVO)	O	OSCAR
C	CHARLIE	P	PAPA
D	DELTA	Q	QUEBEC (KWIBECK)
E	ECHO	R	ROMEO (ROHMEO)
Letter	**Word**	**Letter**	**Word**
F	FOXTROT	S	SIERRA (SEE AIRRAH)
G	GOLF	T	TANGO
H	HOTEL (HOE-TEL)	U	UNIFORM
I	INDIA	V	VICTOR
J	JULIET	W	WHISKEY(WISKEY)
K	KILO(KEELO)	X	X-RAY
L	LIMA(LEEMA)	Y	YANKEE
M	MIKE	Z	ZULU

CHAPTER 4
BATTLE PROCEDURES AND DRILLS
SECTION 13 – **PREPARING FOR BATTLE**

General

1. Battle procedures and battle drills come into play at company level for streamlining and telescoping actions required to be taken by various commanders and units during battle. The procedure aim at quick and timely translation of commander's orders for successful, steady and continuous actions, for the execution of the mission, assigned to a unit, in battle.

2. In order to launch a successful operation, the following actions are necessary:-

a. Reconnaissance.

b. Planning.

c. Issue of orders.

d. Deployment of troops.

e. Organizing fire support.

f. Administrative support.

3. The aim of battle procedure and battle drill can only be achieved, if every soldier knows the part he has to play. It is only through constant practice that the units will be able to attain a high standard in affecting the required deployment with:-

a. The least delay.

b. The minimum orders.

c. The optimum efficiency.

Groups

4. Before an operation takes place, reconnaissance is carried out, a plan is made and orders are issued for its execution. During this process, commanders of units and subunits involved are included in one or both groups i.e., the R and O Groups.

R Group

5. R Group includes the commander and those individuals whose presence is essential to him during his reconnaissance and planning. If too many people are included, they may be seen by the enemy, resulting in casualties and loss of surprise.

6. The company R group will usually consist of the company commander with his communications, if necessary, and his escort and runner. Occasionally, the representative of any supporting arm in his area may join the group. The platoon R group consists of the platoon commander and his runners.

O Group

7. O group comprises those to whom the commander gives his orders, composition of this group is as under:-

a. *Company O group*

(1) Company R group.

(2) Platoon commanders.

(3) Platoon runners.

(4) Weapon Section Commander.

b. *Platoon O group*

(1) R group.

(2) Section Commanders.

(3) Number 1 of Platoon anti-tank weapons.

Echelons

8. The first line transport consists of three echelons:-

a. *'F' Echelon.* These are the vehicles essential to fight the battle.

b. ***'**A' Echelon.* Vehicles readily available to support the troops taking part in the battle.

c. ***'**B' Echelon.* Transport carrying stores and equipment that will not be required until there is a pause in the fighting.

d. The battalion commander will centrally control 'F' Echelon and probably 'A' Echelon. 'B' Echelon will normally be under the control of brigade or division.

9. There are times when some of those usually included in the O group may be on other duties or their role may be such that they can safely be briefed later. In this event, the company commander must give his orders to such personnel afterwards as soon as he can do so.

Layout Group

10. When a unit is required to move from a concentration area to an assembly area or in case of other similar moves, a group known as the layout group is added. Its function is to precede the main body, to layout an assembly area and to see that the unit moves into these areas in the right orders without confusion and delay. The group is normally commanded by one of the platoon commanders and contains representatives from all platoons and weapon section. It should also have some personnel deputed to control the traffic and sign posting.

Main Body

11. All the personnel other than the layout group form part of the main body. This group is commanded by the company commander/company Second-in-Command as circumstances permit.

Battle Procedure

12. The basic sequence of the battle procedure to be followed depends on the nature of the operations and availability of time. Following are two types of battle procedures:-

a. *Quick Battle Procedure.* In all operations of war, if the time available is less the following sequence will be generally applied:-

(1) Issue of warning order.

(2) Quick reconnaissance.

(3) Outline plan.

(4) Coordination and final orders.

(5) Final orders to be issued on ground during coordination.

b. *Normal Battle Procedure.* In case enough time is available the following sequence can be applied:-

(1) Quick reconnaissance.

(2) Warning order including outline plan.

(3) Detailed reconnaissance.

(4) Detailed verbal orders.

(5) Coordination and final orders.

Warning Order

13. The warning order may be issued verbally, by message or by wireless. Its contents will normally include such items as:-

a. A brief statement of the situation.

b. A brief statement of the task.

c. RV and time for O group.

d. The time before which there will be no move of the main body except for reconnaissance parties/layout group.

e. Location of assembly area, if known.

f. Limits on reconnaissance.

g. Any other instructions, e.g., administration or any other restrictions, likely RV for issue of NBC equipment, likely decontamination areas et cetera, in case of NBC threat.

SECTION 14 – **RECONNAISSANCE**

1. When the company/platoon commander makes his reconnaissance, what he looks for depends on the operation to be carried out. Before an attack, he looks for lines of advance and he selects areas for the sitting of major weapons et cetera.

2. Reconnaissance should be planned. The company commander must:-

a. Get all the information available about the situation.

b. Be quite sure that he knows the aim of his reconnaissance.

c. Decide if there is a vantage point in his area of operation.

d. Consider the time available. He must give himself sufficient time to make a plan after his reconnaissance and before issuing his orders. He must also remember that each of his platoon/section commanders must have time to make their own reconnaissance and give out their orders.

3. The company commander must pay attention to his personal concealment before making a reconnaissance in the forward area. He must exercise extra precaution, in the wake of latest Ground Surveillance Radars, while approaching too close to the enemy/objective area.

SECTION I5 - **PLANNING**

The Appreciation

1. When a person is faced with a problem, he thinks out possible solutions and decides which one suits, him best. In the Army, commanders are taught to deal with tactical problems in this way, thinking them out in a logical sequence so as to arrive at a suitable plan. This process is called making an appreciation. With practice, it becomes instinctive.

2. Before getting into specific details about the modalities of appreciation, it is vital to understand that in order to solve a problem or achieve an aim how the appreciation is carried. Appreciation is a process, which involves certain factors in order to achieve that aim or the final result. Those factors are:-

a. Ground and Weather.

b. Enemy situation.

c. Own situation.

d. Time and Space.

3. As mentioned earlier, appreciation leads to the final plan. Ground and weather is the foremost factor, which helps in attaining basic plan. The plan with its complete ingredients is made till end by considering this factor. After this, the appreciation is carried forward. Other factors only help in either modifying or adding some features in order to make that initial plan composed and solid enough to defeat the enemy.

4. The sequence of making an appreciation is:-

a. Decide what is the aim and if it has been given to you, grasp its essentials and bear them in mind throughout.

b. Weigh up the factors, which affect the attainment of the aim.

c. Decide the courses in relation to enemy capabilities/reaction.

d. Decide which course is best to adopt and make a plan accordingly.

5. At company level, it would be a rare occasion, when a company commander is required to carryout an operation independently. While working within the framework of a battalion, the company commander would not be required to carryout a formal appreciation. In this case, a suggested quick appreciation, both for attack and defence are given in succeeding paragraphs.

6. At the same time, the requirement of a formal appreciation by a company commander cannot be ruled out. The format of formal appreciation Attack and Defence are given at Annexes “B” and “C”.

The Aim

7. At company level the aim is normally the task given by the battalion commander e.g., “capture TEKRI ASP”.

The Factors

8. Anything which affects the achievement of the aim is a factor. The factors normally considered are ground, enemy situation, own situation and time and space. But at company level, ground assumes principal importance.

Deductions

9. While weighing up a factor ask yourself the question “so what”? This will test its relevance and give you a logical deduction from it. For example:-

a. *Statement of Factor.* There is no cover between JHUND and NULLAH BEND.

b. *Question.* So What?

c. *Answer (deduction).* I must use smoke while covering the distance.

10. **Surprise and Deception**. At company level, it may appear inapplicable, but in the future battlefield the company commander would be required to consider the elements of surprise and deception in his plans.

11. **Attack (Encounter Battle)**. Here, as the company commander is in close contact with the enemy, he is confronted with a fast moving situation and must make an immediate plan. Basically he requires fixing the following:-

a. Enemy strength and location.

b. Forming Up Place (FUP).

c. Route to FUP.

d. Location of the fire base.

12. Therefore at company level the factors mentioned in paragraph 6 above may be grouped and discussed as follows:-

a. Enemy and Own Situation

(1) *Enemy*. Strength and location.

(2) *Own.* Strength.

b. *Ground.* The company would normally have come under effective enemy fire from the front and so this approach may be safely discarded, The ground should therefore be divided to left and right approaches and discussed under the following headings:-

(1) *FUP*

(a) Location.

(b) *Cover*. For deployment.

(c) *Obstacles*. Between FUP and objective.

(d) *Distance*. To Objective.

(2) *Route to FUP*

(a) *Description.* Of route.

(b) *Distance.* To FUP.

(c) Obstacles.

(d) Likelihood of Enemy interference.

(3) *Fire Base*

(a) Location.

(b) Cover.

(c) Range.

(d) Direction of fire in relation to direction of attack.

c. *Time and Space*

(1) Objective to be captured as soon as possible. . … Minutes

(2) Distance from O group RV to company RV … metres… Minutes

(3) Distance from company RV to FUP……metres.

(4) Unforeseen. 5 – 10 Minutes

(5) Time in FUP. 2 Minutes

(6) Total.

(7) Time for own orders. _____

(8) Time now is . . . . . .

(9) H hour.. . . . .

13. **Defence**. In defence the company will normally be taking up defence within the framework of the battalion. Therefore, company commander will start his appreciation after receiving battalion commander's outline plan. At company level the factors may be discussed as follows:-

a. *Enemy and Own Situation*

(1) *Enemy Situation*

(a) Location.

(b) Composition.

(c) Likely Date/Time of Attack.

(d) Likely capabilities.

(e) Incorporation of latest surveillance means.

(2) *Own Situation*

(a) Own Composition.

(b) Location of Forces on Front, Left and Right of own position.

(c) Status of NBC protection and preparation.

(d) Task/location of GLHMG and 12.7 mm AA Gun.

(e) Task/location of Bakhtar Shikans (BS), if allotted.

(f) Location of Company Recoilless Rifles and MGs.

(g) Details of Fire Support available, Artillery, Battalion mortars, 60mm mortars/MG, DFs, DF (SOS) tasks in your company area.

(h) Detail of Obstacles. Being planned by battalion, which affect your company.

(j) Location of Company HQ and Company Aid Post.

b. *Ground.* A company will normally be responsible for only one approach (i.e., the centre approach) as the left and right approaches will be looked after by the neighbouring platoons and companies. However, in case of a company of the flanking battalion of a brigade it may be responsible for two approaches. Therefore, it may be said that a company will have one (or two main approaches and two (or one) subsidiary approach to guard. While considering the ground factor, the company commander will therefore look for main and alternative weapon positions for defending the main and subsidiary approaches respectively. To do this he should discuss ground under the following headings:-

(1) Description of Approaches.

(2) *Dominating Important Tactical Grounds.* Which dominate the approach. Within this he should look for:-

(a) *Section Posts.* While selecting these, consider attack from the left, front and right. Cater for the following:-

i. Mutual Support.

ii. Depth.

iii. Which DF task are to be fired.

iv. At which point on ground will the MG/ riflemen open fire.\

v. The places where the enemy is likely to take-up positions for fire and movement. (Mine them).

(b) *LMG Positions.* Which give:-

i. Defiladed position.

ii. Enfilade fire.

iii. Good fields of fire.

iv. Camouflage and. concealment.

v. Mutual support.

vi. Alternative positions to cover the subsidiary approaches.

vii. Local protection to the recoilless rifle/LMG/ 60 mm mortar detachment.

viii. GLHMG.

ix. BS.

x. NBC protection aspects.

(c) *Rocket Launcher Positions.* Which:-

i. Cover likely tank approaches.

ii. Cover gaps between the company's recoilless rifles.

(d) *Rifle Trenches.* Which:-

i. Protect the LMG positions.

ii. Protect the recoilless rifle/MGs/ RL positions.

(e) *Location of Company HQ.* Where:-

i. Casualty evacuation can take place.

ii. Ammunition replenishment and supply of rations is easy.

iii. Local protection is available.

(f) *Location of Company Command Post.* Where:-

i. Control of the battle is achieved best.

ii. Local protection is available.

(g) Location of Observation Post/Listening Post. From where:-

i. Sentries can observe enemy movement.

ii. Listening post can give early warning of enemy movement in the area.

(3) *Obstacles.* Both natural and artificial, which can delay the enemy's advance (These would be in addition to those already planned by company/battalion).

(4) *Likely Enemy Assembly Area/FUP.* In case these have not already been registered as DFs, the battalion commander should be requested to have them recorded as DFs.

c. *Time*

(1) Position to be ready by

(2) I have the following time available:-

(a) Day Hours.

(b) Night Hours.

(3) I require the following time:-

	Work	**Day Hours**	**Night Hours**
(a)	Taking of ranges to reference points and clearing fields of fire.		

	Work	Day Hours	Night Hours
(b)	Digging		
(c)	Overhead cover.		
(d)	Camouflage and concealment.		

(4) I have/have not enough day/night hours.

Note: If time is short then some day work may be done at night and vice versa. Also priority of work will have to be laid down. In case of hasty defence overhead cover may take last priority.

14. There might be a situation where a company will have to operate independently; in this situation the format of company commander's appreciation will follow the same format as that of the battalion commander. Ground factor in this case will precede the enemy and own situation.

Courses

15. In attack you consider approaches and in defence the possible layouts of your company i.e., 2 up or 3 up. Courses are discussed under the following heading:-

a. Advantages.

b. Disadvantages.

16. Course Adopted. After this select the course to be adopted (approach in attack and lay out in defence).

Plan

17. Once a decision has been made on the best course, a plan can he made. This plan will be the basis for orders, which the company commander will give to his O group. The plan should be made according to the following headings:-

a. Mission.

b. Execution

(1) General Outline.

(2) No 1 Platoon

(a) Task

(b) Reorganization/Arc of Responsibility

(c) No 2 Platoon

(d) No 3 Platoon

(e) Any Supporting weapons/Arms

(f) Coordinating Instructions.

SECTION 16 - **ORDERS AND BRIEFING**

General

1. Within the infantry battalion, orders are usually given verbally. Commanders must be prepared to receive orders over the wireless. At each level, they extract from the orders they receive all matters affecting their command so that they can pass them on to their subordinates.

2. Orders are given so that all concerned know:-

a. What their commander aims to do.

b. How he intends to do it.

c. What part they themselves have to play.

3. Orders will be given clearly and confidently. They should always be given in logical sequence. There is a standard form of doing this, which helps to ensure that nothing important is left out and makes orders easier to follow. Orders should be as short as possible. The orders as far as possible should be taken down in a field sketch form (see Section 17).

4. When a company commander goes to a battalion O group to receive his orders, he must take his notebook. To save time, he should write down the headings, which his warning order suggests will be needed.

5. Before the company commander begins his orders, he should check that complete O group is present and that all can hear him clearly. He should not allow interruptions during orders but at the end he should give time for everyone to check his notes before asking, “Any questions”.

6. Adequate protection arrangements must be made for the O Group, this will assume added importance in case the orders are given in the close proximity of enemy. Normally one section strength is considered appropriate for the protection of O Group. The section designated for protection can be from any of the platoon, however, it may be appropriate to select from the forward platoons. The section would have better understanding of the forward area. The section can move with the platoon commander of respective platoon.

7. Orders will always be given in the standard sequence, which is:-

a. Landmarks.

b. Situation.

c. Mission.

d. Execution.

e. Administration and logistics.

f. Command and signal.

Landmarks

8. These are given from LEFT to RIGHT and repeated anticlockwise. Clock Ray method should be used in explaining the landmarks.

Situation

9. In this paragraph the enemy and the friendly forces are explained separately as under:-

a. *Enemy Forces.* Known enemy posts, weapons, minefields et cetera.

b. *Friendly Forces.* Company orders should give the battalion mission, activities of flanking company, supporting arms and support weapons. Platoon/Section orders should give the platoon section mission and information about flanking sections/platoons and supporting arms. This heading should also include various attachments and detachments.

Mission

10. The company mission is the task given to it by the battalion commander. The platoon mission is the task given to it by the company commander. It will be

given as a brief clear statement of what the commander intends to do, not how he intends to do it. The mission will be stated as in the following examples:-

a. Capture Clump.

b. Hold the footbridge.

c. Deny the position until…….hours and there after withdraw to village area…….

11. The mission is the most important part of the orders. If subordinates have been told clearly what their commander wants to do, they can make an intelligent effort to do it. But if they do not understand his intentions, things are sure to go wrong.

Execution

12. The execution section should state briefly and clearly how the mission is to be carried out:-

a. It should begin with a general outline. For example, No. 1 platoon commander giving orders for an attack might say, “The platoon will attack two up Left No. 2 section, Right No. 1 section and follow up No. 3 section”.

b. Detailed tasks should then be given to each platoon and the weapon section.

c. Detailed tasks for all supporting arms, attachments and detachments and weapons in location should be included.

d. Coordinating instructions, which are instructions common to more than one subunit, are then given. They may include:-

(1)	Timings	Leave assembly area. Leave FUP. H hour.
(2)	Assembly Area	Location. Action. Order of march on leaving.
(3)	FUP	Location. How marked. Action.
(4)	Start Line	Location. How marked.

Administration and Logistics

13. In this paragraph the commander should say if there are any changes in the normal equipment of the company, platoon or section. Any essential details of ammunition supply, medical and feeding arrangements will also be given.

Command and Signal

14. In this section, the company commander must include:-

a. Where he will travel or be.

b. Where company headquarters will be.

c. Any pre-arranged signals by light or through other means.

d. Channel for wireless set.

e. Nicknames and passwords.

Questions

15. The commander will answer any questions after the O group have had time to study their notes. He should then confirm important points by asking questions.

Synchronization of Watches

16. In the end having asked the questions, the commander must synchronize the watches of the O group.

SECTION 17 - **OVERLAY ORDERS**

General

1. In the field, it is easier and simpler to receive/issue verbal orders on a field sketch or on an enlargement for the following reasons:-

a. No special skill is required. The recipient is only required to prepare a simple field sketch of the area in which a particular operation is taking place.

b. Under battlefield conditions it is easier to read and understand a sketch than lengthy written orders.

c. If the commander who had received the orders on a field sketch/ enlargement becomes a casualty, the person taking over from him will find it easier to understand the orders from a sketch.

d. It is generally difficult to read the handwriting of another person particularly when the individual has taken down anything in haste and under adverse conditions.

Requirements

2. The recipients of the orders on a field sketch/enlargement must be proficient in the following:-

a. Military conventional symbols.

b. Drawing of simple and quick field sketch.

Important Points

3. It is important to streamline the procedure and the usage of the field sketch/enlargements for the delivery of orders. A detailed description of the same has been given in the "The Staff Duties in the Field -1995 (GSP-1620)". However certain guidelines are given in the following paragraphs:-

a. *Limits of Sketch.* Before giving orders, the commander should explain the left, right and the farthest limits of the sketch. He should then give a few minutes to the O group for preparation of the sketch.

b. *Landmarks.* Landmarks will be given in the usual manner from left to right and repeated anticlockwise. In addition to this, the commander may also inform the O group regarding his intention of giving particular landmarks, For example, if he gives an orchard as a landmark, he may also say that it is the objective of No. 2 platoon. At company and platoon level, small landmarks, which have some importance in the operation must not be missed out.

c. *Mission.* Mission should be taken down in writing in the space left at the bottom of the sketch.

d. *Situation.* It is to be dealt with as follows:-

(1) *General Outline.* Should be recorded in the sketch where it is intended to have the FUP by making the necessary symbols of respective subunits. The direction of attack should be shown by drawing lines to their respective objectives by placing the arrow in the middle of the objective.

(2) Sub heading of task for each subunit in turn should be dealt with as follows:-

(a) *Grouping.* By making the necessary symbol of supporting weapons/arms under the symbol of the respective subunit.

(b) *Objective.* By making the symbols of the subunits on the objectives which they have to capture/ defend.

(c) *Arc of Responsibility.* By making lines with arrows at the ends on the objectives covering the respective areas of the subunit.

(d) *Reorganization.* Marked by dotted lines.

(3) *Fire Support Plan.* The following method is to be adopted:-

(a) *Timed Programme.* By drawing the symbol, of the type of fire on the objective. Actual timings in relation to H-hour should be written inside/outside or in the margin to be indicated by an arrow.

(b) *On Call.* By writing target number/nicknames inside the symbol of a particular type of fire where it is intended to be brought down.

(c) *Weapon Fire*. To be shown by symbols at the location of these weapons and then by drawing a line with arrows at the ends pointing at the targets which they have to engage.

(4) *Coordinating Instructions*. These instructions will be recorded as follows:-

(a) *Timings*. By writing timings with their respective symbols. In case there is no military symbol for a

particular thing, for example assembly area, then the time should be written as follows:-

Assembly area — 1200 hours

(b) *FUP/SL/H Hour*. FUP will be written under symbol of SL and H hour at the top of it.

(c) *Order of March.* By either making symbols in assembly area in the required order or by writing it down in the space left at the bottom of the sketch.

(d) *Boundaries*. By drawing appropriate symbols.

e. *Administration and Logistics.* If a symbol exists then the location will be shown by the symbols, failing which, the necessary points will be written on the sketch.

f. *Command and Signals.* Necessary points will be recorded as follows:-

(1) *Location of Headquarters.* By military symbols they are located or likely to be located.

(2) *Success Signals.* In writing on the sketch.

(3) *Channels for Wireless Sets*. In writing on the sketch.

The Verbal Orders

4. The detailed specimen is given in GSP-1796 Battalion Commander's Aid de Memoirs.

Landmarks

5. Landmarks will be given in the usual manner from left to right and repeated anticlockwise. In addition to this, the commander may also inform the O group regarding his intention of giving particular landmarks, For example, if he gives an orchard as a landmark, he may also say that it is the objective of No. 2 platoon.

Sequence of Verbal Orders

6. Following sequence is to be followed:-

a. *Situation*

(1) *Enemy Forces.* The following enemy positions have been located:-

(a) One LMG at each of the following places:-

i. BUMP.

ii. HILL.

iii. FOREST.

(b) One troop of 60 mm mortar just North of BUMP.

(c) There is an artillery Observation Post on RIDGE.

(2) *Friendly Forces.* Battalion attack in two up. B Company is on our left.

(3) *Attachments and Detachments*. In direct support P Battery ex 99 Field Regiment.

b. *Mission*. Capture BUMP-HILL.

c. *Execution*

(1) *General Outline.* Attack will be in two up. Left No. 1 platoon, Right No. 2 platoon and depth No. 3 platoon.

(2) *No. 1 Platoon*

(a) *Grouping.* One machine-gun.

(b) *Objective*. BUMP.

(c) *Reorganization*.

i. On the objective.

ii. Arc of responsibility, cover NW and NE.

(3) *No. 2 Platoon*

(a) *Grouping.* One machine-gun.

(b) *Objective*. HILL.

(c) *Reorganization.*

i. Approximately 100 metres North of the objective.

ii. Arc of responsibility, cover North and NE.

(4) *No. 3 Platoon.* Depth. Behind No. 2 platoon. Be prepared to assist forward platoons. Reorganization in area HOUSE.

(5) *Mortars*

(a) *On Call.* HILL 3 minutes HE Normal Nickname SPANISH LADY.

(b) *Defensive Fire.* BUSH. Nickname GREEN VALLEY.

(6) *Recoilless Rifles.* To provide one detachment to No. 1 platoon and one detachment to No. 2 platoon.

(7) *60 mm Mortars.* Will move with Company HQ and provide fire support by observation where required.

(8) *GL HMG.* ..……...

(9) *BS.* ……..

(10) *12.7 mm AA Gun.* ……....

(11) *Artillery*

(a) Will smoke area in front of RIDGE—HILL from H minus 10 minutes to H hour.

(b) One battery CLUMP, H to H plus 10 minutes, HE – Normal.

(c) Defensive Fire. RIDGE, Nickname ROUGH SEA.

(12) *Coordinating Instructions*

(a) *Assembly Area*. WOOD.

(b) *Forming up place.* SOUTH OF TRACK.

(c) *Start Line.* TRACK.

(d) *H hour.* I200 hours.

(e) *Boundaries.* ROAD inclusive to BRAVO Company.

d. *Administration and Logistics*. Regimental aid post in wood.

e. *Command and Signal*

(1) Company headquarters in forming up place. Will move to HOUSE after capture of the objective.

(2) *Success Signals*

	Subunit	Wireless	Very Light
(a)	Number 1 Platoon	BLACK TENT	WHITE
(b)	Number 2 Platoon	RED SHOES	RED
(c)	Number 3 Platoon	GREEN BELT	WHITE/RED
(d)	Coy	LION KILLED	GREEN/ GREEN

Briefing

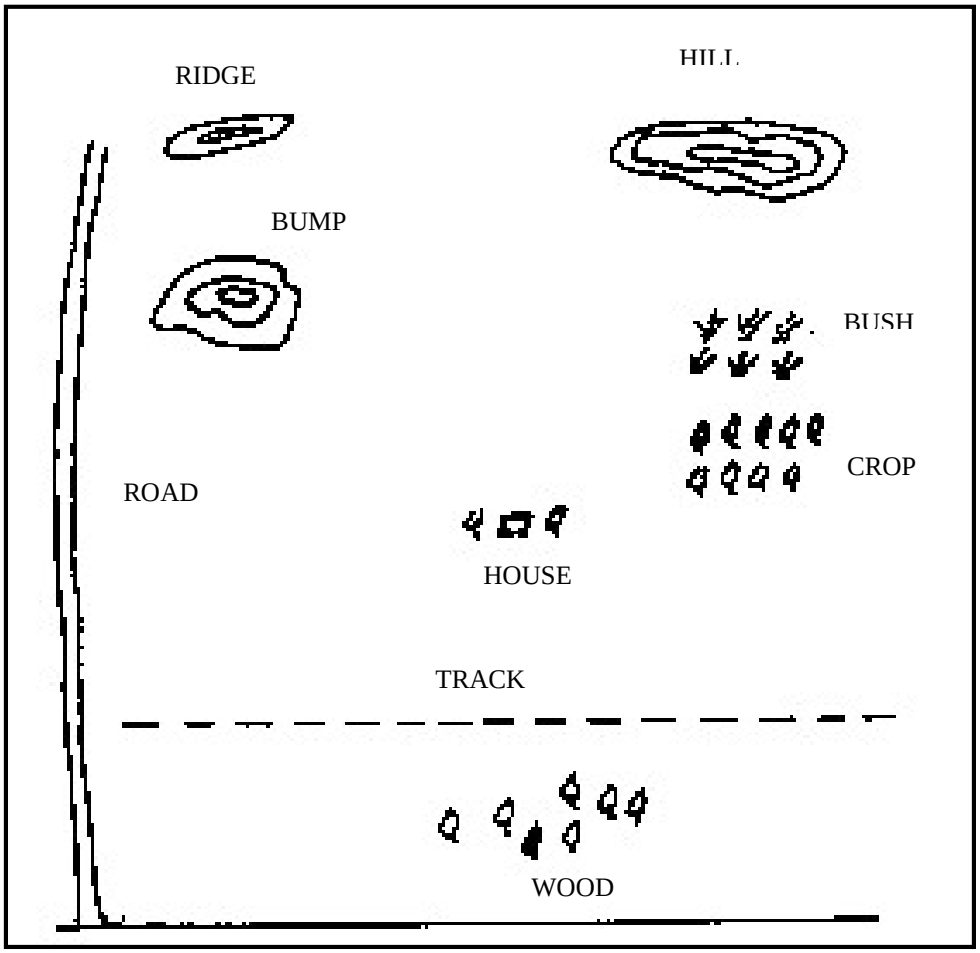
7. There is always a risk that leaders may become casualties early in the battle. All Non Commissioned Officers and men should therefore know not only what they must do themselves but should understand the higher Commander's intentions. Good briefing will result in men going into battle with great confidence and higher morale because they will know the ground they are to go over, understand the purpose of their movements and appreciate what is going on around them. If there is time, the platoon commander should brief his men before every operation unless the company commander himself does this. Briefing should follow the following sequence:-

a. Ground.

b. Situation.

c. General Plan.

8. Men should be briefed with the aid of a rough sketch or a field sketch. The model must be realistic in scale and should show only essential features and details. The way in which the briefing officer brings the model to life is important. He should stick to those details, which directly affect the way his men must act to carry-out their task. He must be brief to hold their attention throughout and he should exclude executive details, which must be given to the O group, except when time is so short that the platoon has to move off on its task almost at once. Collective briefing is no substitute for normal orders but it should cut down the time spent on the 'Situation" paragraph.



(Not to Scale)

Figure 1 – Sketch of the Area of Operation

CHAPTER 5

BATTLE CRAFT

SECTION 18 **– GENERAL**

Theory of Fire and Movement

1. The primary aim of the infantry is to close with the enemy and kill him; to do this, infantry must move. They may be able to move under cover of the ground but the enemy will usually select positions which, as far as possible, give no ground cover to the attackers**.** The induction of Night Observation Devices

(NODs) and Ground Surveillance Radars (GSRs) have increased the enemy observation capability manifold. It follows therefore, that attacking infantry must be made cautious of the effects of NODs and GSRs. They must know that environment at night would be almost similar to that of day. Any abnormal movement/activity will be picked up by the enemy from a far distance. The attacking infantry therefore must know the characteristics of enemy's observation gadgetry and make efficient use of their fire and movement to make the enemy keep his head down and to make their own advance possible.

Individual Training

2. This combination of fire and movement is the basis of platoon and section tactics. It calls for a high standard of individual training, especially in:-

a. Physical fitness.

b. Characteristics and operation of NODs.

c. Marksmanship even with NODs Sights.

d. Personal camouflage and concealment.

e. Weapon handling.

f. Moving with rifles and LMGs.

g. Firing from cover.

Teamwork

3. Fire and movement is, above all, a matter of teamwork, which results from good leadership, a thorough knowledge of the platoon weapons and their application to ground.

SECTION 19 - **SECTION FORMATIONS**

General

1. Section formations are used to facilitate fire and movement and to retain control of command over his section. It will depend on a number of factors discussed in subsequent paragraphs.

Formations

2. There are five basic section formations. A brief explanation and a diagram is given with each formation. Symbols used in the diagrams are:-

(R)	----	Rifleman
(L1)	----	LMG Number 1
(L2)	----	LMG Number 2
(SC)	----	Section Commander
(S2)	----	Section 2IC

3. **Single File**. This formation is used for moving over restricted terrain or when visibility is so reduced that control becomes extremely difficult. It is useful for control during the early stages of a night attack and for following linear features such as hedges, ditches and walls. In Figure 3 the section commander is shown in the third position but he can also be at the head of the section if necessity demands.

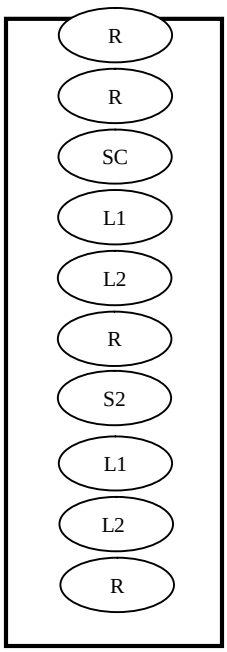


Figure 2 - Single File Formation

4. **File**. This is the primary formation for movement and helps the section commanders to control their sections in close country. In this formation the section can deliver a large volume of fire to the flanks but only a limited amount to the front. It also presents a concentrated target to the enemy. See Figure 3.

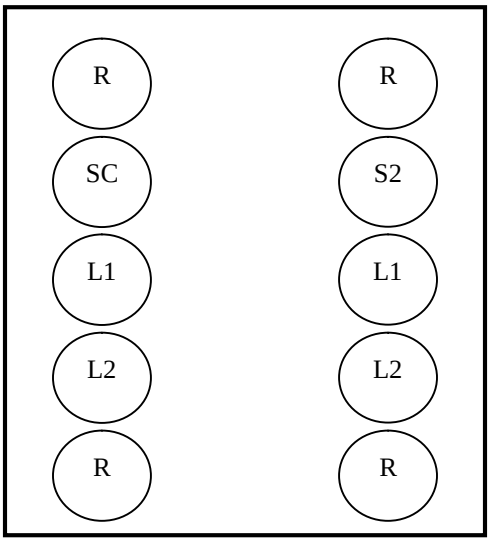


Figure 3 - File Formation

5. **Arrowhead**. This is a good formation for crossing open country and is not vulnerable to frontal fire. See Figure 4.

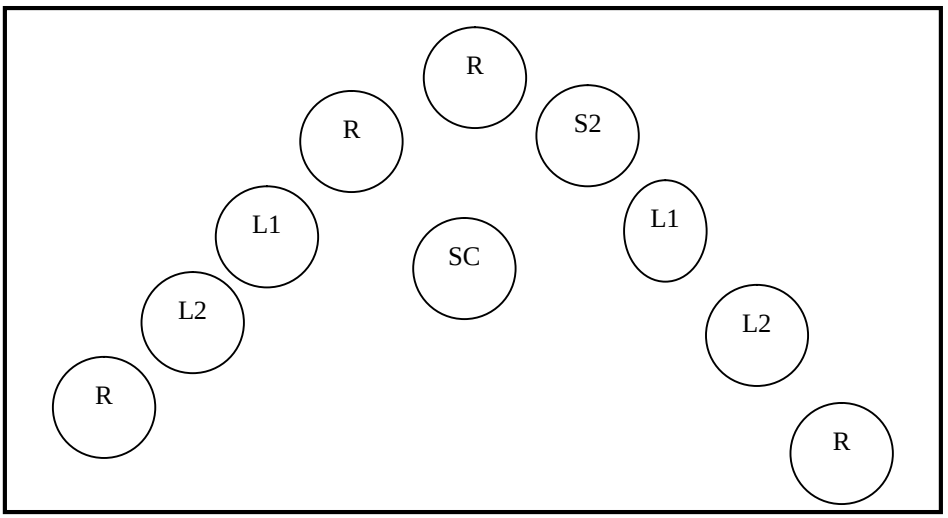


Figure 4 – Arrowhead

6. **Diamond**. This formation is good for control and is not vulnerable to enfilade fire. It gives all-round observation and protection. Each individual in the diamond formation must be able to see the next man. See Figure 5.

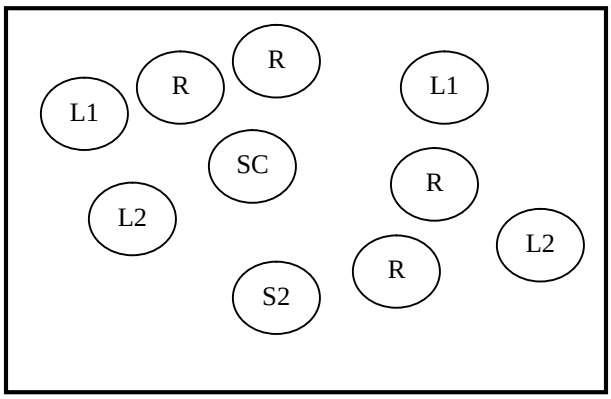


Figure 5 - Diamond Formation

7. **Extended Line**. This formation provides for the delivery of maximum fire to the front. It should be slightly broken in groups so that it is less vulnerable to fire from the flanks and group commanders still retain control over their respective groups. See Figure 6.

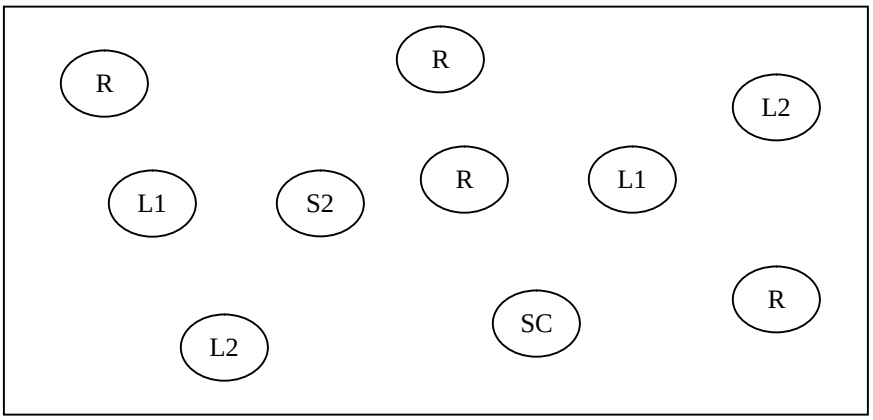


Figure 6 - Extended Line Formation

8. These formations can be varied to suit different conditions and ground. The section commander should be appreciating the ground all the time and must alter his formations, as often as necessary to conform.

Choice of Formation

9. Sections will deploy when within range of enemy small arms fire. The formation adopted will depend on:-

a. Ground.

b. Visibility.

c. Night vision capability of the enemy.

d. Direction from which enemy fire is expected.

e. The need for control by the section commander.

f. The need for producing the maximum fire with minimum delay.

Intervals

10. It is not possible to lay down the interval between soldiers since this will depend on the ground but as a general rule, the dispersion of the whole section on the battlefield should be up to the limit of control. As a guide, in open country the interval between soldiers should not be less than five metres by day. When under fire, troops tend to bunch together instinctively and this must not be allowed. Distances between groups cannot be laid down except in the actual assault, both groups must be within voice control of the section commanders.

SECTION 20 - **PLATOON FORMATIONS**

1. There are three tactical formations that a platoon can adopt in action. These are:-

a. One Up ----- See Figurer 7.

b. Two Up ----- a reversal of “One Up” formation.

c. Three Up ----- when all the three sections are moving abreast.

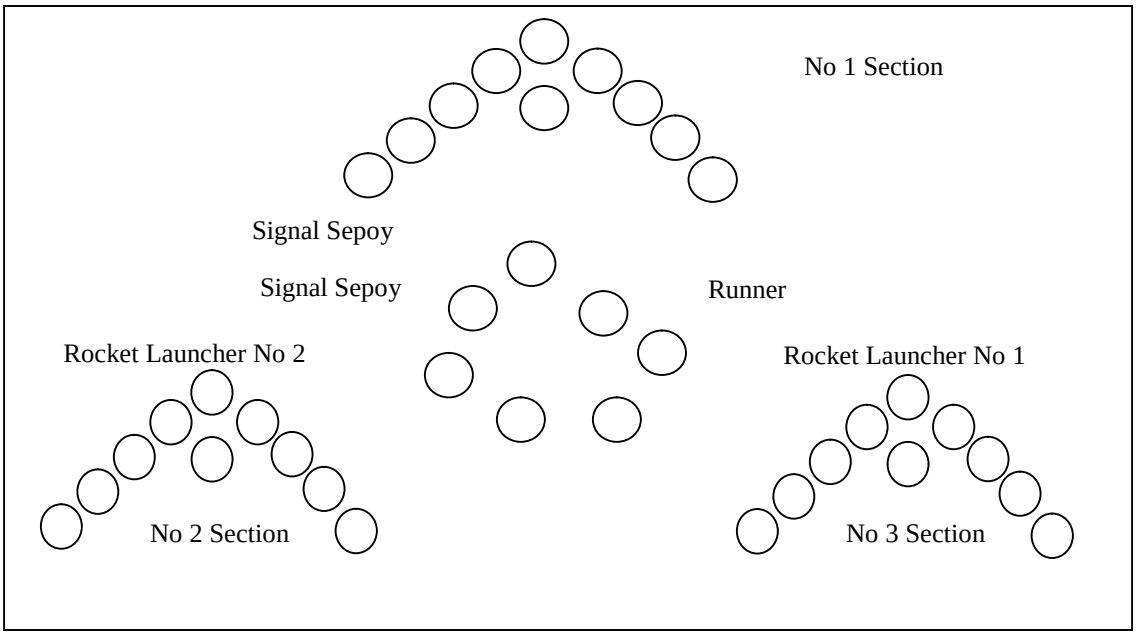


Figure 7 - Platoon in One Up Formation.

2. As to which of these formations should a platoon adopt will depend upon the need for control on the ground and the degree to which the platoon commander will require to manoeuvre. In an advance against an unidentified enemy position it is best to adopt “One Up” formation because it enables the platoon commander to keep the bulk of his platoon back for manoeuvre against the enemy when his position is located. On the other hand, the “Two Up” formation is more suited to deal with an enemy position already located because it facilitates fire and movement. However, it must be clearly understood that once contact is gained with the enemy, no matter what formation is adopted by the platoon initially, it will vary at different stages of the approach towards the enemy as the platoon carries out fire and movement.

3. Position of men in sections is as in Section 19.

SECTION 21 - **COMPANY FORMATIONS**

1. There are three tactical formations that a Company may adopt in action. These are:-

a. One Platoon Up --- When one platoon is moving ahead followed by the remainder of the company.

b. Two Platoons Up --- When two platoons are moving abreast followed by company headquarters and the third platoon.

c. Three Platoons Up --- When all the three platoons are moving abreast.

2. The formation adopted by the company commander will depend on the following:-

a. Need for control.

b. Enemy information and the degree of opposition.

c. Degree of search.

d. Terrain.

e. Speed of movement.

f. Composition of the company with attachment and detachment.

g. Enemy's night vision capability.

SECTION 22 - **FIELD SIGNALS**

General

1. Amidst the noise of battle when subunits are deployed (or when traveling in lorries or other vehicles) voice control will be impossible. A system of simple signals is therefore essential. (When on foot) these signals will be preceded by a short blast on the whistle to help attract the attention of those to whom the signal is addressed. Similarly under Nuclear, Biological and Chemical (NBC) environments, there would be a dire need to give an early warning to own troops

about such attacks. Therefore certain NBC Alarm Signals are also necessary to be practiced at all levels.

Signals with the Hand

2. The following are the standard signals (Annex "D"):-

   a. *Deploy.* Either of the arm fully extended over the head and waved slowly from side to side; the hand to be open and to come down as low as the hips on both sides of the body. If deployment to either flank is wanted, the commander will point to the flank concerned after completing the signal.

   b. *Advance or Follow Me*. The arm is swung from rear to front below the shoulder.

   c. *Halt*. The arm is raised, palm to the front, straight above the shoulder.

   d. *Go Back or Turn Round*. The arm is circled above the head.

   e. *Close or Join Me*. The hand is placed on top of the head, the elbow to be square to the right or the left according to which hand is used. If it is required to close on a flank the leader will point in the required direction before dropping his hand.

   f. *Quick Time*. The hand raised in line with the shoulder with the elbow bent and the arm extended and closed a few times from elbow joint.

   g. *Change Direction (Right or left)*. The arm is first extended in line with the shoulder. A circular movement is then made, on completion of which the arm and body should point in the required direction. When troops are halted the above signal means change position (right or left).

   h. *Incline or Turn (Right or Left).* The body is turned in the required direction and both arms extended in line with the shoulder and pointing in the required direction.

   j. *Take Ground (Right or Left).* Hand brought to the shoulder with the fist clenched, and the arm extended sharply in the required direction two or three times. May also be used to get MT vehicles off road, when practicable, to avoid enemy aircraft attacks.

   k. *Last Order Completed.* The salute, followed by the hand raised vertically above the head, hand open and fingers together.

l. *Indicate Enemy's Presence.* The arm fully extended towards the side of enemy's presence; parallel to the shoulder and hand's thumb pointing towards ground.

m. *All OK Report*. The hand raised in line with the shoulder with bent elbow, thumb pointed upwards while fist closed.

n. *Indicate Obstacle*. Both hands taken above the head and a cross made; the fists of both hands to be closed.

o. *Show Direction of Attack.* Indicate with the help of left or right hook of a boxer and then point towards the target area; left or right hook will also indicate the flank to be attacked.

p. *Indicate Bound/RV/Firm Base*. Raise fists of both hands above the head while keeping these over each other.

q. *Search the Area.* Fist taken to one of the eyes and then the area to be searched indicated with the help of open hand.

r. *Lay Ambush.* Take one hand closer to the face with all five fingers apart.

s. *Throw Grenade*. Show as lobbing a grenade with empty hands.

t. *Break or Rest*. Keep both hands on waistline with close fists and give forward and backward movement to the arms from elbows.

u. *Showing Commander's Rank.* Indicate officer or JCO by keeping hand on the shoulder and hold right hand wrist in case of a Company Havildar Major. For a section commander, show three stripes on the right arm with left hand.

v. *Indicate O Group*. Make a circle with the help of thumb and one finger and raise the hand in line with shoulder.

w. *Double or Increase Speed (MT).* The clenched hand moved up and down on either side of the body between the thigh and shoulder.

x. *Start Up (MT).* Circular movement of the hand as if starting an engine.

y. *Mount (MT).* Two or three slight movements of the hand upward with the hand (palm upper most).

z. *Slow Down (MT).* The arm extended to the side level with the shoulder, palm downwards and hand moved slowly up and down with the wrist loose.

aa. *Lie Down or Dismount (MT).* Arm taken away from body; two or three slight movements with the open hand and towards the ground while keeping the palm downwards.

bb. *As You Were, or Switch Off (MT).* The arms extended downwards with the hand open, and waved across the body parallel to the ground.

cc. *Form Line (MT).* The arm waved horizontally from right to left and back again as though cutting with a sword, finishing with the delivery of a point to the front.

dd. *Form Close Column (MT).* The hand raised perpendicularly above the head and lowered and raised several times.

Signals with the Rifle

3. These signals may be used by scouts, sent ahead of their sections. Care should be taken that the signal cannot be seen by the enemy. The following are standard signals made with the rifle (Annex "E"):-

a. *Enemy in Sight in Small Numbers.* The rifle held above the head at the full extend of the arm and parallel with the ground, muzzle pointing to the direction of the enemy.

b. *Enemy in Sight in Large Numbers.* The rifle held as in the previous signal, but raised and lowered frequently.

c. *No Enemy in Sight*. The rifle held up to the full extent of the arm, muzzle uppermost.

d. *Enemy Centre of Resistance Holding up Advance.* (Indication to cooperating tanks). Place headgear on a rifle and point in the direction from which fire is coming.

Signals with the Whistle

4. The following whistle blasts are used:-

a. *The Cautionary Blast (a short blast).* To draw attention to a signal or order about to be given.

b. *The Alarm Blast (a succession of alternate long and short blasts).* To turn out troops from camp or bivouacs to fall-in, or to occupy previously arranged positions.

c. *Enemy Aircraft in Sight (a succession of short blasts).* Since this signal will often be inaudible, a visual signal will also be used to attract attention, i.e., both arms held above the head and the hands waved. On this signal, troops either get ready to fire, open out or take cover, according to the orders in force.

d. *Enemy Aircraft Attack Ended (two long blasts repeated at intervals of five seconds).* On receipt of this signal all troops resume previous formations. Troops which have been firing will recharge their magazines before moving off.

e. *Start up and Mount or Switch Off* (one long blast).

f. *Dismount* (two short blasts).

Gas Alarm Signals

5. Theses signals are made either with a wooden rattle or with a hand operated siren. These are:-

a. *Local Alarm*. Gas sentries give a local alarm whenever NBC agents are discovered. Any suitable device may be used which produces a sound not easily confused with other encountered in combat. The improvised devices could be wooden rattlers, bells, metal triangles, iron pipes and empty shell cases et cetera. A succession of short blasts with whistle may also be blown for local alarm.

b. *General Alarm*. A continuous blast on the hand operated siren will indicate that a NBC attack (other than with aircraft spray) of sufficient intensity to affect those outside the hearing of the local alarms, has taken place.

c. *Spray Alarm*. A series of short blasts on the general alarm siren will indicate the presence of aircraft spray.

d. *All clear*. Three long blasts on the whistle repeated at intervals of about five seconds, as for “air attack ended”. This indicates that the area is free of gas, and all those within the area (for whom the alarm is sounded) may test for gas, and if satisfied that gas is not present, may remove their respirators.

6. The following are special gas alarm signals for MT on the move:-

a. *Gas Alarm*. A motor cyclist, wearing his anti-gas respirator, will ride down the column, sounding his horn. At night he will ride

down the column and will flash his torch on to his respirators to draw attention. Drivers will halt and respirators will be put on. The column will not resume its march until ordered.

b. *All Clear*. A motor cyclist will ride down the column wearing his respirator in the alert position. If on the move, the column will halt and respirators will be placed in the alert position. The column will not resume its march until ordered.

SECTION 23 – **GROUND AND COVER**

1. In open country, the problem is how to find cover, in close country, there is the difficulty of finding positions with good observation and fields of fire. In Attack or defence, the skillful use of ground can help to gain surprise and save lives. Much practice is, therefore needed in developing an eye for ground. Ground should be appreciated under the following headings:-

a. Fire position.

b. Observation positions.

c. Cover from fire.

d. Cover from view.

e. Cover from Night Vision Devices (NVDs) and Surveillance Radars.

f. Obstacles.

g. Approaches.

Types of Cover

2. Cover from view is often not cover from fire, especially if the move to cover has been seen by the enemy. Concealment from enemy air and ground observation including NVDs is the essential for primary means of gaining surprise. Some of the main types of cover are:-

a. Undulating ground is the least obvious form of cover; when skillfully used, it protects from direct fire and gives no ranging mark to the enemy. It also provides adequate cover from NVDs.

b. Beds of streams and ditches give good cover from view and often from fire as well. However, there is always a danger that the

enemy may pay special attention to them; they may be mined or booby-trapped and precautions against ambush must be taken.

c. Bushes give cover from view but not from fire. In open country they may make good ranging marks for the enemy.

d. Standing crops give cover from view but movement through them can generally be detected.

e. Woods and thick vegetation give cover to men and vehicles from enemy air, ground observation and NVDs**.** They give some protection from small arms fire but HE bombs and shells will explode in the branches of trees and will cause heavy casualties unless troops are dug in and have overhead protection.

f. Buildings and walls afford concealment and protection from small arms fire and shell splinters. When isolated they make good ranging marks for the enemy.

Dead Ground

3. Ground, which a soldier cannot see from his position, is called dead ground. Platoon and section commanders should be able to recognize ground, which is likely to be dead to the enemy. Ground can only be described as dead in relation to the position of an observer as in the example. “The track junction is dead to the enemy present in the farm house”. Troops under cover or in dead ground are safe from enemy observed fire but not from indirect fire. These areas are always likely to be covered by listening posts, standing patrols or selected by the enemy as defensive fire tasks for his artillery and mortars.

Common Mistakes

4. The wrong use of ground may lead to casualties and loss of surprise; some common mistakes are:-

a. Carelessness by officers, JCOs and NCOs while making a reconnaissance, such as unfolding a map in the open or not using a covered approach to an Observation Post.

b. Unnecessary movement in a position overlooked by the enemy.

c. Using conspicuous landmarks such as isolated trees, bushes or houses.

d. Halting troops near road or track junctions or other mapped features, which are always registered as, targets by the enemy.

e. Bad track discipline.

f. Failure to guard against enemy's NVDs/NODs, GSRs and air observation.

Maps and Air Photographs

5. Maps and photographs should be used together to obtain the best picture of ground. The two aids are complementary as is shown by listing the advantages and limitations of air photographs:-

a. *Advantages*

(1) Are more up to date.

(2) Give more details.

(3) Show the size and shape of features accurately.

(4) Allow gradients to be seen in relief with a stereoscope.

b. *Limitations*

(1) Complete geographical cover almost impossible.

(2) Expensive to produce.

(3) Scales vary.

(4) Details of heights not given.

6. Only the topographical information given by air photographs needs to be understood. The interpretation of the details of enemy defences is the task of the experts. Very little time may be spent in mastering the theoretical knowledge of map reading but a great deal of practice is required. Subunit commanders must also master the use of the prismatic compass and the protector. Navigation is a science and never a guess. A junior leader must have complete trust in his compass; this only comes with practice.

SECTION 24 - **SELECTION OF LINES OF ADVANCE**

General

1. The best lines of advance are those which provide the best positions for observation and fire on the enemy, and fire throughout. These two requirements usually conflict. Good fire positions and Observation Posts are normally on high

ground. Covered lines of approach are usually on low ground. A further difficulty is that it may not be easy to keep direction. These factors must be considered whenever a line of advance is being selected. In general, it is better to cross open country early in the advance if this is possible, and to use the covered approaches near the enemy positions.

2. The selection of lines of advance of a platoon is largely a matter of map reading. The best method of doing this is to follow the keyword "Ground":-

G Ground

R Ridges

O Observation

U Undergrowth

N Non-passable

D Defilade

3. After deciding on the area to be studied, which must be considerably wider than the area to be occupied or crossed, apply the headings as follows:-

G *Ground*. Get a rough idea of the characteristics of the area. Is it high rolling-down land, flat, low lying fields, open or close country?

R *Ridges.* Take a chinagraph pencil (any other colour except blue) and mark on the talc with a single line the highest part of all ridges and spurs (i.e., the watersheds).

O *Observation*. Mark any particularly good viewpoints or detached features with a circle.

U *Undergrowth or cover*. Study the location of villages, woods, belts of trees, scrub or rough ground.

N *Non-passable*. Study the obstacles, woods, villages, rivers, canals and railway lines.

D *Defilade*. The covered lines of approach to any point and the areas that afford cover will now be easily picked out.

4. A clear picture of the ground as a whole, its possibilities and its disadvantages will now have been formed in the mind. Clean the talc and make up your plan.

5. The study of the ground carried out by such means may often give the probable enemy positions and action as well as our own. It will be easy to remember and read the ground from the contours without the markings. A clean talc will be required on which to record the plan.

Reconnaissance for Lines of Advance

6. A man making a reconnaissance must know what he is looking for and how to look for it. His reconnaissance must be confined to essentials and must be made from the nearest point from which these essentials can be seen. Reconnaissance for lines of advance can be divided into:-

a. An endeavor to locate the enemy positions.

b. A decision on the line of advance it is intended to follow.

7. It is very often unwise to decide on the whole of a line of advance from the original reconnaissance point. The ground may be quite different when it is reached and fresh decisions may be necessary at each bound. The NCO required to lead his section round a flank may merely be told the general direction he is to take. He should be trained to use ground himself and he must be relied upon to use his own skill and judgment. It is fatal to detail a lengthy line of advance to a NCO. It is unlikely to be the best line, and unforeseen circumstances may soon require a deviation from it.

SECTION 25 - **DIRECTION KEEPING**

1. It is often difficult to keep direction, particularly at night, in fog or in close country. To make a detour to obtain concealment or to avoid an obstacle is liable to throw leaders off the correct line of advance.

2. The following are some of the means of keeping direction:-

a. The use of compass, map, Global Positioning System (GPS) and air photographs.

b. The use of a rough sketch copied from the map or air photo.

c. By keeping two prominent objects in view.

d. By using a series of easily recognisable landmarks each visible from the previous one.

e. By using the stars; the sun and the moon can also be used if their movement is thoroughly understood.

f. By memorizing the route from the map (or air photo). Points such as distances, contours and the direction of streams will all help.

g. The back view should be remembered. Patrols and others who may have to find their way back should look back at regular intervals to note landmarks behind them for their return journey.

h. In exposed districts, trees tend to grow away from the direction of the prevailing wind. Moss may grow on the leeward side of the trunk.

j. Direction marks may sometimes be left on an outward journey (pegs, small piles of stones or the blazing of trees).

k. If the route is being walked by day with a view to guiding by night, men must take particular notice of skylines and objects that they will be able to recognize by night.

l. All NCOs should be able to tell the points of the compass by use of the hands, watch and the sun.

m. In desert, knowledge of the layout of dunes will help in keeping direction.

SECTION 26 – **MOVEMENT**

General

1. Movement in the face of the enemy should be covered by fire. This does not mean, however, that it is impossible to move unless a heavy weight of fire is coming down on the enemy. The most important part of an attack is the movement towards the objective. Supporting fire is one of the aids to that movement. A knowledge of how to move and how to use ground for movement is essential to enable troops to close with the enemy with minimum casualties, especially under enemy's enhanced night vision / surveillance capabilities.

2. Normally troops in action will move at walking pace. In the attack, however, they will work their way forward by fire and movement in short bounds,

doubling, walking or crawling between bounds as dictated by ground and effectiveness of enemy small arms automatic and artillery fire. It must be remembered, however, that doubling and crawling are tiring and should be reserved for critical situations.

3. When it is necessary to cross an open space, such as a gap in a hedgerow, it is best for the whole section or group to double across it together before the enemy has time to fire. When wider gaps are under observation, it may be necessary to filter men across by crawling in ones or twos at irregular intervals.

Movement by Night

4. Darkness gives protection against aimed fire and enables surprise to be gained. However with the induction of NVDs and GSRs, these advantages have been considerably reduced**.** Movement by night therefore requires considerable practice, particularly in the crossing of obstacles.

5. **Differences Between Night and Day Operations**. The main remaining differences between night and day operations are that during night:-

a. Enemy can bring in aimed fire by using NODs/NVDs fitted on weapons. Machine-guns can fire on fixed lines.

b. Sight is largely replaced by hearing; the avoidance of noise is therefore of the greatest importance and ears should be used rather than eyes.

c. Objects are generally visible only on skylines; this increases the importance of low ground both for movement and observation. However, use of NODs/NVDs will mean a cautious movement in the low ground as well.

d. The difficulty of maintaining direction makes previous daylight reconnaissance important.

e. Control is difficult. Close formations are therefore very necessary. However, dispersion at night is equally important as by day, due to presence of NODs/NVDs.

6. Particular attention should be paid to the following points at night:-

a. Equipment should be tested beforehand, to see that it does not rattle; empty sandbags wrapped round equipment and tools will check empty sound.

b. Orders and messages should be given in a whisper just loud enough for them to be heard by the recipient. A simple system of signals should be arranged.

c. Particular care must be taken in crossing gaps where movement may show against the skyline.

d. The stars, the moon and landmarks silhouetted against the sky should be used to keep direction.

e. Commanders must be well forward for control of pace and direction. The company/platoon commander should always be in the lead at night. They must practice using a compass at night.

SECTION 27 - **SELECTION OF FIRE POSITIONS**

1. The selection of fire position calls for protection against NBC attacks and knowledge both of the characteristics of weapons and the use of ground. The requirements of fire positions vary with the task; for example, in the attack, it should be easy to advance from the position; in defence, concealment and a covered approach may be more important. The section commander must be looking all the time for a position where his section can take cover when it comes under effective enemy fire or NBC attacks.

2. To select the best fire position calls for imagination. Sometimes it may be necessary to use trees, walls or rooftops to produce any fire effect at all. This may result in plunging fire but this disadvantage can be overcome by accurate shooting. Cleaverly concealed positions will surprise the enemy with their fire and protect the occupants from enemy observed fire.

SECTION 28 - **FIRE CONTROL**

1. In battle, fire control is the task of the section commander and his Second-in-Command. Without fire control, much of the value of good fire position will be lost. To control his section’s fire and obtain the best fire effect, the section commander must know:-

a. How to locate and indicate targets?

b. How to estimate range accurately?

c. What weapon to use?

d. What type of fire to order?

e. How to position himself so as to control the fire of his section? He must keep a constant watch on the ammunition supply and maintain an adequate reserve.

2. The simple aim of a fire order is to bring fire on to the enemy as quickly and frequently as possible. The most difficult part of any fire order is usually the indication of the targets. This is particularly true in the attack; in defence, every man in the section knows and is used to the ground around him and reference points and relevant ranges will have been given verbally.

3. In the attack or advance, the ground is usually unknown and sections are often on the move; ranges have not been given; the enemy may be dug-in and well concealed and so targets may be hard to pick out. Too often in training, everyone assumes that the section commander will be the first to see the enemy and at once gives his fire orders. In war, however, it is the soldier who first sees the enemy, as often as not. Soldiers can rarely be expected to give a clear verbal indication quickly so tracer may have to be used. Tracer is a valuable means of indicating targets in the attack.

4. It is most important that every soldier realizes the necessity of fire control so as not to waste ammunition. Firing for morale is wrong. In a fire and movement operation, it may be necessary to fire without the enemy being seen but this fire must only be minimum necessary to make the enemy keep their heads down and to maintain fire domination over them. There is a clear distinction between fire control in the advance and in defence.

a. During an advance, an individual will open fire from the shoulder or the hip on enemy within range.

b. In defence, however, no one will open fire except on the specific order of his subunit commander. This order must not be given too soon otherwise the position will be compromised to the enemy and many hours of hard work spent in concealing it will be squandered in a few seconds.

SECTION 29 – **SMOKE**

1. Smoke is used to screen movement from enemy observation or aimed fire. It provides effective cover from all types of NODs/NVDs except belonging to thermal imaging system. It must therefore be used to guard against effects of night

vision**.** It may also be used to distract the enemy's attention. For all practical purposes, smoke will only be used in the attack because it favours the attackers and hampers the defence.

2. At platoon level, smoke can only be produced by smoke grenades which lasts for very short periods. If required for long periods, assistance from the company weapons section will be asked.

3. Smoke is a valuable weapon but it has certain limitations which must be understood:-

a. Weather conditions must be suitable. It is virtually impossible to lay an effective smoke screen in a high wind. A wind blowing at right angles to the smoke screen will also enhance its duration and effectiveness. When there is no wind at all, a great deal of ammunition may be needed.

b. Smoke affects other supporting fire; it usually makes aimed fire support impossible.

c. The amount of smoke ammunition carried is comparatively small.

CHAPTER 6
PROTECTION

SECTION 30 **– GENERAL**

1. Every Commander is responsible for the protection of his men. By protection is meant the steps, which a commander takes to safeguard his command against being surprised and to conceal his positions from the enemy.

2. No unit is secure unless it is protected in all directions from which attack may come, whether from the front, the flanks, the rear or the air. Even when it is some way from the enemy with other troops in front or on the flanks, a unit is liable to be attacked. It should, therefore always be ready to fight and protect itself. Furthermore, security precautions against enemy agents, saboteurs and fifth columnist must always be maintained.

3. This chapter deals with the following forms of protection:-

a. Protection at rest.

b. Protection on the move.

c. Protection against air attack.

d. Protection against Nuclear attack.

SECTION 31 - **PROTECTION AT REST**

1. Whenever a company is halted for any length of time or is in a rest area or defensive position, alarm posts must be allotted. They will be sited to meet ground threat from any direction and, if possible, be concealed from air observation. In a rest area, shallow trenches should be dug, particularly if the stay is likely to be prolonged, to obtain protection from splinters of aerial bombs or shells. All ranks must know where their alarm posts are and practice alarms must be carried out to ensure this. Every man must carry his weapon and ammunition at all times.

Sentries

2. Sentries will always be posted for the local protection of anybody of troops to give early warning of enemy movement or attack and to check the identity of visitors or of suspicious persons loitering nearby.

3. A JCO or NCO will post sentries and make sure that they know and understand their orders. They will be posted to cover the most likely approaches and must be in contact with the NCO on duty.

Orders for Sentries

4. The security of the force depends on the alertness of its sentries. Their orders must, therefore, be carefully thought out so that nothing is omitted, left to chance or liable to misunderstanding. The orders should be given clearly. A section commander will normally brief his whole section at one time.

5. Following should be known to the sentries:-

a. *All Sentries*

(1) What ground to watch?

(2) What to do if anyone approaches?

(3) The password.

b. *Sentries in the Forward Area*

(1) The direction of the enemy.

(2) Where their own neighbouring posts are.

(3) The name of landmarks in front of them.

(4) The patrols that may come through or anywhere near their posts.

(5) The signal for them to fire the LMG laid on a fixed line if they are sentries on LMG.

Challenging

6. The standard challenging procedure will always be followed; failure to do so may result in casualties being inflicted on our own troops, particularly returning patrols. The section commander will always be alerted if anyone not expected approaches the position and, if the situation warrants, he will "stand to" the section; The sentry will do the challenging. The challenge will be given quickly and quietly at a range: which will guarantee the infliction of casualties if necessary but not close enough for the post to be rushed. If the order to halt is not obeyed, it is repeated and if still not obeyed fire must be opened immediately. The section commander must ensure that his whole arc remains covered and that all his men do not concentrate on one incident.

7. The correct challenging procedure is as follows:-

	Action by sentry	**Action by people challenged**
a.	Says "Halt—Hands up" just loud enough to be heard. '	Standstill and put up the hands.
b.	Says "Advance", or, if more than one person has approached, 'advance one".	One man moves forward to the sentry, still with his hands up
c.	Says "Halt" at a suitable distance from where he can identify by means of password.	Standstill, with his hands up.
d.	Gives the password, e.g. , "**SLING**".	Gives the counter password, e.g. "**GHARI**".

	Action by sentry	Action by people challenged
e.	"Pass friend".	The individual checked moves alone towards the sentry, where he is recognized by the section commander. Thereafter, he sits with the section commander/sentry and identifies his remaining colleagues. When all are identified they are allowed to proceed to intended direction.

Password

8. The use of the password must be the last means of identification when other methods have failed. The password will always be in two parts for example challenge BARISH, counter sign AADMI. Passwords are changed daily.

Length of Watches

9. Watches may be as short as half an hour but will not exceed two hours. The shorter watch may be used when a company is working in extreme climatic conditions. The longer watch is normal in completed positions to allow a worthwhile rest between watches. When platoon sentries arc formed by sections in turns, sections should not change responsibility too frequently otherwise there may be confusion over relief. When the section responsibility does change, the platoon commander or platoon havildar must supervise the relief.

Sentries in a Defensive Position

10. The number of sentries normally required by a platoon occupying or preparing a defensive position is:-

a. *Before contact*

(1) *By day*. Normally one ground/air sentry for the platoon but this is subject to local conditions and weather.

(2) *By night*. Two for the platoon. Reliefs of night sentries will be staggered to ensure that one is always fresher than the other. This method also ensures that at least one sentry is always adapted to darkness.

b. *In contact*

(1) *By day***.** One for each section.

(2) *By night*. Two for each section.

11. While the position is being dug, sentries must be sited far enough to enable them to listen unimpeded by the noise of digging but close enough to give warning easily. Sentries are required to give warning of the approach of hostile aircraft. The best way of giving the alarm in the platoon area is by whistle or, if silence is essential, by pulling on a line laid from the sentry to the main position. Roving sentries should not be used.

Section LMGs by Night

12. When in contact, section LMGs must always be manned at night. The two sentries posted in each section at night can do this. The section will thus be divided into three shifts each doing a tour of duty for three hours at a time. The section LMGs will nearly always be on a fixed line or fixed arc. If a LMG is covering the front of another section, the second sentry must watch the section's own arc. So as not to disclose its position, the LMGs should not fire bursts unless the position is rushed. The sentries should also be equipped with Night Vision Devices. Full use of Ground Surveillance Radars should be made.

Sentries in Rest Area

13. Sentries should be posted to cover the most likely approaches, particularly any tracks leading into or near to the rest area. When a sentry has to go into the open, for example to examine vehicles or personnel, he must be covered by another sentry who is concealed, Listening posts or may be laid on tracks leading to the area in place of normal sentries. Also:-

a. *By Day.* Sentries should be posted as far out from the force as possible to give early warning but not out of earshot nor out of sight if the country is close.

b. *By Night.* Sentries may be placed well out but they must have a quick, secure, silent and certain method of contacting their force.

Alarm Scheme

14. If firing begins or the alarm signal is given, every man should go to his alarm position. Thereafter, there must be no further movement until “stand down” is ordered. This means that anyone moving during the period of the alarm is, likely to be enemy. There will be no firing at night until the enemy is a certain target.

15. It is also pertinent to mention that the enemy may not attack physically but may send Remotely Piloted Vehicles, to get the information regarding own likely moves and intentions.

SECTION 32 - PR**OTECTION ON THE MOVE**

Marching

1. When a company is moving in an area where the enemy’s location is unknown and the ground is not completely open, scouts may be employed in front and on the flanks. They must not be allowed to mask the fire.

2. The aim of the scout is to see without being seen. He should use his rifle only in self-defence or in the defence of other men.

3. Scouts work in pairs. They move forward from one objective to another by bounds. When the objective for the next bound has been selected, the leading scout chooses his route to it and moves forward as rapidly as possible while the other remains in observation. When the first scout reaches his objective, he signals, the other to come forward. The process is then repeated. The reasons for this procedure are:-

a. The second scout can cover the advance of the leading scout with fire and help him withdraw if surprised by the enemy.

b. If the leading scout gets into difficulties, die other can call for assistance.

4. Objectives selected for bounds should give a good view ahead and be suitable for signalling back. The distance ahead that scouts move depends on the nature of the country. A short distance in open country is useless unless they can see more than the rest of the platoon. The platoon commander must maintain strict

control over the movement of his scouts so that he can halt them if necessary in order to close up before sending them ahead again.

5. If contact with the enemy is imminent or the scouts give warning of his presence, the platoon should deploy into normal battle formation with one or two sections up as point sections depending on the terrain and frontage required to be covered by the advance guard.

In Military Transport along Roads

6. Bad military transport move discipline may cause delay and congestion in a whole column making it vulnerable to air attack and shelling. Vehicles will normally move in groups of about four or five according to the size of subunits. They should either be same type or belong to the same subunit. Each group should be commanded by an officer/JCO/NCO with a commander detailed in each vehicle. Groups should travel apart if possible to make them a less worthwhile target to enemy aircraft.

7. The rules of Military Transport march discipline are:-

a. The vehicle commander is the senior rank in the vehicle. He is detailed before moving off and is responsible for the conduct of the driver and other passengers. He must travel in a position from where he can see out of the vehicle in every direction, including the rear. If he cannot see out to the rear, he must detail someone in the back of the vehicle to do this. The vehicle commander must follow the route on a map. If the vehicle in front is automatically followed, part of the group may get lost. At unexpected halts, he must get out and go forward to find out and, if possible, rectify the cause of the delay. If all vehicle commanders do this, the group will not be delayed behind one broken down vehicle.

b. The convoy density is the interval, which must be kept between vehicles. This will be laid down for any move in convoy to avoid vehicle bunching and also to prevent the column taking up too much road space**.**

c. *Maximum Speed.* The running speed laid down will be as fast as is thought safe and practicable. When in convoy this will be the speed of the slowest vehicle. There will also be a maximum speed to enable delayed vehicles to make up ground. This will usually not be more than about 8 km per hour greater than the running speed and must not be exceeded.

d. Double banking and overtaking must not be allowed except on the orders of the military police or of officers controlling traffic. Individual vehicles, which have dropped behind, must wait for the column to halt before regaining their positions in the convoy.

e. *Halts.* Times and duration of halts will be laid down for the move. Whenever possible, vehicles should move from harbour to harbour. These harbours, such as woods, will be pre-selected from maps, air photographs or knowledge of the route. Ideally they should provide cover off the road for halted vehicles. They do not have to be any special distance apart. If harbours are not available, vehicles should park clear of the road whenever possible and be camouflaged. Every precaution must be taken at halts to conceal vehicles from air observation. Sentries must be posted and traffic control men detailed where necessary.

f. *Hoods.* Hoods should be removed partially or wholly so as to allow fresh air to pass through the vehicle and also to increase observation of the air sentry.

g. *Breakdowns.* Vehicles which breakdown must be parked off the side of the road and display a signal. Usually a flag, to show other vehicles the cause of delay. If the breakdown is likely to be only temporary, a sentry must be posted on the opposite side of the road to act as traffic policeman and wave on the rest of the convoy. If the breakdown is permanent and the vehicle blocks the road, it must be pushed clear even at the risk of damage to it.

h. The common practice of tying down the LMG with angle irons of the vehicles should be curbed. This is not only ineffective but also is dangerous for own troops. Instead, the use of the pedestal mounts should be advocated.

SECTION 33 - **PROTECTION AGAINST AIR ATTACK**

General

1. Speed is an outstanding characteristic of air attack. Only a few seconds may elapse between the first appearance of the aircraft and the end of the attack. The effectiveness of the defence would therefore depend on:-

a. The speed at which warning of the attack can be given.

b. The time taken after the alarm is given to issue and execute orders for movement or fire.

c. The skill, steadiness and fire discipline of the troops.

2. Infantry protect themselves against enemy air attack by:-

a. Dispersion.

b. Concealment.

c. Prompt aggressive action.

Concealment and Formations

3. On the line of march, concealment may be difficult by day but casualties can be minimised by marching dispersed on either side of the road.

4. When deployed, it is best to adopt irregular formations in such small groups that the target, if observed, do not seem worth attacking. Troops should be in the shadow when possible. In the open they should lie down, keep still and not look up.

5. When occupying a defensive position, the careful sitting of posts and the use of available cover or scrim netting can achieve concealment from aircraft but not from the modern camera. The careful study of air photographs will show all newly turned soil and tracks even if made by only one man. This does not mean that the need for concealment is over, as it is still most important to conceal our positions from ground and visual air observation. Troops must use old tracks whenever possible. New tracks, if made, should be designed to fit in with existing tracks. The enemy may well find it difficult to tell if a position is actually occupied or not.

Use of Small Arms against Aircraft

6. Enemy aircraft will try to achieve surprise by flying low and concealing their approach. The speed of low flying aircraft makes them difficult to identify, but as enemy aircraft and their habits become known, troops must become proficient in rapid aircraft recognition. Effective observation is therefore most important. Air sentries should be posted in the most suitable positions to see the approach of enemy aircraft and yet be able to give prompt warning. They must listen as well as watch. Their duties are tiring and a strain on the eyes, so frequent relieves are necessary. Sentries must carry a whistle or other efficient warning signal.

Actions when Attacked

7. Small arms fire will normally be opened by LMG teams detailed for this role. Riflemen, who are caught in the open or can otherwise fire, should do so to keep the aircraft high. All other troops should be dispersed and must take cover immediately, preferably below ground and remain so until the “All Clear” signal is given.

8. The number of the air sentries to be detailed depends on the ground, but should not be less than three for each company. The direction area, which each sentry is to watch, must be carefully coordinated. Sentries must continually scan the sky, especially in the direction of sun, towards low hills, woods, et cetera; they must also listen for the approach of aircraft, even during a short halt of a unit, if time permits, sentries should be posted on high ground to get a clear view of the horizon. The duties of the sentries are exhausting and entail great strain on the eyes. Arrangements, therefore, must be made for constant relieves.

9. As recognition of low level aircraft and inadequacy of local warning system continue to be the handicaps, the right time to engage a hostile aircraft is when its hostility has been positively confirmed and when it is within effective range of the weapons being used.

10. **Tactics and Techniques Employed by Enemy Aircraft**. In order to determine hostility, soldiers must understand the tactics and techniques employed by enemy aircraft. An aircraft engages a ground target in two different ways:-

a. *Stand Off Technique.* The latest generations of fighter aircraft are capable of attacking without physically going on to the objective area. This can be done by firing precision weapons from a long distance (out of the small arms range). Efforts should be made to utilise the long-range weapons like ANZA to engage such targets.

b. *Dive-in Technique*

(1) *Approach.* Enemy aircraft operating in reconnaissance and ground attack, both roles, will usually fly at low level and at high speeds to avoid detection and interception while approaching the target area. In a typical ground attack, the hostile aircraft may, on arrival on the target area, continue flying at low level as if to pass to a flank of the target, maintaining the course until reaching a pre-arranged pull-up point.

(2) *Delivery.* On arrival at the pull-up point the aircraft will climb steeply .From the top of the climb, with the target still to its flank, the aircraft will "turn in" for the attack usually about 90° in plane view and will make a dive for the target. The position of the point of weapon release will vary with the speed of the aircraft, the angle of dive, the height of attack and the weapon used.

11. **Types of Attack**

a. *Reconnaissance.* Fighter reconnaissance is generally carried out at speeds up to Mach 1 and at heights of 15 metres and above. Targets on the ground recognised by the pilot, if important, may be attacked immediately, either by the reconnaissance aircraft or by an accompanying ground attack aircraft.

b. *Cannon.* Cannon attacks may consist of pull-up from a low-level approach to between 610 metres (2000 feet) and 915 metres (3000 feet) followed by a dive on the target. Angles of dive vary between 15^0 degrees and 30^0 degrees and speed at the time of firing is around 725 kilometres per hour.

c. *Rocket.* Rocket attacks are usually delivered after a pull-up to between 915 metres and 1220 metres from low level approach, followed by a dive on the target. Angles of dive vary from 20^0 to 40^0 degrees and the height of rocket release varies from about 180 metres to 245 metres above ground level. Base of the dive is around 150 metres and weapons release speed is around 725 km per hour.

d. *Napalm.* Napalm attacks are delivered at speeds around 480 km to 510 km per hour from heights below 60 metres. After initially spotting the target, a long, low and level approach is made to the target with no initial pull-up if the target is clearly visible.

e. *Dive Bombing.* Dive bombing attacks may be carried out by fighter ground attack or light bomber aircraft. A dive angle from 45^0 to 60^0 is called a steep dive, while a shallow dive varies between 30^0 and 45^0 degrees. Dives may be started at any altitude, but usually commence after a pull-up from a low-level approach to a height between 1525 metres and 2440 metres. Weapon release speed is around 800 km per hour but the base of dive varies up to 610 metres or so, depending upon the size and type of bomb.

12. From the above it will be seen that during rocket, cannon and napalm attack the hostile aircraft will be within range of small arms fire.

13. **Transport/Assault Aircraft and Helicopters**. More and more transport, assault aircraft and helicopters are being used for logistic support, visual, photographic or radar reconnaissance, troops assault and transport purposes. Their speed will be less than Mach 1. In the case of helicopters, some may also be equipped with cannons, rockets or air-to-surface missiles.

14. **Deployment of Small Arms in Anti-Aircraft Defence**

a. *Principles of Anti-Aircraft Defence by Small Arms.* The following are the main principles to be observed in the use of small arms in Anti-Aircraft role:-

(1) *Speed.* It is essential that fire is opened with utmost speed for successful engagement of an aircraft as it comes for a very brief period within the engagement range. This requires a system of warning the troops of the approaching hostile aircraft and laying down before-hand whether the troops are to open fire on their own initiative or not.

(2) *High Rate of Fire.* For the same reason as given in sub paragraph above, it is essential that the fire must be brought to bear on the target at maximum rate to increase the hit probability.

(3) *Maximum Number of Weapons.* Maximum fire of all available small arms within the range is necessary to make low flying attacks by enemy aircraft a costly venture.

(4) *Coordination and Control of Fire.* It is essential that the fire must be controlled and all weapons coordinated, otherwise, apart, from the wastage of ammunition the fire will not be effective. Weapons, which are out of range, should not open fire. Fire unit commanders must know whether they can open fire on their own or a higher authority would control it. In the latter case adequate arrangement for the passage of orders is essential.

b. *Early Warning*

(1) *Warning.* The targets in forward areas being invariably of small size and camouflaged, the enemy aircraft will have to manoeuvre in the area for sometime before attacking

them, thus providing enough warning time for the defence to be alerted by the OPs. On sighting an aircraft, OPs should inform all concerned by the fastest means/signals so that all weapons are manned.

(2) The efficiency of the defence will depend upon:-

(a) The speed with which warning of the attack can be conveyed to subordinate commanders.

(b) The promptness with which warning can be followed by executive orders to fire.

(c) The skill and fire discipline of the troops.

c. *Techniques of Deployment*. In infantry localities most of the weapons deployed in ground role are also given the secondary role of anti-aircraft defence. These are adequate to cover all approaches and to bring effective fire on the hostile aircraft. Forward localities should have a bias toward ground defence where as depth localities may have a bias towards anti-aircraft defence. Each MG/LMG must be given a specific arc of fire (primary arc). The arcs should not be greater than 100 degrees and should interlock within the effective range of the adjacent gun.

15. **Target Selection and Distribution of Fire**. Attack will usually be by several aircraft and it is desirable that:-

a. Aircraft which are attacking should be engaged first rather than those aircraft which have already made the attack.

b. Each machine-gun should give priority to a target which is in dive in the primary arc allotted.

c. Where aircraft are close together fire will be distributed to engage as many as possible.

d. All arms engage aircraft within their range irrespective of the primary arc and approach.

16. **Time of Engagement**. To avoid shooting at own aircraft by mistake, and giving away own position by premature firing, an aircraft may be engaged when it is involved in hostile act i.e. while diving and delivering weapons on to target, passing over the target and pulling up after the attack. It is also at this time that the aircraft would be within the small arms range. Slow moving aircraft may be engaged when the enemy markings are clearly recognised and are within range.

17. **Correcting Fire**. The fire will be corrected with the help of tracers.

18. **Fire Control**. Premature opening of fire may disclose the existence of a vulnerable point or vulnerable area. Fire therefore must be controlled. The control of fire should be kept under section commander.

19. **Engagement Restriction**. Ground attack aircraft will only be engaged when involved in attacking a position. Transport aircraft, light aircraft and helicopters will be engaged on orders from the company commander who will be responsible for establishing the identity of the aircraft before opening fire.

20. **Air Defence under Adverse Conditions**. If a defended locality which is underground attack, also comes under air attack, only the troops from the neighbouring localities not directly engaged in ground attack should engage the hostile aircraft. Troops under ground attack will fight back the attacking troops.

21. **Protection on Move**

a. Air sentries will be posted in every vehicle with an all-round view.

b. MGs/LMGs will be mounted in every group of vehicles.

c. A convoy, when attacked from the air, will disperse off the road simultaneously engaging the hostile aircraft with the MG mounted for this purpose. If going off the road is not possible the convoy will continue moving at a faster speed engaging the hostile aircraft. This will require development of good fixing techniques and extensive training on the part of the troops.

Logistic

22. **Likely Ammunition Expenditure in Anti-Aircraft Role**. The ammunition expenditure for anti-aircraft firing can be worked out on the basis of a locality coming under air attack and the engagement time. An aircraft attacking a battalion locality in defence will generally come under effective small arms fire of a company only. These weapons are expected to fire approximately the following number of rounds in one engagement of five seconds:-

a.	Per Rifle	- 20	These figures are for one
b.	Per MG1A3 (LMG Role)	- 100	engagement only. The ranges

c.	Per MG1A3 (MG Role)	- 100	of these weapons in anti-aircraft role are 500 metres.

Methods of Working out the Lead

23. **Standard Lead**. The engagement time being short, there is no time to work out the lead at the time of engagement. Standard leads for direct crosser aircraft should, therefore, be known to all concerned. Leads are worked out in terms of lengths of the aircraft. For example if the lead worked out from a particular aircraft is five all weapons should engage the aircraft five lengths ahead of it. Method of calculating the lead is given at Figure 8. No lead is applied when firing at a diving or receding aircraft if the line of aim or the motion of the aircraft coincides with the direction of the bullet's flight.

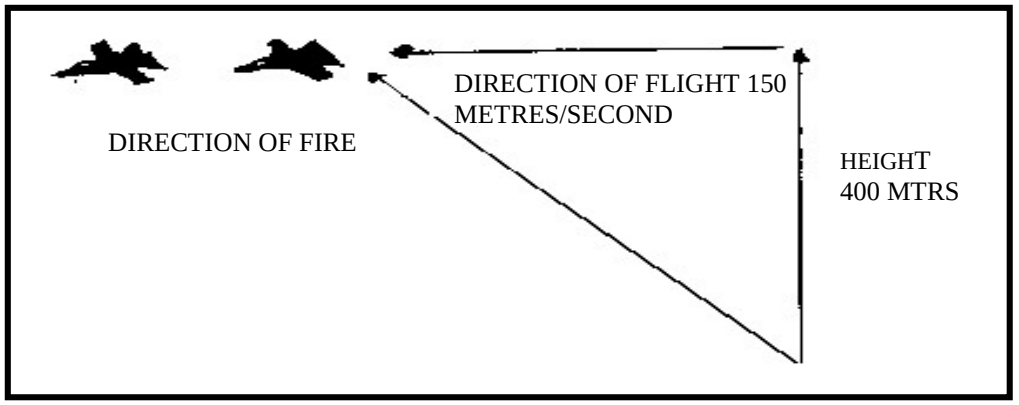


Figure 8 - Method for Working Out the Lead.

24. **Available Data**

a.	Speed of the aircraft	=	150 metres/second
b.	Altitude of the aircraft	=	400 metres (Distance from the ground)
c.	Length of the aircraft	=	15 metres
d.	Time of flight of the bullet	=	0.6 sec to cover 400 m

25. **Method of Working out**

a. The aircraft flying at the speed 50x0.6=100 metres of 150 m (500 feet)/ second will cover/(500X0.6=300 feet) in 0.6 seconds.

b. Length of the aircraft 15 metres (50 feet).

c. No of leads=90/15 (300/50) 6 aircraft.

26. Allowances for firing at various ranges are calculated before hand and summarised in a table to be memorized by the firer. Fire adjustment is effected by observing bullet tracers. To ensure a reliable engagement of targets, fire must be controlled in such a manner as to have tracers passing somewhat higher than and ahead of the target.

27. Firing methods are different for fast flying targets. In such cases it is impossible to fire with aiming allowance, since the allowance will have to be too large. Therefore barrage fire is used. A barrage fire is concentration of fire delivered vertically on a pre-selected spot over some reference point on the target flight route. The target should cross this point during its movement. Fire is conducted at maximum rate for 3-4 seconds. The probability of destruction of an air target with such a barrage is higher when ground fire is well organized and as massed as possible.

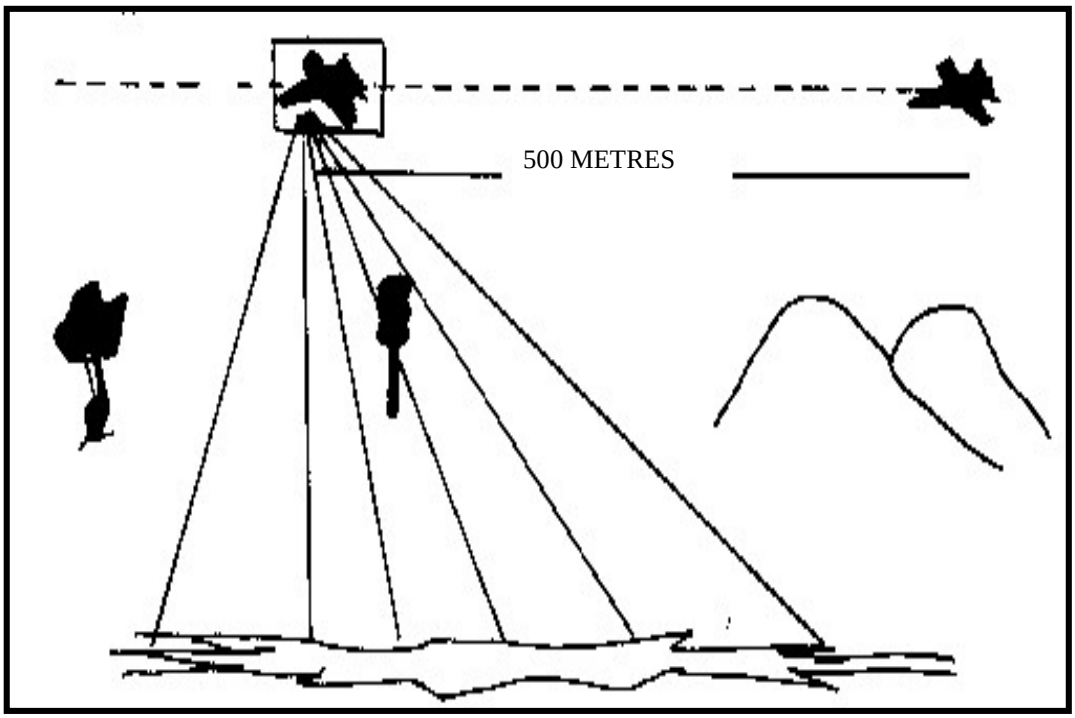


Figure 9 - Chart of Barrage Fire Over a Reference Point. Reference Points 1, 2& 3

Defence against Airborne Attack

28. **Method of Attack**. The sequence of a large scale airborne landing may be as under:-

a. A period of heavy air attacks with the aim of neutralising air defences, destroying morale and softening up the area of the dropping zone. This period may vary from several days to a single concentrated attack just prior to the landing.

b. The initial landing by paratroops preceded by path-finders if available.

c. The build-up by paratroops, gliders or air transported units or formations, which will reinforce and exploit the initial gains.

29. On the other hand, small parties of paratroops may be dropped surreptitiously to seize headquarters and defiles, cause confusion and disrupt communications. Such parties would remain in their dropping zone for a few minutes only as they would not require any heavy equipment.

30. Airborne units are most vulnerable when they are being dropped and before they have had time to collect their heavy equipment and organise into units and subunits. They must therefore, be attacked and destroyed before they have had time to reorganise themselves. It is the duty of all troops to take immediate action against enemy airborne troops, if necessary without orders, two or three sections arriving on the dropping zone within ten minutes of the drop, have a better chance of success than a battalion group arriving an hour later.

31. As it is not feasible to cover all likely dropping zones, the following special counter/measures will be taken:-

a. Early warning system.

b. *Observation Posts*. A good observation system will serve to check exaggerated and inaccurate reports.

c. Good signal communications to pass information.

d. Mobile columns, strong in automatic fire power, are held in reserve particularly in rear of the defence. Light armour if allotted is particularly suitable for employment in an anti-airborne role.

e. Close protection troops for the defence of vulnerable points. Care will have to be taken that troops are not dissipated over a large area.

f. Company should be capable of and prepared to defend itself.

g. Other passive measures as anti-airborne obstacles, camouflage, concealment and the subsidiary use of dummy installations.

32. The demoralizing effect of an airborne attack can greatly be reduced by prior preparations of plans and the means to counter it. Within the means available, all commanders will plan for the defence of their area.

33. **Reporting of Drops**. To assess accurately the size of the force dropped and its location, observers should count the number of transport aircraft (if possible, noting down the type of aircraft) and take a bearing of the place where the troops are being dropped. Thus, bearing from two or more observers will give the exact location of the dropping zone, whilst the number and type of transport aircraft will indicate the maximum number of troops that could have been dropped.

SECTION 34 – **PROTECTIVE MEASURES AGAINST NUCLEAR STRIKES**

General

1. Whether taking protection from friendly or enemy delivered weapons, the units must always balance their nuclear protective posture against the requirement to perform the assigned tasks. Formation and unit Standing Operating Procedures (SOPs) should lay down the protective measures required to be adopted in a nuclear environment under different tactical roles.

Protection

2. **Individual Protection**. Protective measures for the individual can be divided into precautionary measures, protection against the immediate effects of an explosion and protection against residual effects.

3. **Precautionary Measures**. At all times when a nuclear (and or chemical) threat exists, NBC protective clothing less the respirator should be worn. The respirator should be at hand ready for immediate donning if the NBC general warning or local alarm given. However, continuous wearing of full NBC clothing will slow down the activities of all troops particularly those unaccustomed to wearing it. If an attack is considered likely all troops should endeavour to remain under cover, e.g. in shelters or cellars, in armoured vehicles or in slit trenches with overhead cover.

4. **Action against Immediate Effects of an Explosion**. Irrespective of whether an individual is under cover or in open, the immediate action drill on sensing the explosion's blinding flash is to close the eyes, drop to the ground, face downwards, put the hands under the body (if they are unprotected) and stay there until the sound of the explosion has finished.

5. **Protection against Residual Effects**. If warning is received, it will be possible before fallout arrives for individuals to put on their respirators to complete full protection and cover all exposed skin. It will give sufficient protection to an individual. This is best illustrated by reference to the protection factor, which is a measure of the degree of shielding from gamma fallout radiation afforded by a structure or vehicle. To give an idea of a few examples of equipment are:-

a.	Shelter with three feet cover	-	200
b.	Tanks	-	15
c.	Artillery (SP)	-	10
d.	APCs	-	3
e.	Thin skinned vehicles	-	2

6. Once the fallout has arrived personnel can become contaminated as a result of radioactive particles, e.g. dust and mud, settling on or becoming attached to the person. Danger from these particles will be greatly reduced, if the following rules are applied:-

a. Remain under cover for as long as possible consistent with the operational requirements.

b. Keep the bare skin covered.

c. Avoid coming into direct contact with ground or objects, which might be contaminated.

d. Avoid eating or drinking in the contaminated area. Cover food and water before fallout arrives.

e. Prevent contaminated dust or other substances from entering the body, e.g. keep cuts and wounds covered.

f. Do not raise dust by running, stamping or moving vehicles unnecessarily.

g. Avoid opening the mouth, if possible; Breathe only through the nose. For this reason do not smoke.

7. **Personal Decontamination**. It will be necessary if the individuals have been in contaminated area or in contact with contaminated equipment. A contaminated person will have radioactive particles in the form of dust, mud or other substances clinging to his skin, clothing and equipment. Because the contamination must be removed. The hands and face should be washed, paying special attention to the hair and fingernails. Monitoring instruments will then be necessary to show if any further decontamination is required, such as changing the clothing or showering.

8. **Equipment Protection and Decontamination**. Every effort should be made to get equipment under cover before fallout arrives, never the less some equipment is likely to become contaminated. Care must then be taken to limit the spread of contamination as much as possible. In time it will decay, but this may be a lengthy process. It can also be sealed, as for instance by burial. However, the most usual way for equipment to be decontaminated will be removing the source of activity. There are two stages for this:-

a. *Operational Decontamination.* Those measures taken to reduce intensity levels to the minimum compatible with the operational situation. This is some times called Rough Decontamination.

b. *Unit Decontamination.* Involving the reduction of radiation intensities to the lowest possible level and requiring special techniques and equipment. During unit decontamination a unit would be non-operational. It is also called Detailed Decontamination. It is not a quick process.

9. **Methods of Decontamination**. Physically removing radioactive matter from where it is a hazard to a place where it can do no harm. The practical method of cleaning an item is by washing with water. However, large quantities of water, preferably under pressure, will be required. Warm water and detergent should be used where possible, as they are more efficient. It should be remembered that the water is then radioactive because of the particulate matter suspended in it and because some radioactive materials are soluble in water. The contaminated water should be led to a soak away and care should be exercised to keep it clear of drinking water sources and from growing foodstuffs and animals. Radioactive dust can be removed from surface to reduce the intensity but this method is unlikely to remove the hazard completely. It is particularly important to remove a thin layer of loose earth from around fire positions in trenches after fallout to reduce the gamma radiation to the man and because the eyes are vulnerable to beta radiation. Vacuum cleaning is a possibility in static installations, but the resulting radioactive vacuum cleaner bag must be disposed off. Removal confers protection through increasing the distance between the source and the body.

Where large items have become radioactive, as may be the case with heavy equipment close to Ground Zero (GZ), they should be set aside in a special compound and the area marked with warning signs.

CHAPTER 7

RELIEF IN BATTLE AREA

SECTION 35 – GENERAL

Introduction

1. Relief in the battle area is one of the common operations in war. While it is in progress, both the incoming and outgoing units have many troops in the forward areas at the same time. This not only increases their vulnerability but if the enemy attacks, command and control are difficult to exercise. In the interest of morale, troop's first introduction to a new area should be conducted calmly and with speed and efficiency.

Definition

2. A 'relief in line' is an operation in which all or part of a unit/formation in defence is relieved. All responsibilities of the relieved unit/formation are taken over by the relieving unit/formation and the relieved unit/formation is withdrawn.

Purpose

3. Relief in line may be carried out for any one or a combination of the following reasons:-

a. When troops have remained committed for a prolonged period and need rest.

b. Reorganization out of contact to replace casualties and equipment losses.

c. When tactical plan requires a periodic relief.

d. When it is desired to conserve the fighting power and efficiency of our troops.

Aim

4. At company level, the aim is to relieve another company without the enemy's knowledge, at the same time maintaining the position intact. The aim is similar at all levels throughout the battalion.

Principles

5. If the above aim is to be achieved, the principles which must be achieved are:-

a. Secrecy.

b. Speed.

c. Silence.

d. Control.

6. **Secrecy**. The enemy must not be given any indication that the relief is to take place or is taking place, otherwise he can be expected to react violently and heavy casualties to our troops may result. The most usual ways of losing secrecy are:-

a. Too much movement and poor battle discipline by the advance party.

b. Bad wireless discipline, especially when the relief is in progress.

7. **Speed**. The shorter the time men of both units are in the position together, the better it is. A speedy relief demands a thorough knowledge of the relief drill by commanders at all levels. In the company, simple but detailed orders be given so that they all understand what they have to do and how to do it. Battle drills and command and control should be good.

8. **Silence**. Night operations call for silence. Noise travels a long way at night and it is the most likely way to lose secrecy and draw enemy fire. The need for silence may even out weigh the need for speed. Arrangements for control must therefore be such that silence is not broken unnecessarily.

9. **Control**. Good control is the basis of secrecy, speed and silence. It will be decisive should the unexpected happen. Careful thought must therefore be given before hand to the preparation of detailed orders covering all such matters as guides, checkpoints, route marking and movement in transport and on foot.

SECTION 36 - **PLANNING AND PREPARATIONS**

Warning Order

1. A company can usually expect to receive its warning order about 48 hours in advance. The company commander is given all the available information by his battalion commander. After passing this to his platoon commanders, he is required to go forward with the advance party consisting of his platoon commanders and runners. Whenever possible, the company and platoons commander should go forward and see the ground in daylight.

Sequence

2. Relief may be carried out in two stages:-

   a. *Stage 1*. Planning

   b. *Stage 2*. Execution of relief

Planning

3. **Timings**. A battalion will need at least 48 hours for proper planning and conduct of a relief operation. It will include:-

   a. Time for commencing and completing the relief.

   b. Hours of night when relief is to be carried out and whether the relief is to be completed in one or more nights.

   c. Priorities for use of routes, if more than one with timings.

4. **Advance Party**

   a. *Composition*

      (1) Company Commander with his runner.

      (2) One representative of each platoon (The platoon commander or platoon Second-in-Command) with runner.

      (3) Signaler.

      (4) Guides.

b. *Tasks*

(1) To get the maximum possible information about the position and the enemy.

(2) To make all the necessary arrangements to enable the relief of the company to be carried out swiftly and smoothly.

c. The company commander should list all the points on which information is needed before he leaves with the advance party. They include:-

(1) *Enemy Forces*

(a) Opposing units/formations.

(b) Known locations.

(c) Minefields, wire and obstacles.

(d) Known defensive fire areas.

(e) Shelling.

(f) Habits.

(g) Patrol activity.

(h) Observation Posts.

(j) Morale.

(k) Probable intention.

(2) *Friendly Forces*

(a) General layout of the company with special regard to flanking platoons.

(b) Detailed dispositions for the platoons including the number of fire trenches for each section.

(c) Section arcs of fire.

(d) Light machine-gun fixed lines and arcs of fire.

(e) Rocket launcher’s arc of fire.

(f) Likely tasks for the mortars.

(g) Positions of trip flares.

(h) Position on the ground of defensive fire and defensive fire (SOS) tasks and how they are called for.

(j) Position and tasks of support weapons.

(k) Range cards.

(l) Maps and air photographs.

(m) Observation Posts and details of patrols out.

(n) Wire, obstacles and minefields and gaps and lanes through them.

(o) Approaches and dead ground.

(3) *Administration and Logistics*

(a) Reserve ammunition.

(b) Defence stores, extra wire and mines.

(c) Casualty evacuation.

(d) Water supply.

(e) Feeding arrangements including reserve rations.

(f) Sanitary arrangements.

(4) *Command and Signal*

(a) Track plan and runner's routes.

(b) Location of neighbouring platoon headquarters, company headquarters and battalion headquarters.

(c) Position of lines, if laid.

(d) Light signals.

(e) The alarm signal for enemy air, gas or nuclear attack.

(f) Inter trench communication system.

(g) Outgoing unit's password.

d. *Reconnaissance by Advance Party*. Reconnaissance must include:-

(1) Existing defences including minefields and other obstacles.

(2) Relief routes.

(3) Assembly areas.

(4) Debussing and turn round areas for vehicles.

(5) Administrative installations.

Platoon Preparation

5. Assuming that if platoon Second-in-Command is moving with advance party, the platoon commander should do the following:-

a. Brief the platoon Second-in-Command, ensuring that he takes:-

(1) The strength of the platoon to enable him to estimate the number of fire trenches required.

(2) A written list of all information required.

(3) A compass.

b. *Inspect the Men and Stores.* It will entail:-

(1) Ensuring that all men have a thorough wash.

(2) Inspecting clothing, necessaries and equipment.

6. Assuming that platoon commander is moving with advance party:-

a. While the platoon commander is away with the advance party, the platoon Second-in-Command must ensure that the normal administrative preparations are completed. These include bathing, provision of canteen supplies, inspections and replacing deficiencies.

b. The platoon commander will do as much of the briefing as possible before he leaves with the advance party. He will not usually be able to return to do any final briefing, which will normally be done by the company Second-in-Command to the whole company. The following must be included:-

(1) An explanation of the ground together with the procedure for relief and the system of checkpoints.

(2) Action if attacked.

(3) What to do and what to expect in the way of lights and shelling.

(4) Emphasis on silence, keeping contact with the man in front and speed.

c. Rehearsals should include:-

(1) Debussing.

(2) Taking over platoon and section localities in detail.

Order of Relief

7. Relief in line is carried out by stages, either rear to front or front to rear. In determining the sequence of relief, both commanders will consider:-

a. The subsequent mission of the troops being relived.

b. Size and type of elements involved in relief.

c. Enemy's capability to detect and react against the relief.

d. Enemy habits and activities.

e. Characteristics of the area of operation.

f. Need to vary the pattern of relief.

g. Time available for relief.

System of Control Points and Guides

8. Four control points in the battalion area affect the company:-

a. *The Debussing Point.* If the initial move is in troop carrying vehicles, this point will be as far forward as possible without giving away secrecy through the noise of vehicle engines. Surveillance capabilities and ranges of enemy's ground surveillance radar should be kept in mind while its selection. The company with its supporting detachments will arrive under the company Second-in-Command. The point will be marked by lights. An officer and a protection party from the incoming battalion will be on duty near by. A guide from the outgoing battalion will be allocated to lead guide from the company group forward or to a dispersal area if there is any hold-up.

b. *Battalion Checkpoint*:-

(1) Companies are checked in as they arrive.

(2) Companies will not halt here except temporarily and guides will lead companies straight to their company checkpoints.

c. *Company Checkpoints.* This is where the advance party, usually the platoon commander and his runner, wait for the platoon. Also waiting at the company checkpoints are:-

(1) The Company Commander.

(2) The other platoon advance parties.

(3) A guide for each platoon from the outgoing battalion.

(4) NCOs in command of support weapons in the company area.

(5) The company havildar major from the outgoing company who will be incharge of this point.

(6) Other important points are:-

(a) The routes from the battalion checkpoint to the company checkpoints may be lit. They will certainly be taped in difficult places.

(b) At the company checkpoint will be the platoon representative with runners and a guide from each out going platoon. They will meet their respective platoons and lead them off independently to their positions.

(c) If any of the platoons spread out during the march from the battalion checkpoint, they will reorganise here before going further. Company headquarters should meet its company commander or Second-in-Command at this point. The outgoing company havildar major can be incharge of the company checkpoint.

(d) The incoming company commander must immediately report to the battalion commander on his arrival at company headquarters.

(e) If the battalion checkpoint is suitable as a company checkpoint it may be so used. Platoon representatives will then be found at the battalion checkpoint.

(f) Company checkpoints are normally provided for forward companies only.

d. *Platoon RV*. This will be close to platoon headquarters. The platoon commander or platoon Second-in-Command must know the route to this RV from the company checkpoint and he must have selected a platoon dispersal area for use in the event of a hold-up. On the move to the platoon RV, the guide should lead and the platoon commander or platoon Second-in-Command should move behind the leading section to reduce the chances of the guide and himself becoming casualties at the same moment. The platoon commander or platoon Second-in-Command should check the platoon through the company checkpoint. At the platoon RV, guides from the outgoing platoon meet each section. There is no halting, sections are led off to their respective positions.

SECTION 37 – **RELIEF PROCEDURE**

Execution of Relief

1. **Move from Rest Area to Debussing Point**. The battalion will usually move from the rest area to the debussing point in transport. It will debus at the debussing point.

2. **Marking of Debussing Point**

   a. The outgoing unit will be responsible for marking the debussing point and for lighting the route forward.

   b. The outgoing unit will mark the route forward, using the incoming unit's sign, which will be brought forward by the advance party.

   c. The incoming subunit will meet the guides from the equivalent subunit of the outgoing unit. It will be led along its route by the guides to the battalion checkpoint.

   d. Companies will move to the battalion checkpoint in single file. This procedure will be laid down in the initial orders and will depend on the degree of darkness and enemy harassing fire.

3. **Action in Platoon and Section Localities**

   a. Before the platoon arrives, the advance party gives the names of the men in each section to outgoing section commanders, arranging with them which trenches they are to occupy. The outgoing section

commander leads the incoming section to their trenches in pairs or fours as the case may be.

b. On reaching the trenches, the incoming men at once occupy shelter trenches so as not to stay exposed above ground. The trenches must be clear of baggages/items belonging to the outgoing section. There will be “stand to” throughout the relief.

c. Both section commanders then go to each trench in turn; the outgoing section commander briefs the incoming troops on their arc of fire and any other important detail, all of which must be noted by the incoming section commander.

d. Next all incoming section commanders report to their platoon commander that they are ready to take over. The platoon commander informs his company commander and asks permission to complete the relief.

e. The two platoon commanders stay at platoon headquarters ready for any emergency while the two platoon Second-in-Commands together go round each section in turn, ordering outgoing sections to withdraw to the platoon RV.

f. The platoon Second-in-Command will report to their platoon commanders and the outgoing platoon commander with his headquarters will move back to the platoon RV. His platoon will not wait there longer than is necessary to ensure that it is complete before moving to the company checkpoint and company RV.

g. After the outgoing platoon has left, the incoming platoon commander will report to his company commander that the relief has been completed.

h. When the relief has been completed, the incoming platoon commander will send two runners to company headquarter. One will stay, the other will come back. This ensures that at least two men know the routes. The runner to come back will be the one who was with the platoon commander in the advance party.

j. The incoming unit will continue to “Stand to” until the company commander orders to “Stand down”.

Communications

4. There are three means of communication available during the relief and they should be used in the following order of priority:-

a. Line.

b. Outgoing battalion's wireless net.

c. Incoming battalion's wireless net.

5. Wireless traffic must be kept at the normal level so as not to lose secrecy. When this cannot be ensured, traffic must be confined to the outgoing battalion's net which is already known to the enemy.

Protection during Relief

6. Debussing points and checkpoints are protected by the incoming troops. Protection for the relief itself is given by reconnaissance and fighting patrols. In case these patrols are provided by the incoming subunits, the patrols must go forward with the advance party and be given time for reconnaissance and briefing.

Change of Command

7. Command normally passes at each level as follows:-

a. *Company*. When two or more platoons, including the forward platoons, have completed relief.

b. *Platoon*. When all three sections have been relieved.

Day Light Relief

8. A relief operation will almost invariably take place at night. The only possible occasions for a daylight relief would be:-

a. When the enemy has no direct observation over the area in which the relief is to take place.

b. When our air superiority over the battle front is such as to preclude the probability of enemy air action.

c. When the nature of the country is so difficult for infantry and motor transport movement that the hours of darkness are not considered sufficient to complete the operation.

d. Relief of reserve companies to speed up actual night relief.

9. Diagrammatic Layout of relief operation is given at Annex "F".

SECTION 38 – **PASSAGE OF LINE**

Definition

1. A passage of lines is an operation in which an incoming unit attacks through another which is in contact with the enemy. Elements of the unit passed through remain in position and support the attacking unit until their fire is masked. Thereafter, they may remain in position, be withdrawn or may undertake another mission assigned to them.

Basic Considerations

2. The following considerations must be kept in mind whenever a passage of lines operation is undertaken:-

a. Adequate time for planning and reconnaissance is required, therefore, units involved must receive early warning.

b. Plan must be detailed, but simple and well coordinated between the units.

c. The unit passing through should come in as late as concentration in the forward areas before the attack, should be for as short a period as possible.

d. Complete secrecy must be maintained so as not to give away surprise.

Planning

3. **Preliminary Action**. The planning procedure of passage of lines operations are similar to those of relief in line. On receipt of warning order requiring passage of lines, the commander and his staff will make early contact with the unit being passed through. The tactical headquarters/command post of the incoming unit will move near to the headquarters of the unit being passed through so as to facilitate the planning and working out details of the passage.

4. **Coordination**. During the planning conferences the following details will be coordinated by the commanders concerned:-

a. Exchange of intelligence information.

b. Exchange of tactical plans including communication layout.

c. Reconnaissance arrangements by elements of units passing through.

d. Security measures to be taken during passage.

e. Selection of areas of passage and arrangements for guides.

f. Priorities for use of routes and traffic control arrangements for guides.

g. Condition or timings when responsibility for the area of operation will be transferred to the force passing through.

h. Extent of tactical and fire support which the unit in place is to provide for the operations of the unit passing through.

j. Extent of administrative support to be provided by the unit being passed through.

k. Utilization of signal communication of the force being passed through. The use of radio communication should be kept to minimum.

5. **Transfer of Command**. The exact time of the circumstances when the command is to pass should be decided mutually between the two commanders. Normally the commander of the force passing through the lines assumes responsibility for all operations in the area at or prior to the time of attack. He may, however, be ordered by the higher headquarters to take over the responsibility at the time of preparatory bombardment or even earlier. On transfer of command the commander making the passage assumes operational control of the troops of the force being passed through, which are in contact with the enemy at the time of transfer.

Priority for Use of Routes

6. Normally the force passing through is allotted priority for the use of routes. Routes marking and traffic control is the responsibility of the force being passed through, in their area till such time as the operational responsibility and command is transferred to the force passing through.

7. **Tactical Support**

a. The unit in contact provides all possible support to the forcepassing through, e.g., breaching of minefields, provision of guides, fire support and other combat support within their capabilities.

b. Due to the problems of control, only the indirect fire means of the unit in contact will, normally be used to support the passing through force.

c. Though it is desirable that the integral artillery of the unit making the passage supports the attack, it may not be necessary, if enough fire support can be made available from the artillery of the formation in contact. In such cases, to avoid undue concentration in forward areas, the artillery of the unit in contact will support the attack initially and the artillery of the attacking unit will be deployed in rear position ready to move forward in order to support the subsequent phases of the attack.

Administrative Support

8. The formation in contact may be required to provide following administrative support to the attacking formations:-

a. Evacuation of casualties and prisoners of war.

b. Use of administrative facilities i.e., supply points, replenishment points, water points, et cetera.

c. Traffic control and route, protection.

d. Civilian control.

Conduct of Passage of Lines

9. Whenever, possible the passing unit moves during the hours of darkness from the rear positions. Timings should be carefully calculated to ensure that units attack at the correct time without having to move to a forward assembly area. This will reduce to the minimum time of maximum concentration in the forward area.

10. Close coordination at all levels is the key to the success of passage of lines operations. The routes which are to be used by the various subunits must be seen in daylight and must be coordinated with the subunits being passed through.

CHAPTER 8
PATROLS

SECTION 39 – GENERAL

Introduction

1. In war a commander cannot plan a successful operation without accurate and up to date tactical information. Patrolling is one of the most reliable means of obtaining this and is carried out by day and by night in all operations of war. Successful patrolling calls for a high standard of individual training, good fieldcraft, confidence, initiative and determination.

Importance of Patrolling

2. Patrolling is a very difficult and important operation of war and troops who are good at patrolling will be generally good in all other operations of war.

3. Successful patrolling is based on thorough preparations and is a big morale booster for own troops, simultaneously lowering the enemy's morale. The successful completion of the task depends upon the patrol leader who is the most important person of the patrol. The rest of the personnel are meant for his help and safety. Hence the task will be the sole responsibility of the patrol leader and he will not delegate this responsibility to his subordinate commanders.

Aims of Patrolling

4. Patrols may be sent out to:-

   a. Obtain information about the enemy or confirm any information received previously from other sources.

   b. Disorganise or destroy enemy troops or material.

   c. Maintain initiative and dominate No Man's Land and gaps between formations and units.

5. Domination of No Man's Land implies winning and keeping moral and physical ascendancy over the area between the opposing forward defended localities. This makes it easier and safer for our patrols to move in this area and to obtain information. Conversely, it makes it harder and more dangerous for the enemy to do so. Gaps between units must be patrolled to prevent large bodies of enemy infiltrating or forming up for an attack unobserved.

Desired Information

6. There are many methods of gaining information but patrolling is the most effective and reliable method. Through patrolling we can obtain the following information:-

a. *Natural and Artificial Obstacles*. This includes information regarding type and conditions of terrain, nullahs, defiles, mine-fields and ditches et cetera.

b. *Enemy Defences.* Extent and depth of enemy defences, location of automatic and anti-tank weapons.

c. *Enemy Movements.* Enemy supply, routes, patrol habits, stand to and stand down timings, tank leaguers and rally areas et cetera.

d. *Enemy Strength and Composition*. The strength and composition of enemy in the defensive position.

e. *Enemy Missile Sites*. Exact locations of the enemy missile launching sites including current locations of the mobile missile launchers if any.

f. *Enemy DFs and DF (SOS).* Likely DFs and DF (SOS) of enemy.

g. *Enemy Fixed Lines*. Fixed lines of enemy automatic weapons.

h. *Enemy Identification Signs*. Those signs which help us to identify enemy composition. These can be obtained by capturing enemy prisoners or material.

j. *Enemy Installations*. The installations like radar, missiles and NBCW storage sites et cetera.

Strength and Composition of Patrols

7. The strength and composition of a patrol will depend upon its role. Generally the factors which effect the strength and composition of a patrol are as follows:-

a. Task.

b. Strength and habits of enemy patrols.

c. Type of terrain, including available observation and cover.

d. Duration for which patrol has to stay out. Whether a firm base will be established or not and whether the patrol is going out during day or night.

e. Strength which can be easily controlled by one man.

f. Minimum personnel required to accomplish the mission should be taken.

Type of Patrols

8. All patrols have the task of providing information, they must also be prepared to fight, either to gain information required or to protect themselves if the information is being gained by stealth:-

a. There are two main types of dismounted patrols:-

(1) Reconnaissance patrols.

(2) Fighting patrol

b. Other types include:-

(1) Escort patrol.

(2) Standing patrol.

(3) Link patrol.

SECTION 40 - **PLANNING AND CONDUCT**

Sequence of Preparation

1. A suggested sequence might be:-

a. Warning order to patrol leader, usually given by the company commander.

b. Commanding Officer/Intelligence Officer/Company Commander briefs the patrol leader.

c. Selection of Observation Post from the map.

d. Patrol leader carries out time appreciation and makes a reconnaissance plan.

e. Patrol leader gives warning order to his Second-in-Command. The following will be included in it (catch word MPDEWARST):-

(1) M-Mission.

(2) P-Personnel.

(3) D-Dress.

(4) E- Equipment.

(5) W-Weapons.

(6) A-Ammunition.

(7) R-Rations/Rehearsals.

(8) S-Security.

(9) T-Time and RV for orders.

f. Any special preparations.

g. Reconnaissance from Observation Posts alongwith supporting arms representatives, if given.

h. Reconnaissance and planning, company commander will usually help in planning.

j. Patrol leader prepares his orders.

k. Prepares model, if required.

l. Shows area to the patrol from Observation Posts. If the ground permits then the whole patrol will be taken together, otherwise they will be taken in small groups or individually.

m. Patrol leader issues his orders. If ground permits this will be done from the Observation Post, the orders will be given off a sketch/model. If the orders are being given off a sketch/model then the sketch/model should be close to the Observation Post. Even if the orders are being given from the Observation Post (OP), the action on the objective will be explained on the model. It is advisable to have a large and clear sketch/model of the target area.

n. Coordination with those sentries through whose positions the patrol will go out and come in and who will be on duty at that time.

o. Day rehearsals.

p. Preparation and inspection.

q. Testing of weapons by actual firing.

r. Administrative arrangements including food and rest.

s. Night rehearsals and final inspection.

t. Patrol action.

u. Debriefing of patrol leader.

v. Detailed debriefing of the whole patrol.

What a Patrol Leader Must Know

2. Before leaving on patrol, the patrol leader should ensure that his orders cover the following points:-

a. All available information about the enemy.

b. Information about our troops such as disposition of forward troops, minefields, lanes, gaps in wire and details of other patrols going out.

c. The aim of the patrol (MISSION).

d. Time out and where; time back and where. (Time restriction on patrol leader only when considered necessary).

e. Any limitations e.g., boundaries with neighbouring units/subunits, activities like raids and ambushes by flanking formations, affecting his choice of route and timings, particularly our defensive fire areas.

f. Fire support, if any.

g. Recognition, password and any special signals.

h. Action to be taken on meeting enemy.

j. Any special subject on which information is required.

k. Any special administrative arrangements.

l. The enemy's NVD/NODs capabilities and limitations in the area where the patrol is going to operate.

Reconnaissance and Planning

3. The patrol leader must:-

a. Check that he has been provided with all the information he can obtain such as maps, air photographs and reports of previous patrols.

b. Plan and carry-out as detailed reconnaissance.

c. Study the ground with the following in mind:-

(1) Routes.

(2) Obstacles.

(3) Landmarks.

(4) Observation Posts.

(5) Dead ground and covered approaches.

(6) Places where ambushes may be met or laid.

(7) Enemy positions, likely position and defensive fire areas.

(8) Effect of the moon or movement light, if any.

d. Check very carefully distances and compass bearings of bounds or legs on routes, and also the timings.

Physical Fitness

4. Men selected for the patrol should be free from coughs and colds and otherwise physically fit.

Patrol Leader's Orders

5. Orders must be detailed and must cater for every contingency. Patrol orders will always be given to the whole patrol. They must be given slowly and by stages, the members will also be allowed to ask questions as the various points arise. After the orders have been given, the leader must be confident that every member not only knows his own job thoroughly but is capable of taking charge in an emergency or even of completing the mission by himself should the need arise.

6. For orders of Reconnaissance Patrol Leader see Annex "G".

7. For orders of Fighting Patrol Leader see Annex "H".

Rehearsals

8. All actions of the patrol must be thoroughly rehearsed before it sets out. The leader must be quite clear as to what he wants to rehearse; vague rehearsals of formation alone are of little use. A daylight rehearsal should be held to practice the following which should be repeated at night:-

a. Order of march and individual positions in all formations to be used.

b. Method of changing formations.

c. Obstacle crossing.

d. Action on meeting the enemy enroute, both when you see the enemy first and when the enemy sees you first.

e. Action at bounds/RV.

f. All signals that are to be used.

g. Action at halts.

h. Action at lights.

j. Casualty evacuation and prisoner escort.

k. Action on the objective.

l. Any other thing which the patrol leader wants to rehearse.

Inspections

9. During day and before setting out on patrolling, the patrol leader must carry-out an inspection of the patrol. The inspection should be carried out according to a proper sequence so that nothing is left out. The points to note are:-

a. *Physical Fitness of the Personnel*. Some extra men must also be kept ready as replacements in case an individual of the patrol becomes a casualty or falls sick before the patrol leaves.

b. Weapons are complete according to the requirement and in working condition. Weapons must be actually fired and tested.

c. All equipment and stores are complete according to the task of the patrol.

d. Men must be checked again after food and rest, and before leaving so that sick personnel may be replaced.

e. All men are dressed properly and the equipment is being carried in the correct manner.

f. No part of the equipment or clothing should produce noise during movement.

g. The sling of rifles and the parts of weapons and equipment which may produce noise are properly fastened.

h. Proper camouflage of shining portions of weapons/equipment, face and hands.

j. No one should have anything which would give away his identity.

k. Water bottles must be filled properly otherwise they will produce noise during running.

Rest

10. It is essential to give rest to the patrol after day rehearsals so that they are refreshed and can do their task efficiently at night.

Planning

11. In a battalion, the battalion Second-in-Command (2IC) is usually responsible for giving instructions to the patrol. The Intelligence Officer is responsible for giving information regarding own and enemy forces, especially information brought back by any previous patrols. The company commander will help the patrol leader in his planning.

12. **Time for Planning**. Planning should be completed sufficiently ahead of time so that the battalion commander or his representative can give detail instruction to the patrol leader. The greatest efficiency results if commanding officer can plan a day or two ahead. The patrol leader should also get sufficient time to study the map/air photos and carry-out reconnaissance. The time required for this will depend upon the mission of the patrol and on how far the individuals of the patrol are conversant with the area. Four daylight hours are considered the minimum for even a simple reconnaissance patrol, while a considerably longer period will be required for more complicated tasks.

13. **Points to be Included in Planning**. The following points must be included in the plan:-

a. Bounds and RV.

b. Final RV/firm base.

c. Division of objective and distribution of groups to each part of the objective.

d. Action on the objective.

e. Withdrawal from the objective to the final RV.

Equipment

14. **General**. The weapons and equipment to be taken and the dress to be worn will depend on the patrol's task. As a general rule, patrols should operate as lightly equipped as possible. Patrols may be cut-off and have to live up in No

Man's Land until the next night they should therefore generally take with them their water bottles and emergency rations.

a. *Weapons*. It is best to take light short-range weapons. Sub machine-guns/MP5, rifles and bayonets, pistols and unconventional weapons such as knives are all suitable. Weapons should be loaded with actions cocked and safety catches applies. Bayonets should be fixed and blackened. Both high explosive and smoke grenades should be taken. Grenade pins must be checked to ensure that they are readily removable but safe.

b. *Equipment*. Web equipment should be reduced to a minimum. Everything must be well fitted, and must not rattle. Steel helmets should not be worn since they make too much noise.

c. *Clothing*. This will depend on climate and weather but should be comfortable, protect the wearer and enable him to move silently without tiring him. In general, the following should be avoided: -

(1) Smooth surfaces likely to shine in moonlight or under flare light.

(2) Colour contrasts.

(3) Stiff clothing; this rustles during movement.

d. *Footwear*. Rubber soled DMS boots should be worn if available.

e. Patrol should also carry the following: -

(1) Compass with luminous dial and lanyard.

(2) Watch with luminous face and lanyard (watch in pocket)

(3) Field glasses: these are valuable at night.

(4) A pair of well oiled wire cutter and tape.

(5) Torch.

(6) A blanket for carrying a wounded or for muffling prisoner.

f. *Special Equipment*

(1) Laser Range Finder.

(2) Global Positioning System (GPS).

(3) Flame Thrower. Depending upon the nature of task either of these or both can be carried.

(4) Mine detector.

(5) NVDs.

g. *Communication*. Teknim (walkie-talkie) sets for communication within the patrol.

h. *NBCW Equipment*. Since, the enemy has the NBCW capability and there is a always a threat of use of NBC weapons. In case it is established that enemy is likely to employ NBC weapons in the area of operation, requisite NBCW equipment must be carried along.

Fieldcraft

15. The patrol should see or hear the enemy before it is itself seen or heard. Frequent listening halts, are essential. The patrol should lie down, so as to take advantage of the skyline and should observe in all directions, the leader using his field glasses and NVD to scan the ground. At night, the ears reveal more than the eyes. The patrol should not move until certain that there is no one else nearby. There halts are made frequently. Between halts, the patrol should move at the best possible speed.

16. For successful fieldcraft it is essential that personnel camouflage and concealment should be of a high standard and troops should know the techniques of night observation and movement.

17. Special attention must be paid to the following:-

a. *Personal Camouflage and Concealment.* For successful patrolling each man should know the art of camouflage and concealment. The colour and outline of hands, face, clothing, helmet, web equipment et cetera should merge with the background. The following procedure should be adopted:-

(1) The hands, face, neck and equipment should be smeared with soot or mud. These are very effective during night time.

(2) In case the steel helmet is being worn with a piece of hessian cloth, put a camouflage net over it with some small branches and grass et cetera to break its shape and kill the shine.

(3) Wrap a string around the small pack (if it is being carried) and put some leave et cetera in the string to break its shape.

(4) To camouflage the weapon either wrap a cloth or put mud on the shining parts. It should be kept in mind that these should not hinder use of weapons.

(5) Look from the side or through the cover and not over its top. If it is unavoidable to look over the top then remember that a straight line should not be broken.

(6) Avoid the skyline.

(7) Keep in mind that in sunlight your own shadow will be quite prominent and will move with you therefore you must make use of shad/cover.

(8) Select a background in which you can easily merge yourself.

(9) Do not make use of isolated or prominent cover.

(10) Make sure that the enemy does not see you either going in or coming out from the cover.

(11) Move carefully.

b. *Observation at Night*. If night movement is to be carried out it is better to observe the area during day time.

(1) For night observation figure eight-method should be used and eyes must be frequently rested, as they get fatigued. If we stare at one object for a long time it appears to be moving. That is why a sentry at night feels that someone is moving towards him and he opens fire without any reason. To avoid this visual deception it is advisable to see the object against a static object like a finger of an outstretched hand.

(2) *Off Centre Vision*. Vision at night entails the use of special night cells in the eye. These night cells are around the day cells and thus make it necessary to observe by looking at a shift angle (about 6-10 degrees) away from the object. This is called off vision. An object that cannot be seen by staring straight at, it may be visible if the observer looks slightly to one side of it. The image thus formed can be retained for a few seconds necessary to memorize the position and to engage a target.

c. *Movement at Night*. Following are the methods of movement at night:-

(1) *Walk*. While walking the rifle should be held in such a position that it can be used immediately when required. Men must be alert while walking. On hard ground the sole of the boot must touch the ground first and balance must be maintained. While walking the rifle may be used either from the shoulder or the hip position.

(2) *Rolling*. If observed by the enemy at a mound et cetera, men must get away quickly. In such a situation rolling is the best method. While rolling, the hands and the weapon must be clasped close to the body. It should be ensured that the weapon is on "safe".

Formation

18. The formation adopted depends upon the size of the patrol, terrain, control, protection and available cover. Formations should be kept as simple as possible. File and single file formation is also correct but Diamond Formation could be very useful for patrolling. Distance between the men is governed by visibility. When contact with enemy is probable, the guides should move ahead of the patrol leader. They will depend upon the degree of darkness. If there is a wireless operator he should stay close to the patrol leader. His head set should be on all the time. He may stay on listening watch or may open up at various timings in order to report progress. He must always switch on, the moment contact is made. Because of the nature of his task it is desirable that he be given a protection man.

19. For formations of Reconnaissance and Fighting patrols see Sections 41 and 42 respectively.

Halts

20. **Action at Halts**. Whichever the formation a patrol adopts, at halts each man will take-up position facing outwards, for all round defence. No noise should be made while taking positions. A better method is to hold the weapon in left hand and search the ground with the right hand for twigs, dry leaves and stones et cetera before taking up a position. The weapons should be quickly brought into firing positions; the moment position is taken up.

21. **Action on Moving after a Halt**. A drill should be set for moving after a halt so that all can be informed silently and that no one is left behind. One

method could be that the patrol leader taps the man on his right; this man taps the man on his right and this continues till the man on the left of the patrol leader taps him. At this time the patrol leader will know that all personnel have received the signal to move and then he gets up and moves.

Control

22. During move the patrol leader should be well forward so that he may control the speed and direction of movement. The patrol leader and his Second-in-Command may wear some identification mark, like a white patch of cloth.

Routes

23. Routes “out” and “in” should be different. They should be broken down into “Legs”, each having a magnetic bearing and a known distance. Legs should be measured from the map to the nearest fifty paces, finishing at or near a feature easily recognizable at night. These places will be known as “Bounds”. Bounds should be known to each individual of the patrol and the distance between each bound should be between 550 to 750 metres. Since these bounds are prominent places which the enemy patrols might also have selected as bounds, so the RV should be selected 100 metres to 200 metres away from the bounds. Here the compass is reset for the next bound. If the patrol should be dispersed, all members should return to the RV. This drill enables the patrol leader to regain control as quickly as possible. When the enemy interference makes this impossible, the patrol will return to the RV behind that particular RV.

24. By night, except perhaps in bright moonlight, the route should avoid prominent cover such as corners and edges of woods, villages, tracks, hedges and deep defiles as these are likely places for enemy ambushes or standing patrols to be located. The patrol should use open ground rather than cover. When moving along sloping ground, the patrol should “counter” round the slope keeping fairly well up but below the crest. Moving along high ground increases the risk of being seen against the skyline. When crossing a ridge, the patrol should crawl and try to make use of any background available.

25. By day, concealment is very important. Routes should be chosen to take advantage of all possible cover.

Night Navigation

26. The best aid to keep direction at night is the compass. Careful study of maps and air photograph will help but the most accurate way of reaching the

objective is to march on chosen bearings, counting the paces taken on each leg. This method of marching in legs is described above. Features and landmarks help in checking position. Other aids which are some times available are the stars, tracers and movement light.

27. It should be kept in mind that during patrolling a separate party should be made for keeping direction and measuring paces. The patrol leader should not be member of this party.

Obstacles

28. All obstacles whether natural or artificial must first be examined by the patrol leader. To avoid confusion, it is best to keep the same pre-arranged order for crossing all obstacles. Movement should not be rushed as this is a time when the patrol is unavoidably spread out and vulnerable. At least one member should be ready to fire his weapon or throw a grenade if the patrol is surprised by the enemy while crossing the obstacle.

29. If a water obstacle is to be crossed using boats or some improvised method, then this must be practiced during day as well as at night.

30. Some methods for crossing obstacles are as follows:-

a. *Crossing a Nullah*

(1) *First Method (off set method).* If the obstacle is to be crossed from a bridge or a crossing place, then the patrol will establish a 'firm base' at a distance from the obstacle. The patrol leader along with another man will search for the crossing place. Then from the map he will read the bearing to that place and set his compass accordingly. He will also measure the distance up to it. The patrol leader will then take the patrol and cross from the pre-selected area. To come back to his route he will put the back bearing on the compass and walk the same number of paces. After this the patrol will continue on its original route. See Figure 10 below:-

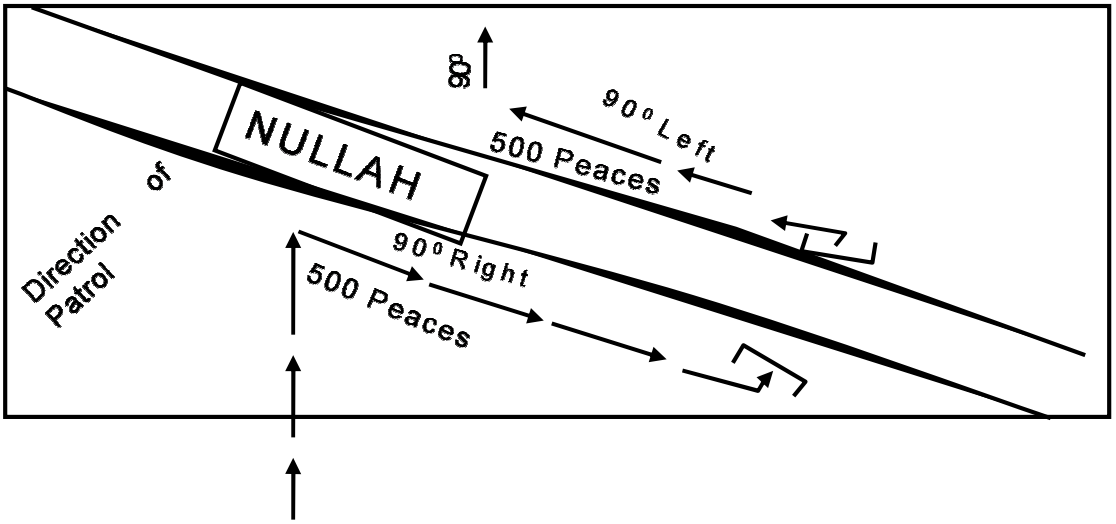


Figure 10 – Crossing a Nullah – First Method

(2) *Second Method*. It is easier to cross the nullah from the same place then the patrol will take-up positions on the home bank. The patrol leader will take half of the patrol across the nullah and take-up position on the far bank. Meanwhile the rest of the patrol, under the Second-In-Command will cover their movement. Once the commander is in position on the far bank, he will send two men who will go and inform the Second-in-Command, who will bring the rest of the patrol across the nullah. The whole patrol will then continue its move.

b. *Crossing a Wire Obstacle*

(1) When the patrol reaches nears the objective, the guide in front will take a light stick and keep moving it in front of him so that he may detect a wire obstacle or the wire of a trip flare. As soon as he spots a wire he goes into position and the rest of the patrol on seeing him, also goes into position. The patrol leader after finding the type of wire goes back and informs the Second-in-Command. After this he comes up again to carry-out reconnaissance of the obstacle. The Second-in-Command, according to a rehearsed drill, briefs the patrol and sends the wire cutting party forward. This party will consist of the following:-

(a) Patrol leader, who will cut the wire (he is already near the wire). He will have the following stores;

hessian cloth or sandbag et cetera, wire cutter and a white tape.

(b) One or two men to hold the wire.

(c) Two men for protection. Guide will do the job of one of them.

(2) *Drill for Cutting Wire*. There are usually empty tin cans with some stones hanging from the wire to produce noise. So, before cutting a wire it should be ensured whether such cans have been used or not. If these are present, then the stones should be removed from them, otherwise a slight carelessness will produce a lot of noise. After this the two men will take-up positions on either side along the wire at a distance of about 5 to 10 metres from the place at which the wire has to be cut. The two men holding the wire, will lie down along the wire, facing inwards, and pull the wire. This task can also be done by one man if the wire is being cut close to the place where it is tied e.g., a picket. The commander will now lie down facing the wire, place the cloth on the wire and cut it. The men holding the wire will slowly bring the loose ends to the ground. It is better to cut the wire close to a picket as only one man will be required to hold the wire.

(3) *Drill for Crossing the Wire*. After creating a passage through the wire obstacle the patrol leader will put the white tape to mark it. He will now send one man back to inform the Second-in-Command and the third in command. The rest of the wire cutting party will cross the obstacle and take-up position slightly ahead on the other side. The person who has gone back brings the Second-in-Command forward and here the patrol leader will hand over the gap to him and cross to the other side along with that man. The third in command will now send the patrol forward one by one and the Second-in-Command will make them go through the gap. Once the whole patrol has gone through, he will close the gap and pick up the white tape. He will then inform the patrol leader, who will check the whole patrol before they resume their move. During rehearsals each man will be explained as to where he has to take-up position after crossing the wire obstacle.

c. *Crossing a Trip Flare*. Trip flares are usually covered by fixed line automatic fire and the aim is to give early warning of enemy's approach. These can be easily camouflaged and easily set-off. On setting off they immediately produce bright light and some smoke, illuminating the area for quite some time. To over come this hazard the guides must have a light stick, which he should keep moving in front of him while advancing. This way the trip flare can be detected before it is set-off. If the patrol leader comes to know about a trip flare, he may adopt one of the following techniques:-

(1) *By–passing the Trip Flare (off set method).* As explained above in drill for crossing a nullah by first method.

(2) *Neutralizing the Trip Flare*. As soon as the guide feels the wire of the trip flare, he goes into position and so does the rest of the patrol. The patrol leader moves forward and tracing the wire reaches the trip flare. He will then insert the safety pin in it and make it safe. It should be kept in mind that neutralizing a trip flare is a difficult task and it would be advisable to adopt some other technique.

(3) *Going over a Trip Flare*. The patrol leader on seeing the wire goes back and informs the patrol. Then he puts a white tape on the ground along the wire and stretches his hand over the wire with a white cloth over it. The Second- In-Command sends the patrol forward one by one and the commander makes them go over his hand. The patrol takes up positions on the other side, according to a rehearsed plan. After every one has passed, the commander will cross the wire and pick up the tape.

(4) *Activating a Trip Flare*. If no other method of crossing a trip flare is feasible then this technique would be adopted. In this method the patrol will take-up forward and tie a string to the wire of the trip flare. This man will also come behind the cover and then pull the string. This will activate the trip flare and if the enemy opens fire, the patrol would be safe behind the cover. This will also help to locate enemy weapons. After activating the trip flare, the patrol should stay behind the cover for about half an hour before continuing their move.

(5) *Action on Setting Off a Trip Flare*. If a trip flare is suddenly set-off, the patrol should move away

immediately, as the position would be covered by enemy's fixed lines. It is better that the patrol should spread out and run towards the RV. Smoke grenades can be used to cover the withdrawal. After regaining Command and control, the patrol will again set-off on its mission. If the trip flare is near when it sets-off, the following actions can also be taken:-

(a) Trip flare is kicked so that it falls down. This will reduce its light.

(b) A blanket or ground sheet thrown over the trip flare will reduce its light for some time.

(c) If the headgear is put over the trip flare it will also reduce the light.

d. *Action on illumination*. Illumination may be done by different methods and for each type of illumination a different technique will be followed:-

(1) *Mortar Flare*. When a mortar flare is fired there is usually enough time to lie down on the ground before it illuminates the area. If caught unaware the best thing is to freeze, as any movement in light would be easily detectable. Whatever the case, one eye should be closed so that after the flare dies down night observation is not lost for some time. While the flare is up, the open eye should be used to observe enemy positions.

(2) *Verey Light*. When a verey light is fired there is no time to go to the ground. It is better to freeze at the spot and close one eye.

e. *Methods to Negotiate Wire Fence:-*

(1) *Manual Method*. It is meant for creating an entry point through the formidable obstacles, stealthily by patrols for their passage: -

(a) *Breaching/Over Coming Border Fence*. The methodology of neutralizing the fence is given the following paragraphs. The specific method for overcoming the fence will be dictated by the prevailing situation. The means for breaking/ overcoming the fence are:-

i. Cutting by insulated wire cutters using rubber gloves.

ii. Partial destruction by artillery.

iii. Blowing up by explosive.

iv. Overcoming by using a plant / tank dozer.

v. Surmounting by using expedients held by infantry for assault across Ditch Cum Bund (DCB) complex.

(b) The searchlights and the energized fence pose peculiar problems while breaching the fence. Therefore, following activities may precede the breaching operation a night earlier or on the same night:-

i. Destruction/neutralisation of search lights by sharp shooters /artillery fire.

ii. Destruction/damage of source of electricity supply i.e., transformer, high-tension wire/ electric poles et cetera.

iii. Illumination of the area of interest to neutralise the use of NVDs by the enemy at night.

(c) *Parties Employment for Crossing:-*

i.	Commander	-	1(Officer/Junior Commissioned Officer/Non Commissioned Officer)
ii.	Cutting Party	-	2 (Numbers 1 & 2)
iii.	Pulling Party	-	2 (Numbers 3 & 4)
iv.	Pillar Reducing Party	-	2 (Numbers 5 & 6)

(d) *Equipment Require for Crossing:-*

i.	Insulated wire cutters	-	7
ii.	Rubber Gloves	-	7

iii.	Rope 1-1/2 inch Circumference (20 Metres long)	-	2
iv.	Picks	-	2
v.	Crow Bar	-	1
vi.	Electric Tester	-	1
vii.	Compass	-	1
viii.	Binocular	-	1
ix.	Night Vision Goggle	-	1
x.	Bamboo - 2.5 Metres Long	-	1

(e) *Drill for Crossing:-*

i. Step -1. Move to checkpoint.

aa. *Commander:-*

(i) He will direct the parties to approach the site through covered route. The parties will halt at checkpoint 400 Ms short of the pre-determined point of breach.

(ii) He will check the equipment of each party, coordinate actions at his own and that of the parties as per the drill and give last minute instructions to the parties.

bb. *Cutting Pulling and Pillar Reducing Parties.* The parties will ensure that they are clear about their task and are properly equipped for the assigned task.

ii. *Step - 2.* Move to the point of breach and passage clearance.

aa. *Commander:-*

(i) He will neutralise the energized electric wire if it is

ahead of fence using bamboo with hook. He will check the fence with the help of electric tester to caution the parties in case the energized wire forms part of the fence. To avoid risk, whether fence is energized or not, all parties will handle wire using rubber gloves.

(ii) On reaching the point of breach, he will indicate to Numbers 1&2 of cutting party the exact location for cutting the fence.

(iii) He will indicate the central row angle iron pillars to pillar reducing party, which are to prepare these for pulling by exposing the concrete plug.

(iv) He will supervise the passage clearance.

bb. *Cutting and Pulling Parties:-*

(i) Cutting party Number 1 on left and Number 2 on right will cut wire using insulated wire cutter initially in the outer row of the fence.

(ii) Pulling party Number 3 and 4 will pull the cut/loose wire towards left and right respectively and will prepare a passage.

(iii) Cutting party Number 1 and 2 will move ahead through the passage and cut central and subsequently inner row wires and similarly pulling

party will pull the wires of central and inner rows.

cc. *Pillar Reduction Party*. The party will prepare the central angle iron pillars falling in the way of the passage for pulling out by loosening its foundation with the help of picks and crow bars, will tie ropes around the pillar and pull it also using additional manpower of cutting party.

Note:- If pillar reduction party is not employed, the pillars may be reduced subsequently with the help of dozer.

(2) *Explosive Method*. It is the most reliable and quickest method various options available are:-

(a) *Option – 1*. Two-stage demolition:-

i. *Stage – 1*. It consists of blowing the gaps through the fence.

ii. *Stage – 2*. Demolition of left over angle iron pillars through the gap.

(b) *Option – 2*. One stage demolition i.e., blowing up of gap though the fence including all angle iron pillars in one go.

(c) *Option – 3*. Explosive means of minefield breaching i.e., Baby Viper/Giant Viper/Mine Clearing Line Charge (MICLIC) can be used where fence is augmented with minefield.

(3) *Mechanical Method*. Dozer/tank dozer/tractor with bucket and backhoe can be used to clear passage though the fence.

(4) *Expedient Method*. Infantry holds a variety of expedients like ladders, planks, life jackets; banglore torpedoes are commonly used during assault operations. Innovative employment of these expedients can prove handy for passage of assaulting/infiltrating troops through the fence. Owing to the large number of such expedients and varying circumstances it is cumbersome to lay down exact

drills/instructions for use of these expedients, which will largely depend upon the inventiveness and ingenuity of the commanders and troops in contact with the fence. Each situation will warrant minor changes in drill and configuration of parties. Therefore, suitable modification may be made in the drill/parties and equipment depending upon the type of expedient.

31. **Action on Being Ambushed**. If surprised, possible courses of action are:-

a. The patrol leader throws a smoke grenade and the patrol runs a pre-arranged distance or follows the leader in the direction he shouts.

b. An immediate physical assault, firing weapons. This is the most practical and appropriate method.

c. Get down, fire weapons and throw grenades until the leader signals on his whistle, then runs as in sub-paragraph ‘a’ above.

d. If enemy opens up from a distance then on orders to the patrol leader, the Light Machine-Gun group will open fire on enemy from one direction and the rest of the patrol will assault from another direction.

Final RV

32. The final RV is selected near the objective and the patrol enters it according to a laid down drill. All round defence is taken-up in the final RV. If there are any changes in the plan the patrol is briefed. After completing the task on the objective the patrol returns to the final RV.

Firm Base

33. A fighting patrol may set up a firm base from which one or more, reconnaissance patrols go out and return. This is done especially in jungle or mountain warfare. It is a help to reconnaissance patrols it they are escorted part of the way out and is useful for casualty evacuation.

Approach to the Objective

34. On approaching the objective, movement will be slow with frequent halts. Any unnecessary noise at this stage might betray the patrol. The following methods of movement are generally used:-

a. *Leopard Crawl*. In this method the man moves on his elbows and knees and is very helpful for low cover. To move forward, use alternately the left elbow and right knee and the right elbow and left knee.

(1) *Leopard Crawl with LMG*. Same as above. Push the LMG with hands as far forward as possible and then drag yourself up to it.

(2) *Leopard Crawl in Pairs.* Two men move together by leopard crawl and take the LMG between them.

b. *Cat Walk*. In this method the man will move forward on his hands and knees. The ground in front must be searched by one hand for stones, twigs and dry leaves et cetera before every move.

c. *Kitten Crawl*. Correct information can be obtained only by going near the enemy. In this method of movement a lot of time is consumed but there is no noise. The man lies down on his stomach, searches the area in front with one hand, then slightly lifts himself on his toes and inside of elbows and pulls himself forward. This is a very tiring method.

d. *Side Crawl*. Lie down on any side, hold the gun from the barrel with your upper arm and put the butt on your lower foot. Move forward on the elbow of your lower arm and your upper foot.

e. *Monkey Crawl*. In this the man moves on his knees and hands. It is a comparatively quicker method of movement and can be used for a cover about two feet high.

f. *Knee Crawl*. If the cover is about waist high then the men should walk on their knees, taking some support by putting the closed fist on the ground.

Fire Support

35. Fire support can be called for over the wireless or it may be pre-planed. Possible fire tasks in support of a patrol include:-

a. Distracting the enemy's attention while patrol crosses an obstacle or approaches the objective.

b. Helping a patrol to extricate itself in an emergency.

c. Support on the objective.

d. If the patrol loses its way then artillery fire can be brought down. The patrol leader may ask for fire on the objective or on a bound. This will help him to find the way. If he cannot find the way like this, then he will get fire on two bounds and take their bearing by a compass. He will then plot the back bearings on the map and get his own position. From this position he can read the bearing to the objective; set the compass accordingly and move towards the target.

Casualties

36. All casualties must be brought back, not only for reasons of morale, but also to deny information to the enemy. Wounded must be attended and made comfortable as soon as possible. A man wounded on the way to the objective may have to be left to be collected on the way back; it may be possible to summon another patrol by wireless to pick him up if he is left by an easily identified feature. When men are wounded on the way back, the problem is not so great. Unless a man is wounded, it is usually possible for him to be carried between two men, sitting on a rifle. If he is badly wounded, a stretcher must be improvised with rifle slings and wood or the available material.

37. A plan must be made for substitution within the patrol of the key members, such as patrol leader, NCO, and radio operator in the event of their becoming casualties.

A Patrol Standing By

38. It may be necessary to have a patrol ready and briefed to follow one or more patrols already out. This can be useful when:-

a. There is a particularly important and difficult task facing a patrol: should the patrol fail or run into serious difficulties, the patrol standing by is sent out to complete the task.

b. A patrol has had casualties and needs help to bring them back.

c. A patrol has taken prisoners and to escort them would stop the patrol completing its tasks.

Debriefing

39. As soon as the patrol returns, the person who briefed the patrol will carry-out its debriefing. Initially only the patrol leader will be debriefed and the patrol will be sent to rest. When the patrol has rested then all the members will be interrogated, in which the patrol leader will narrate all the incidents in the order they occurred. After the patrol leader has told every thing then if any one has any thing to add, will say so.

Patrol Report

40. Although it may be customary for a patrol leader to report verbally to his company commander as soon as he returns, a written report of the patrol must always be made for record purposes. The patrol report will be completed by the officer interrogating the patrol with the patrol leader and will, on completion, be forwarded to the next higher authority.

Contents of a Patrol Report

41. Some of the points which must be covered in a patrol report:-

a. Size and composition of patrol.

b. Task (s).

c. Time of departure.

d. Time of return.

e. *Routes*. Route out and in.

f. *Terrain*. Description of the terrain i.e. dry, swampy, jungle, thickly wooded, high brush, depth of ravines, condition of bridges including types, size, strength and effect on armour, vehicles and foot columns.

g. *Enemy*. Strength ,disposition, condition of defence, equipment, weapon, attitude, morale, exact location, move and any changes in disposition. Reporting priority to be given to nuclear aspects. The time and place where enemy activity was observed are included here.

h. *Map Correction*. Any map correction including significant nuclear damage to terrain.

j. *Miscellaneous Information*. Include aspects of NBCW.

k. *Results of Encounters with Enemy*. Enemy prisoners and dispositions, identifications, enemy casualties, captured documents and equipment.

l. *Condition of Patrol*. Include disposition of any dead or wounded.

m. *Conclusion and Recommendations*. Includes to what extent the mission was accomplished and recommendations as to patrol equipment and tactics.

Signatures of
Patrol leader

Name

Rank
Unit

n. Additional remarks by interrogator.

Signatures of
Interrogator

Name

Rank

Unit

Note: - The report can be in letter form, or it can be made out in manuscript by forwarding units and signalled in message form.

SECTION 41 - **RECONNAISSANCE PATROLS**

Definition

1. These are patrols, of a minimum strength for the task, which gain information by observation and operate generally by stealth. They avoid combat except for self-protection or to take advantage of an unusual opportunity.

Roles

2. The roles in which reconnaissance patrols may be employed include:-

a. Collecting information on features, tracks and the state of the ground.

b. Obtaining details of enemy minefields, other artificial obstacles and the extent of enemy positions.

c. Locating enemy machine-guns and defensive fire areas.

d. Investigating noises made by the enemy, enemy habits and patrol routes.

e. Checking own wire or minefields at first or last light.

f. Acting as listening posts to give early warning of enemy approach and with the ability to call down fire.

Strength and Composition

3. Reconnaissance patrols will usually operate by stealth and therefore, it should have the minimum strength required to accomplish the task. This could be about three to four persons, which would include one officer, JCO or NCO and two to three men for the protection of the commander.

Weapons and Ammunition

4. This patrol may carry rifles and SMGs/MP5A2 with pouch ammunition. In addition the patrol may carry NVD, LRF, mine detector and all the members of patrol should have High Explosive (HE) and smoke grenades.

Dress

5. Dress will be in relation to the weather but should be such which does not produce noise. Shoes should be rubber soled to avoid noise. Web equipment should not be carried and instead of steel helmet they should either wear cap working, beret or cap comforter.

Equipment

6. The patrol should carry only those things, which will be required during their operation. This will include rope, map, compass and binocular, NVD. Rifle sling should be carried out but it should be tied around the waist.

Orders

7. For detailed order of reconnaissance patrol, refer to Annex “G”.

Formation

8. The formation should be such that the patrol leader can easily exercise control. For formations see figure 11 below:-

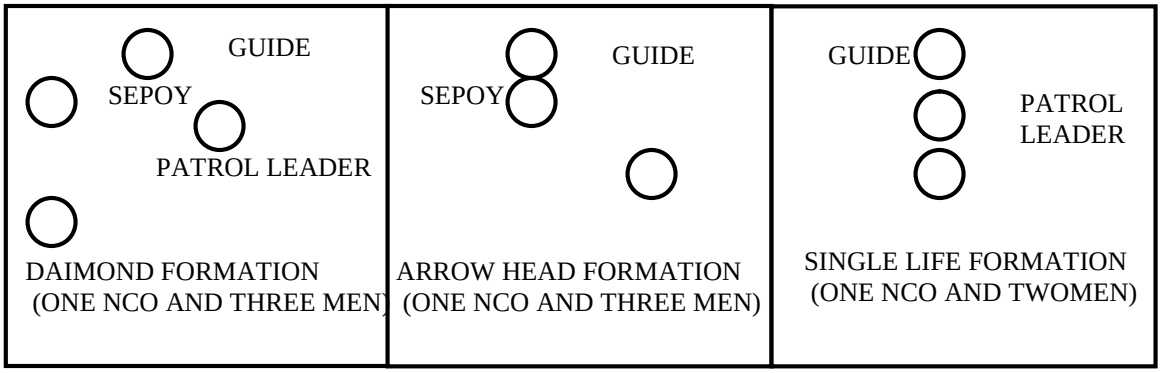


Figure 11 - Formations for Reconnaissance Patrols.

Patrol Action

9. Reconnaissance patrols will leave own position after rehearsals and final inspection. The patrol will move very carefully and on reaching near the final RV it will go into position. In view of the enemy night vision capability, the final RV should be selected at a distance of about 400 metres to 500 metres from the objective. The patrol leader will take one man with him and carry-out reconnaissance of the final RV. If there is no enemy present there, the rest of the patrol will move in. If enemy is present in the RV then an alternate RV should be selected roughly at an equal distance.

10. The patrol leader will have two men in the RV to cover his move. With one man he will start moving towards the objective while observing the area with his binocular/NVD. If the objective is behind some cover and can not be observed from a distance then on reaching closer to the cover, the commander will crawl for about 10 metres and the other man will cover his move. Then he will crawl and the commander will cover him. When he reaches the commander, the commander will observe through the binocular/NVD and then move forward. In this manner they will carry-out reconnaissance of the objective and try to get maximum information using their ears and the NVD/binocular. It is possible that for this task, they will require an hour or more. After this reconnaissance, they will return to the RV. The 2IC will check them using the password. Here the patrol leader will brief the whole patrol on whatever information he has been able to get. He will now take the other sepoy and go for reconnaissance of the objective from another direction using the same technique.

11. If the patrol is of one leader and two men then one man will not be left alone in the final RV but all three will go for reconnaissance. If time is short then patrol leader with one man goes from one direction and the 2IC with the other

man go from the other direction. On reaching the firm base the party which reaches first will check the other party on their return, using the password. Then both parties brief each other. It is essential that all information is known to all the members of the patrol so that if the patrol is ambushed on the way back, any person who reaches own position may be able to give the information.

12. During reconnaissance a patrol may see or hear the following:-

a. *Finding Enemy Position by Fire*. When a weapon fires, the firing sound is called “thump”. The noise produced when the bullet goes through air is called “Crack". The crack will always be heard before the thump as the bullet travels faster than sound. The further the weapon, the greater the time lags between the crack and the thump. By judging this time lag we find out approximately how far the crack is heard we should start counting. The counting should be in such a manner that count of four is completed in one second. Then if the count is up to 4, the distance would be approximately 400 metres. If the count is 6, the distance would be approximately 600 metres.

b. *Noises*. During night the ear is more effective than the eye. A slight noise at night can be heared at a long distance. So a patrol must try to listen to all the noise coming from enemy position so that they can form a rough idea about what the enemy is doing or preparing to do. Some types of noises that can be head are:-

(1) Digging.

(2) Driving a picket.

(3) Changing magazine, loading or unloading weapons.

(4) Changing sentries.

(5) Talking, coughing, sneezing et cetera.

(6) Metallic articles touching each other, like steel helmets, weapons et cetera.

(7) Laying or cutting wire.

(8) Walking,

(9) Filling sand bags.

(10) Loading/unloading vehicles.

c. *Things that can be Observed at Night*.

(1) Lighting a match and glowing end of cigarette.

(2) Lighting a torch.

(3) Light from command post or signal exchange.

(4) Movement on sky line.

(5) Shining of weapons kin moonlight.

(6) Flash produced by firing of weapons.

(7) Back blast of anti-tank weapons.

13. **Actions on the Objective**. Depending upon the size of the objective, the patrol leader will split the patrol into two parties. The parties after moving out from the firm base may select a centre line and move along it, go to a junction point from where they move outwards and carry-out reconnaissance of the objective, again return to the junction point and exchange the information acquired and then finally return to the firm base. They may also go to the extremities of the adjective after moving out from the firm base. Then the parties move inwards to a centre point, exchange the information about the objective then again move outwards to finally withdraw to the firm base.

SECTION 42 - **FIGHTING PATROLS**

Definition

1. These are patrols organized for a particular task with sufficient strength and armament to accept combat.

Roles

2. The roles in which fighting patrols may be employed include:-

a. Denying enemy patrols freedom of action in No Man’s Land.

b. Driving in enemy protective patrols.

c. Interfering with enemy working parties.

d. Carrying out raids.

e. Capturing prisoners for identification purposes.

f. Tank hunting.

g. Laying ambushes.

h. Escorting stretcher parties.

j. To harass the enemy and divert his attention.

k. To escort reconnaissance parties of supporting arms.

l. Securing a foothold over obstacles for further operation by a larger force.

Strength and Composition

3. The fighting patrol may have to fight to accomplish its mission, so it should have sufficient strength for this. This strength is usually from one section to one platoon. The patrol leader may be an officer, JCO or NCO. The strength and composition will always be related to its role and the number, which can be effectively controlled by one man.

Weapons and Ammunition

4. Weapons and ammunition should be given according to the task. It will usually carry LMG, rifle, MP5 and grenades but may also carry rocket launcher, mines, MG et cetera, if required.

Equipment

5. Equipment will be carried according to the task e.g., wire cutter, flame thrower, NVD, binocular, compass, night march chart, rope, white luminous tapes, slings to make stretchers et cetera.

Dress

6. Dress will be according to weather. It may carry web equipment and steel helmet.

Orders

7. For detailed orders of a fighting patrol see Annex “H”.

Formation

8. See Figure 12 below. In case of a fighting patrol of one leader and fifteen men, it could be:-

a. *Navigation Party*. It will consist of four men.

(1) *Guide*. He will carry rifle and a light stick to detect trip flare wires. He will wear a white patch on the back.

(2) *Pacer No.1.* He will carry rifle and some pebbles to help him count paces.

(3) *Pacer No.2.* Same as Pacer No. 1.

(4) *Party Commander*. In addition to personal weapon he will have compass and night march chart.

b. *Action Party No.1*. It will consist of three men:-

(1) *LMG NO. 1*. Will carry LMG.

(2) *Rifle Man.* In addition to the rifle he will carry extra ammunition for LMG, a torch and a blanket.

(3) *Party Commander*. Will carry personal weapon.

c. *Action Party No.1*. Same as action party No.1. Only the rifleman will carry a rope and socks.

d. *Covering Party.* It will consist of three men.

(1) *LMG No.1*. Will carry LMG.

(2) *Rifle Man.* Personal weapon and extra ammunition for LMG.

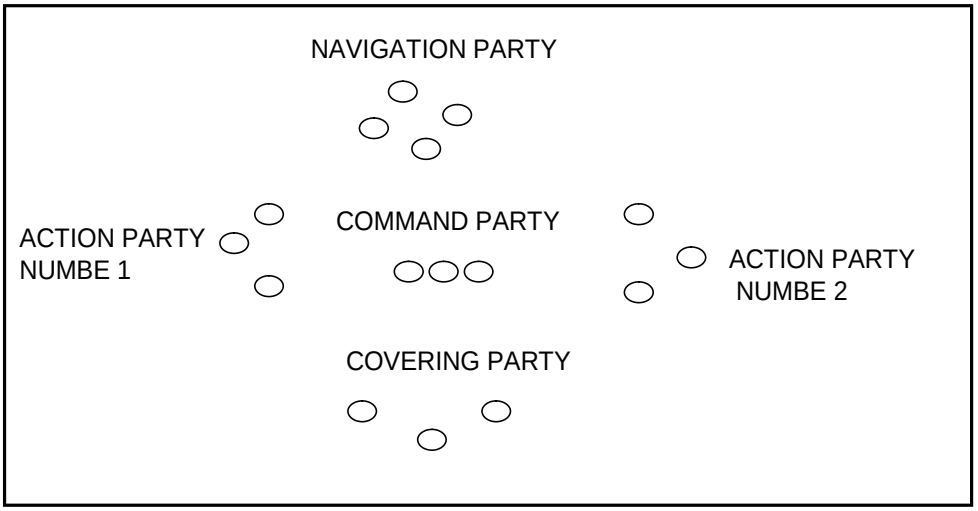
(3) *Patrol 2IC.* Will carry personal weapon.

e. *Command Party.* This will consist of 3 men.

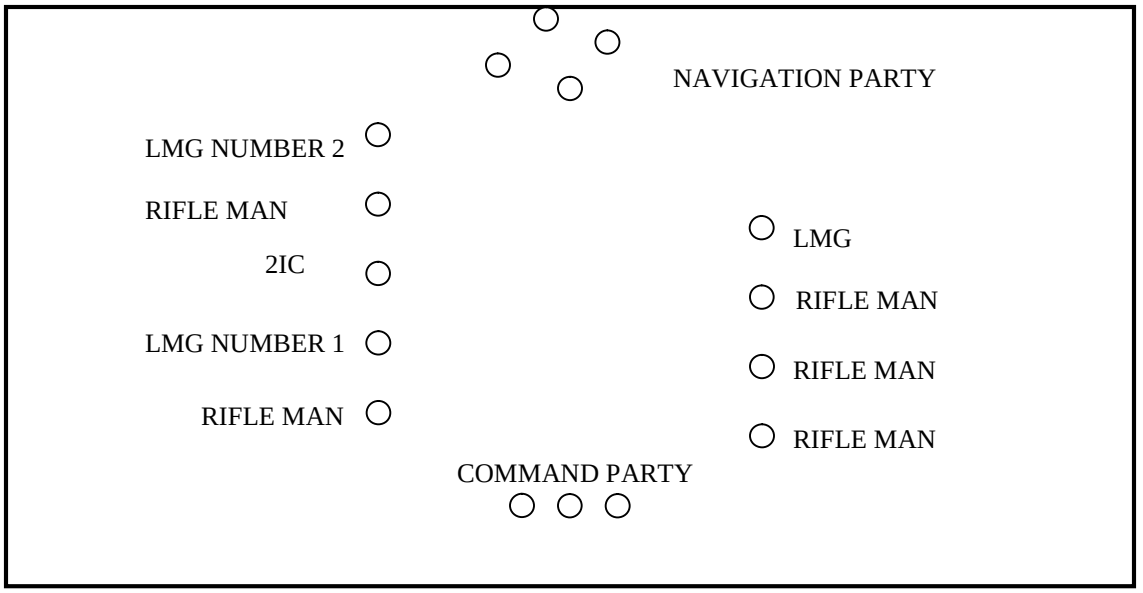
(1) *Patrol Leader*. Personal weapon, map, binocular, wire cutter and morphia injections.

(2) *Signaller*. Wireless set and personal weapon.

(3) *Runner.* Will carry rifle.



DIAMMOND FORMATION



ARROW HEAD FORMATION

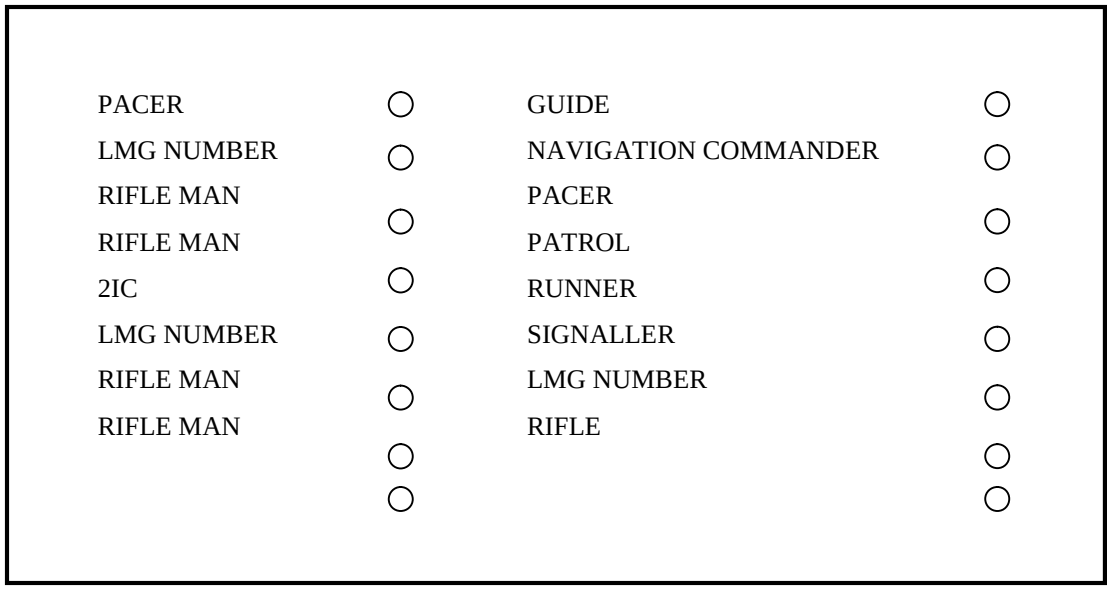


Figure 12 – Fighting Patrol Formations

Action on Objective

9. Patrol should do the following on reaching near the objective:-

   a. Carry-out reconnaissance of the whole area, especially final RV to ensure that no enemy is present.

   b. After a brief reconnaissance, decide how the patrol is going to do its task.

   c. Brief the patrol.

Technique of Capturing a prisoner

10. **Capturing a Prisoner from a Listening Post**. A lot of preparation, time and caution is required to catch a prisoner. It is very difficult to catch a prisoner from main enemy defences and hence we must find out about enemy standing patrols, listening posts, protective detachments et cetera. Reconnaissance Patrols can be used to find their routes, timings, habits et cetera and then a fighting patrol can be sent to catch a prisoner. This can be done by either laying an ambush or by raid. Usually the best place to capture a prisoner is

a listening post. As a guide the following procedure may be adopted for capturing a prisoner from a listening post by a fighting patrol having a strength of 16:-

a. *Formation*. As given in para 8 above.

b. *Parties*

(1) Firm Base party.

(2) Cut-off party.

(3) Action party.

(a) Action group No.1

(b) Action group No.2

c. *Firm Base Party*. During move it will have strength of one party commander and two men but on reaching Firm Base the wireless operator will also join the party. The 2IC will command this party.

d. *Cut-Off Party.* Its strength will be one party commander and three men and two LMG's will be carried. One man will carry a rifle and will act as No.2 for both the LMG numbers.

e. *Action Party.*

(1) *Action Group No.1*. Strength will be one commander and two men during move but on reaching firm base the patrol leader will also join this party.

(2) *Action Group No.2*. Strength during move will be one commander and two men. On reaching the firm base the patrol leader's runner will also join this party.

f. *Weapon and Equipment.* This patrol will carry LMGs, rifles and MP5s. Action party will also carry bayonets or knives. Each man will have a grenade. The other equipment will include compass, binocular, NVD, rope, socks half filled with loose earth, wireless set (PRC-77/walkie-talkie), blanket, torch et cetera.

g. *Action on Objective.* As soon as this patrol reaches the firm base, the patrol leader leaves the patrol with the 2IC and goes for reconnaissance along with all party commanders. After this they come back to take the whole patrol for action, except the firm base party.

(1) *Firm Base Party*. During patrol action this party will have the following tasks:-

(a) Protection of firm base.

(b) On reaching firm base contact main position and inform them of the situation so that artillery guns could be prepared to give fire support.

(c) As soon as surprise is lost this ‘party will call for artillery fire on enemy main positions so that they may not sent any help for the listening post. A code word will be prefixed for this. This artillery fire will be HE mixed with smoke so that the action and withdrawal of the action party can be covered.

(d) To identify all parties on their return by a pass word.

(e) To control the parties coming in and to deploy them correctly in the firm base.

(2) *Cut-Off Party.* During the initial reconnaissance the patrol leader will show this party its place. This will be between the listening post and the main defences. During action it will have the following tasks:-

(a) Put both LMGs at a place where they can effectively fire at enemy main defences.

(b) Open fire as soon as surprise is lost so that no support can come from the main positions and no one from the listening post escapes.

(c) After completion of mission, to cover the withdrawal of action party on the signal of the patrol leader and then withdrawal to firm base.

(3) *Action Party*. Action group No. 1 with the patrol leader advances towards the listening post in pairs, each pairs covering the other. Action group No. 2 moves behind them at a distance of about 15 metres to 20 metres. Their method of movement will be similar to action group No.1. Each pair of action group No.1 is detailed for a sentry. If they can reach the sentry without being detected they will either close his mouth or hit him on the head with a stick a cutting tool or a sock filled with loose earth. If necessary the bayonet or knife may be used. During this the action group No.2 will cover them. If they cannot reach the sentries

undetected, they will keep crawling forward till surprise is not lost. The moment surprise is lost one man from each pair throws a grenade and the other fires at the sentries. Meanwhile action group No.2 will get up and move in the line with actin group No.1. The cut-off party will also open fire on the main position. As soon as action group No.2 reaches action group No.1, the whole action party assaults the post. The patrol leader and another man either overpower a man or knock him out with a blow on his head. As soon as prisoners are taken and the rest are killed, the action party withdraws to the firm base. After covering the withdrawal of the action party, the covering party will also withdraw. The 2IC will identify them and quickly deploy them in the firm base.

11. **Capturing a Prisoner from a Signal Line Party**. In this method the patrol will infiltrate between the gaps of enemy defences, go to the rear and locate a telephone wire. This wire will be cut and the patrol will lay an ambush for the signal line party. This party will usually consist of 2 to 3 men and as soon as they start repairing the wire the patrol will quickly overpower them.

12. **Capturing a Prisoner from a Standing Patrol**. In a standing patrol there will usually be two sentries in different trenches and the rest of the patrol will be resting some distance away. Consequently there will be need for two cut-off parties; one for the main position and the other for the man taking rest. Action will be the same as in catching a prisoner from a listening post.

13. For verbal orders of a fighting patrol for capturing as prisoner see Annex "J".

Reconnaissance by Fighting Patrol

14. At times a fighting patrol may be sent to gain information. The strength will depend upon the situation and task. An example for guidance is given below, where the patrol strength is 16.

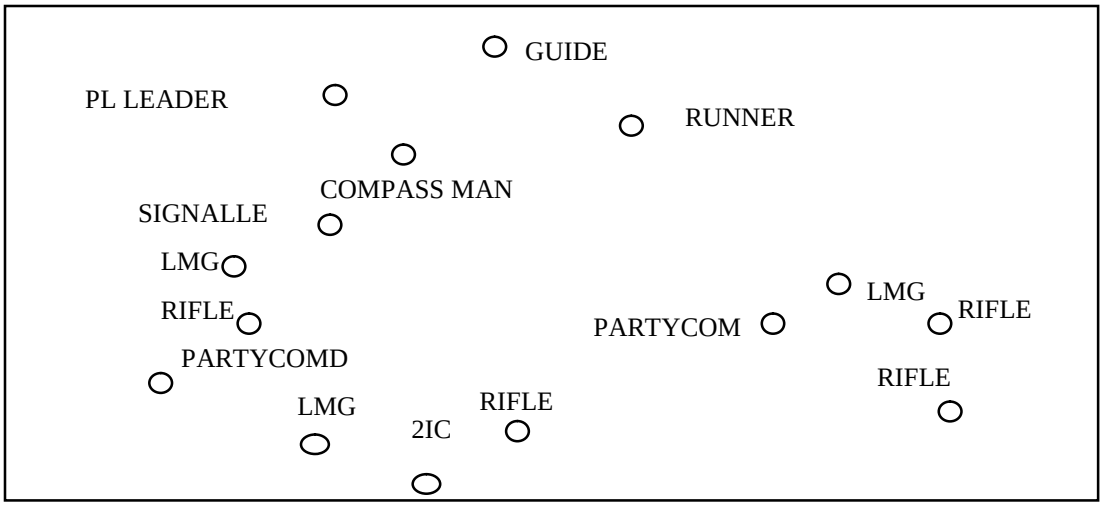


Figure 13 - Reconnaissance by Fighting Patrols

a. *Parties*.

(1) *Firm Base Party*. Strength will be four, including wireless operator. 2IC will command this party.

(2) *Action Party*. Small group of this party can be made according to the requirement. The minimum strength of each group of Action Party should be one commander and three men. The group, which will carry-out reconnaissance of the most important area, will be led by the patrol leader.

b. *Weapons and Equipment*. In addition to the other equipment it must have a PRC-77 for the firm bade party. One LMG, one binocular and a NVD for each group of Action Party.

c. *Action on the Objective (Figure 14)*. If the patrol leader thinks essential, he will send the parties for initial reconnaissance. The left group will carry-out reconnaissance of the objective, the centre group will reconnaissance the centre of the objective and the right group, the right side of the objective. After a fixed time all the parties will return to the firm base. If the leader desires to increase reconnaissance time he will inform the groups on wireless. If any important thing has been seen noticed, the whole patrol will be briefed about it. After this, all parties will return to their places for detailed reconnaissance. After some time the 2IC will call for High Explosive (HE) and illumination shells onto the objective. All groups will make use of the light to observe the objective. As soon

as the light dies down, the group with the patrol leader will fire a burst on enemy post. This will be a signal for all the groups to open fire and attract the attention of the enemy. This action will be done while the groups are under cover from enemy fire. When the enemy opens up, the groups will observe its weapon positions. If enemy opens up with artillery then its DFs can also be located. After a predetermined time, the groups will withdraw to the firm base. If enemy follows, then fire and move will be used. To cover their withdrawal, 2IC will request for bringing HE and smoke fire onto the enemy position. The patrol leader on reaching the firm base will ask for information from all the groups. The whole patrol will be briefed accordingly.

Note: - If the action group does not return after a predetermined time, then the firm base party will return to own position under the 2IC.

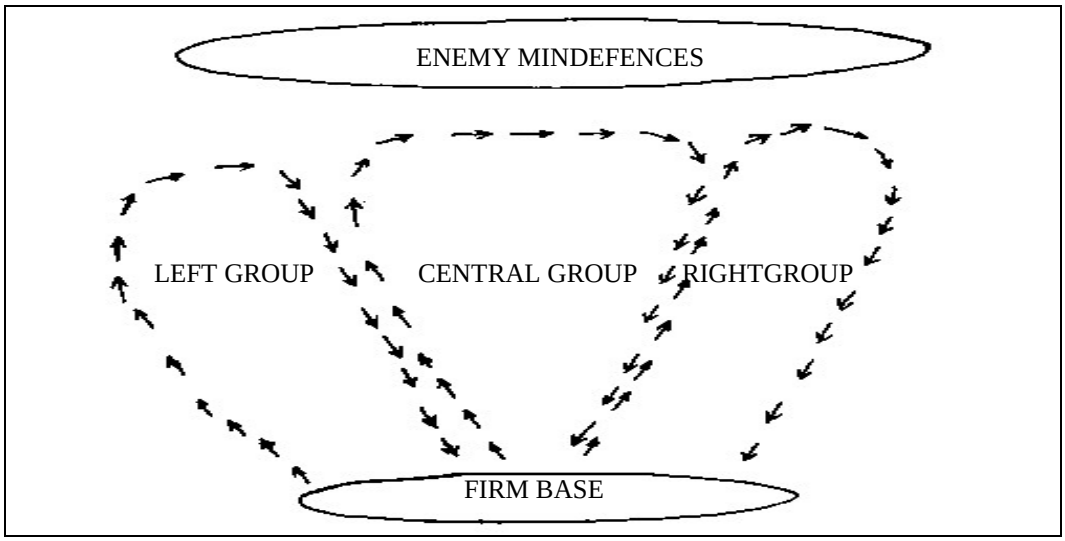


Figure 14 – Action on Objective

SECTION 43 - **STANDING PATROL/ESCORT PATROL/LINK PATROL**

Standing Patrol

1. The strength of a standing patrol will be according to its task. It is usually of a section strength and can be given the following tasks :-

a. To keep surveillance on likely enemy approaches, Forming Up Places (FUPs), defiles, bridges et cetera. It will usually keep under surveillance places, which cannot be observed from the main position.

b. In built-up area it sent to:-

(1) Give information of enemy movements.

(2) Occupy a prominent feature to afford protection to the attacking force and deny enemy observations.

c. To guard own minefield gaps.

d. To guard against enemy infiltrating parties through own gaps.

e. To protect own FUPs.

2. The standing patrols are generally employed ahead of the defensive mine fields. As a guideline they may operate at a distance of 1000-1500 metres from the main defensive position.

3. They may also be provided with suitable communication to include both line and wireless. The patrol may also have additional equipment with it depending upon the nature of task and degree of threat e.g., they may have NVD/NODS, laser range finders, medium/long-range anti-tank weapons and automatics.

Fighting Technique of Standing Patrol

4. The standing patrol also moves like other patrols. After reaching the given area, it will select a place for itself and may also select and alternate place. It will open fire on the enemy and move to alternate position before the enemy can attack. The standing patrol should cause maximum casualties to the enemy and stop him from coming forward. Withdrawal routes should be predetermined. The standing patrol will withdraw only in case the enemy is too strong and even only with the permission of the commander who has sent them. This patrol must have telephone or wireless contact with the main position. If these patrols are sent across the minefields, they must carry few mines to plug those gaps between the minefields which they had used for going across the minefield.

Escort Patrol

5. At times, technical information is required e.g., type of bridges, minefields, water obstacles, routes for tanks et cetera by engineers or armoured

corps. To protect them an escort patrol may be sent. The specialist representative will only accompany the patrol to gain the information required by him and he will not command the patrol. The patrol leader always commands the patrol, even if he is junior. Escort patrol may also be used to protect forward artillery observers. The strength of this patrol is generally one section.

Listening Posts

6. Listening posts will usually have strength of 2 to 4 men each including the patrol leader. Their location is usually 350 to 400 metres ahead of own main positions so that they can give early warning of enemy's approach. Since their tasks are very tiring, they must be frequently changed. If this is not possible than two sentries should be posted at a time so that they can cover the area effectively. They will carry their personal weapons.

7. **Communication.** It is essential to have some kind of communication with the listening posts. A telephone set or PRC-77 may be used. They will use pre-arranged nicknames while giving any information.

8. Only the person who has detailed a listening post can order it to withdraw. It is better that the listening post is sent in front of its own platoon so that they can be easily recognized on their return. Withdrawal routs for them will be pre-determined and the troops through whose position they ate withdrawing will be warned before the listening post withdraws.

Link Patrol

9. At times it may be necessary to guard the vulnerable gaps between units and subunits. Therefore, a link patrol may be planned and sent by either or both of them to ensure that the area is surveilled and entry to enemy infiltrating parties is denied. The difference between a standing patrol and a link patrol is that the former is static in nature and the later accomplishes the task by movement.

CHAPTER 9

AMBUSH, RAID, TANK HUNTING AND ROAD BLOCKS

SECTION 44 – **AMBUSH**

Definition

1. An ambush is a surprise attack from a concealed position on a moving or temporarily halted target. It may include an assault to close with and destroy the

target, or the attack may be by fire only. It does not require the ground to be seized and held. It enables a small unit with few weapons and equipments to harass or destroy a larger and better-armed unit. Ambushes may have as their target men, vehicles or tanks.

Aim

2. The aim of an ambush must always be borne in the mind. This will be:-

   a. Destroy the maximum of the enemy troops, vehicles and animal transport.

   b. Obtain information of intelligence value i.e identifications and documents.

   c. Harass the enemy causing him to use additional troops for protective duties.

   d. Deny enemy patrols freedom of action in no-man's land.

Types

3. There are two types of ambushes:-

   a. *The Hasty Ambush*

      (2) This is an ambush, forced upon a commander by circumstances beyond his control. This type of ambush is normally laid by a fighting patrol, which sets out for a particular task and suddenly comes across an enemy patrol, which it cannot avoid. Success is dependent largely upon the wit and action of each individual member of the patrol who must have had previous training to act correctly by instinct and with the minimum of orders or instructions.

      (3) This type of ambush unless it is very skilfully executed will almost jeopardise the main task of the patrol by giving away surprise and should be avoided by the patrol commander unless it is forced on him by circumstances beyond his control. If this happens every effort must be made to ensure that the ambush is a swift silent affair so that after it, the patrols can carry on with their main task unhindered.

   b. *The Planned Ambush*. Type of ambush that is based on thorough intelligence and detailed planning. The force employed for laying

such an ambush is basically a fighting patrol composed of selected men and a leader.

Categories of Ambush

4. Following are the categories of ambush:-

a. *Point Ambush.* This is the category of ambush in which troops deploy to attack a single killing zone; i.e. complete target can be covered in one killing zone/site.

b. *Area Ambush.* In this category of ambush the troops are deployed to attack multiple killing zones i.e. the complete target cannot be covered in one killing zone/site.

Characteristics

5. Successful ambushes have the following characteristics:-

a. *Surprise.* This is achieved by:-

(2) Silence and stillness.

(3) Intelligent layout, sitting and concealment.

(4) Carefully selecting positions on inconspicuous looking terrain.

b. *Control.* It is difficult to achieve it due to factors like dispersion of the ambush party, need for silence and position of the leader. It can however be achieved by: -

(2) Careful briefing and detailed orders.

(3) Repeated rehearsals.

(4) Field signals.

(5) Positioning of ambush commander.

(6) Discipline.

(7) Individual Training.

(8) Battle Drill.

c. *Fire Power.* As the execution of an ambush is quick and violent, a high concentration of fire is imperative and it is achieved by: -

(2) Good orders.

(3) A good fire plan.

(4) "Shoot to Kill" with rigid fire control.

(5) Maximum use of automatics.

(6) Skilful use of mines and grenades.

d. *Simplicity of Plan*. Simpler the plan, greater the chances of success.

e. *Determination.* Once executed it has to be pressed without remiss and with ruthless determination.

f. *Self Discipline*. Training is necessary to develop the requisite patience and self-discipline. Without them an ambush may fail and cost lives.

Laying an Ambush

6. Having decided on the aim of the operation and the target, the laying of an ambush divides into four main phases: -

a. *Phase –I.* The preparatory phase, which includes the selection of the site, detailed planning, issue of orders and rehearsal of the whole operation.

b. *Phase –II.* The approach phase, which includes movement from friendly lines to the final RV.

c. *Phase –III*. The execution of the ambush, which includes the occupation of the site, carrying out of the ambush.

d. *Phase –IV*. The withdrawal of the ambush force to a selected RV, and, from there, back to own position.

Selection of Site

7. In selecting a site, the following points must be borne in mind: -

a. The size and type of target that is to be ambushed.

b. Method of movement of the target, i.e., on foot or in MT.

c. Time of movement of the target, i.e., day or night.

d. Composition and position of enemy guards or escorts. This will decide strength, arms and equipment.

e. General conduct of the enemy guards or escorts and their habits.

f. Location of enemy reinforcements and other protective arrangements. This may force the ambush party away from what could be an ideal site for an ambush.

g. Tactical considerations, which will include: -

(2) Ground, which will allow the maximum view of the target either from the main position or by observers posted nearby; this will give early warning and enable the ambush commander to make any last minute adjustments to his positions. It must at the same time provide the ambush party with good cover from view.

(3) A good field of fire, normally at a short range to gain maximum effect.

(4) Good lines of withdrawal.

(5) A position, which restricts enemy manoeuvre, e.g., a defile or a road with hills on one side and a drop on the other.

(6) A position which renders the main target out of sight of those who are not intended to be ambushed, e.g., a bend in the road or a track with a spur jutting out.

Planning

8. Following are the planning considerations:-

a. The killing area should be where the enemy is forced to move on a narrow front and where he has to close up to maintain contact and control.

b. Sitting individual positions will vary but the light machine-guns must have priority. Exits must be sealed by fire, mines or natural obstacles.

c. A reserve must be detailed. This must be capable of moving and acting on its own initiative but is initially sited on the same side of the ambush position as the RV to be used on withdrawal.

d. Lookouts must be on watch at both ends of the position; they must be in contact with the main body by means of a cord or cable so that they can give a silent warning. There must also be cleared paths to their positions to allow silent relieves.

e. If the ambush is to last for 24 hours or more, an administrative area is necessary. This must be defendable, out of earshot from the

main position and have lanes to it for silent approach. A local supply of water is desirable. The RV and the administrative area will often, for convenience, be in the same place.

Orders

9. These are issued in two parts. As much as possible are given with the aid of a model before leaving the battalion area. Detailed positions and areas can only be settled on the actual ambush site. For ambush orders see Annex "K".

Occupation of the Site

10. The following points will be borne in mind when considering the occupation of the site:-

a. It is better to occupy the site at night to avoid observation. If occupation has to be carried out by day, it should be done in small parties by a good covered approach, if this is not available, day occupation should be ruled out.

b. Disturb the natural location as little as possible to aid camouflage. Digging, if any, must be camouflaged and the concealment of the whole party must be absolutely perfect.

c. No movement, smoking, coughing or talking must be allowed.

d. Positions of each individual must be so selected that accurate fire can be opened by the maximum number of weapons at the shortest range possible.

e. Always post a sentry at a good viewpoint to give early warning of the target's approach. Some time early warning from an Observation Post with a wireless set can also be arranged.

Execution/Conduct of the Ambush

11. The conduct of point and area ambush will be as following:-

a. *Point Ambush*

(2) The ambush party after reaching at final RV will establish security. The ambush party commander and the subordinate group commanders will conduct the reconnaissance of the objective area from a vantage point and then return to final RV. The security/flank protection elements leave the final RV first and secure the flanks of ambush site. When they are in their position, the rest of the parties leave for their

locations. Once all the elements are in position, the ambush party awaits the target. When the target approaches, the signal elements alert the ambush commander. The signal elements report the direction of movement, the size of the target and any special weapons or equipment carried. Upon receipt of the report the ambush commander alerts the other elements. When the major portion of the target is in the killing zone, the ambush commander gives the signal to start ambush. When it is not required to physically assault and the desired results have been achieved, the patrol leader gives the signal to withdraw to final RV.

(3) On the other hand when the target is required to be physically assaulted/searched, the patrol leader gives the signal to lift or to shift the fire. This is also the signal to start the assault/search. When the assault elements have finished their mission, the patrol leader gives the signal to withdraw to Final RV.

(4) On the signal to withdraw, all the elements withdraw to Final RV. Equipment and personnel are accounted for and the patrol moves back to friendly lines.

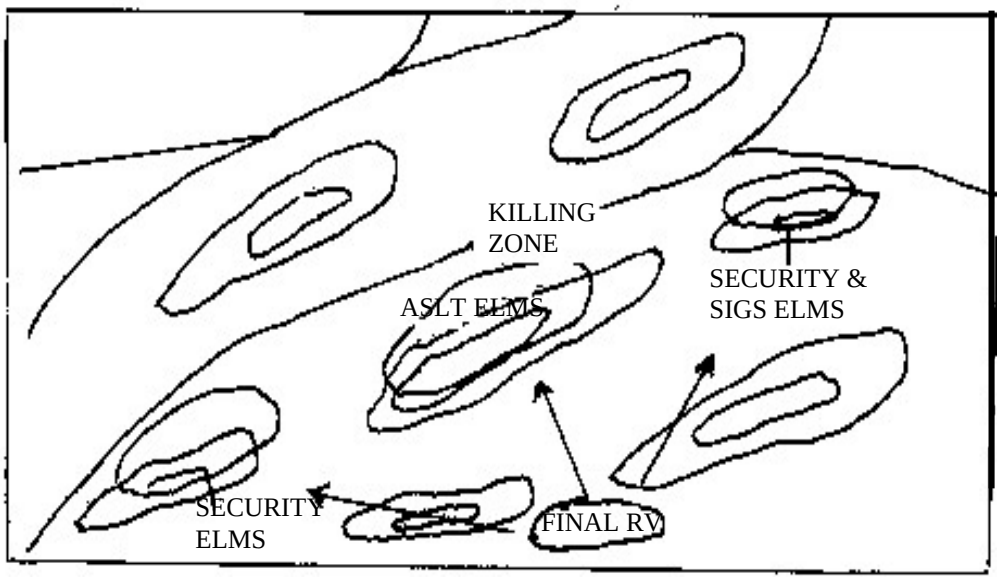


Figure 15 – Point Ambush

b. *Area Ambush*

(2) An area ambush is not normally conducted by a unit smaller than a platoon, the area ambush is required when the target cannot be covered by one ambush site or the size of the target is large enough to be covered in one ambush site. The area selected for such an ambush should offer several suitable point ambush sites. The platoon leader must select one central ambush site around which he can organize the out laying ambushes.

(3) Once the central ambush site has been selected, the platoon leader must determine the enemy's possible avenues of approach to and escape from this site. He, then selects and assigns sites to patrols to cover these avenues. The ambush sites are then selected/established as explained for a point ambush.

(4) Once the sites have been established, all enemy traffic going towards or away from the central ambush site is reported to the platoon leader - who is at the central killing zone until the central ambush has been initiated.

(5) Once the central ambush is initiated, the out laying ambushes prevent any enemy from escaping or entering the area. The actual conduct of ambush is the same as discussed for the point ambush.

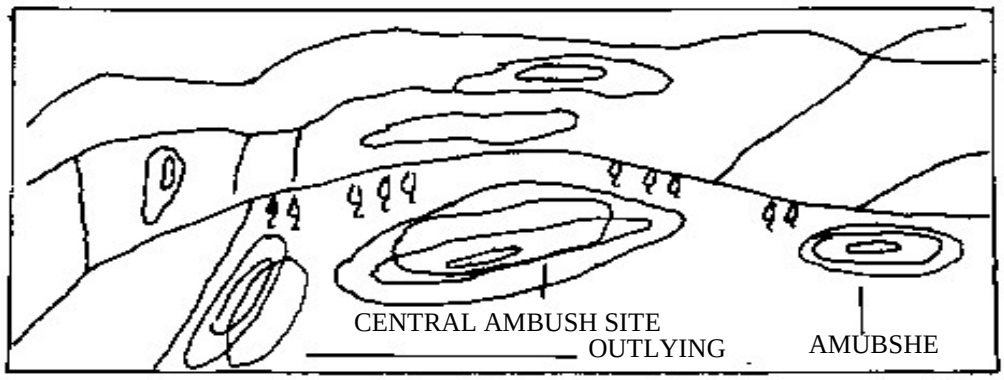


Figure 16 – Area Ambush

12. Following are essential to ensure success in this phase:-

a. Clear understanding of orders by each individual and their strict compliance by them.

b. No talking, coughing or smoking.

c. Patience, if the target does not appear on the scene at the estimated time. The party may have to sit for long hours and wait for the target, if men lose heart and become disinterested; they may fall asleep and be ambushed themselves.

d. Constant alertness for the signal.

e. Bold and swift action on receipt of the action signal.

f. Each individual must concentrate primarily on his own allotted task.

g. Each individual must use the weapon he is told to use.

h. The party must not become involved in a pitched battle.

j. Individuals must watch for the "break off" signal and withdraw from the site immediately after the action is over.

Withdrawal

13. Withdrawal is normally carried out in two stages – firstly from the site to a pre-selected RV at a reasonably safe distance (say from 750 to 900 metres) dependent upon the ground and the enemy dispositions, secondly from the RV to the base.

14. The signal to withdraw can be given by whistle, very light or smoke grenade. Groups then move independently to the RV as fast as possible, covered by a reserve. Time spent in the RV must be kept short to lessen the chances of being outflanked. The withdrawal must be fully rehearsed.

Battle Drill for an Ambush

15. Depending upon the type of the target to be ambushed and the tasks to be carried out an ambush force may be divided into the following groups/parties:-

a. *Action Group*

(1) *Forward Command Party.* This party includes command and communication elements controlling and coordinating the actions of the entire action group. It would normally consist of commander, and the signal personnel/runners.

The party may be conveniently located with main firing party.

(2) *Main Firing Party*. This party is responsible for destroying the advance guard, main body and rear guard. In certain cases the flank protection party may deal with the rear guard. The party should further be divided according to the enemy personnel moving or the number of vehicles i.e.:-

(a) Number 1 vehicle firing party.

(b) Number 2 vehicle firing party.

(c) Number 3 vehicle firing party et cetera.

(3) *Special Task Party*. The use of special task group or killer team in conjunction with the ambush is often desirable. The purpose of this group is to physically move through the ambush site and ensure the destruction of vehicles and material, search enemy dead and vehicles or any other duties considered necessary by the ambush commander. The action of this group must be well planned, rehearsed in detail and violently executed. It is usually of short duration. The nature of the ambush will usually dictate the strength and employment of this group.

(4) *Signal Party.* Two or three personnel with signal set or visual signal aids position themselves on a vantage point on the main axis of advance of the enemy column or convoy. They should be able to inform the command party well in advance so that final preparations for ambush can be made. On receiving the signal the commander would inform the main firing party commander and the road block party commander for final laying of the road block.

(5) *Road Block Party*. The party is responsible to create effective obstacles in front of the moving column and block its further advance. The type of road block would be in accordance with the type of target. Actual laying of the road block would be done after the force commander has received all correct signal from the signal party commander or when he deems necessary. Following are some of the suggested types of road blocks:-

(a) *Conventional*

i. Rocket Launcher/LMG.

ii. Anti-Tank/Anti Personnel mines.

iii. Trip wire/pull switch with explosive.

iv. Mine necklace.

v. Abatis.

vi. Craters with 20 kilograms cratering charge.

(b) *Unconventional*

i. Jumping grenades.

ii. Clay more type improvised mines.

iii. Blocking road by animals, goats, carts et cetera.

iv. Fake barrier check posts.

v. Coal tar drums.

vi. Strong wire stretched between trees.

vii. Burning POL material on the road.

viii. Sharp nails struck through a wooden plank.

(c) *Essentials of an Effective Road Block*

i. Should be able to achieve surprise.

ii. Enemy should not be able to bypass it.

iii. Should be able to cause effective damage on vehicle/personnel in the lead.

iv. Should have an alternative method if primary fails.

v. Should not be obviously located.

vi. Should be covered by fire.

vii. Should be created/placed only when specific target being ambushed is approaching.

b. *Covering Group*

(1) *Rear Command Party*. Can consist of Second-in-Command or third in command or any Junior Commissioned Officers/Non Commissioned Officer along with participating troops or left over personnel with the packs et cetera at the final RV. They would be manning the final RV and "CHECK IN" withdrawing personnel.

(2) *Flank Protection Party*. Two or more parties guarding/ covering the flanks of the main force. The party would allow the target to enter into the ambush site and be trapped without interference and once the action by the action group has taken place, the parties would stop/hold by fire any reinforcements of withdrawing enemy from the ambushed site. They act like a trap door, which automatically closes.

(3) *Covering Party*. Well sited with automatic weapons i.e. machine-guns and 60 mm mortars et cetera to take on enemy in the target area or create effective flanking fire on any reacting/counter-attacking enemy, after own groups/parties have withdrawn. The withdrawal can be checked physically by the party commander or the covering fire could automatically start after the time bracket previously agreed upon. This party further affords a clean break to avoid a running fight.

(4) *Reserve*. Since all parties have been allotted specified tasks and would be briefed, located and positioned accordingly, the commander must have personnel in reserve to supplement a particular party or allot a new task altogether not previously catered; according to any emergent necessity. Hence the party should be located at a distance accessible to the commander and in communication with him. The party would consist of a few personnel depending on the size of the force and the target. They could even be located with covering party if the terrain and communication problems so permit. It may be pointed out that above mentioned groups and parties have been denoted for easy remembrance, they have to be used judiciously according to terrain, tasks and tactical situation.

Difference in Ambush by Day and Night

16. **By Day**

a. Men should be well-dispersed and concealed from ground view as well as view from the air.

b. Fire can be opened at longer ranges, as the target can be seen well.

c. There must be perfect stillness.

d. All parties can act simultaneously with reasonable safety.

e. Supporting fire including artillery fire can be obtained and directed by observation.

17. **By Night**

a. Men must be closer together and fire should be opened at point blank range.

b. Fire must be for a specified period, after which the task must be completed using bayonets or knives.

c. Strict discipline must be observed with regard to noise (talking, coughing et cetera) as this can be heard farther away at night. The same principle applies to lights.

d. Men can take-up positions in the open without much camouflage and can move their limbs more freely. It will however, not be possible, if positions are close to a road and vehicles are using lights.

Proposed Techniques

18. Following techniques are suggested as guideline, the employment of which will depend upon terrain and enemy situation:-

a. *Line Formation*. In this technique the ambush parties are deployed parallel to the target's route of movement (road, track et cetera). This positions the ambush party parallel to the longer axis of the killing zone and subjects the target to the flanking fire. The target is trapped in the killing zone by natural obstacles, mines explosive and direct and indirect fire.

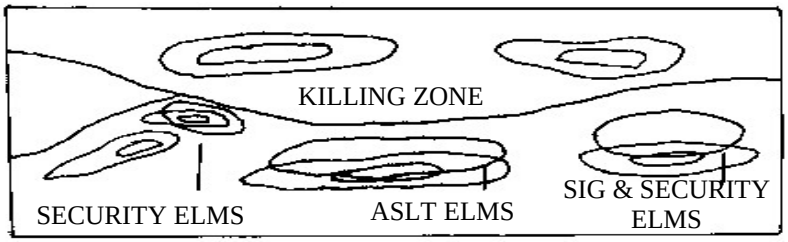


Figure 17 - Line Formation

b. *The L–Shaped Formation*. The L–Shaped Formation is a variation of the line formation. The long leg of the 'L' (assault element) is parallel to the killing zone. This provides flanking fire. The short leg (security/support elements) is at the end of, and at right angle to, the killing zone. This provides enfilade fire, which interlocks with fire other leg. This Formation can be deployed on a strait stretch of a road or track or a sharp bend in a road or track.

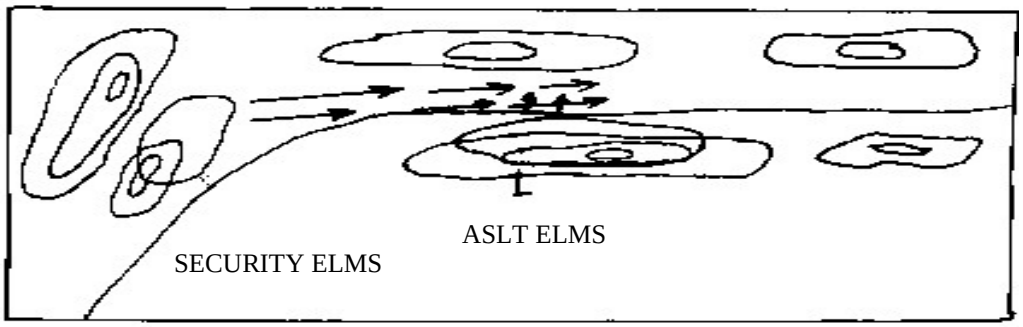


Figure 18 - L–Shaped Formation

c. *V-Shaped Formation*. An ambush party is deployed in two halves on either sides of road having interlocking fire. A third portion of the force is required to cover the withdrawal of assaulting groups.

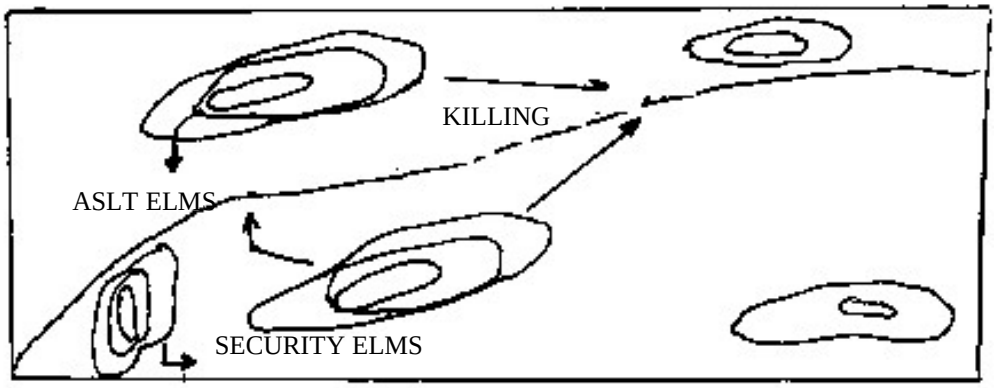


Figure 19 - V– Shaped Formation

Defence against Ambush when Moving in Convoy

19. The best defence against an ambush is not to present an ambush target. This can be achieved by dispersion and frequent varying of formation and size of force; above all by a high degree of alertness. If, despite all precautions, a convoy is ambushed, its reactions must be swift and aggressive. Personnel in vehicles must dismount immediately, take cover and fire. Under no circumstances should cover be taken under vehicles, as they are death traps.

20. When moving in a known guerrilla danger area or a likely ambush area, it is essential that every one should know what action to take, should the column be attacked. The canopies of all troops carrying vehicles must be removed and the tailboards let down. The senior passenger in each vehicle will be responsible that all personnel in his vehicle are alert.

21. The column may be led by armour, or, if there is sufficient armour, it may be distributed down to the column. It may well be that guerrillas or an ambush force will allow the vanguard to pass through and then halt both columns and attempt to deal with each separately.

22. At the first indication of an ambush, vehicles will stop. No effort should be made to drive into the side of the road, as it may be mined. All personnel will debus as rapidly as possible, take-up positions and open fire as ordered in the pre-arranged plan. Offensive action will be taken to destroy the guerrillas/ambush force immediately.

SECTION 45 – **RAID**

General

1. A raid is an attack by a force to accomplish a specific purpose within the enemy position with no intention of holding the invaded objective.

2. Raids of a platoon size will normally be ordered by the battalion commander. In some situations company commander and platoon commander may order and carry-out raids against any opportunity targets. This is likely to happen quite often particularly in the advance. Such raids are normally within the supporting distance of the parent unit and the battalion commander issues instructions covering the purpose of the raid, the fire support to be made available, the size of the force and the time of raid.

Purpose

3. A raid at platoon level may be conducted to:-

   a. Capture prisoner(s).

   b. Capture or destroy specific enemy material.

   c. Obtain specific information on enemy locations, strength, disposition and intentions.

   d. Liberate personnel.

Targets for Raid

4. Trained troops may be given the following targets for raids:-

   a. Bridges.

   b. Railways.

   c. Petrol and oil dumps/supply points.

   d. Airfields.

   e. Important factories.

   f. Power plants.

   g. Communication centres.

   h. Headquarters.

   j. Tank harbours/leaguers.

   k. Gun areas.

   l. Missile launching and missile storage sites.

   m. Any other target considered by the battalion or higher commander.

Characteristics of Raids

5. Raids are characterised by the necessity for quick hard blows, a timely disengagement and withdrawal. Unless carefully planned and skilfully executed, the withdrawal will be the most difficult part of the operation. Security is an important factor since, normally the raiding troops penetrate the enemy positions and are vulnerable to attack from all directions. At times the objective may be beyond the supporting distance of the parent unit. Raids may be carried out during day or night depending upon the mission and the local battle situation.

Planning and Conduct

6. The planning and coordination of a raid should be as detailed and complete as possible. Irrespective of the type of mission, a raid op will usually be divided into the following three phases: -

a. Preparatory phase.

b. Execution phase.

c. Disengagement and withdrawal phase.

7. **Preparatory Phase**. Any raiding operation requires detail preparations by the commander and troops selected for the task. The commander must be provided with:-

a. *Information*. Maximum possible information about the targets/objective from all available sources i.e., maps, air photographs, local population/prisoners and patrol reports et cetera.

b. *Reconnaissance*. Adequate time for detailed reconnaissance. If the raid is to be made at night, the commander should carry-out night reconnaissance in addition to the daylight reconnaissance.

c. *Personnel Selection*. Suitable personnel required for assigned missions should be selected, Physical fitness, skill and capabilities of each individual in relation to the assigned task should be kept in view.

d. *Weapons and Equipment*. The necessary weapons and equipment considered in relation to the nature of the mission. A platoon may require machine-gun support to accomplish its mission. Wherever possible artillery/mortar observers must be provided and suitable means of communications i.e. better wireless sets et cetera. catered for with due regards to the range of operations.

e. *Rations and Clothing*. The climatic conditions, duration and terrain in which the force is going to operate are the main dictating factors for the selection of clothing and rations.

f. *Rehearsals*. Opportunity for rehearsals particularly in stabilized situations. These rehearsals should be conducted on ground similar to the area to be raided; troops carry and use the equipment for the mission.

8. **Planning**. Once the administrative details have been worked out and the necessary information about the target has been obtained then a plan for the entire operation should be prepared. Following points should be kept in mind while planning:-

a. When to move.

b. How to move.

c. Route (s).

d. Hiding places.

e. How and when to conduct the reconnaissance.

f. When to raid the target.

g. Method of withdrawal.

9. **Rehearsal**. The rehearsal for the entire operation should be conducted on similar type of terrain and target whenever possible. Rehearsals clarify the doubts of subordinate leaders, rectify faults and build confidence in the men.

10. **Execution Phase**. This phase covers all the actions from Forward Defended Localities/hideout to the targets and its destruction. Normally the move is carried out in two stages as under: -

a. Move Forward Defended Localities (FDLs)/hide out to the RV and action, in the RV.

b. Move from RV to the target including its destruction.

11. For execution of raids following groups are employed. If there is shortage of men one party should be so deployed that it can perform tasks of more than one party: -

a. *Action Group*. It consists of the following parties:-

(1) *Forward Command Party*. Composed of the commander and his runner and is responsible to control the entire operation.

(2) *Sentry Disposal Party*. Normally two men are employed. One disposes off the sentry while the second covers him. Silent disposal of the sentry will add to success of a raid. In case this is not possible, sentry disposal will be carried out by other means.

(3) *Action Party*. This party is responsible to carry-out the final mission for which the raid has been designed i.e., destruction of a target. The strength of this party will depend upon the size and type of targets.

(4) *Special Task Party*. Any special task which is not undertaken by the above parties, such as destruction of automatic weapons on fixed lines, cutting telephone lines, destroying a wireless set or a command vehicle. These tasks are usually carried out simultaneously in conjunction with other parties.

(5) *Holding Party*. Carries light machine-guns sited at a reasonable distance with a task to bring effective fire on personnel who may directly interfere with the action party.

b. *Covering Group*. It comprises following parties:-

(1) *Rear Command Party*. Normally consists of Second-in-Command to take over in case the commander becomes a casualty.

(2) *Covering Party*. The task of this party is to provide a secure base from which the action group will develop its raid on the objective. It is also responsible to cover the withdrawal of the action group.

(3) *Reserve Party*. Elements not employed in the action or covering groups will constitute this party. This is required to meet any unforeseen situations, which may develop during the conduct of the raid and the rapid withdrawal.

(4) *Flank Protection Party*. To protect the exposed flanks and block reinforcement/escape.

12. The movements of various groups are arranged and coordinated so that all reach their positions at about the same time. This improves the raiding party's capability for decisive action if prematurely detected by the enemy.

13. The principle duties of the raid commander during the conduct of a raid are to decide when to call for, shift or lift supporting fire, to be constantly on the alert for unexpected reactions and to take prompt measures to meet them and to decide when to disengage and order the withdrawal.

14. **Disengagement and Withdrawal**. This is the most difficult phase of the operation and the success of the entire operation depends upon the raid

commander's ability to make timely and correct decision for disengagement on completion of the task and skill of the section/group leaders to move their men with speed to the rallying points and effective retaliation against the enemy reactions after the blow has been struck..

15. A rallying point is prescribed some distance from the objective for the purpose of assembling parties/groups, which have become separated during the raid. A series of rallying points may be prescribed along the route of withdrawal right upto the parent unit area.

16. Parties/groups should be able to cover the withdrawal of each other by providing covering fire, if possible otherwise, this should be arranged within each group/party during their withdrawal from one rallying point to another.

17. **Stages of Withdrawal**. It is normally carried out in following two stages:-

a. *Stage-1*. Firstly from the site to pre selected RV at a reasonably safe distance (say from 800 to 1000 metres, depending upon the ground and enemy dispositions) secondly from the RV to the hide-out or base depending on the distance and time involved. The withdrawal from the site to the RV is generally under the covering fire, so stopping any enemy following the withdrawal. The order for withdrawal must be clear and carried out rapidly. Artillery support to cover the withdrawal can be called if within range. Stragglers if any can make their own way to RV, which should be selected earlier and pointed out to the men enroute to the target. RV for stragglers must, however, be different from the one selected for the main force. At final RV the patrol reorganizes quickly and continues. During reorganization, ammunition is redistributed, casualties receive attention and essential information is disseminated.

b. *Stage-2*. The second stage of withdrawal is more or less similar to that of a fighting patrol returning to its base on completion of its task. It is however, advisable for a raiding force to move faster than a fighting patrol, as the enemy reinforcements, if any are likely to follow up closely.

Factors Affecting Success of a Raid

18. The following factors have a direct bearing on the success of a raid operation:-

a. Morale of raiding force.

b. Leadership.

c. Comradeship.

d. Physical and mental endurance.

e. Good administration.

f. Sound planning based on correct information.

g. Camouflage and concealment.

h. Proper training of troops.

j. Coordinations of actions of various parties.

SECTION 46 - **TANK HUNTING**

General

1. Tank hunting is the art of destroying, immobilizing or putting out of action enemy tanks by individuals or small parties with their own integral weapons.

2. Mobility of the modern tank enables it to pose a threat to defensive position from all directions. As it would not be possible to lay hands on enough anti-tank weapons to cover effectively all possible tank approaches, the use of unorthodox methods to impose caution on the enemy and thus deny him the bold use of his armour would pay handsome dividends. Well trained thank hunting parties can inflict damage on the enemy out of all proportions.

3. Tank by no means is an invincible monster. Troops should be extensively trained to exploit its weaknesses. Apart from proper tank hunting parties forming part of each fighting unit/subunit it is essential that every individual soldier especially the infantry man, should understand the tank hunting techniques and be given intensive practice in their use.

Limitation of Armour and Tank Tactics

4. **Limitation**. To be able to successfully hunt the tanks, it is essential that those connected with the planning and execution of tank hunting have a thorough knowledge of limitations and tactics of tanks. The following limitations of tanks can be exploited to great advantage by the tank hunting parties: -

a. *Comparative Blindness*. Darkness is a great handicap to the armoured troops. Even with the latest infra-red devices, tanks move at a cautious speed. The invisible infra-red equipment has no real search light capability. During day if the crew is forced to close down, from inside they have a limited vision both in direction and degree. The observation within approximately 15 metres around the tank is denied to them. The problems of visibility are further aggravated owing to fog, smoke, mist, sand/wind et cetera.

b. *Sensitiveness to Ground*. Obstacles such as streams or wide ditches, steep hills, marshes, woods et cetera create great problems for tanks. In case of close country, infantry must pave/clear the way for tanks otherwise they will present easy targets. Tanks definitely need help of other arms, particularly infantry. Such as: -

(1) Streams of artificial ditches more than 2.5 metre wide or a vertical of more than 0.9 metre.

(2) Rocky ground or ground with three stumps up to 60 centimetres girth.

(3) Marshes.

(4) Steep and slippery slopes.

(5) Thick woods.

(6) Anti-tank minefields.

(7) Built-up areas.

c. *Vulnerability to Anti-tank Weapons*. Sprocket, suspension, belly of the tank, rollers, sides of the tank engine and the fuel tanks are extremely vulnerable/sensitive spots of a tank, which tank hunting parties can destroy with comparative ease.

d. *Inability to Engage Target Close to Tank.* Though the traverse of the main gun is facilitated by power yet it takes approximately 15 seconds to swing round. The Machine-Gun (MGs fixed) fitted to in the front would obviously take longer time to engage targets. This provides an opportunity to the hunters to close with and destroy the tanks. There is an area of dead arc of 4.5-7 metres above and around the tank which provides safety to tank hunter at a very close range.

e. *Fatigue of Crews*. Travelling and working in cramped positions in the tank tires the crew, who apt to get is careless/off guard when they have operated over a difficult terrain for a long period. Such moments can be exploited with great advantage.

f. *Tank Noise*. It provides early warning to the tank hunters. Some time it assists the hunter to anticipate the direction, likely time of arrival at desired places and also the number of tanks.

g. *Tanks in a Leaguer*. After fighting, tanks often make for a night leaguer. During approach to leaguer and while refuelling, repairing and making preparation for next mission the leaguer is believed to be least organized to locate and fight back intruders. Tank hunters must, therefore, catch the tanks during the period of setting down. In this connection knowledge of the layout of a leaguer (open/close, squadron/regimental et cetera) may help the hunters considerably.

5. **Tank Tactics**. To successfully execute the task, it is imperative that the tank tactics generally employed by the tanks are known to the tank hunting parties so that they can correctly appreciate a particular situation. The following tactics are employed by the tanks during move and engagement of targets:-

a. *Pattern of Deployment*. Tanks generally operate in troop formations getting each others mutual support. While engaging a tank, this aspect of mutual support should be borne in mind. No tank should be considered in isolation.

b. *Position while Engaging a Target*. Tanks normally get into hull down or turret down position before engaging a target, thus gaining considerable immunity from the front.

c. *Infantry Tank Cooperation*. Tanks require infantry protection in close country, built-up area, tall growth and while operating at night.

d. *Action on Tanks Being Ambushed*. Tanks in contact get in position and engage the ambushing parties. Other thanks try to encircle and cut-off the enemy route of withdrawal. In this operation tanks may be supported by infantry also.

e. *Speculative Fire*. Tanks normally bring down fire in area/position which is likely to be used by tank hunting parties. This is done only to satisfy that the area is clear of tank hunting parties, but does not mean that latter have been spotted.

Tank Hunting Weapons

6. The weapons available within an infantry unit including those that can be easily improvised, are adequate for the purpose of successful tank hunting. Points to remember are:-

a. *Small Arms*. Small arms can be effectively used to force the crew to close down. This will restrict their visibility and put them at a disadvantage.

b. *RPG – 7 Rocket Launcher*. A well-concealed RPG–7 will be able to destroy any tank up to 180 to 275 metres.

c. *Rifle Anti-Tank Grenade No 94 (Energa) and M 31*. These are short-range weapons. Penetration power of Energa is 25.40 centimetres and that of M31 is 2.032 centimetres of steel. Their ranges are 70-90 metres and 90-120 metres for moving and stationary tanks respectively. Their disadvantage lies in short range and low rate of fire.

d. *Hand Grenade-HE 36*. When thrown inside a tank, it is deadly for crew. It can be done when cupola is open.

e. *Molotov Cocktail*. It consists of an average size bottle containing sulphuric acid (H_2SO_4) and patrol at 1: 2 ratio. An ordinary cellophane tape or a length of cinema film is wrapped around it. On impact, when the bottle breaks, acid burns the film and ignites the patrol. It spreads burning liquid over the tank and sets it on fire.

f. *Number -77 and 80 Smoke Grenades and Smoke Generators*. These are used for the following purposes:-

(1) To isolate the target tank from supporting tanks.

(2) To facilitate extrication of own parties.

However, the use of smoke grenades needs careful consideration as smoke is a double edged weapon and its premature use may give away surprise.

g. *75 mm Recoilless Rifle*. This weapon, being portable, can be successfully used by the tank hunting parties at 400-600 metres. Its main disadvantage lies in back blast.

h. *60 mm Mortar*. It can be used to inflict casualties on the tank crew and enemy infantry. It can also assist extrication of tank hunting parties by using smoke bombs.

j. *Bakthar Shikan Missile*. It can be profitably employed against tanks and vehicles concentrations of the enemy. These are easily transportable and can be conveniently employed mounted/ dismounted against enemy armour. The effective range of this missile is 3000 metres depending upon the visibility and nature of the terrain. It can be effectively used at night with night vision sights designed for this weapon system.

k. *Anti-Tank Mines*

(1) Use of mines imposes great caution on the enemy tank movement. A tank disabled by anti-tank mines is an easy target to deal with.

(2) Anti-tank mine necklace is used to halt or immobilize a tank moving on a road in defile. It consists of mines/ explosive joined together with a cable. The interval between mines is about 0.5 metre depending upon the width of enemy tank tracks. The mines are coiled up and concealed on one side of the road. The free end of the cable is held on other side of the road. The mines are pulled on the approach of enemy tanks.

(3) Mines should also be laid to prevent tanks getting off the road.

l. *Man Pack Flame Thrower*. This can be used effectively for scorching the tank crew. Fuel tanks should receive special treatment as it may cause an immediate conflagration. Flame directed against air inlets will help in sucking out oxygen from the inside of a tank and cause off choking effect on the crew.

m. *The Platter*. This is a shaped charge using a flower-pot and a platter big enough to cover the open top of it. When filled with explosive and detonated, it can be effective up to 15 metres away. The fuse must be placed in the exact centre of the explosive, the charge face should be concave and the explosive pack this tightly against the end of the container with stand off equal to the diameter of the container. To determine the amount of explosive the following would help:-

Diameter of container	*Grams (Pounds) of explosive*
Less than 5.080 centimetres (2 inches)	226.8 grams (0.5 pounds)

5.080 to 12.7 centimetres (2 to 5 inches)	453.6 grams (1 pound)
More than 12.70 centimetres (5 inches)	907.02 grams to 1.134 kilogram (2 to 2.5 pounds)

PLATTER FLOWER POT EXPLOSIVE

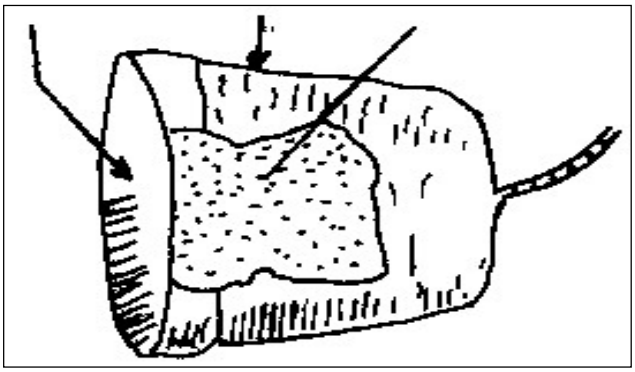


Figure 20 - The Platter

o. *Satchel Charge Expedient*. A satchel charge made out of TNT and at least 2 kilograms (4 pounds) of C-3, C-4, C-5 can be placed in or on the engine compartment of a tank, under the turret ring or jammed behind a drive sprocket to destroy or damage an armoured vehicle.

p. *Crow bars and Wire.* A crow bar inserted into the sprocket or idler wheels will halt a tank. Wire if thickly laid can also immobilize a tank.

Organization of Tank Hunting Patrol

7. Tank hunting party can function either in the form of a patrol when it approaches the target within killing range or in the form of an ambush when the target approaches the killing area. Depending upon the form, the party will be organized differently and will use different techniques to accomplish mission, as explained in the subsequent paragraphs.

8. **Tank Hunting Patrol**

a. *Main Group*. The patrol is divided into the following two groups:-

b. *Killing Group*. It is further divided into two parties of two men per tank. The patrol leader will command this group. This party must be equipped with anti-tank weapons.

(1) *Covering Group*. Supports the action of killing group with fire. This group will have small arms and light automatics. The patrol Second-in-Command may command this group.

c. *Planning and Preparation*

(1) Maximum information about the target should be obtained.

(a) Location and number of enemy tanks.

(b) Pattern of deployment.

(c) Protection arrangements.

(d) Likely time of stay in area.

(e) Routes to and from the target area.

(f) Enemy Observation Posts (OPs) interfering with operation.

(2) After obtaining information the tank hunting and covering parties are suitably prepared. Following actions should be completed before launching of a party.

(a) Division of responsibility of group.

(b) Issue of orders.

(c) Detailed briefing.

(d) Weapons checked and tested.

(e) Rehearsal, if time permits.

(3) While planning following must be covered:-

(a) Weapons and ammunition to be carried and by whom.

(b) Route out.

(c) Time out.

(d) Objective area.

(e) The target.

(f) Final RV (or release point).

(g) Action on objective.

(h) Break away signal, if required.

(j) RV for withdrawal.

(k) Route in.

(l) Time in – latest.

(m) Order regarding casualties.

(n) Be prepared tasks.

Tank Ambush

9. The technique is like a normal ambush, expect in the composition of main weapons and the possible variation of groups. The whole convoy has to come into the ambush site before the action commences; therefore the selected site should have enough length. Tank ambush may have to be hasty or planned:-

a. *Main Groups*. The party is divided into the following group.

(1) *Covering Group*. This party will be responsible for stopping and destroying the leading elements. The group like the cut-off group should be equipped with both anti-tank weapons and small arms. They will also require anti-tank mines et cetera to block the forward move of the target particularly when there is more than one tank to be ambushed.

(2) *Killing Group.* This is the main destroying party and is equipped primarily with anti-tank weapons. Their task is to destroy all tanks that have come into the ambush area and have possibly, stopped because of the action of the covering group. The ambush commander is with this group.

(3) *Cut-off Group*. Fires at the rear elements and destroys those tanks troops who try to escape. This party should be equipped with both anti-tank and small arms weapons.

(4) *Reserve*. This may or may not be necessary, depending on the size of the target and the place the ambush is planned.

b. *Tasks.* As opposed to the tank hunting patrol, the tank ambush lies in waiting for the target to come into a pre-selected area. Therefore, for the success of the ambush, it is important that the correct route on which the enemy tanks are likely to approach

should be determined or be known. The ambush site itself should be outside the view of enemy observers.

10. **Tank Hunting in Various Operation of War**

a. *Advance*

(1) The enemy while trying to gain speed is likely to overlook well concealed weapons. Jolting in tanks and bumpy rides add to the problems of observation by the crew.

(2) Leading tanks kick up dust as a result of which following tanks are denied full view of the area.

(3) Any tank which develops faults or bogged down will be left out of the fighting/supporting troop, and thus present an ideal target for the tank hunting party.

b. *Defence*

(1) Static tank in forward defended localities can be destroyed by parties after infiltrating into the enemy position.

(2) Ambush en-route to leaguer offers maximum chances of success.

(3) While in leaguer the tank may be found really prone to attack; there is hectic activity, noise, refuelling, replenishment, briefings et cetera.

(4) *Tank in Area Defended Localities*. Ambush in area deep behind will cause a great scare. This is true to such tanks, which may be operating on separate axis.

(5) When deployed in close country or built-up area, tanks are vulnerable and hence are attractive target.

(6) At night fixed line role will force the crew to be conscious only of their front – this could be exploited.

c. *Attack*

(1) Tanks while supporting an attack get in hull down position. These may then be found separate from each other by 200-400 metres. This is the time to attack the tanks, which may result in failing of enemy’s attack during its launching stage.

(2) *Tanks in the Forming Up Place.* When in support of day attack and particularly when waiting to join after dawn, the tanks offer a good target.

(3) *Tanks on Move to Leaguer after Capture of Objective.* This opportunity should be kept in mind because it offers many chances of a good kill.

(4) *Attack by Night*. Our potential enemy has developed night fighting capability and is likely to use his tanks in attack during night. This threat poses hitherto encountered problem for the tank hunting parties. The following action can be taken to deal with this situation: -

(a) An ambush is laid on the possible approaches, ensuring no laxity whatsoever is the concealment of the firing positions.

(b) Sufficient number of 60 mm mortar flares and vary light pistol are carried.

(c) As soon as tanks enter the trap, area is lit up by flares et cetera and the tanks are engaged with short-range anti-tank weapons without wasting even a second, followed by a quick and prompt withdrawal by the ambushing party. It must be remembered that flares, vary light rounds would give away not only surprise, but may also reveal some of the ambushing positions. Hence to ensure that the enemy is given no or minimum of reaction time, full weight of the fire must be brought to bear upon the tanks immediately on firing of flares.

d. *Withdrawal*. Tanks operating with rear parties offer good opportunity for ambushes

Attacking a Leaguer at Night

11. **General**. Tanks in a leaguer are concentrated at night. The armoured unit has to carry-out refueling and replenishment during night. Unless protected by infantry they are not safe in a leaguer, and they are a good target for artillery and air. The pattern of a leaguer and points to remember are discussed in the following paragraphs.

12. **Tank Leaguer**. Tank hunting parties must be fully conversant with the layout and defensive aspects of a leaguer. It could be a box formation, square or rectangular. Some important features of a leaguer are:-

a. Tank up on the perimeter with the turret guns facing outwards, thereby covering the leaguer from all directions.

b. Some of the tanks will be actually manned and ready for action, are known as “guard tanks”.

c. The leaguer will have an outer ring of defence, where dismounted machine-guns are deployed on all courses. Machine-guns fire on fixed lines and provide interlocking fire all round the leaguer.

d. There will be night patrolling along perimeter and also some listening posts at 100-200 metres away from the leaguer.

e. B vehicles will be parked under protection of tanks.

f. There will be at least one entrance and an exit.

13. Prior to actual attack, a base should be established at a safe distance from the leaguer and the reconnaissance parties sent out to determine the information regarding organization, extent and approaches to the leaguer. Before attacking tanks, sentries should be disposed off silently (Commando action).

14. Though an enormous fire-power is available within the leaguer, it cannot be utilized for obvious reasons. The approach to leaguer will be facilitated by these factors and the confusion once generated is quite likely to totally disorganise the command. If fire of weapons like mortars is brought in this chaotic situation, the damage to personnel and B vehicles can be very great.

SECTION 47 - **ROAD BLOCKS**

General

1. Whenever a company/platoon is located in an area which is entered by roads or tracks along which enemy tanks and vehicles might approach, a road block should be established. This may consist of anti-tank mines but when these are not available, a road block must be improvised from such materials as fallen trees, rubble or destroyed vehicles.

2. **Characteristics of Road Block Site**

a. *Blocks the Avenues of Approach*. The road block should be difficult to by pass.

b. *Takes Advantages of Natural Obstacles*. Road blocks can be established along the side of a steep hill across streams, marshes, or ravines to take advantage of natural obstacles. Artificial obstacles can then be easily constructed to reinforce, supplement and tie-in with the natural obstacles. This helps to create an effective obstacle with minimum effort.

c. *Easily Defendable*. The position should provide adequate observation to cover obstacle and approach to it by fire to prevent it from being breached. Further, it should provide good fields of fire to the defender and be relatively in accessible to the attacker.

d. *Gains Surprise.* Troops in artificial obstacles should be concealed until it is too late for the enemy to react effectively. Mines, demolitions and catering charges are examples of effective, easily concealed obstacles. Obstacles may be located around a sharp bend in the road, just over the crest of a hill, or when a road passes through a heavily wooded area to gain surprise.

e. *Good Routes to the Rear*. This facilitates resupply and a rapid withdrawal. A slow or disorganized withdrawal of the road block force can lead to its destruction by pursuing enemy.

3. **Types of Road Blocks**

a. *Permanent Blocks*. When the road is not required for use by own troops, it may be closed by one of the permanent obstacles.

b. *Movable Blocks*. Usually, roads have to be kept open as long as possible and movable obstacles, capable of rapid erection, must be provided. These however, require considerable time for initial preparation.

c. *Improvised Blocks*. When there is little time for preparation, full use must be made of local materials to improvise blocks. Examples are:-

(1) *Farm Carts*. Road blocks may be improvised from farm carts, motor cars, lorries et cetera, loaded with stones/concrete or other heavy material. When placed in position, their wheels must be damaged or removed. It is

essential for them to be firmly anchored, otherwise they may be scattered by advancing tanks.

(2) *Tree Barriers.* Bushy top trees with heavy branches and thick foliage should be selected if possible, because the branches reduce the tank's speed and foliage act as a screen. The trees should be fallen to lie on top of one another across the road with their branches pointing towards the enemy's approach. The trees should be fallen in such a way so that the trunks remain attached to the stump. To affect this, no cut should be made on the side of the tree towards which it is to fall. This type of road block can be used only on roads which are to be blocked permanently.

4. **Subsidiary Road Blocks**

a. Plain wires stretched across a road at a height of 1.14 metres will unseat and sometimes decapitate motor cyclists. Fencing, telegraph or telephone wires may be used stretched tight and attached firmly to trees, telegraph poles, walls or fences. The wire should make an angle of between 45 to 90 degrees with the line of the road.

b. Steel wire ropes of 10.16 centimetres "circumference" stretched across a road will stop a tank if suitably fastened. However, they may be severed by machine-gun fire or loosened by two or three repeated pushes by the tank; they do not form an effective tank obstacle.

c. Barbed wire and wire concertina can stop wheeled vehicles. A series of these blocks placed about 10 metres apart should be used. The ends of adjacent coils must be wired together and the obstacle lightly anchored at the sides of the road.

Sitting

5. Suitable sites are in defiles where the road passes between woods, large ditches, thick hedges, banks or buildings. A road block should be so sited that it creates difficulties for enemy drivers in following manners:-

a. To see the obstacle until they are close to it. This is done by sitting it round a corner or over the crest of a hill.

b. To turn their vehicles round.

c. To drive off the road and move cross-country.

6. It may often be necessary to organise road blocks so that friendly vehicles can be allowed to pass through them. A block of this kind consists of two overlapping portions with a movable barrier such as a concertina wire knife in between.

Cover by Fire

7. A road block must always be covered by fire. Normally this should be provided by one section with the platoon anti-tank weapon. The rocket launcher and at least one light machine-gun should be sited to cover the road on the enemy side of the block. The rest of the section should be positioned to protect it and to engage enemy vehicles with fire from different angles.

Sentries

8. Except in forward areas or when in ambush, a sentry should normally be posted at the block itself to identify vehicles as hostile or friendly and in the latter case to act as traffic policeman.

Sign Posting

9. When mines are used in the construction of a road block, prominent notices, must be set up on our side to prevent friendly vehicles from running over them.

CHAPTER 10

INFILTRATION

SECTION 48 – GENERAL

Aim

1. The aim of this section is to explain infiltration and the techniques involved, at company level within the framework of a battalion.

Definition

2. Infiltration is the technique of undetected movement used in conjunction with the several forms of manoeuvre. The infiltrating forces move by individuals or small groups/subunits to a previously designated RV/hide out in the enemy rear. During the movement; individuals/small groups/subunits seek to avoid detection and engagement by the enemy. By this mean an adequate force

compatible with its assigned mission may be deployed into the enemy's rear. In fact the enemy may possibly know the infiltration attempts, he must, therefore, be deceived about the size of the force and its mission. Once the force is assembled, it proceeds to execute its assigned mission.

Purpose and Tasks

3. The purpose of infiltration is to deploy strong enough forces in the enemy's rear for the execution of one or more of the following tasks in conjunction with any of the tactical manoeuvres launched by the main force:-

a. To destroy/neutralize selected enemy localities, gun areas, tanks, headquarters et cetera.

b. To seize important tactical features such as bridges, defiles, high grounds, centres of communication et cetera.

c. To disrupt, delay or hold movement of enemy supporting units and reserves.

d. To stage deception for operations in other sectors.

e. To harass and interdict his rear areas including administrative installations.

f. To obtain and communicate information about the enemy and terrain.

g. To direct the fires of long-range weapons and air.

h. To capture and hold important tactical grounds like bridges et cetera for an eventual link up.

Conditions which Favour Infiltration

4. Conditions which help in the infiltrating of a force to the enemy's rear are:-

a. *Enemy Deployment and Activity*. Gaps in enemy defences, lack of surveillance devices, and poor patrolling render possible undetected movement of infiltrating forces. Infiltration can best be used when the enemy is deployed on a wide front with large gaps between his localities. For example defence in deserts.

b. *Terrain*. Thick-forested areas, hilly terrain, standing crops, broken and difficult ground restrict enemy observation and surveillance

and afford better concealment during move. The rougher and more difficult the terrain, the more it merits consideration for infiltration.

c. *Weather*. Conditions of reduced visibility, darkness and adverse weather conditions help in infiltration.

Basic Techniques

5. A force may infiltrate:-

a. In small parties/groups or individuals.

b. In subunits or large bodies.

c. Combined infiltration, “a small and a large force”.

6. **Small Group Infiltration**. Infiltration in small groups is suitable where enemy deployment is close with narrow gaps. It has following merits and demerits:-

a. *Advantages*

(1) Better chances of successful infiltration by major part of the force even where movement of one or more parties may be detected/prevented by the enemy.

(2) Easier to infiltrate even through the enemy defences which are strong and have smaller gaps.

(3) Easier to conceal small groups.

b. *Disadvantages*

(1) Several small infiltrating groups when operating over enemy territory may present many targets.

(2) Movement is difficult to control.

(3) Collection of the force at the RV becomes difficult.

7. **Infiltration by Platoon/Company Size Groups**. Infiltration larger bodies is suitable where enemy positions are widely dispersed having wider gaps, or when enemy defences are weak and his patrolling and counter-infiltration activities are poor. Hilly terrain, jungle and broken country would be ideal for such type of infiltration. However, the advantages and disadvantages of this technique are as following:-

a. *Advantages*

(1) The commander has complete control at all the times.

(2) Changes in the plan are possible.

(3) Less number of targets is presented; this affords less chances of detection by the enemy.

b. *Disadvantages*

(1) A large force may be difficult to conceal.

(2) Detection of force and interference by the enemy may jeopardize the whole operation.

(3) Moving through one or two lanes where individuals or small groups may have to follow each other would take considerable time.

8. **Combined Infiltration 'a Small and a Large Force'**

a. *Advantages*. The infiltrating troops can use the best method for moving through any area, thereby enhancing security. It enjoys the advantages of both methods.

b. *Disadvantages*. There is a need for detailed coordination. Inevitable mishaps may well create problems for the commander.

c. *When Adopted*. This method is used frequently and is adopted when conditions of the enemy defences and terrain vary from forward to his rear areas.

d. *Method*. The infiltrating troops may start with small party method. If forced by enemy and terrain, they may then switch over to the large force method or vice versa. In some situations they may adopt both the methods simultaneously. In either case parties get together at RV and may either form a large force or else split up into small parties for subsequent operations.

Means of Infiltration

9. There are three means of infiltration:-

a. Air.

b. Water.

c. Land.

10. **Air**. It is usually the most practical and rapid mean of infiltration. Air lifting of troops through standard troop carrier aircraft may be difficult to operate due to unfavourable air situation, non-availability of airfields and navigational difficulties. However airlifting of troops by helicopters would be ideal for reason of flexibility, quick build-up, accuracy of delivery and operations in near darkness conditions with reduced chances of detection.

11. **Water**. Infiltration by water includes the use of surface and under water crafts.

12. **Land**. Land infiltration is best accomplished under conditions of limited visibility and over difficult terrain. Land infiltration has greater chances of success when the enemy lines are over-extended.

Basic Considerations

13. **Intelligence**. Detailed information about the ground and enemy dispositions is essential for the success of the operation. Information is necessary to enable selection of objectives, routes, RVs, to decide upon the strength and grouping of the force, to plan fire support and deception. The bulk of such information is secured through intensive patrolling, air reconnaissance/ photographs and detailed study of maps. Information required includes details of following:-

a. Gaps in enemy's defences.

b. Patrolling activity and presence of any surveillance devices.

c. Depth of enemy's defences.

d. Enemy strength and dispositions in the objective area.

e. Location and strength of reserves, and their reaction capabilities.

f. Ground cover and concealment, dead ground, natural and artificial obstacles, tracks and routes.

14. **Offensive Attitude**. The operations of an infiltration force should be planned as an offensive. Defensive attitude or concepts such as suicide squads generally result in heavy losses without achieving any decisive results. Boldness in planning and execution is a prerequisite for success.

15. **Speed**. Rapidity of movement combined with stealth and secrecy are the essence of infiltration tactics. Speed is gained by discarding unnecessary heavy equipment and by use of light and essential elements. Commander and troops

must have the greatest latitude in matters of dress and weapons. Every gap in enemy's defences must be rapidly exploited. Maximum use should be made of local resources.

16. **Surprise**. It is achieved by operating in hours of darkness, rain and inclement weather and denying the enemy the following information:-

a. That you are infiltrating.

b. From where you are infiltrating.

c. Time of infiltration.

d. Strength of infiltrating force.

17. **Security**. An infiltrated force is vulnerable to defeat in detail. Its security against enemy reactions, during the period it is isolated and detached from the main force, needs detailed consideration. Adequate fire support and administrative backing for the duration of operation is essential. Plans should also provide for an early link up, unless it is planned to exfiltrate the force after completion of its task.

18. **Selection of Objectives**. Objectives must be selected with great care, and should be those which contribute directly to the accomplishment of the overall mission. As a matter of principle, organically vital spots like depth positions, blocking positions against reserves, gun/missile areas which will hurt the enemy most and induce psychological responses are selected as objectives. Their capture should be possible within the resources available to the infiltrating force.

19. **Control**. Infiltration, which is conducted by small groups, during periods of darkness and often across difficult ground, in which ultimately these groups have to assemble, is a difficult operation to coordinate and control. Thorough planning and necessary rehearsals both by day and night on a similar possible piece of ground are must in order to ensure effective coordination and control. Following rehearsals must be carried out a long with thorough briefings and comprehensive orders:-

a. Formations.

b. Movement techniques.

c. Action on enemy contact.

d. Action at bounds/RVs.

e. Action at danger areas.

f. Action in the hide out.

g. Action on the objective.

Phases/Stages of Infiltration

20. An operation by infiltration is planned in the following five stages:- Preparatory stage.

a. The passage through enemy defences.

b. The assembly of the force in enemy's rear and preparation for the accomplishment of mission.

c. The execution of mission.

d. The link up or exfiltration.

Timings

21. Operation by infiltration is inevitably extended over a period of time during which movement, assembly of force, accomplishment of mission and exfiltration or link up actions are executed. This demands careful planning of timings. The following factors need consideration in deciding upon timings:-

a. Period of poor visibility or hours of darkness are essential for the unobserved movement of the infiltrating force. Their assembly and reorganization would also have to be completed at night. Movement by infiltration is generally very slow. It would be better to allow the movement to be completed on open schedule, since it will be very difficult for groups to adhere to timings for reaching bounds and RVs. Rushing them through is likely to invite trouble. Movement may include stalking when passing through or near the enemy positions.

b. For the operation of strong forces deep in the enemy rear, movement should start in the early hours of darkness.

c. If the force is to exfiltrate on completion of its mission, timings should be so worked out as to enable completion of exfiltration during hours of darkness. The force may exfiltrate the same night or on subsequent night (s). In such a case due consideration will be given to the selection of suitable hideouts.

d. The timing of the main attack or link up and other operations of friendly forces must be closely coordinated with the timing of the operation by the infiltrating force. Infiltrated forces may start

execution of its operation simultaneously, or soon after or before the attack by the main force.

e. Timings and conditions should be varied for each operation. A routine alerts the enemy.

f. After having obtained the necessary intelligence, the following planned time schedule can serve as a guide-line:-

(1) Starting from daybreak, 24 hours for reconnaissance, planning, orders and rehearsals down to sections.

(2) 12 hours for administration, rest and move forward of the force near to the forward defended localities, compatible with security.

(3) Infiltration should then begin in the early hours of night as far as possible.

Lanes

22. Within the area selected for infiltration, the infiltrating force moves through gaps in enemy defences. These gapes are designated as lanes. Gaps selected should be of sufficient width to allow the movement of assigned group or units by stealth. If movement is to be conducted in small groups, a series of lanes may be required for their infiltration. As far as possible lanes are selected through uncovered defiles, dead ground, and difficult terrain. More often the required lanes may be through natural or artificial obstacles. Use of engineers in the earlier stages of the operation will be made to facilitate movement through these lanes. This, however, may be more time-consuming and risky as regards the security of the infiltrating force. Therefore, it would be ideal to use the engineers in collecting intelligence about existing lanes and passages through obstacle before selecting/allotting lanes.

23. In making the decision whether to use single or multiple lanes, there are several things, the commander should consider:-

a. Movement on a Single Lane

(1) Makes control easier.

(2) Makes navigation easier.

(3) Increase the chances of detection of the entire force.

(4) Gets the force to its hide out faster.



Figure 21 - Company Moving on a Single Infiltration Lane

b. *Movement on Multiple Lanes*

(1) Makes control more difficult.

(2) Makes navigation difficult.

(3) Decrease the chances of detection of the entire force.

(4) Slows the movement of the force to its hide out.

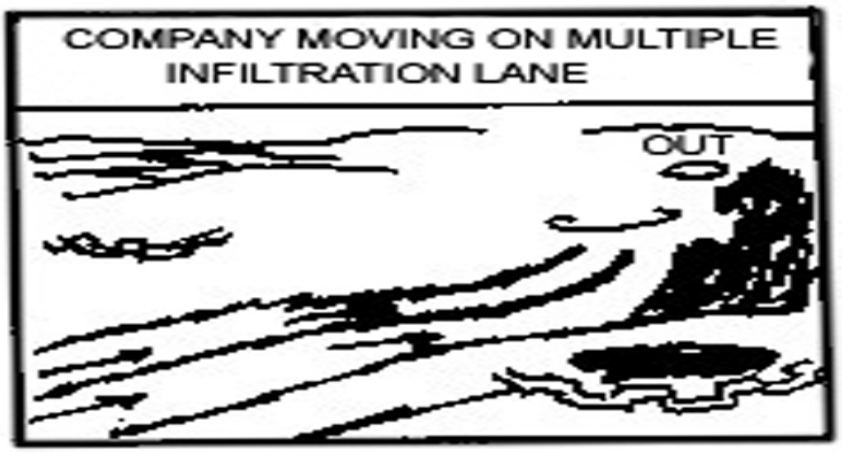


Figure 22 - Company Multiple Infiltration Lanes

24. Lane (s) selected should have following pre-requisites:-

a. Avoid enemy positions.

b. Have cover and concealment.

c. Ease control and navigation.

d. Avoid obstacles and danger areas.

e. Lead to RV and hide out.

RVs

25. A RV is designated, where the groups infiltrating on multiple lanes, will assemble behind the enemy lines and move as a force to hide out. Alternate RV(s) is also laid down, to be used in case the force is unable to assemble at designated RV due to enemy's presence or action. RV should:-

a. Be easy to find.

b. Have cover and concealment.

c. Be defensible in all directions.

d. Be away from likely enemy routes for example roads, tracks et cetera.

Control Measures

26. Detailed and well elaborated control measures are essential to follow the operations minutely and to direct actions of various groups throughout the operation. Infiltration lanes, RVs and units/subunits are allotted nicknames in order to facilitate communication and control. Order of move is carefully planned. Checkpoints may be laid down to enable subunits to report progress of movement and to ensure coordination of fire with the move.

Communications

27. Elaborate communications are provided within the force and with formation headquarters, supporting arms, and other cooperating units. Mainly radio will be used. Radio conversation with the infiltrating force or its groups may not be possible, as it would compromise security. For communication within and outside the force, therefore, relies on the use of pre-arranged codes, nicknames, such as:-

a. Tapping/knocking on the mike.

b. Blowing into the mike.

c. Pressing hand switch to produce sound of clicks.

d. Bird and animal calls.

e. Use of line bedding or strings within groups.

f. Visual signals such as arm and hand signal, infra-red devices and flashlight with coloured lenses. Sound signals and flares are avoided.

28. To minimize conversation, the main body which is relatively secure may ask questions from its groups/leading patrols. For example main body inquiring a patrol “Did you follow Head Lights” (the planned route) and the patrol may reply “No,” by two clicks on mike switch or “yes” by four clicks. Numerous codes like these can be planned. Radio massages may be transmitted in the blind with no call signs.

Grouping

29. The most suitable grouping for move as well as for the accomplishment of mission is to be specified. The strength of the force including individuals from technical arms and fire support must be adequate for the task allotted.

Deception

30. Infiltration against an alert enemy particularly if he is equipped with devices for detecting move requires careful use of deception, diversionary measures, and electronic countermeasures. Since the enemy is likely to be equipped with infra-red illuminating devices/radar, proper use of cover and field craft as for daylight movement is necessary in order to avoid detection at night.

Cover Plan

31. In order to suppress any noises during movement, a cover plan employing artillery and other fire may be prepared. Such a plan is generally restricted to the normal activity of the weapons and equipment, so as not to alert the enemy. Fire employed may be utilized to contain selected enemy localities where necessary and to prevent movement of enemy patrols. If the enemy is equipped with surveillance devices, heavy artillery and mortar will help in destroying/disrupting such devices.

Fire Support

32. Units making an infiltration usually carry weapons and portable infantry heavy weapons including anti-tank weapons. Artillery fire support is therefore, essential when surprise is lost, contact is made or when needed to support the

mission. Detailed fire support plans for the mission as well as to cover the movement are planned. Smoke should also be planned on known enemy positions to conceal friendly movement or to help break contact. Maximum close air support is made available at the objective area. Normally a sizeable force is infiltrated to the assigned objectives, which are within range of artillery from the main position. Where necessary guns may have to be moved forward to roving positions to provide fire support at extended ranges. Wireless sets, artillery observer and forward air controllers may accompany the force. Close liaison and coordination should be maintained with the Air Force, if in support.

33. Suggested method of using supporting arms/weapons are:-

a. *Armour*. Tanks may be used in outflanking and diversionary roles.

b. *Artillery*. Medium and heavy artillery may be able to provide fire support from its original positions. Field artillery may have to be employed from roving positions if out of range.

c. *Mortars*. These should be moved up to the forward defended localities. Additional 60 mm mortars should be carried along.

d. *Rocket Launchers and Anti-Tank Guns*. These may be man handled. They would be required to protect the firm base or an objective when counter-attacked by enemy tanks. Rocket Launchers can be made to their best use in raids/ambushes.

Coordination

34. Following are essential:-

a. Coordination between various groups/units infiltrating and own troops in the forward localities.

b. Coordination between the infiltrating troops and force with which to link up. Some distinguishing signs should be arranged to avoid own troops being confused with enemy.

c. Coordination for calling fire support during the passage through enemy defences and during the accomplishment of the assigned task.

d. Coordination for calling air support/supply.

Exfiltration

35. In certain situations link up may not be aimed at or the main force may fail to link up. Therefore, in all cases before the infiltration starts, detailed planning for exfiltration is carried out and orders issued at all levels.

Patrolling

36. To prevent detection and enemy interference, it is essential that no man's land is dominated by intensive and aggressive patrolling before the operation is launched. Prior to the start of the infiltration, friendly patrols are detailed to prevent enemy from approaching the area selected for infiltration.

Administration

37. Administration of infiltrating force presents major problems. Detailed and thorough consideration is essential not only for the success of the mission but also for the security of the force:-

a. The force must, as far as possible, move self-contained for the duration of the operation; till link up is achieved or till its exfiltration. Generally everything may have to be carried on man. At times it may be possible to take animal transport.

b. In some areas it would be possible to live off the land and on captured ammunition provided the bores of the weapons tally or weapons and stores captured are in sufficient quantity.

c. Arrangements should also be made for air supply including use of helicopters in case the operation is prolonged.

Methods of Handling Wounded, Dead and Prisoners

38. During the infiltration, it may be hard to evacuate casualties without jeopardizing secrecy and security. Casualties can be concealed with two soldiers having medical supplies who can be collected after completion of mission. The dead are concealed and left to be recovered later. It is best to secure and leave prisoner under guard at bounds. Moving the wounded, dead and prisoners to hide out is dangerous when trying to avoid detection.

SECTION 49 - **CONDUCT OF INFILTRATION**

Preparatory Stage

1. Preparations for infiltration are similar to other forms of operations except in the conduct of infiltration where certain points require additional emphasis. Having collected necessary information about the enemy and terrain, plans are prepared and finalized:-

a. Troops are detailed for the operation.

b. Reconnaissance is carried out and orders issued.

c. Decision is taken on weapons and equipment to be carried.

d. Administrative details are worked out and load tables prepared.

e. Briefing and rehearsals are held.

f. May send spotters in the enemy territory to keep an eye on selected routes a night before, if possible.

g. Patrols sent to dominate no man's land in the areas of gaps.

h. Move to forward assembly area(s).

2. Since troops have to operate over unfamiliar terrain, detailed briefing and rehearsals are essential. Where available, air photographs are carefully studied. Troops must be explained in detail, movement plan, actions on various contingencies and on the objective. Briefing is done on accurately prepared sketches/sand models and troops rehearse all their actions under as real conditions of terrain and enemy as possible.

3. Coordination with holding troops must be made before hand for provision of guides, who will assist the infiltrating force for passage through own forward defended localities and obstacles.

4. Commander must study the notes in great details and memorize them. Sketches, notes or marked map will not be taken on mission behind enemy lines.

5. All troops must check their weapons, equipment, ration and medical kits et cetera**.**

Passage through Enemy Defences

6. Having sent forward patrols to dominate no man's land and to prevent enemy interference, infiltrating units/subunits/groups, move up to the enemy's

forward defended localities along assigned routes. On reaching close to the enemy forward defended localities the force moves through selected lane as planned. However, technique of infiltration may have to be changed if the enemy presence realized.

7. **Spotters**. If the spotters have been sent forward, the infiltrating parties, contact spotters in their way and may even be guided by them. Patience is the catchword for spotters.

8. **Guides**. Reliable civilian guides who know the routes and are familiar with the terrain are a definite asset and should be utilised wherever possible.

9. **Grouping**. Grouping for movement through the lanes depends upon terrain, number and size of gaps, enemy dispositions and his security measures. The whole force may move through one lane or it may split itself into groups and move through two or more lanes. For a company size force the grouping up to the RV should be made into platoons, which may be further divided into sections or even smaller groups.

10. **Staggered Timings**. To reduce the size of moving columns, sub-elements in a column on one lane may be divided into several echelons that move at staggered timings. For example a platoon may move on one lane and have its sections move separately with fifteen minutes interval between the first and second section and ten minutes interval between second and third section.

11. **Lanes**. When using more than one lane, each group/subunit infiltrates through its allotted lane. As soon as safe lanes of suitable size are reported by the leading groups, the main force follows using these lanes. However, when accurate information is available before hand, it may be possible to dispense with reporting by the leading elements, thus saving time.

12. **Reconnaissance Patrols**. Various techniques may be adopted to reconnoiter the gaps and to move undetected. It is possible that friendly patrols previously may have been sent through the enemy's defensive positions or the spotters are located in advance. In such cases personnel from the patrols/spotters may be employed to guide the subunits or groups. In other cases small reconnaissance patrols equipped with wireless sets, may be detailed to reconnoiter each lane. Patrols move well ahead of the subunits/groups, reconnoitering gaps and ensuring that these are wide enough for undetected move of the groups/sub-units or unit, which they lead. The ideal then would be for the patrol to return and lead the main body through the gap. But time may seldom permit this. If so the patrol may send a pre-arranged signal to the main body to follow.

13. The patrol must inform the main body of any deviations from the planned route due to presence of enemy, obstacles, trip wires or other obstructions and particularly of the enemy locations found by them. These must either be marked by the patrol on the ground or preferably the patrol should guide the main body for passage through these obstructions.

14. **Actions on Detection by the Enemy**. Units, subunits and groups take all precautions to avoid detection by the enemy and if detected, avoid decisive engagement. Procedures are laid down in advance for action in case enemy localities or patrols are encountered. Troops must know how to silently dispose off enemy sentries. This may, however, be done as a last resort in order that no time is wasted enroute and security is not endangered, while the force is on its way to accomplish the mission. With a high standard of field craft, observation at night and night navigation, it is possible to penetrate through enemy defences without being detected by his sentries. The infiltrating force uses movement techniques based on the likely hood of enemy contact and halts occasionally to watch and listen to the enemy activity. When a halt is signaled every one freezes in place, stays quite, looks and listens. What ever may be strength of a party, it should try to avoid enemy position or patrol. The party may be stronger than the enemy, but once they are seen, their mission is partially jeopardized.

15. **Fire Support**. Fire support may be employed prior to the start of infiltration in order to neutralize enemy surveillance capability. During the movement, fire support is used to impede movement of enemy patrols, to suppress noise of movement and for deception in areas away from the planned lanes of infiltration. This however, will have to be carefully decided by weighing the disadvantages of alerting the enemy who may otherwise be complacent.

16. Often a lane may be found blocked by the enemy. In such an eventuality the following alternatives are open to the subunit or group using the lane:-

a. It may split into smaller groups/individuals.

b. It may be withdrawn to follow a safer lane, provided time bracket permits this. A platoon size well-trained force should be able to crawl through enemy trenches in certain situations.

17. **Civilians**. Hostile civilians met enroute should not be allowed to continue on their way but should be held under custody until the completion of mission. The best would be to place them at an unknown place after heavy intoxication or killing them at the spot.

18. **Roads and Tracks**. Use of tracks and roads should be avoided until well behind the enemy localities and that too also with extreme care and caution. It is necessary to make arrangements for early warning of any traffic to enable the parties to get off the road/track and take cover.

19. If a soldier in the infiltrating force speaks the enemy's language, he should be positioned at or near the front of the column, to tackle a situation in case the force is challenged by enemy Observation Post or patrol.

20. If the infiltrating force is dispersed, it reassembles at the last bound. The force waits until a set number of men return or for set time period, before continuing the move. If the commander does not return, the senior most man present takes the command.

21. Parties should leave no traces on the routes they follow.

Assembly of the Force

22. Having infiltrated through the enemy positions, the groups then move to the RV for assembly. Care is exercised by all groups in approaching the RV. To ensure that the RV is not occupied by the enemy, patrols may be sent forward well in advance to carry-out reconnaissance. The main body may then move in, if the patrol signals that the RV is clear of the enemy. If the enemy is found occupying the RV then employing the same procedure, the groups move to an alternative RV. Necessary guiding arrangement to collect all individual groups at one place in the RV should be made by the elements arriving at the RV first. It is recommended that a contact point is designated close to the area of RV where the groups send reconnaissance patrol for further guidance to the RV. Normal passwords and signals should be used for recognition/identification purpose.

23. Despite all arrangements, if the movement has been carried out in small groups, one odd group (s) may be found missing, because, either their movement was slow or they could not locate the RV. Plans must exist for their guidance, rejoining or exfiltration.

24. Before the force moves from the RV to its objective/firm base/hide out reconnaissance patrol should be sent to get the latest information about the terrain, enemy situation and the objective area. Infiltrating force commander should give this responsibility to the subunit, which is expected to reach the RV first. A lot of time will be saved by this arrangement, because by the time the whole force assembles at the RV; patrol would be back with the required information. For example if a company size force is infiltrating then the company commander

should delegate this responsibility to the platoon which is likely to get to the RV first.

Execution of Mission

25. Having assembled, the force then proceeds to its objective area to execute its mission. A force may be assigned more than one task in which case it proceeds to accomplish its second task or be prepared mission, in a similar manner, on successful completion of first task.

26. **Fire Support**. Maximum fire support, including close air support, is employed for the operation. To facilitate night operations on objective, which may not have been, reconnoitred earlier, Night Vision Devices or illumination may be used with the help of own artillery. While illuminating the enemy area due consideration must be given to the secrecy and security of own force. Illumination may also be employed in other areas for deception.

27. **Defence of Hide Out**. When the infiltrated troops have to stay in the objective area or other areas behind the enemy lines for any length of time they must occupy a strong defensive position with maximum speed. It is essential that an area with natural protection is selected. The enemy is likely to react violently. Natural anti-tank protection would therefore be sought and a strong anti-tank defence laid out including anti-tank mines. It would be necessary to have well planned active air defence.

28. Troops infiltrated for harassing and interdiction missions over an extended period would seek security by movement and going into various hide outs. They avoid decisive engagement with strong bodies of enemy troops.

29. **Administration of Infiltrating Force**. When infiltrating forces operate to a great depth behind enemy lines, and remain there for extended periods, their maintenance is carefully planned. Where possible, supply by air/helicopter is arranged. The force may also live on captured supplies or local resources. Animal transport, if taken with the force, may be used for meat.

Link-Up

30. Simultaneous attacks are launched by the main force on the flank or front of the enemy to penetrate in his defences, and to link-up with the infiltrated force as early as possible. To avoid a firefight between friendly troops at the time of link-up, the forces use pre arranged recognition signals, both visual and sound. In addition to this, link-up points will be pre-designated on the ground and should be

known both to the link-up and infiltrated force. In case a link up operation is not planned, or is rendered unsuccessful, the force may exfiltrate back to the main position or be evacuated using helicopters. Prior planning is a pre-requisite for exfiltration, no matter whatever the situation may be.

31. Specimen of infiltration order is attached as Annexture “L”.

SECTION 50 **– EXFILTRATION**

Definition

1. Exfiltration is a process of reaching own lines from the target area behind the enemy lines using infiltration methods.

General

2. Exfiltration is the most difficult operation. It may not be possible to airlift such troops from the enemy territory. After completing their mission, or for some reason when the link-up does not take place, the infiltrating force may have to fight their way back. The tactics and requirements of exfiltration are similar to those for infiltration. They may have to exfiltrate in even smaller parties than during infiltration to avoid capture by the enemy which will at this stage be more vigilant. There is an added requirement for maintaining a high degree of vigilance when the exfiltrating force closes in to own forward defended localities. Close coordination with the forward troops, timings and routes of exfiltration as well as identifications are necessary.

Circumstances Necessitating Exfiltration

3. Exfiltration may be carried out due to following reasons:-

   a. When the infiltration operation fails.

   b. When the main operation fails and the link-up does not take place.

   c. When there is a change in the overall plan of operation and the infiltration operation is called off.

   d. After the completion of the mission.

Methods

4. The methods and requirements of exfiltration are identical to those for infiltration:-

a. Normal patrol tactics, which is movement in small groups according to the availability of the terrain.

b. The entire exfiltrating force moves from one hide to the other and break their way through the enemy lines supported by own troops. This method may be adopted when the terrain offers adequate cover and there are gaps in enemy position.

Conduct of Exfiltration

5. As soon as the exfiltration becomes inevitable, the infiltrated force commander should immediately manoeuvre his force to escape, while the enemy opposition is still thin and spread out. Surplus equipment, which can not be carried, may either have to be destroyed or burried and to be retrieved if possible at a later stage.

6. If immediate escape is not possible, movement to and occupation of the nearest important tactical ground would be the best course until the time of favourable opportunity for a break- through is available. This break through should best be carried out in the hours of darkness.

7. If the force is not in a position to resist the enemy till the availability of a favourable opportunity, the commander should divide his force into small groups and order them to exfiltrate through the enemy lines at night or hide in the area till the situation permits to exfiltrate.

8. When a pre-planned exfiltration is ordered after the completion of the mission, the force exfiltrates in pre-arranged order over predetermined routes through a series of previously selected RVs. Should the enemy organise a close pursuit some element of the exfiltrating force should assist the rest of the force by fire and movement.

Coordination

9. The following points need special attention:-

a. The planned routes/lanes and time of exfiltration in relation to anticipated tactical situation.

b. Passwords and identification arrangements during exfiltration especially with own troops in the FDLs.

c. Assistance required by the exfiltration force in terms of supporting fire and arrangements of feints.

d. Coordination with patrol commandos or any other friendly parties operating in the area.

CHAPTER 11
FIGHTING IN BUILT-UP AREA

SECTION 51 – CHARACTERISTICS AND REQUIREMENTS

General

1. In our operational environment, especially in Punjab, the border between Pakistan and India runs mostly through populated areas. At places some of the villages are located right on the border and in most of the cases; these are generally in the line of main defensive positions. Tactical importance of these villages certainly offers a good line of defence being relatively at height and also controlling the entire available communication infrastructure in that area. These tactical advantages demand that these villages be included in our defensive positions. Besides these advantages, political considerations also demand that these must be defended and their fall to enemy's hand should be avoided as far as possible. Properly organized defence in built-up areas will force the enemy to commit large forces and inflict heavy losses on him.

2. Fighting in a built-up area must be considered an integral part of every soldier's training. Such fighting is not a job for the specialist. It is essential that all troops should have thorough understanding of the problems with which they will be faced when fighting in built-up areas. Built-up areas are often easy to defend and costly to attack. Cover from fire and from view enables small bodies of determined troops to hold out for long periods.

Characteristics

3. Characteristics of fighting in built-up areas are considered in relation to their effect on the following:-

a. Maintenance of movement.

b. Application of fire.

(1) Direct fire weapons.

(2) Indirect fire weapons.

c. Communications.

d. Logistics.

e. Leadership.

4. **Maintenance of Movement**

a. Heavy aerial or artillery bombardment creates rubble, which may make streets impassable.

b. Since the available cover is set in straight lines, all movement in the open can be observed unless it is concealed by smoke, dust, or darkness.

c. Smoke is effective for concealing movement or for surprise and deception but remains effective far longer than in open terrain and may restrict own observation or movement.

d. Movement has a third dimension, that is, height. This is usually the most effective direction for working within an area, from roof-tops, downwards.

e. In order to avoid open streets, movement may have to be from building to building through breached walls.

f. Large scale movement, even by infantry, is seldom possible so fighting is reduced to subunit actions.

g. Pattern of movement changes constantly in major cities and calls for skill and initiative on the part of junior leaders.

h. Difficulties of command and control are intensified.

j. Movement by night pays greater dividends than in other types of terrain in built-up areas but the problems of command and control are increased.

k. Progress within built-up areas is generally much slower and may be as little as a few hundred metres in 24 hours.

l. Mines and booby traps can be used most effectively in buildings and rubble to further slow down movement.

m. All vehicular movement, wheeled and tracked, is canalized by streets and lanes.

n. Concealment and cover are available to both sides.

o. Streets and lanes that allow movement usually constitute ready made fire lanes and killing zones.

5. **Application of Fire**

a. *Direct Fire Weapons*

(1) Observation is limited and so are fields of fire except along streets and alleys.

(2) Location of enemy weapons is difficult because sound is magnified and creates echoes between buildings.

(3) Mutual support is generally within the section itself or between sections of a platoon.

(4) The effectiveness of weapons, like the machine-gun, is limited due to problems of manhandling inside buildings and limited fields of fire. Short-range automatic weapons are more effective.

(5) The recoilless rifle is ideal for use against buildings and bunkers but is difficult to manhandle and is vulnerable to enemy snipers. Rocket launchers and energa grenades are, therefore, more useful.

(6) Snipers are most effective in this type of terrain.

(7) Flame throwers, both vehicle mounted and man packed are effective in setting fire to buildings and for getting enemy out of bunkers and basements. Indiscriminate setting of fire to buildings must not be permitted.

b. *Indirect Fire Weapons*

(1) Only high trajectory weapons can engage enemy localities behind buildings. Mortars are, therefore, very useful but have limited effect against modern concrete structures.

(2) Observation is rather limited even from high buildings.

(3) Location of enemy guns and mortars is difficult because of magnified sound, echoes and screening of locating devices.

(4) After the break through, artillery support is restricted because of limited observation, trajectory of guns and danger to own troops.

(5) Air observation is generally necessary for finding good targets for artillery, but targets will appear only for short duration and can be easily lost.

(6) The use of artillery fire within built-up areas must be related to the movement plan because heavy artillery concentrations can block streets and make further progress more difficult.

6. **Communications**

a. Intervening buildings will create screening and more reliance will have to be placed on runners, dispatch riders, and liaison officers and at lower levels on visual signals. Where radio is used, sets must work from roof tops.

b. Radio relay will also have a restricted use due to lack of clear optical paths.

c. Line communication will be the mainstay but lines will be liable to interruption due to falling debris and burning buildings.

d. Vehicles with heavier sets may not be able to negotiate the narrow streets or alleys.

e. Existing civil communications may have to be utilized with their attendants accepting security risks.

7. **Logistics**

a. Normal logistic requirements will have to be maintained but with a greater emphasis on small arms ammunition, water and evacuation of casualties.

b. Traffic control problems are increased by the constant changes in direction of traffic due to blocking of new streets by own or enemy actions.

c. Evacuation, control and feeding of civilians will pose a major problem. This must be viewed in the light of running utility services, spies, fifth columnists and saboteurs.

d. The distribution of supplies and ammunition will be magnified.

e. Supplies may have to be dispersed and stocked in defence to cater for possible isolation of areas.

f. Supply and distribution of water will initially have to be planned for, as enemy may contaminate local sources.

g. An increased number of casualties can be expected. A greater number of stretcher-bearers will be required.

8. **Leadership**. Street fighting is a dangerous business. It is physically and mentally exhausting and success is measured in metres and building by building. In these circumstances much will depend on the initiative and standard of leadership at section, platoon and company level.

Requirements

9. At company/platoon level, the requirements of village clearing and street fighting are:-

a. Control.

b. Balance.

c. Speed.

d. Thoroughness.

10. The company/platoon commander should make a simple plan. Which can be achieve by:-

a. Choosing short bounds and limited objectives.

b. Having firm bases from which all movement in the open can be covered by fire.

c. Clearing his objectives thoroughly.

d. Reorganising rapidly.

Nuclear, Biological and Chemical

11. A nuclear attack in cities/towns will seriously disrupt the system of supply and evacuation. The houses may collapse and the street may get blocked thus restricting the movement. The control of refugees will further aggravate an already difficult situation. Such attack may lead to increased casualties.

12. Effects of Chemical and Biological attacks will persist for longer duration because of confinement of the area. The stocks/facilities will thus be made non-operative for longer duration.

SECTION 52 – COMPANY/PLATOON IN DEFENCE IN A BUILT-UP AREA

Basic Considerations

1. Important considerations to be borne in mind while planning defence in built-up areas are:-

a. *Control*. Lack of communication would isolate subunits to fight independently.

b. *Observation*. There may be a few high buildings, which could be used for long-range observation, but generally speaking the visibility of the immediate front may be restricted by buildings/houses for which engineer effort may be required.

c. *Field of Fire*. Mutual support between posts is difficult and often impossible. Fighting is largely at close quarter. Hence greater and effective use of short range weapons is required. Also locality should be of a limited size.

d. *Difficulty of Locating Fire*. Is increased by the echoing effect of fire amongst buildings.

e. *Movement*. Movement within the built-up areas is either from roof tops or through the walls by making holes in the inter-connecting walls. Necessary reconnaissance must be done to make maximum use of both and denying their use to the enemy.

f. *Cover and Concealment*. If it is possible and proper then tank hides are organized to mislead and surprise the enemy by occupying mud houses, whose walls can be demolished for action or a hole made for the use of the tank gun.

Layout

2. Although the decision whether or not to take-up a defensive position in building, and if so which building to occupy, will not normally rest with the company/platoon commander, but he may sometimes have to make this decision,

therefore, he should understand the factors which must be considered. The company/platoon commander will have a definite task to perform and this must of course be his first consideration. Fields of fire must be adequate and it may be necessary to reject an otherwise suitable building because the enemy can get close to it under cover.

3. Generally speaking, built-up areas can stand up to a heavy bombardment from artillery and from the air, provided that the buildings are reasonably strong and not too inflammable. The most durable buildings are those of steel and concrete, the next best are stone buildings.

4. **Selection of Building or Houses for Defence**

a. The first pre-requisite for a building or a house to be selected for defence is that it must fulfil the battlefield requirements. It entails that it should dominate an approach, keep an obstacle under effective fire, provide depth to other buildings or act as a pivot for employment of the reserves. In addition the selected house should fulfil the following requirements:-

(1) It should be strong enough to withstand artillery, mortars and machine-gun fire.

(2) Should have a cellar for providing protection against aerial bombing, to hold stock of ammunition supplies and to serve as a shelter for sick and wounded.

(3) Should have a clear field of fire and observation (approximately from 50 to 100 metres).

(4) Should not be under observation of a dominating building occupied by enemy from where he can bring accurate and effective fire on it.

(5) It should have open space and courtyard in front where trenches could be prepared for protection against bombing.

(6) Should not be so conspicuous and front building in the street. A house located behind the other buildings is more suitable for defence.

(7) Should be mutually supporting with other buildings.

(8) The building should be suitable for all-round defence i.e. it should be capable to beat back an attack from any direction during day or night.

b. After the selection of buildings the defence should be considered from the enemy point of view. This should include the best direction of attack from enemy side and the tactics, which the enemy is likely to employ for clearing it. This will enable the defender to decide about the suitable location of the loopholes, wires, booby traps warning devices and other defensive measures. Loopholes are made on both sides of the walls in case the building is situated in the middle of the street.

5. **Sitting of Weapons**

a. Weapon sites may be found in the building itself, on the roof or in the grounds outside. Any weapon sited within the building must be kept back from the windows or holes through which they are to fire. Automatic weapons must stand on a surface, which will not give off dust when they are fired. Weapons sited inside a building will have narrow field of fire. It is the weapon sited outside the building, which will have the main fields of fire. Soldiers manning these weapons have a covered line of withdrawal into the building to enable them to fight on for as long as possible.

b. At least two and if possible more light machine-guns must be sited to give each other mutual support. The area over which their fire interlocks should be the platoon's selected killing ground.

c. Single rifleman will be posted as snipers on roofs and under the caves of buildings. The others should be posted in pairs outside or inside the building to cover all approaches. They should have plenty of grenades, which are easy to drop on the enemy below.

d. Any natural obstacles should be improved with wire and mines. Streets and lanes should be barricaded and covered by fire.

6. **Selection and Preparation of Loopholes**

a. A number of loopholes should be selected to cover every passage so that the firer can fire from some other loopholes before enemy action, once he feels that the enemy has discovered the first position. Normally those loopholes are most effective and useful which are made at the ground level and in the corner of a building. In addition the loopholes should fulfil the following requirements:-

(1) As far as possible loopholes should be made at the unexpected places.

(2) There should be sufficient space for the firer, to be comfortable and the muzzle of his rifle should not protrude beyond the wall.

(3) The loopholes used by the enemy before the capture of the buildings should not be utilized.

b. The following precautionary measures must be taken to avoid identification and destruction through loopholes:-

(1) Loopholes located close to the ground should be covered with wire mesh for protection against enemy grenades.

(2) Remove all sources of light behind the loophole to avoid detection of the shadow of firer's head.

(3) Ensure that loopholes do not eject smoke.

(4) Place sand bags or fix iron plates around the loopholes to make them bullet-proof. If sand bags are placed on the upper story they ensure that the roof does not cave in due to extra load and as a precautionary measure place a support under it.

(5) Place a sand bag behind every loophole to keep the rifle on it.

(6) Loophole should be large enough to throw a hand grenade out and make a mouse hole in the wire mesh covering the loophole.

(7) Fix the glasses in one or two loopholes if available. This will enable the firer to keep the important routes in range and observe outside moves while remaining safe in a secured position.

(8) A few loopholes should be made inside the building (in the walls, ceiling and floors) with a view to deal with the enemy entering the house.

(9) Some loopholes should be made towards the narrow lanes and stairs for the bayonets to strike the shoulders of enemy passing in front of them. These loopholes may be at a height of approximately one metre from the ground.

(10) Mouse holes covered with wire mesh or iron bars may be used to fire at the enemy while they are passing near it.

(11) Oblique firing holes be made near the narrow lanes to block enemy entrance to various rooms.

(12) Firing positions must be kept well behind the loopholes, if the enemy sitting on the roof of a building can fire downward.

(13) Make some holes for deception purposes on conspicuous places of the building from where you do not want to engage the enemy.

(14) Camouflage the loopholes in such a way that the enemy is unable to detect them. Firing holes should be given an irregular shape and fix a canvas mesh or a perforated zinc plate. Cut it in such a way that the camouflage material coincides with the joint of the bricks and then paint it to blend the bricks and holes. You will be surprised to see that it is transparent from the inside and difficult to identity from the outside.

(15) Conceal the loopholes in the outer wall, using a bush, shadow of a ladder or wooden log.

Control

7. Section commanders must be able to control their sections even if they have to move from one part of the building to another to do so. Within limits, men must be allowed to open fire at their own discretion. Any limits which the platoon commander or higher authority may lay down, must be made clear to everyone.

Sequence of Work

8. A lot of work is needed to put buildings in a state of defence. Some of the tasks involved are more important than others. The order of priority also depends on the time, tools, stores and labour available, but as a guide the following sequence is suggested:-

a. Clear field of fire.

b. Knock glasses out of windows.

c. Nail chicken wire over window frames to keep out enemy grenades.

d. Block all unused doors, windows and chimneys with sand bags and mattresses to stop the enemy getting in.

e. Put up barricades in rooms well back from windows. Weapons can be fired form these and they will give some protection from grenades thrown by enemy house clearing parties.

f. Make loopholes in windows or walls to cover approaches. Sand bags should be used but some skill is necessary in sitting as obvious loopholes in a bare wall are easily located and neutralised by the enemy.

g. Shore up ceilings with timber and sand bags.

h. Wire dangerous approaches.

j. Make mouse holes as necessary between rooms and houses to allow freedom of movement inside.

k. Get rid of inflammable material, turn off gas and electricity and fill the bath and available pans with water.

l. Make small holes in upper floors to enable grenades to be dropped on enemy below.

m. Remove or block stairs with wire or nailed boards.

n. Make a few man sized holes to allow men to go up and down between floors using knotted ropes or rope ladders.

o. Put barbed wire inside windows angled to the floor to discourage enemy from driving in.

p. Dig a crawl trench from a mouse hole on the ground floor to serve as an emergency exist or to give access to trenches outside.

q. Convert cellars or dig shelters with overhead cover.

r. Dig positions outside if necessary for platoon anti-tank weapons. Alternative positions outside buildings are valuable and may also serve to deceive the enemy.

The Advantages and Disadvantages of Village Based Defence

9. **Advantages of Village Based Defence**

a. Villages provide ready-made cover and protection to the defender. They also provide excellent observation.

b. Generally surrounded by sprawling wheat/paddy fields, they provide better fields of fire all round, more so when the crops are not so high.

c. It is difficult for the attacker to pin-point the exact location of weapons on the periphery, or in depth. Tanks can be easily hidden and can fire through holes in walls, thus they cannot be pinpointed and destroyed. Alternately tanks can stay on the rear of the village or somewhere in the centre and can come out to the flanks and engage the assaulting troops.

d. Defender enjoys flexibility owing to the facility of quick protected movement within the locality, using prepared passages.

e. Company/platoon attack has to converge on a narrow frontage, thus allowing greater concentration of defender's fire, and better chances of breaking up the attack.

f. Neither attacking company nor the supporting tanks can sweep over the defence, the attacker have to first gain the foot hold and then fight through the village.

10. **Disadvantages of Village Based Defence**

a. Defended locality is conspicuous; it is easy for the attacker to determine its location, size and extents.

b. The objective, being well defined, can be subjected to accurate artillery shelling. Most of the villages being made of mud or sun baked bricks with narrow streets; it is easier to raze it to the ground by artillery fire.

c. For the same reason, the attacking company/platoon has less chances or going astray by day or night.

d. Most weapons have to be sited on the periphery, the fields of fire in depth being very restricted. These peripheral weapons can be easily subjected to shelling and smoke to cover the attack.

e. The defended locality being in the nature of an island, the attacker may be able to isolate it by getting behind it and by cutting off its routes of withdrawal or reinforcement.

f. Mutual support is restricted.

Conduct of Defensive Battle

11. The Defensive Battle is conducted on the following pattern:-

a. *Action of Protective Detachments*. The conduct of Defensive Battle by the protective detachments in built-up areas is generally the same as in other area. Following elements are employed:-

(1) Reconnaissance elements.

(2) Covering troops or the advance position.

(3) Screens.

b. *Battle of Main Defensive Position*

(1) Action of posts on outer perimeter.

(2) Action by defensive locations.

(3) Counter-attack.

SECTION 53 – **SECTION IN DEFENCE IN A BUILT-UP AREA**

General

1. A section is a basic fighting unit of a battalion; therefore, section commander must be thorough and clear about the actions of his section, particularly in a built-up area. He must give detailed orders to his section before going for an operation.

Defence Inside Building

2. **Tasks of Section**. Section in defence can be employed on following tasks:-

a. To deny access to the enemy, at least to impede the enemy.

b. To contain enemy penetration/to eject enemy penetration by counter-attack.

c. To provide maximum protection from the effects of enemy fire.

Construction of Fighting Positions

3. There is no uniform layout for fighting positions inside buildings. Imagination and ingenuity of mind should be used. Locations and configuration of fighting positions are dependent on:-

a. The tactical situation.

b. The protective value of walls and ceilings.

c. The availability of construction material.

4. Every fighting position is required to have:-

a. Field of fire (for engaging enemy outside, but also inside the house).

b. Protection from all directions, including bottom and top, both against fire and debris.

c. Good access routes.

5. In most cases fighting positions will be constructed for only one soldier, however, they will be arranged to be mutually supporting. Construction material could include tables, chairs, chests, settees, doors, boxes, sandbags and ammunition boxes.

6. The position of the section sentry should be so selected that he must be able to observe the enemy's probable avenue of approach. Section commander's position should be such from where he can control his major weapons and also main fighting.

7. **Examples of Fighting Positions**. Following are some of the examples of fighting positions: -

a. *Fighting Position behind a Barricaded Window*. Figure 23(a)

(1) The rifleman is camouflaged by window barricade.

(2) He can also engage targets inside the house (kneeling positions).

(3) Protection against small arms fire from below by sand or dirt-filled ammunition boxes.

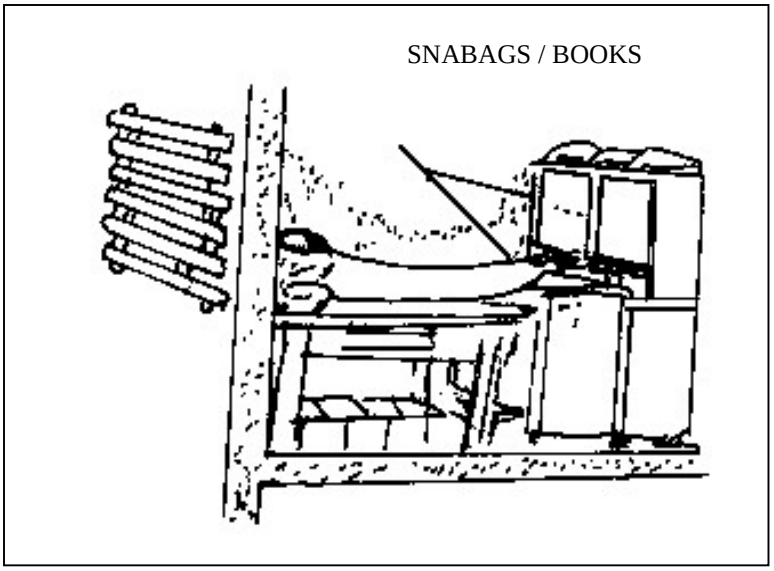


Figure 23 (a) - Fighting Positions behind a Barricaded Window

b. *Fighting Position in Depth of a Room.* Figure 23 (b).

(1) *Advantage.* A rifleman cannot be observed in depth of a room as the smoke by firing is not visible through windows/doors.

(2) *Disadvantage.* Field of fire is restricted and there is a need to establish position for firing through several windows if possible.

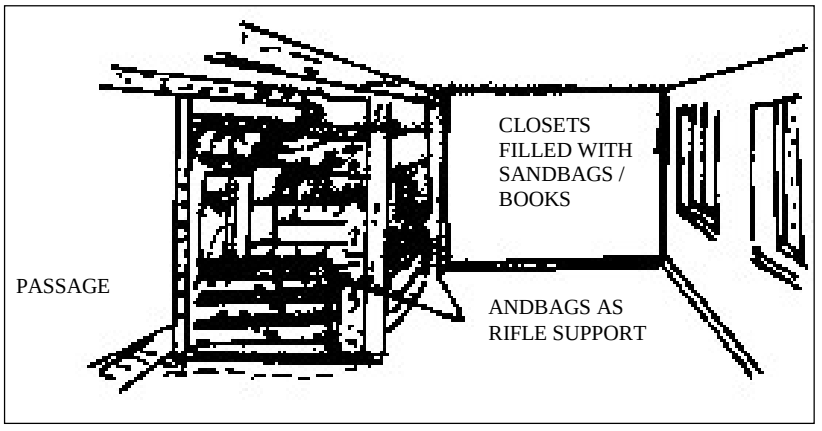


Figure 23 (b) - Fighting Position in Depth of a Room

c. *Corner Loophole.* A corner of a building provides good field of fire but this type of loophole is visible from a distance, therefore, it should be covered through some means, for example bushes et cetera. Figure 23 (c).



Figure 23 (c) - Corner Loophole

d. *Loophole for a Rifleman.* In order to offer a wider field of fire, the loophole for a rifleman should be expanded from inside as shown in figure 23(d).



Figure 23 (d) - Loophole Expanded from Inside

e. *Fighting Position behind an Interior Wall with Fields of Fire through Several Windows.* The fighting position as shown in figure 23 (e) offers an excellent concealment and protection to the rifleman. The cover to the top and to the sides has been omitted in the figure 23 (e)

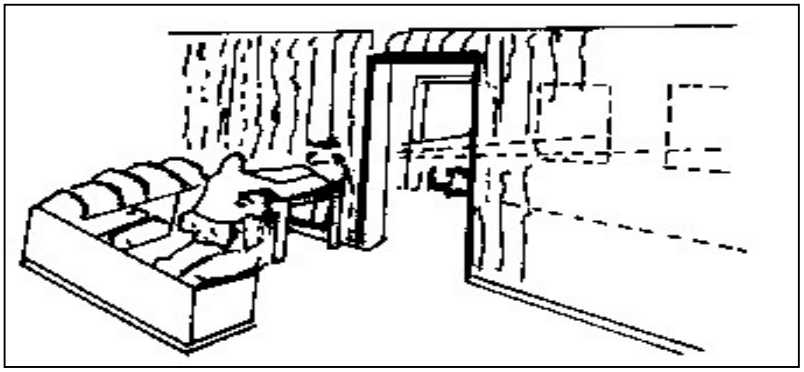


Figure 23 (e) – Fighting Position behind an Interior Wall

f. *Alternate Positions*. For every position there must be an alternate position from which the same task can be carried out. Following should be considered for alternate position: -

(1) Normally it should be located in an adjacent room/on the same floor.

(2) It must have a covered and concealed access route.

Constructing Obstacles

8. After the fighting positions have been selected and prepared, the section should try to create some obstacles inside the house to deny enemy's entry into the house. These obstacles must be covered by fire. Following are some examples of constructing obstacles inside the house: -

a. *Concertina Wire Fence in Conjunction with a Yard Gate.* Figure 23 (f). The concertina wire: -

(1) Must be covered by fire (if required, by an adjacent squad).

(2) Should be reinforced by booby traps, if possible.

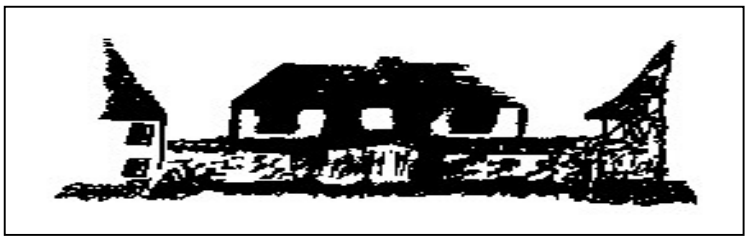


Figure 23 (f) - Concertina Wire in Conjunction with a Yard Gate

b. *In Doors*. All the doors/windows should be locked from inside and barricaded with the help of tables, beds et cetera. Booby traps above the door will further ensure the security as shown in figure 23(g)

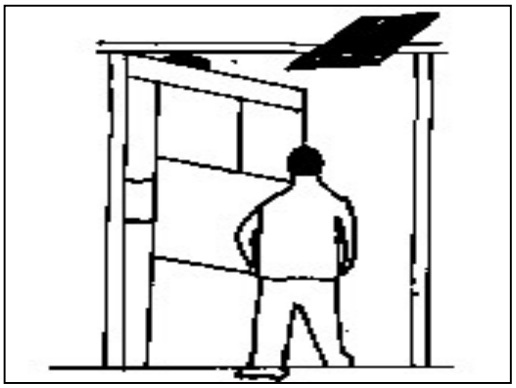


Figure 23 (g) - Booby Trap above the Door.

c. *Window Barricading*

(1) Figure 23 (h) shows a window with wire mesh as protection against hand grenades.

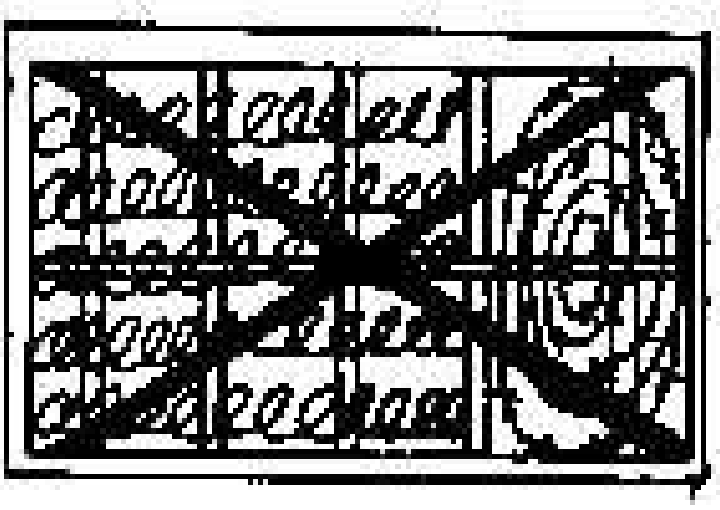


Figure 23 (h) – Wire Mesh



Figure 23 (j) – Windows with Sandbags

(2) Figure 23 (k) shows the windows barricades with boards. It is important to reinforce the boards against small arms. Avoid regular pattern and try to alternate barricaded and non-barricaded windows. Cover non-barricaded windows by fire.



Figure 23 (k) – Windows Barricades with Boards

d. *Stairways Blocked by Wire.* Stairways as indicated in figure 23 (l) are blocked by wire. The blocking effects may be enhanced by booby traps.



Figure 23 (l) – Stairways Blocked by Wires

Blocking Entries

9. Ensure the removal of all potential entry aids as early as possible. Trees in the vicinity of a building should be blocked by wire to prevent climbing. Rooms or adjacent buildings should also be denied to the enemy by booby traps.

Construction of Hand Grenade Throwing Holes

10. While constructing the holes for grenade throwing, it is useful to shape the opening obliquely downwards and to line the hole with a stove pipe section as shown in figure 23 (m)

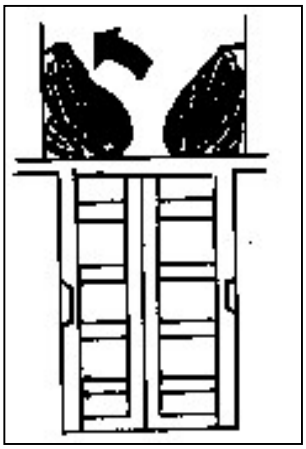


Figure 23 (m) – Hand Grenade Throwing Holes

Mouseholding through Ceilings

11. The purpose of mouse holding through the ceiling is to permit friendly movements despite blocked/barricaded stairways. Ladders, boxes, chests in combination with a shelves and wooden stairs could be used as climbing aids. These aids must be removed/destroyed once the enemy has entered the house. Figure 23 (n).

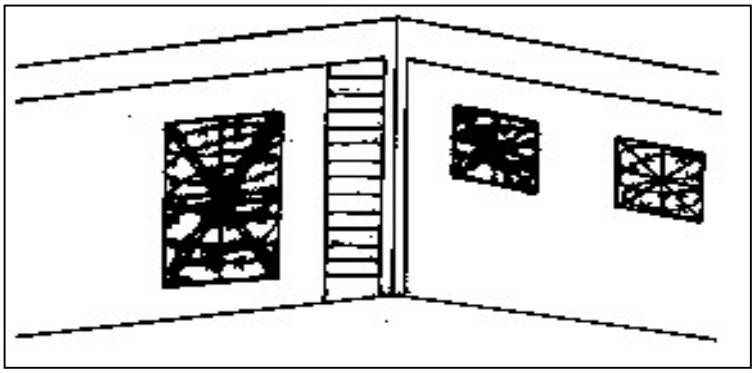


Figure 23 (n) – Mouse holding through Ceilings

Ceiling/Floor Reinforcement

12. The purpose is to enhance the protection of rooms having firing positions. The materials could be wooden or steel beams, thick boards, heavy doors et cetera. Figure 23 (o)

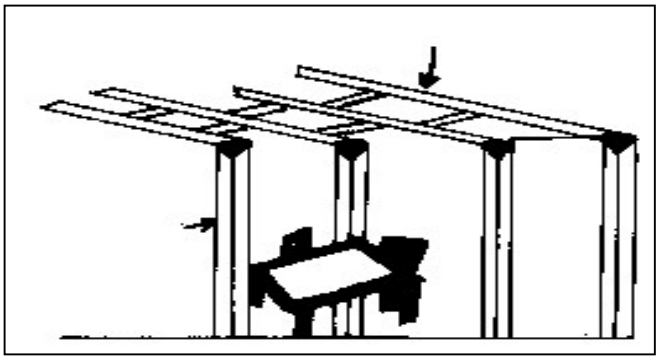


Figure 23 (o) – Ceiling/Floor Reinforcement

Sitting and Construction of a Shelter

13. The section requires a shelter for protection against weapons effects and building collapse. This is normally designated by the platoon commander but it

can also be selected by the section commander. If it is selected by the section commander then following points must be kept in mind:-

a. Use basement rooms which are completely underground and surrounded by as many structural walls as possible. These are concrete or masonry walls with a thickness of more than 30 centimetres.

b. The ceiling must consist of a minimum of 15 centimetres of reinforced concrete.

c. Small rooms provide more protection.

d. If required, shore up the ceiling with beams.

e. The shelter must have an emergency exit.

Fire Fighting Measures

14. Some measures are as follows:-

a. Pre-stock water and sand in every room for fire fighting.

b. Cover flammable floors with a layer of sand.

c. Secure basement doors against flammable liquids by raised thresholds, for example with sandbags.

d. Easily flammable materials like curtains, carpets and furniture must be removed from rooms, which have fighting positions.

e. Shut off gas lines.

f. Remove the fuzes/switch off electrical power before the battle.

Supply

15. Pre-stock supplies in all rooms where soldiers are committed especially:-

a. Ammunition and grenades.

b. Rations and water.

c. Fire fighting materials.

Lines of Communications

16. Contact with platoon commander should be made through telephone and wireless. Within the squad, contact can be established with rope/pull wires.

Alerting and Occupation of Fighting Positions

17. Rehearsals for rapid occupation of fighting positions should be carried out as soon as possible. These rehearsals must be carried out in darkness as well.

18. **Defence Routine**

a. Upon completion of all work, the squad sentry will observe his assigned sector while the rest of the squad alternates between rest and personal care and hygiene.

b. However, you must ensure continuously that:-

(1) All fighting positions are further improved.

(2) All weapons are ready to use at all times.

(3) All soldiers are informed of all changes in the situation.

(4) The operational readiness of section is further enhanced by repeated alert and counter-attack rehearsals.

Fire Control

19. Following should be kept in mind to ensure effective fire control:-

a. Assign each soldier a mission for the employment of his weapon to ensure that he can also engage targets independently.

b. Issue pre-arranged signals to ensure that you can assume fire control again after your soldiers have been engaging targets independently.

c. As a rule the section commander must ensure that his men do not start firing without his prior command.

Counter-Attacks

20. If the enemy has seized part of section position then by concentrating the remaining section, either penetration should be contained or counter-attack should be launched with uncommitted elements before the enemy has time to consolidate. But a point to note is that section commander must lead the counter-attack himself.

Defence Outside Buildings

21. During his advance to contact, the enemy will try to avoid the main streets

and primarily advance through gardens, backyards and parks instead. Therefore, the platoon commander may commit one of his sections outside buildings to cover these approaches. The likely firing positions could be in gardens/parks, behind walls, in rubbled areas and inside stacks of wood et cetera.

Construction of Fighting Positions

22. Fighting positions in parks and other open places are built-in accordance with the principles applicable for the defence in general but in rubbled areas or similar terrain sectors, there are no fixed rules for fighting positions. Here, own imagination and ingenuity of mind should be used. Some of the examples of fighting positions are shown in figures 23 (p) to 23 (r) below:-

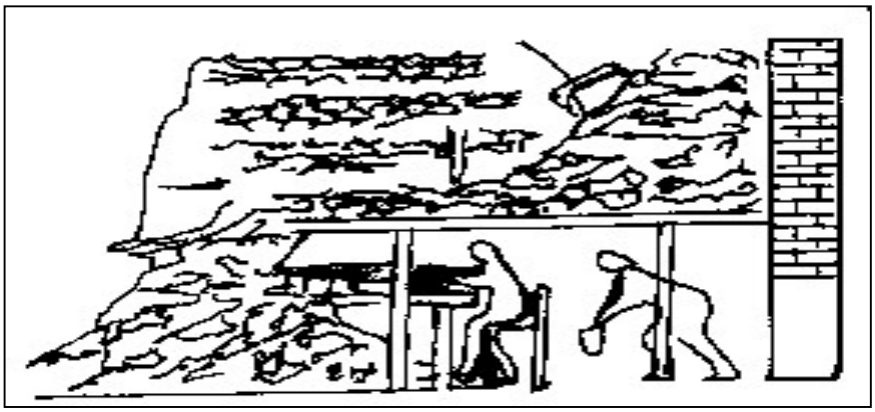


Figure 23 (p) - Fighting Position in a Partially Preserved Building

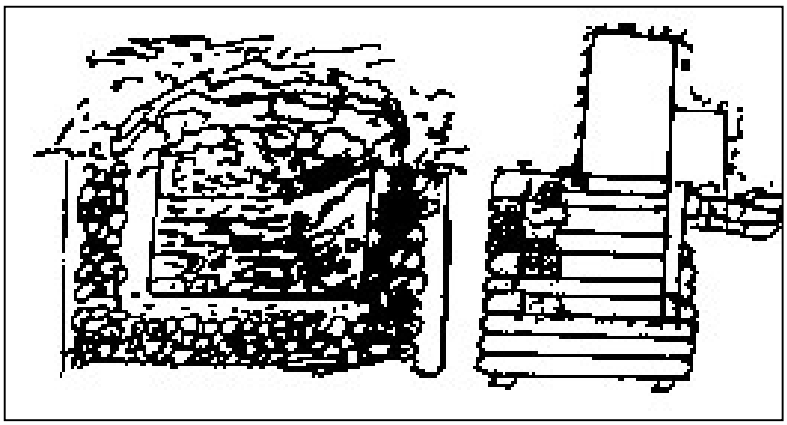


Figure 23 (q) - Fighting Position in a Stack of Wood



Figure 23 (r) - Fighting Position between Concrete Slabs

23. The principles outlined in para 3 above for preparation of defence inside buildings apply correspondingly for all other preparations for defence outside building.

SECTION 54 – **GENERAL - ATTACK IN BUILT-UP AREA**

Basic Considerations

1. **Need to Search the Enemy**. Unlike enemy dispositions in a defended location in the open, enemy dispositions in a built-up area do not conform to any standard pattern. One building may contain a single sniper or a whole platoon. Another may appear completely unoccupied and yet it may not be so. For an attacker there is no way of knowing for certain, except by going in it to find out. He is, thus forced to systematically comb the whole area and root out the enemy room by room, floor by floor and building by building.

2. **Need of Intimate Covering Fire**. The enemy is well concealed and motionless. The attacker is exposed and his movement between rigid and still structures of straight lines are very conspicuous. The enemy has the advantage of being the first to pull the trigger. The attacker does not even know from which direction to expect the bullet. It is therefore, essential to cover the attacker

movement with fire. This covering fire has to be prompt and accurate. It has to cover the movement of very small groups, even individuals.

3. **Capture Objective Bit by Bit**. Once the attacking troops get dispersed inside buildings, it becomes difficult to differentiate between the fire of own and enemy weapons. If a large number of attacking troops are inducted into the battle at one time, at the same point, they would soon get inextricably involved with each other, often with disastrous results. The attacker, therefore, has to work his way forward in small parties in a methodical manner capturing the objective bit by bit.

4. **The Need to Indicate Progress**. The most important requirement is that all elements of attacking troops must be fully aware of each other progress. It is particularly important when infantry is supported by tanks. The technique of indicating progress must be common and be known to all e.g., coloured smoke, or to hang blankets behind the windows, et cetera.

5. **Requirement of Hand Grenades and Small Arms Ammunition**. These will be required in large quantities and appropriate arrangements must be made for the replenishment.

6. **Need of Selecting the Most Secure Approach**. The attacker has to select some or all of the under mentioned routes so that the most secured approach is adopted to avoid unnecessary casualties: -

a. Roads, except highways.

b. Orchards and courtyards parallel to the roads.

c. Roofs of houses.

d. Through the walls of the houses, by making holes in them.

e. Under ground routes.

Requirements

7. **Depth**. Will ensure the maintenance of momentum of the attack.

8. **Short Bounds**. In an attack, towns are normally divided into sectors, sub sectors and sub-sub-sectors and are dealt with as such. Platoons and sections are given objectives within these sectors, sub-sectors and sub-sub-sectors as the case may be. Objectives must be limited and bounds short. Sometimes only a house at time will be cleared, a fresh plan being made for each successive bound. To clear

a street against stiff opposition, it will generally be necessary to employ at least a platoon on each side. If a smaller force is used, the attack will loose momentum.

9. **Domination of Likely Killing Areas**. Open spaces, particularly main streets and squares may be covered by enemy fire; before crossing these, arrangements will always be made for covering fire. No section should move into the open unless a covering fire group is supporting it. There must be one "leg on the ground".

10. **Thoroughness in House Cleaning**. Houses and out-buildings must be searched throughout, including cellars and roofs. Furniture likely to conceal "mouse holes" be removed.

11. **Consolidation of Every House Captured**. Every house must be consolidated after its capture. All round defensive positions must be taken-up and snipers posted on the roof. The house thus becomes a firm base for further advance. Ammunition, especially grenades, must be replaced and casualties dealt with. It should be realized, however that when a whole town or village is being cleared systematically, there will be less emphasis on all round defence at each step than on all available fire power being used to support the advance, but it is still important to prevent infiltration back into areas that have been cleared.

Careful and Simple Planning

12. It is vital that the planning of an attack against a built-up area shall be carried out in the greatest detail. All possible information must be gained from air photographs, maps, street plan, civilians and other sources. The plan must be simple and every man must know exactly what to do, if not, there will be a chaos when a battle has been launched in to the streets of a town.

Administration

13. **Ammunition must be Available**. Dumps will be formed behind the assault troops as they advance, ammunition being delivered by transport of ranks to company sectors and then carried forward by previously arranged parties.

14. **Equipment Should be Light**. Small packs, if worn and used for carrying grenades, should be slung and kept in places by a belt. Leather jerkins and toggle ropes are useful. Smoke grenades, owing to their toxic effect, are very effective for driving the enemy out of cellars.

15. **Plan for the Disposal of Casualties**. Casualties should be left in houses. Stretcher-bearers must always follow up assaulting troops and evacuate casualties to pre-arranged collecting posts. All troops should carry shell dressings as well as field dressings.

SECTION 55 – TECHNIQUE OF TAKING DEFENCE BASED ON VILLAGES

Introduction

1. The likely technique of defence used by enemy would be to create a complex of mutually supporting village localities, with the open areas in between serving as killing ground. Where a village happens to be large enough, the defended locality may be entirely based on its periphery, with only screens and patrols outside. Tanks can be skilfully sited on the periphery by driving them backwards into house, where they remain completely concealed, and from where they only have to move forward a little, if at all, to repulse any attack.

2. Orchards, tube-wells, graveyards, mounds or other ground of tactical importance in between the village will probably be held lightly to prevent penetration, or as protective detachments unless the feature happens to be large enough to be turned into a proper defended locality, in which case it would become part of the complex.

3. **Enemy Defensive Measures**

a. Defensive measures will probably include a series of strong points sited in depth through out a built-up area; heavily defended from within and covered by fire from neighbouring buildings.

b. Streets will be blocked. Blockades and other places likely to provide cover will be covered by enemy fire from nearby houses or other points of vantage and used as a killing ground.

c. Enemy tanks may be dug into houses or used to support a counter-attack.

d. Long-range anti-tank weapons may be sited both inside and outside buildings and places inside streets to fire at point-blank range into main streets. Short range anti-tank weapons may be sited in upper storeys of houses.

e. Inside buildings, weapons will be sited well back form the windows, fields of fire being sacrificed to achieve surprise. Staircases may be removed altogether alternatively, heavily by fire and obstructed by wire. Rooms may be mouse holed, the holes being hidden by furniture, and barricaded to prevent grenades being thrown into them.

f. Certain parts of a town may be comparatively free from the enemy. If a town is completely free from the enemy, beware of booby traps. Generally speaking heavily defended towns are not booby trapped as this would be a hindrance to the defenders themselves.

Implications for Attack

4. Basically the problem is that of capturing and clearing a small built-up area, mostly of mud houses. Yet, the operation does not amount to fighting in built-up areas, for the following reasons:-

a. The mass of the built-up areas happens to be small, involving a sharp engagement of relatively short duration rather than long engagements which ensues in a town or a city made of brick, concrete and steel structures.

b. For the same reasons, it is possible to employ technique, to get behind the defenders at company level.

c. In a large town the defender plans to draw in the attacker, so as to involve him in house to house fighting. But the smallness of the village/built-up area is used more for its protection rather obstacle value.

d. In a town the defender finds it difficult to use his tanks effectively. In a village it is quite easy for the defender to get his tanks in position by knocking down mud walls, the enormous firepower of these tanks at close rang can be devastating for the attacking infantry and armour exposed in the open.

5. There are two other features which have to be taken into account by the attacker. Firstly, the fact is that villages usually lie within range of long-range small arms from each other. Often they happen to be situated close enough to be developed into a mutually supporting complex of village strong points. The area in between villages may be flat fields with some crops trees or small orchards (latter serving as defensive outputs) the village strong point command excellent killing area all round.

6. Secondly, the villages in the forward area are likely to be covered by a defensive anti-tank or mixed minefields, sited if possible, within small arms ranges. In addition to this the village will normally have a protective minefield around it, mainly of anti-personnel mines. This protective minefield, meant primarily to break-up the assault, may not be deep, but it may be backed by wire obstacles and a concealed ditch to act as a pitfall for assaulting infantry.

Technique of Attack

7. Villages provide very good all round observation and field of fire and village strong points command excellent killing area particularly to the front. Coupled with other means of strengthening a position, i.e., mines, wire, ditches et cetera; it would be difficult to plan an attack during the hour of daylight. For the same reason frontal attacks will be very costly indeed. Therefore all attacks which should be aimed at the flanks and rear of the enemy position must be planned during the hours of darkness.

8. Depending on the magnitude of enemy minefields and number of lanes to be cleared, the engineers require to make these lanes. It is after lanes have been made that the battalion F echelon transport will be able to cross over and join the assaulting company/platoon for further exploitation of their success on the objective. Since the clearance of lanes and passages of F echelon transport, et cetera through them require time, therefore, the attack must be launched as soon as possible after last light.

9. The minefields from of enemy position will be covered effectively by small arms/artillery fire, particularly in this case, as stated earlier ideal field of fire and observation would be available. The weapons located on the frontal periphery of the village in particular, will seriously interfere with any breaching operations on the minefields, unless these weapons are effectively neutralised in the first instance. Therefore, to provide opportunity for the engineers to work on clearance of the lanes without any interference from enemy direct small arms fire, it is essential that the attacking company/platoon takes action to deal with these weapon positions during the earliest stages of the attack.

10. Since the enemy strength would be concentrated more often than not on the periphery of the village, the company/platoon commanders plan their attack, direction of assault, strength relative to enemy dispositions, phases and fire support et cetera, in relation to the nature of periphery itself. This of course does not mean that the inner sections of the village will by any means be ignored even if these are occupied by the enemy. Once perimeter buildings of the village are

secured, particularly those to the flanks and rear, the enemy's position in the area as a whole is bound to become untenable after some time.

11. The company/platoons will have to assault by walking across the minefields, with the 'Forming Up Place' preferably in a safe area short of the minefield. Where it is intended to infiltrate in small parties, gaps through the minefield would have to be made secretly on that or the previous night.

12. The fire effect of heavy weapons and supporting the company attack join the attacker till such time that lanes have been cleared, can be put to optimum use by positioning these weapons short of enemy minefields, suitably, to neutralise enemy weapon position located on the frontal periphery of the village.

13. Due to the dominating nature of enemy dispositions, reinforced by ditches, wire and protective mines et cetera, it will not be possible for the company/platoon/ sections to move on to the objective in the normal assault formations and in 'one group'. The assault must progress by 'fire and movement' from house to house properly controlled and in sections/platoon groups, which obviously calls for a high standard of training. Rocket Launchers and energa grenades will be used to demolish houses or make holes/passages for further penetration.

14. The artillery fire support must be prepared meticulously by the battalion on the suggestion of company commander and artillery observer, and much greater degree of accuracy in fire is necessary then in ordinarily cases. Furthermore, the artillery fire support will have to be for a longer duration due to a longer distance, between forming up place and the objective (depth of minefield and the distance between far edge of the minefield and the objective).

15. In order to deal with enemy defence particularly in the closing stages of the assault, the company/platoon/section must ensure carriage of following demolition and other stores:-

a.	Pole charges	for destruction of bunkers pill boxes and houses.
b.	Shaped charges	for same as in a above.
c.	Limpets	for tank hunting parties.
d.	Banglore torpedoes	for clearing strong/points/barbed wire obstacles et cetera.

16. In order to easily unhinge the enemy position in the village, it is better to threaten it from more than one direction at a time. The combination of one flank and rear attack is feasible from command and control point of view. The infiltration of a smaller group to the rear; which may be a fighting patrol – platoon/section strength, with the aim of firing into the enemy position during the attack will pay greater dividends.

Problem of Conducting Attack

17. The main problem of conducting an attack are as follows:-

a. *Platoon Problems*. All attack in built-up areas resolve themselves into a series of independent platoon operations, each against one house (or part of house). For each attack, a platoon requires three groups, probably section for the following tasks: -

(1) To clear the house (or part of it)

(2) To form a firm base from which to support and protect the house clearing group.

(3) To provide a reserve to meet the unexpected situations. A platoon can, if the opposition is light, deal with the clearing of a whole street, but bearing in mind the need for avoiding street, it is normal for platoons to fight their way forward through the blocks using roof tops, sewers and the back gardens and by mouse holing from house to house. A platoon frontage should never exceed 100 metres and if possible should be confined to the width of one house.

b. *Streets*. Any enemy expelled by platoon from the house should be destroyed by low trajectory weapons sited to fire down the streets. Thus, street are used both as boundaries between platoons and section, and also as killing areas for our own weapons. Later, they can be opened up as axis for unit supplies. Against a small enemy, they may be used as the actual thrust lines for an attack.

c. *Covering Fire*. If it becomes necessary to cross open spaces, adequate covering fire, including smoke, must be arranged. No section should ever move into the open until other troops are in position to give covering fire. There must always be one leg on the ground.

d. *Thoroughness in House Clearing*. Houses and out buildings must be thoroughly searched, including the cellars and roofs. Furniture likely to conceal ‘mouse holes’ must be removed.

e. *Firm Base and Reorganization*. Every house must be consolidated after its capture and put immediately to a state of all round defence with snipers posted on roof. The house thus becomes a firm base. This is vital because: -

(1) Fresh attacks in built-up areas will only succeed if they are launched from firm bases.

(2) Unless re-organization is through and each house and bound is made firm as the attack goes forward, the attack will be badly balance to meet a counter-attack and the enemy may infiltrate back into cleared areas.

18. **Platoon Drills for Clearing Small Villages**. A single platoon may be given the task of clearing a small hamlet or group of farm buildings with a single street through it. A battle craft which can be applied to both tasks is similar to that of a platoon clearing a street with the following variations:-

a. *Ambush Party*. A few men under a JCO, preferably the platoon Second-in-Command, should be detailed to go well to the rear of the village and take-up a position on the enemy’s main line of retreat. They must be in position before clearing begins. This party must not be seen and be well clear of the buildings since once the enemy realizes that he is trapped, he will fight harder in his positions.

b. *Fire Section*. The fire section will cover the main street. The platoon may have to fight to get the fire section into position and clearing must not start until the fire section is in its correct place. If necessary the fire section should advance by fire and movement to cover the clearing section as they move through the village.

c. *Clearing Section*. Each house may present a slightly different problem to the clearing section. Two things will influence the section commander when he is planning where to enter the house and how to clear it, the sort of house he has to clear, and the fire support available to him from the covering fire group and fire section. When killing areas have been laid down, the clearing section must try to enter and clear in the way not likely to drive the enemy into them.

d. *Signals*. The platoon commander must give a pre-arranged signal to indicate that the village is clear and that it is safe for his own men to come out into the street.

e. *Platoon Headquarters*. The platoon commander should locate platoon headquarters in the area of the fire section with the task of protecting the rear of the fire section and providing reinforcements whilst he himself should find a good Observation Post from where he can "read the battle" and send reinforcements where they are most needed. He should coordinate the clearing and not remain permanently with one section. Platoon drill clearing a small village is given in figure 24.

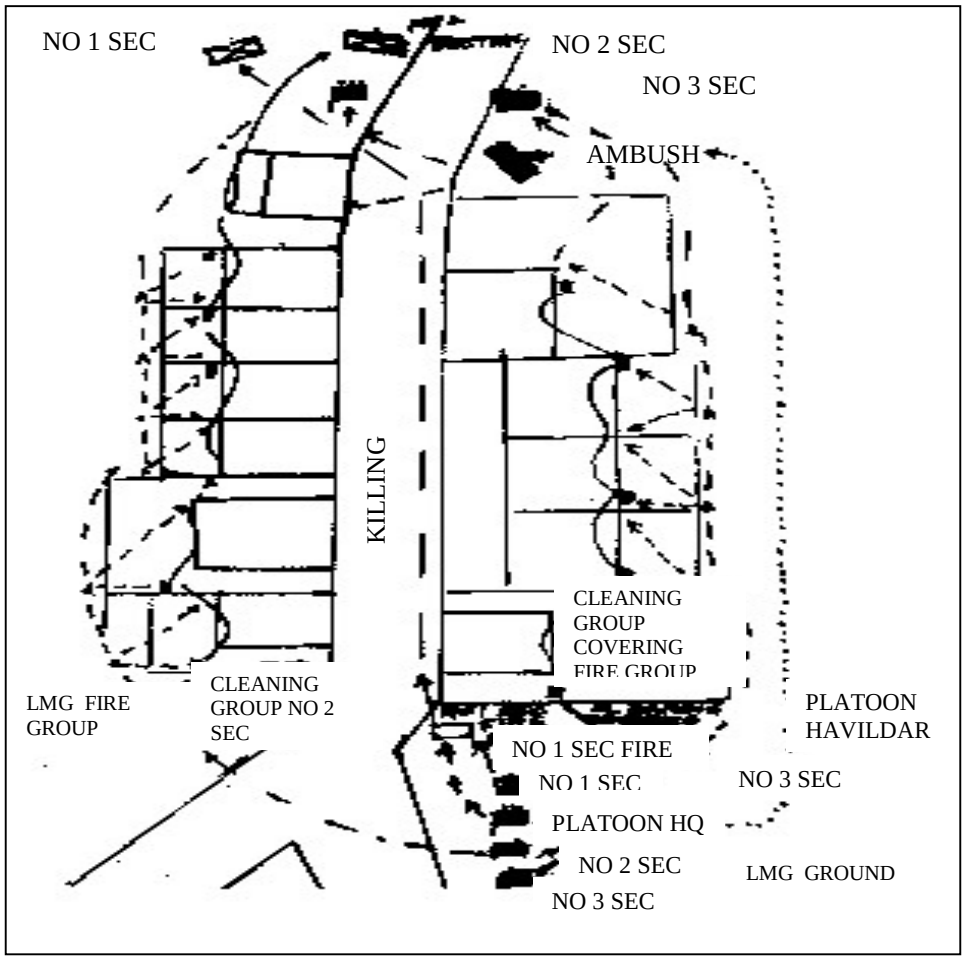


Figure 24 – Platoon Drill for Clearing a Small Village

19. The following general points should be noted:-

a. Civilians left in a village in friendly territory will usually go to ground in cellars and will only prove an embarrassment when the fighting is over. Hostile civilians will be shot if carrying arms and otherwise treated as surrendered enemy.

b. Although all available grenades must be given to the clearing groups, men must not become too "grenade minded". If they throw a grenade into every room they will soon exhaust their supply and will have none when they are most needed. This is a matter of battle discipline.

c. Mosque towers, et cetera may be used as an enemy Observation Post. Mosques, et cetera must be cleared.

d. Smoke may help in approaching some buildings. However, it should be used with care as it may help the enemy to escape.

SECTION 56 – **ATTACK BY SECTION IN BUILT-UP AREA**

General

1. Attack with a section strength on or within a built-up area are normally conducted in the frame-work of a platoon/company. This means that a section can be employed independently or in most of the cases as part of platoon/company operation, such as:-

a. Assault group or clearing group.

b. Covering group/smoke screening team.

c. Ammunition resupply team.

2. Whatever the role a section is required to play, it has to be planned and prepared with special care; section commander must exercise full control over his section.

Stages of Attack

3. The attack will be planed in the following three main stages:-

a. Gaining a foothold or movement to contact a built-up area.

b. Penetration into the built-up area.

c. Reduction of resistance.

Gaining a Foothold

4. Before closing up with the built-up area, the attacker must seize terrain features outside the built-up area which dominate the approaches leading into the town. This may be done by the platoon/company. The section while moving to contact or inside a built-up area must keep following points in mind:-

a. Section should move in single file formation.

b. Avoid detection, but still move swiftly.

c. Exploit direct and indirect friendly fire support, as well as all cover and concealment.

d. Watch out for obstacles and booby traps.

e. Within the built-up area avoid open streets and areas. Use covered and/or concealed approaches. If streets can not be avoided then move close to the buildings in single file.

Penetration into the Built-Up Area

5. In order to clear a built-up area, every house is to be entered. An entry at ground floor level is made difficult by barricades or by booby traps. It is tactically favourable to enter at upper floors because fighting inside the house is easier from there.

Technique of House-to-House Fighting

6. **Basic Considerations**

a. *Need to Search the Enemy*. There is no way of knowing whether a particular building is occupied by enemy or not, so each building must be searched thoroughly.

b. *Need of Intimate Covering Fire*. To cover the movement of small groups or even individuals, maximum use of covering fire must be made.

c. *Capture Objective Bit by Bit*. The objective must be captured bit by bit so that own troops do not involve firing at each other.

d. *The Need to Indicate Progress*. It is most important requirement that all elements of attacking troops must be fully aware of each other's progress. The techniques of indicating progress must be

common and known to all for example coloured smoke or to hang blankets behind windows et cetera.

e. *Requirement of Hand Grenades and Small Arms Ammunition*. These will be required in large quantities and appropriate arrangements must be made for the replenishment.

Section Drill for Clearing a House

7. **Organization**.

a. *Clearing Group*

(1) Section commander.

(2) Bomber.

(3) First entry man.

(4) Second entry man.

(5) Lookout man.

b. *Covering Group*

(1) Section Second-in-Command.

(2) First light machine-gun team.

(3) Second light machine-gun team.

8. **Task of Covering Group**. The following are the tasks of covering group:-

a. Establish fire base and cover the movement of the clearing group by:-

(1) Neutralizing enemy weapons which can fire on the clearing group.

(2) Deception measures.

b. Cover, as far as possible, the flank of the house to cut-off the enemy within.

c. Be prepared to join the clearing group when called up by the section commander.

9. **Action by Clearing Group**. The section commander, having selected the approach/route and the point of entry into the house, moves forward with the bomber to take-up a position from where to direct and cover the movement of the

entry men towards the point of entry. The clearing group either adopts the technique of ground level entry or high level entry. Detailed actions are as under:-

a. *Ground Level Entry*. In this case, the following techniques are adopted:-

(1) Entry men approach the point of entry as fast as they can, using whatever cover is available. The point of entry may be a door, a window or any other opening made into the wall.

(2) Before entering the house, it would be advisable for the entry men to first throw in a grenade and then dash inside as soon as it explodes, before the enemy inside has had time to recover.

(3) Once inside, the entry men get away quickly from the point of entry and stand with their backs to the wall covering the rest of the room and any doors.

(4) On a signal from the entry men (or as a result of observation) section commander and the bomber follow up the entry men. Section commander makes a plan for clearing the remainder of the house and issues order for the next move.

(5) The lookout man takes position at the point of entry into the house, action as a sentry and watching for signal. He also serves as a guide or a runner for liaison with the covering group and the platoon headquarters.

(6) The remainder of the clearing group moves out of the room to clear rest of the house. Section commander leads, followed by the bomber and two entry men.

(7) Before entering the next room, the section commander might fire a burst through the closed doors and then kick it open while the bomber quickly lobs a grenade in the room. As soon as the grenade explodes, the section commander rushes in, gets clear of the door and turns quickly with his back to the door. The bomber enters next, followed by two entry men in succession. If necessary, one of the entry men can be left behind in the first room to watch other doors.

(8) If there is a stairway going up, it must be guarded by one of the entry men against the possibility of being surprised by enemy from above, while the ground floor is being cleared.

(9) When the section commander feels that he has gained a secure base in the building, he will call up the covering group to hold it and if necessary, to assist the clearing group in searching the building further.

(10) If the house has more than one story, the clearing group will go up the stairs (using fire and movement) to clear the floor above. Entry should again be preceded by fire. Great care will have to be taken to ensure that own troops are not hurt by own fire. Clear orders and rigid drill will be necessary. Section commander may lay down that no grenades will be thrown except on his orders.

(11) After the first house has been cleared, it would be advantageous to cross over and secure the roof of the next house, if adjoining to each other. It is much easier to fight one's way from top to the bottom of a house than vice versa.

b. *High Level Entry*. The following techniques are used: -

(1) Where possible this method should be applied even to the first house, because it is easier to go from top to bottom of a house than vice versa. With a little resourcefulness a way can usually be found of climbing up to the roof from outside. Grapple hook with knotted rope is extremely useful for this purpose. See Figure 25



Figure 25 – Knotted Rope and Grapple Hook

(2) By joining several lengths of the knotted ropes and attaching a grapple hook at one end, a man can scale a high

wall, or swing from one building to another, or gain entry in an upper storey window.

(3) If it is not possible to reach the roof from outside, entry should be made at the highest possible point in the building. The floor at which entry is made should be cleared first. After this the house is cleared to the top and from there down to the ground floor.

(4) In Pakistan and neighbouring countries houses in the interior of cities are built close together and their flat roofs offer excellent ground for rapid and deep progress of the attack. Roof top movement, however, is exposed and demands good field craft. Having secured the roof of a house the attackers must clear the house from roof downward before moving to the next house.

(5) Basic drill is the same as for clearing the house upwards. The entry men are the first to climb up and secure foothold in the room. They are followed by the section commander and the bomber. Together they climb the roof. The section commander might move one of the light machine-guns of the covering group on to the room immediately after it is secured.

(6) After securing the head of the stairway, the lookout man (or the light machine-gun group) is posted to guard the rear, while the section commander leads the way down, followed by the bomber and the entry men, using fire and movement technique. All the floors and each room are searched the same way as described before. An important precaution is to keep the staircase covered all the time against the possibility of enemy from downstairs coming up.

SECTION 57 – **INTEGRAL WEAPONS/SUPPORTING ARMS**

Integral Weapons

1. **Rifle**. It is most useful for sniping and also when bayonet is to be used.

2. **MP5/SMG**. Extremely useful for Close Quarter Battle (CQB). It is effective against bolts and locks.

3. **LMG/MG**. It should be sited to cover important avenues of approaches. When fields of fire are restricted, this weapon can be used singly rather than in pairs. In defence these should be placed at ground level to cover the approaches with grazing fires. The position of these weapons should be well protected by barricading and mining.

4. **106 mm RR & RPG-7**

a. The RRs should be jeep mounted.

b. The RR should be sited on the periphery of built-up area to cover all tankable approaches.

c. RR & RL should cover street crossing.

d. The flanks should also be covered.

e. RL should be employed to destroy individuals on tanks.

f. RR and RL can be used in anti-personnel role against small enemy parties.

g. In attack, RR should strengthen the fire base.

h. In attack they destroy enemy automatic weapons at upper storey.

j. Make loophole entries for move.

k. To fire at the corners of houses behind which enemy weapons are suspected.

l. To breakdown heaps of rubble in the road ways, to destroy enemy MG posts and to scatter the mines covering these posts.

5. **Mortars**

a. In attack initially provide pre-arranged fire support.

b. Engage targets on roof tops using super quick fuzes.

c. Heavy mortars can be employed to blast the top floors by using delayed action fuzes.

d. Smoke screens.

e. In defence mortars can engage close DFs.

f. Fire support during counter-attack by isolating enemy positions.

g. Effective against enemy reserves and likely FUPs.

6. **HE Hand Grenades**

a. Hand grenade is one the most essential weapon to clear rooms, cellars et cetera.

b. In attack it is almost a must to throw a grenade before entering into an uncleared house/room.

c. In defence throwing of HE grenade can destroy enemy clearing parties.

d. Smoke grenades are required to cover the move through exposed parts.

e. Smoke grenades can be effective means of signalling.

7. **Energa Grenades**

a. When fired from windows can effectively knock-out enemy tanks.

b. Can be used to knock-out enemy automatic weapons.

c. Effective in making entry holes through doors and windows.

d. Can be an effective weapon of tank hunting party.

8. **Man Pack Flame Throwers**. Flame thrower is a portable weapon comprising three fuel tanks fitted with a harness and a firing gun. On firing, the weapon emits a flame of fire which has a maximum range of 50 metres. The weapon can have a considerable demoralizing effect on the enemy. It can be effectively used through the loopholes of bunkers and doors/windows of the buildings to cause casualties to the enemy troops and set fire to the material inside.

9. **Bakhtar Shikan**

a. Covering gaps between depending localities on broad front against tank threat.

b. The primary targets for TOW/BS are enemy armoured vehicle concentrations. It will be very rarely used against the secondary targets, which may be single tanks, heavily fortified bunkers, pillboxes and gun emplacements. It is used in its secondary role only when it is very essential and when other available weapons fail to accomplish this role.

10. **GLHMG**

a. It can neutralize enemy weapon emplacement upto 800 metres.

b. It can be used against enemy troops concentration upto 1750 metres.

c. It can damage enemy armour upto 600 metres.

d. It can also be used against enemy helicopters in special circumstances.

11. **12.7 mm HMG**

a. *Ground Role*

(1) It can be used on anti-personnel role.

(2) It can be employed for defence against pillboxes.

(3) It can be used against armour vehicles.

b. *Anti-Aircraft Role*. It can effectively be used against enemy slow moving aircraft.

Supporting Arms

12. **Armour**

a. *Defence*

(1) Usually employed in small groups such as troop in support of a company.

(2) Tanks will be deployed to stop enemy tanks forward of the main position and strong anti-tank defence along the tank approaches.

(3) Single tank can engage the attacking enemy tanks through loopholes.

(4) Tanks may be positioned on the edge of the built-up area in mutually supported positions with concealed routes of withdrawal.

(5) Tanks must be employed in over watch of barricades and obstacles along the periphery of the built-up area and in depth.

b. *Attack*

(1) Tanks are employed to isolate a built-up area and cutting the routes of withdrawal.

(2) Provide fire support (including smoke) to assaulting infantry.

(3) Make loopholes with direct fire for infantry to clear the building.

(4) Destruction of buildings with direct fire.

(5) Neutralization of enemy tanks, RR and MGs holding advance of assaulting infantry.

(6) Tank dozers can be employed to clear unmined rubble.

13. **Artillery**

a. *Defence*. Pack artillery may be manhandled to the upper stories or into courtyards and used in strong points.

b. *Attack*

(1) Is of less help to assaulting troops but is used to soften up an enemy held area before attack.

(2) Heavy artillery should be utilized during attack, to destroy strong points and buildings.

(3) Artillery can be employed by sealing off a sector by harassing/engaging the approaches leading to it.

14. **Engineers**

a. *Defence*

(1) Make internal communication.

(2) Work inside the houses.

(3) Strengthening cellars.

(4) Construction of anti-tank barricades.

(5) Laying mines and booby traps.

(6) To build command posts.

(7) Maintenance of water and electric facilities.

b. *Attack*

(1) Clearance of mines/booby traps and anti-tank barricades.

(2) Clearance of rubble to allow the forward moves of vehicles.

(3) Demolition of buildings.

(4) Maintenance of water and electric facilities.

CHAPTER 12

THE ADVANCE

SECTION 58 – INTRODUCTION

Aim of Advance

1. To obtained a position of tactical advantage for development of further operations.

Occasions when a Force Advances

2. Advance may be undertaken at following occasions:-

a. Advance before contact with the enemy.

b. Advance to follow up a deliberate enemy withdrawal.

c. Advance to follow up an enemy withdrawing off balance.

Definitions

3. **Axis**. The line given in the general direction of movement astride which a formation or unit moves.

4. **Bounds**. A tactical feature of importance, present both on the map as well as on the ground, used for coordinating movement during an advance. Troops, except armoured units, will halt at bounds unless otherwise ordered. As the enemy is likely to be encountered on bounds, it is essential to secure them before passing through. Therefore, the forward troops, i.e., point section/platoon, must always approach a bound secured. After securing the bound, the forward troops will halt, inform the next senior commander and await his orders. Bounds are designated by nicknames.

5. **Control Points**. A point designated on the ground either to control movement or as a reference point for reporting location of friendly units. This point should be a conspicuous feature such as a cross road, village, bridge which lies in the zone of operation. It is designated by a number consisting of 3 digits and is shown on maps and traces with a circle round it i.e., 301. Numbers allotted to control points should be in jumbled form and should not follow a consecutive sequence. Control points may be used in any of the following ways:-

a. **Reporting Own Position**.

(1) *Company Commander*. "Hullo one, report location, over".

(2) *Platoon Commander*. "One, my location one thousand metres north east of control point 302".

(3) *Indication of Assembly Area/Forming Up Place.* "FUP, control point 341. Route via control point 306".

6. **Report Line**. A line having no tactical significance (such as a lateral road or railway) on which units or formations report but do not halt. It is given a nickname. When the leading scout of the point section crosses the report line, the point platoon commander reports this to his company commander e.g., "HULLO one, **BADAL CHAMAK OVER"**.

7. **Start Point**. A point on the route of advance which units pass at a specified time. When moving on more than one route it will be necessary to have a start point for each.

8. **Start Time**. The time at which the leading scout of the point section crosses the start point.

9. **Degree of Search**. This is the distance on either side of the centre line, which must be searched by the forward troops during their advance. This will normally be done by observation but, in case the ground does not permit this, then a physical search will become necessary.

Basic Considerations

10. The basic considerations in planning and execution of the advance are:-

a. *Offensive and Bold Actions*. Offensive action consists of rapid exploitation of opportunities, determination in execution, maintenance of momentum, retention of balance and timely employment of reserves. At the battalion level advance should be executed so

boldly and aggressively that the enemy is thrown off balance from the very outset. This will be possible only if: -

(1) The details of the plan and concept of the operation are known to all concerned well before the commencement of the advance.

(2) The commanders move well forward to take appropriate action in time.

(3) The deployment drills and battle procedure of a unit are perfect.

(4) Commanders at all level should visualize correctly the enemy likely opposition enroute and own reactions and plans made well before hand to deal with such contingencies. Time and resources permitting out flanking manoeuvres around the enemy positions may also be considered. Such movements are likely to cause psychological imbalance on the enemy holding up the advance on main axis thereby compelling him to abandon the position, resulting in the seizure and maintenance of initiative.

b. *Maintenance of Momentum.* By continuous and relentless offensive action, the enemy is prevented from carrying out an organized withdrawal, which is turned into a rout and his power to resist breaks. Speed is often the dominant factor. Minor opposition unlikely to threaten the line of communication is generally by-passed.

c. *Simplicity of Plans.* A simple plan, with the battalion commander's intention clearly stated, enables junior leaders to use their initiative.

d. *Balanced Grouping.* The company should be correctly balanced in order to be able to react to unforeseen eventualities. Grouping should be flexible so that the order of march can be changed, without loss of time to meet new situations. Grouping will depend on: -

(1) Aim.

(2) Terrain and visibility conditions.

(3) Likely enemy opposition.

(4) Resources available.

(5) Any special requirement; river crossing or mine breaching et cetera.

e. *Good Battle Procedure*. This will ensure that action is taken without delay. Commander moving well forward with their groups and ready to receive order will ensure that troops are not drifted into battle piecemeal.

f. *Good Communications*. In advance to contact reliable wireless communication is very essential for passing back timely information and for receipt and issue of orders promptly.

g. *Detailed Reconnaissance and Forward Planning*. Detailed reconnaissance and acquisition of maximum information regarding terrain and enemy will enable the company commander to get a clear picture which will help him in forward planning and maintenance of initiative.

h. *Concentration*. By rapid concentration of the main body, artillery fires and air attacks, enemy can be surprised, threatened and out manoeuvred.

j. *Speed*. To obtain speed it is necessary to: -

(1) Advance on a wide front to gain information and exploit quickly the enemy's weak points. In a day light advance, the leading troops in an open terrain should clear a minimum area of 800 metres on either side of the axis.

(2) Have balanced grouping, adequate control and determination to push on.

(3) It is better not to restrict the advance to roads and tracks. If the going is good tanks and light anti-tank elements move cross country to reach certain objectives ahead of the main guard.

k. *Quick Reaction*. The overall rate of advance is governed more by quick reaction in dealing with the enemy when he is encountered than on under haste between contacts. As a yard stick in encounter battle a company may take from 30-45 minutes and a battalion from 1-2 hours under normal conditions.

l. *Use of Air*. Day light advance in our conditions presupposes that we have achieved local air superiority over the area of operation.

Air, if allotted, must be used for interdiction to cut-off and disrupt the retreating enemy which will assist in maintaining the momentum. Close support and air reconnaissance will be of great assistance. Air control team must, therefore, be kept well forward to ensure its expedient use.

Order of March

11. The order of march depends on terrain, enemy situation and mission. In open country, tanks may lead the advance. In close country, or through defiles, infantry will lead the advance. In former case tanks move by bounds, tactically deployed; in latter case tanks may move in file on the road; in both cases at their own speed. The FOO will move with the Company R group, while the MFC may move with company or platoon R group.

Battle Procedure and Drills

12. The battle procedure and battle drills will be:-

a. *Receipt of Battalion Commander's Orders*. Actions by the company commander will be: -

(1) Studies map and carry out quick appreciation.

(2) Issue warning order, verbally.

(3) Prepares to leave for battalion O group's RV.

b. *Receipt of Company Commander's Orders*

(1) *Platoon Commander*

(a) Receives warning order from company commander verbally.

(b) Issue own warning order to section commanders verbally.

(c) Briefs platoon 2IC.

(d) Ensures his wireless set is netted.

(e) Prepares to leave for company 'O' group RV.

(2) *Platoon 2IC*

(a) Is briefed by the Platoon Commander.

(b) Collects unnecessary stores and hands them over to CQMH.

(c) Ensures issue of meals, tea, water and ammunition.

(d) Inspects weapons, ammunition and equipment of platoon.

(3) *Section Commanders*

(a) Receive Platoon commander's warning order.

(b) Brief their sections.

(c) Collect unnecessary stores and hand them over to Platoon 2IC.

(d) Ensure collection and issue of meals, tea, water and ammunition to their sections.

(e) Inspect weapons, ammunition and equipment of their sections.

(f) Prepare to go for Platoon Commander's 'O' group RV.

c. *Reconnaissance and Orders*. Action by:-

(1) *Company Commander*

(a) Receives Battalion Commander's orders.

(b) Carries out his reconnaissance mostly off the map.

(c) Prepares his plans.

(d) Proceeds to RV and issues orders. If possible will shown whatever ground is visible to his 'O' group and then issue orders on a model/map or sketch.

(2) *Platoon Commander*

(a) Receives company commander's orders.

(b) Carries out his reconnaissance and issues orders to section commanders.

(c) Briefs platoon 2IC if he has not attended the platoon 'O' group.

(3) *Platoon 2IC*

(a) Is briefed by the platoon commander and ensures that all information is passed on the men.

(b) In case the platoon is to act as point platoon, he collects additional PRC-77 sets at the scale of one each for the section and one for the platoon commander. He ensures that the sets are properly netted.

(c) Coordinates with the CQMH regarding arrangements for meals and water.

(4) *Section Commanders*

(a) Receive platoon commander's orders and give orders to their own sections.

(b) In case additional PRC-77 sets are being issued, then they collect one each from the platoon 2IC and ensure their netting.

(c) Ensure their sections are properly fed and given adequate water.

d. *Advance*. Action by:-

(1) *Company Commander*:-

(a) Ensures company crosses start point at given time.

(b) Controls the advance of his company.

(c) Informs battalion headquarters about progress of advance.

(d) Keeps his sub-ordinate commanders informed about the general progress of advance.

(2) *Point Platoon Commander*

(a) Ensures his platoon crosses the start point at the given time and reports this by nickname to his company commander.

(b) Controls the advance of his platoon.

(c) Keeps the company commander informed about progress of the advance by reporting on crossing each report line.

(d) On meeting enemy opposition he either:-

i. Clears it with a platoon attack, or

ii. Makes a fire base for the company's attack.

(3) *Follow up Platoon Commander*

(a) Hands over his platoon to the platoon 2IC.

(b) Moves with the company commander's 'O' group.

(c) Joins his platoon for the encounter battle.

(4) *Platoon 2IC*

(a) Keeps himself in picture and takes over if the platoon commander becomes a casualty.

(b) Controls the movement of the platoon headquarters and the follow up sections.

(c) Ensures distribution of meal, tea and water.

(d) Ensures replenishment of ammunition and evacuation of casualties after each encounter battle.

(5) *Point Section Commander*

(a) Ensures his section crosses the start point at the given time and reports this by nickname to his platoon commander.

(b) Controls the advance of his section.

(c) Keeps the platoon commander informed about progress of the advance by reporting on crossing each report line.

(d) On meeting enemy position he either:-

i. Clears it with a section attack, or

ii. Makes a fire base for the platoon's attack.

(6) *Follow Up Section Commanders*

(a) Hand over sections to their 2IC.

(b) Move with the platoon 'O' group.

(c) Join their sections for the encounter battle.

SECTION 59 – **GROUPING AND ORGANIZATION**

Organization

1. The advance is organized as under:-

   a. Covering troops.

   b. Advance guard.

   c. Main body.

Covering Troops

2. Functions. It is a tactically self-contained force operating at a considerable distance from the main force, stationary or moving. It may be in front, on flanks or towards the rear of the main force to develop the situation early and to defeat, deceive or delay the enemy forces till the main force is in a position to deal with it. It should be prepared to fight any type of action necessary for the success of its mission.

   a. *Composition*

      (1) *Mobile Troops*. Operate ahead of advance guards, are organized around mobile elements most suitable to area of operations and perform the following tasks:-

         (a) Reconnaissance on a wide front and provide early information of enemy movement, strength et cetera.

         (b) Clear minor opposition, or may by pass it (only with the permission of higher commander).

         (c) Secure features of tactical importance.

         (d) Prevent the advancing force being surprised or committed unnecessarily.

      (2) *Flank Guards*. Watch exposed flanks and give early warning of enemy approach.

Advance Guard

3. This will normally be a balanced force of all arms including armour, if available artillery to give fire support and detachment of engineers to deal with obstacles. Advance guard moves in front of the main body and provides security from the front. The advance guard provides a force strong enough to clear minor

opposition and to cover deployment of the main body. The advance guard for its own security sends forward a vanguard. The remaining portion of the advance guard is called main guard.

4. **Vanguard**

a. *Composition*. It will depend on the task, nature of terrain, availability of troops and the likely enemy actions. A suggested composition of the vanguard company is as under: -

(1) One rifle company.

(2) One section battalion mortars.

(3) One section from assault pioneer platoon.

(4) Detachment from regimental police and intelligence section.

(5) Air Control Team if allotted.

(6) Anti-Tank Guided Missiles if allotted.

(7) Tanks, if allotted to advance guard.

(8) Elements of LAT, if allotted.

(9) One forward observation officer from direct support battery.

(10) Engineer's reconnaissance party for move only.

b. Likewise a complete company group termed as "follow up company" is also ready to push through and take over as vanguard, if necessary.

c. *Tasks.*

(1) To drive in minor opposition and thus prevent unnecessary delay of the battalion.

(2) To cover the advance of the main guard and protect it against surprise and enemy ground action from the front.

(3) If the opposition is strong to be driven in then to secure ground of tactical importance and to cover the deployment of the main guard.

5. **Points to be Decided by Vanguard Commander**. Whatever may be the type of advance, a vanguard commander's appreciation must include following points:-

a. Likely enemy resistance and its effect on grouping.

b. Detailed tasks for each platoon/weapon section et cetera.

c. Grouping for point platoon and the remaining vanguard.

d. Detailed tasks for supporting arms.

e. Fire plan, including pre-selected targets.

f. Bounds, report lines and control points (give by battalion commander) to assist in maintaining control.

Main Body

6. The main body comprises the bulk of the force fighting power. Units of the main body are so positioned as to permit flexibility for employment after contact with the main enemy force.

7. The advance guard moves steadily along the axis, whereas the move of the rest of the units is controlled by leap frogging. Each unit is moved from one harbour to another harbour and given a clear run. The advantage of this system is:-

a. Order of march can be varied.

b. Presents less attractive target to enemy air.

c. Is less tiring.

8. **Company acting as Flank Guard**

a. *Task.* The mission of a flank guard company is to protect the main guard from hostile ground observation and from surprise ground attack from the flank. In the event of an attack from the flank his task requires action which will permit the uninterrupted passage of the area by the main guard of main body which will give the main guard time to deploy.

b. *Composition*. Usually flank guard company is given additional supporting arms such as tanks, FOO, engineers and elements of LAT. It may also take material such as anti-tank mines and demolitions for constructing obstacles.

c. *Formation*. Adopted by flank guard company depends on terrain, availability of parallel routes, expected enemy action and the method of move by mechanical transport/foot.

d. *Control*. Close liaison is maintained with the main guard by patrols and radio.

e. *Action of Flank Guard*. Depending on terrain and enemy situation the flank guard company may march parallel to the main column or it may occupy a series of defensive positions on the flank. When marching it operates in a manner similar to that of vanguard. When occupying a series of the defensive positions it covers routes of expected enemy approach until the main guard/main body has passed. It constructs road blocks and obstacles along ridges, rivers and defiles et cetera. In this type of the operation, it occupies its positions by a series of leap frog movements.

Company as Part of Main Guard

9. When part of the main guard, the company conducts its march in accordance with the battalion commander's orders. The company marches fairly compact on one side of the road. The distance between individuals, groups and sections/platoons depends on the nature of enemy air threat. F echelon would normally be grouped and move under battalion arrangements.

Reorganization

10. Immediately upon taking an objective, the vanguard company reorganizes itself to repel a counter-attack. Following actions are taken:-

a. Platoons and weapons are placed to cover enemy avenues of approach to the front and flanks of captured objective and digging starts.

b. Patrols are sent and Observation Posts (OPs) are established.

c. Fields of fire are adjusted.

d. F echelon and supporting weapons are brought forward.

e. Key personnel, if casualties, are replaced.

f. Ammunition is redistributed.

g. Casualties and ammunition reports are sent back.

h. Ammunition is brought forward.

j. Casualties are taken near axis for evacuation.

k. Identification of enemy units is reported and prisoners are brought near the road for collection by battalion intelligence personnel.

11. After reorganization the company is regrouped into an effective team with control re-established and plans completed to continue the advance or support further operation of the battalion.

Passing through – Follow Up Company

12. This may take place under the following circumstances:-

a. When the vanguard company has fought an action and has suffered casualties.

b. The vanguard company has covered a reasonable distance and changeover by fresh troops is desirable (usually done after securing a bound).

13. In the former case the battalion commander who is watching the battle would decide whether the same company should continue or the follow up company should take over. He will accordingly warn the follow up company. In the latter case, such warning would have been given in the initial orders for the advance and the follow up company would be prepared for it. In both cases the follow up company should have issued anticipatory orders regarding grouping and orders of march in the event of taking over as vanguard. Same procedure is followed while passing through of follow up platoon.

14. Following procedure will assist in an orderly take over with speed:-

a. On receipt of warning the follow up company moves up and lessens the gaps with the vanguard (yet maintaining a reasonable distance).

b. The follow up company commander with his R and O groups moves well ahead of his company to meet the previous vanguard commander.

c. The old vanguard commander, along with any elements, which might be required to be grouped with the follow up company, should come near the axis. Necessary guiding arrangements would also be made by the old vanguard company.

d. The previous company commander briefs the new vanguard commander about the tactical situation. The follow up company commander then quickly briefs his platoon commanders who brief their section commanders about latest tactical situation. The company then resumes advance without wasting time.

SECTION 60 – **INTEGRAL WEAPONS/SUPPORTING ARMS IN ADVANCE**

Battalion Mortars

1. Mortars are placed well up in the advance to enable the commander to overcome light resistance quickly and effectively. It is normal to allot mortar platoon to the vanguard company. An MFC will accompany the vanguard commander or the point platoon commander. He should be in communication with mortar positions by wireless. When adequate fire support is available mortars may not be used to conserve ammunition.

60 mm Mortar Section

2. The 60 mm mortars will usually be placed in support of point platoon. The section normally does not take part in preparatory firing, when adequate fire support is available. During the attack the mortars fire at targets holding up the advance. After the objective is captured the section protects the reorganization of the company and assist in breaking up enemy counter-attacks.

3. As they have a limited range they should be placed well forward in the order of march so as to engage targets without loss of time. Since they can also fire indirect the possibility of their integration with battalion mortars and co-ordination of their fire by MFC should also be considered.

4. **Anti-Tank Recoilless Rifles**

   a. Protect column against enemy armour.

   b. On reaching a bound or temporary halts these should be deployed ready to engage enemy armour.

   c. Aid in clearing minor opposition.

Bakhtar Shikan

5. TOW/BS is ideally suited for open country where it can exploit its maximum range. TOW/BS cannot be fired while moving. The TOW launcher has to be stationary to be able to acquire, track and destroy a target.

12.7mm MG

6. Due to its long-range and more penetration power, it can be effectively utilized against enemy bunkers/stiff resistance.

7. The weapon can also be used in anti-aircraft role, but will be picketed along the axis of advance.

GLHMG

8. It can be effectively employed as following:-

a. It can neutralize enemy weapon emplacement up to 800 metres.

b. It can be used against enemy troops concentration up to 1750 metres.

c. It can damage enemy armour up to 600 metres.

d. It can also be used against enemy helicopters in special circumstances.

Assault Pioneer Platoon

9. Normally a section will be allotted to vanguard. Since the strength of this section is only one NCO and five OR its role and employment would be very limited.

10. **Communication**. For quick response and employment in indirect role the section may be given PRC-77 sets in addition to the integral walkie-talkie sets. Additional sets for this purpose may be given by the battalion to the vanguard.

Armour

11. Tasks of armour in advance are:-

a. To lead the advance.

b. Provide flank protection.

c. Form part of mobile troops and carry-out reconnaissance in front.

d. In close support to the advancing force.

12. The allotment of armour to vanguard will depend on:-

a. Suitability of ground for armour.

b. Nature of enemy position.

13. In open country tanks may lead the advance. In close country or through defiles, infantry will lead the advance. In former case tanks move by bounds, tactically deployed. In latter case tanks may move in a file along the road, in both

cases at their own speed. When Tanks lead the advance the distance between the tanks and the leading platoon is based on the following guide lines:-

a. Minimum distance as such that the leading infantry is outside the effective small arms fire of enemy.

b. Maximum distance is up to the wireless set range.

Artillery

14. Task of artillery is to neutralise enemy action so that infantry/armour can assault with minimum casualties and to protect assaulting troops on the capture of objective. FOO will be moving with vanguard to control and direct fire.

15. Main requirements to accomplish the above task are:-

a. Major portion of artillery, including medium, should be moved well forward.

b. Once contact is imminent, leading troops must always be covered by some guns deployed on the ground. This is done by leap forging of forward batteries.

Engineers

16. Engineers should move well forward with engineer reconnaissance parties moving with the vanguard. The main tasks for engineers in the advance are:-

a. Overcoming natural and artificial obstacles.

b. Improving communications.

c. Mine clearing.

d. Demolitions to protect a flank (s).

e. Construction of forward airstrips.

f. Water supply.

g. To clear scatterable mines laid by enemy.

LAT Battalion

17. Elements of LAT battalion may be operating in front of point platoon (s). The platoon commander may contact these element, if nearby, to obtain information and such support as they can render without jeopardizing their own

mission. May lead the advance ahead of tanks or be used for flank protection. Some other roles are:-

a. To from part of covering troops thereby leading advance, it may act as mobile troops or flanking guard.

b. Provide fire base for attacking troops.

c. In mobile operations seizing opportunity against minor opposition.

d. Establish liaison/contact between two elements/forces.

Air

18. When available, can assist the advance in the following ways:-

a. Counter air operations.

b. Interdiction to disrupt enemy movements.

c. Reconnaissance-both photo and visual

d. Close air support.

SECTION 61 – **PLANNING AND CONDUCT OF ADVANCE**

Planning

1. The commander should appreciate from the study of his map and information available, where he is likely to meet the enemy and what tactical ground he must secure in that event. Having been informed of this, his subordinate commanders would be mentally prepared to meet the enemy and deploy early so that the enemy is not able to delay the advance. Based on his appreciation a commander would decide upon his objectives, missions for subordinate units, grouping and order of march, routes/axes of advance, protection/security plan, time limits for various actions and how to control the battle.

2. **Command and Control**. To avoid delay the commanders with their R groups move well forward; O groups following close behind. Good communications both forward and rear are essential. Wireless is the primary means, duplicated by runners. Control is also affected by laying down objectives, bounds, report lines, control points, axis and centre line.

3. **Speed**. This means speed both in planning and execution. This is not achieved by hurrying or going in the battle at the double, but by:-

a. Quick decisions.

b. Maximum determination leaving initiative with junior leaders.

c. Balanced grouping.

d. Locating and exploiting enemy weak points.

e. Good control.

f. Determination at all levels to press on. An odd fire by a sniper or small delaying party should not slow down the advance. Such elements should be dealt with vigorously and driven back or by passed if permitted.

4. **Grouping**. This would depend on the type of country, enemy and the possible delaying positions. At all times the force must be sufficiently well balanced to ensure speed in forward movement, reduce vulnerability to an ambush and increase speed of reaction to an opposition.

5. **Order of March**. Correct order of march ensures that various groups are moving at places from where they can be immediately committed, if the need arises. A suggested order of march of company group in open area is as shown:-

One Platoon Up With One Section Up In Open Country

	Remarks
Scouts	75 to 150 metres ahead of point section.
Point Section	Approximate gap of about 150/200 metres between point section and platoon R group. Platoon R group should be outside enemy effective small arms fire range.
Platoon R Group	Platoon commander.
	Runner, Signaller.
	Commanders of Special Weapons (RR, BS, 60 mm

	mortars, GLHMGs, 12.7 mm MG if allotted).
Platoon O Group	Section commanders, RL Commander.
Number 2 Section	-
Platoon Headquarters	Platoon headquarters less R group. Platoon 2IC, RL Numbers.
Number 3 Section	-
Section GLHMG/ 12.7 mm MG	-
Company R Group	Company commander, runners, signallers, FOO, MFC
@Intelligence Detachment	@Do not form part of R and O groups.
Company O Group	Platoon Commanders, RR commanders.
@Detachment Regimental Police	-
Number 2 Platoon	-
Section RR	-
Company Headquarters	Company Senior JCO, Company Havildar Major, Stretcher Bearers
Section Assault Pioneer Platoon	May be behind number 3 platoon
Engineers Reconnaissance party	May be moved up and down according to requirement. May be behind.

Number 3 Platoon	3 Platoon.
Company F Echelon Transport	Moves in bounds from available cover to cover.

Notes: a. For reasons of command and control the section moves in groups under their group commander.

b. Distances between individuals within a group of the point section are 6 to 10 metres keeping in view the enemy air threat.

c. Distances between sections/platoons et cetera depend on terrain vis-à-vis need for control.

d. Formations vary according to the ground/tactical situation.

Gaps

6. Size of the gaps between groups, in the order of march, is calculated in terms of time and NOT of space. Gaps will vary with the speed of units, the type of country, the visibility and the likely enemy opposition. The gaps are such that when the point section gains contact with the enemy, each succeeding element can deploy without serious interference from the enemy. However, the distances are not so great as to prevent each element from supporting the element in front of it.

7. When the forward elements contact the enemy and are preparing for their attack, the rear elements close in slightly, taking advantage of these gaps, so as to be readily available to the next higher commander in case the actions of the forward elements fail. The rear elements, therefore must not stop as soon as the forward elements make contact, but should close in, still maintaining a reasonable distance from the enemy, so that their manoeuvre is not restricted by the direct fire of the enemy. In an open country with good visibility the leading company may be strung on the road for as long a distance as 1100 metres to 1400 metres and battalion may be spread over 3 – 5 kilometres.

8. **Advance in Mechanical Transport**. When the advance is by mechanical transport and since the gaps are to be in terms of time, various components will be moving with bigger gaps in between. The advance is led by the leading platoon of vanguard and the rear echelons move from harbour to harbour. It may however be noted that the point platoon within itself will be fairly close, as it must be able to

embus and get into action on foot; here it is marching time and not vehicle speed that determines the gap.

Reconnaissance

9. The leading elements must report the following information, which will be of value to higher commander:-

a. *Terrain Conditions*. To include roads, other means of communication, going and obstacles, natural and artificial.

b. *Enemy Positions.* Strength and weakness of enemy positions.

c. *Enemy Movement.* Particularly enemy moves which are likely to threaten the flanks of advance.

d. *Engineer Information.* Such as, water obstacle, bridges, their conditions, mines et cetera.

Action on Meeting Opposition

10. During the advance initially the opposition will be light and then gradually it will stiffen and minor engagements will be fought to drive in enemy delaying detachments. In the first case the speed is essential. In the second case all available effort is necessary to maintain the momentum of advance.

11. As soon as the forward elements of a vanguard gain contact with the enemy, the commander must move forward, contact the point platoon commander, get information, (the point platoon commander, who if heavily committed is not called to the rear to report situation) make a quick appreciation to decide whether the opposition can be dealt with by the leading troops or it will be necessary to commit the whole company group. He should watch the proceedings but should not interfere with the junior leaders when the situation is well within their capabilities.

12. The vanguard commander would be kept informed at all times. He will be so placed in placed in the line of march that he can carry-out reconnaissance immediately. Meanwhile no time is wasted if correct battle procedure is followed for deployment of troops.

13. When effective small arms fire by enemy is received, the leading elements return fire and manoeuvre to neutralise the resistance. It is extremely important that the leading elements react rapidly to this situation.

14. **When Resistance is Strong.** When the leading elements can not overcome the opposition their actions will be to:-

a. Pass back information.

b. Deploy on a wide front and locate enemy flanks.

c. Occupy any tactical features on the flanks, which are not occupied by the enemy.

d. Close towards the enemy and engage him with fire.

e. Keep a watch on the enemy, locate his weapons and weak points.

f. Assist the deployment of the vanguard/main guard and be prepared to support its attack by forming a fire base.

15. Once the point platoon has fought an action, the vanguard commander must immediately decide whether the same platoon should continue as point or another platoon should take over.

Bypassing Opposition

16. **Pursuit**. Commanders may be required to outflank or bypass an opposition. This calls for initiative on the part of advance guard commander who must be given information about the opposition he is to bypass. He must pass this on to the vanguard. Information must be sent back about the enemy by-passed and arrangements made subsequently by the advance guard to deal with him. Vanguard cannot bypass a position without permission.

17. **Deliberate Advance**. In a deliberate advance, it is profitable to attack any enemy strong point at once to disrupt the enemy defence and accelerate our own advance. Any one of the companies may be ordered by the battalion commanders to by-pass opposition, provided, by doing so he is gong to gain speed. The following points should be kept in mind while deciding to by-pass:-

a. Terrain.

b. Enemy opposition.

c. The company which bypasses does not outrun the fire support of the battalion.

d. The move forward of the main guard is covered and there are some elements on the main axis i.e., enemy is contained frontally.

e. The main guard retains the capability of clearing the opposition and linking up with the bypassing force, at the earliest.

Carriage of Infantry

18. In advance, infantry may move:-

a. On tanks, if allotted

b. In armoured personnel carriers (APCs) if available.

c. On foot.

19. Infantry carried on tanks is very vulnerable and impedes tank action. Infantry should not be carried on leading tanks, which must remain ready to engage the enemy. Armoured personnel carriers are the most suitable form of carriage. They give some protection and have good communications. They are also less tiring and have a good cross country performance.

Degree of Search

20. This is the frontage to be covered by mobile troops when being employed in a protective role. Unless otherwise ordered the vanguard is responsible to clear the axis of advance of all enemy effective small arms fire. The advance guard on the other hand, must ensure that the move of the main body along the axis is not interfered by enemy long-range automatic/mortar fire.

Formation

21. Formation adopted by vanguard/point platoon/section will depend on terrain, visibility and nature of expected enemy opposition. In case, the axis of advance is dominated by features (adjacent to the axis) these must be searched. Certain elements of vanguard should move along them.

a. *Scouts*. Move astride axis with a distance approximately 50 to 75 metres between them. The second scout is approximately 75 to 150 metres ahead of the rest of the point section. These distances depend upon ground and visibility.

b. *Point Section*. Moves by groups and not as individual. Distance between individuals/within a group and between the groups would depend on terrain, visibility and enemy threat.

c. *Point Platoon*. Moves fairly dispersed yet maintaining tactical unity.

d. *Rest of the Vanguard*. Moves astride the axis but not deployed. Distance between individual/groups depends on degree of enemy air, artillery threat.

e. *Precautions*

(1) No move on the road but away from it.

(2) Take positions tactically while halting.

(3) All personnel and vehicles arrange for protection against enemy ground, air and artillery fire.

SECTION 62 – POINT **SECTION IN ADVANCE**

Preparation for Battle

1. Personal camouflage; break up the outline of the steel helmet, haversack and shoulders using scrim, net and local foliage typical of the country over which the section is to operate.

a. Check that weapons are clean and serviceable and sufficiently oiled for the type of country and climate. Set sights at 200 metres.

b. Check that ammunition is clean and that magazines and grenades are properly distributed. Flank men in open country may need smoke grenades.

2. Preparation for Battle

a. Personal camouflage; break up the outline of the steel helmet, haversack and shoulders using scrim, net and local foliage typical of the country over which the section is to operate.

b. Check that weapons are clean and serviceable and sufficiently oiled for the type of country and climate. Set sights at 200 metres.

c. Check that ammunition is clean and that magazines and grenades are properly distributed. Flank men in open country may need smoke grenades.

Section Commander's Orders

3. The section commander will number-off his men before giving them his orders under the following headings:-

a. *Ground:* To include reference points if possible**.**

b. *Situation*

(1) Enemy forces.

(2) Friendly forces; details of platoon formation and task.

(3) ***Mission***. The section mission.

c. *Execution.*

(1) Route, if applicable.

(2) Section formations.

(3) Flanks for the LMGs.

d. *Administration and Logistics*. Any necessary information from the platoon commander's orders.

e. *Command and Signal*. any necessary information from the platoon commander's orders.

Reference Points and Anticipatory Orders

4. In advance to contact, the section commander will be on the lookout as he moves for:-

a. New reference points for fire orders. He will usually pick two or three within about 400 metres range in a wide arc on his front. He will describe these to the section as they advance; each man will either acknowledge with a hand signal or shout "not seen".

b. Position, which the section can take cover, if it comes under effective enemy fire. Whenever, possible, the section commander will point out such positions in anticipatory orders such as "If we come under effective fire. LMG group take cover in scrub, rifle group in line of that bank". Some such indications will certainly be necessary in country where there is little cover, or on a steep forward slope.

5. Unnecessary shouting should be avoided at this stage. It only gives away own locations to the enemy.

Action on Contact

6. Sections must be trained to continue the advance in spite of the noise of fire directed at someone else and regardless of stray rounds amongst them.

7. Effective enemy fire in this situation is of enemy small arms fire which would cause heavy casualties if the section continues on its course. Most men instinctively drop to the ground when under fire. This actin is generally wrong

because the enemy usually opens fire when his target is in a place offering little or no cover. The best course is to run, thereby making a difficult target. This action would lead to loss of control by the section commander unless properly controlled. The following drill will ensure that control is not lost by the section commander, but if lost, is regained quickly:-

a. As soon as the enemy opens fire, and if the fire is effective every man will run to the nearest cover or that pointed out by the section commander in his anticipatory order. No one will run more than 20 metres or so unless running to the cover pointed out by the section commander.

b. Every man will dive or drop into his cover and at once crawl some way so that the enemy cannot have his sights on him when he reappears.

c. Every man will move up to a position of observation; any individual out of earshot of the section commander must crawl close enough to be able to hear his voice.

d. Any one who actually spots the enemy will return the fire, with tracer if available, without waiting for an order from the section commander.

e. In principle, when contact has been made, no man in the section should be idle. Everyone should be either:-

(1) Observing; or

(2) Firing if the enemy is seen; or

(3) Moving to a new position of observation; or

(4) Moving to a new fire position.

f. Bunching must not be allowed. Apart from the LMG numbers when necessary, no one in the open by day should ever be within five metres of is nearest neighbour.

g. This drill may be summarised as follows: -

(1) Take cover as soon as the enemy fire is found to be effective.

(2) Dash-Down-Crawl-Observe-Sight-Fire.

Location of the Enemy

8. Location of enemy fire is usually not easy. Failure to locate the enemy means that the section will be unable to move without suffering heavy casualties except under cover of smoke. It may lead to loss of initiative by the section and halting of the platoon advance.

9. There are three stages in locating the enemy:-

a. *Observation*. Look in the area from which the thump comes; the time between the crack and the thump gives a clue to the range. Look for movement, smoke or any thing unusual. If nothing is seen after about thirty seconds or so, it is unlikely that the enemy will be located by looking.

b. *Fire*. The section commander will give a fire order to a couple of riflemen to fire two shots into likely cover. The rest of the section will keep a careful watch on their arcs of observation. If there is no answering fire, the section commander should try some other likely target.

c. *Movement*. The section commander will order two men to get up and double forward about ten metres to different covers. He might do this again if no fire is drawn the first time. If the enemy is there, he must be well trained not to fall for these old tricks and begin firing at such poor targets-a man getting up and dashing ten metres is a very hard target to hit. If there is still no enemy reaction; the section commander must carry on with the advance.

Target Indication

10. If any soldier locates the enemy before his section commander, he will load tracer in his rifle, shouts out loudly “watch my tracer” and fire, until the section commander gives a fire order on the strike of the tracer. This method of target indication will only be used if tracer ammunition is available.

Winning the Fire Fight

11. As soon as the section commander knows the enemy’s disposition, he must give a fire order to bring sufficient weight of fire on the enemy to neutralize them. If one or more men who have spotted the enemy have begun firing, the section commander regains control by shouting “stop” before his fire order.

12. While winning the fire fight, the section commander must make his battle appreciation. Having won the fire fight, the section commander must keep the initiative by continuing to bring fire down on the enemy while his section closes for the assault.

Battle Orders

13. These will be as brief as possible. There is no need to mention the enemy position if this is known to all concerned. The mission is to capture a particular feature or clear the route of advance. Battle orders will specify the direction from which fire and movement will take place.

Execution of the Assault

14. When attacking by day the section will move forward to and through the objective in groups by fire and movement, each group will alternatively support to move forward of the other group by covering fire. The angle between the two, i.e., the LMG providing covering fire and the assaulting elements would be as wide apart as ground permits. The assaulting group can widen the angle by swinging outwards as they assault.

15. Number 2 of the LMG is responsible for watching the group commander and listening to his orders and also for watching the movement of the advancing element so that covering fire can be given at critical moments and can be varied from slow to rapid as necessary. This helps to conserve ammunition and ensures that covering fire is given when it is most needed. Number 2 will fire his rifle when LMG has a stoppage or magazines are being changed as fire must be maintained while the advancing elements are moving in the open. These are the only occasions when Number 2 fires his rifle.

Reorganization

16. Once the assault has been made, the following reorganization will take place:-

a. LMG(s) will rejoin the assaulting elements immediately at the double, if not accompanying the assault.

b. Section commander will:-

(1) Order a search of the area of the objective for any enemy wounded or hiding. Riflemen detailed to search will be covered by other riflemen.

(2) Check the positions of the two groups, allot arcs of observation and detail reference points.

Making a Fire Base

17. In case the point section cannot overcome the enemy position by its own fire and movement, the point section commander will make a fire base and try to find out the following:-

a. Strength and extent of the enemy position.

b. Exact location of enemy positions, especially of his automatic weapons.

18. To achieve this, the section commander will do the following:-

a. Spread his section over 100 to 150 metres, using fire and movement.

b. Draw enemy fire by firing at suspected enemy positions (on the flanks of those positions which have already opened fire).

c. Draw enemy fire by moving his riflemen in pairs, over short bounds towards the flanks of those enemy positions which have already been located. This movement must be covered by the fire of our own LMGs.

19. He then passes this information to the platoon commander by personal contact (the platoon commander comes forward in this case), by field signals or by PRC- 77.

SECTION 63 – **POINT PLATOON IN ADVANCE**

Signals

1. The platoon commander will arrange a whistle signals code for use when he cannot make his intentions clear by hand signal or voice. The following code is suggested:-

a. *Section Identification*. One short blast for Number 1 section two for Number 2 and three for Number 3. On hearing his section

identification signal, section commander concerned will look in towards the platoon commander and acknowledge.

b. *Order for Detailing the Fire Section*. This signal would follow the section identification signal. The following signals may be used: -

(1) A series of short blasts (the alarm) –"You will be fire section".

(2) A series of long blasts- "I am going left flanking".

(3) A series of short/long blasts-"I am going right flanking".

Control

2. Platoon headquarters will usually split as follows:-

a. Platoon R group.

b. Platoon headquarters less R group under platoon 2IC.

c. In certain circumstances, such as an imminent tank threat, the platoon commander may keep the rocket launcher team forward with him.

3. The platoon commander must always be able to exercise the control by whistle, hand signals or wireless.

Formation

4. The more frequently used formation by a platoon in the advance is "On Up". However, the formation will depend on such factors as terrain, enemy situation, composition, degree of search et cetera.

5. Point platoon commander must keep control of his section. When advancing along the axis of a road in country where there are thick hedgerows, he must therefore have his complete section on one side of the road.

6. The platoon commander will ensure that no part of his platoon is ever beyond the potential support of some other part of it.

Platoon Commander's Orders

7. Before starting the advance, the platoon commander must give detailed orders to his platoon, covering the following:-

a. *Landmarks*. Given from rear to front.

b. *Situation.*

(1) *Enemy Forces*. To include the enemy's strength, location and likely delaying position.

(2) *Friendly Forces*. To include the company mission, flanking troops and their tasks and covering troops (if any).

(3) *Attachments and Detachments*. Only those concerning your platoon.

(a) *Under Command*. "Section 60 mm Mortars."

(b) *In Direct Support*.

(c) *In Support*. "P Battery 201 Field Regiment and Battalion Mortar Platoon".

c. *Mission*. "Advance along Road Lahore-Barki with a view to securing area upto road nullah junction as soon as possible".

d. *Execution*

(1) *General Outline*. The advance will be in . . . . . phase as follows:-

(a) *Phase 1*. Secure upto area……..

(b) *Phase 2*. Secure upto area……..

(2) *Number 1 Section*. Havildar Rashid.

(a) *Task*

i. Act as point section in phase 1.

ii. Advance behind platoon headquarters in phase 2.

(3) *Number 2 Section*. Lance Havildar Habib.

(a) *Task*

i. Act as follow up section in phase I and be prepared to take over as point section on orders.

ii. Act as point section in phase 2.

(4) *Number 3 Section*. Naik Mulazim.

(5) *Rocket Launcher*.

(6) *Weapon Section Detachment*. (If any).

(7) *MFC*. (If given).

(8) *Coordinating Instructions*.

(a) Time to leave Assembly/Concentration Area.

(b) Start Time.

(c) Start Point.

(d) Axis of Advance.

(e) Bounds.

(f) Report Lines.

(g) Control Points.

(h) Degree of Search.

(j) Order of March (with gaps in terms of distance).

e. *Administration and Logistics*

(1) *Ammunition.* "Pouch ammunition will be carried".

(2) *Food*. "Haversack lunch will be carried".

(3) *Water*.

(4) *Casualties*. Give instructions for dead and wounded.

f. *Command and Signals*

(1) *Location of Company Headquarters*. In the order of march.

(2) *Location of Platoon Headquarters.* In the order of march.

(3) *Instructions for Wireless*.

(4) *Nicknames*. Give their meanings also.

(5) *Signals.* The whistle signal code mentioned in paragraph 1a above, if it is being used.

Reaction to Point Section Coming Under Effective Enemy Fire

8. This drill will be carried out in three stages. In the first stage, as soon as the point section goes to ground, the whole platoon will prepare for offensive action as follows:-

a. The platoon commander with his party will go forward to a position from which he can see what is happening and report contact with the enemy to his company commander by wireless.

b. The platoon 2IC will keep the rest of the platoon as far forward as possible without involving them in the point section's battle. The platoon 2IC must understand that if he allows any of this part of the platoon to be pinned down under enemy fire, he will spoil the chance of successful platoon fire and movement.

9. In the second stage, as soon as the platoon commander realizes that the point section cannot overcome the enemy using its own fire and movement; for example if he hears two enemy LMGs firing, he will make a battle appreciation and act as follows:-

a. Send his runner to the platoon 2IC/O group with a verbal message giving: -

(1) RV for O group.

(2) RV for platoon.

b. Order the point section to act as fire section and indicate the flank from which the platoon will assault.

c. Report the situation by wireless to the company commander.

10. In the third stage:-

a. As soon as the platoon commander's runner/wireless message arrives, the platoon 2IC will organise the move of the rest of the platoon to the platoon RV. On arrival, takes up a position of all-round defence.

b. Simultaneously, the platoon commander moves to the O group RV and, as soon as the assault section commanders and the Rocket Launcher NCO/sepoy have assembled, he will issue his battle orders.

Flanking Attack

11. The platoon commander will give battle orders to his 'O' group. The only headings necessary are:-

a. *Landmarks*. Give the bare minimum landmarks to include the objective, forming up position and route to it.

b. *Enemy Positions*.

c. *Mission.* "Capture area raised ground as soon as possible".

d. *Direction of Attack.* "Attack will be left flanking".

e. *Position of Fire Section.* (If not already clear). "Number 1 section in area broken ground".

f. *Forming Up Place*. "Jhund".

g. *Route to Forming Up Place*. "Along the nullah-bridge-Jhund".

h. Any Variation of the Battle Drill.

j. Reorganization.

k. H hour.

l. Fire Support Plan.

12. At the end of his order, the platoon commander with his O group will rejoin the rest of the platoon in the platoon RV where he will brief the platoon 2IC. If the ground permits, the point (now fire) section must continuously work its way forward and/or to a flank so as to:-

a. Widen the angle of covering fire.

b. Keeps the enemy guessing as to the direction of the main assault?

c. Prevent the enemy from pinpointing the positions of individuals in the section.

The Assault

13. The leading section for the move to the assault position is the section which has been moving on the flank chosen for the assault; the left rear section therefore, lead when the platoon is left flanking and the right rear section when it is right flanking. The platoon commander will not lead this move unless he doubts the ability of the section commander to get to the right place:-

a. In the Forming Up Place, when left flanking is ordered the leading section forms up on the platoon commander's left and when right flanking is ordered it forms up on his right.

b. In the move to the objective, platoon headquarters will move behind the leading section. In the assault, the platoon commander will be accompanied by his runner while the rest of the platoon headquarters moves in rear under platoon 2IC. If no task is given

to the rocket launcher, it should go with platoon headquarters on the flanking move.

c. When moving to the objective, if the platoon meets opposition from further to the flank, the drill is for the leading section LMG groups to drop off to neutralize it.

d. When moving to the objective, if the platoon meets opposition actually in the covered approach, this must be cleared before the attack can continue. The drill is for the leading section to act as a point section advancing. If this section cannot overcome the opposition, the platoon commander will order it to act as another fire section while he assaults the new opposition with his third section.

Variations

14. Variations are:-

a. Putting in a quick flank attack with one section, holding the third section in reserve.

b. Stepping up the fire power of the point section with an extra LMG or even another complete section, in this case the platoon 2IC will usually be ordered to take control of the supporting fire.

Making a Fire Base

15. In case the enemy position cannot be cleared by the point platoon, its commander will deploy it into a fire base for the company's attack. This will be made by moving the other two sections in line with the point section, one on either side, using fire and moment. In this manner the platoon will cover an area of approximately 450-550 metres. While the fire base is being made, the platoon commander can make use of artillery/mortar fire, including smoke, to keep the enemy's head down/blind him.

16. The vanguard company commander will come forward and gets a briefing from the point platoon commander regarding the enemy's strength, location and flanks. He will then make his plan and inform his platoon commanders of their role in it. This may be on the following lines: -

a. Company will attack in two phases from the right.

b. *Phase 1*. Number 2 and Number 3 Platoons will capture area up to the roads while number 1 platoon will provide fire support by

observation. The weapon section will be in Number 1 Platoon's area. Number 1 Platoon Commander will control their fire also.

c. *Phase 2*. Be prepared to capture area bound on nickname "CHALEES CHOR". Alternative signals will be red/green verey light.

d. *Forming Up Place*. Nullah bend.

e. *H Hour*. 0800 hrs.

17. In case the point platoon is required to put in an assault in phase 2 of the company attack, it will get up from its fire base location and move straight in to the attack. The flanking section will provide converge towards the centre section so as to reduce the frontage to reasonable limits.

Reorganization

18. Once the objective has been taken; the platoon will take-up a defensive position, which may either be on the objective itself or forward of it, having all round fields of fire. The reorganization may be carried out in a number of ways, some of which are:-

a. *Alternative 1*

(1) The leading section in the order of march takes the centre, on the original axis of advance.

(2) The second assault section goes straight across the objective to the far flank.

(3) The fire section will take the remaining exposed flank.

(4) Platoon headquarters rallies in the centre.

b. *Alternative 2*

(1) The leading section, reorganises in the centre.

(2) The second assault section, reorganises on the inner flank.

(3) The fire section reorganizes on the outer flank.

(4) Platoon headquarters rallies in the centre.

19. Immediately after the attack the platoon commander will go round the position to check the sections arcs, especially the LMG arcs of fire.

20. The platoon 2IC will check casualties and ammunition expenditure and will arrange casualty evacuation, re-distribution of ammunition holdings and subsequently replenishment of ammunition. To facilitate this, the section commanders should be ready with their casualties and ammunition states and, as soon as they see the platoon 2IC coming their way, they should shout out their casualties and ammunition figures. This will quicken up the reorganization process.

21. The platoon 2IC will report this to the platoon commander who will report by wireless to the company commander.

22. During advance to contact, the chances of a counter-attack by the enemy are rather remote. Also, it is not normally intended to stay in the captured area for long. In fact, as soon as the objective has been captured, the follow up platoon will probably pass through and resume the advance as the new point platoon. The old point platoon will now move its casualties on to the axis so as to facilitate their evacuation/burial by the RAP/subedar major following behind. As soon as the company's F echelon vehicle comes up, the platoon 2IC will quickly arrange replenishment of ammunition and the platoon will resume the advance as the last platoon of the company.

SECTION 64 – **ADVANCE BY NIGHT**

General

1. **Aim**. The aim of these notes is to high light the problems involved in night advance and suggest means to overcome them.

2. Night advance not only affords notable advantages in view of the vulnerability of own troops to enemy air but also offers opportunities for success, achieving surprise and exploiting natural effects of darkness. Night advance, like any other night operation, produces a psychological effect on enemy out of all proportion to the strength of the force engaged as the senses are strained, the imagination tends to exaggerate threats and dangers, sounds and movements seem sinister, the size of force is difficult to gauge and calculations to time and space are often faulty. Conversely all these disadvantages equally apply to advancing force but success awaits that side which has learnt to overcome darkness and boldly turns it to advantage.

3. **Problems**. The problems common to all night operations are equally applicable to night advance. These are:-

a. Maintenance of direction.

b. Movement-slow.

c. Command and control.

d. Evaluation of situation.

e. Effective employment of fire support means.

Basic Requirements

4. These are the same as for day advance but great stress must be laid on:-

a. Simplicity of plan i.e., well defined objectives, avoiding complicated manoeuvres and adherence to simple battle procedures.

b. Best possible daylight reconnaissance, where possible. This, however, may not be always possible therefore greater reliance will have to be placed on study of maps and air photos. Arrangements for direction keeping and traffic control should also be more thorough.

c. Detailed preparation of fire support plans including use of illuminating devices.

d. Planning and preparation should be more detailed than by day. An appreciation of likely delaying positions and enemy capabilities there will have to be made. All possible eventualities should be considered and catered for.

e. Communications should be duplicated.

Factors to be Considered for a Night Advance

5. Following are some of the important factors to be considered:-

a. *Pressure on Enemy*. Not to allow the enemy respite and take advantage of the hours of darkness to move back and organise his next position. It may be necessary to advance both by day and night to keep the pressure.

b. *Tactical Situation*. To take advantage of a tactical situation which is unlikely to last until the next day?

c. *Surprise*. Infantry on its own or supported by tanks can often advance successfully by night and may achieve considerable surprise the next day. Use of armour will achieve shock-action effects.

d. *Degree of Visibility*. In case of higher formations involving movement of wheels, et cetera, the phase of moon will also be considered.

e. *Wear and Tear*. Tanks and vehicles need time for maintenance in long advance and all troops need rest after a day long advance. Supply forward should be the aim.

f. *Passage of Defiles*. Safe at night when observation is denied to enemy.

g. *Ground*. Both tracks and wheels require relatively good going for night movements.

h. *Enemy*. The degree of enemy air threat during day.

j. *Morale*

(1) Although as much ground may be won in the first few hours of daylight as in a night's hard march, the pressure on enemy by night will affect his morale considerably.

(2) The morale of own troops also tends to be high during advance.

k. *Mission*. Mission given by the higher authorities.

6. **Conduct of Advance By Night**

a. *Grouping and Order of March*. Normally infantry would lead. In case of light opposition and good going tanks may lead closely supported by infantry.

b. *Distance. T*he distance between individuals/groups/subunits will be fairly closed. No man should be out of visibility distance of his neighbour. From point section till the tail of vanguard, it may be only 450 metres or so.

c. *Gaps*. Gaps between subunits/groups would be considerably reduced. The reduced gaps would however restrict manoeuvre (which is limited at night) and may also bring rear elements under enemy small arms fire. To avoid this, the main guard should follow the vanguard with a gap of not less than 400 to 450 metres.

d. *Frontages*. The advance would be carried out on a much narrower frontage than by day. Bigger frontages would be difficult to control.

e. *Formations*. The formations at night will be more compact than by day. The point platoon may move with two sections up in extended formation. Rest of the company may move in single files astride the axis.

f. *Degree of Search*. This again would be limited. Point platoons should search up to maximum visibility distance. The following platoons should also be made responsible for a flank each.

g. *Command/Control*. As the gaps would be smaller and formations compact therefore commander would be moving well up. Following will assist in command/control:-

(1) Headquarters moving on centre line and commanders wearing distinguishing signs.

(2) Communications duplicated-radio and runners.

(3) Bound and reports lines at shorter intervals.

(4) Emphasis on steadiness in probing forward rather than speed.

(5) In view of strain short leap frog by point platoon/vanguard.

h. *Connecting Files/Link Groups*. These are men placed in between the gaps to maintain contact. They should be furnished by the larger unit to keep contact with the smaller unit e.g., in case of vanguard by the main guard. The number of connecting files/groups places between the subunits would depend on space interval, the terrain and visibility. Connecting files/link groups pass messages and signals received from the vanguard/main guard minimum strength of a connecting file should be two men one watching to the rear (main guard) and one to the front (vanguard). The vanguard may also employ connecting file in between the gap with the point platoon.

j. *Reconnaissance of Area*. For moves at night it is essential that the route and general going is reconnoitred, where possible, and plans are made to overcome minor/major obstacles enroute. A study of air photos and maps of the area, especially of possible delaying positions must be made before hand. In plains of Punjab where there are too many tracks, proper arrangement for direction keeping, route marking, guiding and night marching should be made.

Aids in Keeping Direction in Cross-Country Moves

7. The following aids will help in keeping direction:-

a. *Compass*. To maintain direction of march.

b. *GPS*. To maintain direction and measurement of distance.

c. *Pacers*. To measure distances on foot.

d. *Millimetre*. Should be checked for accuracy.

e. *Tracers*. Fired by any weapon.

f. *Pre-planned Artillery Concentration*. Can be fired on call to illuminate predetermined reference points.

g. *Flares*. Can be dropped by aircraft.

h. *Movement Light*. By shining searchlight against low clouds.

j. *Tape*. For route marking.

Action on Meeting Enemy

8. The assessment of enemy strength is likely to be much more difficult by night. However, the flash of the enemy weapons and the type of weapons fired would indicate his location, strength and flanks. The commander will have the following alternative to deal with the situation:-

a. A quick frontal assault.

b. Out manoeuvre the enemy and go to his rear.

Frontal Attack

9. Frontal attacks, otherwise not advocated, may, on occasions, be the simplest and quickest method of clearing the opposition. Command and control would be simpler and there would be no loss of direction. The inadequacy of own fire support to close with the enemy would be compensated by the cloak of darkness providing cover. The disadvantage of enemy coordinated fire would be considerably reduced because it would be limited to fixed lines and defensive fires only.

Flanking Attack

10. Since manoeuvre at night is difficult, battle drills would have to be evolved and practised. Unless there is moonlight or the area is illuminated it

would be necessary to have bearings already worked out for possible attacks. A set of bearings worked out for a point would generally be applicable to most situations e.g., the manoeuvre element moves at 90º to a flank for a certain distance and then again at 90º to a flank for a certain distance and then again at 90º parallel to the axis till it reaches the FUP. From these onwards the actions would be those of a night attack. Following will assist in such manoeuvre:-

a. Tracks/nullahs leading towards objective may be used part of the way but not near the objective as they may be mine/covered with fire.

b. Selected routes should be along low ground, where possible to avoid silhouetting.

c. Night march charts should be prepared before hand.

d. Drills should be evolved so that held up troops do not fire at own manoeuvring force. Following will be useful in indicating target/exercising control: -

(1) Firing tracer.

(2) Radio contact.

(3) Firing by manoeuvring force.

(4) Light signals.

11. Due to the difficulties of control, normally this manoeuvre would be restricted to company level in dark nights and to battalion/regimental level in moonlit night or by artificial illumination.

Identifying the Objective

12. The aids in knowing the extent and depth of the objective would:-

a. Flash of enemy weapons.

b. Own weapons firing tracers indicating the extent of the objective.

c. Own artillery shelling on the objective.

d. Use of flares.

Flank Markers

13. Light with shutters to produce different coloured lights can also be carried on flanks of the manoeuvring force and adjusted so as to be visible to own troops

but hidden from the enemy. Flank marking light from held up troops would also be useful. Too complicated system should be avoided.

Reorganization

14. The reorganization would be the same as for any night attack. Arrangements for guiding F echelon and follow up troops passing through should be made for a quick resumption of advance.

Villages/Woods Enroute

15. These should be searched if lying close to the axis. In case they are away from the axis then they may be by passed. Normally leading troops should not by pass opposition at night as they can fall into a trap.

16. **Covering Troops**

a. *Mobile Troops*. Their employment without infrared devices/attle-field illumination would be very restricted. When employed they would operate at lesser distances with sufficient proportion of infantry for protection.

b. *Flank Guards*. May be used in moonlit night. Better employed by picquetting lines of approach rather than moving parallel to the main column. May need some infantry for protection.

Harbour Drills

17. When sufficient distance has been covered or daylight move is to be avoided then the force may go to a harbour area completing the move before first light. Harbour parties should in that situation, move well forward and drills for getting in and out of the harbour should be perfected.

Precautions against Ambush

18. Effective search by vanguard/main guard and other security arrangements would considerably reduce the chances of enemy ambush. However, counter-ambush drills must be well practiced.

Infra-red/Night Vision Devices

19. Since in a future conflict infra-red/ Night Vision Devices are likely to be used both by us and our potential enemies, it is therefore important to know the implications of its use.

a. *Implications*

(1) Infra-red sources may be detached by the enemy who is himself suitably equipped.

(2) Continuous viewing puts a considerable strain on the operator.

(3) Infra-red images have unusual shadow grading and perception of depth is difficult.

(4) It is affected by rain, fog et cetera.

b. *Application*

(1) It should be used with caution because of detection.

(2) Smoke has the same effect on infra-red lights as on daylight.

(3) Searchlight may be used to blind enemy infra-red lights.

Night Advance – Point Section

20. The drills for night advance and encounter battle at section level are:-

a. *Stores*

(1)	Lamp aiming	-	3
(2)	Arm band white	-	1
(3)	PRC-77/Walkie-Talkie Set	-	1
(4)	Very light pistol/shooting pencil	-	1
(5)	NVD	-	1

b. *Advance Phase*

(1) *Deployment*

(a) Extended line.

(b) Physically spread as follows: -

i. *Dark Night*. 100 metres i.e., 10 metres between each man.

ii. *Moonlit Night*. 140 metres i.e., 14 metres between each man on the axis.

(2) *Section Commander*. To move on the axis in the centre of his section. One sepoy runner with an aiming lamp to be readily available to him. Colour of aiming lamp to be red. Section commander to wear armband.

(3) *Flank Men*. To have an aiming lamp each. Colour of aiming lamps to be green.

(4) *Encounter Battle Phase*

(5) *Reaction to Effective Enemy Fire*. Hit the ground.

(6) *Location of Enemy Flanks and Weapons*. Section commander must exert his command and do the following:-

(a) Order LMGs to open fire 50 metres left and right of the extreme left and right weapons already located (by virtue of their flashes). If the enemy flank weapons open up in reply to this speculative fire then the LMGs are to fire a further 50 metres to the flanks till such time as the flanks have been located.

(b) Order riflemen to open fire on the flashes of enemy weapons, which had originally opened fire with a view to causing casualties/keeping the enemy head down.

(c) In case enemy does not open fire from his flanks as envisaged in (a) above, then the section commander must do the following: -

i. Order his LMG teams (number 1 and 2) to move 5-10 metres towards their respective flanks and forward. At the same time these teams should make aggressive noises (e.g., **YA ALI, NARA-E-HAIDRE** et cetera) so as to attract enemy's attention and draw his fire.

ii. If even this fails then he must fire a verey light round and move two men teams to

either flank and forward under the light of this flare while the rest of the section opens fire in the general direction of the enemy.

iii. If no fire is returned, the point section must move forward by fire and move technique and approach closer to the enemy.

Night Advance-Point Platoon

21. The drills for night advance and encounter battle at platoon level are as follows:-

a. *Stores*

(1) *Sections*. Same as for point section less verey light pistol.

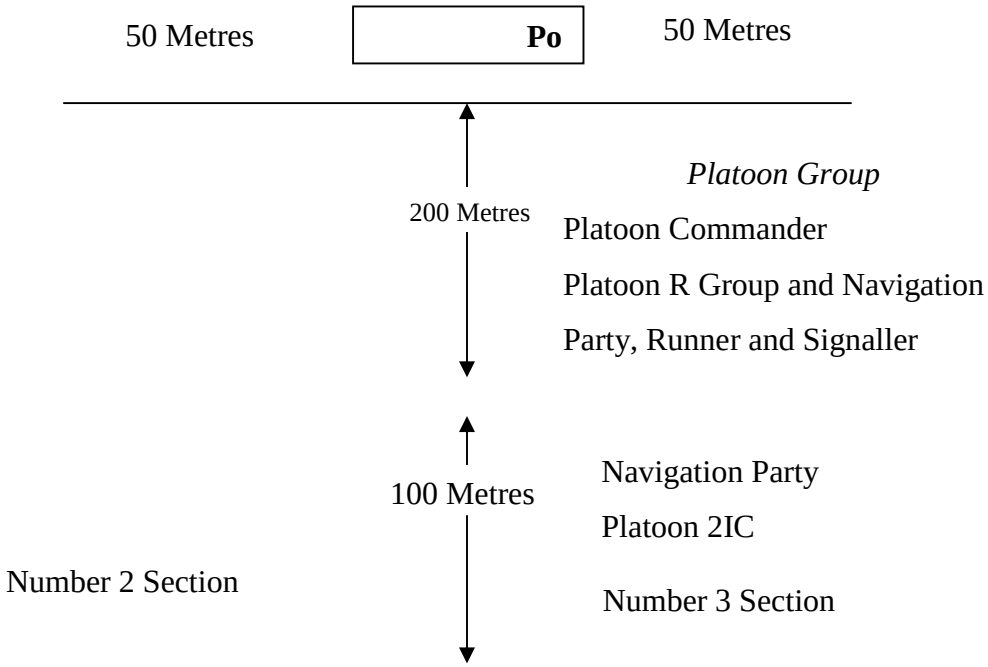
(2) *Platoon Headquarters*

(a)	White patch (30 x 30 centimetres)	-	1
(b)	Walkie-talkie Set	-	1
(c)	PRC-77	-	1
(d)	Compass	-	1
(e)	GPS	-	1
(f)	Map and map case	-	1
(g)	NVDs	-	4
(h)	Torch	-	1
(j)	Blanket	-	1
(k)	Pebbles/stones for counting paces)	-	1 Set of counters
(l)	Lamp aiming.	-	1

b. *Advance Phase-Deployment*

(1) *Point Section.* Extended line 50 metres astride axis depending on degree of visibility, nature of the ground and the degree of search required.

(2) Remainder of Platoon. As shown below:-



(3) R Group and Navigation Party. To move together.

(4) *Section Commanders*. To move with their sections.

(5) *Platoon Commander*. To wear white patch 30 x 30 cm on his back. It should be secured by tying in shoulder epaulets and with belt.

c. *Encounter Battle Phase*. Reaction to point section coming under effective enemy fire: -

(1) Platoon halts.

(2) Point section tries to locate enemy weapons, flanks and determines the distance of the enemy from own position. Point section marks enemy's flanks with green lights and forms a fire base. Section commander despatches runner to

brief platoon commander. Runner carries red light. Platoon commander also sends report on PRC-77 if communication is through.

(3) Platoon Commander moves forward with R group and navigation party towards red light once he sees it switched on and does the following: -

(a) Meets runner from point section, takes back verey light pistol, and receives briefing on following: -

i. Distance of enemy from fire base.

ii. Distance of fire base from present location of platoon commander.

iii. Platoon Headquarter will be indicated by red light, (to be carried by platoon commander's runner).

(4) *Move to FUP*

(a) Platoon commander sends his runner to fetch platoon (for left flank attack left rear section will lead and vice versa).

(b) *Calculations*. The example below is worked out for a left flank attack: -

i. If extent of enemy's left flank is 50 metres from the axis add 600 metres for moonlit night and 400 metres for dark night. For dark night distance will be 400 metres + 50 metres = 450 metres. Convert this into paces @ 120 paces = 100 metres. Total distance in paces is 540 paces. Take bearing of axis and minus 90 degree (for right flank attack, add 90 degrees). For example if bearing of axis is 120 degrees then 120 degrees-90 degrees = 30 degrees. Now march on a compass bearing of 90º for 540 paces and stop. Platoon 2IC to be incharge of navigation party. The second signaller should count paces.

ii. Calculate distance of enemy from present location (the place where point section

runner meets platoon commander) and convert it into paces at the scale of 120 paces = 100 metres e.g., if enemy is 200 metres from fire base and runner has walked back 70 paces the distance to enemy will be 240+70=310 paces. Add 100 paces to this to ensure that enemy's depth locality is properly engaged. Therefore, total distance is 310+100=400 paces. Add 90º to bearing calculated in 1 above, 90º+ 30º =120 (i.e., bearing of axis). Now march for 410 paces on a compass bearing of 120º and stop. The platoon commander who is leading will now be standing on the centre line of his FUP.

iii. The leading section is moved left of platoon commander so that its right most man is next to him. Similarly the second section (right assault section) deploys right of platoon commander with its left most man next to him. Section commanders stand in centre of their sections. The right most man of the right assault section wears an aiming lamp on his right side (away from the enemy and towards his own fire base). This light will be of yellow colour and will be switched on only after the FUP has been crossed.

iv. The platoon now turns full right and marches towards its objective on a bearing of 120º + 90º = 210º for a distance of 540 paces. The navigation party walks in centre of the platoon and approximately 10 metres ahead.

v. Platoon Headquarters to walk in the centre of platoon behind the navigation party.

(5) *Assault Phase*

(a) Platoon to march on compass bearing led by navigation party.

(b) *Fire Support*. This is to be provided by artillery, mortars and point section. Artillery and mortars will

stop firing according to fire support plan. Point section will cease firing on signal from platoon commander. Suggested signals are: -

i. Nickname on wireless.

ii. Yellow light of flank marker of right assault section to act as guide for point section commander and fire to be stopped as soon he judges the assault troops to be in danger from the fire of his section.

iii. If verey lights are being fired by either side, then progress of assault troops becomes visible to own fire base and fire can be stopped by observation.

iv. If no verey lights are being fired, platoon commander can fire a verey light to show objective to assault troops. This verey light can be taken as signal for fire base to cease firing. NVDs and weapons infra-red/night vision sites can also help in determining the time of cease fire as these will be available to the fire base commander and weapons crew.

v. Assault to be sounded from as near the enemy as possible in case flank move has not been detected, otherwise assault when 100 metres from the enemy.

(6) *Reorganization Phase*

(a) Reorganization to start only after objective has been cleared.

(b) The left assault section to deploy left of axis.

(c) The right assault section to deploy right of axis.

(d) Platoon headquarters to deploy 50 metres in rear on the axis and switch on red light facing rearwards.

(e) Point section to come forward on seeing success signal and deploy 50 metres behind platoon

headquarters astride axis. Red light of platoon headquarters to act as a guide.

(f) Once reorganization is completed, forward sections are moved 20 metres away from axis on respective sides so as to allow the follow up platoon to pass through.

Supporting Arms

22. **Armour**

a. While at night the enemy air threat is much reduced, the use of tanks can have a considerable effect on the enemy's morale. Even a few tanks may achieve this; in darkness the noise of tank movement is deceptive as regards numbers and directions. The feeling of uncertainty to which tanks give rise at night can be great, especially among inexperienced troops. They would however, need close infantry protection and due to limitations mentioned below their employment would be limited.

b. Advantages of using tanks at night are:-

(1) Speed and momentum of tank fire-power at relatively short range may prove decisive and the tanks will also be available to meet enemy counter-attack during reorganisation.

c. *Limitations*

(1) Problems of control.

(2) Inability of tank commander to clearly see own infantry.

(3) Vulnerability to short range anti-tank weapons.

(4) Difficulty of accurate target acquisition time.

d. *Night Movement*. It is difficult without using infra-red lights or tanks searchlight. Infantry can assist by guiding/route marking. Suitable formation is line ahead on road and staggered line cross country.

e. *Engaging Targets*

(1) Target indication is difficult at night therefore drills for indicating target must be perfected. Drills for stopping tank fire should also be practiced.

(2) In full moon tank commanders/gunner can engage target up to 450 metres/550 metres.

(3) *Direct Illumination*. Uncontrolled use of searchlights would be dangerous. When firing with the help of the tank's own spot light the glare of the beam affects the gunner's sight. Results are better when firing with the help of other tanks with their spot light deployed on flanks.

23. **Artillery**. During a night advance the artillery is confronted with the following problems:-

a. *Deployment*. Due to restricted visibility and absence of dayligh reconnaissance and deployment of guns becomes difficult and takes a much longer time. Following steps can be taken to overcome this problem:-

(1) Guns should be deployed well forward before last light in order to gain maximum range initially. If this is not possible then the reconnaissance parties should reconnoiter forward areas during daylight.

(2) For subsequent redeployment without daylight reconnaissance the following actions could be taken:-

(a) Gun areas should be kept as close to the main axis as possible. These must be coordinated with the higher headquarters beforehand.

(b) Survey troops of division locating battery should move well forward to provide survey detail to regimental reconnaissance parties for deployment of the guns.

(c) Elaborate traffic control arrangements.

(d) To minimise redeployment, maximum use be made of long-range guns.

(e) Large scale build-up of artillery is not possible at night. Leap frogging would be carried out within the regiment culminating in a regiment build-up.

b. *Observation*. The adjustment of fire and gathering intelligence by the observer becomes very difficult due to restricted visibility. To overcome this difficulty the following measures can be taken: -

(1) The advance guard and the follow up battalion must be provided one observer per company because changing/ replacement of observers at night is problematic. It wold not be possible to employ static Observation Posts due to restricted visibility.

(2) Adjustment of fire by observation can be facilitated to some extent by the use of illuminating rounds-hence there is a need for having more illuminating rounds during a night advance.

(3) Gathering of maximum information of likely targets during daylight by resorting to map reconnaissance, air photos and aerial reconnaissance.

(4) Since targets can not be accurately located at night, artillery fire would be directed to area target, which would be searched by fire.

(5) To cover the area targets effectively use of larger calibre guns should be preferred.

24. **Engineers**. Tasks of engineers are same as for day advance. Engineers should move well forward with engineer reconnaissance parties moving with the vanguard.

25. **Light Anti-Tank Battalion**. May lead the advance ahead of tanks or be used for flank protection. Tasks otherwise are same as for day.

26. **Battalion Mortars**. As the targets would not have been previously registered by fire, mortars can provide fire support by applying a searching shoot to the predicted position of the target. Whilst such a shoot should produce effective fire on the target, a large part of the ammunition fired will not fall on the target and will be wasted. For safety reasons also fire can not be brought closer to own troops.

27. **Assault Pioneer Platoon**. May be grouped with the vanguard or follow immediately behind it to clear mines/booby traps on the main axis without delay.

Company Weapons

28. Fire support from platoon/company integral weapons would need proper control. Machine-guns can provide effective fire support by night with the night vision sight. Flares could also be used to illuminate the area. The disadvantage of

using lights and flares to own assault troops should also be considered. When necessary, smoke can be used to screen own movement and blind the enemy. The methods of controlling the fire of own troops in the fire base in relation to manoeuvre force should be properly coordinated with the assault troops.

29. Recoilless rifles, rocket launcher and 60 mm mortar's fire support would also be quite effective.

Administration

30. **Transport**

a. Drivers must be fully trained to drive vehicles at night and to carry- out all routine maintenance and refuelling without the aids of lights.

b. In night moves there is a natural tendency for vehicles to be driven nose to tail. This makes them vulnerable to ambush or if suddenly illuminated are subjected to intense fire

c. Drivers must have sufficient practice in: -

(1) Night Navigation.

(2) Formations for night driving.

(3) Crossing obstacles, negotiating ditches, bank, headges, and minefields gaps.

d. Battalion standing orders must include: -

(1) Action in harbour/assembly area.

(2) Breakdown drills.

(3) Action if ambushed.

(4) Action on becoming lost.

e. In areas like Punjab water channels are met frequently impeding the mobility of vehicles. It should be a drill to carry picks, shovels and a few wooden logs/pieces in each vehicle to assist in crossing a water channel or improve the route.

31. **Recovery/Repair**. These would be both slow and difficult therefore all equipment should be tested before the advance commences.

32. **Medical**. Evacuation of casualties would be slow. Precautions must be taken to ensure that casualties are not lost. Troops should also have a high standard of first aid training.

33. **Traffic Control**. Guiding and traffic control arrangements should be good to avoid confusion and loss of direction.

CHAPTER 13

ATTACK

SECTION 65 – INTRODUCTION

General

1. This chapter provides doctrine for employment of the rifle company in attack. Conditions of terrain, climate and special operations may dictate modifications to techniques and procedures for attack. However, the doctrine for attack remains the same.

Definitions

2. Some of the attack definitions which must be understood are as follows:-

a. *Attack.* Attack is an operation designed to so disrupt enemy forces that their will to resist is destroyed and ability to execute a coordinated defence is defeated. This is achieved by fire and movement, aimed at inflicting on the enemy such losses in personnel and equipment or to so deprive him of ground of tactical importance, that he is forced to surrender or to withdraw, in the process of which his destruction as an effective force can be completed.

b. *Deliberate Attack.* A carefully planned and executed offensive action, in which various elements of a command are employed in such a manner as to utilize their power to the greatest advantage to the command as a whole.

c. *Counter-Attack.* Offensive action by a part or all of a defending force against an attacking force for the purpose of restoring the original positions and trapping and destroying the hostile force at a point particularly favourable which is carried out by a pre-designated part of the force.

d. *Local Counter-Attack.* An attack organized by the local commander to destroy enemy forces which have penetrated or threaten to penetrate his defensive position, generally using a pre-arranged plan.

e. *Diversionary Attack.* An attack, wherein a force attacks, or threatens to attack, a target other than the main target, for the purpose of drawing enemy attention away from the main effort.

f. *Feint Attack.* An attack which simulates intense activity of a large scale attack and keeps the enemy guessing about its weight and time.

g. *Holding Attack.* Attack designed to hold the enemy in a position, to deceive him as to where the main attack is being made, to prevent him from reinforcing the elements opposing the main attack, and to cause him to commit his reserves prematurely and at an indecisive location. Also called containing attack.

h. *Attack by Infiltration*. When infiltration is the main feature of an offensive operation, it will be described as an attack by infiltration.

j. *Main Attack.* The principal attack or effort, into which the commander throws the full weight of the offensive power at his disposal, directed against the main objective of the campaign or battle.

k. *Mopping Up*. The liquidation of remnants of enemy resistance, in an area that has been surrounded or isolated, or through which other units have assembled/passed without eliminating all active resistance.

l. *Exploitation*. Exploitation means taking full advantage of success in battle and follow up initial gains.

m. *Spoiling Attack.* A tactical manoeuvre employed to impair or delay a hostile attack. It is normally launched while the enemy is in the process of forming up or assembling for an attack. Usually employed by units in defence or in withdrawal by an attack on enemy assembly area or Forming Up Place (FUP).

n. *Assembly Area*. As assembly area is the place where attacking units/subunits, organize for an attack. An ideal assembly area is free from enemy observation and harassing fire and has good entrances and exits for transport. Here company, platoon and section commanders make contact with their opposite numbers of

supporting arms, if not already done so. Communications are checked, ammunition issued, weapons examined for their serviceability and reorganization stores sorted out. Troops are given, if possible, hot tea or meal particularly in case of a night attack. Orders are rechecked.

o. *Forming Up Place (FUP).* From the assembly area assault troops move to the FUP and deploy in the battle formation in which they have to advance up to the objective. The forming up place needs to be easily recognisable, well sign-posted, lit for a night attack if necessary and not under enemy observation. It is from here that troops break cover at H hour to advance towards their objective. The company/platoon commander will here make certain decision that he is in his proper place in relation to his supporting tanks.

p. *H Hour*. H hour is the time at which leading elements of the assault troops cross the start line. All timings for an attack are given in their relation to H hour.

q. *Start Line*. Start line is the line which leading elements of the assaulting troops cross at H hour. It must be clearly recognisable. It may be artificial, that is, taped, or natural such as a road or a narrow water channel. It should, if possible be at right angle to the line of advance and must be secure from enemy interference. It must not be behind any obstacle. The forward edge of the forming up place can be the start line, especially for a night attack.

r. *Assault Line*. This is a line used to coordinate the final deployment of attacking troops in preparation for night assault against an enemy position. It is located as close to enemy positions on the objective as attacking troops can move; before becoming dangerously exposed to own supporting fire. Artillery will cease to engage forward enemy positions when the attacking troops reach this line; it may then engage targets in depth/on flank. There will be no assault line in a daylight attack. Troops carrying out the assault will continue their fire and movement through the objective. There may, however, be occasions when the progress of attack is slowed down or even temporarily halted by accurate enemy fire. In such circumstances artillery et cetera fire will be brought to bear on known enemy dispositions. In order to take maximum advantage of this supporting fire, the assaulting troops must get as closer to it as safety conditions permit, and the moment the fire is lifted they, will assault.

Basic Considerations

3. Whatever the type of attack the following basic considerations must receive attention:-

a. *Initiative*. The company/Platoon/Section commanders must maintain initiative to dominate the enemy by active ground reconnaissance. All efforts should be made to continue pressure on the enemy and not allowing him to rest until attack is finally launched.

b. *Selection of Objectives*. It should be ensured that the objectives selected are the positions which will hurt enemy the most and their capture will neutralize him. The objectives should be easily identifiable as well as their capture must be possible within the time and the troops employed.

c. *Impetus*. The impetus of the attack must not be allowed to die down. This will entail immediate attack by follow up platoons/ sections once the phase-1 troops have captured the objective. Delay in attack of subsequent phases will give reaction time to enemy to evict own troops which may result into failure of attack.

d. *Fire Support*. The company/platoon attacks must be supported by fire up to and beyond the objective. Supporting weapons of the battalion must be moved forward as soon as possible according to the progress made. Tanks, if attacking along with the company should move as close as possible to own artillery fire and provide intimate support.

e. *Intelligence.* The company/platoon commanders should at all times get maximum information about the enemy locations, weapons and obstacles, which would help him in capturing the objective. This should be done by use of patrols before the attack and also getting whatever information is available with the battalion.

f. *Speed*. The time available to the enemy to prepare his defences must in all cases be kept to a minimum. The weaker the defence the less time is needed for preparation of an attack, but the urgency to get on and capture the objective must be the predominant consideration for company and platoon commanders.

Types of Attack

4. Basically attacks may be classified into two types:-

   a. *Quick Attack.* It is launched against weak defences to take full advantage of the unpreparedness of the enemy. It is undertaken when information about the enemy situation is correspondingly inadequate, which is countered by corresponding increase in fire support for attack and surprise achieved by speed. The quick attack is an extension of the advance, in that the attacker soon after contact, exploits his initial success, breaks into and disrupts the enemy position. In the quick attack some degree of risk can be taken and a greater degree of decentralisation will be necessary. Speed is an essential feature and will be obtained by having well established/rehearsed deployment, battle drills and a simple plan.

   b. *Deliberate Attack.* This type of attack is normally mounted against strongly prepared and well coordinated enemy positions. The assaulting force may be restricted in movement, but this disadvantage is usually overcome by integrating fully, the fire support afforded by all available armour, artillery and air. More time is spent on detailed reconnaissance, elaborate preparation and detailed coordination.

5. Within the above main types, attacks may take following different forms:-

   a. *According to Timings of H Hour*

      (1) *Day Attack.* Day attack means that the attacking troops cross the start line after first light and before the last light.

      (2) *Night Attack.* Night attack means that the attacking troops cross the start line after last light and before first light.

      (3) *Dawn Attack.* Dawn attack means that the attacking troops cross the start line just before or at first light.

      (4) *Dusk Attack.* Dusk attack means that the attacking troops cross the start line just before or at last light.

   b. *According to Timing of Fire Support*

      (1) *Silent Attack.* Silent attack is characterized by the absence of preparatory bombardment before H hour. Silent attack is resorted to when the relative strengths in fire power and terrain conditions favour attainment of surprise.

Silent attack does not imply total absence of fire prior to the attack. A complete fire plan is prepared but remains on call and is brought into effect on call, the moment surprise is lost.

(2) *Noisy Attack.* Noisy attack is defined by the incidence of preparatory bombardment of enemy defence before H hour. Noisy attack is resorted to when information about hostile weapons and relative strength in fire power favour preparatory bombardment.

SECTION 66 – **STAGES OF ATTACK**

General

1. An attack may be divided into following stages:-

   a. Preparatory.

   b. Assault.

   c. Destruction.

   d. Exploitation.

2. These stages are closely interlinked and may overlap. For example, there will be many occasions when the assault, the destructive and the exploitation stages will overlap.

Stage 1 – Preparatory

3. **General**. Sound and swift planning is important since improperly conceived or delayed plans increase the vulnerability of the company and jeopardize success. Emphasis is on speed and aggressiveness.

4. **Company/Platoon Plan**. Following matters will have to be decided while making a company/platoon plan for attack:-

   a. Objectives with inter subunit boundaries (provided these are easy to identify).

   b. Layout in assembly area and FUP.

   c. Details of tank support, if any and arrangements for infantry tank target indication.

d. Fire support (from artillery, own base of fire and tanks).

e. Method of crossing obstacles.

f. Attack formations.

g. Action in case either or both the assault subunits get pinned down.

h. Technique of fighting through the objective.

j. Limit of exploitation.

k. Reorganization.

5. **Actions of Leaders**

a. *Company Commander*

(1) Company commander normally receives battalion warning order for an impending attack, which facilitate early planning. The warning order is usually fragmentary and may contain information such as the time of attack, mission, objectives, preliminary plan for employment of elements of the battalion, information pertaining to the issuance of the battalion attack orders and administrative instructions necessary for proper preparation. The company commander then normally issues a warning order to his subordinates, giving them essential information concerning the attack.

(2) In actual combat operations, use of written operation orders at company level and below may be infrequent. Objectives, control measures, time of attack, et cetera may accompany an oral operation order on simple overlay.

(3) The plan of attack consists of a scheme of manoeuvre and a fire support plan. These are inter-related and developed concurrently. In developing his plan of attack, the company commander is mindful of the need to accomplish his mission in the shortest possible time and with the fewest casualties. He strives to develop a plan which is simple yet complete, provides flexibility, achieves surprise, directs the major effort against the principal objective and minimises the vulnerability of the company.

b. *Platoon Commander*

(1) The company order or warning order for an attack issued to the platoons is usually a brief fragmentary order containing the time of attack, the company's mission and necessary administrative instructions.

(2) On receipt of the initial company attack order, the rifle platoon commander immediately issues a warning order to his platoon and begins the battle procedure for attack. He will move forward to meet the company commander at the time and place set for the issuance of company orders. During the absence of the platoon commander, the platoon 2IC and section commanders supervise the continuing preparation for the attack, which includes turning in or drawing of equipment, ammunition and rations, checking camouflage, checking and cleaning of weapons, setting battle sights, et cetera.

c. *Section Commander*. On receipt of the platoon commander's orders, he studies the terrain in detail, paying particular attention to the planned route, landmarks on which to guide his section and known and suspected enemy positions. He plans actions of his section from the time it departs the assembly area to re-organization phase on the objective. He plans in detail his assigned tasks and actions at danger areas along the route and consolidation.

6. **Scheme of Manoeuvre**

a. Company commander's manoeuvre elements are his rifle platoons. He formulates the scheme of manoeuvre by considering his mission and by designating the main and supporting attacks, the direction of attack, the objectives to be seized and the employment of the reserve, if any.

b. Each manoeuvre element is given specific mission prescribing its role in the scheme of manoeuvre. Missions assigned to attacking platoons are usually designated as terrain objectives to be captured.

7. **Fire Support Plan**

a. The company commander develops the plan for fire support concurrently with the scheme of manoeuvre and in as much details as time will allow. In developing his plan, the company

commander considers the use of all fire support to include organic and all other indirect fire weapons.

b. In developing his plan for fire support, the company commander considers:-

(1) The enemy situation, dispositions and terrain.

(2) The nature of enemy dispositions and the effects desired.

(3) Capabilities of organic and non-organic weapons.

(4) Availability of ammunition.

(5) Time for fire delivery and duration.

(6) Timed and on call fires.

(7) Use of smoke.

(8) Means of communication.

c. He plans fire support to neutralize known and suspected enemy positions which would interfere with the rapid movement of manoeuvre elements to seize and consolidate the objective. Fire support plans are prepared to block avenues of approach for enemy counter-attacks. Since available fire can not always engage all targets simultaneously, the company commander plans to neutralize targets which would have the greatest effect on the accomplishment of the mission. The fire support plan normally provides the greatest concentration of fire on enemy positions in the objective just prior to the assault.

d. Fire support from a unit in contact may be made available to the attacking company passing through or around its position. The reserve platoon may be assigned the mission of providing fire support to the attacking platoons. In either case, close control and coordination are mandatory.

e. While incorporating the above weapons/supporting arms in the company/platoon's fire support plan the following points must be kept in mind:-

(1) Movement between the start line and the objective will be covered by artillery and mortars up to the following distances from the objective.

(a) 120 mm Mortars. 450 metres.

(b) Medium Artillery. 360 metres.

(c) Battalion Mortars. 275 metres.

(d) Field Artillery. 180 metres.

(2) Beyond the minimum safety distances of artillery/mortars, the assaulting troops will have to rely on the following: -

(a) Tanks in close support (if available).

(b) Recoilless rifles/MGs/60 mm mortars in the base of fire or near the start line.

(c) The platoon rocket launcher.

(d) Volleys of energa grenades.

(e) Smoke grenades.

(f) In the very end the platoon will have to fight its way into and through the objective by fire and movement.

f. *Smoke*. Smoke can be effectively used to reduce hostile observation and conceal the attacker's movement, to mark targets, and to serve as visual signals.

g. *Communication*. The company commander ensures the communication is adequate to control all types of available fire support during the attack. Control measures should include code designation of on call targets, and visual signals to be used in shifting, lifting or calling for fire as appropriate. Communication security must be exercised in the planning and building phases of an operation to prevent enemy intelligence personnel from gaining information through radio traffic analysis.

h. The plan for supporting fires at rifle platoon level is similar to that of company level. The plan includes the use of all available fires, organic and non organic of the company and higher units. If available supporting fires are inadequate to support his scheme of manoeuvre, the platoon commander may request additional supporting fire from the company commander, or he may use a rifle section to provide additional intimate supporting fires. The platoon leader's plan for supporting fires includes: -

(1) *Initial Employment*. The platoon commander designates general firing positions and assigns target areas. His decision on the employment of weapons is influenced

primarily by observation, fields of fire and weapon characteristics.

(a) *Machine-Guns*. If positions near the start line afford observation and field of fire over the route to the objective, machine-guns may occupy these positions just prior to H hour. If portion of the route is obscured from the start line, the best supporting fire may result if one machine-gun occupies a position near the start line while the other accompanies the manoeuvre element. If observation and fields of fire do not cover the route of the manoeuvre element from positions near the start line, the platoon machine-guns should accompany the manoeuvre elements.

(b) *Anti-Tanks Weapons*. These will move by bounds and accompany the manoeuvre elements. If no armour threat exists, anti-tank weapons are used to fire on grouped personnel, bunkers and similar targets.

(2) *Employment During the Assault*

(a) Machine-guns and anti-tank weapons should be in a position to cover the movement of the rifle sections in the assault. Where possible, forward positions that permit fire through gaps in the assault or on the flank of the assault should be used to provide maximum supporting fire during the assault and continue to fire until masked by the advance of friendly troops. When fires are masked, these weapons displace, usually to new supporting positions. One or more machine-guns may accompany the rifle sections in the assault.

(b) *Displacement of Machine-Guns*. Plans are made for machine-guns to displace by echelon to new positions when their fires become masked by the advance of friendly elements or when control becomes difficult. Displacement by echelon ensures close and continuous supporting fire for the assaulting elements.

(3) *Consolidation*. General locations from which to cover likely avenues of enemy approach are designated on the objective. Routes for displacement of weapons on the objective should permit rapid movement and early occupation of positions on these selected locations. Temporary positions are selected for machine-guns permitting flanking fire across the front of the platoon, whenever possible, unless the terrain or the threat of a likely approach to the position makes another position more important. Concurrently, the construction of defensive positions within the objective should be expedited.

(4) *Control*. The platoon commander plans to keep his organic and supporting weapons within communication distance so that he can control their fires throughout the attack. If this is not feasible, the platoon 2IC or section commander may assist in this control.

8. **Control Measures**. Control measures ensure teamwork and coordination between and within supporting fires and manoeuvre units. They minimize the possibility of defeat in detail by ensuring the controlled application of combat power. Control measures as applied by the company commander permit his platoon commanders freedom of action within prescribed limits necessary to ensure progress of the attack in a desired manner. Control Measures at Company level are:-

a. *Time of Attack*. The time of attack is the time when the leading elements of the attacking company cross the start line.

b. *Assembly Area*. An assembly area is where preparation for the attack takes place. Within the assembly area the company commander designates dispersed platoon assembly areas where the attack orders are issued, maintenance and supply are accomplished and the organisation for combat is completed by subordinate commanders. Assembly areas should be positioned and prepared for all round defence to the extent the situation and time permit. They should also provide concealment, room for dispersion, suitable routes forward and security from ground or air attack. When possible, they should be beyond the effective range of enemy field artillery.

c. *FUP*. It is the last position which is occupied by the elements of the company to allow any final coordination and deployment into initial attack formation. It should offer cover and concealment

from enemy observation and direct fire, be easily recognised on the ground and be large enough to accommodate the company. A halt in the forming up place should be for a short duration. Any unnecessary delay needlessly exposes the unit to enemy fires and reduces the degree of surprise which could otherwise be achieved.

d. *Start Line*. This is a line generally perpendicular to the direction of attack, easily recognizable on the ground and as close to the enemy as possible. It should not be under observed fire by enemy's direct or indirect fire weapons. Fire support for attack is related to the time when attacking troops cross the start line.

e. *Boundaries*. Boundaries between platoons may or may not be designated by the company commander in an attack/offensive operation. In most situations, the company commander is able to observe closely and control adequately his attacking platoons, precluding the need for boundaries. Platoon boundaries are used when:-

(1) Terrain and limited visibility severely reduce the commander's ability to observe and control his attacking platoons.

(2) Coordination of fires of two adjacent units would be facilitated.

(3) There is a requirement to control converging forces.

f. *Axis of Advance*. An axis of advance indicates the general direction of attack for a unit. It should follow the well defined terrain features of an avenue of approach, such as series of ridge lines, hilltops or roads extending to the objective area. The company commander attacks along his assigned axis as rapidly as conditions permit and does not allow small enemy forces to delay the seizure of the objective. The company commander may manoeuvre his platoons and place his fires freely on either side of his axis, as necessary, to avoid obstacles, to engage the enemy or to bypass enemy forces of such strength as not to threaten his security or jeopardize the accomplishment of his mission. The company commander ensures that such deviation does not interfere with adjacent units and that his elements remain oriented on the objective.

g. *Direction of Attack.* A direction of attack is a specific direction which the main attack follows. The terrain along the direction of attack must be cleared of effective enemy resistance.

h. *Phase Line*. A phase line is an easily recognized terrain feature normally perpendicular to the direction of advance and is used to control fires and unit movement and to limit the advance of attacking elements. Company reports arrival at (and some time clearance of) phase lines.

j. *Objectives*. The company objective is usually a key terrain feature, or it may be an enemy force. When seizure or destruction of the company objective requires the employment of more than one platoon, the objective is clearly divided into platoon objectives to draw responsibility. When he deems it necessary to divide his assigned objective, the company commander directs the main attack towards the decisive terrain or positions, and he directs secondary attack towards objectives which contribute to the overall accomplishment of the assigned mission. The platoon commanders of the attacking platoons further divide their platoon objectives into specific section objectives. Objectives should have as many of the following characteristics as possible:-

(1) They should be easily recognizable on the ground.

(2) They should facilitate convergence of effort.

(3) They should be within effective range of the battalion mortars.

(4) They should be of a size the company is capable of defending upon seizure.

(5) The number of objectives selected should be the minimum required to accomplish the mission and facilitate coordination and control.

k. *Positioning of Commanders*

(1) Company commander should position himself where he can best control the entire company and influence its action.

(2) *Platoon Commander*.

(a) During the assault, the platoon commander should position himself in between and to the rear of the

assaulting sections while the platoon 2IC should move with the follow up section. However, there will be times when a platoon commander will have to get up and physically lead the assault. But this should be an exception rather than a rule.

(b) As a platoon commander has restricted wireless communication with his sections, he makes adequate use of his voice for conveying orders to his section commanders. In the heat of battle nothing is likely to assure a soldier more than hearing clear crisp orders given confidently, loudly and as orders by his commander.

9. **Security**. Security measures in the attack are designed to protect the company by providing sufficient warning and permit reaction time to counter the enemy threat. The company provides for security against enemy air and ground action. The degree of security that a company commander deems necessary depends on the probable need, degree of threat and proximity of adjacent friendly units. In turn, platoon commanders are responsible for the security measures to prevent the enemy from determining the time and place of attack and to cover the company's preparation. During the attack, security is gained by effective employment and positioning of forces, rapid and aggressive movement, control of key terrain, use of line communication, discipline, and exploitation of all surveillance means.

10. **Reorganization, Consolidation and Continuation of the Attack**

a. The company commander plans the required reorganization and consolidation of the objective and includes appropriate instructions in the attack orders. The plan is necessarily tentative, but must be complete. It is changed as required upon seizure of the objective.

b. Consolidation is the organizing and strengthening of a newly captured position to secure it against a counter-attack. The plan for consolidation includes areas to be secured by rifle platoons. Upon the seizure of the objective, initial emphasis is on a hasty defence posture to preclude a successful enemy counter-attack. Platoon commanders immediately despatch patrols forward in the direction of the enemy. Their mission is to obtain information on the enemy and relay it back. How far forward they will go will depend on the terrain and the situation.

c. Reorganization is the restoration of the order of the attacking company. It is accomplished by restoring the chain of command, evacuating and replacement of casualties, replenishing or re-distributing ammunition, moving of command post and other actions necessary to prepare the company for further operations.

d. The company commander includes a plan in the initial company attack orders for a limited attack to exploit success after reaching the objective. This plan is based on a detailed map and aerial photograph study, ground or air reconnaissance and intelligence and orders received from battalion. The plan includes the employment of the rifle platoons.

11. **Actions in Assembly Area**. On arrival, the company/platoon commander must:-

a. Post sentries (Ground/Air).

b. Site and rehearse alarm posts and supervise digging in if this has been ordered.

c. Carry-out a final inspection of the company/platoon to make sure that every man is fully prepared for battle.

d. Supervise or order cleaning of weapons, supervise changing of magazines.

e. Test wireless, if ordered.

f. Rehearse formations and order of march for the move to the FUP.

g. Battle order equipment to be taken off if there is time for the men to rest.

h. Supervise feeding, if time allows.

12. **Movement to the FUP**. The company moves forward from the assembly area to the FUP. The departure from the assembly area is timed so that minimum time is spent in the FUP and the company can cross the start line at the prescribed time. The order of march should facilitate deployment in the FUP into the initial attack formation as appropriate.

13. **Actions in the FUP**. The following actions normally take place:-

a. Assault troops deploy in assault formation.

b. Troops lie down.

c. Objectives are shown to the sections.

d. Weapons are cocked and bayonets fixed.

e. Last minute briefing, if necessary, is done.

f. Normally a company should not stay in FUP for more than five minutes. Troops move out so as to cross the start line at H hour. In the event, the start line is the forward edge of the FUP, then the troops will leave at H hour.

14. **Orders**. The layout of a company and platoon commanders orders are given at Annexes ‘M’ and ‘N’ respectively.

Stage 2 – The Assault

15. **General**

a. Once launched, the attack is conducted aggressively until the objective is seized. Attacking company moves forward as rapidly as possible to keep the enemy off balance and allow him no time to react. Enemy weaknesses are exploited. Leaders at all echelons retain flexibility of action and exhibit initiative by adjusted plans as required by the situation.

b. The company commander position himself at a place, from where he can best influence the action of the company by controlling and efficiently using his supporting fires, his personal leadership and the timely and effective employment of his chain of command to ensure accomplishment of the mission. He makes a continuous appreciation of the situation and is ready with the possible courses of action open to him. He keeps his platoon commander’s and battalion commander informed at all times. When a situation develops which is beyond the capabilities of company commander he requests assistance from the battalion commander.

16. **Advance From the Start Line**

a. During the movement from the FUP to the objective enemy mine field is likely to be encountered. In such an eventuality the forming up place will be located on the home side (own side) of the enemy mine field. If lanes through the enemy mine field have not been cleared before hand by the engineers for the sake of surprise and secrecy, infantry will have to assault through the mine field. There are the following two methods of doing so:-

(1) *Assault through the Minefield in Extended Formation.* Advantages and disadvantages are:-

(a) *Advantages*

i. If jumping mines have been combined by the enemy with other mines, there will be less casualties.

ii. In case automatics and artillery opens up there will be lesser number of casualties.

iii. There will be no requirement for a fresh forming up on the enemy side of mine field and thus there will be no waste of time and loss of control.

(b) *Disadvantages*

i. There are likely to be more casualties by anti-personnel mines.

ii. Casualty evacuation will be extremely difficult and time consuming.

(2) *Assault through the Minefield in Single File.* Advantages and disadvantages are:-

(a) *Advantages*

i. There are likely to be less casualties from the anti-personnel mines.

ii. Better command and control.

(b) *Disadvantages*

i. There will be requirement for a fresh forming up on the enemy side of the mine field.

ii. Since fresh forming up is required which may be under enemy artillery and automatics, lot of valuable time will be lost.

iii. There will be more casualties due to enemy artillery and automatic fire.

iv. Morale will be adversely affected by casualties sustained under the eyes of the troops.

b. The first duty of the assaulting platoon is to maintain the momentum of the attack. The principle of fire and movement must be observed. The attacking platoons must be organized into assault and follow up sections. The assaulting sections should advance close behind the covering fire, provided by tanks, artillery, battalion and company weapons, with the purpose of destroying the enemy before he can recover from its effects. Where covering fire from supporting arms is not available, platoon and company integral weapons should be used to help the forward movement. Having closed up with the objective under the covering fire the attacking platoons assault the objective and go through it at the maximum possible speed.

c. If the platoon is subjected to artillery or mortar fire along the route, it should quickly move through or around it.

d. When enemy resistance is encountered short of the objective, fire is returned immediately by those groups in position to fire, organic and attached supporting weapons engage the enemy and the platoon commander may request additional supporting fire as required. Depending upon the situation, it may be possible to by pass it. By passing is considered only when the enemy cannot be neutralised or eliminated and cannot affect accomplishment of the mission. When such action is taken, it should be reported to the company commander and battalion commander.

e. At the time when own artillery fire lifts to the depth objectives, the assaulting troops should expect a heavy volume of fire of all types from the enemy positions. As a result, own troops are likely to suffer casualties and the advance may be temporarily held up. An assaulting platoon is likely to face one of the following situations:-

(1) *Situation 1*. The attack is successful and the objective is captured.

(2) *Situation 2*. (Figure 26) One of the assaulting sections is held up while the other one is maintaining steady progress. The held up section should under no circumstances become passive. It should try to continue its attack by fire and movement i.e., while one group fires the other group rushes forward to the next fire position and vice versa. In order to

help this section, the platoon commander should use his rocket launcher, the recoilless rifles/MGs/60 mm mortars in the base of fire, smoke and volleys of energa grenades. In case the section cannot advance even by this method it should then make a base of fire and support the attack of the other assault section. The platoon commander may now consider committing his follow up section from a flank to capture the objective of the held up section.

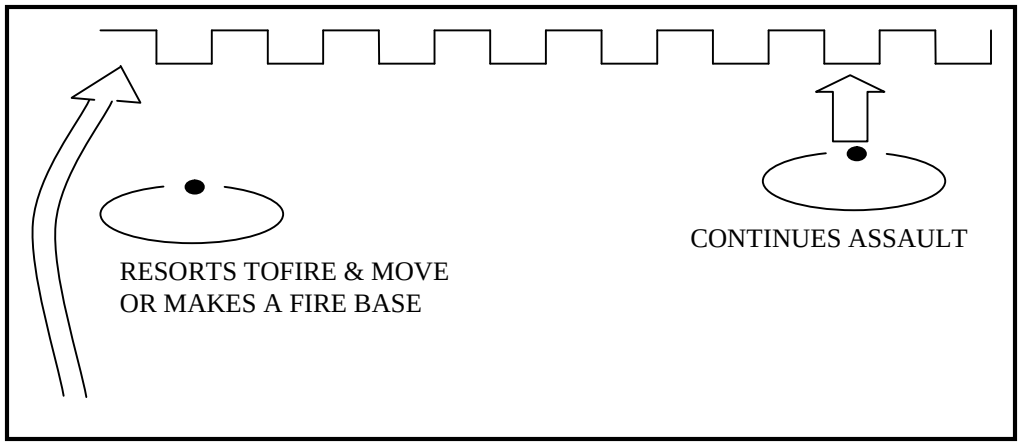


Figure 26 – One of the Assaulting Section Held Up – Situation 1

(3) *Situation 3.* (Figure 27) Both the assaulting sections are held up and the attack is not making any progress. In this case, the held up sections should resort to fire and movement. The platoon commander can take the following steps to restore the forward movement: -

(a) Locate the enemy post which is causing the delay.

(b) Make use of smoke.

(c) Fire the rocket launcher/energa grenades.

(d) Use the fire of the nearest tank.

(e) Report situation to the company commander on the wireless set and ask for artillery fire (in case safety distance of 180 metres is available). Otherwise

request for 60 mm mortar/recoilless rifle/machine-guns fire from the base of fire.

(f) In case none of the above succeeds in restoring the momentum of the attack then he must deploy his leading sections into a base of fire and under cover of smoke lead his follow up section into attack from a flank.

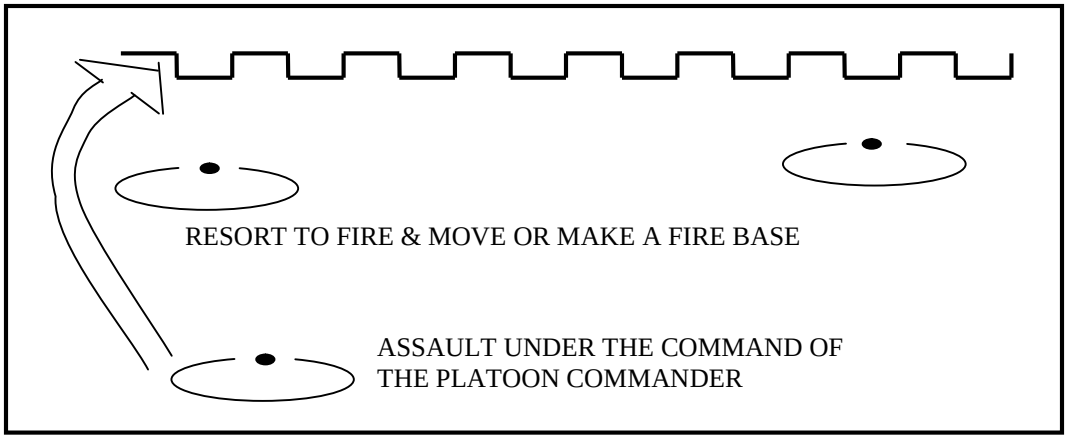


Figure 27 – Both the Assaulting Sections Held Up – Situation 2

f. *Mines and Wire Obstacles*. Following guidelines are suggested for movement through and across these obstacles when encountered during assault:-

(1) *Movement through the Minefield*. In case the enemy succeeds in stopping the attack while own troops are still in the minefield, they will have no option but to resort to fire and move. This has its risks but is certainly the lesser of the two evils.

(2) *Breaching the Wire Obstacles.* In case the enemy erects formidable wire obstacles like the triple concertina fence, the assaulting troops will be unable to rush through it. Also, as wire obstacles are normally sited close to the defensive position, their breaching will perforce have to be done at a very vulnerable stage of the attack. The following drill is suggested as a guideline: -

(a) *Base of Fire*. The assault sections deploy into a base of fire near the wire. The platoon rocket launcher and if possible, the company recoilless rifles/machine-guns/60 mm mortars, must also thicken up this fire.

(b) *Smoke*. The section commanders throw smoke grenades to blind the enemy.

(c) *Bangalore Torpedo*. Under cover of the smoke screen and the fire of the assaulting sections, two men (preferably platoon representatives) from each section, crawl/rush forward and place a bangalore torpedo each inside the wire obstacle. In order not to mask the fire of the leading sections, these pioneers must place the bangalore torpedoes towards the flanks of their respective sections.

(d) *Withdrawal*. They cover their withdrawal by throwing another smoke grenade each.

(e) *Movement through the Wire*. The platoon commander must decide whether the follow up section or the leading assault sections will pass through first. At this stage it may be advisable to move the follow up section first so as not to disturb the base of fire which has been created by the leading sections. Also, the leading sections are likely to have suffered fairly heavy casualties and may not be in a position to assault the enemy trenches. However, this will have to be decided on the spot.

17. **Employment of Follow Up Troops**

a. The company commander commits his follow up troops to ensure accomplishment of the mission. He uses these troops in the subsequent phase of attack to maintain the momentum and exploit the advantage gained by platoons having attacked earlier. The third platoon may be kept behind to follow up and deal with any contingency arising.

b. It would be normally the third platoon in a company which will be kept by the company commander uncommitted earlier to be utilized when the need arises.

c. During the attack he must keep his platoon commanders fully informed so that can plan for anticipated employment.

d. Similarly the company commander must pass all relevant information to the battalion and ask for the fire support or any other assistance required.

Stage 3 – The Destruction

18. **Fighting Through the Objective**

a. There will be no assault line in a daylight attack. Troops carrying out the assault will work their way through the objective and only that section will resort to fire and move which has come under accurate and heavy fire making further progress difficult. The others will continue the assault.

b. Once the enemy Forward Defended Localities (FDLs) has been cleared, the task of fighting through the objective begins or what is commonly called the dog fight, when the infantry destroys the enemy without remorse or regret.

c. Since, at this stage, no artillery support is available, the company will have to fight its way forward by making the best use of integral weapons. If tanks are available, good infantry tank cooperation will be necessary. The company must destroy enemy anti-tank weapons while the tanks provide support with their main guns and machine-guns.

d. A company may either have to fight through an objective frontally or roll it from a flank.

19. **Reorganization**

a. This is the process by which detailed control of the company is regained and the company is firmly established in its new locality, ready to repel any enemy counter-attack. The detailed arrangements must be pre-planned and included in the initial orders. Anti-tank detachments and machine-gun section commanders may have accompanied in the attack so that they can make an early reconnaissance during the initial reorganization. The company commander should lose no time in going to his platoons to coordinate the defensive layout on the objective. When F echelon transport comes, the platoons must be prepared to help the gun crews to get their weapons and ammunition into position. If

captured enemy positions cannot be used, junior leaders must exert all their drive to ensure that their men dig in rapidly.

b. Patrols ordered by the company commander must be sent forward to give local protection during digging and early warning of enemy preparations for counter-attack. At least one light machine-gun group per platoon should be sited forward in the role of sentry.

c. Tanks which have been supporting the attack will undertake anti-tank defence until the infantry anti-tank weapons are sited and coordinated.

d. Casualties will be collected and cleared by stretcher bearers. A systematic search of the battlefield must be made to ensure that no wounded man is overlooked.

e. For reorganization, selected defensive fire (DF) tasks for artillery and mortars to break up enemy counter-attacks will be arranged beforehand. The company commander confirms their suitability during reorganization and if necessary he may suggest additional defensive fire tasks.

20. **Actions After Reorganization**

a. After an attack, there is a natural tendency for a reaction to set in. This is dangerous as the enemy may counter-attack and there is much to be done. The company may be called upon to send out aggressive patrols to help dominate no man's land and more work will be necessary on the defensive positions. On the administrative side, attention will be given to such points as cleaning of weapons, ammunition replenishment, feeding, personal hygiene and all the normal actions taken in defence.

b. The company commander is also responsible for the protection of any tanks, anti-tank guns, Bakhtar Shikans or MGs which are ordered to occupy positions in his company area and for any other assistance they may need without prejudice to his own defence requirements.

Stage 4 – The Exploitation

21. In the even of success, the company commander must use his initiative to seize tactical features ahead of his original objective. Such action may have important results, in that it may upset enemy plans for counter-attack and oblige

him to make detailed reconnaissance to determine the extent of penetration on his front. Exploitation therefore helps our reorganization and hampers enemy interference with it. The company commander must understand from his battalion commander's orders the extent of exploitation allowed to him which will depend upon a be prepared mission type of task given to the company.

Battle Procedure and Battle Drill

22. For details of battle drill and battle procedure of attack, refer to annex 'O'.

SECTION 67 – **EMPLOYMENT OF INTEGRAL/ SUPPORTING WEAPONS/ELEMENTS IN THE ATTACK**

Recoilless Rifles

1. **General**. These weapons must be used boldly. They should be fully incorporated in the fire support plan, engaging known enemy targets from the general vicinity of the start line or from the base of fire (if being established). They must join the company on the objective for reorganization.

2. **Before and During the Attack**

   a. The protection of assembly areas and the forming up place where the threat of attack by armour exists.

   b. To give close support either on timed programme as part of the fire support plan engaging known enemy targets from the general vicinity of the start line and/or from where good cover is available or give fire "on call".

   c. Take on opportunity targets i.e. enemy anti-tank weapons and automatics et cetera.

3. **Reorganization**

   a. As soon as the objective has been captured, the recoilless rifles should take part in the reorganization phase. Because of their mobility, fire power and long-range, they will be of great assistance in beating back enemy tank and infantry counter-attacks. They will also be able to move forward of the objective and support rifle companies in their be prepared missions.

b. Recoilless rifles deploy immediately on arrival on the objective, to cover the most likely tank approaches. Rifle company commanders allocate areas and tasks to recoilless rifles bearing in mind the ground, likely approaches for enemy tanks, the final positions of their platoons and pattern and type of existing obstacles.

c. Immediate deployment of short range anti-tank weapons is of equal importance. They should be sited in relation to the positions planned for recoilless rifles. When allotting tasks to recoilless rifles during the attack, ammunition requirement for the reorganization phase and be-prepared mission should be borne in mind.

d. *Suitable Targets*. Recoilless rifles are essentially direct firing weapons. Ideal targets for them therefore are:-

(1) Tanks moving or stationary.

(2) Known weapon positions.

(3) Pill boxes, bunkers and concrete emplacements.

e. The recoilless rifle can be effectively used in a built-up area to destroy strong points, provided open space for the back blast is available.

Machine-Guns (MGs)

4. It is very awkward to carry the tripod of the MG1A3 in the assault. Also, in case MG fire is desired, it will be very difficult, when caught under enemy fire, to fit the gun onto the tripod and then open fire. Even if this is done, the MG will present a big and lucrative target to the enemy. Therefore, the following three alternatives are suggested:-

a. The MG must be carried in the assault in its LMG role and the tripod be brought up in the F echelon during reorganization.

b. The tripod to be carried by the follow up section. In this case it will be carried along but will not be used till the reorganization stage.

c. All three tripods of the company to be grouped with the base of fire (if it is being established).

5. The main tasks of MGs in the attack are:-

a. Give close and intimate support to assaulting troops by moving with the assault echelons.

b. Protect the flanks of assaulting troops.

c. Assist in speedy reorganization.

d. Take part in be prepared missions.

6. **Employment**. The employment of the MGs during various phases of attack viz before, during and on reorganization are:-

a. *Before Attack*

(1) May take part in the initial fire plan.

(2) Protection of assembly area and forming up place.

(3) Protection of flanks.

b. *During Attack*

(1) May provide protection to flanks of the attacking troops.

(2) Assist the infantry during the actual assault.

c. *Reorganization*

(1) Because of their mobility, accuracy, fire power and long-range, MGs will be of great assistance in beating back enemy counter-attacks.

(2) Their tentative tasks and location would have been included in the company commander's initial orders. These will however, be adjusted on arrival on the objective depending on the enemy, own situation and ground.

(3) Quick coordination should be done and instructions for any adjustments passed on the wireless.

(4) MGs organic to attacking rifle companies must be able to assist the infantry by close and intimate support.

(5) MGs should conform to the fire and movement pattern of their companies thereby providing maximum fire support during the actual assault. This pattern of deployment will also ensure engagements of enemy weapons, which have remained un-disclosed prior to the actual attack.

40 mm GLHMGs

7. Most of the tasks as spelled out for the MG1A3 in the para 6 are also performed by the GLHMGs however owing to their longer range and fire effect;

they can be suitably employed in the fire base also. During reorganization stage these can be employed effectively to beat back enemy counter-attack.

60 mm Mortars

8. These also have accurate direct firing capability and their characteristics must be kept in mind while incorporating them in the fire support plan. They may either be located near the start line or be grouped with the base of fire. In case they are located near the start line then an effort must be made to provide them with wireless communication with the company commander.

Battalion Mortar Platoon

9. Mortars should be so sited as to be able to provide fire on the objective and beyond. During the attack, the Mobile Fire Controller (MFC) locates himself where he can best assist the company commanders. He may remain with the company commander or he may go where he can best observe the advance of the attacking platoons. In either case, he should be able to call mortar fire in an emergency. Throughout the attack, the MFC makes recommendations to the company commander on how supporting fires can be best employed.

10. Mortars generally engage area targets. The most effective fire is observed fire, but during periods of poor visibility, the mobile fire controller should be prepared to deliver fire against dangerous areas from maps.

Bakhtar Shikans

11. Following assistance can be rendered to the attacking troops by the battalion ATGM Platoon:-

a. Protect assembly area and FUP against armour attack.

b. Flank protection of own attacking troops.

c. Can be effectively employed in base of fire/fire base.

d. Holding a firm base for development of own armour operations against enemy.

e. Assist in reorganization phase and help in beating back enemy counter-attack.

f. Assist own troops in exploitation phase.

Assault Pioneer Platoon

12. **Tasks**. The main tasks will be:-

a. Clearance of route for 'F' echelon vehicles.

b. The reconnaissance and marking of enemy minefields.

c. The use of minor assault demolitions (e.g., the pole charge and the bangalore torpedo).

d. The destruction and removal of obstacles.

e. Repair and maintenance of routes.

f. Use and operation of viper lane charge.

13. **Mine Field Clearance and Use of Explosive**. The platoon may be required to provide a number of lanes through a minefield for the assaulting infantry. The platoon may also be employed to destroy wire obstacle by the use of the explosives carried in the platoon. It will also assist in the assault of strong points and defended buildings using the pole charges.

Artillery

14. Artillery caters for counter bombardment, covering fire and defensive fire. In covering fire, timed programme from start line will be the basis of fire support for deliberate attacks. The support should be a mixture of timed concentrations and concentrations "on call". Lifting of artillery fire can be indicated by "all gone" procedure. Unallocated targets will be engaged by Forward Observation Officers (FOO) accompanying the attackers by observation. If there is a good Observation Post from which the forward observation officer can observe the whole company area of attack, he may remain in it until the objective is reached. He will be in touch with the company commander on wireless.

Tanks

15. Supporting tanks link up with the assaulting infantry in the assembly area. The company/platoon should know the capabilities of tanks allotted to them and the commander's name and specially how to recognise and communicate with them. A company in attack may work with a troop of three tanks. Tanks and infantry usually move to the forming up place on separate routes unless the tanks are lifting the infantry.

16. In closed down tanks, the crew cannot see well or hear small arms fire. It is therefore often necessary for enemy posts holding up the platoon to be indicated to the tanks before they can shoot at them.

17. As tanks attract enemy fire, sections should not deploy too near to them. Whether tanks or infantry lead depends on the opposition and the type of country. If tanks reach an unexpected minefield and cannot advance until a gap has been cleared, the infantry must advance alone with the tanks giving them supporting fire as best as they can. Normally in close country, the infantry will lead, in open country, the tanks will lead.

18. Infantry can be carried on tanks but not into battle. They usually dismount before the tanks deploy. Before infantry mounts the tanks, this must be rehearsed in mounting, where to sit, dismount and what to do if the enemy engages the tanks with artillery or small arms fire when they are aboard.

LAT Elements

19. At times, depending upon tactical situation, battalion may be allotted few LAT elements from the LAT battalion to provide assistance in the attack. These elements if allotted may be employed in following roles:-

a. To strengthen the fire base.

b. Join the assaulting troops during reorganization stage to counter enemy armour threat during counter-attacks.

c. Protection of the exposed flanks of attacking troops.

d. Destruction of bunkers and pillboxes.

Engineers

20. If allotted, engineers elements perform following tasks:-

a. To construct, repair and maintain tracks for the assaulting troops.

b. Assist assaulting troops by removing obstacles such as mines and booby traps, which may hinder the progress of attack.

c. Assist the battalion/company commanders in the planning of attack by giving advice on engineer related matters.

d. Destruction of enemy bunkers and other fortified structures.

e. They can also assist and participate in attack as normal infantry if situation so demands.

SECTION 68 – **NIGHT ATTACK**

General

1. The advent of Night Vision Devices (NVDs) has enhanced the night fighting capability manifold. Since it is easy to use these devices when stationary, these are more advantageous to a defender as compared to the attacker. The induction of NVDs has not yet necessitated a complete change in the existing doctrine of night attack. However, there is a need to recognise the impact of these devices on night operations and modify our drills and procedures to neutralise their effect.

Impact of NVDs on Night Attack

2. NVDs enable the enemy to observe, recognise and effectively engage targets, comparatively at longer ranges. Some of the effects of NVDs on night attack will be as under:-

a. Enemy can determine the location, size and nature of our activities at longer ranges, enabling him to take effective counter-measures.

b. Enemy will be able to carry-out coordination and exercise better command and control at night.

c. Enemy will be able to bring observed defensive fire of all types, on the attacker, up to about 1000 metres.

d. Artillery observers, equipped with night observation devices (NODs) will be able to effectively engage targets up to a range of 1200 metres. This makes the movement from assembly area to Forming Up Place (FUP) and the process of forming up vulnerable to observed enemy fire.

e. Availability of NVDs on tanks/anti-tank guided missiles and recoilless rifles will facilitate the defender to neutralize/destroy targets at comparatively longer ranges.

f. Element of surprise has been reduced considerably due to enemy’s new ground surveillance capability of accurate detection of movement and concentration of troops/formations upto a depth of 40 to 60 kilometres.

g. Night Vision capability will enable armour to operate at night in close support role facilitating more rapid and potent counter-attacks.

h. NVDs will assist commanders at all levels to observe and control move of troops during all stages of attack.

3. **Guiding Principles**. Following guiding principles govern the planning and conduct of night operations:-

a. *Simplicity of Plan*. Unless the plan is simple, (free from too many phases and actions involving change of direction) control will be difficult. Objectives must be limited and well defined.

b. *Best Possible Information and Reconnaissance*. Commanders down to section level must have seen their objectives and the ground they are to traverse, during daylight, so that they recognise them in darkness.

c. *Detailed Preparation*. To ensure that troops set on to their objectives and reorganise in darkness, thorough preparations such as fixing easily recognisable reference points and taking compass bearings of objectives from forming up place are essential.

4. **Advantages of NVDs in Attack**

a. Night attack with NVDs would pay greater dividends.

b. Greater frontage and depth can be maintained in the attack.

c. Sustaining the weight/pressure of counter-attack for a longer time.

d. It will be possible for an advancing force to follow up its day light success at night.

e. Speed of operations at night will be faster.

f. Attacker can get accurate fire support from the fire base due to the NVDs fitted on weapons.

5. Following assume added importance for a night attack:-

a. Time for detailed planning and preparation.

b. Detailed information of terrain, enemy and expected weather conditions.

c. Secrecy and surprise.

d. Detailed orders and simple manoeuvre plan.

e. Easily identifiable objectives.

f. Well trained units and subunits.

g. Use of camouflage, concealment and cover to neutralize the effects of enemy NVDs.

h. To be out of the detection range of NVDs, FUP should not be less than 1000-1200 metres from objective.

j. Adverse weather conditions like rain, dust, mist and haze/fog reduce the effectiveness of night vision devices.

k. Phase of the moon would also be an important consideration in determining the time of attack.

l. Time required for logistic support would be an important consideration.

6. **Measures to Neutralise Enemy Surveillance Devices**. It must be clear to us that the surveillance and Night Vision Devices do not turn night into a day, and as such have certain inherent limitations which if exploited intelligently with other counter-measures can offset the advantage gained by the enemy. Some of the measures which can be adopted to neutralise the enemy surveillance devices are:-

a. *Move to Concentration/Assembly Area*. Move to concentration area/assembly area is likely to be picked up by battle-field surveillance radars at a distance of 20 kilometres and by infantry radars once the troops are within 2-3 kilometres from the forward defended localities. Though the artillery battle-field surveillance radar's have an effective range of 20 kilometres but due to built-up areas, jungles/vegetation, dunes and raised ground, it may not be always possible to detect the movement at maximum ranges. However, in order to conceal our movement, following changes/ measures are suggested:-

(1) Movement forward to assembly area to be conducted as late as possible so that minimum possible reaction time is available to the enemy even if this move is detected at maximum ranges.

(2) Deception at the highest possible level should be planned and means provided to accomplish it. Battlefield surveillance radars can not clearly distinguish between, personnel and animals moving in the same area, similarly these can not identify the types of vehicles of the same size. This weakness may also be exploited intelligently to own advantage.

(3) Equipment to jam enemy radars opposite the area of interest during critical periods should be acquired.

(4) As far as possible always resort to night movement.

b. *Selection of Time of Attack.* Bad weather conditions and poor visibility reduce the effectiveness of surveillance devices. Therefore, attack should be launched during adverse weather and poor visibility conditions.

c. *Selection of FUP*. Enemy is in a position to detect the movement with infantry radar and Night Vision Devices and can direct effective fire on the troops, therefore:-

(1) Covered approach to FUP should be selected and used even during night.

(2) As far as possible area providing cover from radar and Night Vision Devices should be selected for FUP.

(3) Artillery fire should be planned to cover the movement and blind the Night Vision Devices by the extensive use of smoke and flares especially in the assault stage of attack.

d. *Optimum Utilization of Terrain*

(1) *Covered Approach*. Where terrain permits and is of broken nature the folds of ground will provide enough cover from ground observation even if the enemy is making use of night vision devices. Besides most of the surveillance radars have only a line of sight capability and it cannot penetrate folds of ground. Therefore, ground which provides cover to attacking troops should be preferred even if the approach is long and difficult.

(2) *Use of Vegetation*. Depending on the area of operations, dense undergrowth, trees and heavy foliage will serve as cover against enemy Night Vision Devices and radars.

e. *Field Craft*. Since night observation has greatly improved, therefore, laying proper emphasis on field craft will be essential even during night.

f. *Maximum Utilization of Own Night Vision Devices***.** For acquisition of target and effective employment of own supporting arms as well as navigation, own Night Vision Devices must be put

to maximum use. This calls for high standard of training of own troops in handling and use of night vision devices.

g. *Jamming*. Jamming the enemy sensors and radars with electronic countermeasures and destroying them with supporting fire.

Types of Night Attack

7. There are two types of night attack:-

a. *Noisy*. The noisy attack uses all available fire support to cover the assault as in day light attack.

b. *Silent*. A silent attack is launched to obtain advantage of surprise and is generally resorted to when relative strength and terrain conditions do not favour a noisy attack. Fire support plan for a silent attack is made as for the noisy attack but the fire is opened only when surprise is lost.

8. In both types of attacks, the aim is to maintain secrecy as long as possible, reducing the time available for enemy reaction. The degree to which secrecy can be maintained is usually the factor which determines whether attack is to be noisy or silent. For a noisy attack, deception measures must be planned to conceal the strength, direction and timings of attack. NVDs environment has reduced the chances of achieving surprise; therefore, all attacks in future may have to be planned as noisy attacks.

9. Factors affecting choice of objectives at night are:-

a. Impaired visual contact and difficulties in keeping control as it is difficult to use NVDs while on the move.

b. Difficulties in the effective applications of unplanned fire support at night.

c. Coordination of fire and movement between sections and platoons during the hours of darkness because every one will not be using NVDs.

d. Difficulties in providing effective fire support from the base of fire. Although NVDs have reduced the difficulty to an extent.

Reasons for Launching Night Attack

10. Briefly, the conditions in which a night attack is given preference over a day attack are:-

a. Achieve a greater degree of surprise.

b. Facilitate the passage of obstacles.

c. Minimize the effects of aimed fire.

d. Avoid enemy fixed line defensive fires.

e. Complete the success of a day attack, which has not reached its final objective.

f. Maintain the momentum of attack.

Reasons for Failure of Night Attacks

11. Failure can often be ascribed to:-

a. Loss of direction.

b. Difficulty in getting up supporting weapons, especially recoilless rifles and Bakhtar Shikans.

c. Loss of commanders and key personnel.

d. Confusion on reaching the objective.

e. Loss of wireless sets, thus making it difficult in calling for fire.

f. Delays resulting in the arrival of troops on objectives in daylight.

g. Faulty positions taken up in darkness during reorganization.

Stages of Night Attack

12. **Preparatory Stage**. This will include the reconnaissance of enemy positions and obstacles, the assembly, forming up and positioning of supporting weapons and the deception plan. The sequence of action in this stage will be as follows:-

a. *Reconnaissance, Planning and Orders*

(1) Company/Platoon/Section commanders and if possible even men should know and memorise their routes and objectives. Compass bearings of objectives should be taken from FUP. All such actions will take place by day. If not properly controlled, it will give away the intention to the enemy. Therefore, a concealed place should be selected for this purpose and small groups guided to it in turn.

(2) Orders should be given early so as to allow enough time for junior leaders to complete their reconnaissance and orders in daylight.

b. *Actions in Assembly Area*. Action in assembly area should include:-

(1) Dispersal and return of troop carrying vehicles (if used in move from the concentration area).

(2) Digging in, if necessary.

(3) Posting of air sentries.

(4) Hot meal, if possible.

(5) Sanitary arrangements.

(6) Arrangements for move to forming up place, order of march, guides et cetera.

(7) Final check up.

c. *Move from Assembly Area to FUP*

(1) Will normally be in single file in the order subunits are to form up in forming up place. To save time, troops may move in file formation.

(2) Platoon/section commanders must keep complete control over their men. Troops should move closed up and without making any noise. Commanders generally move at the head of their units/subunits.

(3) The move may or may not be covered by own artillery fire. It will depend on the situation.

d. *Guiding Arrangements from Assembly Area to FUP*. The route(s) should be clearly marked and if necessary guides should be provided or passive night signs used. In case of armour infantry combined operation, route marking and the marking of FUP is the task of infantry, accompanied by liaison officer and guides of the tank subunit taking part. It can only be done after a thorough reconnaissance. Marking should be done just before the arrival of troops.

e. *Choice of FUP and Start Line (SL) for Night Attack.* Following points must be kept in mind:-

(1) FUP/SL be selected out of effective range of enemy direct firing weapons. Enemy direct firing weapons will be effective at longer ranges due to NVDs; therefore, the distance of FUP in plain is to be kept from 1000 to 1200 metres. In case of broken ground and built-up areas, this distance may be reduced.

(2) Covered places if available should be selected for forming up after ensuring that they are not occupied by enemy.

(3) Route to FUP should be prominent but concealed from enemy observation.

(4) Marking signs should not be displayed so prominently as to give away the location.

(5) An alternative FUP must also be selected and made known to all ranks. Move to alternative FUP should be rehearsed.

(6) Domination of no man's land around FUP must be ensured with the help of holding troops where available.

(7) Depending upon the tactical situation, follow-up companies may be kept at a distance of 500-750 metres behind the forward companies when the latter is in the FUP with minimum marking. However, when the assaulting companies leave the FUP and move towards the objective, the follow-up companies move forward to the SL and remain there till the leading companies have completed their phase 1, or as ordered.

f. *Marking of FUP and SL*. Directional help within an FUP is provided by the centre line and SL itself. Security considerations rule out elaborate use of even passive night sings and tape during infiltration or flanking attacks, involving long night marches through or round the enemy FDLs. It will be necessary to plan an easily recognizable RV, preferably a natural feature, near the selected FUP and SL. In the absence of natural or artificial feature, minimum help of marking material should be taken. From the RV, subunits will be guided by the FUP marking party up to the centre line of FUP and onward to their respective places. Marking in the FUP should be minimum possible. Move in and occupation of FUP will require a very high standard of training.

g. *Protection of FUP*. The ability of the enemy to observe at night will not allow elaborate arrangement for protection of FUP. As an

essential requirement the process of forming up should be completed within minimum possible time to reduce detection and reaction time by the enemy.

h. *Actions in FUP*

(1) Marking by the forming up place marking party before troops arrive.

(2) RV for guides meeting companies and platoons.

(3) Layout down to sections.

(4) Orders on lights, talking and smoking.

(5) Arrangements for casualties.

(6) Navigating party to check up bearings et cetera.

j. *Navigation Party*. May consist of two or three men including one NCO from the intelligence section. This party moves along the centre line on compass bearing. The NCO handles the already set compass while the other two count paces the distance would have already been worked out off the map and converted into paces. The subunit commanders must keep in touch with the navigation party and ensure that the assaulting troops keep peace with it.

13. **The Assault Stage**

a. Troops will cross the start line, which in many cases will be the forward edge of the forming up place and only halt long enough to check the direction and formations.

b. The selection of H hour, when darkness is necessary during the attack, depends on the time required to complete the operation, including the reorganization before daylight.

c. The formation of the assaulting troops will vary according to the ground and degree of darkness, but normally sections should be in extended line and well dispersed to avoid enemy's observation and fire.

d. On reaching within assaulting distance, troops will charge on to the objective and complete its capture as soon as possible.

e. *Fire Support*. A noisy attack will be supported by a timed fire support plan from 'H' hour onwards. Fire support plan for silent attack will be prepared as per noisy attack but brought into effect 'on call' once surprise is lost. In any case FOOs/MFCs will

accompany the assaulting companies. The use of a pre-arranged nickname along with the serial number of task table by attacking commander/FOO/MFC on wireless would be enough to convert a silent attack into noisy one.

f. *Aid to Direction Keeping*

(1) Thorough reconnaissance by day at all levels and maximum use of NVDs/GPS.

(2) Reference points, compass bearings, pacing and firing of tracer bullets by own weapons on the objectives at intervals.

g. *Command and Control.* This is more important at night. Company/Platoon/Section commanders must maintain a tight grip on their men and ensure that contact with flanking sub units is not lost at any time.

h. *Illumination*. Once our attack is detected the enemy may make extensive use of illuminating devices and turn night into day. In case this happens and the enemy succeeds in bringing down accurate small arms fire on own troops, causing heavy casualties, the attacking troops will have no option but to resort to fire and move. However, while resorting to this technique the assault troops should keep the following in mind:-

(1) This should be done only as a last resort when everything else has failed.

(2) Periods of darkness must be made use of to rush forward.

(3) Fire and move should only be done if there is sufficient light (by artificial illumination/moonlight) to prevent own casualties.

(4) Bounds must be kept small so as to help in achieving safety.

14. **Destruction Stage**

a. Once the enemy Forward Defended Localities has been cleared, the task of fighting through the objective begins or what is commonly called the dog fight, when the infantry destroys the enemy without remorse or regret.

b. Since, at this stage, no artillery support is available, the company will have to fight its way forward by making the best use of the

integral weapons. If tanks are available, good infantry tank cooperation will be necessary. The company must destroy enemy anti-tank weapons while the tanks provide support with their main guns and machine-guns.

c. A company may either have to fight through an objective frontally or roll it from a flank.

15. **Reorganization Stage**

a. Reorganization is more difficult at night. Unit/subunit commanders may be uncertain of the exact location of their troops which should be well coordinated positions to meet the counter-attack. The main points which can contribute to successful reorganization are:-

(1) Good communication which should be duplicated.

(2) Every man must know the details of the plan.

(3) Clear and detailed orders must be issued to get supporting weapons and F echelon forward. RVs and guides should be organized. When an obstacle has to be crossed, this movement will be controlled by higher headquarters.

(4) Commanders should ensure that troops take-up position in close vicinity (or a little forward) of their objectives, that patrols are sent ahead, and listening posts and sentries positioned.

(5) The reorganization plan must be known to all ranks before hand, including the arrangements for DFs and the nicknames by which they are to be called. Approximate platoon/section areas and their tasks must be given before hand.

(6) Company/Platoon commander must be well forward on the objective.

(7) Artillery and mortars should be prepared to produce a smoke screen at first light, if required.

(8) Readjustment must be made at first light if necessary.

(9) Recoilless rifles and machine-guns which have come up during darkness must be adequately protected in their platoon localities.

b. As in the deliberate attack by day, assault company commanders will exploit forward with patrols as soon as their objective have been secured. Such patrols must go forward as soon as possible to give warning of enemy preparations for counter-attack. They must be quite clear about the task they are required to perform.

c. The company will reorganise in the area originally laid down by battalion commander, subject to any adjustment dictated by the ground or the location of the enemy trenches.

d. If the enemy's trenches are not to be occupied, it is very important to begin digging without delay. When the platoon commander sites new trenches he must guard against a natural tendency to put them too close together. As a precaution, he should roughly pace the distance between trenches and sections. He must also allot arcs of fire for each trench. If necessary, any positions which are still too close together must be readjusted at first light.

e. It is important to site section sentry positions (usually two men with the light machine-gun) forward of each position and away from the noise of the digging.

f. The importance of immediate digging in on the objective is again emphasized. Every soldier must be made to understand that only by this means can he protect himself from the enemy's fire, which will certainly be brought down on the captured objective. It is also essential to take full advantage of the cover afforded by the remaining hours of darkness.

g. The platoon must be prepared to help machine-gun sections or anti-tank detachments located in its area for the reorganization phase. Their presence will strengthen the platoon position and may prevent it from being overrun.

h. *F Echelon*. It is essential that units work out a simple and effective drill for the coming forward of F echelon vehicle and the distribution of the ammunition and stores carried in it to the various sub units. The following is a suggested drill at company level:-

(1) *RV*. Prior to the attack the Company Havildar Major (CHM) must be given an RV to which he must report with the F echelon, on the receipt of a given signal.

(2) *Guides*. Guides from the forward platoons must report to this RV straight after the attack so as to receive and guide the recoilless rifles into their respective platoons.

(3) *Carrying Parties*. Each platoon must send back a carrying party of one Non Commissioned Officer (NCO) and one or two sepoys per section. In the first trip they should carry ammunition and in the second trip, digging tools. (Till the digging tools are brought up the platoons should make use of their entrenching tools). In case a recoilless rifle is to reorganise with the platoon, its jeep may be used for carrying forward the ammunition and digging tools of the platoon.

(4) *Signal Line Parties*. Each platoon should send the spare signaller with the carrying party to fetch a telephone and to lay the telephone wire. A simple method of doing this is to fix one end of the wire of a signal drum with the terminals of the company signal exchange and to keep unrolling the drum on the way back to own platoon. In this way as soon as the signalman reaches his platoon headquarters, line communication gets established.

(5) *Drill for Unloading the Vehicles*. The Company Havildar Major (CHM) should ensure that the vehicle is parked behind some cover and that there is no bunching around it. He would order the carrying parties to deploy at a distance of approximately 50 metres away from the vehicles. He should then call the signallers first and hand over their requirement to them. After that he should call one platoon carrying party at a time and give them whatever they need.

SECTION 69 – **REDUCTION OF STRONG POINT**

General

1. In an attack on a prepared position, pill boxes, concrete emplacements and fortified dugouts and bunkers will be met. They will usually be sited so as to be mutually supported by fire from each other and from the surrounding fire trenches. They will often be protected by wire and mines. The rifle company will have to deal with such defences, though usually with support from tanks and

sometimes aircraft. On occasions, however, less formidable types of pill boxes will have to be taken by the company with its own weapons using smoke and high explosive (HE) grenades and anti-tank weapons but against a strong position an assaulting team of sappers and infantry is necessary, and the attack will often have to be made at night.

2. Strong points, as the name implies present, peculiar problems to the attacker. Our adversary has constructed a large no of strong points, along the border which help the defender to provide pivot of manoeuvre for strong mobile forces held in the rear or assist in denying specific areas of ground to the attacker.

Aim

3. This section aims at laying down the techniques and drills for reduction of a strong point.

Definitions

4. **Strong Point**. A key point in a defended position, usually strongly fortified and heavily armed with automatic weapons, around which other positions are grouped for its protection. A strong point may also be organized in isolation to guard or deny free use of an approach (line of Communication) to the attacker and hinder his operations. Such a position is well stocked in all respect. It is capable of fighting and sustaining itself for specified period, even when by passed.

5. **Weapon Emplacement**. A prepared position for one or more weapons or pieces of equipment for protection against hostile fire or bombardment and from which they could execute the tasks assigned to them.

6. **Bunker**. Essentially a weapon emplacement, having space for storage of ammunition and rest of crew.

Types of Strong Points

7. Strong point is essentially a perimeter defence, organized in depth, based on a number of mutually supporting localities/positions. Terrain has lot of influence on its layout and staying capability. Main types of strong points which could be encountered in our operational area(s) are as follows:-

a. *Mountain Area*. Major operations in the mountain develop along the main axis and avenues of approach. The defender seeks to deny the use of main axis and supply routes by turning the important

heights and dominating features into strong points. Limited approaches are available to the attacker due to the height and ruggedness of terrain.

(1) *Advantages*

(a) Diminished or no role of armour in attack.

(b) Need for great physical efforts by the attacker.

(c) Artillery and air support are relatively less effective.

(d) Logistic support of attacker becomes cumbersome and slow.

(e) Defender commands the terrain by observed fire.

(2) *Disadvantages*

(a) Unwanted and often larger gaps than necessary have to be accepted.

(b) Mutual support between strong points is often difficult to achieve. Fields of Fire are less than desirable and fire is often plunging.

(c) Positions are usually more defined for enemy observation.

(d) Once lost, counter-attack force faces difficulties.

b. *Plains of Punjab*. As the terrain does not pose any restriction on move except for some area where natural or artificial water obstacles restrict the move, the strong points have lesser credibility. However, in order to deny a particular approach to the attacker for sustained operations, these may be organized on main axis around dominating features, built-up area, and natural/man made obstacles. Strong points in the plains will present a continuous front of impregnable defence, leaving, no or little gap in the defended localities.

(1) *Advantages*

(a) Defender has the advantage of fighting on ground of its own choosing.

(b) Mutual support between strong points and weapon positions are excellent.

(c) Good fields of fire are available.

(d) Ingeniously sited positions can be easily concealed and may not even be detected in photo reconnaissance.

(e) Artillery can be employed creating devastating effects against troops (in the open) even when they have penetrated own defence.

(f) Lost positions can be regained with relative ease by quick counter-attacks by mobile reserve.

(2) *Disadvantages*.

(a) Employment of napalm attacks and gunship stand off close support can be disastrous.

(b) Conventional artillery and air support can be used to maximum effect in support of the attack being a pin point and well recognised target in the area.

(c) Armour can be fully utilized against strong point defence in fire support role.

c. *Desert/Semi Desert Terrain*. Due to lack of communication infrastructure, difficulty in cross-country move and scarcity of water, the major operations are developed along the existing roads and tracks which serve as axes of maintenance. The defender seeks to deny the maintenance facility by organising defence on strong points basis in vital areas based on dominating sand dunes, built-up areas and available drinking water resources. Due to vastness of the area and paucity of men and resources, strong points in the desert may not be mutually supporting, thus presenting gaps for the attacker to manoeuvre.

(1) *Advantages*

(a) Vital communication centers converted into strong points would force the enemy to reduce them at all cost.

(b) Strong points/pivotal defence organized in depth would retard the speed of offensive operation thus provide enough reaction time to the higher commander.

(c) Employment of reserves for counter-attack/counter penetration tasks would be easier within/outside the

strong point premises in desert environments because of sufficient room for manoeuvre available.

(d) Even if by passed by the attacker, it would act as a continuous thorn till it is physically cleared.

(2) *Disadvantages*

(a) A huge amount of defence stores would be required to build strong points in order to have enough sustenance.

(b) Because of the vastness of the terrain, these can easily be by passed, which might have psychological effects of isolation on the troops building the strong points.

Characteristics

8. **Strengths**. A strong point:-

a. Dominates the area all around effectively by observation and fire.

b. Has mutually supporting bunkers within itself and draws/provides mutual support from/to the adjoining localities/strong points.

c. Has mines and wire obstacle around it. Ditches around dominating features may also be created, if the terrain and resources permit.

d. Has its integral depth and when grouped with other strong points, provides or seeks depth to/from others.

e. Subject to availability of troops and resources it may be organized on outer and inner perimeter basis.

f. Is well stocked with ammunition, rations, water, petrol, oil and lubricants and other necessary equipment to cater for requirement of a specified period, under well prepared dugouts.

g. Is capable of staying and fighting for a specified period even when by passed.

h. Serves as a pivot for a counter-attack by a combine arms force.

j. Has ability to readjust position when attacked and to launch local counter-attack to regain the lost locality.

k. Is provided with adequate artillery, tank, air and anti-tank support.

9. **Limitations**. A strong point defence:-

a. Is static in nature and lacks requisite aggressiveness.

b. Consumes bulk of troops in forward location's to organise all round defence, thereby leaving little force to act as reserve. Depth localities may not always be very strongly held.

c. In order to meet multi directional threat, it leads to dissipation of resources in all directions, thereby reducing concentration of fire at any one point.

d. Heavy concentration of troops, weapons, vehicles and dumps in a limited area, present vulnerable targets to heavy artillery fire, aerial bombardment including napalm and chemical attacks. Destruction caused to the dumps will seriously affect the staying power.

e. Can be pinpointed by extensive aerial photography and use of other means for gaining intelligence.

f. Can be deceived about the actual point of attack by launching feints or carrying out fixation.

g. Poses problems in evacuation of causalities, when bypassed or surrounded.

h. Defender in the strong point is likely to be nervous when surrounded/bypassed or cut-off.

Reduction of Strong Point

10. **Problems of the Attacker**. Irrespective of the limitations and weaknesses of a strong point, an attacker has many problems to face in its reduction. Some of the main problems are:-

a. Non assailablity of flanks as the locality has been prepared for an all-round defence compels attacking troops to resort to attack by penetration.

b. The clearance of security Zone/out posts in likely to give away intentions.

c. Enemy has good domination of the area all around by observation and fire. Close reconnaissance of the strong point is difficult.

d. Additional troops are required to surround/isolate the strong point, besides those physically attacking the objective.

e. Passage through the minefield and negotiation of ditch/bund (where existing) seriously hinder the movement to the objective and is likely to cost heavy on men and material.

f. The tanks cannot be used in close support role.

g. Command and control, especially at night will be difficult.

h. Special equipment is required for crossing the ditch, scaling high bund and destruction of the bunkers.

j. Re-adjustment within the strong point and from adjoining localities will have to be checked physically by the troops and through extensive artillery fire.

11. **Planning Considerations**. Attack on a strong point poses numerous problems for the attackers, which can only be overcome with thorough planning and preparation for the operation. This being a complicated operation requires certain special considerations, other than those applicable to the conventional attack. Special considerations that must, receive attention are:-

a. *Detailed Reconnaissance and Planning*. Before conceiving a plan for operation, it is essential that detailed information on the layout of the defence, weapon emplacements, location of reserve and obstacle system et cetera are obtained to determine weaknesses in the defence. For gathering of the information, all available means at unit and higher level must be employed. Plan should then be carefully worked out to decide mainly on the following: -

(1) Direction of attack.

(2) Technique of negotiating obstacle.

(3) Bunker bursting.

(4) Fire support.

(5) Deception.

b. *Psychological Pressure*. In defence based on strong points, the defender becomes over cautious about his security and sensitive to being cut-off. Attackers should therefore ride on the nerves of the defender to put a psychological pressure on him thereby reducing his effectiveness for giving a stiff resistance. This can be obtained by isolating, fixing, destruction of bunkers, and dumps with the preponderance of fire prior to the launching of attack. The defender is likely to give up fighting if the force in strong point is effectively besieged.

c. *Selection of Objective*. Objective selected should be such, which should not only facilitate reduction of the localities in the strong point, but also give least resistance from the enemy direct firing weapons, Often it may not, however be possible. Besides this, it should also permit establishment of effective base of fire for destruction of the bunkers hindering advance of the assault troops.

d. *Speed*. After the initial breach in the defence has been created, the following up elements should be quick to exploit the success and carry the fighting to the depth objectives. All efforts should be made to beat back the counter-attack which in this case will be much quicker. Any delay in exploitation of the initial success may put the operation in jeopardy. Speed in the operation can be attained by carrying out detailed planning, following proper drills and procedures and displaying determination in execution.

e. *Deception*. As the defender in the strong point possesses ability to readjust position, to beat back the attack, and launch counter-attack to regain the lost post, it is vital that he is deceived with regard to the point of attack till the last. Deception plan should therefore, be carefully worked out, by carrying out fixation of the force, firing at places other than being attacked and resorting to other activities. These measures will obviously demand additional troops/ resources which preferably should be other than engaged in actual assault.

12. **Stages of Attack on Strong point**

a. Planning/preparatory stage.

b. Assault/Dog fight stage.

c. Reorganization stage.

13. This operations vastly differs from a normal conventional attack because of the following essential differences:-

a. *Planning/Preparatory Stage*

(1) An attack on a strong point is a complicated affair because planning for the attack would entail integrated planning for crossing a water obstacle/fairly effective, a tank ditch or a twin water channel.

(2) This type of operations would mostly involve minefield breaching/negotiating operation. In addition to the

defensive minefields, the irregular pattern and liberal use of nuisance and protective minefield by the enemy would force the attacker to evolve some new methods/devices for developing/making paths for the assault troops. The tactics/techniques of every time running through minefield will take heavy toll on assault troops well before they close with their objective.

(3) As opposed to normal attacks in this situation, the attacker has to force his way through a series of wire obstacles and then tackle fortified bunkers/pill boxes.

(4) Each strong point enjoys effective mutual support from the neighbouring strong point. In addition, it has the capability of fighting even if by passed/isolated.

(5) There being very little, if any, gap between various locations, enemy defence do not exposes them to out flanking manoeuvre of the attacker.

(6) Such attacks require very elaborate and detailed information about the enemy dispositions, preparations have to be more thorough and actual plans, assault technique and drills will have to be rehearsed many times to attain perfection.

(7) Additional outside fire support such as air, armour, artillery, and Light Anti-Tank is required for reduction of strong point. Actual reduction of bunkers/strong point being the task of infantry/engineer teams calls for very intimate engineer support.

(8) The existing authorisation generally is inadequate for such a task. A company/platoon therefore, has to be beefed up by additional manpower, weapons, stores/equipment et cetera.

b. *Assault/Dog Fight Stage*

(1) In the assault/dog fight stage, it is essentially platoon/ section commander’s battle; therefore, it calls for very high standard of infantry leaders training.

(2) This operation has peculiar problems of command, control and coordination as in a raid operation.

c. *Reorganization Stage*

(1) Such type of operations would require greater coordination at subunit level for speedy reinforcement/replacement so as to be able to face the enemy's attack.

(2) The troops would be subjected to heavy pounding by enemy air/artillery, which would impede the process of reorganization.

14. *Passage through the Minefield*. Present doctrine recommends passage through the minefield in extended line or file formation, with emphasis on later, depending on the density of minefield and enemy frontages. Authorisation of pneumatic mine shoes for complete battalion, modern devices for creating lanes through minefield, blowing gaps in wire obstacle, negotiating ditch/bund and reduction of any bunkers serve/provide us with multiple options. Some viable options are being discussed in succeeding paragraphs.

a. *Option 1-Passage Through Minefield in File Formation Using Mine Shoes*

(1) In this option, the battalion forms up with two companies up, each company forming up in file formation, with two platoons up. Third platoon will form the follow up platoon for its parent company, complete strength of the leading two companies will wear mine shoes. Navigation cum Obstacle Group will lead each single file.

(2) At H hour, the Navigation cum Obstacle Groups will cross start line and lead their respective platoons in single file on the previously worked out bearings. On arrival at the far side of minefield, leading section of the platoon takes position and starts engaging enemy bunkers.

(3) The follow up platoons will stay on the home side of minefield till the leading platoons have negotiated the ditch. It is important to prevent the follow up platoons from getting involved into the battle of leading wave, prematurely. When the leading wave has crossed the ditch and has started bunker clearance, the follow up platoons move through the minefield with mine shoes on and reach far end of the minefield.

(4) When all the three platoons of the leading companies have crossed the ditch, the four lane clearing parties (1x Plastic

Foam Sprayer and 1x STEP SAFE each) initiate the STEP SAFE at previously worked out bearings to clear lanes of 300 metres length. After having done this the lane clearing parties start moving on these lanes. This process will be repeated every 300 metres till the other end of the minefield is reached. Portable heavy weapons of the companies which have not been employed in base of fire move on these lanes keeping a distance of approximately 50 metres from the lane clearing parties. Elements of battalion headquarters can use one of the lanes.

(5) At this moment, the follow up companies await orders in the forming up place. These companies will not wear the mine shoes. As soon as the battalion commander assesses that the forward companies have completed their fighting on the objective, he asks the follow up companies to move up through the safe lanes and stay behind the water obstacle, ready to start Phase 2 operation.

(6) Move of men and material will continue through these lanes till such time the engineers have breached minefield and created safe lanes for vehicles and heavy equipment.

b. *Option 2-Assault Through Minefield in Extended Line*

(1) In this option, the battalion forms up with two companies up and each company with two platoons up (Total 12 sections up) in an extended line. The leading wave personnel will wear the mine shoes. The leading wave will have navigation party at platoon level.

(2) The leading wave will assault through the minefield in extended line at an approximate frontage of 500-600 metres.

(3) Follow up platoons of the leading companies stay on the home side of the minefield till the leading wave has crossed wire obstacle, The follow up companies and portable heavy weapons not being used in base of fire will also use these lanes.

c. *Option 3-Passage Through Minefield in File Formation without Mine Shoes*

(1) In this option each leading company will form up in file formation with two platoons up and the third platoon will

form the follow up platoon. At battalion level, a total of two files. Navigation cum-Obstacle group will lead each single file.

(2) Follow up platoons, companies and portable heavy weapons will use these very lanes later on to cross the minefield.

15. **Groups for Reduction of Strong point.**

a. Breaching Group.

b. Fire support group.

c. Assault Group

d. Follow up group.

16. **Breaching Group**. Comprises four to six personnel from assault pioneers or engineers to breach the wire obstacle for the assault group. They perform their tasks under cover of supporting fires using bangalore torpedoes and wire cutters to create the gap.

a. *Personnel/Equipment*

(1) *Commander*. NCO (Engineer/Pioneer) with map, compass and wire cutter.

(2) *Pioneers*. 3/5 with prodders, wire cutters, two bangalore torpedoes and 200 metres white tape for marking gap at night.

b. *Tasks*

(1) In the preparatory stage, reconnaissance of the selected area for the gaps. Prepare demolitions and take compass bearing to the selected sites.

(2) Move up to the selected site under cover of supporting fires, and through the protective minefield, lay the tape to indicate the gap for subsequent use.

(3) Make more than one gap in the wire by using wire cutters and bangalore torpedoes.

17. *Fire Support Group*. This is formed by personnel from platoon/ company/battalion support elements as under:-

a. *Personnel/Equipment*

(1) Commander - Platoon 2IC/Section Commander (number 3 section).

(2) 1xLMGs - 2 (from number 3 section).

(3) 1xMG - 2

(4) 2x106/75 mm Recoilless Rifles – 8

(5) 2x60 mm Mortars _ 4

(6) Detachment ATGM (if available) _ 3

b. *Tasks*

(1) Carry-out reconnaissance and site all weapons of the group.

(2) Take compass bearing to loop holes of bunker to be engaged by each weapon.

(3) Brief and rehearse individual crews of supporting weapons.

(4) At the time of attack ensure destruction or neutralisation of assigned bunkers.

(5) Assist the assault group by firing on position outside the bunkers, which threaten the manoeuvre.

(6) On capture of location, re-deploy to strengthen the position and counter enemy reaction.

18. **Assault Group**. The assault group is divided into two sub-groups i.e. Alfa and Bravo, each having a section strength. The two sub-groups are further sub divided into two teams each. The method adopted is to capture and destroy one/two mutually supporting bunkers, with one team assigned to each bunker. This method of capture is easy and subsequent bunkers are captured by leapfrog from bunker to bunker, preferably from the blind side. On completion of its mission, this group supports attack on depth bunkers by the other group. Details of equipment, tasks and grouping are as following: -

a. ALFA GROUP (Number 1 Section)

(1) Personnel/Equipment

	Personnel/ Equipment	*Number 1 Team*	*Number 2 Team*
(a)	Commander	Section Commander	Section 2IC

	Personnel/ Equipment	*Number 1 Team*	*Number 2 Team*
(b)	2xLMG	2 (1xLMG)	2(1xLMG)
(c)	Riflemen	2	2
(d)	Rocket Launcher numbers	2	2
(e)	Each individual carries two grenades and one anti-personnel mine		

(2) *Tasks*

(a) Carry-out reconnaissance and rehearse the operation in their particular task

(b) Fire on area adjacent to the bunkers, which may affect their capture. This task would be performed by one of the two teams.

(c) Initially 1 and 2 teams capture and destroy two forward most bunkers using grenades, demolitions and Rocket Launcher. After capture of near two bunkers in the depth by BRAVO Group, team number 1 of this group may be assigned to capture the bunkers further in depth.

(d) Assault and kill enemy in bunkers or in communication trenches in the vicinity.

(e) Support assault of BRAVO Group.

(f) Nominate riflemen as reserve for special weapons and rehearse them in their use. These personnel will take over the personal weapons in case the personnel of these weapons become causality.

(g) Prevent enemy reinforcement.

b. Bravo Group (Number 2 Section)

(1) Personnel/Equipment

	Personnel/ Equipment	*Number 3 Team*	*Number 4 Team*
(a)	Commander	Section Commander	Section 2IC
(b)	2xLMG	2 (1xLMG)	2(1xLMG)
(c)	Riflemen	2	1
(d)	Flame Thrower	2	
(e)	Each individual carries two grenades and one anti-personnel mine		

NOTE:- One rifleman from number 4 term has been detailed to carry/use the flame thrower. Additional LMGs will be taken from number 3 section.

(2) *Tasks.*

(a) Capture line of bunkers immediately in the depth of ALFA Group objective/by employing number 3 and 4 teams alternatively.

(b) Support ALFA Group by fire.

(c) All other tasks of ALFA Group.

19. **Follow Up Group**. (Cut-off and Reserve Group). This group is also divided into two teams (number 5 and 6). The organisation, task personnel/ equipment are as following:-

a. Personnel/Equipment

	Personnel/ Equipment	*Number 5 Team*	*Number 6 Team*
(1)	Platoon Commander's Wireless Runner	-	

	Personnel/ Equipment	*Number 5 Team*	*Number 6 Team*
(2)	Team Composition	Section Commander Platoon 2IC	Section 2IC
(3)	1xLMG	2	
(4)	Riflemen	2	2

b. *Tasks*

(1) Follow up success of assault group.

(2) Augment or reinforce any one of the groups.

(3) Destroy any additional bunkers in depth or any undisclosed bunkers.

(4) Act as reserve for unforeseen eventualities.

(5) Form block, to prevent enemy reinforcement from the rear.

(6) With the help of pioneers destroy bunkers, which are not required for subsequent use.

NOTES:- 1. 1xLMG of the support group.

2. This group may be augmented by weapons from the support group, if and when necessary

Conclusion

20. Attacking a defence based on strong point is a very difficult operation. It needs meticulous planning; dedicated training at all levels and extremely strong will to do it. While making a strong point, enemy shall endeavour to create as many hurdles and obstacle as possible to retard the attacker's movement and frustrate his will to fight. The attacker has to devise most unconventional ways and means to overcome enemy's strong point avoiding major losses to troops.

SECTION 70 – ASSAULT ACROSS WATER OBSTACLE WITH HIGH FAR BANK

General

1. This section should be read in conjunction with "Infantry Battalion in Battle 2000". The characteristics of enemy defence on a major water obstacle have been covered in the above pamphlet. The relevant aspects of the attack applicable to company/platoon/section commanders are discussed in this section.

Problems of Attacker

2. The company and platoon would be confronted with consecutive obstacles in this operation before it finally captures the objective at far bank. These obstacles are as follows:-

a. Enemy mine-field of generally 600 metres to 900 metres depth depending upon the duration/length of night in that particular period of the year.

b. Wire obstacles next to the water line at the far end of minefield.

c. The wet gap with varying width.

d. Scaling of the high far bank, clearing and destruction of the bunkers.

Planning

3. Some of the important aspects of planning relevant to understanding the techniques and drills to be followed by platoons and sections are discussed below:-

a. *General*. As these operations would be conducted with proper planning and coordination at battalion and higher level, it is understood that all the arrangements regarding the fire support, equipment, stores and administration will be already made and the necessary coordination carried out. The company and platoon commanders should be fully in picture regarding the time of attack and development of bridgehead.

b. *Assault Frontages*. The assault frontages will be slightly more as compared to normal attack to cater for the easy swimming across the water obstacle. However, it will be ensured that the attacking

troops land on the far bank in proper waves and the command and control is not lost.

c. *Gathering of Information*. After receipt of the orders, the company and platoon commanders must gather all information regarding the route to be followed through the minefield, enemy positions, deployment and the place where they would be breaching the wire obstacle and carrying out crossing. All available information in the form of patrol reports, maps, air photos and civilian reports should be studied.

d. *Preparation and Rehearsals*. As this operation is to be conducted in the hours of darkness, and with meticulous coordination, the techniques enumerated later in this section must be thoroughly rehearsed and all preparations done to the last detail. The earmarked troops for navigation and blowing of the wire obstacles and destruction of the bunkers must be well rehearsed on the model and if possible on the ground. The carrying of the equipment, stores and ammunition would need earmarking of personnel in the platoons and sections and all eventualities catered for. It should be kept in mind that each section is capable of assaulting one enemy bunker and a platoon is able to clear a group of three bunkers which would normally be one section position of the enemy. Similarly, a company is capable of capturing an en platoon spread over 275 metres of front. These should be taken as guide line for planning but may not be rigidly followed.

e. *Fire Support*. Apart from the fire support provided by the battalion, tanks and artillery, the company machine-guns should accompany the assault to provide intimate support. Maximum utilization of Recoilless rifles, MGs including GLHMG and 12.7 mm MG, Rocket Launchers and energa grenades be made by carrying these in the assault.

Conduct

4. The conduct of operation by company and platoons will be as follows:-

a. *Forming Up by Company*. The company is going to deploy in the same formation in which they are going to cross the minefield. Possible layout of a company in forming up place is given in annexes with minor variation in grouping.

b. *Crossing of Minefield*. Crossing of minefield in single file is preferred over extended line. The main reasons for adopting file formation are the lesser casualties due to mines and the small arms fire and the command and control. File formation will have following additional advantages:-

(1) Anti-mine shoes (if worn during mine crossing) are a protection against blast type anti-personnel mines and not against the trip wire mechanism which could be linked with jumping/fragmentation type of mines. In file formation, leading man in each single file would move with feeler stick and thus, the rest of the persons would not be exposed to the trip wire.

(2) Comparatively speaking the command and control problems for the attack in the file formation are far less acute than in the extended line while passing through the minefield, particularly at a stage when under fire from positions covering the ditch and the minefield.

(3) File formation, especially at section level, facilitates the move through various obstacles including minefields in conformity with the section's ultimate objective across the ditch i.e, one enemy bunker. The section is divided into three specialized groups primarily for tackling one bunker each and move in file formation will facilitate this.

(4) Though minefield does not create a funnel effect on move of the attacker but the wire obstacle and the ditches create the funneling effect. If the section or the platoon moves through the minefield in extended line then they will be required to break the extended line formation and converge on to gaps created in the wire obstacle for the section/platoon to funnel through. Generally, the same situation will be repeated while negotiating the ditch. Use of file formation at section level in the minefield will thus increase the speed of the operation.

c. *Assault Formation and Frontages*. Considering the basic requirement of providing depth in attack and reaction capability of the enemy, a company should normally assault with two platoons up, unless the situation demands otherwise. At platoon level, however, keeping in view the deployment pattern of the enemy on the far bank, all three sections should attack simultaneously so that

they concurrently tackle all the three bunkers of one enemy section. The third platoon of the company could then pass through the leading platoon and expand sideways. Since a frontage of approximately 275 to 300 metres is normally assigned to enemy platoon in defence along such an obstacle with inter platoon gap of about 150 metres, it can be inferred that a leading assault company may be given an area of about 300 metres on the bund to clear.

d. *Crossing Over Wire Obstacle.* Blowing the wire obstacle is the responsibility of assault pioneer and the navigation group. The assault pioneer and navigation group reaches the wire obstacle and blows a gap through which the rest of the sections pass. After performing this task they dash forward to the near bank and provide fire support to the assaulting sections.

e. *Crossing the Water Obstacle.* The moment gap is created in the wire obstacle, the section rushes through that and gets into the water. In case of flowing water, entry is made from some distance upstream to cater for the drift, which is roughly twice the width of the water obstacle; each individual has a life jacket to enhance the swimming capability. Persons in each section swim with their commanders or 2ICs, as per their grouping. Provision of intimate fire support is very important at this stage. As the base of fire and the artillery would have been lifted by now. The provision of rocket launcher, in each assault section is one of the solutions to this problem. Rocket Launcher along with assault pioneer and navigation group will be engaging the enemy loop holes or the bunkers at this moment so as to keep enemy's head down.

f. *Scaling of High far Bank.* Far bank is scaled with the help of bamboo pole ladders, which are installed against the bank by the assault troops. The two groups of the section i.e., assault group and cut-off group climb the far bank under their respective commanders and have sufficient expedient with them to do so. High far bank is always steep and difficult to climb. Attacking troops must identify the dead areas and try to scale the bund at the top most speed under the covering fire of navigation group of the third section/platoon located on the home bank and the base of fire. To climb the slippery patches due to wet boots and uniform, the entrenching tools may be used to dig and create the foothold. Grapnels if used will be a great help in scaling the high far bank. Once the two groups have climbed up, the navigation group joins their respective sections.

g. *Bunker Clearance*. Each platoon has to deal with three enemy bunkers .e, one per section. Out of these three bunkers one of them would be RCC bunker having a steel door et cetera. The important aspect to keep in mind is that all the bunkers are cleared from the rear to avoid cross fire. Clearing group immediately gets behind the bunker. It clears the bunker by approaching it from the rear with the help of explosive and energa grenades. Rocket launcher also joins this group to either blow off the steel door or immediately cater for the tank threat from the rear. Assault group after crossing the wire obstacle and ditch, rushes up the slope of the bund, firing at the loop holes of enemy bunkers and lobbing smoke grenades into them. The group then moves beyond bund to shoot any escaping enemy or to stop any enemy reaction. Once the door has been blown, hand grenades are again lobbed in the bunker, followed by the suppressive fire so that any remaining defender is neutralized.

h. *Reorganization*. In order to forestall enemy's counter-attack, the reorganization on the objective must be quick and effective. The most expected direction of a counter-attack is either from a front or a flank. The company and platoon commanders should ensure coverage of these directions. The captured portion of the bund must be physically occupied but only by an adequate number of troops and the remaining troops should occupy positions ahead of the obstacle.

j. *Replenishment of Ammunition/Stores*. Keeping in view the number of obstacles which have to be crossed, the assaulting troops only carry the pouch ammunition. Due to the high rate of ammunition expenditure, the leading sections and platoons should be quickly replenished with ammunition/stores. Though this will be done under battalion arrangements, but the sections and platoons must send their guides to promptly collect it on the far bank at a predesignated place.

Conclusion

5. Physical fitness, thorough training, rehearsals and motivation of troops will ensure the success of such operation. Junior leaders must have excellent command and control over their sub units to take them right up to the final objective.

SECTION 71 – **ASSAULT CROSSING OF A WATER OBSTACLE**

General

1. This section should be read in conjunction with Section 6 of pamphlet "Operations of War, Volume II, Offensive Operations 2001". The crossing of a water obstacle, if strongly opposed, will be a difficult and complicated operation; it will seldom be carried out by a battalion at its own. The subject, therefore, is only dealt with in outline in this section.

2. **Problems Facing the Attacker and Important Points**. The problems are as follows:-

a. *Clearance and Securing of Home Bank.* This necessitates following actions:-

(1) Reconnaissance.

(2) Planning.

(3) Clearance and Consolidation.

b. *Reconnaissance for Assault Across*. The requirement would be the reconnaissance for: -

(1) Approaches to the obstacles – suitable for vehicles and foot infantry – to include cover.

(2) Various areas required for multifarious purposes.

(3) Objective (s), terrain, enemy deployment, weapons and strength.

(4) Suitable crossing sites.

(5) Approaches coming in/going out of objective area.

c. *Security*. The maintenance of secrecy will be a big problem as assault across water obstacle is a major deliberate operation requiring detailed reconnaissance planning and preparation. The security of the operation will be achieved by the secrecy and deception. This should be worked out, in detail.

d. *Assault Across and Consolidation*. This requires very meticulous planning and preparations, which are dealt with in detail in subsequent paras.

e. *Direction Keeping*. It is a serious problem at night while crossing a water obstacle, particularly in poor visibility due to

darkness or foliage, when the other bank cannot be seen, when there are no prominent landmarks, when the current is strong and crossing has to be diagonal and when it is silent crossing and assistance of fire cannot be utilised to maintain direction. In such cases, it is, suggested that initially a group of path finders precede the first wave, who should place dim coloured lights facing the start line, to be switched on and off after a specific time after H hour. This will be only possible if other side is not held by the enemy. If other side is held, the path finders should proceed with the first wave for markings.

f. The most important points which will facilitate the success of assault across water obstacle are:-

(1) Secrecy.

(2) Deception.

(3) Speed.

Basic Requirements

3. Basic principles of the attack are:-

a. Clear orders and briefing based on sound planning and intelligence.

b. Simplicity

c. Surprise.

d. Speed and determination in execution.

e. Rapid exploitation of success.

Stages of an Assault Crossing

4. The operation will normally be divided into nine stages, some of which will overlap each other:-

a. Reconnaissance.

b. Appreciation and plan.

c. Detailed planning and orders.

d. Assembly and preparation of troops and crossing equipment.

e. Advance to the water way.

f. The assault.

g. Establishment of bridgeheads.

h. Construction of bridges and crossing of main body of the force (reserve battalions or brigades).

Reconnaissance and Planning

5. Information is needed on:-

a. Enemy defences, including mines on both sides of the water obstacle.

b. Details about the obstacle:-

(1) Width, depth, current and type of bed.

(2) Banks, height and slope.

(3) Islands, fords, weirs and sand banks.

c. Subsidiary obstacles, ditches or fences.

d. Approaches to and exits from the water for wheels and tracks.

e. Suitable crossing places for boats and rafts.

6. The sources of information are no different in this operation form any other; patrols, Observation Posts, maps, air photographs and prisoners of war et cetera. the engineers can normally supply fairly detailed information about the obstacles, which can be checked or enlarged on by patrols. Joint patrols, infantry and tanks or infantry and engineers, will frequently be necessary. Care must be taken not to disclose the intended crossing places and reconnaissance of the obstacle may be controlled by brigade or division and patrol activity kept along the whole front as part of a deception plan.

Advance to the Waterway

7. Areas to be used by assault troops are given in Figure 28:-

a. *Concentration Area*. Out of enemy artillery range, if possible where troops can rehearse for coming operations.

b. *Brigade Assembly Area*. One assembly area per brigade is usually required. Here battalion vehicles, other than those crossing under brigade arrangements, after the initial bridgehead has been secured, are assembled.

c. *Battalion Assembly Area*. One will be needed for each assault battalion, not far from the obstacle. Here the battalion makes final preparations before moving to the forming up place (s).

d. *Boat Off – Loading Point*. If the assembly area is some distance away from the obstacle and it is possible to take transport forward of it before the assault without prejudicing security, a boat off – loading point is selected to which the vehicles carrying the assault boats are driven before being unloaded.

e. *Checkpoint*. This will ensure that companies move forward in the correct order at the right time.

f. *FUP*. Where assault troops get into their assault formations; in this case, the formations in which they enter the water. When the home bank of the obstacle is secure, the water line is the start line.

g. *Crossing Places*. When troops put their boats in the water.

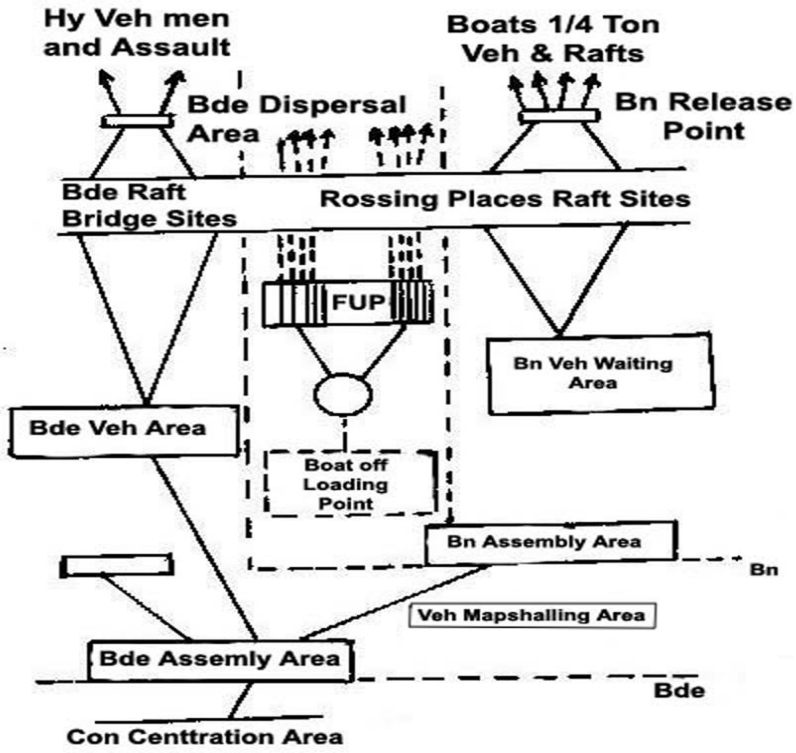


Figure 28 – Assault Crossing of a Water Obstacle

Vehicles

8. Vehicles of assaulting battalions are collected in vehicle marshalling area near battalion or brigade assembly areas and called forward in order of priority to battalion or brigade rafts or bridges, according to their classification. The organization required includes:-

a. *Vehicle Waiting Areas.* To ensure an even flow of vehicles over the bridges and ferries, it may be necessary to organize vehicle waiting areas between assembly areas and the water obstacle. These areas should not contain more than a small group of vehicles at any one time lest they present good targets for enemy artillery fire and aircraft.

b. *Release Points.* These will be organized on the far bank. Each assaulting battalion will normally require one, where vehicles will be met by company guides and taken to their respective reorganization areas as in a normal attack.

c. *Brigade Forward Release Areas*. Areas at which vehicle crossing under brigade arrangements can be met by unit guides. Normally one of these areas is required per brigade.

Duties of the Bank Group

9. The smooth and orderly progress of the crossing depends largely upon the efficient direction and control of movement of troops and vehicles between the assembly areas and release points on the far bank. These duties are undertaken either entirely or in part by a bank group, specially formed for the operations. Usually a bank group is formed by a separate unit. An assaulting battalion may have to supply some of the men needed to help in assault crossing and may sometimes have to form its own bank group.

10. The main duties of the bank group in an operation where a battalion does not provide its own bank group are:-

a. Directing and controlling movement of personnel and vehicles into and forward from assembly areas.

b. Close supervision of the crossing, which entails the upkeep of an even flow of traffic across the obstacle by making full use of all available bridges, rafts and crossing place.

c. Protection of routes to crossing place.

d. Off-loading assault equipment and helping to manhandle it forward to the forming up places.

e. Guiding troops to the crossing places.

f. Organizing each crossing place including protection.

g. Operating the boats, rafts and outboard motors; ferrymen are provided for assault companies, at least.

h. Organizing and staffing medical posts, recovery posts and prisoners of war cages needed near the obstacle.

j. Organizing and staffing the assembly areas and other areas.

k. Keeping the next, higher commander informed of the progress of the build-up.

The Assault and Establishment of Bridgehead

11. The plan cannot succeed unless all ranks are absolutely determined to get over the obstacle and to secure their allotted objectives. All ranks must be carefully briefed on maps, models and air photographs so that they can act intelligently if the battle does not go according to plan.

Fighting and Waves

12. Fight is a term normally used on brigade level and above. Within the battalion, normal attack terms are used. A and B companies cross at H hour, C company at H plus 20, et cetera. companies, under certain circumstances, may have to cross in two or more waves, or by shuttle service (following up companies).

13. **Objective**

a. Must be limited, particularly at night.

b. Primary objectives are those positions form which the enemy put aimed small arms fire onto the crossing places.

c. Secondly, objectives are those positions which overlook the obstacle and from which the enemy can direct artillery fire onto the crossing places.

H Hour

14. The time at which the first wave of assault boats enter the water.

Fire Support

15. This is for a normal attack, but the following points are emphasised:-

a. If silent crossing is attempted, fire support must be on call.

b. Flat trajectory weapons are useful for:-

(1) Flank protection.

(2) Shooting in assault troops by firing from a salient to the flank.

(3) For close protection of the crossing places.

(4) Neutralizing enemy defences, particularly close to the far bank.

(5) Tracer fire from the machine-gun is a useful aid to direction keeping.

c. Full use must be made of the fire power of troops already holing the home bank. In particular, their recoilless rifles can provide flank protection to the initial landings where required.

d. The fire of the tank guns and machine-guns can be utilized to provide fire support. This necessitates reconnaissance, selection of fire positions and move of tanks to the fire positions keeping in consideration the requirement of secrecy and silent attack.

Mopping Up

16. This must be thoroughly carried out by forward troops and not left to the follow up companies. Bypassed enemy will continue to fight and may seriously impede both follow up companies and the rear to the troops who have gone through them. The delay imposed by mopping up in front must be accepted.

Anti-Tank Defence

17. It may be sometime, before recoilless rifles and tanks can reach forward companies in the bridgehead. Platoon and section anti-tank weapons are, therefore, needed. Anti-tank defence calls for guns to be put across the obstacle as early as possible. Arrangements may be made to carry certain amount of anti-tank mines by the assaulting troops, which should be laid immediately on arrival for the protection of the bridgehead.

‘F’ Echelon Transport

18. Priority for crossing for most operations:-

a. Recoilless rifles.

b. Ammunition.

c. Forward observation officer’s vehicle as applicable.

d. Commander, medical and signal vehicles.

Administration

19. After a battalion has gained its bridgehead, it may be temporarily cut-off by enemy counter-attack and shelling of the bridges, therefore:-

a. An adequate reserve of all requirements, particularly ammunition must be put across the obstacle with the assaulting troops, on the man or in ‘F’ echelon.

b. Arrangements must be made to hold wounded and prisoners of war for a time until they can be evacuated.

c. Men must carry enough food and water to prevent it from becoming a high priority requirement for next 24 hours.

CHAPTER 14
DEFENCE

SECTION 72 – **INTRODUCTION**

General

1. This chapter provides doctrine for the employment of the Rifle Company in defensive operations. Conditions of terrain and climate as well as intensities of warfare may dictate modification of techniques and procedures for defence; however, the doctrine for defence remains the same.

2. The aim of defence is to prevent, resist, repulse or destroy an enemy attack, with a view to creating a situation, favourable for taking offensive action against enemy.

3. **Purpose**

a. Causing destruction and attrition on to the enemy force(s) thereby, degrading its capability to continue the offensive/attack.

b. Holding specific piece of terrain and denying gain or entry into an area.

c. Conditioning the enemy for an attack by own forces. This may be done by weakening, delaying, dividing, canalising or massing the enemy in an area suitable for own attack.

d. Economising in one area in order to concentrate superior forces for an offensive action elsewhere. This aim is usually to be achieved within specified dimensions of time and space.

4. **Definitions**

a. *Defended Area*. An area of operational responsibility containing a group of mutually supporting localities organized in depth. Any enemy, who penetrates into it or passes round the flank, will be engaged by the fire of all weapons that can be brought to bear on him.

b. *Defended Locality*. An area of ground organized for all round defence. The general line of the foremost localities will be known as the line of the Forward Defended Localities (FDLs). Enemy penetrations will not be accepted into a locality. Enemy that does penetrate will be attacked by fire, by local counter-attack, or by a deliberate counter-attack as ordered by the higher commander.

c. *Defended Post*. A position held by a small subunit such as an infantry section or platoon. Posts are grouped together in mutual support to form a defended locality.

d. *Original Positions*. The fire trenches of sections and other weapons selected within given platoon area; dug and prepared to engage the enemy in the first instance.

e. *Alternative Positions*. One or two positions to which a weapon team or a section moves within the platoon area to carry-out another task or in some cases the same task. It will normally be at-least 200-300 metres away from the original positions so that enemy artillery fire directed on the original positions does not neutralise the alternative positions.

f. *Counter Penetration.* The action taken by a defender to halt penetration of his defence by the enemy and to destroy by fire enemy forces which have penetrated between his defended localities.

g. *Counter-Attack.* Attack by a part or all of a defending force against an enemy attacking forces, for such specific purpose as regaining ground lost or cutting off or destroying enemy advance units, and with the general objective of denying to the enemy the attainment of his purpose in attacking. In sustained defensive operations, it is undertaken to restore the battle position and is directed at limited objectives.

h. *Local Counter-Attack*. An attack organized by the local commander to destroy enemy forces, which have penetrated or threaten to penetrate his defensive position, general using a pre-arranged plan. This may be organized at battalion or even company level.

j. *Area of Responsibility*. The ground allotted to a unit or sub unit on which it will plan and fight its battle. It is delineated by boundaries.

k. *Important Tactical Ground*. Ground, the capture of which by attacker will render the defender incapable of fighting a successful battle in his area of responsibility.

l. *Mutual Support*. Within a company defended locality, platoons will give mutual support to each other with small arms fire. Thus gaps between platoons are covered by fire. Within a platoon, section position will be so sited that enemy assaulting any one of these can be engaged directly with small arms fire from at least one another. Thus they are mutually supporting and are closely knit than platoons within a company.

m. *Arc of Fire*. An area of ground, for which a fire unit or a weapon is responsible, and within which, it will engage targets. An arc of fire must not be confused with field of fire, which is the area in which it will be possible to fire effectively.

(1) *Primary Arc of Fire*. It is the arc, which is the basic task or responsibility of a fire unit or a weapon.

(2) *Secondary Arc of Fire*. It is the arc, which is the subsidiary task or responsibility of a fire unit or weapon, taken on only when not engaging the primary arc of fire.

n. *Field of Fire*. The area which a weapon or a group of weapons may cover effectively with fire from a given position.

o. *Fixed Line*. A weapon is said to be laid on a fixed line when it is so laid that its fire can be directed to a predetermined area regardless of darkness or fog (under all conditions of visibility). The gun is usually fixed on a certain arc of fire unless it is required for engaging opportunity targets.

p. *Enfilade Fire*. When the longer axis of the beaten zone of the fire of a weapon covers the longer axis of the target it is said to be enfilade fire.

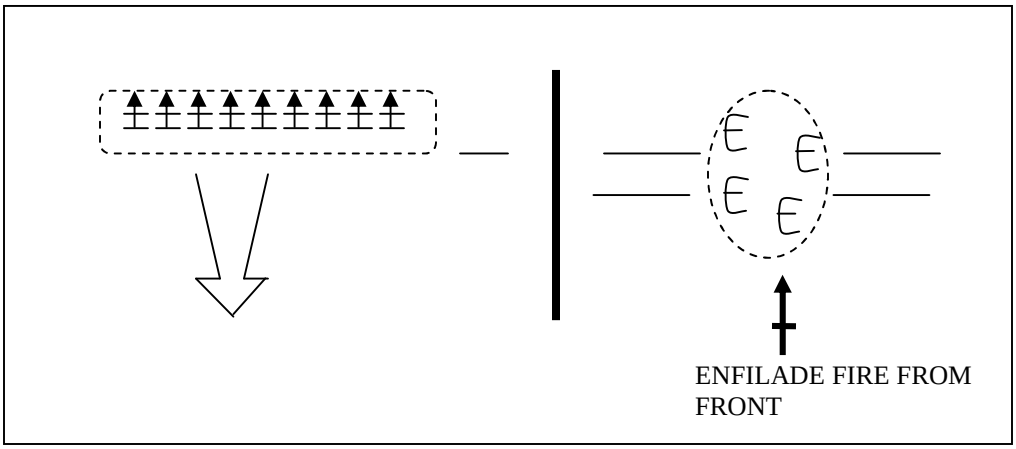


Figure 29 - Enfilade Fire from a Flank

q. *Defilade*. A weapon which is so placed that it is covered or protected from the enemy fire to its own front is said to be defiladed.

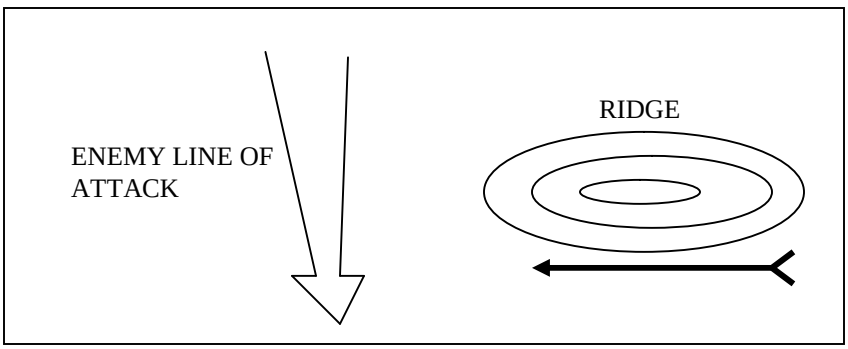


Figure 30 – An Anti-Tank Gun Defiladed from the Enemy.

r. *Overhead Cover*. It is a cover from view and is generally obtained by sitting defences under natural cover such as hedgerows and trees. Where natural cover is not available, camouflage nets, CGI sheeting, timber planking and brushwood, may be used.

s. *Overhead Protection*. Protection provided to occupants of field fortifications against air burst and splinters. It is normally provided by 45 centimetre of loose earth, placed overhead supported on CGI sheets or timber planks et cetera.

t. *Bunker*. A dug out with narrow port/holes for firing weapons. It is provided with overhead protection against splinters of a near miss, variable time (VT) or airburst shells.

u. *Pill-Boxes*. A small low fortification usually made of concrete, steel or filled sandbags, which houses weapons such as machine-guns and anti-tank weapons, and may be deemed to withstand the direct hit of a particular weapon.

v. *Fire Trench*. A field work, dug below the normal level of ground and designed to enable one or more soldiers to use their weapons (rifle, pistol MP5 or light machine-gun) effectively from cover.

w. *Slit Trench.* A field work, dug below ground level, designed to provide protection during short halts or against an air threat, when fire trenches or weapon pits are not necessary.

x. *Listening Posts*. A listening post normally consists to two to four men and it is mounted during night to give early warning of enemy movement. Listening posts are located to cover likely enemy approaches. They should have cover, concealment and good routes of withdrawal. Due to strenuous nature of their task, listening posts should be relieved as often as possible. If frequent relief is not possible then double sentries are posted to ensure proper vigilance to prevent surprise.

y. *Defensive Fire (DF).* Pre-arranged fire which can be brought down quickly on actual or suspected enemy movement. It will include:-

(1) Tasks close to defended localities to break-up the enemy attack.

(2) Tasks in depth to disorganise enemy preparations just before and during his attack.

z. *Defensive Fire (SOS).* Defensive fire (SOS) tasks cover the most vulnerable approaches, usually close to the position. The guns will remain laid on their defensive fire (SOS) tasks when not otherwise engaged in other fire tasks.

aa. *Harbour Party*. In defence this party is incharge of laying out the assembly area and guiding the main body into it. It is normally commanded by Officer Commanding Combat Support Company and comprises at least one guide per rifle platoon.

bb. *Vital Ground.* Ground the capture of which by the attacker will render the defender incapable of fighting a successful defensive battle in that sector.

Defensive Doctrine

5. The defensive doctrine envisages seizure of initiative from the enemy after inflicting maximum losses by skilful use of the ground, obstacles and firepower. It includes employment of protective detachments for early warning, deception, delay, disorganisation and harassment of the enemy; main positions for causing attrition and denying ground of tactical importance, and the reserves to eject or destroy the enemy by offensive action. Although positional in nature, the doctrine is characterised by the offensive spirit and aggressiveness, which implies maximum use of aggressive patrolling, raid, ambushes and tank hunting parties. Effective domination of no man's land and creating a sense of insecurity in the enemy is the firepower and manoeuvre.

Fundamentals of Defence

6. Detailed discussion of the fundamentals of defence can be found in GSP-1355, “Operations of War Volume I, Defence – 2001”. Some of the fundamentals which more often influence planning, layout and conduct of defence at Company, Platoon and Section level are discussed below:-

a. *Enemy*. The company commander must first assess the nature of the enemy threat, his strength, composition, disposition, his capability and intentions.

b. *Terrain*. At Company, platoon and section level the area for the defence will in most cases be given by the battalion commander or any other commander ordering the defence. The Company, Platoon and Section Commanders should still carry-out a detailed analysis of their allotted areas so as to identify the likely enemy approaches, the areas which can provide the best observation (because the

observation is essential for providing information of the enemy and adjusting indirect fire), areas affording cover and concealment, the areas which need creating/erecting obstacles and the areas which need special surveillance measures such s dead ground. Cover and concealment should never be sacrificed for fields of fire; 100 metres is minimum, acceptable field of fire, but 270-500 metres is desirable for optimum use of automatic weapons.

c. *Frontages*. The Company frontage will be dictated by the area allotted to it. However it should be emphasised that platoon and section frontages are not unduly stretched in the interest of command, control and cohesiveness of the defences. The criterion of platoon/section frontages should be that the commanders are capable of exercising control by a word of mouth. The area should be defended by firepower and not by manpower. Gaps between platoons and sections should be effectively covered by fire, minefields and link patrolling.

d. *Mutual Support*. Ideally when the fire of one subunit sweeps the front of the other subunit the localities are said to be mutually supporting. This may however not always be possible at company level. In such cases the gaps between platoons should be so covered by direct fire that despite the penetration enemy can not mount an attack on any one platoon without being seriously interfered with by the other platoons. Within the platoon however, complete coverage of the front of the other section by direct fire of the adjoining section should be ensured.

e. *All-Round Defence*. Although the defence is primarily designed to repel the enemy attack along the most likely approach, the company, platoon and section commanders should remain prepared to defend themselves against attacks from any direction. This is achieved by careful initial dispositions of platoons/sections, skilful preparation of fire trenches and prior planning/preparation of alternative positions both for platoons/sections and automatic weapons. Platoon is the lowest level at which subunits should be sited for all-round defence.

f. *Depth.* Adequate depth in defence is essential to absorb the weight of enemy attack, prevent out flanking and gaining time for countermeasures apart from denying observation and reconnaissance of the defended area. While effort is made to achieve depth in the company defences, it may have to be scarified at platoon level duplication in favour of mutual support over

extended frontages. When depth cannot be achieved in the initial dispositions due to wider frontages depth positions should be planned and pre-sections not directly engaged by the enemy provided the situation so dictates.

g. *Aggressiveness*. Maintenance of aggressiveness pays rich dividends in any defensive battle. It is maintained by:-

(1) Aggressive patrolling to dominate no man's land.

(2) Carrying out raids.

(3) Laying of ambushes.

(4) Effective use of tank hunting parties.

(5) Spoiling attacks.

(6) Counter-attacks.

h. *Security*. Necessary steps to avoid tactical surprise must be taken. At company/platoon level this is done by establishing Observation Posts, Listening Posts and Standing Patrols, in addition, screens and other protective detachments are deployed ahead of the main defences by battalion commander. Security is also achieved by camouflage and concealment, communication security, track discipline and minimum day movement in the defences.

j. *Surveillance*. The commander must ensure that en is kept under constant surveillance during all phases of the defensive battle. Optimum utilization of available surveillance resources is essential for which a comprehensive plan should be made by the commander.

k. *Administration*. A defensive battle generally last for a considerable period of time and demands a high degree of determination and will to fight from defenders. This requires in addition to leadership, adequate supply of ammo, food, water, sleep and needs care.

Classification of Defence

7. **According to Technique**

a. *Area Defence*. A type of defence where the bulk of the defending force is deployed in selected tactical localities with the task of preventing enemy penetration and destroying enemy attacks by

fires in front of the position, and destroying him through counter-attack.

b. *Mobile Defence*. In this, primary reliance is placed on the use of mobile elements and fires for offensive actions. In mobile defence minimum forces are employed around nodal points which are the villages or the areas where different approaches in the desert converge. It may be developed into strong points/fortifications. Such nodal points are normally selected in the forward areas and strong mobile reserves are held in the rear. The decisive battle in this case is fought within the defended area accepting a temporary loss of some of the ground. The enemy is so contained within the defended area that his destruction is facilitated.

8. **According to the Degree of Preparation**

a. *Deliberate Defence*. It is normally organized when out of contact with the enemy or when contact with the enemy not imminent and time for development is available.

b. *Hasty Defence*. Normally organized while in contact with the enemy or when contact is imminent and time available for development is limited.

Echelons of Defence

9. Defence is echeloned for following purpose:-

a. Force the enemy to break his attack into phases, so that the defensive position cannot be overrun in one continuous attack. The more the time taken by the enemy to over run the defensive position the higher will be his attrition through defensive fire.

b. Allow the defender to estimate the direction of enemy's main and secondary attack so that the attacker's initiative can be counter-attacked by re-disposition of defensive firepower and the reserves.

c. Create opportunity for launching of the reserves for ultimate destruction of the attacker.

10. To be able to achieve the above object, the defence is organized into the following echelons:-

a. *Protective Detachments*

(1) Reconnaissance Elements.

(2) Covering Troops

(3) Advance Positions.

(4) Screens.

(5) Patrols.

b. Main Positions.

(1) Forward Defended Localities.

(2) Depth Localities.

c. *Reserves*

11. The actions of a company/platoon/section will depend whether it forms part of the Protective Detachments, Main position or the Reserve. In order to carry-out their roles effectively in each of the echelons the company/platoons/ sections must know purpose of their employment and understand the techniques to achieve that purpose.

SECTION 73 – **PROTECTIVE DETACHMENTS**

Types of Protective Detachments

1. Protective detachments are of the following types:-

a. Reconnaissance element.

b. Covering Troops.

c. Advance positions.

d. Screens.

Reconnaissance Element

2. Reconnaissance elements perform following tasks:-

a. Keep a constant watch on enemy movement/actions.

b. Pass back information.

c. Harass the enemy.

3. These troops will normally be provided by Light Anti-Tank battalion/corps reconnaissance regiments.

Covering Troops

4. Covering troops are mainly composed of mobile elements. These troops may be assigned some or all of following tasks: -

a. Delay the enemy in order to gain time for the development of main positions.

b. Contribute to the destruction of the enemy as it advances.

c. Gain information about enemy.

d. Prevent enemy from obtaining information about defensive position.

5. Delay is imposed by taking up successive defensive positions along the likely enemy approaches, making full use of natural and artificial obstacles. Positions are pre-planned but not pre-prepared. Therefore, to allow time for the next position to the prepared, withdrawal between positions is carried out by leapfrogging the various elements of the covering troops.

6. Task of the infantry company in covering troops is to take-up defensive positions along the successive lines of delaying positions. The layout of these defensive positions is similar to those in the main position but the sequence of their organization and preparation and the conduct of battle will differ for the following reasons:-

a. Time being very short, detailed reconnaissance by company/ platoon commanders before the troops arrive on the ground, will not be possible. Coordination will, therefore have to take place while the troops are preparing the positions. This method is similar to that adopted in the withdrawal.

b. Weapon pits are sited to engage the enemy at longer range and the fire is also opened at longer ranges to force early deployment of the enemy.

c. Larger frontages may have to be occupied.

Advance Position

7. Advance position is used when covering troops are not being employed. This had comparatively smaller force, consisting mainly of standard infantry reinforced with elements of LAT battalion. To ensure that they cover the front effectively they are given adequate number of automatic and anti-tank weapons.

Owing to limited mobility, advance position is reasonably close to the main position.

8. The task of advance positions is to delay the enemy in his attempts to gain possession of localities, which would assist him launching attacks against the main position.

9. The delay is imposed by taking up successive defensive positions along natural obstacles. These positions, which are sited along the likely enemy approaches, are pre-planned and pre-prepared. Their layout and organization is therefore similar to those in the main position. However, they may not necessarily be prepared by the troops, fire is opened at longer ranges to force early deployment on the enemy, and the position is vacated just before the enemy attack is due to arrive on the position.

Screens

10. Screens are Company/Platoon/Section size outposts established laterally with intervals in front or on the exposed flanks of the main defensive position. Screens should be deployed only as far forward from the main positions as is necessary to deny the enemy observation and reconnaissance of the main positions and its obstacles. Screen should be deployed on the likely enemy approaches on defensible ground, which provides good observation and covered routes of withdrawal. Screens may also be employed by division/brigades on seemingly unlikely approaches away from the main defence for surveillance and early warning.

11. Situation may permit a brigade/battalion commander to deploy a company/platoon size screen to exploit advantages of a piece of ground, which may otherwise not be falling within the defensive position. For example an isolated knoll about 600 metres away from the forward defended localities of company/battalion can be occupied by a reinforced platoon strengthened by minefields et cetera so that enemy is forced to attack the screen position before developing attack on the main positions. Deployment of such screens will give added strength to the main positions.

Tanks

12. The tasks of screens irrespective of the size are as follows:-

a. Deny close observation of the main positions and its obstacles.

b. Provide information about the enemy.

c. Inflict casualties upon the enemy.

Composition

13. Troops for screens will normally be drawn from the depth companies/ platoons holding the main positions. As such their number and size is kept as small as possible, so that maximum numbers of troops are available for preparing and manning the defences. The considerations for deciding the strength of the screen are:-

a. Frontage of the defence which it is screening.

b. Period for which deployed.

c. Direction and quantum of threat.

d. Defensibility of the ground where located.

14. Owing to the demanding nature of the task, company/platoon commanders should be very selective in choosing the personnel for screens. Depending on the size, it will often be desirable to detail company/platoons/sections as a whole for screens, ensuring that the platoons/section commanders so selected are brave, imaginative and reliable. A suggested composition of a screen is as follows:-

a. Company with artillery/mortar observer and communications.

b. *Company*. With all or some of the following additional weapons/ personnel:-

(1) Extra tripod of MGIA3.

(2) 106/75 mm recoilless rifle.

(3) 60 mm mortars or artillery/mortar observer.

(4) GLHMG.

(5) 12.7 AAMG.

(6) Bakhtar Shikan.

(7) Communication.

c. *Platoon*. With all or some of the following additional weapons and men:-

(1) Tripod MGIA3.

(2) Rocket launcher.

(3) 60 mm mortar detachment or mortar observer.

(4) Bakhtar Shikan weapon systems.

(5) GLHMG.

(6) Communication.

15. Mines may also be laid in front and on the flanks of the screens if not already deployed in the minefield. Availability of mines and time / resources for laying will have to be considered.

16. **Control and Deployment**. Screens are controlled by the authority who orders it. Screen commanders should therefore be clearly told about the headquarters, which will order its withdrawal. In most cases, it will be the battalion headquarters however, when screens are deployed away from the battalion on order of the brigade/division headquarters, command channels should be clearly indicated. General area of the screen will be given by the commander ordering the establishment of the screen, yet the screen commander will have to select the most appropriate position for his troops/weapons. Major criteria for a good screen position are, clear observation and covered route for withdrawal. In order that larger areas are covered, platoons may have to be split in section posts, but sections should neither be split nor deployed over extended frontages. Capabilities of quick disengagement should always be retained while sitting a screen. A screen should have a number of alternative fire trenches to quickly switch fire from one direction to another.

17. **Conduct of Screen Battle**. The battle of screens can be divided into three stages i.e. preparatory, contact and withdrawal. In all stages however, continuous passage of information about the enemy is most essential. Various actions of the screen during different stages will be as under:-

a. *Preparatory Stage*. This is the stage when attacker is assembling his forces and making preparations for the assault. During this stage the screen should:-

(1) Gain information about the enemy strength and concentration areas by careful observation.

(2) Gain information of the areas of enemy interest by watching the patrol and reconnaissance activity.

(3) Keep enemy patrols away from the main position by judicious engagement with minimum essential weapons.

(4) Plan, prepare and rehearse withdrawal of the screen. Withdrawal plan should be coordinated with concerned subunit in the main positions.

b. *Contact Stage*. This is the stage when the attacker sets of to contact and assault the main positions. During this stage the screens should: -

(1) Engage the enemy with available fire at long-ranges.

(2) Call for artillery/mortar fire.

(3) Force the enemy to deploy for attack.

(4) Make preparations for withdrawal.

(5) Ensures that it does not get/inextricably involve with the enemy.

c. *Withdrawal Stage*. This is the stage when the screen having done its job is ready to fall back in the main position. During this stage the screen should:-

(1) Seek permission for withdrawal.

(2) Pass back all available information about the enemy.

(3) Continue exerting pressure on the enemy with all available fire.

(4) Inform all concerned regarding its withdrawal.

(5) Withdraw to the main positions at the very last moment.

18. It should be noted that withdrawal of a screen should be along a covered route and cannot be carried out without the permission of the commander who had ordered establishment of that screen. In case the communications break, screen commander may withdraw at his own provided he has suffered 50% casualties or the time up to which he was required to stay has elapsed.

SECTION 74 – **PLANNING OF DEFENCE**

General

1. Before planning the defence, the company/platoon/section commanders will have received orders from the superior commanders. These orders will always give the grouping and tasks of the subunits concerned. The orders in most cases will include:-

a. Situation.

b. Addition or reduction in resources.

c. Area of responsibility.

d. Line of FDLs.

e. Special tasks if any.

2. Within this framework, the sequence of events for planning of the defence will be as follows:-

a. A quick analysis of the tasks and all available information.

b. A quick reconnaissance either from map or ground to determine important tactical ground and major approaches into the area of responsibility to decide the out line plan which should include: -

(1) Platoon/section localities.

(2) Obstacle plan in general.

(3) Location of anti-tank weapons in general.

(4) Defensive fire tasks.

(5) Surveillance and security tasks.

c. Issue of out line plan.

d. Detailed reconnaissance, carried out concurrently by company, platoon, section and supporting arms commanders to site platoons/sections and position supporting weapons.

e. Coordination of defences by company/platoon commanders to ensure that important ground is held by closely knit company/platoon localities and that it fits in the defensive layout of neighbouring subunit/unit. Coordination at each level follows the coordination of immediate superior commander.

f. Issue of confirmatory orders and briefing.

3. **Defence Planning is Continuous**. During and after the formulation of the basic plan, the company/platoon commanders concurrently consider contingency plans and means for making an easy transition to an alternate plan if the situation so requires.

Considerations for Evolving the Defence Plan

4. **Company Commander**. Company commander is the lowest level where more than one problem is to be tackled. The magnitude of problems decreases as the level goes down. It is therefore necessary for the company

commander to analyse the situation in its entirety to evolve a sound defensive plan. Although no formal appreciation is required at company level a detailed consideration of all factors and sound conclusions are necessary. Some of the aspects which a company commander must consider are:-

a. *Ground and Enemy*. The company commander must carry-out a thorough study of the map and ground reconnaissance and evaluate the terrain from the viewpoint of enemy capabilities. It is advisable to see the ground from the enemy side and identify the weak spots. From the study of the ground the company commander should be able to:-

(1) Identify the approaches leading into the area of responsibility, pin pointing the most likely tank approach and divide the area into platoon sectors ensuring that responsibility of defending an approach is not divided between two platoons.

(2) Recognize the ground of tactical importance and decide which of these are to be held initially and with what strength.

(3) Assess the value of natural obstacles and examine the need for their improvement. If none is available see the feasibility of creating one.

(4) Locate likely assembly areas for the enemy and pick them up as defensive fire targets.

(5) Identify dead grounds and take measures to cover them.

(6) Examine the need of security and decide upon the location of security/surveillance out posts and Observation Posts.

(7) Select a suitable location for the company command post and company headquarters.

(8) Select suitable location for 60-mm mortar sec, RRs, GLHMG, and any other weapon given to him from battalion headquarters.

(9) Select counter penetration positions.

b. *Own Situation*. The company commander now carries out assessment of the tasks wherein he matches his resources with the requirements dictated by the ground and decides:-

(1) Localities which must be held initially and those which can be earmarked as alternative positions.

(2) Whether his resources are adequate or he must request for more. (Weapons and mines only).

(3) What strength could be used as out posts?

c. *Time and Space*. Initiative being with the enemy, time is generally short for preparation of defences. Even if enough time is available the tasks are so numerous that nothing short of a carefully worked out time schedule will ensure orderly preparation of defences and adequate rest for the men. From the time and space consideration, the company commander works out the timings for movement (if movement is not being done under the battalion), time required for reconnaissance and orders at all levels and the priority of work. Although most of the tasks in organizing a defensive position are carried out concurrently, the situation may require that priority be attached to some. The company commander specifies the sequence for the preparation of the position and any special precautions to be taken regarding camouflage. The normal priority of work is as under:-

(1) Organise protection and surveillance (Observation Posts, out posts and patrols).

(2) Position recoilless rifles and machine-guns.

(3) Clear fields of fire, remove objective masking observation and determine ranges to probable target locations or reference points.

(4) Prepare open weapons emplacements and fire trenches.

(5) Establish line communication.

(6) Emplace obstacles and lay mines.

(7) Improve primary position to include overhead cover and camouflage.

(8) Prepare alternate and counter penetration positions.

(9) Prepare communication trenches.

(10) Prepare latrines and rest areas.

5. **Platoon Commander**. Considerations of platoon commander will depend on whether he is operating independently or under the company. In case

of the former a platoon commander will consider all the points usually considered by the company commander in addition to those which he considers while operating under the company. A platoon commander will usually get from company commander the grouping and tasks of the platoon in addition to the boundaries of the area of platoon responsibility. In most of the cases, company commander will also indicate generally the location of recoilless rifles, machine-guns and any other support weapon allotted to the platoon. The task of the platoon commander is then greatly simplified. He however, considers the ground in relation to his resources for the following:-

a. What are the suitable defiladed positions for the recoilless rifles, machine-guns and any other support weapon grouped with the platoon such as anti-tanks.

b. Where the rocket launcher should be sited.

c. Which are possible section localities so that the support weapons are adequately protected while retaining effective control?

d. Which is the suitable location for the platoon command post for effective control on all sections? (There will be no separate platoon headquarters).

e. Which are suitable alternate positions for weapons and sections?

f. How fields of fire can be improved with minimum clearance effort?

g. How the dead ground if any will be covered?

h. How best observation can be achieved. Where should Observation Post be located?

j. How best security can be achieved. Where if any should be the out post(s)?

k. How best can camouflage and concealment be achieved?

6. **Section Commander**. After receipt of orders from the platoon commander, the section commander decides on the number and type of fire trenches depending on the strength and task of his section. Although the tasks of section commander are more of supervisory nature, he should consider the following aspects:-

a. How the fire trenches should be designed to be effective all round?

b. How can he be achieving best camouflage and concealment?

c. Where the excavated earth should be dumped?

d. What should be the arcs of fire and which landmarks should he use for indicating the same?

e. Where and how much area is to be cleared for fields of fire?

f. Where should he locate himself to exercise effective control?

7. While planning the defence, company, platoon and section commanders should invariably keep the fundamentals of defence in mind and interpret the guidelines given in paragraphs 4, 5 and 6 objectively within the frame work given by the superior commanders.

Surveillance and Security

8. Every commander is responsible for security of his command which includes protection against espionage, sabotage, subversion and surprise by the enemy. In the overall defensive layout security is provided by the protective detachments, yet the possibility of enemy action despite the protective detachments cannot be overruled. Hence the company commander establishes a surveillance system to operate day and night. It includes:-

a. *Observation Posts*. Normally one or two Observation Posts in a company area will be sufficient. A Company will more often have one artillery and one mortar observer in its location. Best use of these Observation Posts which will be located on vantage points (high ground, houses, tops of trees) should be made by supplementing them with own manpower. Battalion intelligence section personnel if available are best suited for this duty. Alternatively a few intelligent and educated soldiers from the company can be earmarked. Every Observation Post should be manned round the clock and be equipped with Observation Post log, map, binocular, NVDs, compass, watch, whistle and communication (telephone/wireless et cetera). Ground surveillance radars if available may be employed on such Observation Posts. Observation Post should carry-out systematic surveillance of the area of responsibility by dividing it in sectors and record every information may it be minor, locate enemy positions on map, trace route of vehicular movements and record bearings of all flashes given out by enemy guns/mortars.

b. *Listening Posts*. These are deployed at night ahead or on the flanks to provide early warning of enemy approach. These are meant to compensate for the loss of visibility of the Observation Post. Listening posts should be equipped apart from weapons, with

binoculars, NVDs/NODs and communication means. The personnel selected for listing posts should be intelligent, dedicated and well trained particularly in night vision and listening techniques. These should preferably be non-smokers. Listening posts should fire only in self-defence. They should gain as much information as possible by unobtrusive observation and noise identification.

c. *Sentries*. Single sentry by day and double sentries by night will always be posted all along the front and depth localities. As a rule, sentries will be posted on automatic weapons. If a machine-gun is located in a section position sentry should be posted on this machine-gun in preference to the light machine-gun next door. Usual tenure of duty of a sentry is 2 to 3 hours beyond which efficiency sharply decreases. At night tenure of duty of the pair of sentries should be staggered so that one sentry at a time is changed and continuity of surveillance is maintained. Depending on the ground, a one sentry per section and one for company headquarters should be adequate. During day, in addition to normal ground sentries, one air sentry per platoon is also posted. All sentries should be equipped with binoculars and communication means while the air sentries should also have whistles.

d. *Patrols*. Apart from the patrols sent ahead of the defences in accordance with battalion/brigade patrol programme, each company plans 'Link Patrols' for surveillance of the inter platoon and inter company gaps at night. These are three to four men patrols moving according to a planned schedule and making contact with the neighbouring locality to keep the gap under effective surveillance. They move, halt, listen and move again.

9. The above are some of the most commonly used methods for ensuring effective surveillance and security in defence. There is however no end to human ingenuity and various other methods can be devised/improvised. Company Commander should pay special attention to properly organizing surveillance in their companies because more often ill planned surveillance activities result in avoidable fatigue to men due to lack of sleep. An unvigilant sentry is worse than not having one.

Sitting

10. **Weapons**. The sitting of the automatic and anti-tank weapons is the most important part of the planning, as their tasks provide the framework around

which the defence of the company/platoon is built. Therefore machine-guns, Bakhtar Shikans and recoilless rifles should be sited first and their gaps covered by light machine-gun and rocket launchers teams. Riflemen should then be positioned for local protection. Machine-guns and recoilless rifles are sited by the company commander:-

a. *Machine-Guns and Grenade Launcher Heavy Machine-Guns (GLHMG).* The integral machine-gun and additional weapons like grenade launcher heavy machine-guns in the platoon are sited to cover the most dangerous infantry approach into the platoon area. The aim is to place a curtain of fire between assaulting enemy and own positions. The company commander may order that machine-guns interlock their fire with an adjacent platoon. Machine-guns are generally sited singly in defiladed position and are protected by the section deployed in the area. If employed in pairs, they should be far enough to avoid being neutralized by one incoming mortar or artillery shell. The ground should provide grazing field of fire. In addition to original position the platoon commander should select alternative positions. When not firing, machine-guns are laid on their primary task. At night machine-guns are laid on fixed lines.

b. *Anti-Tank.* The anti-tank layout will form the framework of the main defences. The company commander having considered the likely tank approaches will select localities and allocate his recoilless rifles accordingly. While sitting anti-tank weapons, balance should be struck between different anti-tank weapons, which vastly differ in ranges. Small anti-tank weapons such as energas and rocket launchers apart from being sited in the main position can also be employed ahead of the defences in pre-selected areas, where the tanks are likely to take cover soon after hitting the protective minefield or engagement by a recoilless rifle. Anti-tank weapon positions should provide oblique fire at the tank approaches, cover routes for ease of movement to alternative positions and indirect fire positions. Their location and direction of fire are coordinated with other anti-tank weapons located within the platoon/company area. The anti-tank weapons can also be employed against enemy automatics, vehicles, and grouped personnel. Alternative positions should be prepared for each recoilless rifle and Bakhtar Shikans but occupied only when absolutely necessary or ordered by company/platoon commander:-

(1) *Sitting of Anti-Tank Weapons.*

(a) At times the energas and Rocket Launchers can be sited together, with 106 or 75 mm recoilless rifles covering them from the rear, both for provision of depth and to be in a position to extricate them, should the situation so demand. This will produce the following results:-

i. The long-range of 106/75 mm recoilless rifles will not be fully utilised.

ii. The tanks will be given a free run-up to the effective ranges of Rocket Launcher and Energa.

iii. The tanks will have a very short run, say 200/300 metres, from the point, from where the fire of all anti-tank weapons become effective and the position on the ground of the short range anti-tank weapons. Tanks will cover this distance of 200/300 metres, even at the speed of 25 kilometres per hour, in approximately 24 seconds, which will encourage the tanks to “charge” and overrun the short range anti-tank weapon positions. It will reduce very much the number of rounds, which can be fired at the tank before it over runs the position.

iv. Once the short range anti-tank weapons, which are in maximum number in a battalion, are over run (and these are the positions where the remaining infantry will also be positioned), the long-range anti-tank weapons will be divorced from the infantry and the short range anti-tank weapons, therefore, the long-range anti-tank weapons will be more easily made ineffective later on by the tank/infantry combined teams of the attacking enemy.

v. In such type of sitting the depth of fire is reduced very much.

(b) The following techniques therefore are suggested: -

i. *Technique 1*. The weapons are sited so as to be able to engage enemy at maximum effective ranges. The short range anti-tank weapons to occupy temporary positions at a distance of 500/600 metres ahead of main position. Some long-range anti-tank weapons may also be positioned along with short-range weapons in temporary positions. The weapons open fire according to their respective effective ranges. Thereafter they carry-out a controlled withdrawal run only to the main position under the cover of the weapons in the main position. When the tanks advance further, again they have to first go through the fire curtain of long-range plus-short range weapons. That way, the tanks will be under continuous fire for a distance of 1,000 metres or more.

ii. *Technique 2*. Only some long-range weapons are sent ahead in temporary positions, which after having engaged the enemy, withdraw at suitable time to main defences. All weapons, as in technique. 1, are sited to be able to open fire at maximum effective ranges and the short range weapons giving "Depth of Fire" to the fire of long-range weapons. The engagement of the tanks starts at the maximum effective ranges by the long-range weapons then the short-range weapons. The net effect is that as the tank is getting close to the main defences, it is getting in thicker and thicker anti-tank fire.

(c) At times, it is quite likely that certain approaches are absolutely impassable for the tanks whereas other approaches are good tank going. In such a situation, the anti-tank weapons including short range weapons may be concentrated in the threatened area by denuding un-threatened area.

Alternative and secondary positions should be prepared for all anti-tank weapons. This improves their effectiveness.

(d) Judicious, intelligent and clever use of mines and other natural and artificial obstacles will further assist.

(2) However, the difference in the ranges of the energas and the rocket launchers is negligible where hundreds of metres are involved, when a comparison is drawn vis-à-vis the tank gun, with the result that to site them as explained above, would not be sound. Therefore, the rocket launchers and energas would be sited together, with 106 or 75 mm recoilless rifles covering them from the rear, both for the purpose of provision of depth and to be in a position to extricate them, should the situation so demand.

(3) The number of weapons so employed, will depend on the ground, the threat and the availability. However, the guiding principle should be that the number of anti-tank weapons should be more than the expected tank threat without denuding the main defences altogether.

(4) When the advancing enemy tanks hit the protective minefields or are under the effective fire of RRs from the main defensive position, they would instinctively manoeuvre to get into nearby suitable hull down positions. In this general area pre-selected sites are manned by tank hunting parties equipped with energas/RPG-7. They take on the tanks from the flanks and destroy them at short ranges.

c. *Light Machine-Guns and Small Arms.* The platoon commander must allot overlapping area of responsibility to sections and light machine-guns within the platoon area allotted by the company commander. Light machine-guns must have a fixed line and general area of fire. To cover the platoon area by small arms fire during all conditions of bad visibility the platoon commander must ensure that:-

(1) Range cards are prepared.

(2) Proper surveillance practice is carried out with NVDs so as to identify the enemy movement with ease.

(3) Erect aiming sticks for the weapons, which do not have NVD, so that firer can put across his fire in correct direction.

d. *Defensive Fire (DF).* To help the company/platoon hold its position, certain defensive fire tasks for artillery, 81 mm and 60 mm mortars are planned. Defensive fire in depth is primarily the responsibility of the artillery and will normally be planned at brigade and higher level. The purpose of defensive fire, particularly the close defensive fire is to destroy and disorganize the attacking enemy and break the assault. Close defensive fire tasks are normally planned in front and on the flanks of the defensive positions but there is no restriction to selecting targets within the defensive position where penetration is anticipated. Such targets should be selected keeping in mind the safety needs of own troops.

e. As many targets as are necessary to cover the whole area of responsibility should be picked up for defensive fire. Usually six targets per firing unit (battery/mortar platoon) are recommended. Mortar fire is particularly suitable to cover dead ground and enemy positions behind a crest. 60 mm mortars are ideal for engaging enemy in the close vicinity of own troops. Defensive fire is normally brought on the targets through the artillery/mortar observers but the same can also be asked for by platoon commanders on command net. In such situation the platoon commander must furnish the following information to the company headquarters:-

(1) Enemy strength.

(2) Enemy location: a grid reference or target location in relation to a known Defensive Fire (DF).

(3) Direction in which the enemy is moving.

(4) Time at which the target is seen.

11. **Troops**. Having decided upon the location of major weapons, the company commander plans to position his platoons. Normally the company commander should strive to defend the area of responsibility with two platoons forward and one in depth. However, if the frontages are over extended there is no restriction on using all the three platoon forward. In such cases the disadvantages occurring from lack of depth should be fully weighed. Rifle platoons will usually be deployed three sections up. Only in rare cases, mostly in detached roles, the

platoon may be sited two sections forward and one in depth. A section will have three or four fire trenches. The section commander could either stay centrally to control both the light machine-gun groups or be close to one detailing the section Second-in-Command with the other light machine-gun group.

12. Tendency to split up sections and platoons into groups of two men each, and string them across the front at considerable distance from each other in a bid to cover ground with men is to be resisted. This is generally undertaken because of threat of enemy infiltration. Such a deployment renders the defence insecure and ineffective. Not only will the enemy succeed in infiltration but also succeed in taking a few isolated men as prisoners.

13. It would be more effective deployment if rifle platoons occupy the best defensive ground in their area of responsibility and cover the rest by fire. The deployment of the section would be governed, among other factors by the requirements of effective control by the section commander. The section could be grouped into teams, each team occupying two trenches of two to three men each. The distances between these trenches may be between 5 to 20 metres depending on the nature of ground, and observation.

14. There is also a tendency to seek depth in defence at section and platoon level. Weapons sited in echelons within a section are normally masked by the presence of friendly troops immediately in front and thus cannot be effectively employed. The fire effect of the sub unit at the critical point of the attack is thus reduced. An effective deployment could be to site the weapons of a section in an irregular line so that they cover the front, curving on the flanks to cover gaps. To achieve all-round defence capability, trenches for one of the groups of the section could be prepared at a short distance in depth facing the rear and to be occupied only in case of need. Even at platoon level in defence, where ground, frontage, and other conditions demand, we may consider the feasibility of deployment with three sections up. The depth in defence could be organized at company level where the line of weapons in depth can have a clear field of fire.

Fire Support Plan

15. Fire support planning is the responsibility of the company commander while advice/assistance can always be obtained from the artillery observer and supporting arms commanders. Fire support planning in defence involves the preparation of coordinated fire plan which provides for effective defensive fire regardless of visibility conditions, the aim being to destroy and stop the enemy in front of the FDLs. Company/platoon commander will normally have the following types of fire available to them:-

a. *Indirect Fire*. Provided by artillery and 81/60mm mortars. 60 mm mortars are normally used as direct firing weapons but they can also be used for indirect fire if terrain/situation so demands. In case these are used for firing in indirect role, DFs are also selected for these as for the 81 mm mortars. Company commander may at times, selects DFs for 60 mm mortars for use within his company. Indirect fire is planned in the form of defensive fire tasks. Defensive fire tasks are coordinated at battalion/brigade level. Hence the company/platoon commanders are only required to select suitable defensive fire targets and submit to the battalion headquarters. After finalization of the "Defensive Fire Task Table", the company commander will receive a copy of the same, which will indicate the target numbers and fire units allotted to each target. Company/platoon commanders should ensure that relevant targets and their numbers are known to their men and are even memorized for quick engagements. Every one in the company should remember the no of defensive fire (SOS) by heart and know its location on ground. Following points should be borne in mind while selecting targets for defensive fire:-

(1) The enemy is brought under fire as soon as his approach is detected.

(2) Enemy is held under increasingly heavy volume of fire as he approaches close to the defended localities.

(3) Enemy is subjected to concentrated fire even after penetration in own defences.

b. *Direct Fire*. Is provided by infantry weapons and tanks if allotted. While machine-gun and light machine-gun fire is planned to form a mesh of grazing fire in front and on the flanks, recoilless rifles, Bakhtar Shikans and tanks are left free to engage any targets in their area of fire which may be in range. Rifles and rocket launchers supplement opportunity targets within range. Every effort should be made to hold the fire till the enemy comes in the killing zone otherwise not only the effort will be wasted but the entire position will be given out without achieving the desired results. Drills for opening and stopping fire particularly at night should be evolved so that wasteful firing by all weapons/localities can be avoided. Company/platoon/section commanders should evolve and rehearse procedures/drills by which fire of the neighbouring locality can be brought to bear upon the enemy approaching the adjacent locality.

c. *Anti-Aircraft Fire*. See chapter 6 for details.

16. Fire support plans are coordinated with obstacle plans.

Obstacle Plan

17. Defence, as far as possible should be based on a natural obstacle. Natural obstacles such as deep-water ways are always the most economical and generally the most effective. Others, such as ditches, sunken lanes and thick hedges if improved and strengthened may prove useful. However, when natural obstacles are not available, artificial obstacles such as minefields are planned to canalize, restrict, delay or stop the enemy ground movement. Obstacles in front and on the flanks of the defensive position are planned and coordinated by the highest headquarters responsible for the defence of an area.

18. Defensive minefield will be planned by formation headquarters while protective mine field will be planned by the battalion commander. Defensive minefield will be laid by formation resources while protective minefield will be laid by integral and additional resources allotted to a battalion. Defensive minefield should be covered by anti-tanks weapons and MGs while protective minefield will be covered by small arms. While planning minefield, following coordination should be carried out:-

a. Requirement of patrol lanes, giving number width and location and alignment after having considered frequency and duration of use.

b. Location of listening posts, so that passage for occupation and withdrawal could be provided.

c. Dead ground, so that density of the minefield can be increased in these places and also for making arrangements for covering these areas by fire and surveillance.

d. Possible hide-outs for tank hunting parties so that passage can be provided and space left for occupation. Also see para 10(b).

e. Areas where far edge of the minefield becomes out of range of observation.

f. Arrangements for closing the gaps and lanes within the minefield after withdrawal of screens/listening posts. Either screens or listening posts should carry few mines or responsibility of closing such gaps/lanes should be given to the troops located in the closest vicinity.

19. **Mines**. Even if natural obstacles are available, mines will have to be used in either of the following ways:-

a. Improving or strengthening of the natural obstacle.

b. Filling gaps between obstacles.

c. Laying minefields in depth to contain penetration.

20. Well sited and carefully concealed minefields covered by fire are effective obstacles. Rifle company/platoons may be called upon to lay minefields. Mine laying details are contained in Field Engineering and Mine Warfare Pamphlet No. 5 GSP-1167.

21. **Wire Obstacles**. Wiring is a quick way of providing an obstacle to enemy infantry or improving an existing one; a little is better than none at all. The object of wiring is to slow down the enemy's advance and to give a better chance of killing him with fire. Wire must be sited so as to check the enemy outside the grenade throwing range of the defended localities. Wire not covered by fire is useless. Wire should be sited as far as possible to help achieve surprise. For this reason, it should be well hidden in such places as sunken lanes, hedgerows and long grass/crops. For details refer to Field Engineering and Mine Warfare Pamphlet No. 2A (Obstacles).

22. **Booby Traps**. Not to be sited within wire obstacles except on specific orders from the battalion commander. They must then be carefully recorded.

23. **Trip Flares**. Trip flares are used in defence to give early warning of enemy approach. They are usually sited along wire obstacles, minefields, and hedgerows running towards the position and covered by fire. The flares burn brightly for about a minute and clearly silhouette men within 25 metres and up to 45 metres if they are between the flare and the defenders.

Counter-Attack Plan

24. Counter-attacks at company level will be rare because the main task of a company in defence is to repel or stop attack and contain the penetration if the attack is partially successful. Therefore bulk of the company's strength will be committed in reinforcing the forward platoons, or occupation of alternate/counter penetration positions to contain the penetration. Also when the enemy has penetrated a company position held by two platoons, his strength will in most cases be more than what a depth platoon can overcome by counter-attack. However it does not preclude the possibility of a counter-attack at company level altogether. The size of enemy penetration and condition of his troops after the

penetration at times may provide good opportunities of a quick counter-attack by the depth platoon of the company in defence. Counter-attacks should therefore be planned as limited objective attacks designed to destroy or eject the enemy from an area of penetration and to regain the lost portion of the defended locality with a force not committed in fire fight.

25. Counter-attacks should be planned and integrated in the plan of defence right from the start. For planning a counter-attack, the company commander assumes likely penetrations and if such penetration includes ground of tactical importance, plans an attack through a flank by the depth platoon. The plan consists essentially of control measures, such as:-

a. Objective.

b. Start line.

c. Route.

d. Fire support plan.

e. Code words/Nicknames.

26. All subordinates should be informed of counter-attack plans, which should invariably be rehearsed.

Command and Control

27. Command and control is exercised by line and radio communications and personal visits. In order to be readily available in the battle area, a command post should be selected in the FDLs quite distinct from the company headquarters:-

a. *Command Post*. The command post should be located in such a manner that apart from being protected, it affords the best view of the company defences and the area in front and to the flanks including the approaches. It may be necessary to establish more than one command post to provide observation over the entire company area. Although the command post is the company/ platoon commander's battle location, he goes where he is needed keeping the company headquarters informed of his location and always remain in communication with the battalion head-quarters. Only the company/platoon reconnaissance group stays in the command post. Platoons are responsible to dig the company command post if located in their area.

b. *Company Headquarters*. It is normally located in line with the depth platoon close to the main route leading to the company area.

It should preferably be located in a defiladed position concealed from air and ground observation. The company headquarters normally provides its own security with headquarters personnel. Additional security may be obtained by locating it near the depth platoon. The company aid post will be established in the company headquarters for temporary collection and care of casualties awaiting evacuation. Prisoners of war may also be collected and held near the company headquarters until evacuated. Company senior/junior commissioned officer assisted by the company havildar major is responsible for organization and preparation of the company headquarters.

c. *Communications*. In defence the company primarily uses the line communication while radio and runners are the other alternatives.

(1) *Line*. Telephones lines are provided as follows:-

(a) Battalion exchange to company headquarters by battalion signal platoon.

(b) Fire orders line to artillery observer by the direct support field regiment.

(c) One omnibus circuit through all platoons, Observation Post and important weapon positions to company headquarters/ command post by company signallers.

(2) All telephone lines should be properly buried and regularly maintained by the company signalers.

(3) *Radio*. Following radio sets are available in a company:-

(a) One PRC-77 set for company commander on battalion net.

(b) Four PRC-77 sets on company command net having the three platoons as substations.

(c) One spare PRC-77 set for use by patrols, Observation Post or replacement of any set.

(d) Twelve walkie-talkie sets for use by platoon and section commanders.

(e) Pre-arranged pyrotechnic and sound/visual field signals are also useful for communications as alternative means.

Battle Procedure and Battle Drills

28. Detailed battle procedures and drills are attached at Annex ‘P’.

Administration

29. In defence the administration at company/platoon level consists of supply of food and water; replenishment of ammunitions; evacuation of causalities/ prisoners of war and replacement of battle casualties of weapons and equipment.

a. *Supply of Food and Water*. Except in case where a company is detached for independent role, food will be cooked at battalion 'B' echelons either brigades or independent. Whatever the case, company quarter master havildar (CQMH) will be responsible for provisioning. Food, water and mail along with any other item needed by the company will be delivered at the company head-quarters twice in 24 hours, which is after the last light and before the first light. Platoons collect their share of food and water from the company headquarters at specified timings.

b. *Replenishment of Ammunition*. Ammunition is replenished from rear to front. Company havildar major is responsible for maintaining sufficient quantities in the company F echelon vehicle and send to platoons the required quantities either through the stretcher bearers returning after leaving the casualties at the company headquarters or the parties coming to collect food. If none of these is possible, company havildar major should get the delivery of ammunition arranged himself with the help of company headquarters personnel.

c. *Evacuation of Casualties/Prisoners of War*. Own casualties and prisoners of war are evacuated from front to rear. Stretcher bearers bring the casualties from forward trenches to company headquarters while a few persons or the walking wounded are detailed as escort with the prisoners of war, up to the company headquarters. Stretcher-bearers while returning to the forward platoons also carry ammunition boxes with them.

d. *Replacement of Battle Casualties of Weapons/Equipment*. Damaged weapons/equipment is immediately replaced from the depth localities and replacement asked from the battalion headquarters, who take similar action immediately and later demand the same from divisional ordnance company.

SECTION 75 – PREPARATION OF A DEFENSIVE POSITION AND ROUTINE IN LINE

General

1. Efficient conduct of the defensive battle largely depends on how well the position has been prepared and how sound are the battle drills/standard operating procedures (SOPs) of the company. A well-trained company would normally complete preparation of the defensive position in 24-36 hours depending on the degree of enemy interference.

2. This section lays down some of the important duties of various personnel in a company and a few standard operating procedures, which if followed may prove helpful. There is however no rigidity about these standard operating procedures and can be modified to suit particular requirements. Duties and standard operating procedures applicable in a certain area/formation should invariably be contained in the standing orders for war maintained by every battalion/formation.

Duties of Commanders

3. **Company Commander**. Apart from command and control of the company and attached troops; planning and conduct of the defensive battle, the company commander is also responsible for smooth functioning of the company in day to day matters. Some of the important duties/task of a company commander are as under:-

a. Sitting platoon localities and company headquarters.

b. Sitting company support weapons (Recoilless Rifle, MGs and 60 mm Mortars) affiliated weapons (tank and missile), company Observation Post and out posts.

c. Coordination between platoons, important weapons and neighboring company.

d. Allocation of priorities and distribution of work to platoons.

e. Briefing/debriefing of patrols and interrogation of prisoners of war.

f. Ensuring that all orders/instructions are understood by all concerned.

g. Ensuring timely and accurate submission of reports and returns.

h. Inspection of platoon positions at least once a day and arms and equipment twice a week.

j. Ensuring proper administrative arrangements.

4. **Platoon Commander**. Platoon commander is an important link between the company commander and the men, because it is the platoon commander who is always in communication with the company commander. Platoon commander by virtue of his being present with the men can really contribute to the successful conduct of the defensive battle. In addition to what has been mentioned earlier, the platoon commander's duties/task include the following:-

a. Sitting of section localities and platoon command post.

b. Sitting of light machine-guns and rocket launcher including marking of fire trenches after personally examining the arcs and fields of fire.

c. Coordination of fire between sections/light machine-guns.

d. Equitable distribution of work between sections and platoon headquarters.

e. Briefing of the platoon personnel regarding the plan and conduct of the defensive battle. Briefing/debriefing of listening posts.

f. Ensuring that orders/instructions issued by the company headquarters are fully understood by all and implemented in the true spirit.

g. Passage of timely and accurate information to company headquarters.

h. Inspection of platoon positions, arms and equipment daily.

j. Programming duties, rest, bathing and washing.

k. Calling defensive fire and ordering opening of fire.

l. Ensuring that ranges have been measured.

m. Ensuring that no unnecessary movement takes place in the platoon area.

n. Ensuring that the runner knows the location of each section and company headquarters.

5. **Section Commander**. As soon as the troops arrive in the position and before they begin to dig, the section commander must show them their

dispositions and fire tasks; this ensures that they know how to deal with any immediate enemy attack. In addition the section commander is responsible for ensuring the following:-

a. All men clearly understand their task and all orders/instructions passed by the platoon/company commander.

b. All weapons are so placed that each man can fire on the ground allotted to him. The section commander must site each fire trench personally after examining the field of fire from a prone position.

c. Track discipline is strictly maintained.

d. Ensures that there is no slackness in work and the tasks are completed in time.

e. Proper camouflage and concealment has been carried out before day break.

f. Range cards are properly prepared. Section commander may have to prepare range cards himself.

g. Ranges have been measured/paced.

h. Every man recognises the important reference points.

j. Arrangements for laying the automatics on fixed lines exist and that required weapons are laid on fixed lines just before last light every day.

k. Sentries are properly equipped, briefed and posted by him or by the section Second-in-Command. Division of responsibility between the two is necessary.

l. Making arrangement for inter trench communications by night preferably by use of line beddings.

m. No one smokes or talks in the defensive position at night, and no movement in view of the enemy is carried out by day.

n. Clothes and equipment are not littered in the area.

o. Rest area is close to the position and properly made particularly against snakes.

p. Anti malaria precautions and other hygiene measures are observed by each individual.

q. Sick men are promptly reported upon.

r. No one leaves the position without prior permission of platoon/ company commander.

Duties of Second-in-Command

6. **Company Second-in-Command**. Appointment of an exclusive company Second-in-Command does not exist in the Table of Organization and Equipment (TO&E) of an infantry battalion for performance of routine peacetime duties. However, in the battlefield environments, senior most platoon commander in the company will take over the responsibilities of company Second-In-Command to assist the company commander in planning and taking over the company in case company commander become a casualty. It is therefore, necessary that company tactical headquarters be located in the area of senior most platoon commander or the platoons be so sited in defence that senior most platoon commander is available to the company commander at a short notice. Following duties will be performed by the senior most platoon commander while acting as company Second-in-Command:-

a. Remains available to the company commander in the process of planning and coordination.

b. Ensures organization and efficient functioning of the company headquarters and the company command post in the absence of company commander.

c. Takes over as company commander in case company commander becomes a casualty.

d. Inspecting stand to of platoons other than those being inspected by the company commander.

e. Leading important patrols if ordered.

f. Ensuring that the Observation Posts function efficiently and manning the same periodically.

g. Organising routine in line.

h. Any other task assigned by the company commander.

j. Manning the company headquarters/command posts in the absence of company commander.

7. **Platoon Second-in-Command**. Platoon Second-in-Command will be responsible for:-

a. Bringing the platoon to the platoon defensive area from the company RV.

b. Preparation and organisation of platoon command post.

c. Organising administration of the platoon.

d. Lead important patrols/listening posts if so ordered.

e. Man platoon command post in the absence of the platoon commander.

8. **Section Second-in-Command**. Apart from remaining prepared to take-over command of the section in case the section commander is a casualty, the section Second-in-Command will be responsible for:-

a. Assisting section commander in completing the section tasks.

b. Leading patrols and commanding listening posts if so ordered.

c. Posting sentries.

d. Commanding designated light machine-gun group.

e. Relieving section commander for rest and bathing et cetera.

Duties for other Appointments

9. **Company Senior Junior/Commissioned Officers**. He is basically responsible for administration, morale and welfare of the men of the company. His duties are:-

a. Ensuring timely and proper delivery of food and water.

b. Ensuring that latrines have been dug and adequate facilities for bathing and washing are available.

c. Ensuring that men receive mail regularly and system of despatch of private letters is functioning properly.

d. Assisting company commander in censoring letters if censorship has been ordered.

e. Ensuring that sick soldiers are not kept in the platoons unnecessarily and that they receive timely medical aid.

f. Handling wounded and dead and ensuring their proper evacuation/burial.

g. Visiting platoons to ensure efficient functioning of administration and looking for symptoms of low morale.

h. Visiting B echelon to check on its proper functioning.

j. Any other task assigned by the company commander.

k. Ensuring that men receive their pay and adequate arrangements for sending money orders are available. Also ensure that soldiers have completed their nomination forms and family allotment forms.

10. **Company Havildar Major**. He is primarily responsible for the company F echelon and alternates his duties with the pay NCO. He stays in the company headquarters unless otherwise ordered. His duties are:-

a. Organisation and preparation of company headquarters defensive position including camouflage, concealment and stand to.

b. Briefing all runners and ensuring that company/platoon runners know the location and routes to platoon command posts and Observation Post.

c. Ensuring timely and adequate supply of ammunition and other defence stores. He should replenish company F echelon truck from battalion A echelon truck without loss of time.

d. Organise sentry duties at company headquarters.

e. Assisted by pay NCO in handling Prisoners of War particularly in their safe custody and evacuation.

f. Be on the look out for signs of any loose discipline amongst troops by visiting platoons occasionally.

g. Earmark car park and ensure track discipline and proper camouflage of visitor’s vehicles.

h. Check soldiers going out of and coming in the defensive position and inform company commander.

j. Establish check post, see para 27.

11. **Company Quartermaster Havildar**. He stays with the company B echelon and is responsible for:-

a. Organising of proper functioning of company B echelon including its security, camouflage and concealment.

b. Delivery of food, water and mail at the appointed time and place.

c. Provisioning of all quartermaster stores, less ammunition, but including POL to the company/platoon on demand.

d. Arranging for armourer and vehicle mechanic services as and when needed.

e. Arranging for washing of clothes by the washerman. He collects clothes in platoon bundles and delivers to the washerman at B echelon and washed clothes in similar manner while delivering food.

Company/Platoon Tasks

12. In the preparation of a defensive position company/platoon has to carry-out a number of tasks. Unless these tasks are equitably distributed, the completion of defence in the stipulated time may be difficult. Tendency to draw manpower from the platoons on minor pretexts should be ruthlessly curbed. Company/ platoon tasks may include:-

a. Sentries.

b. Patrols.

c. Digging.

d. Wiring.

e. Mine laying.

f. Setting trip flares.

g. Helping supporting detachments in platoon locality.

h. Battalion/company tasks, regimental aid post, command posts and Observation Posts.

13. Much of the effort can be saved if NCOs and headquarter personnel also help in digging and priorities laid down by the company/battalion commander are followed strictly.

Battle Drills (SOP)

14. Good battle drills or SOPs are means of achieving efficiency. Actions which do not vary with ground, enemy situation and time can be made standard and practiced in training. Simple battle drills or SOPs help in the initiation of correct action by subordinates in the absence of orders and are thus of great help in operational environments. While most of the SOPs are normally contained in the standing orders for war, a few simple ones are given below for guidance.

15. **Alarm Scheme**. While the position is being dug, there must be an alarm scheme, which must be rehearsed. Troops take-up firing positions in or near their own fire trenches depending on the progress of digging on the given signal. Men working away from the section position are also told what to do in case of alarm. Normally they should rush back to their section position. At least one light machine-gun per section should be placed behind a cover to bring fire immediately.

16. **Digging**. Apart from officers and junior commissioned officers, who have other jobs to perform, every one should take his share of digging. Fire trenches of appropriate type should be selected and dug. Various types of fire trenches are given in Annex 'Q' but these are for guidance only and should not be followed if they do not fit in the situation/ground. Platoon/section commanders should invariably measure range of important and easily identifiable landmarks from their position by pacing so that the enemy is engaged accurately.

17. **Digging Time**. Any yardstick for digging times is liable to variation owing to the following factors and the motivation of the men:-

a. Type of ground.

b. Aids and stores available.

c. Weather.

d. Day or night.

e. Ease of concealment.

f. State of training.

18. However, the following are planning times for infantrymen digging without any explosive or mechanical assistance:-

Serial	*Work to be completed*	*Planning Times*	
		Easy Soil	*Difficult Soil*
a.	2 men fire trench	½ -2 hours	2-4 hours
b.	Additional overhead protection over fire trench	1½ hours to 2½ hours	3 – 5 hours
c.	2 men fire trench with overhead protection and shelter	3-5 hours	5-9 hours

19. **Clearance of Filed of Fire**. Only those bushes, plants or obstructions should be removed which affect observation, and fields of fire in the arc of the

weapons. Indiscriminate removal of bushes for the sake of fields of fire and cutting of trees/branches for overhead protection will affect the camouflage and concealment adversely.

20. **Personal Equipment**. During digging personal equipment should be disposed as follows:-

a. Items needed for fighting such as weapons, pouches and steel helmets must be kept in front or beside the trench within easy reach and taken with the man whenever he leaves the section position. Automatics and rocket launchers will not be taken instead rifle of number two will be taken in such cases.

b. Other items of personal equipment must be concealed in a place where they can be found easily in dark and where they will not be covered by soil.

c. After the trench is ready, items of personal equipment not needed for battle and the digging tools should be kept inside the trench.

Priority of Work

21. The sequence of work will vary according to the situation. The following table should be taken as a guide:-

Serial	*Task*	*Stores*	*Remarks*
a. Digging	(1) Automatics and sentries are placed to cover likely area of attack.	Picks, shovels and camouflage nets.	Initial area will be selected by company/ platoon.
	(2) Site mounds or high parapets as required for protection from direct fire.		Section commanders while exact location will be selected by the respective weapon commanders.

Serial	*Task*	*Stores*	*Remarks*
	(3) Plan in advance, the position of shelter trenches and communication trench.		
	(4) Decide on concealment and track plan.		
	(5) Clear immediate fields of fire.		
	(6) Dig fire trenches to a depth of one metre.		
	(7) Clear full fields of fire.		
	(8) Complete fire trenches and parapet.		
	(9) Camouflage trenches.		
b. First Improve-ment	(1) Revert if necessary.	Angle iron pickets or timber as supporting strut, the structure being held up by sandbags or ammunition boxes.	
	(2) Construct overhead protection elevated over fire trench.		
	(3) Dig shelter bay.		

Serial	*Task*	*Stores*	*Remarks*
	(4) Camouflage shelters.		
	(5) If rain is expected, dig catch-water drains above positions.		
	(6) Dig drainage sumps and slope floors.		
	(7) Make ammunition recesses in fire trenches.		
	(8) Begin crawl trenches to join up fire bays as necessary.		
c. Further Improve -ments	(1) Continue and deepen crawl trenches.	Duck boards or planks and small boxes.	
	(2) Improve drainage and flooring if necessary.		
	(3) Construct sentry embrasures, if necessary, and camouflage.	Half-round embrasure or culvert sections, sandbags and pickets.	

22. **Mine Laying and Wiring Parties**. Mine laying and wiring parties which often work near the enemy must take the following precautions:-

a. They must have local protection.

b. They must work quietly.

c. As few men as possible should work on enemy side of the fence.

d. Arms, pouches and respirators, if issued, should be at hand for immediate use.

e. Men not working should be disposed so as to give added protection.

23. **Communications**. The main means of communications are:-

a. Radio.

b. Line.

c. Light signal.

d. Runner.

e. Field signals.

f. Whistle.

g. Voice.

24. The radio link to the company commander is so important for the conduct of defence that radios must be protected and remote antennas should be used to improve communications. When line is laid, it must be buried and if possible duplicated, so that if one line is destroyed by gunfire an alternative route exists. Line must always be duplicated by radio communications.

25. Light signals can be easily missed and are not entirely reliable. They must be understood by the section commanders and sentries. Defensive Fire (DF) can be called for by light signals. Control of illumination will be centralized at company headquarters. Runners must know the routes between section positions and platoon headquarters and to company and battalion headquarters. They must observe strict track discipline. However, when battle is joined, runners are liable to become early casualties.

26. Platoon and section commanders must ensure that their men can be stood to silently in case of alarm and that sentries are able to alert their section commanders quickly and quietly. Some form of wire or string must be positioned from trench to trench within a section area and from the section commander's trench to the platoon commander's trench. A system of tugs can be used to pass the alarm and other signals. A whistle can usually be heard over the noise of battle and can be a valuable means of communication. While wearing the NBC suits, only field signals will be used to convey the massages.

27. **Check Post**. Forward company deployed astride a road/track must establish a check post on the road/track in line with the depth platoon displaying. “No vehicles beyond this point" sign. If possible a barrier should be erected. Regimental police personnel if available should be used at this check post. Company havildar major is responsible for sitting the check post.

28. **Movement in the Defences**. No unnecessary movement within the defensive position or to and from the defensive position particularly during day should be allowed. When absolutely necessary, following procedure should be followed:-

a. *Commander*. Should invariably be accompanied by wireless set operator and the runner. Seconds in command remaining present in the command post knowing at all times knowing the location of the commanders.

b. *Others*. Use only the specified routes, always carrying their weapons, ammunition and steel helmets. Movement of single man should not be allowed at night.

29. **Camouflage and Concealment**. Camouflage of the position is freshened every day before first light. Fresh tracks if made are obliterated as soon as possible. Metallic objects are not allowed to remain littered in open. Tents/ bivouacs if used are pitched after last light and removed before first light. Returning patrols are given the tasks of locating weaknesses in the camouflage.

30. **Report and Returns**. Reports and returns required by the battalion headquarters are contained in the standing orders for war. Some of the most common report and returns are given below which are normally passed on telephone. Formats for written reports can either be found in “Staff Duties in the Field 1995, GSP – 1620” or “Administration in the Field, GSP-1044”:-

a. *Situation Reports.* Twice daily at timings laid down by the battalion headquarters.

b. Special Situation Report. On occurrence.

c. Battle Casualty Report. Daily in the evening.

d. Discovery of Blinds. On occurrence.

e. Ammunition Expenditure Reports. On occurrence.

f. *Vehicle Casualty Return*. On occurrence.

g. *Fighting Strength*. Daily in the evening.

h. *Sick State*. Daily in the evening.

j. *Prisoners of War*. On occurrence.

k. *Captured Material*. On occurrence.

l. Patrol Reports**.** On occurrence.

m. Hostile Air/Artillery Shelling/Mortaring Report. On occurrence.

Routine in Lines

31. Having prepared the defensive positions, the defender has to wait for the initiation of hostilities by the attacker. This period of waiting may at times extend to weeks and months. It is, therefore, necessary to follow a routine whereby the troops do not become stale and high degree of battle worthiness is maintained at all times. The routine in line will consist of the following activities:-

32. **Information**. Between battles, the constant requirement is to obtain information and at the same time deny information to the enemy. The usual ways of doing this are:-

a. Observation from Observation Posts and by sentries.

b. Patrolling.

c. Shelrep/Mortreps.

d. Denying information by:-

(1) Good camouflage, concealment, track discipline and strict control on movement in the defences.

(2) Strict fire discipline.

33. **Arms and Equipment**. Due to constant handling/use under unfavourable conditions, arms and equipment develop defects and deteriorate rapidly. In order to maintain the highest degree of battle efficiency, following measures should be taken:-

a. Daily cleaning and maintenance of arms. Time for cleaning automatic and heavy weapons to be staggered.

b. Periodical test by firing in rear areas, and bore sitting of recoilless rifle and rocket launchers.

c. Periodical visit by unit armourer, equipment and boot repairer, tailor and vehicle mechanic/electrician for inspection and repair.

d. Periodical inspection by company/platoon commanders of arms and equipment to ensure serviceability.

34. **Inspections**. Inspection by company/platoon commanders help to ensure a high standard of readiness. Routine inspections should include arms, ammunition, communication equipment, fire control instruments, tools, clothing, personal cleanliness of troops and the general cleanliness of the area.

35. **Rest**. No one can give his best for long without sleep. Company/platoon commanders must always try to see that they themselves and their men get enough rest. The only way to ensure this is to organize rest in the same way as any other duty is organized. Rest will mostly be taken in day time in properly prepared rest areas. Rest areas should be adequately comfortable and close to the positions.

36. **Hygiene and Sanitation**. By virtue of prolonged stay in the defence, hygiene and sanitation generally suffers and diseases spread. To keep the troops in good state of health company/platoon/section commanders should devote extra attention to this aspect, particularly when the stay prolongs. Following point should be borne in mind:-

a. Latrines should be located adequately away from the positions but not so far away that troops avoid using them. These should be properly dug and frequently changed.

b. Bathing and washing arrangements tied up preferably in the area of depth platoon/company headquarters. If water is not available in the area small parties should be sent to the nearest water source in the rear according to a programme for this purpose.

c. Company quartermaster havildar should collect dirty clothes and ensure quick delivery after washing by the washerman located in B echelon.

d. Company senior junior commissioned officer should visit platoon areas periodically and arrest any signs of bad sanitation.

e. Sick soldiers should be provided with prompt medical attention.

f. Anti malaria and anti snakebite precautions should be strictly followed.

g. Regimental Medical Officer should inspect the troops and company/platoon areas periodically to check on health of troops and sanitation of the area.

37. **Stand to**. Stand to timings are given by the battalion headquarters. However, the company/platoon commanders can order their company/platoon to stand to in order to check that the troops are fully alert to meet an attack. Matters which the company/platoon commanders should check on stand to are:-

a. Arcs of fire.

b. Fixed lines.

c. Sentries challenging procedure and passwords.

d. Range cards.

e. Communications.

f. Storage and cleanliness of equipment.

g. Dress.

h. Camouflage and concealment.

j. Weapons and ammunition.

k. Cleanliness of men and area.

l. Welfare of each man by a quite word of encouragement.

38. The duration of stand to and its timings should be varied. Stand down must be staggered within sections i.e., all sections must not stand down at the same time.

SECTION 76 **– CONDUCT OF DEFENCE**

General

1. The conduct of the defensive battle will depend more on the enemy actions, and concept of the commander of the main positions. However there will always be three distinct phases of a defensive battle:-

a. *The Preparatory Phase*. This is when the attacker is assembling his force and making preparations for the assault and the defender is preparing his defensive positions.

b. *The Assault Phase*. This is when the attacker is attempting to break into and through the main positions and the defender seeks to repel the attack, seal off any penetration and stabilize the situation.

c. *The Counter-Attack Phase*. This is when the enemy attack has lost its impetus and the moment has come to restore the position or to resume the offensive.

The Preparatory Phase

2. This phase is characterised by intense activity for the defenders who will be striving to complete the preparation of their position, gaining maximum information about the enemy, delaying and harassing him to gain moral ascendancy. Much of the activities of this phase have been covered in sections 73 and 74, while a few additional activities are given below:-

a. *Collection of Information.* This will be done by patrolling, continuous observation, commando operations, ground and air reconnaissance and air photos. All these activities will be co-ordinated at battalion/brigade level but the rifle company/platoons will be responsible for carrying out patrolling and commando operations. Daily observations will be submitted to the battalion headquarters usually in the form of situation reports, where these will be collected and interpreted. Company/platoon commanders should remain mentally prepared to send out patrols and plan the preparation work accordingly.

b. *Domination of No Man's Land.* Despite the protective detachments operating ahead of the main positions, enemy patrols will break into gain information. This information will have to be denied to the enemy by dominating unoccupied area ahead of the main positions by patrols and snipers in such a way that movement of enemy patrols and reconnaissance parties is made hazardous. This activity will also be coordinated at battalion/brigade level.

c. *Stay Behind Parties*. During preparatory phase, stay behind parties will be sent across the main defences/international border. These will give early information about enemy's intentions and at the same time can cause some destruction.

3. During the preparatory phase the battle is fought entirely ahead of the main positions by a combination of the following:-

a. Protective Detachments.

b. Demolitions, nuisance mines and route denial.

c. Air reconnaissance and interdiction.

Settling Down

4. Detailed planning, final adjustment of positions and coordination of fire can be made only when troops are on the ground. Battalion and company commanders will personally inspect every position and the sitting of every supporting weapon. Artillery and mortar tasks must be checked and, whenever, possible, registered without revealing the plan to the enemy. Obstacles must be inspected and the fire tasks covering them perfected.

5. Every commander and man must know the plan and his own job; he must have confidence in himself, in troops and weapons in support of his defence. Commanders at every level must be satisfied that they cannot be surprised by day or night, and that enemy reconnaissance patrols cannot approach or observe their positions. When the situation permits, commanders must go out in front of their positions to see themselves weaknesses in defence from the enemy points of view, as this may disclose a vulnerability in their plan, which has been overlooked.

6. All means of intercommunication must be checked. Arrangements for maintenance of line must be practiced. As far as possible, telephone wires should be buried enough. The system of administration must be adjusted as necessary. It must cover all detachments from supporting arms in the locality.

7. **Harassing Fire**. The objects of harassing fire are:-

a. To cause enemy casualties.

b. To lower his morale.

c. To disturb his rest.

d. To interfere with his administrative routine.

e. To interfere with the concentration of men and material.

8. Harassing fire will include the fire of all weapons. Requests for day and night fire programmes are made by battalions and coordinated not below brigade level. Stereotype methods, which allow the enemy to predict when and where he will be hit next, be avoided.

9. During the preparatory phase the battle is fought entirely ahead of the main position by a combination of the following:-

a. Protective detachments.

b. Demolitions, nuisance mines and routes denial.

c. Harassing fire.

d. Deceptive measures.

e. Air reconnaissance and interdiction.

The Assault Phase

10. The assault phase begins when the enemy's advance has been halted by the main position. At this stage practically all the protective detachments except a few patrols will have fallen back and most of the lanes through minefields and other obstacles closed. After contacting the main position enemy's patrolling and reconnaissance activities will increase. He may also launch a series of probing attacks to gain information and feed the strength of the defences. All these activities should be firmly and decisively frustrated by application of minimum firepower preferably from positions other then the original. It will be difficult to differentiate between probing and the main attack. Company/platoon commanders should therefore, exercise strict fire control so that they do not give away their positions or run out of ammunition at the critical time.

11. Following measures may be taken to beat off an attack:-

a. Call for DF as soon as the enemy is sighted or positive information of his assembly at some place is available. DF should not be called if the enemy sighted is in small number.

b. Open fire with a few weapons (GLHMG, MG and RR) when the enemy comes in range and watch his reactions. At night fire should not be opened unless enemy's approach is confirmed by patrols/ listening posts.

c. If the enemy takes cover or withdraws, it may not have been the proper attack. Change the weapon position if it has been positively located by the enemy, otherwise continue occupying it. Changing of positions is not necessary at night.

d. If the enemy presses on, continue intermittent engagement with selected weapons (GLHMGs/MGs) and call for appropriate DFs, while holding fire of rest of the weapons. If enemy approach is confirmed, continue fire on fixed lines in short bursts and illuminate the area intermittently to watch enemy's progress.

e. When the enemy reaches 500 metres line, open up with LMGs also and continue engaging till the enemy is forced to take cover or withdraw. Riflemen keep engaging opportunity targets while recoilless rifles and rocket launchers engage tanks/vehicles and

grouped personnel as they come in range. Artillery/Mortars DFs continue with appropriate adjustments.

12. Despite the defender's best efforts, a determined enemy will at some stage be able to press on his assault and some forward positions will probably be penetrated or overrun. A platoon or even a section locality, which may be intact wholly or partially, must continue to fight and bring concentrated fire on the company so long as it can, even though it may have been by passed or isolated. In such a situation it may be necessary to shift weapons to alternative positions. After the initial onslaught has passed, such localities will provide the base for readjustment/reinforcement of the defence or launching the local counter-attack. If the platoon position is penetrated, the adjacent positions must hold fast and counter/contain the penetration by fire and readjustment of defences. Then if the situation permits, depth platoon will be launched either to occupy counter penetration positions or eject the enemy by a local counter-attack. All readjustments in weapon positions/defences will be done under orders of the company commander.

13. The decision whether to launch counter-attack or occupy counter penetration position will be dictated by the following factors:-

a. *Importance of the Area Penetrated*. It may often happen that the enemy gets possession of an insignificant area either deliberately to deceive the defender or just to occupy territory because the opportunity existed. Company commander should be careful to use his depth platoon only on those areas which are important to his mission and not dish away his strength on unimportant objectives.

b. *Size of Enemy Penetration*. The extent of the area occupied by the enemy and quantum of troops in that area is yet another consideration which will dictate the feasibility or otherwise of launching the counter-attack. While strength of the enemy may not be a serious disadvantage, the physical extent of the area, may be a factor against launching a counter-attack.

c. *Time*. If the enemy has had sufficient time in the captured area to reorganize and dig in, a counter-attack may be futile. Unless the counter-attack can be launched within an hour of enemy occupation of an area, it will more often be doomed to failure.

d. *Vulnerability of Troops*. Troops after leaving their prepared positions will be most vulnerable to enemy artillery and small arms fire particularly in day time. Company commander should consider

this aspect as against the advantages likely to accrue by launching the counter-attack.

e. *Weakening of the Defensive Positions*. Removal of troops from depth localities will weaken the defensive position and may lead to speedy disintegration of the defences particularly by day. Company commander should carefully weigh the need for counter-attack against the possible dislocation and weakening of the defences, especially when the counter-attack is not one of those planned and rehearsed.

14. It must be borne in mind that the success of a counter-attack is not dependent upon the numerical superiority but on speed/violence with which it is launched. A handful of men with even limited fire support taking advantage of the confusion that will prevail amongst enemy ranks at the end of an attack, can drive out an adversary many times larger than their size.

15. All along the assault phase, platoon/company commanders should keep their superior headquarters fully informed. Timely passage of accurate information is necessary for initiation of appropriate action by the next higher commanders.

Counter-Attack Phase

16. The counter-attack plan is conceived and integrated in the overall defence plan of the battalion commander, right from the start. The battalion commander must keep a flexible mind and carefully decide the correct moment for launching his counter-attack force, to ensure decisive results.

17. The success of local counter-attack at battalion level depends upon the speed and surprise. The emphasis therefore, should be on speed rather than the quantum of force.

18. The form of local counter-attack most likely to succeed is spontaneous attack with bayonet against local and small scale penetration of forward localities. Such an attack depends on the circumstances, at the time and the initiative of the local commander.

19. Troops, however, are required to be instantly available for immediate local counter-attack against enemy penetration. Where possible, certain platoons should be earmarked in the company plan for counter-attack roles, but platoons so earmarked will usually have the primary role of holding a part of the battalion locality. Once they become involved in the firefight to defend the locality, they

are not likely to be able to launch a counter-attack. Smaller the force to be launched, the quicker the counter-attack will be conducted, an inspired rush by a few courageous men is likely to met with more success than a carefully prepared but a delayed counter-attack.

20. It should not be a matter of course for a battalion commander to counter-attack with a depth company. If a commanding officer appreciates that a threat is purely local, there are occasions when a depth company, not yet committed to the battle, can be used effectively to counter-attack and restore the situation. Depth companies are committed to the defence of important ground, and to commit them to counter-attack in the face of serious enemy threat may weaken the defence by abandoning ground of tactical importance. Therefore commander must take appropriate decision as the situation dictates.

21. It is neither possible nor prudent to lay down the unit commander's exact capability of local counter-attack against various dimensions of the enemy penetration. Since speed and vigor are the essence of a local counter-attack, hence a bold battalion commander may be able to restore even an exceedingly adverse situation. Certain guidelines regarding the reaction ability of company/battalion commander under some situations are given below:-

a. Loss of section or two in a forward company locality should be restored by the company commander by launching a quick counter-attack with depth platoon. Since the enemy penetration is being checked by the depth section of the platoon under attack, the company commander has reasonable chances of success against such penetration. Counter-attack against the loss of an entire platoon locality by the company commander would be a difficult proposition. He must, however, take measure to check the penetration by adjusting the defence/shifting into alternative positions to facilitate the counter-attack by the battalion commander.

b. As above, the battalion commander may launch a counter-attack against the capture/penetration of up to one or two platoons in any one of his forward companies. Since the loss of an entire forward company would entail induction of fresh troops by the enemy, the chances of success of a local counter-attack against such a situation (being late) would not be compatible. The battalion commander, therefore, may decide to occupy a counter penetration position to check further build-up/exploitation by the enemy. Availability of a troop of tanks, however, considerably; adds to the counter-attack

capability of the unit commander and can be effectively employed to eliminate such penetration.

c. Once the platoon/company commander has stabilized the situation after carrying out readjustment of defences, and the enemy's penetration contained, the company/battalion commander must decide to immediately launch his counter-attack before the enemy follow up troops are inducted. The company local counter-attack in all probabilities must be mounted within 30 minutes. Similarly, the battalion counter-attack must be launched within an hour, if it has to produce tangible results.

Counter Penetration

22. **Definition**. Counter penetration is defined as physical action taken by a defender to halt further penetration into his defence by the enemy and to cause maximum casualties/attrition by fire to enemy force which has penetrated his defended localities.

Concept of Counter Penetration

23. Counter penetration will generally precede a counter-attack by the next higher echelon. Careful planning and timely occupation of counter penetration positions will not only limit enemy penetration but will also contribute towards the success of counter-attack.

24. A counter penetration position is a deliberately prepared position which is sited in sufficient depth. It should not get simultaneously involved with the battle of forward defended localities. On the contrary, it should be adequately displaced to induce an additional phase in enemy's operation, thereby depleting him of his strength to continue the operation and also to disrupt his time and space dimensions. Thus, allowing time for own counter actions against him. The main philosophy and advantages that can be accrued from occupation of counter penetration position are:-

a. To stop further enemy penetration.

b. To disturb and finally defeat the enemy attack plan.

c. Restrict space for further enemy operation and his subsequent phases of attack.

d. To provide opportunity to the next higher commander to launch counter-attack.

25. **Readjustment vis-à-vis Counter Penetration**. It is pertinent to clarify that readjustment should not be mixed up with counter penetration. Readjustment is within the defences for which alternative positions are dug. Readjustments are normally carried out before the assault or any penetration. Whereas, counter penetration positions are occupied only when the enemy has over run a portion of forward positions or has achieved penetration within the defences. Occupation of counter penetration positions by the local commander will facilitate launching of counter-attack by the next higher commander.

26. **Counter-Attack vis-à-vis Counter Penetration**. Due to obvious advantages, the foremost choice of any company commander should be to launch a counter-attack. However, if opportunity of launching the counter-attack is lost then he can resort to counter penetration. Some of the reasons which may force a commander to go for counter penetration include:-

a. The enemy has achieved a sizeable penetration which is beyond the capability of a local counter-attack.

b. Lifting of troops for local counter-attack may seriously jeopardize/weaken the defences.

c. Time and space dimensions do not warrant launching of a counter-attack.

27. **Occasions**

a. Enemy has achieved penetration, which is beyond the capability of local commander to eliminate by counter-attack.

b. To stop enemy's move/flow for destruction by the counter-attack force.

c. The time and space dimensions do not warrant launching of counter-attack by the reserves.

d. As part of overall design of battle to commit enemy infantry and separate his armour for destruction piecemeal.

e. There may be occasions when appreciation of a commander goes wrong and enemy achieves complete surprise, for which a commander is not prepared. In such an eventuality, troops will have to be moved to an unprepared counter penetration position. For occupation of an unprepared counter penetration position, commander along with his subordinate commanders will have to move earlier than the main body. They will carry-out a quick reconnaissance and tasks/areas will be given to subordinate

subunits there and then. In such case, company guides will accompany the O group to guide their companies to their locations on arrival. Central guidance arrangements will have to be made by the battalion. Time permitting, troops must dig down, otherwise troops will have to take position in open to contain enemy penetration. If tanks are made available for such a task, they could immediately be deployed ahead of counter penetration position to gain maximum time for preparation. Moreover, by this time, composition of the attacking enemy will become clear and tanks could then be deployed to guard against the likely armour threat.

Conduct

28. Within the framework of an infantry battalion, counter penetration positions are planned to be occupied by depth companies or by troops in areas likely to be least committed in battle. These companies when asked to perform a counter penetration role would continue to perform their original task with minimum of one third of the force in situation.

29. Once the enemy attack materialises and the forward locations have been over run, the battalion commander will be in a position to correctly assess and visualise the size and magnitude of threat. He can judge and decide whether the depth platoon(s) of forward companies will be able to halt the attack for sometime or not. If in his opinion, the depth elements will be able to check the penetration, then he must put his counter-attack plan in execution, otherwise, he orders occupation of appropriate counter penetration position(s) to facilitate the brigade counter-attack. Although this is a difficult decision but still the battalion commander due to his foresight and experience should be able to take a correct and bold decision.

After the Battle

30. It is a normal tendency on the part of troops to relax immediately after a battle. This must not be allowed to happen. The platoon commanders must immediately make a round of their platoons and take stock of what needs to be done. Some of the points which may require their immediate attention are as follows:-

a. *Replenishment of Ammunition*. Till this is done, redistribute the existing stock and ask riflemen to hand over their ammunition to the LMG numbers. Also strip every dead/prisoners of their ammunition.

b. *Replacement/Shifting of Weapons*. This may become necessary either because on LMG of a forward section has been knocked out of action or if it is felt that the enemy has pinpointed the location of one of the recoilless rifles/MG positions. In the first case, the platoon commander should replace the LMG with one from the depth section and should put in an urgent demand to his company commander to send him a replacement. In the second case, he may have to move it to an alternative position. For this he will need the company commander's permission, who in turn, will have to clear it with the battalion commander (only if a recoilless rifle is concerned).

c. *Adjustment of Manpower*. In case any one of the section has suffered very heavy casualties, the platoon commander may have to redistribute manpower amongst the sections.

d. *Evacuation of Casualties*. Once the casualties have been evacuated, the dead must be collected and either sent back to the battalion or be given as decent a burial as is possible. Shaheeds should be buried as "amanats".

e. *Remaking of Bunkers*. In case a bunker has been destroyed by enemy shelling or tank fire it must straight away be made serviceable by clearing its loopholes. However, the platoon commander must decide whether he wants to remake the bunker in its present location or at a new location.

f. *Morale*. Although own troops may have won a resounding victory, yet the loss of a section mate will invariably bring sadness to a soldier's heart, therefore, the platoon commander must go and talk to his men, comfort them, console them and ensure that depression does not set in.

Night Operation

31. Sentry duties should be organized on section basis. Personnel of the machine-gun and recoilless rifle or any other supporting weapons within the company area will be grouped by the company commander along with the rifle sections so that the duties are better spread. As far as possible day and night dispositions should be the same.

SECTION 77 – **THE FIRE PLAN IN DEFENCE**

General

1. The aim is to destroy and stop the enemy in front of the Forward Defended Localities (FDLs). It is the responsibility of the battalion commander to provide the maximum weight of fire, both anti-personnel and anti-tank, on the approaches to his locality. This is planned with the advice of the commander of all supporting arms. Super-imposed on the fire plan is the counter bombardment organization.

Platoon Weapons

2. Platoon positions should, if possible, be so sited as to have a field of fire up to 400 metres. Riflemen should be trained to shoot with confidence up to ranges from 300 to 400 metres. Though their most effective killing ranges are 300 metres, and at times, it may well be necessary for them to start destroying an enemy advancing in large numbers at ranges up to 400 metres.

Recoilless Rifles (RRs)

3. A rifle company has two RRs, which are allotted/sited in the locations covering the most threatened approach leading into the defences. Distribution/ allotment of RRs is specified by the company commander in his verbal orders. Responsibility of sitting RRs primarily rests with the company commander in consultation with the platoon commander in whose area these are sited. RRs can effectively engage the targets at a distance from 1100 to 1300 metres. In defence RRs can be effectively employed to:-

a. Destroy the assaulting armour and other vehicles.

(1) Separate armour from infantry.

(2) Owing to their long-range, provide fire cover to minefields and other obstacles.

(3) Can be used at anti personnel role at longer ranges.

b. Provide anti-tank and anti-personal fire support during counter-attack.

c. If sited towards the flanks, cover inter company/platoon gaps.

40 mm GLHMG

4. GLHMG is an extremely effective weapon system against troops/vehicle concentrations. It is so sited that it should defend the most important piece of ground in the company defences. It can destroy targets up to a range of 1750 metres. In defence these can be used as under:-

a. Opens up at longer ranges to disrupt assembly and forming up of assaulting troops before the attack.

b. Ideally suited while launching spoiling attacks.

c. During assault stage, its fire can be combined with other long-range anti-tank and automatic weapons to break the momentum of assault.

d. Provide fire support during counter-attacks.

e. Covers inter company and inter platoon gaps by fire.

f. Can be used while conducting raids and ambushes.

12.7mm MG

5. Can be employed in following roles:-

a. To provide anti-aircraft protection during day. It is quite effective against slow moving aircraft and helicopters.

b. Can engage ground targets and soft vehicles when employed in ground role.

c. Provide fire support during counter-attack phase.

d. Cover minefield and other obstacles.

e. Can be used effectively in raids/ambushes.

Artillery and Mortar Defensive Fire

6. The aim of defensive fire is to break an enemy attack before reaching the assault line. When possible, this will be done by Forward Observation Officers (FOOs) controlling fire by observation:-

a. *Defensive Fire Tasks*. To be able to engage an enemy attack when observation is impossible e.g., in close country, bad visibility, or by night, the commander must pre-arrange defensive fire tasks to cover every possible enemy axis of approach into his

position and also in depth to catch the enemy in possible assembly areas, forming up places and defiles. Close defensive fire tasks should be as near to our own forward positions as safety to our own troops permits.

b. *Defensive Fire (SOS) Tasks*. To get the quickest response to a call for defensive fire, one defensive fire (SOS) task can be selected for each troop/battery or regiment, or in the case of 81 mm mortars, for each platoon. The task chosen as defensive fire (SOS) will normally be the defensive task on the most vulnerable approach. Guns allotted to these tasks remain laid on them whenever they are not otherwise engaged.

Drill for Artillery Defensive Fire Planning

7. The battalion commander, advised by his direct support field battery commander will select close defensive fires in consultation with his company commanders.

8. The battery commander will enter the battalion defensive fire support plan on the task table (PAFZ-6047), have it approved by the battalion commander, and forward a copy each to his regiment commander, adjutant and battery command post.

9. The direct support regiment commander, on behalf of the brigade commander, will, if necessary, coordinate defensive fires near inter battalion boundaries. The brigade commander in consultation with regiment commander, will select defensive fires in depth as necessary, and will initiate or suggest any grouping of targets. He may also recommend the desirable density of fire on selected defensive fires, particularly in the case of grouped defensive fires. The regiment commander will prepare the task table (PAFZ-6047) containing the coordinated brigade defensive fire support plan and forward one copy to divisional artillery headquarters and one to his regimental headquarters.

Armour

10. The tank is designed primarily for aggressive operations and it should not be employed in a purely static defensive role. All defences are, however, temporary and the fore-runner of further offensive operations; it must, therefore, be aggressive. The role of tanks in defence is mainly, therefore, taking part in counter-attack for which they should generally be retained centralized and not split up throughout the positions as local support for units and sub units of infantry. However, when the enemy has temporary superiority, it is probable that

some of the tanks available may be split up, allotted to companies and used defensively, but it is essential that the number so employed should be reduced to a minimum.

11. Penetration by an attacker into any defensive position under modern conditions is frequently inevitable. The defensive position must, therefore, be in depth with suitable areas selected for launching of counter-attack. The majority of the tanks will be held in reserve as part of a counter-attack force.

12. The following factors must be considered while allotting armour in support of infantry units and sub units:-

a. The enemy.

b. Ground, type of country, vital ground.

c. Infantry layout.

d. Tanks available and their layout.

e. Quality of the tanks compared with those of the enemy.

f. Number and quality of own anti-tank guns and mines.

Light Anti-Tank

13. Light anti-tank elements, if allotted, may be employed on the following tasks:-

a. They may be given an independent locality to defend for a limited period when other troops are not available.

b. They may be employed to reinforce or thicken up the battalion anti-tank defence. But it must be borne in mind not to dish them out to platoons and sections in penny packets.

c. Depending on the level of the defensive planning, they may be employed in the reconnaissance elements or as part of the advance positions/screens.

d. They may be employed to cover gaps in the defended localities.

e. May be assigned the task of dealing with any enemy heliborne threat.

f. May be given under command the counter-attack force. In this role they may support the counter-attack from a base of fire, may physically accompany the assaulting infantry or may be employed on flank protection of the counter-attacking force at a higher level.

Air Support

14. In defence, a company commander is less likely to receive direct air support than in attack. The air effort would normally be supplied to indirect support planned at a higher level. However, air support should be considered when planning or rehearsing counter-attack roles. The battalion commander should assist the divisional counter bombardment officer with reports of enemy shelling and with flash and sound bearings; this information may lead to reconnaissance aircraft locating hostile battery positions which may be attacked by our own guns directed from the air or by ground attack aircraft. For this purpose, at least one set of cloth panels must be held by each battalion or if possible by each company for indicating targets to aircraft.

SECTION 78 – **REVERSE SLOPE DEFENCE**

General

1. **Use**. A reverse slope defence is one organized on the portion of a terrain feature that is masked by a crest from enemy direct fire and ground observation. All or any part of the forces on the FDLs may be on the reverse slope, depending on the terrain in the area to be defended and the situation. In mountainous and desert terrain where hills or the dunes are of similar pattern and height, occupation of the forward slope of the next hill/dune from the attacker side also falls in the category of reverse slope defence. In this case, minimum essential elements are deployed ahead of the main defences on the first hill/dune to act as security elements and rest of the force is deployed in the main defensive positions organized on the forward slope of next feature. A successful reverse slope defence depends on control of the crest either by fire or physical occupation. Reverse slope defence may be advantageous when any of the following conditions exist:-

a. When the forward slope is untenable because of enemy fire.

b. When the forward slope has been lost or not yet regained.

c. When the terrain on the reverse slope gives better fields of fire than the forward slope.

d. When necessary to avoid occupying disadvantageous terrain such as a dangerous salient or re entrant.

e. When possession of the forward slope is not essential for observation.

f. When necessary to assist in achieving deception and surprise.

2. **Advantages and Disadvantages**

a. *Advantages*. In a reverse slope defence, the forward elements of the battle area are protected from enemy ground observation and direct fire weapons. Enemy indirect fire weapons normally deliver unobserved fires, which reduces their effectiveness. When the enemy has been deceived about the true defence disposition, he may advance to close contact before he has discovered the battle area. The defender may effectively employ deceptive measures such as dummy positions on the forward slope. These advantages permit greater freedom of movement, more detailed improvement of position, ease of supply, and rest for the troops.

b. *Disadvantages*. The major disadvantages are the difficulty of maintaining observation of the enemy and the restricted range for direct fire weapons. Without observation, the effectiveness of friendly direct and indirect fires is limited. Obstacles and minefields on the forward slope can not be covered by direct fire weapons. If the enemy seizes the crest, he can have the advantage of attacking the battle area moving downhill, while a counter-attack to eject him must move uphill.

Organization of the Reverse Slope Defence

3. **Disposition of Troops**. The reverse slope is generally organized according to the fundamentals of defence. Special considerations include the following:-

a. Observation and protective elements to include FOO are established on or just forward of the topographical crest to give observation over the unit's entire front. These groups may vary in size from two men to a rifle section (usually from the depth platoon). They may be reinforced with machine-guns, anti-tank weapons and tanks. At night, these groups are strengthened to prevent infiltration and surprise.

b. The forward platoons are located within effective small arms range of the crest. They are situated to permit maximum fire on the crest, the approaches around the crest, and the forward slope of adjacent terrain features. A desirable location for the depth platoon is the crest of the next high ground to the rear within supporting range of the forward elements.

4. **Fire Support**

a. MGs and other automatic weapons are placed where they can deliver the most effective surprised fire on the enemy as he crosses the crest. Their sectors of fire provide maximum coverage between the FDLs and the crest. MGs arcs of fire are designated as in the forward slope defence.

b. In the reverse slope defence, anti-tank weapons may be positioned in the depth platoon area to have better advantage than when defending the forward slope. Firing positions on the next high ground behind the forward position often give better fields of fire than positions in the forward area.

c. The depth platoon in a reverse slope defence can often provide more effective fire support for forward platoons.

d. Final protective fires from indirect fire weapons are normally placed along the topographical crest or forward crest of the hill to deny it to the enemy. When the FDLs are as much as 500 metres behind the crest, it may be more advisable to locate the artillery and mortar fire closer to the FDLs than to the crest. In this case, targets are planned on both sides of the crest to keep the enemy under fire as he advances toward the planned artillery and mortar fire.

Conduct

5. The conduct of a reverse slope defence generally parallels that of a forward slope defence. The observation and security groups give warning of an enemy approach and delay and disorganise him with long-range fires. To give a better coverage of the front during periods of darkness or poor visibility, warning and illuminating devices such as trip flares, searchlights, infrared equipment, night vision devices, and mines are used extensively. As the enemy forces the observation and security groups to withdraw, prearranged targets are fired on the forward slope to disrupt the enemy attempts to mass for an assault. Automatic weapons with the security groups are withdrawn first so they can occupy their primary firing positions before the enemy reaches the crest. Direct fire weapons within the battle area withhold their fires until suitable targets appear. The fires of grenadiers are allocated and used on forward slopes and crests. As the enemy crosses the crest, artillery and mortar fires commence. If the enemy makes a penetration, the counter-attacking force destroys or ejects him and re-establishes the security group on the crest.

SECTION 79 – **PLANNING AND CONDUCT OF SPOILING ATTACK**

General

1. A spoiling attack is a tactical manoeuvre employed to impair or delay a hostile attack. It is normally launched while the enemy is in the process of forming up or assembling for an attack. It is executed to:-

   a. Destroy a portion of enemy force.

   b. Throw the enemy off balance.

   c. Deny the enemy surveillance of the FDLs.

   d. Discourage and disorganise all further attempts attack own defensive position.

   e. Gain time and force enemy to disperse his strength for providing protection to his launching area.

Essential Requirements

2. Following points must be kept mind:-

   a. The success of spoiling attack will depend upon the initiative and violent action by the junior leaders. No physical assault on the enemy is envisaged. Only attack by fire is advocated.

   b. The plan will have to be well rehearsed and fire support coordinated. At times the troops may have to be kept in the close proximity of the likely enemy FUPs and assembly areas.

   c. The decision to commit a force in spoiling attack must be carefully considered against the possible piecemeal loss in light of overall mission. At company level, up to a section strength may be adequate.

   d. The plan for spoiling attack will normally not be initiated below company level.

   e. Before launching spoiling attack, it is important to detect, identify and confirm enemy intentions as early as possible.

Early Detection

3. Deliberate effort must be made at all time to detect, identify and pinpoint enemy echelons of attack as early as possible. Some of the means are:-

a. Observation Posts (Intelligence and Artillery).

b. Listening post/standing patrols on likely avenues of enemy approaches.

c. Patrolling.

d. Radio Interception.

e. Air Observation Posts.

f. Interrogation of prisoner of war and civilians of the area.

Conduct and Planning

4. Deliberate pre-planning is essential for launching a successful spoiling attack. Therefore it is essential to:-

a. Earmark troops and commander for each contingency.

b. Rehearse movement and actions during day and night.

c. Integrate all available fire support, for each action.

d. At company level, up to a section with automatic weapons (MGs & GLHMG) 60 mm Mortar, RPG-7 et cetera may be employed. At times standing patrols may be assigned to take on the spoiling attack.

SECTION 80 – **HOSTILE SHELLING, MORTARING AND BOMBING REPORTS**

Shelling and Mortaring

1. An army in the field is supplied with an elaborate counter bombardment organization, whose sole task is the accurate location and neutralization/ destruction of enemy hostile batteries. The organization can only be fully effective if every man understands the importance of submitting quick and accurate reports. All shelling or mortaring et cetera must be immediately reported, irrespective of whether the shells or bombs are falling in the platoon area or not.

2. When possible, sentries should be provided with a pencil and paper, compass and a copy of the standard index letter code given below, for the purpose of making quick notes.

Index letter Code for Hostile Shelling, Mortaring and Bombing Reports

3. In order to speed up the transmission of reports, the following letter code will be used, preceded by the appropriate code word:-

SHELREP (for enemy artillery fire)

MORTRP (for enemy mortar or rocket fire).

BOMREP (for enemy air attack).

a. From (unit; use current call sign for artillery units or address group for other units).

b. Position of observer (grid reference preferred; encode if sent by wireless. May be omitted by artillery units if position of observer has previously been reported).

c. Grid or magnetic (state which) bearing of flash or sound or groove of shell (omit for aircraft).

d. Time from.

e. Time to.

f. Area shelled, mortared or bombed. May be sent either as: -

(1) Bearing (in degrees or mils) and distance (in metres) from observer.

(2) Grid reference (encode when B is sent in code).

Note: Where method in (a) above is used, maximum accuracy possible is essential.

g. Number and nature of guns, mortars, aircraft or other means of delivery.

h. Nature of fire (registration, bombardment, harassment, et cetera. Omit for aircraft).

j. Time of flash to bang (omit for aircraft).

k. Damage (optional, not normally used. In code if sent by wireless).

4. **Security**. The messages will always be transmitted in clear except:-

a. A (Unit) - for which the current call sign will be, used by artillery units and current address groups in other cases when the message is sent by wireless.

b. B (Position of observer) - which will be encoded when sent by wireless, except when the originator is on the move, e.g., in the case of a reconnaissance patrol.

c. F (Area shelled, et cetera) - which if sent as a grid reference, will be encoded when sent by wireless, except when B is omitted and when it may be sent in clear.

d. K (Damage) - which will be encoded if sent by wireless.

e. Cases when the originator considers that the conditions prevailing warrant a higher classification.

5. If any heading is unknown or is not applicable, the letter is omitted.

6. **Enemy Air Attacks**. For the sake of simplicity, hostile air attacks are reported on the same form as **SHELREPS**, but their purpose and disposal are quite different.

7. **Example**. The following is an example of a **SHELREP**:-

SHELREP (.) A (.) KZ (.) B (.) 467898 (.) (ENCODED) (.) C (.) MAGNETIC 350 DEGREES FLASH (.) D (.) 0803 (.) E (.) 0808 (.) F (.) 463892 (.) (ENCODED) (.) G (.) FOUR FD GUNS. (.) H (.) REGISTRATION (.) I (.) SEVEN HE (.) J (.) 10 SECONDS.

SECTION 81 – **COMPANY/PLATOON COMMANDER'S CHECKLIST**

1. Given below is a suggested checklist for a company/platoon commander.

a. *Warning Orders*. Have you issued this?

b. *Verbal Orders*

(1) *Before*

(a) Have you written down headings for receiving your senior commanders' orders?

(b) Are you carrying your orders card?

(c) Are you carrying the defence appreciation format?

(2) *After*

(a) Are you clear on the location of MGs/GLHMG's of the company?

(b) Are you clear on the location of both the recoilless rifles and bakhtar shikans if allotted?

(c) Are you clear on the location of the other sub unit?

(d) Are you clear on the artillery/battalion mortars/ 60mm mortars/MG DF and DF (SOS) tasks in your area?

(e) Are you clear on the location of minefields and wire obstacles in your area?

(f) Are you clear on the location of battalion/company headquarters, regimental aid post and company aid post?

c. *Reconnaissance*

(1) Have you carried out a thorough reconnaissance of your area?

(2) Have you seen your area from the enemy's point of view?

(3) What are the likely enemy's approaches?

d. *Deployment*

(1) Which ground dominates them by observation and field of fire?

(2) While selecting platoon & section positions have you catered for the following:-

(a) Frontal Attack by the Enemy:-

i. Mutual Support.

ii. Depth.

iii. Which DF tasks are to be fired?

iv. At which point on ground will the MG's/ LMG's/riflemen open fire?

v. When you open fire where is the enemy likely to take-up positions for fire and movement?

(b) Left flanking attack by the enemy. (Same as for frontal attack).

(c) Right flanking attack by the enemy. (Same as for frontal attack).

e. *Rocket Launchers*. Have you selected positions for the rocket launchers to:-

(1) Cover the gap between the recoilless rifles?

(2) Cater for all likely tank approaches?

f. *Anti-Tank Weapons/MGs/LMGs/GLHMGs*

(1) Are these sited in defiladed positions?

(2) Have you selected alternative positions to cater for an attack from either flank or rear?

(3) Are they mutually supporting?

(4) Do they provide enfilade fire?

(5) Have you selected alternative positions to cater for an attack form either flank or rear?

(6) Have you selected their fixed lines?

g. *Company Headquarters*. Have you selected this from the point of view of security, command and control?

h. *Platoon Headquarter*. Have you sited this at a place from where casualty evacuation and replenishment of ammunition can be carried out easily?

j. *Command Post*. Have you sited this to ensure your effective command of the defensive battle?

k. *Obstacles*

(1) Are you laying any minefields? If so then have you decided the following:-

(a) Location to include length and depth.

(b) Type and density.

(c) Who will lay them?

(d) When will work start?

(e) Approximate time when work will finish?

(f) Is this minefield covered by fire?

(g) Are you laying any wire obstacles? If so then have you decided the following: -

i. Location to include length and depth.

ii. Type and density.

iii. Who will lay them?

iv. When will work start?

v. Approximate time when work will complete?

vi. Are the wire obstacles covered by fire?

l. *Trenches*

(1) How many trenches will you dig in each section?

(2) How many trenches will you dig in platoon headquarters/ company Headquarters?

(3) How will you make over head cover and over head protection?

m. *Administration*

(1) Have you selected suitable rest areas for your subunits?

(2) Have you selected suitable sites for your latrines?

(3) Have you tied up the feeding arrangements for your company/platoon?

(4) *Ammunition*

(a) Do you have enough ammunition?

(b) Have you tied up ammunition replenishment procedure for your company/platoon?

(5) *Washing.* What will be the washing and bathing arrangements for your subunits?

n. *Communication*

(1) Has your runner seen the location of battalion, company Headquarter and the other two platoons?

(2) How will you communicate with your platoons and sections?

o. *Movement.* What tracks will you use?

p. *Signals*. What will be the signals for opening fir:-

(1) During Day?

(2) During Night?

q. *Security*

(1) How many sentries will you keep while the position is being dug?

(2) *Check Posts/Radars/Listening Post/Standing Patrols/Tank Hunting Parties*. Have you decided the following:-

(a) Locations for all the security/surveillance elements?

(b) Strength and which section will provide?

(c) Routes to and back?

(d) Time out and in?

(e) What arrangements have you made to close the minefield gaps/lanes after withdrawal of standing patrol/listening posts?

(f) What is your anti-aircraft defence plan?

SECTION 82 – **DEFENCE IN BUILT-UP AREAS**

General

1. Subject to availability of manpower, defence in built-up areas can be organized in two forms:-

a. *Perimeter Defence*. An ideal method, when defence is organized in continuous concentric rings on the basis of strong points each mutually supporting wherever possible. A number of reserves are placed by sector for counter-attack. This indeed, is only possible if adequate manpower is available.

b. *Defence Based on Number of Localities*. As problems of defence are accentuated by restricted fields of fire, which mitigates effective mutual support, the disadvantage can be offset by organizing defence in a number of defended localities, between which infiltration may be accepted with a view to isolating and

destroying the attack. Due to restricted field of fire, the localities must be of limited size and will normally be held by subunits. A reserve for counter-attack at all levels is essential and depending on the availability of troops should be as strong a possible.

Basic Considerations

2. Important considerations to be borne in mind while planning defence in built-up areas are:-

a. *Control*. Lack of communication would isolate subunits to fight independently.

b. *Observation*. There may be a few high buildings, which could be used for long-range observation but generally speaking, the visibility of the immediate front may be restricted by buildings/ houses for which engineer effort may be required.

c. *Field of Fire*. Mutual support between posts difficult and often impossible. Fighting is largely at close quarter. Hence greater and effective use of short range weapons. Also locality should be of limited size.

d. *Difficulty of Locating Fire*. Is increased by the echoing effect of fire amongst buildings.

e. *Movement*. Movement within the built-up areas is either from roof tops or through the walls by making holes in the inter-connecting walls. Necessary reconnaissance must be done to make maximum use of both and denying their use to the enemy.

f. *Cover and Concealment*. It is possible and proper that tank hides are organized to mislead and surprise the enemy by occupying mud houses, whose walls can be demolished for action or a hole made for the use of the tank gun.

Planning

3. Planning of defence in built-up areas require a detailed preparation and judicious use of all available resources on the basis of priorities. In general the planning would involve the following:-

a. *Reconnaissance*. A thorough and detailed knowledge of the area is essential and can only be gained by detailed reconnaissance and a study of air photographs, maps and town plans. All short cuts, concealed entrances to buildings, possible detours and Observation

Posts should be known to all ranks. Routes for use by reserve must be reconnoitred and movement down these routes practised. In addition, the reconnaissance should include discovering of available material which can be collected.

b. *Liaison with Civil Authorities*. Whether friendly or otherwise the civil efforts must be mobilised and exploited to the fullest.

c. *Deceptions*. In front of the forward defending lines, there should be a screen of small mobile detachments, possibly located on the outer perimeter of the town or village. Their tasks will be to give the impression that certain areas are being occupied as main defensive positions. When, in fact, they are not and do contest the enemy's initial bridgehead into the built-up area. Their strength will depend on the size of the town and the force provided for its defence. Withdrawal routes of these parties must be reconnoitred and also the intermediate positions which can be taken up before the final withdrawal into the defended localities. Orders must state clearly the degree of resistance to be shown and who is to control the withdrawal.

d. *Provision and Storage of Ammunition, Supplies and Water*. All necessary arrangements must be made to cater for all possible needs.

e. Efficient traffic control and marking of routes and buildings.

f. *Defended Localities*. Section and platoon post will be grouped to from defended localities, which in turn may be organized as components of a large defended locality or area. In villages, and small towns, perimeter defence is often possible as distances between forward defending lines and reserves are not great. In larger towns, manpower will probably be insufficient to defend the whole perimeter; it will, therefore, be necessary to concentrate on the defence of selected sectors such as main routes into the area and various key points which must be denied to the enemy and which can easily be adopted for defence. The actual sitting of the posts will take into considerations:-

(1) *Field of Fire*. To cover the main killing areas in streets and open spaces.

(2) *Protection*. Bearing in mind the ease with which attacking tanks can demolish all the strongest houses.

(3) *Concealment.* This will lead to a variety of positions being selected. Sometimes posts will be sited in open spaces between houses. At other times, better positions may be obtained by occupying the ground floors of selected buildings with sniper positions on the roofs above. Infiltration between the localities held may have to be accepted and to counter this; each locality must be self-supporting and stocked for defence. The defence, however, must always be aggressive and plans must be made to use reserve against enemy who may infiltrate.

g. *Strong Points*. Large strongly constructed buildings in each defended locality will be converted into strong points and held as the hard core of the defence. From these, fire will be brought to bear on the main approaches to the locality. The main streets and open spaces will be blocked by barricades, wire obstacles and mines, and these will be covered by fire from the strong points. Whenever possible strong points should be mutually supporting and sited in depth; otherwise the enemy will have little difficulty in isolating and dealing with each in detail.

h. *Reserves*. It is essential that the defence is flexible as well as resolute. A mobile reserve must be maintained at all levels. Ultimately, a mobile reserve may have to hold a locality or post in order to meet a changing situation, but its primary role will be to undertake the following tasks:-

(1) Cutting off and destroying enemy penetration by moving to an area from which enemy movement in the open can be engaged.

(2) Restoring a post or locality that has been over run by destroying the enemy in it. Routs for the movement of reserves can be prepared by ‘mouse-holing’ through blocks of houses.

SECTION 83 – **DEFENCE OF BUILDINGS**

Defensive Capabilities

1. A defended house, including its interior and roof must be capable of all round defence. It should, if possible, be inconspicuous. It must be capable of standing up to bombardment. Priority of durability is:-

a. Steel or reinforced concrete frames.

b. Stones.

c. Bricks.

d. Wood – to be avoided whenever possible.

Reconnaissance of a Building

2. Points to look for are:-

a. Observation Posts.

b. Fields of fire with particular reference to the main task.

c. Loop holing required.

d. Strength of building against small arms fire and bombardment.

e. Too many Entrances and Exits are a source of weakness.

f. Communication with adjacent houses and fire trenches in the garden or outhouses, if any.

g. Alternative positions.

h. Water supply – drinking and fire prevention.

j. Material available in the building and a rough estimate of work required to prepare it for defence.

k. Dressing station.

l. Cook house.

m. Latrines.

Local Protection

3. By day, and some times by night, one or two positions may have to be occupied outside to prevent the enemy getting near enough to neutralize defence inside the buildings.

4. Sentries should, if possible, be posted outside the building, especially at night. Fire trenches should be dug for protection of sentries against bombing and as possible alternative positions. By night, small patrols formed by the combined garrisons of buildings of a defended locality may on occasions be used to provide additional protection. Dummy positions may mislead the enemy in his selection of lines of advance.

5. Protection should be afforded inside the building by:-

a. Sandbags, mattress, cupboards, et cetera. instead of walls, particularly those with loophole, and floors which are not bullet proof.

b. All entrances not required for use should be barricaded. Those not barricaded must be prepared for blocking at short notice. Staircases, and some times passages, may require blocking. It may be possible to remove some of the stairs out of a staircase.

c. Communications between rooms and to the roof and cellars must be considered.

d. If sandbags or other protective material are laid on floors, it may be necessary to shore up the ceiling of rooms or cellars below.

e. Water and sand must be available for dealing with fire. All bath basins and buckets should be kept full against burst of mines or fire. Gas and electricity should be turned off at the mains and inflammable material not required should be removed.

f. It may be necessary to demolish certain parts of a building to remove danger from falling masonry or fire to provide overhead protection from rubble once the building has been strengthened.

Sitting of Weapons

6. All weapons should be sited well back from window. Sandbag walls should be erected in occupied rooms, well back from windows. The glass of all the windows should be removed from defended house and from adjacent houses in the vicinity.

7. In general, machine-guns should fire from lower floors to avoid plunging fire; sniper should find positions in upper floors and on roof.

8. Methods of concealment and deception include dummy loopholes, use of lace curtains over window and dummy figures at apertures not in use.

9. Wire netting or similar material should be put over ground floor windows which affords protection against hand grenades.

SECTION 84 – **DEFENCE ON EXTENDED FRONTAGES**

General

1. At times a battalion may be required to deploy on extended frontages when the approaches to be held are widely apart or due to resources constraint. This deals with the problems of defence on an extended front and lays down guidelines to meet such an eventuality. Suggested Frontage are given at Annex 'R' and its Appendices 1, 2 and 3.

2. Problems of Extended Frontages

   a. *Command and Control*. It is affected due to limited signal resources and large distances.

   b. *Protective Detachments*. These will have to be greater in number and mobile, thus would tax the resources of the commander and may affect the strength of main defensive position.

   c. *Reserves*. Size, composition and location of reserves for timely reaction along the extended frontages will be a serious problem. Troops may also be required to take care of the enemy heliborne and air borne threat.

   d. *Insufficient Artillery Support*. Large frontages also create the problems of locating, observation and concentration of artillery fire.

   e. *Increased Threat of Infiltrations*. Large frontages and big gaps will increase the possibilities of infiltration by the enemy.

   f. *Problem of Surveillance*. The absence of any electronic means available with the ground force and limited support of army aviation poses serious problems of surveillance of gaps in the defence.

   g. *Mutual Support*. Since frontages will be extended, the gaps between defended localities will increase, thus direct mutual support will be difficult. Mutual support in such situations will be in terms of the ability of the next higher commander to readjust positions or to reinforce the threatened locality in time to deny the attacker his objective(s).

   h. *Administration*. There will be an increased requirement of administrative support.

Fundamental Considerations

3. Even with extended frontages, the fundamentals of defence remain operative and must not be disregarded. However, while planning the defence on extended frontage, a definite choice has to be made between presenting a continuous front or holding only selected areas in strength with gaps between them.

4. **Enemy**

a. The commander through his intelligence staff/agencies and protective detachment must first assess the nature, quantum, capability and direction of enemy threat in his area of responsibility.

b. Protective detachment employed should be strong enough to inflict maximum delay and attrition on the enemy.

5. **Terrain**

a. The defensive position should enable the defender to canalise the enemy into predetermined areas where he can be destroyed by fires and employment of reserves.

b. In this case the key areas are held in sufficient strength and the enemy is forced into the predetermined killing areas.

6. **Frontages**

a. The limiting factors are terrain, command and control, resources available and the capability of defence at night.

b. Frontages of company, platoon and sections cannot be extended, mainly due to limitation of command and control. However, the frontage of an infantry battalion may be extended by increasing inter platoon and company gaps.

c. Additional positions should be dug at platoon, company and battalion levels, which could be occupied within the defensive location once points of attack have been ascertained.

d. Large gaps between defended localities should be accepted at a level where adequate reserves are available to counter the enemy manoeuvre to exploit the gap(s). Such reserves are generally available at brigade level and above.

e. Light anti-tank (LAT) elements within battalion/brigade defensive locations should be a given sizeable area (relatively less important) to guard.

7. **Depth**

a. With wide frontages, some penetration of defensive position is inevitable. But to prevent the enemy from exploiting the penetration beyond own reaction capability, selected locality in depth should be held firmly.

b. Depth being an important requirement of defence, lightly held positions in depth must have prepared positions for occupation at short notice from the neighbouring locality.

c. Depth at company and battalion level must always be retained. A battalion may, however, normally deploy three companies up.

8. **Mutual Support**. Defensive position consists of a number of mutually supporting defended localities, and the mutual support at the unit/subunit level is defined as "the ability of adjacent localities to engage the enemy in the intervening gap by direct fire". But mutual support in this case can be defined as "the ability of the next higher commander to readjust positions or to reinforce the threatened locality in time to deny the attacker his objective(s)".

9. **Reserves**. At battalion level, it will be difficult to have reserves that are uncommitted. Depth companies may be given the tasks of occupation of counter penetration, positions reinforcement or even launching of quick local counter-attack. Reinforcement at company and platoon level will normally be by weapons.

Planning of Defence

10. The sequence of planning in case of extended frontages remains the same as for area defence. Planning and preparation for defence on extended frontage is a process that should start, and a major portion completed, well before the outbreak of hostilities. A "hasty defence" on extended frontage should be avoided. When sufficient time is available, additional positions at platoon, company and battalion level should be dug which can provide immeasurable flexibility in the conduct of battle specially on a broad front.

Organization and Layout

11. Depending upon the number of approaches leading into the battalion area, the main position will consists of compact company/platoon defended localities

with large gaps in between them. Various combinations/options suggesting frontages for platoon, company and battalion are attached as Annex 'R' and its appendices. These may be applied judiciously according to the varying situation.

Anti-Tank Defence Plan

12. Keeping in view the increased frontages and the threat, the recoilless rifles, Bakhtar Shikans and other anti-tank resources should be judiciously employed and integrated with obstacles. The main defensive position will be built around the anti-tank defence plan. Biases of the obstacle should also be anti-tank oriented.

Communication

13. Signal resources of battalion signal platoon will have to be augmented for effective communication.

Surveillance

14. A comprehensive surveillance plan, integrating all available resources, has to be planned, and executed just like a fire support plan, catering both for day and night surveillance. Field radars four per battalion should be catered for.

Fire Support Plan

15. No change is required from normal procedure except that the DF plan will be made according to sectors and battalion mortars may have to be split and made mobile for quick deployment. Artillery resources will have to be augmented for effective fire support over the extended frontages.

SECTION 85 – **DEFENCE ALONG A WATER OBSTACLE/CANAL/ ANTI-TANK DITCH**

General

1. A series of irrigation canals/water channels exist parallel to our eastern borders which are effective obstacles to an attacker. Where these do not exist, anti-tank ditches have been dug for the same purpose. However, incorporation of these in the defensive position creates some peculiar problems of their own and force the defender to deploy the force in linear fashion.

2. This chapter deals with peculiar problems of defence along a canal/anti-tank ditch and lays down guidelines for effective organization and conduct of defence along such obstacle.

Types and Characteristics of Obstacles

3. **Canals**. There are three main types found in Punjab:-

a. *Link Canals*. (Figure 31A). They were constructed as a result of the Indus Basin Water Treaty for transfer of water from the three western rivers. They are generally not used for irrigation and are the 'incut' type, which is below the ground level. Both the banks are raised by 1-2, metres above the ground level. There width varies from 90-125 metres and depth from 3-5 to metres. Water level may be from 2 to 4 metres. Width of the bank may be as wide as 3 metres.

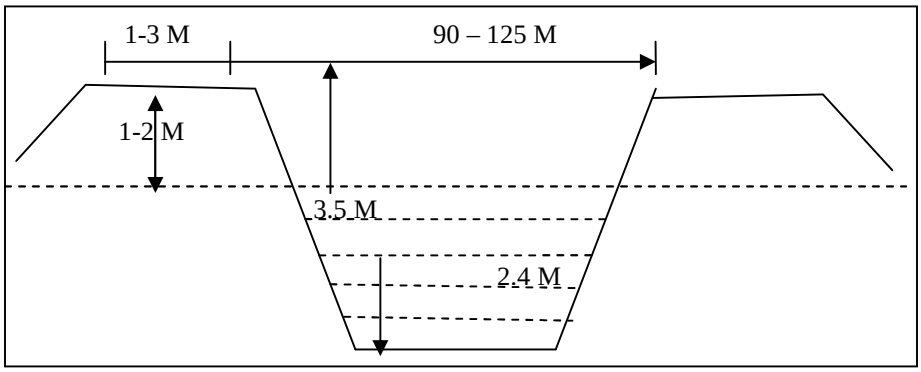


Figure 31 A – Link Canal

b. *Irrigation Canals*. (Figure 31B). They are the in-filled types and raised above the ground level to facilitate flow of water into irrigation channels and minors. Width varies from 60 – 90 metres. Banks are higher than the ground level to prevent flooding of the surrounding area, width of the bank varies from 3 to 5 metres.

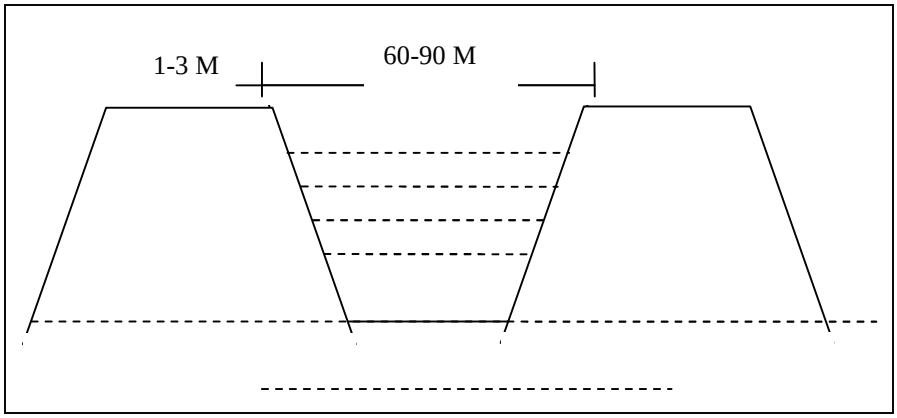


Figure 31 B – Irrigation Canals

c. *Combination of the First Two Types*. (Figure 31 C). They are partially 'incut' and partially 'infilled'. These include some of the link canals also used for irrigation or even some of the irrigation canals. Width varies from 60-90 metres. Depth varies from 3 to 5 metres. Their banks are 2-3 metres higher than the ground level and as wide as 3 metres or more on both sides/ (In case of any of the above canals that are located close to the international border and are to be used for defence purpose, their home banks are further raised by 1-2 metres mainly astride the major enemy approaches to give better observation and fields of fire.

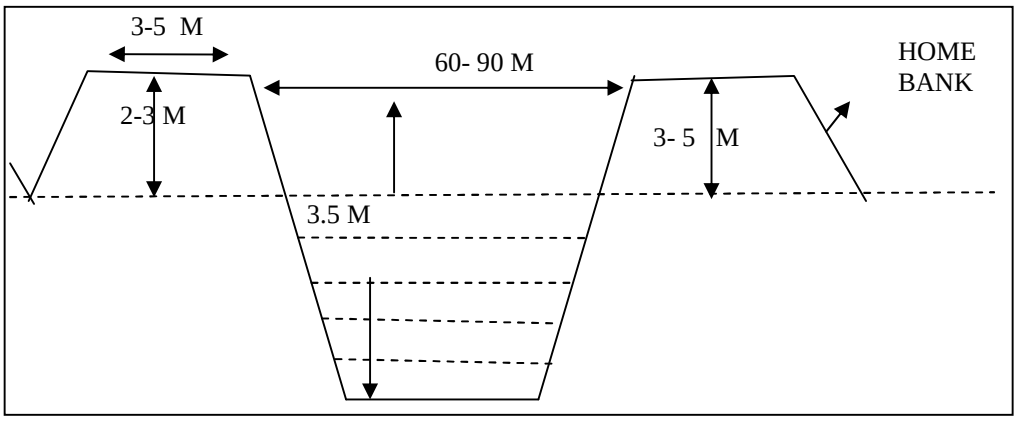


Figure 31 C – Combination of Irrigation and Link Canals

d. There are some other characteristics of these canals which are common to all:-

(1) *Water Level 2-4 Metres*. May be increased or decreased depending on the season and the fact as to who controls the head works and regulators.

(2) *Bed Mostly Loamy*. Can be forded by tanks in some cases.

(3) *Brick Lining*. In some cases brick lining of canal banks has been done.

(4) *Gradient*. Mostly steep where banks are brick lined.

(5) *Current 2 to 3 Knots*. In case of link canals it may be 4 to 5 knots.

(6) *Approaches/Exits*. Marshy at most places due to escaping water channels, particularly when flowing to full capacity.

(7) *Observation and Fields of Fire*. High banks provide good observation and fields of fire but these are poor at some places due to elephant grass and villages. In case both the banks are of equal height, the observation towards the front/and the flank is considerably restricted. Although the field of fire may be clear, yet the fire effect from across the far bank is reduced to some extent due to plunging fire. The problem is more acute if the far bank is slightly higher than the home bank.

(8) *Concealment*. Limited cover from air is available due to lines of trees on both sides. Positions on the canals can not be concealed form the enemy due to their prominent and regular layout.

(9) *Bridges*. Class 9/12 on tracks connecting population centres, class 40/70 along main roads or regulators and siphons. Also there are foot bridges, some of which can take jeeps.

(10) *Lateral Movement*. Good service tracks exist on both sides for all traffic, limited only by low classification of culverts. Cross country movement in areas close to canals is restricted due to numerous escape channels and marshy areas. However, the service track along most of the canals exists either on the home/far bank or between the bank and

the water line restricting their use for lateral move during active hostilities.

(11) *Tactical Features in the Vicinity*. Other than high banks of the canals, numerous villages on both sides may be the only features of tactical importance, which effect observation and field of fire

(12) *Obstacle Value*. The amphibious, fording and bridging capabilities of our potential enemy have considerably reduced the obstacle value of canals.

e. However, the vertical banks and marshy approaches/exits would require expedients and/or preparation. Brick lined canals with water flowing 1-1.5 metres below the bank would be a problem for enemy tanks for entry and exit. While expedients like assault track way, PSP and summer-field track are available with the enemy for negotiating marshy patches. Dozing of canal banks particularly of the 'infilled' type would flood the area of approaches and exits, creating further problems. However, our enemy has been experimenting and developing canal crossing techniques. Banks as deep as 6-9 metres may have to be cut through for approaches and exits. The distinct possibility of vertical envelopment by our enemy in future, further reduces the obstacles value of canals, for initial phases of an attack operation, the obstacle in that case becomes more relevant to enemy build-up later.

4. **Ditches (**Figure 31 D)

a. *Width*. These are normally of 24 metres width but depending upon the type of country, their width may vary from 6 – 30 metres.

b. *Depth*. 2-3 metres.

c. *Bed*. Dry or marshy when filled with stagnant water. These may be full or half filled.

d. *Banks*. Only home bank is high up to 2-3 metres, which is known as "BUND". Width of banks is 1-2 metres.

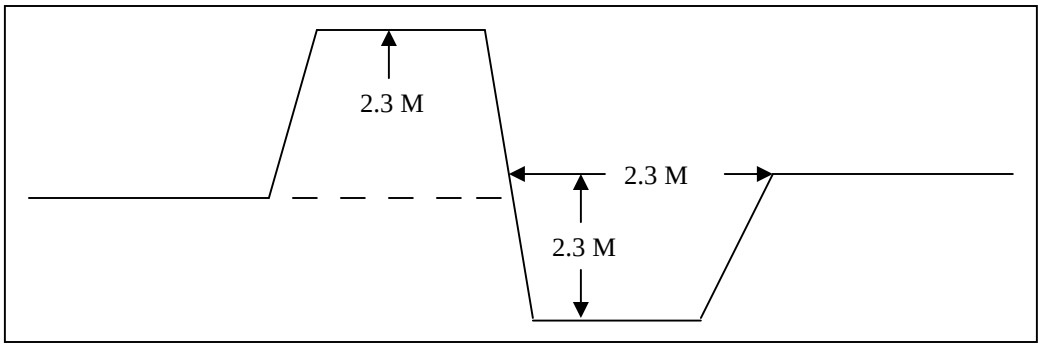


Figure 31 D – Ditches

e. *Gradient*. Is steep except where filled due to lack of maintenance.

f. *Approaches/Exits*. Good-cross country or where tracks/roads are available.

g. *Concealment*. Is difficult except behind the "BUND" on home side.

h. *Bridges*. Nil except foot bridges. However, crossing places are deliberately left along tracks/roads.

j. *Tactical Features*. Only the "BUND" on home side and villages in the vicinity has tactical importance.

k. *Lateral Movement*. Is possible along service tracks.

l. *Obstacle Value*. Ditches of 24 metres width or more are effective anti-tank obstacles, if well maintained. But the width must be increased from time to time with the increase in enemy'' bridging capability.

Planning and Layout of the Defence

General

5. The sequence of planning of defence along a canal/ditch is the same as for area defence. This should be studied in conjunction with the pamphlet GSP-1355 Operations of War (Volume 1) Defence 2001 and GSP- 1808 Infantry Battalion in Battle, 2000.

6. In the succeeding paragraphs, only the points of difference have been discussed.

Protective Detachments

7. In the type of defence where normally little space is available ahead of the main position or where space is available but is restricted due to obstacles and build-up areas, precluding the use of strong mobile protective detachments like covering troops, it would be normal for the protective detachment to act as yet another defensive line/position ahead of the obstacle (canal/ditch) on which the main defences have been based. At battalion/company level, mask positions for some major weapons should be occupied ahead of the obstacle to deceive the enemy about the location of main defences and to cause casualties. Elaborate arrangements should be worked out to ensure their safe fall back to the main defensive position.

8. **Organization**. To execute their task, the protective detachment will include some or all of the following components:-

a. Armour and anti-tank resources including Bakhtar Shikans.

b. Most artillery in range is in support.

c. Substantial infantry element and adequate engineer support to prepare the obstacle plan.

d. Aviation support for observation, surveillance and command and control.

Main Positions

9. **Organization and Layout**. Depending upon the number of approaches into the divisional area, the main position will consist of compact battalion/ company defended areas/localities, with gaps in between them. Battalion reserves will be employed to destroy enemy forces which are able to cross the canal and to restore the canal line by counter-attack. In some cases where the situation gets out of control they may be used for counter-penetration role.

10. **Infantry Company in Defence**. Sitting of Forward Defended Localities:-

a. *Ditches*. In case of ditches too close to the border, there is not choice in most places. The bunds have been so sited that there may be no advantage in going ahead of these for sitting of FDLs, unless some dominating ground is available. However, some mask positions, besides protective detachments, must be sited forward of a ditch to achieve some measure of surprise on the enemy.

b. *Canals*. A Commander on the spot is the best judge to decide whether to site the forward defended localities forward, behind or astride a canal. A commander is required to make use of the canal for more effective defence in view of its natural strength. In case he is to occupy defensive positions ahead of the canal, he will, therefore, be occupying lower ground level than the canal. He will have elements across the canal who will pose tactical and logistic problems especially during the battle. If close to the borders, defences should be sited on the obstacle itself. The bulk of troops should be located on the higher bank or far bank if of equal height. The forward elements after giving a battle should fall behind the obstacle, eventually to provide further depth. Their secure fall back should be planned in advance and all arrangements should be coordinated during the planning stage.

11. **Frontages**. A battalion with three companies up it may cover up to 5000 metres. A company with three platoons up may be deployed between 1500 to 2000 metres.

12. **Depth**

a. Greater depth than in normal deployment is required to balance lateral extension of the defended area. This does not imply committing bulk of troops to defensive positions sited in depth along the axis of advance. There should be more alternative prepared defensive positions in depth to which uncommitted troops or those from least threatened sectors can be moved rapidly.

b. In our environment it may not be possible to achieve standard depth at all levels. Following measures should, therefore, be adopted.

(1) Retaining depth at company/battalion level at all costs.

(2) Use of obstacles and lightly held positions in depth.

13. **Mutual Support**. In defence of canals mutual support within and between the forward positions will be easier to achieve and for depth positions move of troops to reinforce in time, the threatened locality, with troops from the other positions will also be easier. But routes for the purpose must be improved and maintained.

14. **Reserves**

a. At battalion/company and below it will not be possible to have uncommitted reserve. Depth troops may counter penetrate, reinforce or even launch quick local counter-attacks. Reinforcement at company and platoon levels will normally be by company/ platoon weapons.

b. Reserves should normally not be employed forward of the obstacle unless the problems of crossing, restricted area of manoeuvre and high risk of its premature commitment have been fully evaluated. Brigade and battalion reserves may be employed forward of the obstacle only when bulk of the troops located are holding important tactical ground.

15. **Fire Support Plan**. No change is recommended from normal procedure.

16. **Anti-Tank Defence Plan**

a. When forward defended localities are ahead of or on the canal/ ditch, some anti-tank weapons in the FDLs are needed to shoot enemy tanks.

b. When on the home side with obstacle system in front, bulk of these weapons would be sited in depth. Some should also be earmarked as reserve for counter-attack and for counter penetration in anti-tank role where the breakthrough is likely to take place.

c. In both cases, sufficient weapons should also be deployed ahead of the FDLs with protective detachments and mask positions. Villages provide good area for mask positions and tank ambushes.

d. When far bank is also high, those few located in FDLs should be sited on far bank.

e. Emplacements/bunkers of recoilless rifles and bakhtar shikans must be of low silhouette (below ground level if possible) and well concealed, with alternative weapon positions.

17. **Obstacle Planning**

a. *Minefields*

(1) Ideally there should be minefields on both sides of the canal/ditch irrespective of which side of the obstacle the bulk of troops is located. This would, however, impose

serious restriction to manoeuvre own reserves. It is, therefore, recommended that minefields on home side of the obstacle should only be laid in gaps. However this observation does not apply to protective minefields.

(2) Minefields in depth must also be planned and laid particularly along tank approaches/exits, across obstacles and in front of counter penetration positions, leaving areas for counter-attack.

(3) Gaps and lanes left in the minefields for own troops must be mined when no longer required before the enemy can exploit these.

(4) In case of ditches also minefield is better sited on the far side as well as some in depth.

b. *Wire Obstacles*. Double concertina wire and low wire entanglement must always be used in conjunction with minefields preferably on both sides of the obstacles.

c. *Under Water Obstacles*. Picketed wire entanglements or floating but anchored zebra wire may be used in the canals as well as ditches approximately 6 inches below the water surface. Concertina wire on the home end of the water line about 15 centimetres above the water surface will be very effective. Floating mines may be used, if available, planning must include creation of under water obstacle when the far bank has been lost to the enemy.

d. *Flame Throwers*. Flame throwers sited on the home bank will create a ball of fire with napalm as the enemy assault comes in. Plans must be made to send a sheet of burning oil from upstream to the area where enemy crossing is taking place.

Field Defences

18. While constructing field defences along the water obstacles or ditch/bunds following guide lines will be useful:-

a. Canals especially infilled type or the ones which are partially in cut and partially in filled are special structures and are sensitive to any earth work disturbance. Excavation done at random may breach the canal and result in unwanted inundation. Before carrying out any construction or excavation on the canal banks, a liaison should

be kept with the Irrigation Department for the technical advice. During war, it will be good practice to have earth filled gaps available down to platoon headquarters level, to cater for situations where Irrigation Department does not approve the excavation inside the bund.

b. Concrete bunkers with overhead and frontal protection should be made for major weapons only. The roofs and walls should be so designed so as to take direct hit from field guns. Steel doors and windows over loopholes should be provided which can be locked from inside. This will guard against enemy throwing hand grenades or using pole charges.

c. A bunker constructed at ground level inside the bund will be more stable, than the one in the middle or upper half of the bund. But this must be seen in conjunction with observation and field of fire available for effective fire effects. Construction of bunkers is recommended to be made in the centre of the bund. Following advantages will accrue due to this positioning:-

(1) It will reduce plunging fire.

(2) It would be more stable than the one constructed in the upper half and more economical as compared to the one constructed at the base.

(3) Profile of the bunker will not break the sky line, hence easy to conceal.

d. Weight of the bunker and the super imposed load of the bund may cause collapsing of ditch. This would enforce leaving of space between the base of the bund and the edge of the ditch. A ledge of 3-5 metres is recommended to guard against this eventuality. Ledge can also be used by drag lines for subsequent maintenance of ditches.

e. Approaches to canals being invariably marshy, may require use of road expedients for transportation of construction material to the site.

f. Route in and route out and communication trenches for defences on home side of the canal, ditch or bund should be made as in other areas because these will run in the area of the bund on the home side which would generally be higher than the far side. However, for defences sited in bund on the far side these can either be excavated laterally inside the bund when its width technically

permits, or made of sand bags and duly camouflaged when digging through the bund is not feasible. The communication trenches are made on the reverse slope in case no space in provided between the obstacles and the bund. In case space is available these could be made in that space. Furthermore, the forward slits must have digging approaches from the rear, instead of forward, digging expanding rear wards. As the earth excavated should further reinforce the top of the bund to give it more sustenance.

g. Keeping in view the threat from vertical envelopment, necessity of having loopholes in the rear may the shape of an arc to cover the area having an overlap with the adjacent bunker.

h. Firing bays should be as designed as would enable the weapons to fire effectively without the muzzle being projected out in the open. This can be achieved by having platform behind the loophole to support the weapons and a projection ahead of and over the loophole. Such projection will also help in the concealment of the bunker by filling earth on it.

j. Where a choice is offered between construction of linear vis-à-vis kidney shaped bunds constructed in a zig zag/broken pattern, the latter should preferred as they offer the following advantages:-

(1) Will provide better deployment positions.

(2) Will be difficult for the enemy to pin point and engage the bunkers.

(3) Camouflage and concealment will be easy.

(4) Will be difficult for the enemy to pickup slit trenches in the folds on zig zags.

k. Dummy bunkers or dummy firing bays be constructed for deception at intervals. This will deceive the enemy about the location of actual ones and even if located, will divide the enemy fire.

19. Home banks of canals which are level or lower than for bank should be raised for better observation and field of fire. Vegetation on the far bank should be controlled during peacetime so that fire lanes can easily be cut through during emergency. Service roads on canals are on the canals bank which may not be used during hostilities. It is, therefore, necessary to construct tracks behind home banks of operationally important canals.

20. **Bridges**. Bridges being the most suitable and likely areas for assault crossing, should be defended as strong points including those in depth.

21. **Location of Command Post**. Company command post should be sited in the area of depth platoon in case company is deployed on the obstacle in two platoons up. However, the company commander must visit/supervise the battle from any of his forward platoons area under the following situations:-

a. When his personal presence is warranted to influence the battle.

b. When launching a counter-attack.

c. During the preparatory stage and lull periods.

22. Platoon command posts per force have to be located within their areas. If no suitable sites in depth are available these may have to be on the bank/bund. An alternative command post in depth may be prepared for the purpose.

23. **Surprise and Security at Company Level**. The following measures will be useful:-

a. Protective detachments and mask positions must endeavour to deceive the enemy regarding location of main position.

b. Field defences must be proof against enemy artillery, air and direct firing weapons as much as possible.

c. Communication trenches laterally and rearward should be a regular feature of such defences.

d. All movement on the banks/bunds should be restricted to darkness.

e. Major weapons should have alternative bunker/positions.

f. Tunneling and low silhouette of weapons emplacements on the canal bank/bund.

g. Use of night detection means and silent sentries should be made.

h. Intensive patrolling, listening posts at night and mask position will enhance security and surprise.

24. **Administration at Company Level**

a. Traffic control and recovery arrangements at each bridge site in the area will be needed and these will be under the battalion arrangements.

b. Company aid posts may have to be located forward if the bulk of troops are on the far side. In addition, an ambulance car with medical staff on home side of bridge and boats/rafts as alternative means of crossing are required at each site. In view of the change in role, casualty evacuation of protective detachments should be given special and separate attention.

c. Feeding and other arrangements for daily maintenance of troops may have to be undertaken on both sides of the canal and movement across restricted to hours of darkness.

d. Administration within platoon/sections for washing/drying of clothes, bathing, rest, arms cleaning et cetera during day time should be well organized and carried out in concealed places and reverse slopes.

The Conduct of Defensive Battle

General

25. Defensive battle in this case should essentially be conducted in accordance with the existing doctrine as laid down for area defence.

26. A few particular aspects, however, need special consideration:-

a. *Preparatory Phase*. In the preparatory phase, maximum possible enemy including armour must be destroyed forward of the obstacle by a combination of concentration of all available fire including air and coordinated use of protective detachments and obstacles system. These must be suitably organized to achieve that end. Lanes and gaps in the minefields must be closed in time.

b. *Assault Phase*. During this stage:-

(1) Reserve demolitions must be fired in time.

(2) Gaps in main defences to be occupied by withdrawing elements, if not already occupied by task forces.

(3) Shift troops from less threatened areas to areas of likely enemy breakthrough to achieve concentration of force and flexibility.

c. *Counter- Attack Phase*

(1) Brigade and battalion reserve may counter-attack forward of obstacle in rare cases and only if bulk of troops are located on the far side to hold important tactical features.

(2) Otherwise, reserve at all levels must be used to quickly destroy all serious penetrations on the home side of an obstacle. Normally commanders are likely to get sufficient reaction time to move their reserve if they have been reading the battle correctly and anticipatory moves have been undertaken.

(3) Where counter attack fails, or is forestalled, further penetration must be halted at all cost by counter penetration and all available means including reserve anti-tank weapons, reserve of engineer troops and mines, air and artillery fire till the higher commanders reserve become effective.

Mask Positions at Company Level

27. **Definition**. "A weapon position selected ahead of main defensive position from where the weapon can perform the same tasks as from original position. A position which is prepared to deceive the enemy about the original position/layout of weapons in the main defence. After completion of task, weapon is moved back to its original alternative position".

28. **Situation Warranting Employment of Mask Position**. A clever blend of mask position with dummy position will pay dividends. A weapon team can occupy mask position in any of the following situations:-

a. When it is desired to deny information to enemy about the original location of weapons in main defence.

b. During day time, weapons are place approximately 300-500 metres ahead in mask position to gain additional range in order to engage enemy at longer ranges thus gain time/cause attrition.

c. To occupy a dominating piece of ground ahead of original position.

d. These positions can also be meshed with depth position and or towards the flanks when require.

29. **Strength Composition of Mask Position at Company Level**. This will be commensurate with the type of weapons being employed in mask position. Complete crew of the weapon will move to the mask position.

SECTION 86 – DEFENCE AGAINST NAPALM BOMB

Aim

1. The aim of this section is to provide information on the capabilities and limitations of napalm bomb and the defensive measures against its effect.

Effects of Napalm Bomb

2. Napalm bomb may contain 450 – 700 litres of gelatinized fuel, weighing 400 to 500 kilograms. Fighter or fighter - bomber aircraft usually carry two bombs, one under each wing.

3. On impact the bomb produces a fireball and spreads thickened fuel over an elliptical area approximately 100 metres long and 40 metres wide. The fireball will burn for 6-10 minutes with reduced intensity. The temperature generated is as high as 1000-1200° C. Some types of napalm thickened with polymerio substance will also omit pungent fumes which irritate the eyes and respiratory system.

4. The general effects of napalm bomb on various types of targets are as under:-

a. *Troops*. Personnel in open are extremely vulnerable to napalm attack. Those in bunkers or trenches with overhead cover are less susceptible to the fire hazard. However, they suffer from heat and effects of hot gases. Psychologically, the fire may have a great demoralising effect, unless the troops are well-trained in defence against napalm bomb.

b. *Tanks, Vehicles and other Weapons*. Napalm is very effective against armour, vehicles and weapons. The thickened fuel will stick to the surface and may be drawn into the surface; the damage may not be serious. Once the napalm gets inside the tanks or vehicle, it may set fire to the ammunition, patrol et cetera causing serious damage. Barrels of guns or other weapons maybe damaged by the intense heat, if exposed to the sufficient quantity of burning fuel.

c. *Buildings and Other Structures*. Napalm is most effective against structures, containing inflammable material such as wood and straw. Against reinforced concrete building, the effect of Napalm is negligible. Brick and masonry structures, such as permanent fortified positions, factories, bridges, tunnels et cetera, cannot be easily penetrated and are thus less susceptible to napalm.

d. *Machinery and Installations*. Non combustible machinery and tools may be twisted out of shape by the heat. Radar and radio equipment may be seriously damaged by burning of combustible parts.

5. **Targets**. Following targets are suitable for napalm attack:-

a. Concentrations of troops in the open.

b. Vehicles, including armour.

c. Gun and weapon positions.

d. Supplies or equipment concentrations.

e. Command posts.

f. Airfields, aircraft, and radar installations.

Limitations of Napalm Attack

6. Napalm Bomb Attack has the following limitations, which can help in defensive measures against such attacks:-

a. The attacking aircraft has to fly on a constant course up to the targets.

b. The aircraft has to fly at a relatively slow speed of 500 to 550 kilometres per hours.

c. The attack on the target is delivered from a height of 15-60 metres.

d. The duration of effects of napalm attack is relatively short.

Protective Action

7. The greatest impact of napalm attack is its psychological effect. The demoralisation and confusion caused may result in abandonment of defensive measures resulting in great damage. Well trained and motivated troops, displaying calmness and presence of mind, will minimize damage to personnel and equipment and continue to perform the assigned mission. A thorough

indoctrination of the individual soldier in the capabilities and limitations of napalm and efficient defensive measures is essential to prevent unnecessary damage.

8. When subjected to napalm attack, protection from the effects of incendiary material is provided by the following:-

a. Shelters and dugouts.

b. Trenches with overhead cover.

c. Natural cover, such as ravines, ditches and pits.

d. Stone or brick buildings.

e. Locally available non flammable material such as green grass, ranches or thick clumps of trees.

f. Tanks and APCs, when closed down, offer temporary protection.

g. Short time protection can be provided by overcoats, groundsheets and capes et cetera, wrapped over the body.

9. Following preparations/actions by the troops in their respective fields of activity would serve to reduce the effects of napalm attack:-

a. Well dug and camouflaged positions.

b. Overhead cover, especially for ammunition pits and command posts.

c. Burying of telephone cable.

d. Coordinated sitting and effective use of all weapons in anti-aircraft (AA) role.

e. Knowledge of correct fire fighting drill to minimise damage to personnel, weapon and equipment.

f. Use of fire extinguishers and other improvised means such as water, mud, earth, and canvas et cetera for putting out fire.

g. When engulfed by fire, the tank crew should keep the cupola closed, jettison external fuel tanks and drive out of the fire area at an acute angle.

h. Areas where fires are easy to start, such as ammunition, petrol vehicles etc should be isolated to prevent the spread of fire.

j. Good knowledge for providing prompt first aid to the victims of burns.

Active Defence

10. Aircraft making napalm bomb attack are highly vulnerable to ground fire due to the limitations given in para 9 above. Even if a 'hit' is not achieved, a barrage of fire form automatic small arms fire will make the task of accurate delivery of napalm bomb more difficult. A high standard of training in the use of machine-guns and small arms in AA role, therefore, assume a great significance. Good AA discipline will make the job of the pilot hazardous one.

11. **First Aid**. When napalm attack has been delivered, the first action required to be taken is to extinguish or localise any fires, which may affect personnel, weapons, equipment and ammunition et cetera. Use should be made of all available fire-fighting means, such as water, fire extinguishers, earth, sand and mud or any other local material. An important element of fire fighting is the administration of prompt first aid to personnel affected by burns.

12. Following first aid treatment is recommended irrespective of the degrees of the burn:-

a. Extinguish flame by wrapping the individual in a blanket, ground sheet or by pouring water on him. Putting out fire by water will help in reducing the temperature of burnt surface as well.

b. The individual should lie at rest.

c. Cover the burnt area with a clean sheet. Face should be left exposed.

d. Tropical applications should be avoided, i.e. no ointment et cetera be applied to the burnt area.

e. Hot sweet drinks containing salt should be administered.

f. Pain killing drugs available should be administered, and record of the type and dose given be sent along the casualty.

13. Minimum handling of the casualty especially the burnt surface is important, to avoid further damage to the part and eventual complications.

Training-Specific Measures

14. Following measures should be specially stressed during training:-

a. Thorough troops indoctrination in the enemy's napalm capabilities and limitations.

b. Use of natural cover, when attacked by napalm.

c. Knowledge and use of correct fire fighting drill/equipment.

d. Use of improvised means for putting out fire.

e. Salvage of equipment.

f. Correct siting and effective use of weapons in the AA role.

g. Providing prompt first aid to burn victims.

h. Treatment of burn victim for shock effect.

SECTION 87 – **DEFENCE BASED ON STRONG POINTS**

General

1. The term 'Strong Point' refers to a piece of ground which is strongly defended by a force, from the size of a platoon to a battalion, after having gained strength from the existing terrain features/structures, artificial obstacles and strong defence works. The locality is prepared to face enemy attack from all sides and is expected to continue fighting even when bypassed. The time up to which this locality can continue to fight depends on the logistic stocks available within it. The stocking policy is decided at the formation level and is a part of the overall design of the defensive manoeuvre of that formation.

Aim

2. In the mountains, a strong point simply aims at providing strength to a defending force to hold dominating features, thus denying the enemy, the possibility of carrying his offensive into the rear areas. In the plains and desert, the strong points will have any or combination of the following aims:-

a. Defending tactically important features.

b. Defence of communication centres.

c. Defence of selected points astride the enemy's axis of advance, thus conserving forces and avoiding dispersion (more likely in desert sector).

Concept

3. Any formation which opts for defence based on strong points basically aims at the following:-

a. Forcing the enemy on the offensive to commit maximum infantry for reduction of strong points, on the whole, degrading his infantry.

b. Denying the opening of Main Supply Route (MSR) of the enemy by defending the communication centres, thus imposing acceptable delay and carrying the much needed logistics support to the bypassing spearhead forces.

c. Having brought the enemy offensive to a premature halt, use the strong points as pivots for the mobile reserves in the counter-attack mode.

Prerequisites

4. Following prerequisites must be fulfilled by the force opting for a defence based on strong points:-

a. Enough resources in the shape of defence construction stores and obstacles should be available.

b. Time is available to convert suitable localities into strong points.

c. Trained manpower like engineers are made available for construction work and erection/laying of obstacles.

d. Adequate mobile reserves are available to destroy enemy forces which have bypassed the strong points or to relieve invested strong points.

e. Enough strong points are created in the depth to inextricably involve enemy's infantry within the main defence.

Fortress Defence

5. An extension of the concept of defence based on strong points is the concept of fortress defence. This basically aims at converting major towns and cities in the operational area into strongly defended places using the strength provided by the built-up areas and other such structures. This is a time consuming exercise and requires enormous resources. Since civilian population is invariably affected by such a venture, political considerations have to be weighed before undertaking it. A fortress defence can take the shape of a series of strong points on the periphery of a town or a continuous line of defence works. The enormity of the task necessitates work to be undertaken well in advance during peacetime.

SECTION 88 – DEFENCE UNDER NBC ENVIRONMENTS

Variations of Defence

1. Defence under NBC environments demands certain fundamental variations as compared to the conventional means. The distinct variations are:-

a. *Concept*. A conventional defence aims at repulse, resist and destroy the enemy by conventional means. Whereas under NBC environments, it aims at creating situation, where enemy is forced to present a concentrated target for destruction by NBC Killing Zone, carrying out destruction by nuclear strike and mopping up by counter-attack forces.

b. *Dispersion*. Beyond the company level there has to be dispersion both lateral and in depth to avoid mass casualty under a NBC strike by the enemy.

c. *Flexibility*. Adequate flexibility should be maintained to avoid the effects of nuclear strike by having alternative defence locations and troops hides, where the main defence are occupied during the final stage of the enemy mopping up after the NBC strike.

d. *Cohesion*. Extra emphasis should be laid on command, control and communication to maintain the cohesiveness of combat groups. As under the NBC environment, due to massive destruction of units and subunits, they may get isolated and might have to sustain independently for long duration.

Fundamentals of Nuclear Defence

2. While the basic principles of defence remain valid even under nuclear environment, the following fundamental factors must be kept in mind:-

a. *Survival against Nuclear Strike*. The exact intelligence about the number of weapons, the yields of their warheads and the dept of nuclear delivery, means will indicate the presence or absence of nuclear threat in a particular sector. This will indicate the type of defences, conventional or nuclear, to be adopted by the formation. The type of initial defensive posture is a command decision and will be decided basing on the commander’s appreciation of the enemy intention.

b. *Ability to Withstand Conventional Attack*. When the defence is based on the nuclear posture, it becomes extremely vulnerable to conventional attack by penetration or infiltration. Contingency plans for such a situation must be prepared at the initial planning stage.

c. *Quick Transitional Capability*. At times, the nuclear threat may develop quickly and at other times, an existing threat might disappear due to move of NBC weapons to another section or due to close proximity of enemy troops. Such changes in the nature of threat necessitate quick transition from conventional to nuclear defence and vice versa. Formations posses the flexibility to undergo this transition with minimal delay.

Basic Considerations of Defence

3. **Ground**

a. *Vital Ground*. The NBC weapons have not altered the tactical value of ground in the Land battle. Areas will always contain ground which must be held or dominated by one side or the other. In NBC operations, ground is considered vital when it contains vital installations such as NBC weapon sites or a headquarters. Further, ground can be considered vital in the sense of depth in which to continue fighting. A defender who has been pushed back to the limit of ground, over which he is prepared to fight, must endeavour to regain it.

b. *Obstacle*. The tactics which enables the defender to develop his full strength are a series of physical containments and NBC strikes. They can be most effectively applied when the ground hampers the attack. Obstacles are therefore of first importance in any defensive battle and their effective use will be particularly important in the NBC defence. Whether natural or man made, a major obstacle will give the defender the following advantages: -

(1) It will restrict the enemy's ability to manoeuvre and cause him to hesitate prior to his main assault thereby increase the possibility of creating NBC targets.

(2) It will assist in denying the enemy information about our force deployment in depth.

(3) It will separate the enemy assaulting force from their tanks and artillery weapons.

(4) It canalises the enemy into pre-selected NBC killing zones.

(5) It forces the enemy to concentrate on either side and present the target for NBC strike.

4. Where no major obstacle exists, the defender will endeavour to build an obstacle zone in order to impose delay on the enemy and to force him to present NBC targets. This may be achieved by combining a natural obstacle with an artificial obstacle, or by joining up devastated areas to form a continuous belt. The zone will be developed as a vast obstruction to wheeled and tracked vehicles so that their progress through the zone is canalised into routes selected by the defender. The depth of this zone may extend from 5 to 15 kilometres on either side of the obstacles. Besides the nature of the country, it will be dictated by the protection to own troops in the main defensive position from the projected NBC strike.

5. In NBC environments following frontages to be adopted by units/subunits are recommended:-

Serial	*Unit*	*Frontage*	*Depth*	*Gaps*
a.	Section	150 Metres	-	200 Metres
b.	Platoon	500 Metres	300 Metres	800 Metres
c.	Company	1800 Metres	1500 Metres	3200 Metres
d.	Battalion	7-8 Kilometres	6 Kilometres	3200 Metres

SECTION 89 – **NIGHT VISION DEVICES AND DEFENCE AT NIGHT**

Night Vision

1. Eyes require time to adapt themselves from seeing with day cells in sunlight to seeing in darkness with night cells. The process of change is known as night adaptation, and takes time. As much as 30-45 minutes may be necessary.

2. When night cells are exposed to light they lose their adaptability. The brighter the light the greater the loss, and the longer will be the re-adaptation period. When possible one eye should be kept shut when white light is used.

Defence at Night

3. Despite improvements to night fighting equipment, darkness still gives considerable protection against aimed fire, and its use helps to gain surprise. Moving and fighting at night are more important than ever before. They require much practice. Obstacle crossing requires particular training.

4. The main differences between night and day operations are:-

a. Enemy rifle fire may be un-aimed, except at short ranges.

b. Hearing replaces sight as the predominant natural sense. Therefore silence is of the greatest importance.

c. Enough resources in the shape of defence construction stores are of the greatest importance.

d. Without night fighting aids, objects are generally visible only on skylines. This increases the importance of using low ground for movement and observation in that, as it hides you and tends to 'skyline' the enemy.

e. The difficulty of maintaining direction makes previous daylight reconnaissance highly desirable, especially for company, platoon and patrol commanders.

f. Control is difficult for attacker. Close formations are normal, except on clear moonlit or starlit nights, which will permit more open formations.

g. Although night fighting equipment causes a lot of strain on the eyes of the sentry but with the practice, he can survail and constantly watch the area with more confidence.

CHAPTER 15

WITHDRAWAL

SECTION 90 – GENERAL CONSIDERATIONS

General

1. Withdrawal is an operation of war in which a deployed force disengages from the enemy and executes a planned manoeuvre to the rear or away from it. This may be carried out under one of the following conditions:-

a. In order to disengage from an unsuccessful battle or as a result of defeat.

b. Two avoid accepting a battle under unfavourable conditions and thereby gaining time for creating overall favourable conditions to resume the offensive.

c. To reinforce offensive operations.

2. Whatever the situation, the aim of the force commander must be to maintain an intact front throughout and occupy new defensive position in the rear with as little interference from the enemy as possible. However, no withdrawal should take place without the explicit orders of the higher commander.

Definitions

3. **Rear Parties**. Troops who remain in position in the forward defended localities until the time for final abandonment in order to prevent a quick follow up by the enemy. If the main body is withdrawing out of close contact with the enemy and is not likely to get involved in a running fight, no rear parties may be left behind.

4. **Rearguard**. A rearguard is a balanced force, usually of all arms to cover the withdrawal of a formation and to gain time for it to complete the occupation of a new position in the rear. Whether covering troops are being used or not, a rearguard is usually required to cover the withdrawal of a force and to protect its rear. Rearguard actions differ from covering troops because the rearguards take the form of occupation of defensive positions and do not fight a series of mobile delaying actions as covering troops do.

5. **Covering Troops**. A composite force of all arms whose main task is to provide information about the enemy, protect the main force, harass the enemy and delay him by making him deploy. The covering troops impose delay by fighting a series of mobile delaying actions.

6. **Intermediate Position**. A position designed to delay the enemy advance and to allow more time for the preparation of the new main position in rear. There may be several intermediate positions before the main defensive position, depending upon the delay required and troops available. Distance between the main position and intermediate position should be such that the enemy should be required to move his artillery in case he decides to attack the intermediate position. In case a natural strong obstacle is available it must be used.

7. **Checkpoints**. Checkpoints may be established at all levels such as platoon and company. There object is to enable subunits/units to be checked as they withdraw in order to ensure that all are withdrawn according to plan and that they are clear before positions are finally abandoned.

8. **Rendezvous (RV)**. RVs are used to enable subunit and unit commanders to regain control of their subunits.

9. **Embussing Point**. A point on the route of withdrawal where troops are picked up by vehicles. From here, the withdrawal is conducted in vehicles to save time and effort.

10. **Debussing Point**. A point near the new main defensive position where troops are made to debuss from vehicles. Here they are met by guides who take each subunit to its own respective location in the new defensive positions.

11. **Release Point**. A point at which troops and transport, heretofore moving along one route, are directed to previously determined locations. In withdrawal it may be combined with the debussing point.

12. **Non-Essential Stores/Personnel**. These are those stores and personnel which are not needed for the continued defence of the present position and are likely to impede withdrawal. Examples of these are unserviceable equipment, surplus ammunition, rocket launcher (if there is no tank threat), bedding, cooking utensils, picks, shovels, wire, mines, tents/bivouacs and sick and wounded personnel. These may also consist of those incharge of non-essential stores and weapons. Non-essential stores and personnel are sent back to the new main defensive position, under battalion arrangements, prior to the commencement of the withdrawal. This is normally done after last light.

13. **Reconnaissance Party**. That party, which, during a withdrawal operation, carriers out a reconnaissance of the new main defensive position. It is responsible for sitting and marking the defences in the new main defensive position and for guiding the main body into it. It consists of the Second-in-Command element of the withdrawing force. At platoon level it is normally comprised of the platoon second-in- Command and one representative per section.

14. **Fundamentals of Withdrawal are**:-

a. *Control*

(1) Withdrawal at any level is a difficult operation to control. The highest commander must make a coordinated plan

which will be controlled by a tied programme along the whole front. Orders must be got down to the foremost troops in good time. Finally, if the commander is to contend with the unpredictable events that so often take place during withdrawal, the plan must be both simple and flexible. The following will assist in formulating a simple plan and will ensure a degree of flexibility:-

(a) Avoidance of tight timings.

(b) Early move back of unwanted vehicles and equipment.

(c) Maintenance of units and subunits intact during withdrawal.

(d) Reduction in the number of commanders to be sent back to reconnoitre the new positions.

(2) At platoon level this is achieved by:-

(a) The platoon commander being the last man to leave the defensive position.

(b) Keeping the platoon's RV close behind the platoon positions.

(c) Ensuring that all section commanders know the route back to the platoon and company checkpoints and RVs.

(d) Use of single file formation, with the commander at the head and an NCO at the tail.

b. *Clean Break.* To avoid a running fight and to achieve a successful withdrawal a clean break is essential. A clean break may be achieved by one or a combination of the following actions: -

(1) Maintenance of intact front with a portion of the force and normal activity along the front till the very last minute.

(2) The establishment of one or more intermediate positions.

(3) The use of rear parties/covering troops/rearguards.

(4) Secrecy.

(5) Night withdrawal.

(6) A counter-attack or spoiling attack.

(7) *Security*. To achieve this, leap frogging must be resorted to. While doing this rearward hop should traverse sufficient space to force the enemy to re-deploy at least his field guns and reassemble/regroup his attacking forces so that adequate time is gained and enough attrition is caused on the enemy.

c. *Maintenance of Morale*

(1) A withdrawal entails a greater strain on the morale of troops than any other operation of war. Low moral may turn a withdrawal into a rout. It is important to ensure that every thing is done to counter this. The following will assist in the maintenance of morale: -

(a) *Information*. All ranks must be given reliable information. Lack of it will encourage rumours, more often than not of a depressing nature. A sound system of security is essential. The enemy must also be kept under observation by all means available, particularly against outflanking moves.

(b) *Administration*. Good arrangements must be made throughout the withdrawal for the supply of food, ammunition and petrol.

(c) *Discipline*. Discipline is vital. There must be no relaxing of the normal safeguarding and maintenance of equipment. It may be necessary to abandon certain stores and equipment, but this will only be done on the instructions of the highest commander.

(d) *Leadership*. As always, leadership will take troops through anything. At company/platoon level this is achieved by:-

i. The setting of a personal example by the commanders.

ii. Counteracting false rumours and exaggerated reports by keeping the men informed.

iii. Making full efforts to prevent wounded men from being captured by the enemy.

d. *Gaining Time for Occupation of the New Position*. Adequate time is required for preparation of new defensive position(s). This can only be achieved if all possible measures are adopted to trade minimum space for maximum time. A platoon may be asked to contribute to this effort by being made part of the rear party. Time can be gained by adopting following measures:-

(1)	Having the new position well in the rear i.e., trading space for time. This will depend on the terrain.	This will however be applicable at higher levels.
(2)	Establishment of intermediate positions(s)	
(3)	The use of covering troops/ rearguards/rear parties.	
(4)	Counter-attacks/spoiling attacks.	
(5)	Making full use of obstacles and demolitions.	

e. *Secrecy*. All possible steps must be taken to keep the time of withdrawal secret from the enemy. This is important if a clean break is to be achieved. This will not be easy because the tactical situation will probably make it obvious that a withdrawal is likely. The following will help in maintaining secrecy:-

(1) All abnormal movement to take place during the hours of darkness.

(2) Normal activity along the front to be maintained.

(3) Only those individuals who have to know about the withdrawal in order to start preliminary planning should be told early.

(4) Deception.

(5) Wireless security.

(6) Harassing fire plan to cover the noise of transport et cetera.

f. *Security*. All possible measures must be taken by the higher commander to provide security to the withdrawing force. The company commanders will be informed of the employment of covering troops, rearguards and if necessary flank guards.

g. *Balance*. Balance is essential if a withdrawing force is to avoid running fight or carryout controlled movement to the rear. The withdrawing force must not be caught off balance on the move. At battalion/company level it is essential to have security to the front, flanks and rear at all times.

h. *Intelligence*. Every effort must be made to get early information of the enemy movement. Only accurate and early intelligence will permit the adjustment of plans to minimise enemy interference with our withdrawal and to delay his advance through bold counter strokes. A company/platoon may contribute to this effort by sending out patrols. However, for the most part, it will depend on higher headquarters for any intelligence that is relevant to its withdrawal. Intelligence, pertinent to subordinate/flanking commanders, should passed on to them promptly and in sufficient details.

Timings

15. In withdrawal it is necessary for a commander to lay down certain definite timings in order to coordinate the withdrawal across his front. These are normally of two types:-

a. *Essential Timings*. These are always laid down. They consist of: -

(1) *The Time of Denial*. This is the time up to which the position must be denied to the enemy. Up to this time, the FDLs are fully manned.

(2) The time before which no rearward move may take place except by the reconnaissance parties. In order to arrive at this time, a commander will usually consider the following factors:-

(a) The strength required on the present position in order to deny it up to the time laid down by the higher authority.

(b) The amount of time required to withdraw the troops from the present position.

(c) The degree of secrecy required to prevent the enemy realizing that a withdrawal is taking place.

b. *Additional Timings*. In addition to the above, some or all of the following timings are also laid down:-

(1) The time at which the forward troops may begin thinning out. This is normally a few minutes before the time of denial.

(2) *The Time of Final Abandonment*. This is the time at which the position is finally abandoned by the rear parties.

(3) The time by which all troops will be clear of a line in the rear of the position being abandoned. This is required to enable the supporting artillery and air to know when they can engage the enemy on abandoned positions.

16. **Security**. All possible measures must be taken by the higher commander to provide security to the withdrawing force. The company commanders will be informed of employment of covering troops, rearguards and if necessary flank guards.

Obstacles

17. Obstacles are employed to slow down and canalize enemy movement, to add strength to positions and thus gain time. Because of paucity of engineers and resources, full use of natural obstacles should be made. The correct use of obstacles may impose greater delay on enemy than any other means. Infantry, however, on their feet cannot be stopped for long by any form of engineer obstacle alone. The aim of engineer work must therefore be, to hold up and, if possible, destroy enemy tracked and wheeled vehicles.

Reaction of Civil Population

18. If the withdrawal is made in enemy territory the population should be expected to stay and measures must be taken to guard against hostile acts on their parts. In a withdrawal in friendly territory roads may be blocked by refugees and operations may be seriously handicapped. There will be ample opportunities of enemy fifth column activity. These problems can be solved by cooperation with civil government and use of police.

SECTION 91 – **PLANNING OF WITHDRAWAL**

Planning

1. Following important points must be considered:-

   a. *Ground and Weather*

      (1) Choice of rear main position.

      (2) Choice of one or more intermediate positions.

      (3) Employment of covering troops and rearguard.

      (4) Routes of withdrawal.

      (5) Obstacle and demolition plan.

      (6) Grouping.

   b. *Enemy Situation*. There is a possibility of the enemy seizing the initiative and making the withdrawing force react to his moves:-

      (1) *Surprise and Security.* All possible steps must be taken to keep the time of withdrawal secret from the enemy. This is important if a clean break is to be achieved. The following points will help in maintaining secrecy and achieving surprise:-

         (a) Only those individuals who have to know about the withdrawal in order to start preliminary planning should be told early.

         (b) All movements over and above the normal should take place during the hours of darkness.

         (c) No abnormal action by forward troops till the time of withdrawal.

      (2) Maintaining contact with the enemy and not allowing him a free hand. This will dictate the strength and composition of the covering troops or rearguard.

      (3) *Good Intelligence.* Every effort must be made to obtain early warning of any enemy action likely to disrupt the plan of withdrawal.

   c. *Own Situation*

      (1) Composition and disposition of own troops carrying out withdrawal. It should be covered up to a level of two steps

higher and two steps lower i.e. a company commander must know the details up to brigade level.

(2) Covering troops, rearguard, flank guard and intermediate positions.

(3) *Administration*

(a) Maintenance.

(b) Transport commitments.

(c) Destruction of equipment and dumps.

d. *Time and Space*

(1) Timings

(2) Choice of one or more intermediate positions.

Selection of Position

2. **General**. An infantry battalion/company engaged in a withdrawal may be called upon to do one or more of the following tasks:-

a. Form part of covering troops.

b. Act as rearguards.

c. Establish an intermediate position(s).

d. Prepare new main position.

e. Form the reserve and be prepared to launch the counter-attack.

3. **New Main Position and Intermediate Position**. New main position will normally be laid down by the higher commander. Its distance from the old position will be governed by the time required for its preparation and the amount of delay that can be imposed on the enemy by intermediate position(s) or by other methods. Depending on the time required for imposing delay on the enemy, one or more intermediate positions may be occupied. An intermediate position must:-

a. Be strong enough to compel the enemy to deploy his main force before his advance can continue.

b. Be sufficiently far apart to force the enemy to move his artillery before he can bring it within range of the next position in rear.

c. Be far enough to permit clean break and avoid running fight.

d. Be on ground which is suitable for defence by a minimum number of troops, providing good observation, fields of fire and covered routes of withdrawal.

e. Be behind a natural obstacle, if available.

Orders and Briefing

4. In both orders and briefings the following points must be included:-

a. Full details of withdrawal timings, routes, RVs and position of other troops in rear.

b. Action to be taken in the event of enemy attack. Plans may include alternative routes. RVs or counter-attack to extricate a unit/subunit which has been cut-off or the employment of daylight withdrawal technique.

c. Details of rear parties and their commanders.

d. A warning against leaving behind any thing of value to the enemy such as ammunition, papers, marked maps or stores of any kind.

e. A warning against rumours.

f. Reasons for the withdrawal.

g. The new main defensive positions.

h. A warning to maintain normal activity till the last possible movement.

j. Suggested headings for company withdrawal orders are at Annex ‘S’.

Reconnaissance

5. Two separate reconnaissance are required as following:-

a. *New Main Defensive Position.* The new main defensive position should always be reconnoitred in day time before the withdrawal starts. This is done by the reconnaissance party. In a company/ platoon, this task is usually performed by the platoon 2IC working under the company Second-in-Command. He should take with him one guide per section to meet the platoon when it arrives and guides it to its new position. During his reconnaissance the platoon Second-in-Command should mark the outline of each trench.

b. *Withdrawal Routes*. The company/platoon commander and, if possible, the section commander, should walk and memories the route. Company/platoon commander must select a company/ platoon RV where the platoons/sections will gather before moving to the battalion/company checkpoint. Section commanders must walk and memorise the routes from their position to the platoon RV.

Non-Essential Stores/Personnel

6. Generally the first to withdraw after the reconnaissance parties are the non-essential stores and personnel, often immediately after last light. Considerable thought must be given to their method and time of collection so that secrecy is maintained. The company/platoon commander sends non-essential stores and personnel to battalion/company headquarters at the given time. From company headquarters onwards the company commander arranges their rearward move under the CQMH.

Normal Activity

7. It will not be easy to make a clean break if surprise is lost or the enemy is pressing closely. The company must continue all its usual activities until the last possible moment, this includes normal wireless traffic. The total volume of fire from the company area should not seem to lessen until the position is finally given up. At the same time, any abnormal increase in fire may draw the enemy's attention to the position and lesson the chances of making a clean break.

8. There is always danger of enemy infiltration at the time of withdrawal. Patrolling will therefore be important and reconnaissance patrols will be sent to cover gaps and give early warning of enemy approach.

Communications

9. It is essential for a withdrawing force to maintain communication at all costs. Therefore it is necessary to leave the telephone line intact till the very end. A few minutes before the time of denial, wireless sets should be switched on and their netting checked. A few seconds before the withdrawal starts the telephone should be disconnected and the line abandoned.

SECTION 92 – **CONDUCT OF WITHDRAWAL**

Techniques of Withdrawal

1. **Withdrawal by Night**. By night or during the periods of reduced visibility or in the absence of definite enemy pressure, the rear most units/subunits/depth localities withdraw first and those in contact with the enemy, last. Normal activity is maintained by artillery and rear parties stay until the remainder force has withdrawn. Rear parties must be strong enough to give the impression that the position is held in strength. Night withdrawal helps to offset enemy air superiority. This is often the decisive consideration. In night withdrawal, MGs should stay forward as long as possible to enable advantage to be taken of their firepower. Mortars must be able to respond to calls for fire until the last possible moment. FOO to remain with the companies. Improved night shooting devices raise the threat of enemy tank action at night, and therefore, some anti-tank guns must remain in position until shortly before it is abandoned. Should enemy pressure force a withdrawal at night, daylight withdrawal technique may have to be used.

2. **Withdrawal by Day**. The extrication of forward troops is much more difficult by day. They will probably have to fight their way back and be subjected to air attack if enemy has air superiority. During day, when withdrawing under enemy pressure, units/subunits disengage from the enemy by fighting their way to the rear, passing through or around forces positioned in the rear, which cover the withdrawal of forward units/subunits. Thus by day, the depth companies are the last to leave as they cover the withdrawal of front companies. Similarly mortars, machine-guns, anti-tank weapons and tanks are left with the forward units/sub-units till the last. Tanks are particularly useful for helping to extricate infantry during a day withdrawal.

3. **Day versus Night**. It is; therefore, clear that the battalion can withdraw more easily by night than by day. In the latter case, unless surprise is achieved and there are good covered lines of withdrawal, combined with an obstacle to slow up the enemy’s advance it will be difficult to avoid a running fight and severe casualties.

Sequence of Withdrawal of Infantry Battalion

4. In a withdrawal by night, a battalion in contact with the enemy may not only have to withdraw to the new main position where it may be required to occupy and prepare a forward locality, but also delay the enemy on an

intermediate position. Diagrammatic illustration of withdrawal operation is given at Annex ‘T’. The sequence of events for such an operation might be:-

a. The reconnaissance of the new main position in the rear and the reconnaissance of the intermediate position (s).

b. The intermediate positions will be occupied, possibly by a company group each.

c. Non-essential stores/personnel will withdraw soon after last light.

d. If there is no threat from the enemy’s armour, the RRs and Bakhtar Shikans may be withdrawn along with the non-essential stores. Such a decision can only be taken by a commander on the spot.

e. The thinning out of forward companies and supporting arms in these localities leaving behind the rear parties.

f. Last to leave the area is the rear parties.

Reconnaissance of New Main Position in the Rear

5. As a principle, commanders at all levels must stay and fight the battle in the present position. Second-in-Commands are sent back for reconnaissance and planning of the new main position in the rear. Composition of the battalion reconnaissance party is normally laid down in the battalion standing orders. A suggested composition of a battalion reconnaissance party is at Annex ‘U ‘.

6. A battalion reconnaissance party is normally called upon to perform the following duties:-

a. Reconnaissance of the new main position. Marking of each weapon pit so that troops arriving at night can dig in easily during darkness.

b. Establish release point and detail guides to lead companies and platoons to their localities on arrival.

c. Arrange traffic control and lay telephone lines to all companies/ platoons localities on arrival.

d. Administrative arrangements, i.e, provision of ammunition, rations and petroleum, oils and lubricants (POL) et cetera.

Control Measures

7. Progress is determined by establishing checkpoints at units/subunit levels and control is regained by establishing unit/subunit RVs. It is not always

necessary to have both checkpoints and RVs. The route/routes of withdrawal to checkpoints and RVs must be reconnoitred during the daylight and should be known to all concerned.

8. Functions and manning of checkpoints/RVs and embussing points are as under:-

a. *Checkpoints*. It is an easily recognizable place where routes from unit/subunits converge and through which units/subunits will move on their way to RV. Here troops will not halt but a commander will be required to report when the whole of his unit/subunit has passed through. Platoon and company checkpoints may not always be necessary. They are established when the route is difficult or when distance between forward troops and RV is excessive for control and coordination. A battalion checkpoint is normally manned by naib subedar adjutant. Battalion command post usually stays here. Company checkpoint is manned by pay naik. Battalion checkpoint should have wireless/line communication with the battalion RV. However, three is no need of having wireless or line communication at company checkpoint. At platoon level checkpoints may not be established.

b. *RVs*

(1) Here units and subunits will stop and reorganize before continuing the withdrawal. Routes from checkpoints to RVs should, if necessary, be marked, sign posted (with shaded lights at night) and guides detailed. Battalion RV may be manned either by Subedar Major or combat support company commander. Company RV is normally manned by company havildar major. Battalion RV should have line communication with embussing point and battalion checkpoint.

(2) Communication, wireless/line is not necessary at the company RV.

c. *Embussing Point*. Normally one embussing point will be established for an infantry battalion. It may be located at the battalion RV or behind it. While selecting embussing point, it must be ensured that enough space is available for the vehicles to turn about and, if required vehicles should be able to disperse and wait for the withdrawing troops. In case companies are spread out and

parallel routes are available for transport and the tactical situation permits, battalion commander may decide to have more than one embussing point well in the rear of the old position on company routes of withdrawal. In such a case strict orders must be issued and enforced against the use of vehicle lights. If there is a danger of secrecy being lost, the noise of the vehicles must be covered by harassing fire. This is normally located near the battalion RV and is suitably marked and protected. In order to prevent confusion, vehicles are marked with the number of each subunit. It is manned either by subedar major or the mechanical transport JCO. A separate embussing point is established for the rear parties near their own RV.

Withdrawal of Non-Essential Stores/Personnel

9. Generally first to withdraw after reconnaissance parties are the non-essential stores and personnel, immediately after last light. The non-essential stores/personnel are sent back by the units/subunits to reach the given RV at the designated time where these are loaded into earmarked vehicles and sent back under centralised arrangements by the battalion.

Withdrawal of Companies

10. For smooth and organized withdrawal of their companies, the company commanders will need to know:-

a. Where he is to withdraw.

b. The time when he may begin to thin out.

c. The time by which he must deny his position to the enemy.

d. The position of troops, if any, through which he is to withdraw.

e. The route to the battalion checkpoint/RV

f. The withdrawal plan of other troops, if any, e.g., covering troops, or troops on the flanks.

11. **Withdrawal of Depth Companies**. Depth companies may be withdrawn as companies. However, if there is normally one route of withdrawal of the platoon, may be staggered to avoid confusion while moving to the company checkpoint/RV. Depth companies are normally called upon to send back some troops earlier for the protection of battalion RV/embussing points, such troops when detailed withdraw after closing of these points.

12. **Withdrawal of Forward Companies**. Rearward movement of all elements of the forward companies less rear parties begins at the designated time. Normally the first to leave will be the depth platoon and company headquarters less the company commander with wireless set and runner. Depth platoon starts withdrawing at the laid down time which is normally the time at which thinning out is to start. This platoon after passing through platoon checkpoint (if any) and platoon RV, withdraws to the company RV where it waits for the rest of the company to arrive. The forward platoons less rear parties normally start withdrawing at the laid down time of denial. The company commanders will invariably be the last of all when they have satisfied themselves that every one else has withdrawn from the positions. The company commander takes over control of the whole company at the company RV from where he takes the whole company to the battalion checkpoint/RV/embussing point. During move back, the company must have security at the front, flanks and at rear at all time.

13. **Rear Parties**. Normally a platoon from forward companies and part of weapon section (GLHMG, MGs, 60 mm Mortars and RR) is designated as rear party. The platoon so designated will normally be the one required to move last. Machine-guns are those which have the best fields of fire. Battalion mortars and artillery are available to support them till the last minute. When the thinning out starts, rear parties come under command of the designated commander who is normally one of the forward company commanders. The rear parties have their wireless sets netted to the rear parties commander and all further orders are received from him. The rear party commander remains in communication with the battalion commander. Rear parties are properly briefed by their commander before the withdrawal starts and if possible during daylight so that readjustment of the position to cover the most dangerous approaches leading into the rear is carried out silently and smoothly during the hour of darkness.

14. Even under the heaviest pressure by enemy, the rear parties will stick to their positions and will not withdraw before the time of abandonment. Their withdrawal before the time of final abandonment may jeopardise the withdrawal of main body who may get involved in a running fight. After staying on position till the time of final abandonment, rear parties withdraw on orders of the rear party commander to their own checkpoints/RVs and embussing point, which should be well forward.

15. **Battle Procedure**. Annex ‘V’.

SECTION 93 – **SUPPORTING ARMS IN WITHDRAWAL**

Armour

1. During a day withdrawal, tanks may protect the rearward movement of the companies. They may either be withdrawn with the companies or moved to a position to the rear to support the withdrawal and assist in final extrication from old position. Tanks will also be employed as part of covering troops or rearguards. May be held as the commander's reserve for counter-attack and spoiling attack tasks.

Artillery

2. Artillery should be in a position, as long as possible to provide maximum fire support during withdrawal and to simulate normal activity till the last minute. FOO/MFC should stay with the company/platoon concerned. Apart from normal concentration, artillery may provide smoke screens to disengage companies/platoons in contact during a day withdrawal. Artillery support is required for intermediate positions, covering demolitions, gaining time for reconnaissance and for occupation of rear main positions. If the air situation permits, artillery reconnaissance and air Observation Posts will be of great value in making use of the long-range of artillery, particularly the heavier types.

Engineers

3. Engineer's tasks in withdrawal will include:-

   a. Keeping open rearward communication.

   b. Carrying out demolitions and laying mines.

   c. Defence work in the new main position intermediate position.

   d. Destruction of abandoned equipment railways, civil administration services and food stock et cetera.

   e. Booby trapping the abandoned positions.

Company Weapons

4. During day withdrawal, 60mm mortars and recoilless rifles may leap-frog to the rear with the platoon to which they have been allotted. They must provide maximum fire power over the entire platoon/company area during withdrawal. During night withdrawal, 60mm mortars stay with the company as long as

desirable. The RRs and Bakhtar Shikans if allotted, withdraw along with non-essential stores if there is no tank threat. The noise of loading the recoilless rifles onto their jeeps and of starting the jeeps should be drowned by own shelling. Machine-guns, 60mm mortars and recoilless rifles may all be left with the rear parties.

Battalion Weapons

5. **Bakhtar Shikan**. This is extremely effective weapon system which can be used against armour. Maximum range can be exploited to cause delay and losses to enemy armour. In case there are chances that enemy may follow own withdrawing troops with armour, these are left with the rear parties, otherwise, these form part of non-essential stores and are sent back along with RRs.

6. **12.7 mm HMG**. It can be used on two roles.

a. *Ground Roles*

(1) It is used on anti-personnel role and opens up at longer ranges to force he pursuing enemy to deploy.

(2) It can be employed against armour.

(3) It can be employed at intermediate positions against enemy troops and vehicles. Owing to its long-range and fire effects, this will force the enemy to deploy.

b. *Anti-Aircraft Role*. It can effectively be used against slow, moving aircraft. During day withdrawal, this will be employed to provide anti-aircraft protection to the withdrawing troops.

7. **GLHMG**. GLHMGs may form part of rear parties if there are chances that enemy may follow own troops immediately. These can also be employed at the intermediate positions for their optimum utilization. GLHMG's fire has following characteristics:-

a. It can neutralize enemy weapon emplacement up to 800 metres.

b. It can be used against enemy troops concentration up to 1750 metres.

c. It can damage enemy armour up to 600 metres.

d. It can also be used against enemy helicopters in special circumstances.

CHAPTER 16

OPERATION IN A NUCLEAR, BIOLOGIAL AND CHEMICAL ENVIRONMENT

SECTION 94 – **GENERAL**

General

1. Nuclear, Biological and Chemical (NBC) weapons can cause casualties, destroy or disable equipment, restrict the use of terrain, and disrupt operations. They may be used separately or in combination to supplement conventional weapons. The company must be prepared to fight on an NBC contaminated battlefield. This section prescribes active and passive protective measures to avoid or minimize the effect of NBC weapon.

2. To assist him in planning and conducting NBC operations, the company commander designates an NBC defence team. The team consists of one Junior Commissioned Officer and two Non Commissioned Officers (NCO); one NCO will be the NBC operations NCO assigned to the company. The commander also designates, as required, other special teams such as radiological monitoring and survey teams, chemical agent detection teams, and decontamination teams. There is no laid down number for the teams. However, there should be at least one team per radiac metre and chemical agent detector kit and one company decontamination team.

NUCLEAR ENVIRONMENT

Nuclear Weapon's Effects

3. Nuclear detonations produce four main effects i.e. blast, thermal radiation, nuclear radiation, and electromagnetic pulse. The danger from each of these effects depends on the type of weapon, its explosive yield, the height of burst, the distance from the detonation, and the hardness of the target.

4. **Blast**. Immediately after a nuclear detonation, a high-pressure shock wave develops. It travels away from the point of detonation in all direction at the speed of sound. This shock wave causes most of the destruction created by a nuclear detonation.

5. Strong winds, caused by the passage of the shock wave, propel objects such as tree limbs and debris through the air, turning them into destructive

missiles. Exposed troops and structures are vulnerable to blast effects. Personnel inside structures can be hurt by the collapse of the structures. Persons outside can be hurt by flying debris.

6. **Thermal Radiation**. Intense heat and extremely bright light are formed within seconds of a nuclear detonation:-

a. *Heat*. The intense heat starts fires in buildings and woods. Such fires may spread quickly because of the burning debris scattered by the blast. The heat can also burn exposed skin.

b. *Light*. The light produced by the detonation can cause temporary or permanent blindness. Temporary blindness, may last for 5 to 10 minutes. At night, the loss of vision will last longer because the eyes will probably have adapted to the darkness. However, recovery should be complete in 15 minutes. The light can injure eyes permanently of those troops who are looking at the fireball at the time of detonation.

7. **Nuclear Radiation**. A nuclear detonation produces two types of nuclear radiations i.e., initial and residual. Both types can injure or kill soldiers:-

a. *Initial Nuclear Radiation*. This is invisible radiation emitted within the first minute after detonation. It travels at the speed of light and can damage human tissues and blood forming cells. Since initial nuclear radiation travels so fast, the only way to be protected from it is to be in a protected position before the detonation.

b. *Residual Nuclear Radiation*. This is a radiation, which lasts after the first minute. It consists mostly of neutron-induced radiation and fallout.

8. Neutron-induced radiation is a radiation produced by the neutrons that were made radioactive by the explosion. It exists only on the earth's surface near the point of detonation. It depends upon the intensity and extent of yield of the weapons. The only significant source of residual radiation from an air burst weapon is neutron-induced radiation in the soil in a fairly circular pattern beneath the point of detonation.

9. Fallout is produced when material from the earth is drawn into the fireball, vaporized, and combined with radioactive material and then fall back to earth. The larger particles fall back right away, near the point of detonation. The smaller particles gradually settle down to the earth's surface in the area thousands of

square kilometres. The radiation dose rate of these areas may vary from an insignificant level to a dangerous one.

10. **Electromagnetic Pulse (EMP)**. This is massive surge of electrical power, similar to an extremely strong radio signal. It occurs within seconds of a nuclear detonation and is transmitted through the air in all directions from the point of detonation. EMP can damage electrical (especially solid-state) components of equipment (radios, radars, computers, and vehicles) and weapon systems (TOW/Bakhtar Shikan/Green Arrow Missiles). It does not present a physical hazard to soldiers.

Types of Nuclear Bursts

11. There are three types of nuclear bursts, air bursts, surface bursts, and subsurface bursts.

12. **Air Burst**. An air burst is the detonation of a weapon above the ground where the fireball does not touch the earth's surface. Fallout or radioactive material from an air burst has no military significance unless rain or snow falls through the radioactive cloud and brings the material to earth. Neutron-induced radiation is the major radiation hazard.

13. **Surface Burst**. A surface burst is the detonation of a nuclear weapon at such a height that the fireball touches the surface of the earth. Blast, thermal radiation, and initial nuclear radiation are not as widespread as from an air burst. Induced radiation is present but is masked by fallout. The fallout produced by a surface burst is a dangerous hazard because it can cover a large area with high levels of radioactivity.

14. **Subsurface Burst**. A subsurface burst is the detonation of a nuclear weapon so that the centre of the fireball is beneath the surface of the earth. If the fireball of this type burst breaks through the earth's surface, fallout is produced. Thermal radiation will not be a significant hazard, as it is absorbed by the soil. Blast effects are also reduced, but shock waves passing through the ground or water will extend for some distance, residual radiation will occur in and around the crater.

Nuclear Hazard Warning

15. To warn the company of a friendly nuclear detonation, the battalion normally issues a warning message. The format for this warning is normally prescribed by unit Standing Operating Procedure (SOP) and should contain:-

a. A foreword indicating that the message is a nuclear strike warning.

b. Instructions on what protective measures to take, or order to evacuate the area.

c. The expected time and general location of the detonation.

16. Once the warning is received by the company, the commander disseminates it in the company using the chain of command. He also specifies protective measures.

Nuclear Hazard Alarm

17. The company SOP should prescribe a nuclear hazard alarm and also a signal to indicate that the hazard is no longer present. The standard nuclear hazard alarm is the vocal alarm FALLOUT. The standard signal for indicating that the hazard is no longer present is the vocal signal ALL CLEAR.

18. As soon as a nuclear hazard is detected, the nuclear hazard alarm should be given. It can be given by any soldier detecting the hazard. Once the alarm has been initiated, it must be passed through the company commander and then passed throughout the company as quickly as possible. When the hazard no longer exists, the ALL CLEAR signal is given. This is normally initiated by the company as quickly as possible.

Protection from Nuclear Weapons Effects

19. The best protection from the immediate effects of a nuclear detonation is to take cover in fighting positions, culverts, ditches, or behind hills. Soldiers should face away from the explosion. They must stay down until the blast wave passes and until the debris stop falling. Then they check for (and treat) injuries, damage to equipment and supplies, and prepare to continue the mission.

20. Radiation is the only nuclear effect that remains after a nuclear detonation. It may last for days or even year and cover a large area. Since radiation cannot be detected by human senses, radical equipment must be used to detect its presence. The procedures for radiological monitoring, surveying, and reporting must be prescribed by SOP.

21. If the company must stay in a fallout area, all troops should stay in positions with overhead cover as much as possible. They should cover their mouth and nose with a scarf or handkerchief to keep away from inhaling radioactive particles. The company teams continually monitor the radiation level

in the area. Once the fallout has passed, soldiers brush the radioactive dust off their clothing and scrape the dirt from the area immediately around them. Radiac-metre operators continue to monitor and report radiation levels. All troops should wash themselves and their requirement whenever possible. The time the company may stay in a contaminated area depends on the amount of radiation to which its troops have been exposed, the intensity of the radiation, the protection available, and the needs of the mission.

Monitoring Radiation Levels

22. When fighting in a nuclear environment, the company closely monitors the amount of radiation which its troops have already absorbed (dose) and the amount to which they are being exposed (dose rate). To do this, the company uses the dosimeter and the radiacmeter. There should be at least one radiological monitoring and survey team designated for each radiacmeter. The IM-93A dosimeter does not require a designated team. It is normally carried by a platoon commander.

23. **IM-93A/UD Dosimete**r. This metre indicates the total radiations received by troops. It is the size of a fountain pen and easy to read. Each platoon normally has two dosimetres.

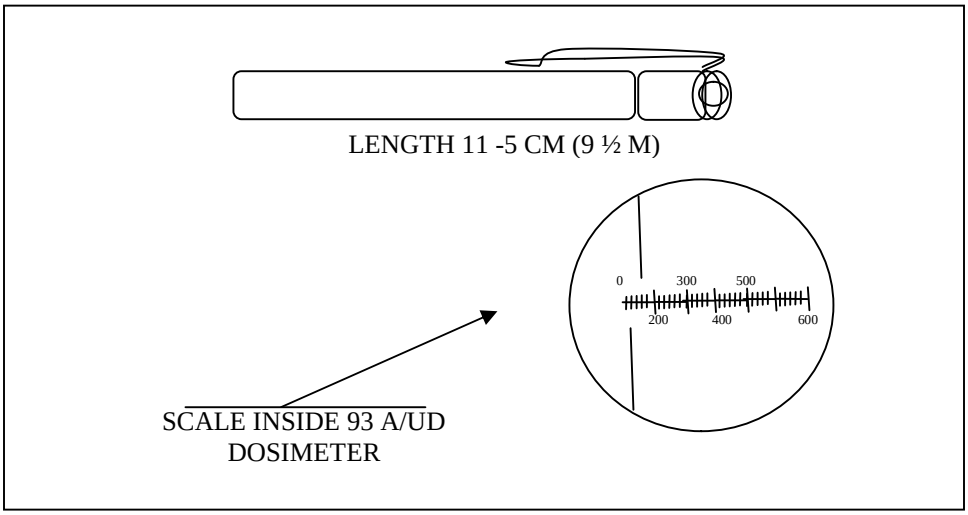


Figure 32 - IM-93-A/UD Dosimeter

Note: For military purpose, one roentgen is expressed in raids.

Two spare dosimeters are in the company headquarters. Platoon commander maintains a record of the time and amount of each reading. The frequency of the

readings and report is set by the company SOP. At prescribed time intervals, the platoon commanders report their readings to the company commander. The company NBC operation NCO consolidates these reports from the platoons and sends the consolidated report to battalion. The format for this report is given at Annex ‘W’.

24. **IM-174A/PD Radiacmeter**. This is used for area monitoring and survey. It measures gamma radiation in units from 0 to 5000 raids per hour.



Figure 33 - IM-174A/PD Radiacmeter

25. Each radiological monitoring and survey team normally consists of two operators (a primary and an alternate). These soldiers must be trained in the use and maintenance of these devices and in techniques of radiological monitoring and survey.

26. Radiological monitoring and survey starts on the order of the company commander or in accordance with the SOP. When a contaminated area is detected, the radiological monitoring and survey teams mark the area with radiological contamination markers. They also record and report, to the company commander, the radiation dose rates and the time and location of each reading. The company NBC operations NCO consolidates these reports and sends the information to battalion, using the NBC 4 report, format of report is given at Annex ‘X’.

27. The radiological monitoring and survey teams conduct either periodic or continuous monitoring. During periodic monitoring, the teams monitor different points within the company area at least once each hour. The teams conduct continuous monitoring when:-

a. The company gets a fallout warning.

b. The company is moving.

c. A nuclear detonation is reported, seen, or heard;

d. Radiation above 1 raid per hour is detected by periodic monitor-ring.

e. As ordered by the commander.

28. The team stops continuous monitoring on order from the commander or when the dose rate falls below 1 raid per hour (expect for units on the move, as they could enter a contaminated area any time enroute).

Chemical or Biological Environment

29. Since enemy forces also have both Chemical and Biological (CB) weapons, the company may have to fight under active Chemical and Biological conditions. Chemical and Biological weapons may be used separately or simultaneously with nuclear or conventional weapons. Regardless of how these weapons are used, the company must be able to survive and continue its combat mission. To ensure this, the company must be trained to meet the NBC standards of proficiency prescribed.

Characteristics of Chemical Agents

30. Chemical agents are poisons, similar to insecticides, but are far more powerful. They are meant to kill or injure humans. They are normally released to cover large areas and may be placed on a target as a gas, liquid, or aerosol. A mixture of agents can be used to cause casualties. These can be disseminated by artillery, mortars, rockets, missile, aircraft spray, bombs, and landmines.

Characteristics of Biological Agents

31. Biological agents are disease-producing germs. These agents may be dispersed as aerosols by generations, explosives, bomblets, missile, and aircraft. Harmful germs may also be spread by the release of infected insects such as flies, mosquito, fleas, and ticks.

Effects on Troops

32. Chemical and Biological agents may enter the body through the eyes, nose, mouth, or skin. They can cause sickness or death. Liquid agents may be dispersed in water and on equipment, terrain, and foliage. The agents may stay for hours or days causing a hazard to unprotected troops.

Effects on Equipment

33. Chemical and Biological agents have little direct effect on the mechanical operation of equipment. Liquid chemical agent contamination on equipment can restrict the equipment's use until it is decontaminated. The company must be prepared to decontaminate its vehicles and equipment. The commander should designate a decontamination team to assist and advice the company on the de-contamination of vehicles, equipment and personnel.

Effects on Terrain

34. Liquid chemical agents may restrict the use of terrain and buildings. The company does not decontaminate terrain; this is normally done naturally by the weather. This may, however, take a long time. Therefore, the company should by pass contaminated areas when possible. When this is not possible, the company may cross contaminated areas after soldiers put on their protective clothing. After crossing a contaminated area, soldiers and equipment must be decontaminated.

Detection of Chemical Agents

35. The senses may not be able to detect chemical agents because most agents are odourless, colourless, and tasteless in battlefield concentration. Therefore, chemical agents must be detected by using the M8 man packed, automatic chemical agent alarm i.e. the ABC-M8 chemical agent detector kit.

Detection of Biological Agents

36. Biological agents are extremely difficult to detect. Soldiers must be alert to any indication that biological agents are being used. Any un-usual occurrences of sickness in troops or civilians should be reported promptly.

Protective Measures

37. The company SOP must prescribe protective measures against Chemical and Biological attacks. Protective measures include:-

a. Warning signals.

b. Protective equipment and clothing.

c. Treatment of casualties.

d. Decontamination of soldiers and equipment.

38. **Warning Signals**. The commander should prescribe a primary and alternate warning signal for alerting soldiers that chemical or biological agents have been used. The warning signals must be disseminated throughout the company and understood by all soldiers. Anyone detecting the use of chemical or biological agents should give the warning signal.

39. **Protective Equipment and Clothing**. A soldier's primary protection against a Chemical and Biological attack is his protective mask. It protects him from inhaling chemical or biological agents.

40. **Chemical Attack**. For full protection against liquid chemical agents, soldiers wear their protective mask and hood, chemical protective over garments, chemical over boots, and chemical protective gloves.

41. Once chemical agents have been employed or while the threat of a chemical attack exists, the company commander must decide whether to keep all troops, or only a certain number, masked and in chemical protective clothing. This is called Mission-Oriented Protective Posture (MOPP). When feasible, the commander specifies the degree of protection before a mission. (The degree of; protection may be prescribed by battalion.) Later, he may direct that the protection be modified according to the threat, temperature, and workload.

42. The MOPP level directed by the company commander will determine what equipment and clothing must be worn and/or used and what precautionary measures must be applied. Therefore, it is essential that the commander and his subordinate commanders be familiar with MOPP concepts. MOPP procedures should be stated in the company's SOP.

43. The following chart shows the requirements for protective equipment and clothing for different MOPP levels.

PROTECTIVE EQUIPMENT AND CLOTHING FOR MOPP LEVELS				
MOPP	**OVERGARMENT**	**OVERBOOTS**	**MASK/HOOD**	**GLOVES**
1.	Worn, opened or closed based on temperature	Carried	Carried	Carried
2.	Same as MOPP 1	Worn	Carried	Carried
PROTECTIVE EQUIPMENT AND CLOTHING FOR MOPP LEVELS				
MOPP	**OVERGARMENT**	**OVERBOOTS**	**MASK/HOOD**	**GLOVES**
3.	Same as MOPP 1	Worn	Worn, hood opened or closed base on temperature	Carried
4.	Worn, closed	Worn	Worn	Worn

44. **Biological Attack**. The best defence against biological agents is strict enforcement of all preventive medical and field sanitation measures along with high standard of personal hygiene.

45. The duty uniform and gloves protect against bites from insects (such as mosquitoes and ticks) that may carry disease-causing germs. Clothing should be buttoned, and trouser legs should be tucked into the boots. Covering the skin reduces the possibility of biological agents entering the body through cuts and scratches and also keeps disease-carrying insects from reaching the skin. Insect repellents and insecticides are effective against most disease carrying insects. High standards of sanitation also improve protection against some insects.

Treatment of Chemical Agents Casualties

46. The casualties from a chemical attack must be treated as soon as possible to prevent further injuries or complications. This treatment includes both first aid measures and decontamination. The symptoms, and first aid steps, for chemical agents are given in the succeeding paragraphs.

47. **Nerve Agents**. The symptoms of nerve-agents poisoning are unexplained runny nose, blurred vision, tightness in the chest, difficulty in breathing, drooling,

nausea, twitching, and convulsions. The injection of atropine and atropine respiration or resuscitation are the first aid measures for soldiers showing symptoms of nerve-agent poisoning. A soldier gives himself three injections of atropine, 10 to 20 minutes apart, when he experiences these symptoms.

48. **Blister Agents**. The symptoms of blister agent poisoning are redness of the skin in 4 to 6 hours, and blisters in 6 to 12 hours, after exposure. These symptoms may be delayed for several hours or days, depending on the type of agent used. There is no first aid for blister agent poisoning other than decontamination. If burns or blister develop after decontamination, soldiers cover the area with sterilized gauze or a clean cloth to prevent infection. The blisters should not be broken. However, if they break, they should be treated as open wounds.

49. **Blood Agents**. The symptoms of blood agent poisoning are increased breathing rate, dull throbbing headache, and nausea. The first aid for blood agent poisoning is to put two crushed ampules of amylnitrite close to the affected soldier's nose. If the affected soldier is still in a contaminated area, the crushed ampules are inserted inside his protective mask. If symptoms persist, this treatment is repeated about every 4 to 5 minutes until normal breathing has returned or until a total of eight ampules have been used. Artificial respiration may be given if breathing becomes difficult or stops.

50. **Choking Agents**. The symptoms of choking agent poisoning are coughing, choking, nausea, and headache. The first aid for choking agent poisoning is to keep the affected soldier still, warm, and comfortable. He should not be moved unless necessary.

Chemical Agents Decontamination of Soldiers and their Equipment

51. A soldier has both an M13 individual decontaminating and reimpregnating kit and an M258 skin decontaminating kit. The M13 kit is carried in the protective mask carrier. There is no specific place to carry the M258 kit. The M13b kit is used by a soldier for decontaminating his clothing, individual weapons, and equipment. The M258 kit is used to decontaminate the skin.

Biological Agent Decontamination of Soldiers and their Equipment

52. Soldiers decontaminate themselves by showering with soap and hot water. Germicidal soaps are used, if available. The nails should be thoroughly cleaned and the hairy parts of the body should be scrubbed. Contaminated clothing is washed in hot, soapy water if it cannot be sent to a field laundry for deconta-

mination. Cotton items may be boiled. Soldiers wash their contaminated equipment in hot, soapy water and allow it to air out.

53. The company is responsible for partial decontamination of its equipment, vehicles, and personnel. Complete decontamination is done primarily under the divisional arrangements at a location specially designated for this purpose. The commander should, however, designate a company decontamination team to supervise and assist in unit decontamination. This team normally consists of six to ten soldiers.

CHAPTER 17

LOW INTENSITY CONFLICT

SECTION 95 – GENERAL

Introduction

1. Low Intensity Conflict in various forms has become a major manifestation of violence short of war. Possibility for troops becoming involved in Low Intensity Conflict is increasing, because when war on large scale is held in check this type of war will occur.

2. Low intensity operations may take the shape of external or internal engagement; however this chapter primarily deals with internal engagements. In the context of internal engagement, envisaged operations may include the counter-insurgency operations, terrorist operations and peacetime contingency operations. It is therefore, imperative that soldiers at all level know their particular role in Low Intensity Conflict.

Scope

3. This chapter covers:-

   a. Concept of low intensity operations.

   b. Category of operations in the context of internal engagement.

   c. Considerations, planning and conduct of low intensity operations at company level.

4. This chapter may be read in conduction with conjunction with GSP-1934, Operation in Aid of Civil Power, GSP-1560 Counter-Insurgency Operation Part-I and GSP-1841 Counter-Insurgency Operation Part-II

SECTION 96 – **DEFINITIONS AND CONCEPT**

1. **Definitions**

a. *Low Intensity Conflict*. It is defined as a politico–military confrontation between contending states or groups. It is below conventional war and above the routine, peaceful competition. It often involves protracted struggles of competing principles and ideologies. Low intensity conflict ranges from subversion to the use of armed forces.

b. *Civil Disobedience*. Active or passive resistance of the civil population to the authority or policies of government, by such means as unlawful strikes.

c. *Civil Disturbance*. Group acts of violence and disorder prejudicial to public law and order. These will necessitate police, and possible, military intervention.

d. *Insurgency*. The terms, guerrilla, revolutionary, terrorist, dissident and rioter are used on occasions to indicate differences in the opposition. When it is not necessary to indicate specific differences, however, insurgent has been used to cover all the roles implied by the foregoing terms. It also includes saboteur, enemy, insurrectionist or rebel, when applicable.

e. *Counter-Insurgency*. Military, para-military, political, economic, psychological and sociological activities undertaken by a government, independently or with the assistance of friendly nations, to prevent or defeat subversive insurgency, and restore the authority of the central government.

f. *Counter-Insurgency Operations*. Operations which own forces maintaining and restoring law and order may have to undertake in support of the established government, in order to counter the threats in an insurgency setting. These threats include civil disturbance, terrorism and organized insurgency, irrespective of whether these are communist, racially or regionally inspired or directed from within or outside the threatened territory.

g. *Counter-Intelligence*. Intelligence activity aimed at destroying the effectiveness of inimical intelligence activities and the protection of information against espionage, of individuals against subversion, and of installations and material against sabotage.

h. *Covert Operations*. Operations which are so planned and executed as to conceal the identity of, or permit plausible denial by the sponsor. They differ from clandestine operations in that emphasis is placed on concealment of the identity of the sponsor rather than on concealment of the operation; i.e. disguised but not concealed.

j. *Guerrilla Warfare*. Military or para-military operations conducted in enemy held or hostile territory by irregular predominantly indigenous forces. Guerrilla warfare will be the main preoccupation of security forces in counter-insurgency.

k. *Insurrection/Revolt and Rebellion*. When subjects revolt, they openly express their dissatisfaction with the established government or its policies. When such an expression is armed and organized, it becomes a rebellion. When rebellion has a large measure of support and aims to overthrow the government, a state of insurgency exists.

l. *Internal Security*. Any military role which involves primarily the maintenance and restoration of law land order and essential services, in the face of civil disturbances and disobedience, using minimum force. It covers action dealing with minor civil disorders with no political undertones as well as riots savouring of revolt and even the early stages of rebellion.

m. *Limited War*. International armed conflict, short of general war. It may be limited geographically, by the scale of forces or by the weapons employed but will be conducted overtly by formations of regular troops.

n. *Sabotage*. An act or an omission calculated to cause physical damage in the interests of a foreign power or subversive organization.

o. *Subversion*. Action taken to undermine the military, economic, psychological, political strength or morale of a nation and the loyalty of its subjects.

p. *Terrorism*. A resort to violence by a dissident faction in order to intimidate and coerce people for political ends. This may manifest itself in sabotage and assassination by individuals or small groups, or in the form of ambushes or attacks on civilians or police by large bodies of terrorists. It can take place in either urban or rural areas.

2. **Concept**. The term low intensity conflict covers a wide variety of military and non military operations other than a conventional/nuclear war. This kind of warfare is resorted to when the resources of at least one of the belligerent are limited or when both parties desire to avoid either the greater risk or greater cost involvement in a more intense effort. With the conflict spectrum, low intensity conflict is distinguished from high intensity conflict and mid intensity conflict. As the definition of high and mid intensity conflict one confined to war between regular armed forces, low intensity conflict corresponds to the threat or use of force, below the level of regular conventional or nuclear conflict. Low intensity conflict covers following range of activities:-

a. *Non-Violent*

(1) Creation of climate of social discontent.

(2) Political and economic destablization.

(3) Psychological war.

(4) Subversion.

b. *Violent*

(1) Guerrilla war.

(2) Armed insurrection to over throw state authority.

(3) Armed insurgency.

(4) Terrorism.

(5) Destabiliztion operations.

(6) Counter-insurgency.

3. **Main Differences between Low Intensity Conflict and Conventional War**

S.No	*Category*	*LIC*	*Conventional War*
a.	Mission	Non-combat/combat	Combat

S.No	*Category*	*LIC*	*Conventional War*
b.	Role	Military role may be indirect and in support of political, economic and informational instruments of national power. Political instrument is dominant.	Direct
c.	Rules of Engagement	Ambiguous	Clear cut
d.	Responsibility	High responsibility and authority to low level commanders (i.e., platoon, company)	Relatively low responsibility to low level commanders
e.	Area of Operation.	Can not be accurately defined	Well defined.
f.	Quantity of force	Individual/small group action	Much More
g.	Friction with Civilian	High and daily	Low and seldom.
h.	Coordination with civil administration /agencies	High and daily	Low and seldom.
j.	Mode of Operation	More reliance on special/unconventional operations	Conventional operations
k.	Conduct of operation	Small dispersed with fulfilling served mans	Large and massive formation concentration on single mission.
l.	Frequency	More frequent to happen	Erupts very seldom
m.	Enemy	Ambiguous	Very well known.
n.	Use of	Not used	Used.

S.No	*Category*	*LIC*	*Conventional War*
	Destruction weapons		
o.	Engagement	Internal/external	External only
p.	Nature	Constructive/Destructive	Destructive only
q.	Tactics	Unorthodox	Orthodox
r.	Target	Less purely military, rather they are psychological and political	Purely military
s.	Contender	States or groups with in a state	States.

SECTION 97 – **BASIC CONSIDERATIONS**

1. Low intensity operation will occur in variety of forms under variety of situations. It is difficult to determine exactly the kind of operation and the environment under which low intensity conflict will occur. However generally, following aspects will be considered while dealing with Low Intensity Operations.

a. *C^4 I.* The mechanisms of command, control, communication (C^4I) in low intensity conflict must be coherent and integrated, including civil-military coordination, to implement policy from strategic to tactical level. Junior leaders and soldiers must have a sound understanding of C^4I structure and arrangements that they might encounter in low intensity conflict. They must understand how brigade and lower units are organized for Low Intensity Conflict, the command and control process and intelligence preparation.

b. *Training*. During Low Intensity Conflict, type of operation and the kind of situations will be manifold and rapidly changing. Special training will be required to deal with all possible eventualities. Special training will include search and cordon, establishing checkpoints and terrorist training, sniping handling of explosive, fire fighting and bomb disposal et cetera.

c. *Information*. Accurate information at right time to the right person regarding terrain insurgents, terrorists and civil aspects is essence of planning and decision making. All sources of information must be activated and integrated to the optimum.

d. *Unity of Effort*. Military efforts must be integrated with other governmental agencies to gain a mutual advantage.

e. *Terrain Analysis*. Normally, the LIC threat is not large force and avoids positional warfare. Most important terrain aspect is that provide security and logistical support. Terrain analysis is used to deny advantage to the threat and generally includes identification of key areas, cover and concealment, population status, logistical sustainability, observation and approaches.

f. *Coordination*. Coordination is must to counter threat capabilities. It involves the integration of supporting elements and functions into all operations. All sources of intelligence must be used, and force protection and anti terrorist procedures must be practised.

g. *Use of Locals*. Soldiers must understand the importance and the operational necessity of winning and retaining " The HEARTS and MINDS" of the local people.

SECTION 98 **– CATEGORIES OF OPERATIONS**

1. Military operations in low intensity conflict fall in following categories:-

   a. *Counter-Insurgency*. It embraces the active field of those measures, which will be adopted by the government and security forces to defeat the insurgents and restore government authority. Hence, counter-insurgency operations will be carried out in any one or more of those areas, where the central governments is directly responsible for law and order or where the provisional government (or even the central government in the worst possible situation of the collapse of the provisional government) have called in the armed forces for help. Counter-insurgency could include:-

      (1) Countering ambushes and attacks on isolated police and military posts installations and communications by forwarded bodies of terrorists.

      (2) Countering guerrilla warfare.

b. *Anti-terrorist Operations.* The anti terrorist category of operations comes between internal security and counter-insurgency operations. When terrorism is manifested by assassination and sabotage either by individuals or small groups, it may represent on advanced stage of a locally inspired and controlled rebellion or it may be the part of the first phase of insurgency; but when formed bodies of terrorists start carrying out ambushes and attacks on civilians and police, it denotes that trouble has some what escalated and must certainly be viewed as an insurgency.

c. *Peacetime Contingency Operations.* These operations include such diverse actions as disaster relief (flood relief, explosions in ammunition dumps, fire fighting at large scale, train accidents et cetera) certain types of counter drug operations and land, sea, and air strikes–characteristics of these operations is the rapid mobilization of effort to focus on specific problem. Usually in a crisis, that is guided at national level.

d. *Miscellaneous Civil Disturbance*. These operations fall in the realm of internal security. It covers any military role, using minimum force, which is concerned primarily with the maintenance and restoration of law and order and essential services in the event of civil disturbances and civil disobedience. These could means dealing with:-

(1) Civil disturbance arising out of labour disputes, racial, area, class or creed antagonism and tension, social unrest or disorder.

(2) Civil disobedience and riots, with or without such political under tones as saviour of revolt or rebellion.

(3) Terrorism perpetrated by individuals and small groups in the form of assassinations and sabotage, especially in populated areas.

SECTION 99 – **INSURGENCY AND COUNTER-INSURGENCY OPERATION**

Insurgency

1. **Characteristics of Insurgent Tactics**

a. Imaginative, unorthodox and flexible operations with over-whelming reliance on the elements of surprise.

(1) Offensive action always, even when temporarily on the defensive.

(2) Night operations to a large extent.

(3) Detailed and meticulous preparation of an attack, besides ensuring superior strength and favourable conditions.

(4) Frequent use of all types of ambush.

(5) Foot mobility of a high order.

(6) Extensive use of mines, booby traps and improvised obstacles.

b. Fortified villages (well constructed and defended).

2. **Insurgent Tactics**

a. *Terrorism.* It is used to demoralize those who are absolutely in order to be fully effective. It is used to demoralize those who are loyal to the government, to extort support from those who are politically uncommitted, and as a means of maintaining discipline and enforcing obedience among individual revolutionary soldiers.

b. *Guerrilla Tactics*. Mao-Tse-Tung had laid down certain guide lines which governs all guerrilla tactics and these are:-

(1) Always seek to deceive and surprise.

(2) Make an uproar in the East and strike in the West.

(3) Act with speed and maintain mobility.

(4) Be as cautious as a virgin and as quick as a rabbit.

(5) When the enemy halts, trouble him particularly by night attacks, to prevent him from resting and to unnerve him.

(6) If the enemy withdraws, pursue; if the enemy advances, withdraw.

(7) When the enemy seeks to avoid battle, attack.

(8) Divide and destroy the enemy; don’t hold places and cities.

(9) In attack ten against one.

(10) Fight only when victory is certain; run away when it is not.

(11) Guerrilla initiative is expressed in dispersion, concentration and the alert shifting of forces.

(12) When the situation is serious, the guerrilla must move with the fluidity of water and the ease of the blowing wind.

c. *Deception*. This is achieved by:-

(1) *False Intelligence*. It is deliberately planted on the security forces by apparent sympathizers, returning prisoners or deserters with prearranged stories, treacherous guides or decoys.

(2) *Exploitation of Treachery*. This means penetrating the security forces with a view to obtaining weapons and information and carrying out subversion and acts of treason.

(3) *Fake Surrenders*. Individuals, groups or even a guerrilla unit may apparently defect and join the security forces. Their aim is to obtain equipment and training prior to re-defection at a critical stage.

d. *Surprise*. This is achieved by:-

(1) Merging with the local population.

(2) Skilful use of ground in any operation.

(3) Speed of approach marches.

(4) Calculated deception.

(5) Diversionary action.

(6) Attacks in areas, which have previously been quite.

(7) Attacks at long distances away from known guerrilla base.

e. *Ambush*. Insurgents will frequently stage an incident deliberately in order to lure the security forces into an ambush.

f. *Infiltration*. Guerrillas make an extensive use of infiltration, because the ground and the nature of operations often permit even large forces to live undetected inside government dominated territory.

g. *Sparrow Tactics*. In the active resistance and subsequent phases, village guerrillas and regional troops split into small groups along roads frequented by security forces in order to:-

(1) Snipe at all who venture outside fortified posts.

(2) Cause delays and thus restrict movement.

h. *Attack*. A deliberate attack against a prepared position is generally undertaken only after meticulous preparation and when sufficient superiority has been achieved. Insurgent commanders are taught to make detailed arrangements under headings such as:-

(1) Preparation, including reconnaissance and dumping.

(2) Breaching.

(3) Fire support.

(4) Assault plan.

(5) Complementary guerrilla attacks elsewhere.

(6) Recovery of casualties and weapons.

(7) Withdrawal plan.

j. *Defence*. The framework for defence can be summarized as under:-

(1) *Observation Villages*. These are located in areas controlled by the government, along lines of communication or on the firings of guerrilla infested zones. Observation villages avoid all action likely to bring reprisals.

(2) *Passive Villages*. A large number of villages around an insurgent base area perform more or less the duties of an observation village, but features such as watch towers, isolated snipping and cleared fields of fire, serve to confuse the troops as if these villages are real centres of resistance. Searches/patrols may encounter booby traps and mines within these villages.

(3) *Fortified Villages*. The defence of the village will be based on three echelons:-

(a) First line using terrain outside the village.

(b) Second line on perimeter of the village.

(c) Third line using dominant features and strong points within the village.

k. *Mines and Booby Traps*. Guerrillas make considerable use of mines and booby traps. The skill and cunning with which mines, and particularly booby traps, can be set by insurgents is of a very high order.

l. *Anti-Tank Tactics*. Some methods are to:-

(1) Mine tank tracks.

(2) Mine crossroads and other turning places.

(3) Place remote-controlled mines in the roadway with the firer in a nearby ditch.

(4) Destroy road surfaces.

(5) Throw smoke grenades, followed by grenades on cords designed to wrap around the guns.

(6) Use light and medium anti-tank weapons.

(7) Hold a strong force near an obstacle to destroy the tank when it slows down.

3. **Phases of Insurgent War**

a. *Preparatory Phase*

(1) Setting up the organization and infiltration into society.

(2) Expanding the organization.

(3) Establishment of insurgent intelligence.

(4) Propaganda and psychological warfare.

b. *Active Resistance Phase-Terrorism*

(1) Terrorism against individuals.

(2) Terrorism against groups.

c. *Insurgency Guerilla Warfare*

(1) Forming of guerrilla units.

(2) Liberated areas and forming of a revolutionary government.

(3) Harassing and weakening government forces.

(4) Political action.

d. *Insurgency Open Affective Phase*

(1) *People's War*. The insurgent movement enters into its final phase and assumes the forms of people's war against the government when it is evident that the balance has clearly swung in favour of the insurgent forces.

(2) *Compulsions of Mobilization Warfare*. Before undertaking regular mobile warfare, the insurgents make absolutely certain that a sanctuary exists for the training of regular forces and that adequate supply of hydrogen equipment are obtainable. The insurgents need for increased supply as well as secure supply lines is great rather crucial at this juncture. They will need advisers in large numbers of improving the quality of their troops and will welcome volunteers from communist/friendly countries.

(3) *Realistic Direction*. The insurgents will only embark upon this final stage of revolutionary/insurgent war after extremely careful deliberation and planning. As a matter of fact the need for it will not be there if the results of previous activities and subversion have lead to the overthrow of the existing government. However, in case this has not been so and the final phase has to be launched, then its aim will be to defeat its military forces and depose the government.

Counter-Insurgency

4. **Planning Considerations**

a. Mission.

b. Enemy.

c. Terrain and Weather.

d. Troops and Resources Available.

e. Time.

f. Intelligence.

5. **Pattern of Counter-Insurgency**

a. *Establishment of Base*. The first requirement for the armed forces to establish a secure base or bases from which aggressive operations against the insurgents can be mounted. Until this has

been done the purely military tasks involved in seeking out and destroying the insurgents cannot be successfully achieved.

b. *Securing and Enlargement of Control Areas*

(1) Once a secure base has been established, operations can be undertaken to secure control areas within which the authority of the civil government can be maintained. Initially these may be carried out from the base itself but as it becomes necessary to deploy greater strength, further forward operations bases must be seized and areas, outwards from these cleared systematically.

(2) The operations to secure and enlarge a government control area, launched from a forward operation base, are known as pacification operations. An area adjacent to the base or existing control area should be selected, and this should be progressively dominated by regular forces carrying out clearing operations, cordons and searches, patrols and ambushes.

c. *Operations in Depth*

(1) Because of the slowness of the operations to establish control areas, the insurgents are liable to have considerable freedom of action elsewhere. If sufficient troops are available, a series of offensive actions, designed to locate and destroy insurgents in adjoining provinces and regions should be mounted simultaneously.

(2) Such operations will not aim at clearing areas but rather, by deep penetration raids, capturing or killing guerrilla leaders, harassing their movement and communication, destroying their food supply and lowering their morale.

6. **Operational Methods**

a. *Patrols*. Their main aims in either pacification or depth operations are:-

(1) *Reconnaissance*. To locate and report insurgent movement.

(2) *Harassing*. To carryout small scale offensive operations or actions against insurgents as and when encountered.

(3) *Standing*. Providing early warning and security in offensive as well as defensive roles.

b. *Ambushes*. Insurgents operations are likely to be disrupted considerably and they will be demoralized much more by well planned and intelligence executed ambushes by the security forces than by any other type of operation. The ingredients of a successful ambush are good intelligence, sound planning, thorough training, self-discipline and sense of presence.

c. *Cordon and Search*. Through this operation an area where small parties of insurgents, wanted persons, arms, ammunition, explosives, equipment, food or documents are suspected to have been concealed, is isolated and searched systematically.

d. *Search and Clear*. This operation is conducted in an area where the insurgents, possible in some strength are suspected to be located. It aims at locating insurgent elements and installations, and destroying them or driving them out of the area.

e. *Fix and Destroy*. Based on specific intelligence, the operation aims at contacting, fixing and destroying enemy units, Headquarters or other installations. Typical examples are attacks on enemy camps or fortified villages.

f. *Counter Penetration*. In these operations security forces are deployed in the defence of a border shared with a country which is assisting the insurgents. These are based mainly on patrol and ambushes.

g. *Special Operations*. These are mounted under special circumstances e.g. to tackle insurgents located in difficult terrain such as mountainous region (without proper communications), jungle, swamp, Special Services Group companies or specially trained troops in small numbers are often more suitable for such tasks.

SECTION 100 – **GUERRILLA WARFARE**

Application

1. Guerrilla warfare may be used by forces, either separately or in conjunction with regular force, to achieve speedy and economical fulfillment of their military, political or economic aims. On the other hand guerrilla warfare may be resorted to when one party to a conflict is inferior to the other in one or more of the essentials of modern warfare i.e., armaments, resources, regular forces or strategic positions.

2. Insurgency and commando operations form part of the guerrilla warfare and counter-insurgency and counter commando operations form part of counter guerrilla warfare.

Guerrilla Objectives

3. It has one of the following objectives:-

   a. To assist the regular armed forces in their operations.

   b. To defend the country as a last recourse when the regular armed forces have been defeated.

   c. To instigate a national action to regain the liberty of a country subjugated by the enemy.

   d. To overthrow a dictatorial or tyrannical form of government.

   e. To harass and weaken the existing government causing it to fall so as to permit the establishment of a new government.

4. **Tactical Aim**. The tactical aim of the guerrilla is three-fold:-

   a. They want to draw enemy forces, in national war in order to weaken his front line so as to destroy him piecemeal, impede his task of pacification and preventing him winning over the population.

   b. They intend to weaken his infrastructure.

   c. To win the support of the people.

5. **Limitations of Guerrilla Military Forces**

   a. *Lack of Mass*. It is incapable of concentrating comba power for any extended period, unable to exploit success and thus compelled to accept some thing short of complete victory. However, if support from outside the country is avail final victory can be achieved.

   b. *Lack of Means*. Mostly guerrillas depend on seized arms and equipment, which is not sizeable in quantity.

   c. *Betrayal*. The guerrilla is vulnerable to betrayal by member of his organization.

   d. *Volunteers without Contract*. Mostly the guerrilla commander leads a group of people who regard themselves as patriotic

citizens. They have no contract to their duties, so many consider family obligations more important than the overall guerrilla cause.

e. *Lack of Training and Discipline*. Generally the guerrilla lacks the formal training and discipline, which results from military training. This deficiency seriously limits the effectiveness of guerrilla operations.

f. *Vulnerability to Psychological Warfare*. The guerrilla who is less motivated is exposed to Psychological Warfare operations directed against him.

g. *Defence and Holding Operations*. Due to limited fire, speed, guerrilla force have normally limited capability for defence or holding operations as compared to regular units.

h. *Radius of Operations*. Due to lack of transport guerrilla operations are restricted in smaller radius.

6. **Phases of Guerrilla Warfare**

a. *Phase 1*. This phase starts in peacetime or immediately on declaration of emergency. During this phase, when the enemy is still strong, resistance operations normally tend to be conducted less openly. Action is generally limited to collection of information, recruiting, training, organization and small scale operations.

b. *Phase 2*. In this phase when the situation changes to favour the guerrillas operations become more overt making large scale actions possible. During this period, the organization is further strengthened by a additional volunteers, individual soldiers and regular army units cut in the course.

c. *Phase 3*. In this phase, guerrilla units are organized of the basis of a regular army. This phase starts when the enemy no longer has he capability of launching all out counter-attacks against the guerrillas.

7. **Counter Guerrilla Action**. Following concepts are basic to the conduct of a counter guerrilla campaign:-

a. *Winning the People*. This consists of the creation of favourable attitudes in the population thus isolating the guerrillas from their popular base.

b. *Total Effort*. Guerrilla struggle being a war of totality can be effectively defeated only with total effort. Ideological and organizational means, political and military campaigns, intelligence and propaganda, persuasion as well as coercion must be skillfully employed by the counter guerrilla forces.

c. *Adoption of Guerrilla Methods*. A conventional army may often make the mistake of dealing with the guerrillas in the classic military pattern. This concept must be disregarded and guerrillas must be fought with guerrilla methods by specially trained units which can be trained and equipped without great cost and without detriment to be regular army.

d. *Efficient Government*. The counter guerrillas must have a viable, dynamic and stable government behind them in order to win.

8. **Strategy of Counter Guerrilla Military Operations**

a. *Initiative*. Counter guerrilla military forces must always maintain initiative. Guerrillas must not be given an opportunity to rest and regroup.

b. *Mobility*. Counter guerilla forces should have mobility superior to guerrillas. This can best be achieved by use of helicopters, wheel/ track vehicles and an effective communication system.

c. *Surprise*. Surprise remains a basic actor against guerrillas. It can be attained by good intelligence and keeping the movement secret.

d. *Leadership*. It is just as important to the counter guerrilla forces as to the guerrillas. Boldness, drive, endurance and aggressiveness are pre-requisite for commanders.

e. *Intelligence*. To achieve success counter guerrilla forces must have a highly organized system of intelligence. Mass organization, police, security forces, local population and other intelligence sources must be fully utilized. Efforts should be made to penetrate inside the guerrilla organization to get the timely crucial information.

f. *Organization Innovation*. An essential element is the counter guerrilla operations. Part of the anti-guerrilla forces must be organized as rangers and commandos who can fight as guerrillas. They must be properly trained and well equipped.

9. **Operations Peculiar to Counter Guerrilla Military Action**

a. *Sweeping Operations*. They are essentially the deployment of a superior military force at attack the guerrillas from all possible directions at the same time. The aim is to wipe out guerrilla bases. The degree of success depends upon the comparative strength of the rival parties, the area involved and the mood of the indigenous population. Follow ups of sweeping operations are 'isolation' in case of failure and "rehabilitation" in case of victory.

b. *Isolation*. It amounts to surrounding the guerrillas area using troops, barbed wire fences, ditches, forts, strong points et cetera,. Communication is strictly controlled; trade is cut-off. Isolation is mainly a weakening process, so that after a certain period of time, the counter guerrillas can launch their attacks again. Without a foreign sanctuary effective isolation can be lethal.

SECTION 101 – **ANTI TERRORIST OPERATIONS**

General

1. Aim of this section is to provide guidance to the troops on tactics and technical to be employment against terrorist.

Terrorist Strategy and Tactics

2. **Terrorist Strategy**. Terrorists indulge in calculated subversive violence for furtherance of their social, religious or political aims. They endeavour to attain one or all of the following objectives:-

a. Causing widespread disorder and panic in the society.

b. Extorting undue concessions for themselves and their comrade's bosses.

c. Forcing the government to resort to strong retaliatory measures, this might be self-destructive and counter productive.

d. Gaining maximum publicity for terrorizing the public and improving bargaining position.

e. Dictating public to cooperate and punishing those elements of society who do not succumb to intimidation.

3. **Terrorist Tactics**. Terrorist resort to following tactics to attain their goals:-

a. *Assassination*. The terrorist groups systematically liquidate the opponents with surgical precision. A reign of terror is let loose so that no one dares oppose them because of risk of life.

b. *Arson*. Loot and arson become the order of the day. Besides life, property is also rendered unsafe. This adds to the anxiety of the general public.

c. *Bombing*. Explosions of varying intensity cause widespread destruction and terror. Normally busy market centres public places and communication networks are targeted to shake the confidence of people and discredit the rule of law.

d. *Hostage Taking/Kidnapping*. To get specific concessions like release of hard-core activists or to extort ransom, hostage taking or kidnapping may be resorted to by the terrorists. These actions improve their monetary condition also.

e. Sabotage. Key installations of the government are targeted and a series of well-planned sabotage activities are initiated which gradually gain momentum.

f. *Hijacking*. It is another effective tool of terror which is employed to extort concessions and gain wide spread recognition and publicity.

g. *Hoaxes*. False alarm is caused in the public by fake phone-calls about explosives having been placed somewhere, likely to detonate at a given time.

h. *Car-lifting/Snatching*. Car-lifting, besides creating a sense of insecurity helps the terrorists to use the snatched/lifted cars in committing crimes and makes handsome addition to their bounties.

j. *Indiscriminate Killing/Firing*. Indiscriminate killing is done to spread scare among the public.

Terrorist Targets

4. Likely military targets for terrorist are:-

a. Weapons.

b. Ammunition.

c. Explosives.

d. Night vision devices.

e. Communications equipment.

f. Chemical equipment.

g. Uniforms and equipment.

h. Command and control facilities.

j. Areas where large groups may assemble, such as shopping facilities, gymnasiums, theatres, barracks, dining facilities, clubs.

k. Warehouses.

l. Transportation centres.

m. Military personnel and their dependents.

Sequence of Terrorist Activities

5. Any terrorist action normally follows the following sequence:-

a. *Preparation and Planning*. The target is selected keeping the following in view:-

(1) The attacks by the terrorists on the institutional structure of the government symbolize their power.

(2) Modern terrorism is inextricably linked with advanced telecommunications and the terrorists always seek maximum media coverage.

(3) The terrorists would prefer to attack such targets, which provide maximum opportunities of success.

(4) It implies selection of the targets, which are vulnerable pressure points of the government or the choke points of the society.

b. *Seizure of Strongholds and Hostages*. After planning the terrorist action, the next stage is to go for the target. In a quick action the terrorists take control of their target and the persons in the target area are taken hostages.

c. *Demands and Deadlines*. After the target areas are fully secured by the terrorists, they try to communicate with the outer world. They start dictating their terms for release of the hostages. Their demands usually specify dead-lines which are repeated off and on.

d. *Escape by Terrorists*. Whether their demands are met or not, the terrorist always have a plan of escape. They keep on looking for

the opportune moment to flee but their condition is similar to that of a cornered cat which reacts violently against those who have trapped her.

e. *Options for Government/Law Enforcement Agencies*. The government/law enforcement agencies cannot succumb to the:-

(1) The ideal outcome of such incidents is the arrest of all the terrorists who are holding hostages without any loss of life or injury.

(2) At times the governments may partially accept the demands of the terrorists.

Military Options

6. To counter terrorism, following military options can be exercised:-

a. *Stronghold Options*

(1) *Perimeter Security*. Deny access to buildings and installations by unauthorized personnel.

(2) *Restricted Areas*. Denial of unauthorized personnel to sensitive areas.

(3) *Pass and Identification System*. Establishment of a system whereby authorized personnel can easily be identified. It should be designed to restrict individuals' entry to authorized areas on need to know basis.

(4) *Storage of Vital Machinery and Equipment*. These are protected against damage by explosions, contamination and theft.

(5) *Emergency Equipment and Plans*. Special equipment like fire-fighting et ceteras is to be kept readily available.

(6) *Informant Nets*. Extensive informant nets are established in both the target areas and the surrounding communities. Such informants are in no way different from the counter-espionage informants or those employed for counter subversion.

b. *Open Air Option*

(1) Aggressive posture against terrorists.

(2) Establish control points throughout the populated area of a town or a city.

(3) To help in the maintenance of law and order.

c. *Ambush Option*

(1) Ambushing is usually the most favourite option available to neutralise terrorists. They can be easily demoralized by well planned and intelligently executed ambushes by the security forces than by any other type of operation.

(2) The ingredients of a successful ambush are good intelligence, sound planning, thorough training, self-discipline, surprise and rapidity and a sense of presence.

Planning and Conduct of Anti Terrorism Operations

7. **General Principles**

a. *Intelligence*. Anti terrorist forces must have a well-organized system of intelligence. Mass communication organizations, police, security forces, local population, media and other sources of gathering intelligence must be fully utilized.

b. *Precision and Rapidity in Action*. Anti terrorist forces have to seize and maintain the initiative always. Terrorist must be kept under pressure, denied the opportunity to rest and regroup, much less to expand.

c. *Surprise*. Surprise is achieved through the following methods:-

(1) Rapidity in planning and execution of an operation/action.

(2) Restricting information on a “need to know” basis.

(3) Segregating security forces from civilians.

(4) Controlling civilian population.

(5) Avoiding obvious changes in daily routine or in disposition before an operation.

(6) Making use of hours of darkness or difficult terrain and bad weather.

(7) Using unexpected, unusual or extraordinary tactics.

d. *Security*. Both the preventive as well as detective security measures are taken to check terrorism: -

(1) Preventive measures include enemy's efforts to gain access to unauthorized information to carry-out sabotage or subversion.

(2) Detective measures are concerned with the detection and investigation of breaches of security or clandestine activeties, detrimental to the security of armed forces or the civil population with which our forces are in contact.

e. *Deception and Destruction.* Deception forms an integral part of all operational and administrative plans. Some of the deception measures are:-

(1) Passing on incorrect information to known terrorist informers.

(2) Planning operations in conjunction with paramilitary forces, which are known to have penetrated into terrorist intelligence systems.

(3) Deception and apparently uncoordinated movements by several scattered units, followed by a coordinated attack on main objective.

(4) Psychological war and propaganda leaflets being dropped or obvious air reconnaissance, air strikes or helicopter movements over an area followed by offensive operations in another area.

(5) Tempting terrorists to go for an attractive target, which is actually a trap.

f. *Communication.* A good communication system is essential to enable all agencies required to be co-opted in the operation, to keep in constant touch with one another.

8. **Stronghold Option**

a. *Immediate Action Plan*

(1) *General.* The immediate action plan should be formed with in one hour after the team has been deployed in the holding areas. It is based on minimal information and usually involves the assault of a stronghold. It must be simple with minimum reliance on speed and fire power.

(2) *Information Required for Immediate Action Plan*

(a) Approach/vehicle route (s) to stronghold.

(b) Plans/layout of stronghold; i.e., blueprint, photo, sketch.

(c) Possible entry points.

(3) *Planning Factors*

(a) *Approach*. Approach to the stronghold, selected should be as direct as possible, using vehicles or helicopters to move fast and to stop as close to the stronghold points of entry, as possible.

(b) *Entry*. A guaranteed method and point of entry must b selected, which puts the Assault Group directly into the reported terrorist location. Following must be ensured:-

i. Have a minimum of two entry points.

ii. Entry is guaranteed. Sufficient methods of entry equipment must be carried.

iii. Terrorist location is directly assaulted.

(c) *Assault*

i. When attacking stronghold, maximum use of fire power, grenades and gas must be used to confuse the terrorists and gain the initiative.

ii. While using gas, ferret cartridge would be more effective than close support due to the lack of smoke.

iii. The whole team should be employed with assault group, assaulting the stronghold and sniper group providing cover or as reserve.

iv. Tactics used are similar to those used a normal stronghold assault.

(d) *Reorganization*

i. During reorganization following must be ensured:-

aa. Separation and guarding of hostages and terrorists.

bb. Search of stronghold.

cc. Protection of evidence.

dd. Avoidance of booby traps.

ee. Evacuation of casualties and medical assistance to the wounded.

ii. Chances of more casualties in Immediate Action Plan require more effective medical cover and evacuation facilities.

(e) *Speed*. Success of the entire operation depends on maximum speed during execution.

(f) *Grouping and Tasks*

(g) *Assault Group*

i. Attempt at least two entries into the stronghold.

ii. Undertake direct assault on terrorists's stronghold.

iii. Clear stronghold and separate terrorists from hostages.

(h) *Sniper Group*

i. Provide covering fire including gas and smoke.

ii. Act as reserve.

iii. Organize reception for hostages.

iv. Cordon stronghold.

(4) *Execution*. During execution following points must be borne in mind:-

(a) It is a direct assault and likely to involve casualties.

(b) It would be implemented hence the team leader may be on his deliberate reconnaissance. Team Second-in-Command and group commanders must therefore be fully prepared for team launching.

(c) Aim must be achieved by ‘shock action’.

(d) A guide could be useful.

(e) It may become the basis of a stronghold option plan.

b. *Deliberate Plan*

(1) *General*. It is not an ideal option for anti terrorist teams since the terrorist would have prepared defence against the assault.

(2) *Planning Factors*

(3) *Intelligence*. As for immediate action plan.

(a) *Approach*. Use of indirect approach, wit cover and concealment and greater chances of closing up undetected with the points of entry should be preferred.

(b) *Entry*. Minimum two entry points must be guaranteed.

(c) *Assault*

i. It must be based on maximum surprise, speed and shock effect.

ii. Use of gas and flash should be made to distract, unnerve and confuse the terrorists.

iii. The entire team should be deployed with assault group, assaulting the stronghold and sniper group providing cover and reinforcement.

(d) *Reorganization*. As for immediate action plan.

(e) *Speed and Precision*. Success of the entire operation depends on maximum speed and precision during execution.

(4) *Tasks*. As for immediate action plan.

(5) *Execution*

(a) Through briefing and extensive rehearsals are necessary.

(b) The success of assault depends on:-

i. Surprise.

ii. Precision and rapidity of action.

iii. Aggression and controlled violence.

iv. Maintenance of momentum.

(c) Assault should be carried out during hours of darkness.

(d) Normal tactical considerations must be fully applied.

(e) Initiation of assault will be undertaken by the Team Leader.

c. *Ambush Option*

(1) *General*. This option entails attacking the terrorists whilst they are moving between two points.

(2) *Planning Factors*. Following factors should be considered:-

(a) Selection of Ambush Site.

(b) *Commentary*. As the terrorists leave stronghold, information regarding the number of terrorists, their dresses and description et cetera should be constantly passed by the left behind/carefully positioned snipers.

(c) *Escort Vehicle*. A police escort vehicle must lead the terrorists's vehicles.

(d) *Preparation of Terrorists Vehicles*. The terrorists's vehicle should be so positioned/prepared that it becomes easier for the teams to operate.

(e) *Ramming Vehicle*. The size of this vehicle must be large enough to block/stop the terrorist vehicle.

(f) *Blocking Vehicle*. If there is a possibility of escape, a blocking vehicle towards the rear of the rammed vehicle may also be used.

(g) *Entry*. Consider different entry points in the terrorists's vehicle and any special tools that may be required.

(h) *Use of Gas*. The use of gas, particularly ferrite rounds or hand held close support grenades will assist the assault.

(j) *Use of Smoke*. Smoke may be used to prevent public/press entering the incident area.

(k) *Rehearsals*. Rehearsals must be carried out before the actual action.

(3) *Grouping and Tasks*

(a) *Assault Group*. The number of assault groups would be dictated by the size of the vehicle involved. They will have following tasks:-

i. Enter vehicle from one direction, but two means of access must be planned in case the door is locked.

ii. Clear the vehicle.

iii. Receive and guard hostages clear of the vehicle.

(b) *Sniper Group*

i. To provide drivers for ramming/blocking vehicles.

ii. To provide cut-off groups.

iii. Firing of smoke/gas.

iv. Conduct running commentary along the route.

v. Act as reserve to assault group.

(4) *Execution*

(a) *Occupation of Site*. Every member must be in position approximately 10 minutes prior to the action. Too early an occupation is likely to create tension and affect the performance of the teams.

(b) *Initiation*. The initiation will be undertaken/ ordered by the team leader in the form of a countdown.

(c) *Assault*. Following sequence should generally be followed:-

i. As the ramming vehicle is moving in, to crash the terrorist vehicle, the assault group moves.

ii. Gas is fired into the vehicle. Snipers may also shoot at the tyres and engine block of the vehicle.

iii. A little before the vehicle has halted, the assault group should enter the vehicle. Shot guns, crowbars and even explosive may be required/used to effect the entry.

iv. Meanwhile the gas should prevent the terrorists from harming the hostages.

v. The vehicle must not be over crowded with men and weapons – clearance should be made in one direction with the Reception Group positioned at the main exit.

vi. Drivers should act as cover/cordon party.

vii. Snipers may isolate the area through firing smoke if the same is required.

(d) *Reorganization*. The reorganization process is similar to as in any other option. In this the hostages are separated from terrorists and cleared from the area quickly, particularly if the gas has been used.

SECTION 102 – **MISCELLANEOUS DRILLS AND PROCEDURES**

1. **Road Blocks**

a. *Aim of Establishment*. Road blocks may be established for the achievement of the following aims:-

(1) Maintain a constant check on movement and traffic.

(2) Apprehend wanted persons and couriers.

(3) Prevent the smuggling of arms, ammunition, explosives and supplies especially food.

(4) Assist the civil authorities in effectively implementing food control measures.

b. *Types of Road Blocks*

(1) *Deliberate*. These may be established on a main road or a highway and act as a useful deterrent to illegal traffic and unlawful movement; but usually these are not likely to produce effective results.

(2) *Snap*. These are used for spot checks and may be placed on any type of road or route.

c. *Principles for Establishment a Road Block*

(1) Concealed Sitting.

(2) Security.

(3) Early Warning Devices.

(4) Communications.

(5) Administration.

d. *Drills and Procedures to be Followed at Road Block*

(1) *Procedure for Searching Persons*

(a) Body, search should include under the arms, the stomach, inside the thighs and the crotch.

(b) Women will be searched only by a women searcher; particular attention needs to be paid to their hair.

(c) Ask to produce driving license.

(d) Checking of documents if any.

(e) Person should be made to stand hands on his head facing toward his vehicle at the time of search.

(2) *Procedure for Searching Vehicles*

(a) Each vehicle must be dealt with separately. It is generally preferable to select a few vehicles for a thorough examination Vehicles awaiting examination, whether cursory or complete, must be held under guard in a waiting area.

(b) Two armed sentries and two unarmed searchers should be used to search large vehicles and buses

whereas an armed sentry and an unarmed searcher should suffice to search a car.

(c) The occupants of a vehicle must be asked to get down and stand well clear of the vehicle. However, the owner/driver of the vehicle must watch the whole search for obvious reasons.

(d) The road block commander will ensure that all occupants of a vehicle are checked and frisked while the vehicle search proceeds.

(e) The vehicles and the driver should be handed over to the police if there remains any suspicion that there is still something hidden, otherwise once the search is complete, the vehicle will be allowed to proceed.

2. **Establishment of Control Points**

a. *Aim of Establishment Control Points*. Its aim will be to control the movement of pedestrians and vehicles and to help in the maintenance of law and order.

b. *Composition*. Normally a control point should not be manned by less than a platoon, although the minimum strength required will vary in accordance with the number of roads that are to be controlled and the number of traffic lanes that exist in each road. Following is the suggested minimum scale:-

(1) *Control Points*. Commander, signaller and runner.

(2) *Barrier Sentries*. One sentry for each barrier and one Non-Commissioned Officer for each road or lane of traffic.

(3) *Covering Party*. A party of two men covering the barrier sentry.

(4) *Standing Patrol*. Wherever possible or required, two men in an Observation Post or on a nearby rooftop should watch adjacent buildings and roofs for bombs and acid throwers as well as the whole area for approaching crowds.

(5) *Reliefs*. Reliefs are necessary for all the above mentioned parties.

(6) *Civil Police*. A civilian policeman should be made available for each lane of traffic, if possible.

(7) *Miscellaneous*. Interpreters and women searchers are also required.

c. *Operation at Control Point*

(1) The commander or his Second-in-Command must be present at all times.

(2) All the pedestrians and vehicles will be channelled through the barriers and checked properly in accordance with the current orders.

(3) The commander will ensure that the area surrounding the control point is patrolled constantly.

(4) The commander must maintain a regular and proper log of all important incidents and events.

(5) The commander must assess very carefully the resultant effect of imposing any degree of control on both traffic situated in a densely populated area, because no matter how efficient the control point is, a crowd of impatient civilians or a traffic jam can quickly build-up.

3. **Picketing**

a. *Aim*. To secure a road for a limited period to provide safe passage to convoy.

b. *Method*

(1) *Advance Guard*. It will clear the axis by bounds ahead, the main body which will establish these pickets on both sides of the road and a rear.

(2) *Rearguard*. It will be responsible for the protection of main body, the passage of the convoy within its area and for calling in the pickets on the completion of the task.

(3) *Movement*. Force will either move on foot or in vehicles and should be supported by armoured vehicles as for as possible.

c. *Distance between Pickets*. Distance between pickets will be determined by the necessity of covering the intervening ground by fire. The distance of pickets from the axis of advance depends on the type of terrain and the minimum time taken/needed for establishing the pickets.

d. *Strength of Picket.* The size of the picket will vary in accordance with the likely threat, the type of terrain and visibility. However, it should not normally be less than an infantry platoon in strength.

4. **Patrolling**

a. *Mobile Patrolling*

(1) *Composition*

(a) Preferably each patrol should be commander by an Officer/Junior Commissioned Officer.

(b) The strength of a patrol should not be less than 2 vehicles (12 men).

(c) *Weapons*

(d) Every individual will carry his personal weapon with ammunition.

(e) Section weapons (MG1A3 fixed onto the vehicles and ready for use) will also be taken along.

(2) *Ammunition*. Each individual will have 300 rounds loaded in the magazines and put in a bag. A box of 1000 rounds with each MG1A3 is also to be carried.

(3) *Equipment*. Following equipment will be carried:-

(a) Updated map of the area.

(b) First aid box.

(c) Every individual to carry a torch.

(d) Night Vision Devices and night sights (if available).

(e) Binoculars and compasses.

(f) Water bottles.

(g) Stretchers.

(h) Picks and shovels (two per vehicle).

(j) Axes (two per vehicle).

(k) Search light fitted on each vehicle.

(4) *Conduct*

(a) Patrol leader will travel in the leading vehicle.

(b) The distance between two vehicles while on the move will be 150 metres and 100 metres during day and night respectively.

(c) Speed of vehicles must be kept slow to be able to observe and take deliberate action on movement.

(d) Patrol will not be adjusted to any other route/task unless ordered by unit commander or necessitated by circumstances beyond control.

(e) No civilians, other than the guides/informers (where necessary) and suspects apprehended, will be made to sit in the vehicles.

(f) All processions/mobs will be avoided. Contact will only be established if specific orders have been given in this regard.

(g) During forced halts due to traffic jams et cetera, both vehicles will try to cover each other and number of sentries will be increased so that all sides of the vehicles are under observation.

(h) In case of a b breakdown, it will be ensured that faulty vehicle is towed to some safe place preferably to a police station/nearest own unit/base.

(j) Patrol leader will exercise proper fire control. Patrol will open fire only in self defence or upon seeing miscreants/dacoits causing damage.

(k) Search lights will be used during night patrolling.

b. *Fighting Patrol*

(1) *Composition*

(a) *Patrol Leader*. Section Commander.

(b) *Strength*. Generally strength of the patrol should be a section. However, the strength may be varied by the battalion/company commanders according to the situation. Each patrol should be accompanied by a minimum of two policemen.

(2) *Equipment/Weapons*. Patrols must carry the following:-

(a) *Personal Weapons*. Maximum/spare magazines (filled) will be carried with each rifle/MP5.

(b) Section weapons each with 1000 belted rounds.

(c) Hand grenades.

(d) *Wireless Sets for Communication*. For communicating with groups and with base/battalion headquarters.

(e) Field dressing.

(f) Water bottles.

(g) At least one stretcher and binoculars.

(h) *Conduct*

(j) Patrol will be divided in two groups.

(k) Two scouts will lead the patrol at a distance of about 100 metres ahead of the leading group. This distance may vary according to the terrain. Distance between individuals within a group should be 3-4 metres.

(l) Scouts to have communication with patrol leader through walkie-talkie sets.

(m) Likely ambush places will be covered by leap and frog method.

(n) Action will be taken on lines mentioned above, apart from the normal measures/procedures adopted by fighting patrols.

5. **Cordon Technique**

a. *Urban Area*

(1) Cordon and search is normally a joint police and military operation, with the aim of capturing wanted persons, arms, ammunition, explosives, documents or supplies.

(2) To stop interference with inner cordon, an outer cordon, is placed at a distance from the inner cordon in the form of road blocks. The aim is to restrain people from joining the inhabitants of the area being searched.

(3) While getting into position, complete surprise should be achieved. An area should be surrounded before the inhabitants realize what is happening.

(4) An important factor influencing the success of a cordon is its exact timings. The area must be surrounded from all sides simultaneously.

(5) A cordon set up at night must b continuous with every man in sight of the next one. It can be thinned out after daylight and maintained by a chain of Observation Posts of two to three men, with patrols covering the gaps.

(6) Approximately a platoon will be required for a cordon of 100 metres in a built-up area. The following is offered as a guide:-

(a) *Cordon Facing Inward*. 15 men per 100 metres.

(b) *Facing Inwards and Watching Rooftops/Windows*. Above the cordon, additional five men per 100 metres.

(c) *Covering Side Roads*. An additional two to three men per 100 metres only for those roads or alleys which lead into the area.

(d) *Area with Large Modern Buildings*. Number may be suitably reduced in such an area.

(7) In order to gain a good field of view a cordon should be as far back as possible and not on the inner kerb on the move behind the cordon in each sector to act as pursuit parties should anyone breakthrough.

(8) *Duration of Cordon*. The duration of a cordon which may have to remain in position for long periods, will depend upon the following:-

(a) Size of the area.

(b) Number of buildings therein.

(c) Density of the population inhabiting the area.

(d) Strength of the personnel available for searching.

(9) *Thinning Out of the Cordon*. The following action will be taken to thin out of the cordon:-

(a) First the escape routes should be wired off and then sentries should be tactically sited, on rooftops/ heights, et cetera, if possible.

(b) Sentries, preferably from one subunit, should be sited in pairs and in depth, and should be in shade during day as far as possible.

(c) During the thinning out process, complete sections should be moved. They can form a tactical reserve and be given rest intervals at a close by place.

(10) *Orders*. These must include details about the following:-

(a) The target area, with roads or open spaces forming is perimeter.

(b) Location and time of RV.

(c) Assembly area.

(d) Routes to the assembly area and from there to the forward RV.

(e) Order of March.

(f) Time for the cordon to be in position.

(g) Orders for the apprehension of the cordon breakers.

(h) Details of the search party organization.

(j) Company areas of responsibility on the perimeter.

(k) Battalion/company cages.

(l) Deception measures.

b. *Rural Areas*. Following are the main variations in the cordon techniques in country side or rural areas:-

(1) *Larger Force*. As the threat of armed resistance is likely to be greater, a larger force will be needed.

(2) *Stronger Mobile Reserve*. In the rural areas the area to be cordoned will be larger than the one in the urban region under a similar situation. Hence, a stronger mobile reserve will be required.

(3) The main problem is to get the cordon into position without alerting the escapers. One of the methods of achieving this

will be to move the cordon into position on foot with a simultaneous approach from different directions.

(4) It will not be possible for the cordon to cover the entire area with troops; therefore, sections and even less may be sited to cover likely escape routes if the situation so warrants. However, the cordon troop should not be out of communication at any time.

6. **Search Techniques**

a. *Urban Areas*

(1) Immediately after the cordon is established, the police will inform the inhabitants of the concerned area through a public announcement that:-

(a) The area (describing its boundaries) is about to be searched.

(b) A house curfew has been clamped and everyone must stay indoor.

(2) A search party should comprise one to three policemen and a military escort. The later protects the former and stops anyone escaping.

(3) *Methods*. The under mentioned drills should be followed:-

(a) All occupants should remain in their rooms while the rooms are being searched. Individuals should be searched first and then their rooms.

(b) It is best to search buildings from top to bottom. Arms, ammunition and explosives may be concealed or hidden. Mine detectors are invaluable for arms detection.

(c) Once a house has been searched, it should be marked so.

(d) Those who are awaiting their turn for search must not be permitted to move into a building which has already been searched.

(e) Cages are a military responsibility which entails erection, establishment, guarding and their

functioning. Buildings if available may be used as part of the cage complex.

(4) *Screening*

(a) This aims at clearing persons of good character and at picking out suspects for higher level screening at the central cage.

(b) At the central cage, the police will interrogate at all suspected persons picked out during the search and decide who should be detained. Troops will escort suspected persons to the central cage and keep them under observation, while they await their turn for interrogation.

(5) *Escort Troops*. Once screening has been completed, wanted persons must be evacuated straight away. If troops are required to do so, then this should be appreciated in advance and escort parties with appropriate transport must be detailed before hand.

b. *Rural Areas*. Following are the main differences between search techniques of rural areas and those of urban areas:-

(1) *Stronger Escorts*. Owing to the increased threat of armed resistance, a stronger escort for the search parties will be needed.

(2) *Prior Clearance of the Area*. The area must be cleared of enemy or potential enemy before the search commences.

(3) *Systematic Search*. If the houses to be searched are spread over a wide area, as may be the normal case, the plan must cater for a systematic search with the cordons closing in.

(4) *Imaginative Search*. Insurgents will generally hide wanted persons and material, very ingeniously in dilapidated places, caves and even in intricate but elaborate tunneling systems. They will also extensively use call-traps, mines and booby traps. Hence, it is imperative that the search is conducted imaginatively and the men chosen for this task are mentally alert, possess sharp and minute observation and are adequately trained for this purpose.

7. **Protection of Road Convoys**

a. *Prerequisite for Successful Countermeasures*. The most effective way of discouraging the insurgents from laying ambushes is to convince them that any and every ambush of their's will always result in a quick, efficient and successful counterstroke. The following techniques should be followed for achieving this result:-

(1) Precautionary measures to forestall such ambushes minimise their effect should be evolved.

(2) Convoys should be organized in tactical groups and provided with adequate artillery support, and also air support if possible.

(3) A mobile reserve should be available.

b. *Precautionary Measures*

c. Intelligence.

(1) Secrecy and deception.

(2) Clearing probable ambush sites.

(3) Patrolling and offensive operations.

(4) Control of vehicle traffic.

(5) Use of local forces.

(6) Use of artillery and airforce.

(7) Mobile reserve.

(8) Efficient communications.

(9) Protection in individual vehicles.

(10) Recovery and repair plan.

d. *Organization of Escort*. Following is an example of an escort based on a LAT company, including a few armoured cars if available:-

(1) *Advance Patrol*. A LAT platoon as a point platoon with a couple of armoured cars together with engineers in transport, or in armoured personnel carriers if possible.

(2) *Close Protection Group*. LAT Company headquarters plus its platoon or platoons supported by infantry and artillery, infantry in armoured personnel carriers if feasible.

(3) *Reserve/Strike Group*. LAT platoon or platoons with the remaining armoured cars, together with infantry mounted in transport, preferably armoured personnel carriers and artillery.

(4) *Helicopter Support*. Helicopter support should be provided to give overhead cover and assist in command and control, wherever possible.

(5) *Grouping*

(a) The armour/armoured cars and LAT elements should be interspersed so that each group includes some of these.

(b) The convoy should be divided into groups, each group comprising between five to ten vehicles with a nominated commander.

(c) The interval between groups should be sufficient to minimise the number of vehicles in any one ambush e.g. up to five minutes intervals between groups.

(d) The distance between vehicles will depend upon the visibility and speed of the convoy. The avoid loss of control it may be as little as 20 metres for a fast moving convoy.

e. *A Suggested Sequence of Action*. Observation of the under mentioned procedure will ensure the best security when moving through a black area:-

(1) Secure the first firm base near the starting point of the convoy and deploy artillery. Assemble the mobile reserve.

(2) Secure the second firm base with infantry moving by helicopter or on foot. Follow it by moving some artillery to the second firm base either by helicopter or by road with armour and infantry protection.

(3) The convoy should now move by bounds in the earlier described tactical groups with the point platoon well ahead, with the airborne FAC and artillery observer overhead, and supported by armed helicopters, if possible.

(4) Assuming the convoy has reached the second firm base, artillery should move there from the first firm base, by air

or by road, with the protective force that was established originally at the first firm base. The mobile reserve would also move to the second firm base, before this whole procedure is repeated over the next 10 to 15 kilometres.

SECTION 103 – EMPLOYMENT OF SUPPORTING ARMS IN LOW INTENSITY CONFLICT

1. **Armour**

a. *Reconnaissance Regiments*. Possible tasks are:-

(1) Direct or indirect fire using main armament machine-guns.

(2) Offensive operations against insurgents, including the reduction of strong points by fire.

(3) Convoy protection for the movement of troops and supplies through dangerous areas and escorts for very important.

(4) Evacuation of casualties under line.

(5) Manning a stop line.

(6) Reconnaissance and confidence patrols.

(7) Snap road blocks.

(8) Cordons – generally employed in the outer cordon.

(9) Route reconnaissance and the checking of critical points for sabotage (e.g. bridges).

(10) Supplementing radio communication networks.

(11) Limited crowed dispersal.

b. *Armoured Units*. Tasks include:-

(1) Providing fire support.

(2) Offensive operations against enemy units, including close support of infantry and the reduction of enemy strong points.

(3) Relief of isolated outposts, and, in some cases, close protection of convoys where the danger of an annihilation ambush exists.

(4) Acting as a deterrent or protection force in capital cities or operational bases.

2. **Artillery**. Following tasks may be carried out by artillery:-

a. Provision of liaison and advice.

b. Defensive fire in support of operational bases and own controlled areas.

c. Support of raids and ambushes. This will require detailed fire support planning.

d. Neutralization of located enemy hides, bunkers and fortifications, their destruction when medium guns are available.

e. Harassing of enemy's patrol bases and supply routes when in range.

f. Provision of artillery intelligence.

g. Air defence when required.

h. Counter bombardment and counter mortar.

3. **Engineers**. The following engineer work is likely to be required in the combat zone. The priority of the tasks will vary with the area and the type of operations:-

a. Construction and maintenance of roads and tracks. This may require additional engineer units and plant.

b. Construction and maintenance of forward airstrips and helicopter landing points.

c. Water supply, which may have a high priority in an under developed area e.g. Sind or Baluchistan.

d. Clearing mined and booby traps. These are amongst the insurgents main weapons and the most difficult to counter in counter-insurgency.

e. Field defences, including bunkers, battle shelters, clearing fields of fire and the fortification of buildings in urban areas.

f. Demolitions, especially of insurgent positions and fortified villages.

g. Construction of obstacles including minefields.

h. River and obstacle crossings.

4. **Army Aviation**. Helicopters could be utilized for the following purposes:-

a. Reinforcement of troops.

b. Close fir support.

c. Reconnaissance.

d. Logistic support.

e. Evacuation of casualties.

5. **Special Services Group**. Special Services Group troops are particularly suited, trained and equipped for counter-insurgency operations. Small parties may be infiltrated or dropped by parachute, including free fall, to avoid a long approach through enemy dominated areas, in order to carry our any of the following tasks:-

a. The collection of information on the location and movement of insurgent forces.

b. The ambush and harassment of insurgents.

c. Infiltration of sabotage, assassination and demolition parties into insurgent held areas.

d. Border surveillance.

e. Limited community relations.

f. Liaison with, and organization, training and control of friendly guerrilla forces operating against the common enemy.

CHAPTER 18

ADVANCE INFANTRY TRAINING

SECTION 104 – GENERAL

Introduction

1. Infantry units are called upon to carry-out raids, ambushes, patrolling et cetera directed against human or material targets, located immediately behind the enemy lines for a limited duration. Such operations demand higher individual skills, physical standards, better group techniques and procedures. Hence, there is a need to have advance infantry training.

Aim

2. The aim of advance infantry training is to make all ranks capable of carrying out raids, ambushes, tank hunting and patrolling immediately behind the enemy lines for a limited duration.

Scope

3. The scope of advance infantry training is to:-

a. Improve individual's skill in handling weapons and equipment to enhance his mental/physical endurance and to develop his will to successfully accomplish the mission.

b. Practice techniques, drills and procedures by which small infantry groups can successfully carry-out the given tasks.

c. Raise the overall standard of infantry units in performing their roles in all operations of war and in all terrain conditions.

Organization of Training

4. In order to ensure uniformity in training and to assist the units in its planning/conduct, the subjects to be covered and the standards to be achieved are suggested in the succeeding paragraphs. These subjects have been dealt with in greater details in the pamphlet "Advance Infantry Training and Operation, 1991 (General Service Publication (GSP) code number 1829) in which, a syllabus for an advance infantry training course of 8 weeks duration has also been suggested. This syllabus can be suitably modified by the units to meet their peculiar terrain and operational requirements. Units may even incorporate this syllabus in their normal training cadres instead of running separate advance infantry training courses.

Subjects

5. The following subjects are pertinent for the conduct of advance infantry training:-

a. Physical training.

b. Field craft.

c. First aid.

d. Mountain craft.

e. Unarmed combat.

f. Survival, escape and evasion.

g. Water crossing expedients.

h. Handling of personal weapons.

j. Navigation, including use of global positioning system.

k. Explosives and demolitions.

l. Air reception.

m. Communication.

n. Conduct of artillery and air shoots.

o. Battle intelligence.

p. Tactics and field exercises.

q. Electronic surveillance devices.

r. NBCW training.

Standards to be Achieved

6. Standards to be achieved are of two categories:-

a. Individual standards (Annex ‘Y’).

b. Unit/subunit standards (see para 8 below).

Individual Standards

7. **Officers, JCOs and NCOs**. Should acquire the standards mentioned at Annex ‘Y’.

a. *Other Ranks*. Should acquire the standards as given at para 5 to 12 at Annex ‘Y’ in the following subjects only:-

(1) Physical training.

(2) Field craft.

(3) First aid.

(4) Mountain craft.

(5) Unarmed combat.

(6) Survival, escape and evasion.

(7) Water crossing expedients.

(8) Handling of personal weapons.

b. These standards at first sight may seem a little high but with constant effort and progressive training these can be achieved even by the average soldier.

8. **Unit/Subunit Standards**. Having attained the requisite individual standards, units/subunits should be capable of carrying out the following tasks:-

a. Raids (headquarters, gun areas, dumps et cetera).

b. Ambushes (mechanical transport columns, foot columns et cetera).

c. Patrolling with particular reference to seeking information about enemy, carrying out reconnaissance with specific aims and capture of prisoners of war.

d. Destruction of bridges/culverts.

e. Holding of communication centres/bridges and preventing enemy from destroying these.

f. Tank hunting.

g. Crossing of water obstacles or negotiating a difficult hill feature/ desert area.

CHAPTER 19

SUPPORTING ARMS

SECTION 105 – ARMOURED CORPS

General

1. The unit of Armoured Corps from which the infantry company is most likely to receive support is the tank regiment integral to the infantry division.

2. **Glossary of Special Terms**

a. *Direct Fire*. When the tank gunner can see the target, observe the shot and make corrections using his sighting devices.

b. *Semi-Indirect Fire*. When the tanks gunner cannot see the target and only the commander can see it or the tank gunner may also see it but cannot observe the strike of round accurately and, therefore, is assisted by the commander in observation and correction.

c. *Indirect Fire*. When neither the tank gunner nor the tank commander can see the target and the fire is conducted by an observer, generally in a similar way as the artillery fire.

d. *Hull Down Position*. A tank is in hull down position to the observer, when only the turret of the tank is exposed to it and the hull is protected by the ground. It is the normal fire position for a tank from which the gunner engages a target with direct fire using his gun sights.

e. *Turret Down Position*. A turret down position is one in which the whole of the tank is protected by the ground from the target except commander, the head which appears over the skyline. This position enables the commander to observe to his front prior to moving forward into a hull down position. It is also possible, depending on the ground to use semi-indirect methods of fire from this position.

f. *Fire Support*. This term is used in tank/infantry cooperation. When employed in this role, the tanks support the infantry by fire on to the objective without physically accompanying it.

g. *Close Support*. In this role the tanks work closely in conjunction with the infantry in an attack right up to and on to the objective, moving at tank speed and using basic tank tactics i.e., fire and movement.

h. *Marrying Up*. Is the process by which arrangements are made in the assembly area, to ensure that: -

(1) Correct subunits of both tanks and infantry are launched into the battle as planned.

(2) Personnel of all arms know each other and plan together.

(3) Methods of target indication are tied up thoroughly.

(4) Communications between tanks, infantry and other supporting arms are working satisfactorily. Time taken to complete the marrying up will depend on the state of training of participating units.

j. *Rally Area*. An area usually in the rear and in vicinity of the objective where tanks assemble after an attack for re-grouping, evacuation of casualties, and quick replenishment, if necessary. Tanks remain under command of local commander to beat off any counter-attack, if it comes. They are ordered to the rally area only

when the local commander is satisfied with the progress of the reorganization on the objective with regard to anti-tank defences.

k. *Leaguer*. A formation which a unit or subunit adopts in the immediate vicinity of the FDLs during or after the days fighting for rest and replenishment. Normally troops will be dispersed, camouflaged and under cover when in leaguer during the daylight and closed together at night. Due to close proximity of the enemy, effective control and security measures are obligatory.

l. *Tank Light*. This is that light when the tank gunner can first recognise his targets through the telescope and is usually 20-30 minutes after the normal first light. Tank last light is when the tank gunner can no longer recognise his target and is usually 20 – 30 minutes before the normal last light.

Characteristics of Tanks

3. The primary capabilities of the armoured regiment are:-

a. Firepower.

b. Protection.

c. Mobility.

d. Flexibility.

e. Shock Action.

4. **Firepower**. The regiment's firepower is derived from the tank main guns and from machine-guns used as secondary armament. The tank is basically a direct fire weapon, used primarily against enemy tanks and other 'hard shell' targets. The gun can, however, also be used for indirect fire like artillery guns to support the infantry in certain situations. The tank machine-guns are used to neutralize or destroy enemy personnel and other "soft" targets including enemy aircraft. This tremendous firepower enables the commander to deliver hard hitting blows where they are most effective.

5. **Protection**. A tank cannot be designed to give absolute protection against attack from all ranges and angles, and yet allows good firepower and mobility. The result is a compromise that affords a reasonable chance of survival at our own battle ranges. However, it provides sufficient protection against enemy small arms, mortars and field guns. This enables the commander to manoeuvre his troops under enemy fire and operate deep into enemy positions. Some tanks are also designed to give some degree of protection against an NBC attack.

6. **Mobility**

a. The cross-country ability of tanks enables them to engage targets quickly from a variety of fire positions and avoid enemy observation and fire. This ability, combined with speed and movement, is vital in all phases or war.

b. Mobility springs as much from the mental attitude of commanders at all levels, and of tank crews, as from the performance of the machine.

7. **Flexibility**. Flexibility is complementary to mobility in enabling a commander to concentrate and switch firepower on the battlefield in order to carry-out one or a number of alternative tasks. It is the product of quick reaction to orders given over reliable radio links.

8. **Shock Action**. The bold use of tanks, exploiting all four of these characteristics to the full, produces shock action. This is a combination of:-

a. Surprise and its consequent effect on enemy morale and reactions.

b. Concentrated firepower in the critical place at the right time. Penetration of the enemy area, especially from the flanks or rear is particularly likely to result in shock action and as a result, to disrupt the enemy's plans and capabilities, and to weaken his will to resist.

9. **Blindness**. The vision from an armoured vehicle is limited when closed down, both in direction and degree. At any moment only one fifth of the surrounding country can be observed. Nothing can be seen at ground level within 9 metres except by the driver immediately to the front. The degree of elevation or depression of the armament which is possible to obtain is limited, giving in effect a "dead arc" around the tank. This "dead arc" is approximately 15 metres for the main gun all round and 5 metres for the bow gun in front within an arc of 45 degrees only.

10. **Sensitiveness**. Tanks are capable of negotiating obstacles but the following impose limitations:-

a. Streams or artificial ditches more than 2.5 metres wide or a vertical step of more than .9 metre.

b. Marshes.

c. Rocky ground or ground with tree stumps more than 60 cm in girth.

d. Steep and slippery slopes.

e. Thick woods.

f. Anti-tank minefields.

g. Built-up area.

11. **Vulnerability to Anti-tank Weapons**. There is NO armoured vehicle that can NOT be knocked out in some way e.g., by mines and anti-tank guns or tank hunters. This vulnerability can be offset only by careful planning and the maximum co-operation of all arms engaged in battle.

12. **Fatigue of Crews**. Cramped positions, noise, vibration, jolting, continuous wearing of headphones, looking through optical instrument, lifting heavy fuel drums and shells impose considerable strain on crews in continuous operations. This causes efficiency to drop greatly and so, adequate rest periods must be planned.

13. **Necessity for Regular Vehicle Maintenance**. This requires time and hard work. A period of at least 4 hours time must be allowed in every three or four days for vehicle maintenance other than normal daily replenishment and checking.

14. **Heavy Administration**. Large quantities of fuel, ammunition and mechanical spares required for maintenance of armoured troops demand much time, transport and labour.

15. **Vulnerability in Close Quarter Fighting**. At close quarters, tanks are vulnerable to enemy short-range anti-tank weapons. In close country, tanks are sometimes unable to see supporting infantry easily or to bring their guns to bear on the target.

16. **Difficulty in Holding Ground**. Tanks must work in close conjunction with infantry if ground is to be held for longer periods.

17. **Darkness and Bad Visibility**. Despite the increasing efficiency of night observing aids, observation at night or in bad visibility will be restricted and the ranges of engagement will be shorter. Also vision from a closed down tank is limited. At any one time, only one-fifth of the surrounding area can be observed. Nothing can be seen on ground level within 5 metres of the tank, except by the driver immediately to the front. The degree of depression of the gun is limited, thus giving a ‘dead arc’ around the tank.

18. **Size, Weight and Noise**. Size and weight may affect a tactical plan by ruling out certain routes. Noise can ruin surprise, but skilful driving or a cover plan will reduce this limitation.

19. **Vulnerability to Air Attack**. Both the armoured and un-armoured vehicles of the regiment are vulnerable to enemy air, particularly the administrative echelons. Although the armour protection of the former gives them some immunity, they can be destroyed or damaged by direct fire, by bombs or rockets. Since the armoured regiment has a poor air defence capability, it must, therefore, rely as much as possible on the protection afforded by the passive measures and secondly, not to engage hostile aircraft until attacked by them.

20. Most of the tactical limitations can be overcome with the help of other arms.

Tank Support

21. Tank support for the infantry will normally take the following forms:-

a. In attack, close support to destroy or neutralize enemy's automatic weapons. The technique of infantry-tank attack may be any of the following, depending on the situation and the terrain: -

(1) *Same Axis*. The tanks and infantry form up in the same FUP and attack the objective while moving on the same axis. While the infantry moves continuously towards the objective, the tanks may either move continuously or may resort to fire and move. When the tanks move continuously, they may either move at infantry speed or may leave the FUP little later than infantry and move at tank speed. In all cases, however tanks and infantry should reach the objective almost simultaneously. This technique of attack is resorted to, when the same axis is suitable, for infantry as well as tanks.

(2) *Parallel Axis*. The tanks and infantry form up in separate FUPs, which however, are close to each other and attack the objective on parallel axis. While the infantry moves continuously up to the objective, the tanks may either move continuously or by fire and move. Attack on parallel axis is resorted to when same axis is not suitable to any one of the two arms and two separate axis which are parallel and close to each other are available for the attack.

(3) *Converging Axis*. The tanks and infantry form up in separate FUPs which are wide apart from each other, and attack on two separate axis which converges on the same objective. In this case, it is desirable that tanks move continuously at tank speed but should time their leaving the FUP in relation to movement of infantry in such a way that both reach the objective almost simultaneously.

b. *Reorganization*. During reorganization, to neutralize such enemy positions beyond the objective, as may be interfering with the infantry reorganization and to help break up enemy counter-attack until the arrival of infantry anti-tank weapons.

c. *In defence*:-

(1) To mount counter-attack(s).

(2) To counter enemy penetration.

(3) To destroy enemy armour by offensive action while lending weight to the anti-tank defensive layout where it is absolutely inescapable.

d. In advance, to support the vanguard.

e. In withdrawal, close support of rearguards/rear parties.

22. Tanks are capable of carrying infantry forward. Although a tank can carry 15 men, when tactically loaded, a troop of tanks can only lift a platoon. Although this is an excellent means of increasing mobility, it suffers from a serious disadvantage that tanks will always tend to draw fire and will not themselves be able to use their main armament. Tanks must, therefore, only be used as a means of transport up to the battle area and infantry must dismount before the tanks come under enemy observation and fire. Infantry will never travel on the leading troops of tanks since they must be free to engage any opposition.

Infantry Assistance to Tanks

23. Infantry is responsible for the protection of tanks from close range anti-tank weapons, particularly in close country and built-up areas where, owing to their poor visibility, tanks are very vulnerable to this form of attack.

Recognition

24. In order to enable infantry to recognize, the tanks with whom they are co-operating, following signs are painted on regimental vehicles, in white colour:-

a. *A Vehicles*. All vehicles are numbered on the turret in Arabic numerals commencing serially with the commanding officer's tank to the last tank of C squadron and other tanks of the headquarters squadron as under:-

(1)	Regimental Headquarters	-	1–2
(2)	A Squadron	-	3–16
(3)	B Squadron	-	17–30
(4)	C Squadron	-	31–44
(5)	Dozers	-	45–47
(6)	Armoured Recovery Vehicles	-	48–50

b. *B Vehicles*. Squadron letter and Arabic numeral combination, number serially from A Squadron vehicles to C Squadron (except the vehicles of headquarters squadron).

c. The following are the numbers of tanks in each troop and the troop commander's tank will be the senior in number excluding the tanks of the headquarters:-

d. *A Squadron*

(1)	Squadron Headquarters	3	-	4	-	-
	Number 1 Troop	5	-	6	-	7
	Number 2 Troop	8	-	9	-	10
	Number 3 Troop	11	-	12	-	13
	Number 4 Troop	14	-	15	-	16
(2)	*B Squadron*					
	Squadron Headquarter	17	-	18	-	-
	Number 1 Troop	19	-	20	-	21
	Number 2 Troop	22	-	23	-	24
	Number 3 Troop	25	-	26	-	27
	Number 4 Troop	28	-	29	-	30
(3)	*C Squadron*					
	Squadron Headquarter	31	-	32	-	-
	Number 1 Troop	33	-	34	-	35

Number 2 Troop	36	-	37	-	38
Number 3 Troop	39	-	40	-	41
Number 4 Troop	42	-	43	-	44

Intercommunication

25. Communication between tanks and infantry is of vital importance. The main means of communication are:-

a. *Direct Speech*. If the infantry is close to the tank, the infantry commander can speak to the tank commander either verbally, by getting up on the tank, or by means of the tank telephone, if separated from the tank, wireless must be used but a quick method of indicating targets known to both tanks and infantry is essential.

b. *Wireless*. All tanks have a wireless set, which can work on the platoon commander's net.

c. *Visual Signals*. Rifle signals, tracer ammunition, verey light, coloured smoke and parachute flares from 60 mm mortars can be used by infantry to indicate targets to the tanks.

Method of Infantry/Tank Target Indication

26. **Aim**. The aim of any method of infantry/tank target indication is to enable the tank commander to identify the target as soon as possible.

27. **General**. When tank and infantry are operating together in operations, enemy targets are observed, located and dealt with automatically by tanks and infantry. The situation may, however, arise when one or the other is incapable of dealing with certain enemy targets, in which cases the assistance of the other may be required. In such cases, there will be a requirement to clearly indicate the targets. The method to indicate the target in such situations is termed as tank infantry target indication.

28. In most situations, infantry will need assistance of the tanks to destroy enemy tanks or machine-guns whereas, the tanks will need the assistance of infantry for the destruction of anti-tank guns, which are in such positions, where the fire of the tanks is ineffective. It must be remembered that in most cases whatever the tanks can observe, it can also be destroyed by them.

Communication Means

29. These are:-

a. By radio.

b. By tank telephone.

c. By personal contact.

d. By flag/hand/weapons signals.

Sequence

30. It is as under:-

a. Attract the attention of the tank/infantry commander.

b. Make him to look in the RIGHT DIRECTION.

c. Give him the RANGE.

d. Describe the detail of the TARGET.

e. Give him the EXECUTIVE ORDER.

f. Observe and correct the fire when target has not been correctly observed/identified/engaged.

OR

(WHEN REFERENCE POINT IS PRE-ARRANGED)

a. As per para 4a above (Attract the Attention).

b. As per para 4b above (Right Direction).

c. Indicate the Reference Point and indicate the target from reference Point, (Give Direction and Range from Reference Point).

d. As para 4d above (Details of Target).

e. As per para 4e (Executive Order).

f. As per para 4f above (Observe and Correct).

OR

(WHEN REFERENCE POINT IS NOT PRE-ARRANGED)

a. As para 4a above (Attract the Attention).

b. As para 4b above (Right Direction).

c. As para 4c above (Give Range).

d. Give the description of Reference Point to be indicated, and 'Name' the Reference Point.

e. Indicate the Target from Reference Point (Give Direction and Range from Reference Point) and also describe the details of the target.

f. As per para 4e above (Executive Order).

g. As per para 4f above (Observe and Correct).

OR

(CREATING AN ARTIFICIAL REFERENCE POINT)

a. As per para 4a above (Attract the Attention).

b. As per para 4b above (Right Direction).

c. As per para 4c above (Give Range).

d. Explain the method of creating the Artificial Reference Point.

e. Indicate Target (Explaining details of Target) from Artificial Reference Point (Give Direction and Range of Target from Artificial Reference Point).

f. As per para 4e (Executive Order).

g. As per para 4f above (Observe and Correct).

Procedure – Using Radio

31. Attracting the Attention

a. The infantry radio sets will be used:-

(1) To offer the target using the tank prefix 99. This may be followed by the tank turret number or designation Alpha/Bravo when only that particular tank is visible to the infantry commander. Examples are as follows:-

(a) Hullo 2 for 99 target over.

(b) Hullo 2 for 99 target for turret 6 over.

(c) Hullo 2 for 99 target for 2 Alpha over.

(d) Hullo 21 for 1 target for 2 Alpha over.

(e) Hullo 21 for 1 target for turret 6 over.

(f) Hullo 21 for turret 6 target over.

Note: - All these examples have been given to fulfil the requirements which are likely to occur due to numerous situations/number and variety of wireless sets or type of radio nets.

(2) To inform a tank commander that an infantry commander is about to move across to contact him either in person or through tank telephone.

b. The use of light or smoke signal is not recommended to attract the attention unless radio/telephone/personal contact is not possible and light or smoke signals are so well pre-arranged that no misunderstanding is possible.

32. **Making the Commander look in the Right Direction**. The infantryman has to imagine himself in the position of the tank and all indications are from that direction:-

a. By pre-arranged reference points – simple and quickest.

b. Using direction of tank's gun barrel. For example: -

(1) Gun barrel – right – 3 O' clock; left –7 O' clock et cetera (clearest method).

(2) Gun barrel quarter/half/three quarter/full right.

(3) Hull – left – 10 O'clock, right – 5O'clock, et cetera (also very clear method but in hull down position, it may not be possible to see the hull).

(4) Axis quarter/half/three quarter/right/left et cetera. (not very clear method; should be used when the other methods are not possible).

c. Firing WP smoke (best); composition: coloured not available with artillery; high explosive, machine-gun burst (poor method – infantry may have to adopt it), mortar (infantry may have to adopt it, better than machine-gun) verey light (infantry may have to adopt it, poor method) for example:-

(1) 2 for 99 firing WP smoke for indication.

(2) 3 for 99 firing mortar smoke for indication.

33. **Range**. The range to the Reference Point being indicated or to the Target, will be as accurate as possible for example:-

a. 2 for 99, target for you over.

b. 2 for 99, your gun axis: right; 2 O'clock; 500, AKELA DARAKHT; left 10 O'clock, 1000 CHAR JHARHIAN; sub say right wali JHARI, destroy, over (The acknowledgement of the message has not been shown in the above example).

34. **Description of Target**. Make full use of physical land marks; describe target in full details; if target is obvious keep description minimum.

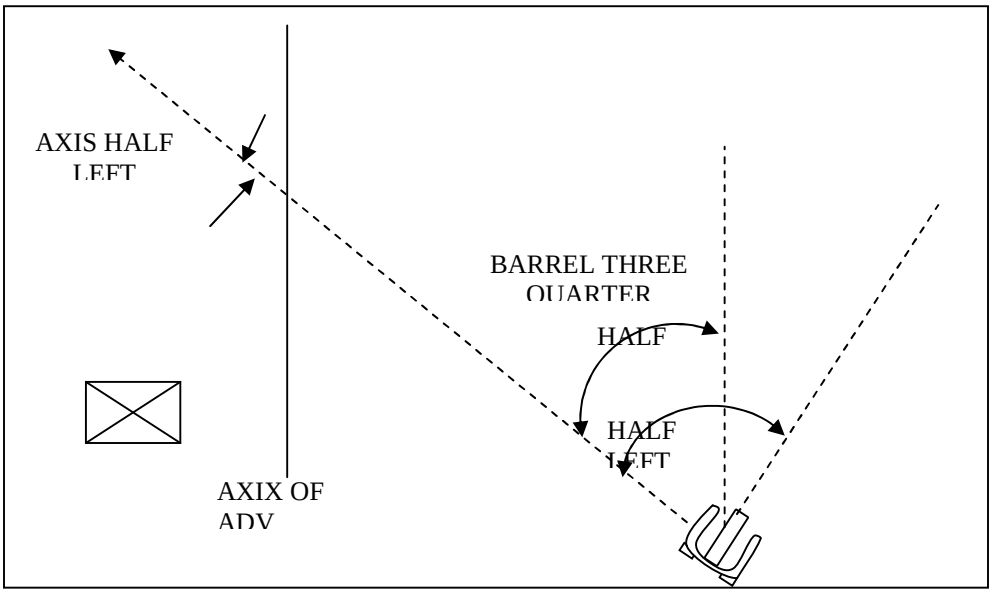


Figure 34 - Infantry – Tank Target Indication

35. **Indicating from Pre-Arranged, Reference Point**

a. Once the tank has been directed to look in the right direction, it is then necessary to direct him to the Reference Point and from the Reference Point to the Target. This can be done either by linking the Reference Point to the target along with the range by some physical land mark on the ground.

OR

b. Where above is not possible, by giving the distance and direction to the target from the Reference Point, in terms of metres LEFT, or RIGHT ADD or DROP. In either case the fullest possible description of the Target and its location should be given, and again the infantryman must endeavour to imagine himself in the position of the tank.

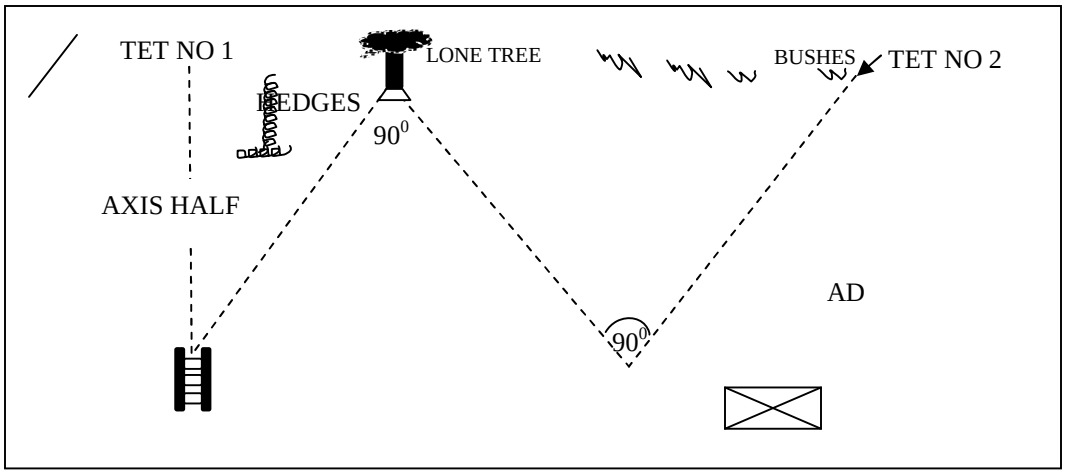


Figure 35 - Infantry – Tank Target Indication

36. **Indicating Form a New Reference Point**. On occasions, it may be necessary to indicate a new Reference Point when one has not been pre-arranged. In this case, first the direction will be given, followed by the approximate range to the Reference Point and lastly a description of the Reference Point. The Reference Point will be named. Once the Reference Point has been identified, then the same procedure as given in paragraph 9 above will be followed in directing the tank on to the target.

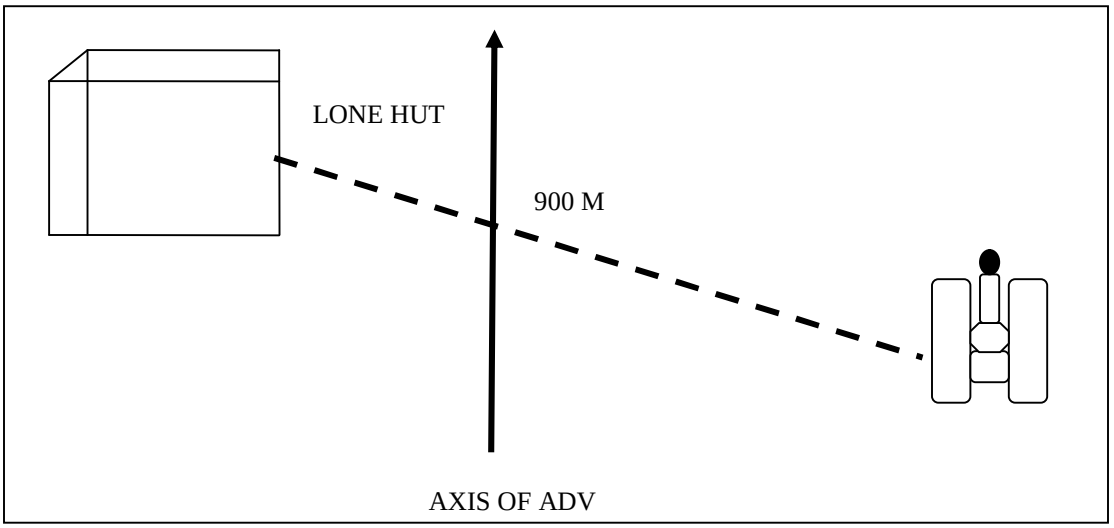


Figure 36 - Infantry – Tank Target Indication

37. **Creating an Artificial Reference Point**. Some times no suitable Reference Point can be found in the target area owing to the nature of the ground.

In such cases, it will be necessary to create an artificial Reference Point, either by the infantry or by the tanks. The infantry will do this by firing a mortar bomb or ask the tank to fire a shell preferably WP or high explosive from the main gun, giving the general direction and range. The point of burst of the bomb shell then becomes the artificial Reference Point and the target is then indicated in the normal way.

38. **Correction of Tank Fire when the Target is Not Identified**

a. If the tank commander fails to see the target or fires at the wrong one, the infantry will:-

(1) Give another description of the target.

(2) Use first short for reference.

b. Correction will be given as:-

(1) Direction by clock rays and range method using first shot as reference.

OR

(2) "Left" or "Right", "Add" or "Drop" in hundreds of metres as the infantry observer sees it on the ground. The tank commander allows for variations in ground when applying gun corrections.

39. **Executive Order or Disposal of the Target**. Once the target has been identified, he should be given the disposal instructions i.e. destroy, neutralize for so many minutes et cetera. The action being taken by the individual who is asking the fire should also be explained e.g., "I am moving round left flank" or "I am remaining behind cover". The decision about the weapon/ammunition will be of tank/infantry commander, who is going to engage the target.

Wireless Procedure

40. The details have been given in the preceding paragraphs for various eventualities. The message below is as an example only:-

Infantry – Hullo 2 for 99 target for you over.

Tank – 99 target over.

Infantry – 2 Axis your gun barrel – right One O'clock-one thousand-TALL TREE-machine-gun-destroy over.

Tank – 99 wilco out.

41. **Method of Tank Infantry Designation of Reference Points**. As given in the preceding paragraphs.

Procedure Using Weapons/Hand Signals

42. These must be pre-arranged and practiced. One should resort to these only when other means are not possible. For other signals see Armour Data and Reference Book.

43. **Attracting Attention of the Tanks**. This is done by the infantryman holding his rifle above his head at the full extent of the arm and parallel with the ground, muzzle pointing in the direction of the enemy. The rifle is moved up and down till such time that tank acknowledges by moving its main gun barrel up and down.

44. **Distance to Target**. After the attention of the tank has been attracted, the infantryman gives distance to target. This is done stretching one arm parallel to the ground in the direction visible to tank, and then carrying on the closing/ opening motion at the elbow. Every close and stretch motion indicates 30 metres.

45. **Nature of Target**. The nature of the target is indicated with the help of thumb, rifle and steel helmet. Fist closed and thumbs pointing down or muzzle of the rifle pointing to the ground indicates enemy anti-tank weapon. The thumb/muzzle of rifle pointing up means enemy machine-gun or any other automatic weapon. Removing the steel helmet from head and showing its top side indicates enemy armour.

The Echelon System

46. **Variations in the Regimental Echelon System**. There are several methods of organising the echelons and many variations in the detailed method of replenishment. The system described in the subsequent paras is one possible method.

47. **First and Second Line Transport**. Logistics at regimental level are concerned with two levels of transport:-

a. *First Line*. This is all unit transport.

b. *Second Line*. This is the transport held by the Supply and Transport Battalion in the formation.

48. **Subdivision of Echelons**

a. *'F' Echelon*. 'F' echelon consists of those men and equipment needed to fight the battle.

b. *'A' Echelon*. This is formed from those men and equipment which must be readily available to support 'F' echelon. 'A' echelon may be further sub divided into:-

(1) *Al Echelon*. This holds only those vehicles needed for 50% replenishment and some fitters.

(2) *A2 Echelon*. This contains the balance of the regiment's POL and ammunition vehicles (50%) together with the LAD and technical stores.

c. *'B' Echelon*. 'B' echelon consists of those men and equipment not required in 'F' or 'A' echelons.

49. 'A' echelon is always formed and is normally sub-divided into Al and A2. At times, 'B' echelon may not be formed.

50. **Reinforcements**. The seven days reinforcements of personnel authorised to the regiment would be deployed in A1 and A2 echelons according to the CO's wishes or formation policy.

Command and Control of Echelons

51. **Overall Responsibility**. The OC Headquarters Squadron is normally responsible to the CO for the work of all echelons.

52. **Command of Echelons and Packets**. Variations in the echelon organisation will dictate who should command the echelons and the squadron packets. There are two essential factors:-

a. Each of these commanders must have their own communications.

b. The tank must not interfere with the commander's primary role. For example, the Mechanical Transport Officer and QM (Technical) is generally considered unsuitable for appointment as A1 echelon commander, since this is likely to interfere with his primary task of providing spares and repair of vehicles.

53. **A1 Echelon**

a. *Summary*

(1) *A1 Echelon Commander*. 2IC HQ Squadron.

(2) *Squadron A1 Echelon Packet Commanders*. Squadron senior JCO.

(3) *Location*. Within the regimental area.

b. *Control*

(1) The A1 echelon is always located within the regimental area and normally centralised under overall control of RHQ. It must therefore be kept as small and mobile as possible.

(2) Squadron A1 echelon packets could at times be decentralised and placed under squadron control. This will ensure that battle replenishment is more immediately available than it would be if the packet had to move from A1 echelon to the squadron area. Less movement may mean less chance of enemy ground or air attack. The disadvantages of placing A1 echelon packets under squadron control are:-

(a) Less flexibility within A1 echelon.

(b) An increase in vehicles in the forward area.

(c) An extra task for squadron headquarters in moving, finding suitable sites for and probably protecting the packet.

(3) If squadrons are given control of their A1 echelon packets, they are located with the squadron headquarter and orders for the senior JCO are passed through the 2IC. However, to facilitate the passage of such orders the SJCO can travel in squadron commander's jeep where a radio set is available on the squadron command net.

54. **A2 Echelon**

a. *Summary*

(1) *A2 Echelon Commander*. OC Headquarters Squadron.

(2) *Squadron A2 Echelon Commanders*. Squadron Quarter Master Defadar (SQMD).

(3) *Location*. Brigade administrative area.

b. *Control*. The echelon is located in the brigade administrative area under the control of the Deputy Assistant Adjutant and Quarter Master General (DAA&QMG), and moves forward to replenish the regiment on orders from the brigade headquarters.

SECTION 106 – **ARTILLERY**

General

1. **Types of Fire Support**. Fire support responsibilities of each artillery unit/subunit towards the subordinate formation and units is specified in the following terms which indicate the degree of certainty with which requests for fire support will be made:-

a. In direct support.

b. In support.

c. At priority call

d. In tactical and technical orders, this is called allotment of artillery support.

2. **In Direct support**

a. An artillery unit or subunit placed in direct support of a subordinate formation or unit becomes responsible to provide guaranteed fire support of its own guns and to arrange for additional fire (un-guaranteed) from other artillery unit/subunit. Normally a close support regiment is placed in direct support of a brigade and its each battery in direct support of an infantry battalion/armour regiment.

b. In order to provide guaranteed fire support, the artillery unit/subunit in direct support establishes liaison and communication with its supported formation or unit and provides observers to it. In order to arrange for additional fire from other artillery unit/subunits, it establishes communications with them.

Artillery organizations and communication cater for these commitments. For instance, in a brigade:-

(1) The commanding officer of the close support regiment advises the brigade commander and the battery commanders are advisers to the battalion/armoured regiment commanders.

(2) Direct communications are established between brigade headquarters and close support regiment headquarters and between battalion/armour regiment headquarters and battery commander and observers on one hand, and between close support regiment headquarters and divisional artillery headquarters and other artillery units on the other hand.

(3) Observers are deployed to cover the battalion/armour regiment frontage within the brigade zone of fire and act as FOO with the assaulting companies/squadrons.

c. In attack, the fire of direct support artillery is normally super imposed in the fire support plan so that it can engage impromptu targets when required by the assaulting troops.

d. It, therefore, follows that the artillery in direct support has to establish a very intimate relationship with the supported formation/ unit that cannot be developed over night. It is therefore, essential that the artillery support and planning should be handled by artillery officer, who are well known to the formation and unit commander, in whom they have confidence and who are well versed in the procedures and customs of their headquarters and units. This is particularly important at brigade and battalion/armour regiment level where the forward infantry and armour unquestionably like to have their ‘own’ gunners with them. Therefore, peacetime affiliations between the supported formations and their close support regiments must be maintained in battle.

e. However, in a battle, situation may arise when the formation commander may consider that the requirement of fire support by a particular brigade is no longer overriding and he may therefore, temporarily place the fire of its direct support regiment at priority call to another brigade or task which may be more important. For the duration, the fire of its direct support regiment will, therefore, remain un-guaranteed to the brigade. However, the advisers, communications and observers will normally not be disturbed.

Although such temporary changes are forced by battle conditions, yet it is always better to revert to original affiliations at the earliest to ensure quicker and more effective artillery support.

3. **In Support**

a. Close support regiments in direct support of brigades answer calls for fire from each other when not firing for their own brigades. In other words, whereas every close support regiment is in direct support of its own brigade, it may be in support of other brigades. In support artillery, therefore, does not provide advisers, communications and observers to the supported formations and units, because this is already provided by the artillery in direct support.

b. Besides close support regiments, general support units are also placed in support of the brigades for augmenting the fire of direct support regiments. Since, general support units normally do not have any direct support responsibility, some of their observers are placed at the disposal of direct support regiments for providing depth to the observation zone or for covering the gaps and inter formation boundaries.

c. Since the number of artillery unit /subunits in support of a formation/unit is generally more than one, and because every artillery unit/subunit is in support of more than one formation/ units, it is necessary to exercise control so that calls for fire are met judiciously. Normally control is exercised by specifying:-

(1) The duration of support, in terms of time and in relation to phase of the operation.

(2) The permissible ammunition expenditure in rounds per gun.

d. Since the fire of the support artillery may not be available when required or may be withdrawn (lifted) any time, therefore it is super-imposed in the fire plan so that its non-availability or withdrawal does not materially affect the fire support.

4. **At Priority Call**

a. An artillery unit/subunit that is not in direct support of any formation but is merely in support of any formation or units may be placed at priority call to a formation or unit or to a specific task. For instance, general support artillery of a division may be in

support of the brigades but at priority call to a particular brigade whose fire support requirement is more than the other or may be at priority call to the divisional counter bombardment officer for counter bombardment.

b. An artillery unit/subunit in direct support of a formation or unit cannot be placed at priority call to another formation or unit. However, at times it may be temporarily placed at priority call to another formation or unit for a specific operation or period of time for which duration; it will no longer be in direct support of its own formation or unit. However, the artillery commanders, observers and communications will not be changed. For example, a close support regiment in direct support of a brigade in reserve during one phase of an attack may be placed at priority call to the brigade in contact with the enemy during that phase.

c. The fire support of 'at priority call' artillery is guaranteed for which it provides communications to the supported formation or unit but normally does not provide adviser and observers, which are provided by the direct support artillery.

5. Artillery supports infantry operations as follows:-

a. *In Attack.* Covering fire.

b. *In Defence.* Defensive Fire or DF.

6. Artillery may also provide the following types of support:-

a. Harassing fire on known and suspected targets to disorganize the enemy and lower his morale.

b. Smoke to:-

(1) Blind enemy observation and so prevent observed fire by his artillery and aimed fire by his small arms and anti-tank weapons.

(2) Screen movement of own troops.

(3) Deceive enemy.

(4) Indicate the beginning or end of a particular phase of a fire support plan.

c. Counter bombardment (CB) to neutralize or destroy enemy artillery and mortars.

d. Illumination by star shell, or indication of targets to fighter aircraft by firing coloured smoke.

Covering Fire

7. To get maximum advantage from covering fire, infantry must move as close to it as safety allows. Before any attack, therefore, the artillery officer should be asked about the safe distance that will vary with the type of gun supporting the attack and the direction from which it is firing. As a guide the safe distance (for 105 mm field guns or 105 mm howitzers) is 200 metres and for medium guns. (130 mm guns or 155 mm howitzers) is 400 metres.

8. Concentrations can be “timed” or “on call”. In an attack, concentrations on timed programme basis are, however, preferable. When timed concentrations are being fired, infantry must maintain the expected rate of advance, otherwise they will either be left behind and the covering fire will lose its effectiveness or they will rum into own artillery fire and thus suffer casualties.

9. Some targets which are likely to interfere at any stage of attack may be placed on call. Concentrations on call be converted into timed concentrations, when required by calling the target number and giving time for which it is to be engaged. Example, “Target U 1546, engage forthwith, 5 minutes, normal”.

10. Usually a judicious mixture of timed and on call concentration will form the covering fire schedule for attack.

Defensive Fire

11. In defence, likely enemy FUPs and approaches to battalion position, and at times even some of own positions are selected as DF tasks. Each DF task is given a number so that fire can be called for and brought down quickly on the target. The most dangerous approach to the battalion position is selected as the DF (SOS) task. When the guns in direct support are not firing, they are always laid on this target. There can be only one DF (SOS) task for one firing unit. In the case of field and medium guns it is a battery and in the case of mortars it is a troop.

12. Company commanders can suggest to CO the DF tasks immediately in front and in the vicinity of their localities but their (DF Tasks) final selection and coordination is done at battalion level, the CO and the battery commander having the final say in this matter.

Responsibilities of Formation or Unit Staff

13. From the above, it follows that closest touch must be maintained by the formation and unit staff with their artillery staff because the deployment areas and movement has to be coordinated with the dispositions and movement of other arms. This is particularly so when problems of security and movement of artillery are involved. The artillery commanders plan but the formation or unit staff has to:-

a. Approve the areas selected by the artillery commanders.

b. Inform subordinate formations or units, where and when different elements of artillery will reconnaissance, survey or deploy within their boundaries.

c. Allot priority to the movement of artillery reconnaissance and survey parties.

d. Arrange protection of artillery.

e. Allot routes and timings when movement is involved.

f. Arrange for the supply and movement of ammunition to firing positions/wagon lines or at ammunition points.

g. Issue instructions for concealment plan.

h. Keep the deployment and registration timings in mind when deciding about the timings of the operation.

Control

14. Artillery fire is controlled by the observers and the battery commander who is at battalion headquarters, all of whom have wireless, and possibly line communications to the guns. During an attack an artillery observer will often move with the attacking infantry in which case, he is called a Forward Observation Officer (FOO). His tasks are:-

a. To act as forward Observation Post (OP).

b. TO ENGAGE OPPORTUNITY targets and modify the fire plan when asked by battalion commander or whosoever has made the plan.

c. To engage DF tasks covering the objective.

d. To pass back information.

15. A field regiment can provide nine observers, so if a battery is in direct support of a battalion, it will have three observers available to it. In defence these observers will be located in forward company/platoon positions. The later must help the observers to get the best positions for their Operations and ensure they are adequately protected.

Target Indication and Engagement

16. Infantry platoon commanders must know how to control artillery fire so that they can take over temporarily if the artillery observer becomes a casualty or if one is not present. Method of indication and engagement of target and of controlling artillery fire is simple. It is fully explained in Annex 'Z'.

Action by Supported Arm Observer

17. Need for indication of a target to the artillery observer on the wireless set will only arise when the concerned supported arm element does not have personal contact with the artillery observer. In such a case following procedure is used:-

a. The supported arm observer will send a special warning order to the artillery observer.

b. He will then indicate the target to the artillery observer by giving:-

(1) Location of the target.

(2) OT bearing. (Observer to target bearing)

(3) Description of the target.

c. Lastly, he will specify to the artillery observer as to:-

(1) What effect is desired from artillery fire.

(2) When and for how long fire is to be provided.

18. **Special Warning Order**. This is sent to attract the attention of the artillery observer and to tell him about the presence of a target e.g.,"target, target, target".

19. **Location of the Target**. This is normally given as a grid reference (without height) e.g., "Grid reference 654529". If the supported arm observer cannot give a grid reference, then he will use one of the following alternatives:-

a. *Reference Point*. A reference point is used which is both obvious and clearly visible, or has been previously agreed to, or both the

parties know a recorded target. The target is then indicated by a target grid correction from the reference point with reference to OT bearing, e.g.: -

”Reference clump. Oscar tango 820 mils. Go right 300, add 200”.

OR

“Reference target M 6998. Oscar tango 4500 mils. Go right 200 add 400”.

b. *Reference Salvo*. In badly mapped or featureless country, the artillery observer may be asked to provide a reference point in the shape of a concentrated six rounds salvo for which he should be given a rough indication of where to place it, such as by reference to the axis of advance or a map square, e.g., “Fire salvo for reference point in square 5525”. The supported arm observer then gives a target grid correction from the salvo. Go left 200, drop 400”.

c. *Tracer or Smoke*. The supported arm may fire tracer or smoke at the target to indicate it to the artillery observer.

20. **OT Bearing**. After giving the location of the target by one method, the observer then sends OT bearing.

21. **Description of Target**. This should include tactical as well as physical description e.g.:-

“A light machine-gun under a bushy top tree”.

OR

“Two hundred infantry along with troop of tanks, forming up for attack in green fields”

22. **Effect Desired**. This indicates type of support required, e.g., neutralization or smoke et cetera instead of HE at the target.

23. **Time and Duration of Fire**. This is used to indicate to the artillery observer the time and duration for which fire is required in one of the following ways:-

a. “Neutralize Now”, effective HE fire is required as soon as possible.

b. “Smoke at 1200 Hours for 3 Minutes”.

c. “Neutralize for 5 Minutes. Report when Ready to Engage”.

24. **Example**. An example of the complete target indication by supported arm observer to the artillery observer is as follows:-

“Target, target, target”.

“Grid reference 659582. Oscar tango 4540 mils”.

“Two machineguns in brown bushes beyond the green fields”.

“Neutralize for 3 minutes now”.

Action by the Artillery Observer

25. On receiving these Orders:-

a. The artillery observer will read back the fire orders received from the supported arms observer.

b. If he identifies the target immediately, he will say “target identified” and will himself carry-out the shoot straight away.

c. If he does not identify the target immediately he will say “wait out”.

d. When he succeeds in identifying the target, he will say, “Target identified” and will carry-out the shoot.

e. If he cannot identify the target at all, he will say “target not identified, send corrections”

f. Whatever the case, he will edit the fire orders received from the supported arms observer and pass them back to the guns over the artillery net so that the first round is fired without delay.

Procedure for Engagement of Targets

26. When the artillery observer reports “target not identified, send corrections”, then the supported arms observer has to take the shoot himself till the artillery observer reports, “Target identified”.

27. Seldom will the first round fall on the target. Invariably the fire will have to be corrected and brought on the target by sending corrections through a process called ranging, which consists of:-

a. Firstly, bringing the round on the line OT by sending corrections for line.

b. Then bringing the round on the target along line OT by sending corrections for range to establish a bracket.

Action by the Supported Arm Observer

28. **Correction for Line**. On seeing the first round, the supported arm observer will measure how much (in degrees or mils) it is to the right or left of the line OT, apply to it the OT factor and convert the angle into metres. He will then order a correction to bring the next round on to the line OT so that a definite observation can be made as to whether the round is "Over" or "Short". If such an observation is not obtained another line correction is ordered.

29. **Bracketing for Range**. If the first round falls on the line OT or near enough to it (in the OT lane) or as soon as a round is obtained on the line OT by sending corrections for line to give a positive indication of it being 'over' or 'short' the supported arm observer starts correcting for range along this line. As it is not possible to measure accurately how far "Over" or "Short" of the target a round may be, a process of bracketing is used. Basically this consists of:-

a. Getting an opening "long bracket" of 800, 400 or 200 metres depending upon the judgement how much 'over' or 'short' the round appears to be from the target. Whenever in doubt give a bold correction of 800 metres should be given.

b. Halving the long brackets progressively until a "short bracket" or 100 metres is obtained.

30. For example, if a correction of "Add 800" produces a bracket, it will be followed by "Drop 400". If this round falls "short" the next correction will be "Add 200" and so on. As soon a 100 metres bracket is obtained, the supported arm observer orders, "Add (or Drop) 50, on target". Should the target be hit before ranging is completed' the supported arm observer will immediately report "On target", and no further ranging will be necessary.

31. At any stage, if the supported arm observer does not see a round fall or burst, he will send the report "unobserved" to the artillery observer or ask for smoke round to be fired.

Action by the Artillery Observer

32. Each time the guns fire the artillery observer reports, "Short" to the supported arm observer and the latter orders a correction. If the artillery observer can see the rounds, he must visualize the OT line of supported arm observer and,

from each correction ordered, try to identify the target. As soon as he identifies the target he will report, “Target identified” and will take charge of the shoot.

33. On receiving the report “Unobserved” from the supported arm observer, artillery observer decides which of the following courses to adopt:-

a. Fire a salvo at the same place.

b. Give a correction that is likely to give the supported arm observer a better chance of seeing the round.

c. Fire a round of smoke.

Fire for Effect

34. When ranging is completed and the supported arm observer reports, “On target”, the artillery observer punishes the target in accordance with the supported arm observer’s initial orders. If, however, the supported arm observer has ordered, “Report when ready to engage” the artillery observer reports, “Ready to engage” and waits for the supported arm observer to order, “Fire”.

35. The supported arm observer must report whether or not the fire is effective and, if necessary, should order a correction.

36. The supported arm observer may, if he wishes, ask for the target to be recorded for further use. He will then be given the target number by the artillery observer.

SECTION 107 – **FIELD ENGINEERING**

1. **Field Defences**

a. *General*

(1) What are the chances of survival on the modern battlefield? Are the odds on or against? The answer is summed up in the motto “Dig or die”. Digging and the intelligent use of materials will transform the better into odds on survival, even taking into account the use of nuclear weapons, but to be effective, the digging and the use of materials must be intelligently done. Practical training based on sound principles and practices is the only way to success. It is a small premium to pay for a very large reward in battle.

(2) Field defences are completed only when all the weapons and troops have been properly located, various weapons positions and shelters prepared and the required obstacles erected around the position for local protection. In other words it encompasses all digging, provision of overhead cover/protection and construction of artificial obstacles.

(3) This paragraph will deal with construction of trenches, weapon pits and shelters only.

b. *Principles of Design*

(1) *Use of Weapons*. Fire positions and weapon emplacements must allow completely effective use of the weapons for which they are designed.

(2) *Protection*. As far as other requirements permit, protection should be provided against all probable hazards short of a direct hit or a close nuclear explosion.

(3) *Speed of Construction*. This is vital for saving casualties.

(4) *Simplicity and Economy*. Designs should be strong and simple, economical in digging and materials, and make use of materials or stores likely to be available.

(5) *Progressive Development*. The design of a defence work should as far as possible allow of its progressive development and improvement to make it either stronger or more habitable or both.

(6) *Concealment*. Designs must be such that the completed work is reasonably easy to conceal.

c. *Nuclear Effects*

(1) Defences designed against modern conventional weapons are in general extremely effective against nuclear weapons. The presence of the nuclear threat does not materially alter the basic principles of design outlined above but it raises certain points of principle importance.

(2) The thermal effect is the greatest potential source of nuclear casualties. Thermal shielding in the form of any opaque and non-inflammable material virtually neutralizes the thermal effect. It will normally be used as an overhead

cover (OHC) placed over loopholes and shelter entrances, et cetera.

(3) Blast and overpressures resulting from a nuclear explosion tend to ‘build up’ against an obstruction. Hence in earth-works above ground level, vertical or near vertical faces should be avoided.

(4) Earth-shock may cause widespread collapse of earth-works, particularly of unrevetted excavations. To reduce the risk of burial, excavations should not normally be dug more than four feet six inches below ground level unless they are very strongly revetted. For the same reason, overhead protection, unless very strongly supported, should not normally exceed 45 centimetres in thickness.

(5) Direct gamma radiation is best warded off by keeping excavations and shelters as small as possible and, in the case of large shelters only by increasing up to 75 centimetres the thickness of the OHP.

d. *Principles and Methods of Construction*

(1) The basic works which are the foundation of most types of field defences are as follows:-

(a) Open trench and pits.

(b) Revetment of trenches and pits when necessary.

(c) Overhead cover when required.

(d) Overhead protection when required.

(e) Drainage.

(2) *Slit Trench*. It is the basic for protection and it is the first step towards provision of OHC and OHP by improvised methods. It may be required for any of the following purposes:-

(a) For protection which will include:-

i. Communication trenches.

ii. Shelter trenches.

iii. Access trenches.

iv. Basis for small improvised shelters.

(b) Fire trench in a tactically sited fire position.

(3) *Revetment*. Trenches and weapon pit will normally be riveted in any of the following circumstances:-

(a) To prevent the walls from falling in poor soil.

(b) In all soils before OHP is added.

(c) To repair a broken down trench et cetera.

(d) To strengthen defence which is taken for long duration.

(e) In all circumstances for excavation deeper than 1.5 metres from ground level to standing level.

(4) *Overhead Cover (OHC)*

(a) Provided by any form of roof which shields against flash and light effects of nuclear air burst. It should not touch the body and be non-inflammable.

(b) Required OHP can be provided over open pits or slit trenches which are meant for sleep or which are meant for protection against air attacks et cetera.

(c) Materials used should be such that repair and replacement can be easily done.

(5) *Overhead Protection (OHP).* These are designed primarily to stop fragments from air bursts but if well constructed it will give a considerable degree of protection against light mortar bombs, flame and nuclear effects.

(6) *Drainage*. In wet weather and wet soil, open trenches and pits must be drained to keep them fit for use. Drainage of any of the following three categories may be employed:-

(a) *Catch Water Drains*. To prevent surface water from entering the pit et cetera.

(b) *Internal Drainage*. To lead water which has entered into one or more low points, this can then be drained by external means or by means of stone filled sumps.

(c) *External Drainage*. To lead water from the low points out of the trench and into natural drainage or external soakage pits.

e. *Types*. The trenches, weapon pits and shelters for infantry are given in Annex 'Q'.

2. **Obstacles**

a. Purposes of Obstacles

(1) Tactical obstacles are those placed in the enemy's way in order to prevent or delay his movements, to divide and disperse his attack, or to shepherd him into positions where he will be particularly vulnerable to attack.

(2) A tactical obstacle may be:-

(a) *Anti-tank*. Against movement of AFVs, both across country and on road.

(b) *Anti-personnel*. To stop, slow down or confuse enemy infantry attempting to overrun, or infiltrate into a defended post or locality.

(c) *Anti-vehicle*. Usually in the form of road blocks designed to stop the enemy's wheeled vehicles where they can be brought under aimed fire; the most usual form of ambush.

(3) Non-tactical obstacles may be required for various purposes, usually one of the following:-

(a) For the protection of important installations against infiltration and sabotage, et cetera.

(b) In a policing operation, to check movements of crowds rioters; or to isolate a street or district of a town.

(c) For administrative purposes such as vehicle checkpoints.

(4) Non-tactical obstacles may often be of the same designs as those in paragraph 2 a (2), (b) and (c) above. But there will not be the same considerations in regard to sitting and concealment and the conditions under, which they are constructed, will be different.

b. *Obstacles to be Covered by Fire*

(1) Obstacles designed to stop the enemy must be covered by observation and fire from the defender's position. An

obstacle which is not so covered may, by giving the defender a false sense of security that is worse than useless.

(2) Obstacles created during a withdrawal for the purpose of delaying the enemy's advance need not always be covered by aimed fire. But unless so covered, their delaying power may be less than expected. Mines and booby-traps will usually be employed to increase delaying power.

c. *Principles of Sitting and Design*

(1) The following are the main principles governing the sitting and design of obstacles:-

(a) *Direct Fire*. An obstacle, if it is to have any sure stopping power must be under the direct observation and fire of the defender. Therefore, the sitting of obstacles and of fire positions is interdependent. Artificial obstacles, such as barbed wire fences, can be sited to suit fire positions, but where natural obstacles are made use of; the fire positions must be sited so as to cover the obstacle.

(b) *Concealment*. The ideal is an obstacle of whose existence the enemy is unaware until he runs into it. It will then cause surprise and confusion because plans will not have been made in advance to deal with it. A large, obvious obstacle may produce confidence in the defenders but if a smaller one can be concealed, it may prove far more effective in the long run. The use of ground to provide shielding for an obstacle may also be important in other respects.

(c) *Strength*. The obstacle should be strong enough to provide real stopping power. The extent to which this can be produced must be balanced against other requirements.

(d) *Irregularity*. Linear obstacles laid out in large and irregular zigzags tend to break up an attack by shepherding the enemy into pockets; they are also as a rule, more easily covered by fire from a flank. The distance between successive belts of an obstacle should be varied. This will make it more

difficult for the enemy to destroy or overcome them.

(e) *Denial of Cover*. No obstacle should be capable of giving effective cover to the enemy. Ditches, banks, et cetera unless under effective fire from a flank must be made unusable.

(f) *Cloaking of Fire Positions*. An obstacle should not point directly to a fire position enfolding it. Dummy obstacles may sometimes be used in order to mislead the enemy as to the positions of defended posts.

(g) *Facilitating the Defence*. Local counter-attack plans must be provided for. Gaps left for this purpose or for movement of our patrols, must be disguised and covered normally by a moveable obstacle.

(h) *Protection against Hand Grenades*. There should be a distance of at least 40 metres between an obstacle and the nearest occupied post.

(2) No one obstacle is likely to meet all the above requirements in full; but these all should all be considered.

d. Types of Infantry Obstacles

(1) Natural obstacles

(a) Effective obstacles:-

i. Deep or fast flowing rivers.

ii. Canals.

iii. Bogs and marshes.

iv. Cliffs beyond easy human climbing.

(b) Partial obstacles:-

i. Thick hedges and creepers.

ii. Existing fences.

iii. Jungle et cetera.

(c) All the above have to be improved and thickened up. Thick jungle must NOT be accepted as impenetrable.

(2) *Artificial Obstacles*. Artificial obstacles are of two types:-

(a) *Improvised Obstacles*. When nothing is available from the standard equipment then something, which is locally available, should be made use of in the best possible way. This will require ingenuity. Some of the suggested obstacles are:-

i. Thorny bushes.

ii. Camouflaged pits and trenches.

iii. Dewheeled trucks et cetera.

iv. Panjis – used in jungle. They are made of a bamboo, which is hardened by heat.

(b) Obstacles Made of Standard Equipment:-

i. Barbed wire obstacles.

ii. Anti-personnel mines and booby-traps (these are NOT dealt with in this chapter).

(c) Combination of natural and artificial obstacles.

e. *Wiring Stores*

(1) *Wires*

(a) Ordinary barbed wire is normally issued in 12.7 kilograms reels, each containing 120 metres of barbed wire.

(b) To run barbed wire off a reel, put a short picket or windlassing stick through the reel. Ordinary barbed wire can be twisted and windlassed or cut with ordinary wire cutters.

(c) Dennert wire is issued only in the form of concertinas. Each concertina weighs 23 kilograms and when stretched out, have a length of 14 to 16 metres and a diameter of about 1 metre. Dennert wire is made of steel and it cannot be twisted or

windlassed, nor cut with ordinary wire cutters. Concertinas may be issued parceled.

(2) *Pickets*

(a) The long angle iron picket is 1.8 metres long and weighs about 5.5 kilograms. The notches are for the attachment of wires. When the picket is driven 45 centimetres into the ground, the uppermost notch is about 1.2 metres above ground level.

(b) The short angle iron picket is 60 centimetres long and weighs about 2 kilograms. When driven in at an angle as an anchorage the uppermost notch is about 15 centimetres above ground level.

f. *Trip Wires*

(1) A trip wire consists of a singe strand f wire fixed to a line of short pickets, or to trees, bushes, et cetera. It should be fixed about 23 centimetres above the ground and tightly stretched.

(2) *Concealment.* Conceal the trip wire where practicable, e.g., in long grass or growing vegetation, and avoid trampling this down as far as possible when laying the wire. In open ground, the wire will be less obvious if laid along a natural line such as the side of a track or edge of a field.

(3) *Pickets*

(a) Spacing of the pickets will depend to some extent on how firmly they can be driven-in, in hard ground, they may be up to 5 metres apart and in soft ground these can be at a distance of 3 metres or less from each other.

(4) Short angle iron pickets are suitable; or use timber pickets 5 to 8 centimetres in diameter and 60 to 90 centimetres long. Use trees, bushes, et cetera, if they exist, instead of pickets.

(5) *Wire*. Ordinary barbed wire is normally used. Plain wire is equally suitable and is fastened in the same way.

(6) *Party*. Four men are a convenient party.

Numbers 1 and 2 carry and drive in the pickets.

Number 3 carries the reel of wire.

Number 4 fastens the wire to the pickets.

(7) Stores for 100 metres of Trip Wire.

	Number	*Man Loads*
Short pickets	30	4
Barbed wire reels, 120 metres	1	1

(8) *Tools*

Sledge hammers (for angle iron pickets)	1	
Mauls (for timber pickets)	1	
Windlassing sticks (1 per man)	4	
Wire cutters	1	
Wiring gloves, pairs	1	1
Total Man Loads:		**6**

g. *Low Wire Entanglement*

(1) The depth of the entanglement from front to rear should be about be about 1 metres. Further belts may be added in depth. The height of the wire may depend on concealment, but in general it should be between 15 to 60 centimetres.

(2) *Concealment.* "Low Wire" is commonly used in scrub or time woodland, where is can as a rule be very well concealed. It is not so suitable for long grass or growing crops since these are bound to be trampled down whilst the wire is being laid and this will show up for at least several days.

(3) *Pickets*

(a) Whenever possible, use trees, bushes, et cetera, instead of pickets.

(b) If pickets are used, space them irregularly at 2 to 4 metres intervals.

(c) Short angle iron pickets can be used but they must be driven in at least 15 centimetres and the maximum height of the wire can then be only 45 centimetres.

(d) Timber pickets about 1 metre long and 5 to 7 centimetres in diameter are suitable and may often be driven in 15 to 30 centimetres, using mauls.

(4) *Wire*. Use ordinary barbed wire (120 metres reels). Fix the wire to the pickets at varying heights between ankle height (15 centimetres) and knee height (about 60 centimetres). The wire should be fairly tightly stretched.

(5) *Party*. For a belt which is 10 metres wide, a wiring party of one NCO and eight men is convenient. They first place the pickets, four driving, four carrying, and working along the belt. They return then to one end and two pairs wire along the belt in echelon, each pair "wandering", over half the belt. One man in each pair carries the reel and the other fastens the wire. The remaining two pairs follow adding cross wires and other wires as necessary over the whole width of the belt.

(6) Stores for 100 metres of low wire entanglement which is 10 metres in width are as under:-

	Number	*Man Load*
Short pickets, angle irons of forestry	200	25
Barbed wire reels of 120 metres	15	15

(7) *Tools*

Sledge-hammers or mauls	4	2
Windlassing sticks	8	2
Wire cutters	2	
Tracing tapes	3	2
Wiring gloves pairs	4	
Total man loads:		**46**

(8) *Time*. Construction time varies considerably according to the ground; it is unlikely to be less than two hours for 100 metres in daylight. Construction of this wire obstacle should not normally be attempted at night.

h. *Single Concertina*

(1) *Party*. 1 NCO and 10 men.

(2) *Time*. Under average conditions, for 100 metres of fence, it is 20 to 30 minutes by day and 40 to 60 minutes at night.

(3) Stores for 100 metres of fence.

	Number	*Man load*
Long pickets	25	7
Concertinas	6	6
Barbed wire reels, 120 metres	1	1

(4) *Tools*

Sledge hammers (or thumpers)	4 (2)	
Windlassing sticks	10	
Wire cutters	2	
Tracing tapes	1	
Wiring gloves, pairs	8	
Total man loads:		**14**

j. *Stores and Marking Out*

(1) *Carrying Stores*. Complete party makes one journey. Numbers 5 to 10 make a second journey.

(2) *Marking Out*. Nos. 1 and 2 under direction of NCO lay tape to mark the line of pickets.

(a) *Fence Construction*. The fence (100 metres) is constructed in one task as follows:-

(3) Numbers 1 and 2 carry and position the 25 pickets along the tape. Distance between pickets should be approximately

4 metres. They then fetch the reel of barbed wire and stand by to fix the fence wire.

(4) Numbers 3 to 6 working in pairs, follow numbers, 1 and 2 and drive in the pickets to a depth of 45 centimetres.

(5) Numbers 7 to 10 carry, extend and place the six concertinas working individually, each man in turn takes one concertina and drops it behind the 3rd, 7th, 11th, 15th, 19th and 23rd pickets. Then working as a team, the four men open out each concertina in turn and drop it over five pickets, i.e., the picket behind which it had been placed and two pickets to either side.

(6) As soon as the first concertina has been placed, numbers 1 and 2 starts fixing the fence wire. The wire is fixed to the pickets just above the concertina and then windlassed to the concertina about midway between pickets. It should be moderately slack when first fixed to the pickets; it will become tighter when windlassed.

(7) *Simplified Drill*

(a) The construction of this fence is so simple that normally no drill is necessary. All that is required is for the concertinas to be extended in line and the minimum number of long pickets driven in to hold them in place and prevent them closing up.

(b) One picket should be driven at the centre of each concertina when extended and one picket at each end of it driven through the end coils of adjacent concertinas. Thus the minimum number of pickets required is two per concertina plus one.

(c) 100 metres of fence can be constructed in 5 to 10 minutes.

(8) For operational night wiring, the following drill was used in the western desert in the 1939-45 War, in particular for the immediate wiring of a captured position: -

(a) Men were lined up in parties of four (two pairs) in front of the FDLs and at 15 metres intervals, covering the length of wire to be laid.

(b) Each Party Carried. Ist Pair. One concertina closed but unfastened

2nd Pair. One sledge hammer and one long picket each.

(c) On a word of command all the parties advance a given number of paces (usually 30) to their front and then halt.

(d) The two concertina men, helped if necessary, extend their concertina outwards both directions and hold on to it. If they could meet the neighbouring concertina so much the better but time is not wasted to achieve this.

(e) The two picket men each drive one picket at his end of the concertina.

(f) Each party then returns to the FDLs independently.

(g) Each operation is usually completed in less than 10 minutes.

(h) Any gaps in the line of concertinas are plugged with further concertinas the following night.

k. *Triple Concertina*

(1) *Party*. 1 NCO and 10 men.

(2) *Time*. Under average conditions for 91.4 metre of fence:-

(a) 40 to 60 minutes by day.

(b) 1 ½ to 2 hours by night.

(3) Stores for 100 metres of fence are:-

	Number	*Man loads*
Long Pickets (including 2 spare)	52	13
Concertinas	18	18
Barbed wire reels, 120 metres	4	4

(4) *Tools*

Sledge Hammers (or thumpers)	2

	Number	*Man loads*
Windlassing sticks	10	
Wire cutters	2	
Tracing tapes	4	
Wiring gloves pairs	11	
Total man loads:		**35**

(5) *Stores and Marking Out*

(a) *Carrying Stores*. Complete party makes three journeys. Numbers 3 to 10 makes a fourth journey.

(b) *Marking Out*. Numbers 1 and 2 under direction of NCO lay tape to mark the line of the first (enemy side) row of pickets.

l. *Cat wire Fence Type 1*

(1) *Party*. 1 NCO and 10 men.

(2) Stores for 100 metres of fence are:-

	Number	*Man loads*
Long pickets	64	16
Concertinas	12	12
Barbed wire reels, 120 metres	6	6

(3) *Tools*

Sledge-hammers (or thumpers)	4	
Windlassing sticks	10	
Wire cutters	2	
Tracing tapes	4	
Wiring gloves pairs	11	3
Total man loads:		**37**

(4) *Stores and Marking Out*

(a) *Carrying Stores*. Complete party makes three journeys; Numbers 3 to 10 make a fourth journey.

(b) *Marking Out*. Numbers 1 and 2 under direction of the NCO lay out tape to mark the line of the first (enemy side) cattle fence.

m. *Cat-wire Fence Type 2*

(1) *Party*. 1 NCO and 10 men.

(2) Stores for 100 metres of fence are:-

	Number	*Man Loads*
Long Pickets	96	24
Concertinas	24	24
Barbed wire reels, 100 metres	9	9

(3) *Tools*

Sledge-hammers (or thumpers)	4	
Windlassing sticks	10	
Wire cutters	2	
Tracing tapes	4	
Wiring gloves pairs	11	3
Total man loads:		**60**

(4) *Stores and Marking Out*. As for the Type I fence, but carrying the stores will require six journeys by each man.

n. *Cat-wire Fence Type 3*

(1) *Party*. 1 NCO and 10 men.

(2) Stores for 100 metres of fence are:-

	Number	*Man loads*
Long pickets	128	32
Concertinas	30	30
Barbed wire reels, 120 metres	12	12

		Number	*Man loads*
(3)	*Tools*		
	Sledge-hammers (or thumpers)	4(2)	
	Windlassing sticks	10	
	Wire cutters	2	
	Tracing tapes	4	
	Wiring gloves pairs	11	3
	Total man loads:		**77**

(4) *Stores and Marking Out*. As for Type 1, but for carrying the stores the complete party makes seven journeys, and numbers 4 to 10 make an eighth journey.

Road Blocks

3. **Sighting and Design**

a. In choosing a site for a tactical road block, the following should be aimed at:-

(1) A site in which it will be difficult for the crew of an approaching vehicle to see the block until close to it, e.g., just on the home side of a blind corner.

(2) A stretcher of road where deviation is difficult or impossible, e.g., between woods, on an embankment, causeway, or bridge, in steep defiles, or on a “shelf” road.

(3) A site where the block and the ground immediately in front of it can be seen and covered by fire from concealed fire position.

b. Tactical roads blocks will nearly always have to be improvised from local resources and must be designed to use such materials as are available. Various methods are described in paras ‘g’ below.

c. To provide an effective anti-vehicle obstacle, the block should be capable of stopping a heavy vehicle such as an armoured car charging it. If the block is sited round a bend as described in para ‘a (1)’ or if mines are laid in front of it, a vehicle will be slowed down before reaching the block itself.

d. Obstacles are usually associated with mines and an enemy who has become “mine conscious” may often be halted by a token obstacle, such as a single concertina fence, laid across the road. If no mines are available, small heaps of rubble placed in a regular pattern on the road surface in front of the block may be equally effective.

e. Likely deviations should be blocked or mined.

f. If a moveable block is required, the gap should, on a wide road, be at right angles to the block.

g. *Method of Construction*

(1) *Hasty Road blocking*

(a) *Wrecked Vehicle Block.* Carts, lorries, motorcars, buses, et cetera may be placed across the road. If time allows, the block can be made more efficient by:-

i. Previously loading the vehicles with earth, stones, or other heavy materials.

ii. Tying the vehicles together with wire or wire rope.

iii. Removing or badly damaging the wheels. If a movable block is required a gap must be left and one or two vehicles (preferably runners) used as a gate.

(b) *Barrel Block.* Tar barrels or 200 litres oil drums filled with earth, rubble, et cetera, make a fairly good anti-vehicle obstacle, there should be at least two rows. The normal oil drum is 60 centimetres in diameter, so one drum for each foot width of road will be required. The drums from which the tops must be cut out, should be placed in position empty and then filled. If the road is wide, a truck will be necessary to transport the earth et cetera, which may then be filled into the drums off the tailboard. If no drums are available containers of similar size may be improvised with wire netting. Such containers, called “gabions” will not hold earth but can be used for containing stones, rubble, broken bricks, et cetera.

(c) *Tree Block.* Falling trees across it can block a road of moderate width. The trees should be fallen so that the trunks will remain attached to the stumps and if possible, lying across each other. To effect this, the cut should be made only on the farther side of the trunk and to about two third of its thickness; the tree must then be pulled or pushed over in the direction it is required to fall. Trees of 30 centimetres diameter are big enough but they must be of such a height that when felled they cover the full width of the road.

(d) *Concertina Block*

i. Concertinas are laid in a series of loops. There should be at least two belts of concertinas about 10 metres apart.

ii. A single concertina fence stretched straight across the road plus mines, real or phoney, may be far better than nothing.

(e) All blocks will be made more effective by the addition of mines. These should be laid in front of the block, in turning places, and in possible deviations.

(f) *Deliberate road Blocking*. For the construction of road blocks required mainly for administrative or policing purposes, there will usually be more time; and a supply of common defence stores will as a rule be available. The most useful of these are:-

i. Sandbags.

ii. Long pickets.

iii. Barbed and plain wires.

iv. Concertinas.

v. CGI Sheeting.

4. **Mine Warfare**

a. *General*

(1) Obstacles play an important role in various operations of war. The commanders at all level endeavour to make use of the natural and artificial obstacles to their advantage. Mines are also one type of artificial obstacle, which can be used to a great advantage particularly during defence and withdrawal.

(2) Mines are explosive devices designed to achieve a specific damage on vehicles or personnel, which come over them. These are also double-edged weapons because these cannot differentiate between friend and foe. These must always be employed after careful planning and coordination between all arms because otherwise these can cause more damage and casualties to own forces and can also restrict own mobility. To be able to derive maximum advantages, it is important that infantry must be fully aware of the various aspects of mine warfare.

b. *Types of Minefields*

(1) *Protective*

(a) Laid to assist the unit laying it in its local close protection.

(b) Sited and laid by the units they protect.

(c) Info abut them passed to higher headquarters.

(d) Laid close to FDLs and covered by small arms and anti-tank weapons of the unit.

(e) Depth normally up to 250 to 300 metres.

(f) Marked and recorded.

(2) *Defensive*

(a) Lay to defeat enemy penetration between units and also to strengthen the defence.

(b) More remote from positions than protective.

(c) Covered by long-range weapons like MGs, RRs mortars and artillery.

(d) Detailed sitting is done by infantry commanders advised by armour, artillery and engineers.

(e) Lay by engineers with the assistance of infantry.

(f) Marked and recorded.

(3) *Barrier*

(a) Laid to block enemy formations in selected areas, especially to the flanks and rear and to deflect his approach into pre-selected battle areas.

(b) Planned with the intentions of concentrating mobile reserves and fire power against the enemy while the enemy is enmeshed with them.

(c) Denial of observation to the enemy and difficulty of breaching it are essential.

(d) Laid by engineers with necessary assistance from infantry.

(e) Marked and recorded.

(4) *Nuisance*

(a) Laid to delay and disorganise enemy and to hinder his use of an area or route.

(b) In defence, laid on enemy approaches and FUP to disorganise his attack. In withdrawal these are laid along enemy route of advance.

(c) Policy for laying is given by division or higher level.

(d) Detailed siting and laying is done by engineers or infantry assault pioneers.

(e) Not marked unless these are likely to endanger own troops operating in the area.

(5) *Phoney*

(a) An area of ground used to simulate a minefield with the object of deceiving.

(b) Once enemy is made mine conscious phoney mine-fields can be used to a certain extent as a substitute

for live ones, thereby, saving mines, labour and time.

(c) Serve as minefield gaps.

(d) Marking et cetera is done like live minefields.

c. *Definitions*

(1) *Mine Cluster*. It is a point on ground, which is mined and is the basic unit of pattern mine laying. It can be only anti-tank mines or anti-tank and anti-personnel mines mixed.

(2) *Anti-tank Cluster*. One anti-tank mine.

(3) *Mixed Cluster*. One anti-tank mine plus a number of anti-personnel mines (usually 2 to 4) within a semicircle of two paces radius.

(4) *Mine Row*. A single line of clusters at 6 paces average interval between centres of clusters. Type of clusters in a row is constant.

(5) *Mine Strip*. Two parallel rows of clusters with 6 paces interval between rows and 6 paces interval between clusters in each row. Types of clusters in a strip are constant.

(6) *Minefield Lane*. An un-mined or demined route through minefield, normally 8 metres wide and marked for passage of friendly forces. Normally the centre line only will be marked using a strand of fish line or tape secured to pickets driven flushed with the ground.

(7) *Minefield Gap*. An un-mined or demined portion of minefield to enable friendly forces to pass through in deployed formation. It is seldom less than 100 metres wide. Not normally marked as a gap.

(8) *Irregular Outer Edge (IOE).* Groups of 8 or less clusters laid between the outer strip and the outer perimeter marking fence of likely enemy approaches to deceive about the beginning and depth of minefield is known as IOE.

(9) *Density*

(a) Average number of mines per metre of frontage of the minefield.

(b) A standard strip of anti-tank mines only produces a density of 1/3 mines per 1 metre of front.

(c) Density of anti-tank and anti-personnel mines is worked out and is given separately.

(10) *Depth.* It is the distance between outer perimeter marking fence and inner perimeter marking fence. It has no relationship with density.

d. Mine Laying

(1) General

(a) There are two mine laying drills for laying pattern minefields:-

i. Engineer's method or normal method.

ii. Hasty method or parallel strip method.

(b) The normal drill will be used by engineers, either with or without assistance from other arms, to lay defensive and barrier minefields.

(c) Parallel strip laying drill will be used for protective minefields, where time for reconnaissance and setting out is short or where the depth of the mine-field must be limited to enable the defenders to cover the whole with fire or to facilitate patrolling. This drill can be used with a minimum of reconnaissance and setting out; in essence, it produces a narrow minefield made up of three parallel strips giving a density of one mine per metre.

(d) Since individual working numbers have the same duties whichever drill is used, it is unnecessary for other arms to learn the normal drill for the occasions when they assist sappers using that drill.

(e) Consequently all arms will be trained in the parallel strip drill only whereas engineers will be trained in both drills.

(2) Parallel strip drill is given at Annex 'AA'.

e. *Reporting and Recording*

(1) *Reporting*

(a) The responsibility for initiating and forwarding tactical reports and the information to be included in each report are as set out in the table below. The reports at serials 1,2 and 3 should normally be sent by despatch rider but may be sent by wireless in code if security permits and speed demands it.

Serial	*Report*	*By whom and When initiated*	*To whom forwarded*	*Details to be included*
1.	**Intention**	Unit or formation commander on deciding to lay a minefield	Next higher formation	1. General area of proposed minefield (by grid reference). 2. Number and width of lanes and gaps. 3. Estimated numbers and types of mines. 4. Estimated starting date and time. 5. Estimated completion date and time.
2.	**Sitting**	a. Laying unit on completion reconnaissance and final sitting. b. Only if materially different from intention	Formation ordering laying	1. Boundaries of minefield (by grid reference). 2. Number, widths and approximate positions of

Serial	*Report*	*By whom and When initiated*	*To whom forwarded*	*Details to be included*
		reports.		lanes and gaps.
		c. Formation ordering laying.	Next higher formation	3. Estimated numbers and type of mines. 4. Estimated starting date and time. 5. Estimated completion date and time.
3.	**Start of laying**	a. Laying unit when laying starts	Formation ordering laying.	1. Time laying started. 2. Estimated time of completion.
		b. Formation ordering laying	Next higher formation.	3. Numbers and types of mines being laid.
4.	**Completion**	a. Laying unit when laying is completed.	Formation ordering laying.	1. A trace on the largest scale map available.
		b. Formation ordering laying.	Next higher formation.	2. Total numbers and type of mines actually laid. 3. Full details of minefield lanes and gaps and of how they are marked.

(b) At engineers headquarters, the intelligence officer should keep a trace on the largest scale possible map of all minefields, own and enemy which are known in the divisional area. This trace should be used to keep the formation trace up to date. The formation should issue the trace periodically to all units concerned.

(2) *Recording*

(a) The minefield record is for recovery purposes and it must therefore be accurately and clearly made out. Units must take a special pride in achieving the highest standard of accuracy in their minefield recording. They should realise that on each record completed the reputation of the units is very much at stake.

(b) The record form should be submitted within 48 hours of laying in triplicate to engineer or headquarters at division whether laid by engineer or other arms. The latter send only two copies, one for retention by Chief Engineer Army.

(c) It is the responsibility of the officer-in-charge of laying to make out and submit the minefield record. The record must be submitted even when the minefield is intended to be purely temporary. Events may make it necessary to leave the minefield in position for longer than anticipated and some other unit may therefore have to lift it.

(d) A complete record form is given in PAFQ-01. As this form and its use is standard with other armies to which the Pakistan Army might be allied, absolutely no variations in its use are permissible.

(e) Instructions for completing the form are as follows:-

i. Maximum length of field to be recorded on one form is approximately 600 metres.

ii. The scale of the sketch must be shown 2.5 centimetres normally equals 50 paces but the scale may be larger.

iii. All landmarks, and there must be at least two, and intermediate markers if used must be shown on the sketch.

iv. The sketch must show Magnetic North, direction of the enemy, approximate trace of perimeter fence, length and depth of the minefield.

v. It must also show magnetic bearings and paced distances:-

aa. From landmarks to intermediate markers if used.

bb. From landmarks (or intermediate markers) to strip markers at beginning and end or rear strip.

cc. From landmarks (or intermediate markers) to lane entrances.

dd. From landmarks (or intermediate markers) to perimeter fence.

ee. For each straight-line section of lanes.

vi. The following magnetic bearings and pace distances should also be shown:-

aa. For each straight-line section of each strip.

bb. Between the beginnings of strips.

cc. Between the end of strips.

dd. The sketch should also show such additional information as the recording officer considers necessary.

vii. When recording minefields laid mechanically, rows should be lettered commencing from the enemy side. The word ‘Row’ should be substituted for ‘Strip’ wherever it appears on the form or in the above instructions.

viii. The form is used for recording nuisance minefields by lettering them and recording them in the strip table of the form. The words for ‘strip’ where it appears in the form.

ix. If time is short, the form must be completed down to the broken line and the sketch must be completed. In the interest, however, of those who may have to lift the minefield, taking advantage of this provision should be very exceptional.

x. When an existing minefield is lifted, the clearing unit will quote the number of the form for cancellation in its completion of task report.

xi. When an existing minefield is changed a new form will be submitted showing the entire field after changes.

f. *Mine Lifting*

(1) Mine lifting is the soul responsibility of engineers. The requirement for mine lifting can be during the operation or at the end of hostilities. Quite often it becomes necessary that mine lifting should be done by only those units, which laid the minefields. There are no standard drills or parties laid down for these operation but some guide lines to save own troops from accidental hazards are given in this section.

(2) First of all the person in charge of lifting must obtain the record of minefield. This is important for the reason that it will provide him the exact data on pattern of minefield, position of various strip, type of mines used with their number in each strip.

(3) In case the minefield has a mixed strip, then there is a need for providing following additional equipment to each man working on it:-

(a) Shoes anti-mine.

(b) Face visor.\

(c) Long prodders 2.5 metres bamboo with 30 centimetres long prodder.

(4) When sufficient time is available, the lifting should preferably be done strip by strip. The OIC should locate the strip marker with the help of landmarks and bearings given in the record performa. He should employ about six men who prod and locate the mines on the strip. Following behind at about 50 metres distance should be another party of about six to eight men which uncovers and disarms the mines [removes the fuze from the mine and puts it about 1 metre away from the mine]. The last party of 10 to 15 men which is not closer than 50 metres from the second party should remove the mines from the pits and take them to the dump.

(5) The fuze for anti-tank mine must be stocked separately from the mine. The pressure cap of the fuze should always be kept away from the body of the fuze till safety disc is used. While carrying the fuze of anti-tank mine (which is also the anti-personnel mine) make sure that:-

(a) The pressure cap is removed.

(b) The fuze is placed in a manner that the detonator is the upper most and no substance is in contact with it.

(c) When carrying these fuzes without pressure caps, do not put them in a haversack or one on top of the other. It is best to carry few of them placed side by side and making sure that the detonator is not disturbed.

(d) Segregate damaged mines from serviceable mines.

(6) While disarming mines, never use force to open the pressure cap. If a mine is in a doubtful condition leave it to be destroyed in situation later but mark it.

(7) Do not use more than one man on a cluster for prodding or disarming mines.

(8) In case a minefield was captured by the enemy and later handed over to own troops and mines have to be lifted from

such a minefield then additional precaution, will be required. Some of the points requiring attention are:-

(a) Enemy is likely to disturb the pattern of rows and strips at few places. This is particularly dangerous in case of anti-personnel mines and lifting becomes difficult.

(b) The enemy may have removed the pressure plates of anti-personnel mines. This situation is particularly dangerous if the pressure caps of anti-tank mines and their fuzes are removed because in such a case the detonator of the fuze is bare and the slightest abrasion by the prodder can actuate the mine. In case of such mines do not try to prod and remove them but destroy them with explosive in position.

(c) The enemy may have put some additional mines. This requires a thorough prodding of the area.

(d) The enemy may have removed some of the mines. Account for the pits and try to search them in the area nearby.

(9) The mines or strips laid in areas which are affected by flood water or streams of canals or rivers and snow generally get buried deeper or drifted from their original position. Ensure proper search and if required use explosive to blast the doubtful places. Try and search the missing mines near the turnings of the stream where these are normally stuck in shrubs, sand or silt.

(10) When the mines are laid for long time then there might be thick growth of weeds, shrubs et cetera on the rows or strips. It is better to burn the undergrowth and then destroy all mines in situ without disturbing them.

(11) In case anti-lifting devices are used, destroy the complete strips by explosive. Do not try to lift mines from the strips where anti-lift devices are used.

(12) Use only those men for prodding and disarming, which have had sufficient training in mine warfare.

5. **Booby Traps**

a. *General*

(1) Booby trap is a trap to entangle a careless soldier. The trap is a device usually of explosive nature aimed directly at the reduction of his morale and his mobility.

(2) The responsibility for the detection and clearance of trap rests primarily with the engineers and assault pioneers but all field units of other arms must be familiar with their use to avoid casualties on first occupation of the buildings, harbour areas vacated by enemy because he is going to use these traps against own troop.

b. *Essential Parts of A Booby Trap*

(1) The essential parts of a booby trap are:-

(a) Mechanism.

(b) Charge.

(c) Method of connecting mechanism to the charge.

(d) Means of concealing and operating the trap.

(2) *Mechanism*. The standard mechanisms used in traps are the different switches but these can be improvised as well. The standard switches are as given below:-

(a) *Switch Number 4, Pull*. Operated when a pull of 2.7 to 3.6 kilograms is applied on the attached wire. Normally used with moving object like doors, windows et cetera.

(b) *Switch Number 5, Pressure*. Operated by pressure between 9.5 to 27 kilograms on the lid of the switch. Used under floor boards, stair cases, furniture et cetera.

(c) *Switch Number 6, Release*. Works when the load on its lid is removed. Minimum safe pressure is normally more than 1.3 to 1.8 kilograms. Can be used under a drawer, behind doors, under boxes or any object, which is likely to be lifted.

(d) *Switch Number 9, Delay*. It is used for delay action in raids behind enemy lines or for delayed

demolitions in withdrawal. Duration for delay ranges from few minutes to a day.

(3) *Charge*. It can be made of any explosive available for use. The quantity of explosive will depend upon the target and the result desired.

(4) *Connecting Mechanism to Charge*. Adapters used with switches which have percussion cap and snout to hold the fuze. From the percussion cap instantaneous fuze connects the mechanism to the detonator in the charge.

(5) *Concealing and Operating the Charge*. It is most important part of the trap and should be so well concealed and camouflaged that the enemy falls in the trap automatically. Enemy habits and weaknesses must be exploited. Following principles are followed in concealing and camouflaging charges:-

(a) *Concealment*. Signs of preparation must be removed.

(b) *Concentration*. Necessary for effectiveness.

(c) *Double Bluff*. An obvious trap to mark the other well concealed trap.

(d) *Variety*. Avoid stereotyped traps for effectiveness.

(e) *Constructed Localities*. Use at places where enemy must pass.

(f) *Inconvenience*. Must cause annoyance to make the enemy reckless.

(g) *Every Day Operation*. Use at places, which are required for every day routine.

(h) *Curiosity*. Use in a manner and at a place, which excites the curiosity and the troops feel excited to disturb things to quench their curiosity.

(j) *Variety*. Employ different methods and places to make detection difficult.

c. *General Precautions*. Exercise greatest care and caution when moving in areas suspected of having booby traps. A careless step will cost your life as well as the life of your comrades. Some of the rules to be followed are:-

(1) Move with eyes open.

(2) Minimum persons to work on a mechanism.

(3) Handle with care.

(4) Do not use force.

(5) Never cut a tight wire.

(6) Take cover while pulling mines.

(7) Leave alone those mechanisms, which you are not familiar with.

6. **Explosives and Demolitions**

a. *Basic Explosives*

(1) *Gun Cotton Slab*. The slab measures 15.5 x 5.centimetres and the centre of each face has a tapered hole designed to hold the standard 28 grams primer. This type of explosive is seriously affected by weather conditions and must be kept in sealed containers. The weight of each slab is 539 grams out of which, 85 grams is water content and 454 grams is explosive content.

(2) *CE-TNT Slab*. Rectangular slab weighing 1 pound. Each slab measures 11.4 x 6.5 x 4.5 centimetres and has two sockets designed to take special one centimetre primer and the slab is initiated by a 28 grams primer. Not effected by moisture as gun cotton explosive.

(3) *PE 3A (Plastic).* It is plastic by nature and is provided in 227 grams cartridges. Each cartridge is 20.3 centimetres long x 3.8 centimetres in diameter. This explosive can be used under water.

(4) *Bangalore Torpedo*

(a) This is a ‘prepared charge’ designed for breaching wire obstacles, the type now issued to infantry units is known as the ‘Torpedo Bangalore, 3.8 centimetres, Mark 1’. The Torpedo consists of a light steel tube 3.8 centimetres in diameter. This tube is filled with HE, and is issued in 1.8 metres sections, each weighing 6.4 kilograms. Each section has a male and female end, and these are fitted with

spring clip joints so that a number of sections can be connected together to make up any required length. A detachable bullet-shaped nose is supplied with the torpedo; this is designed to fit into the female end of the leading section in order to assist the movement of the torpedo along the ground. The male end of each section contains a special built in primer for use in conjunction with detonating cord and a double initiating set.

(b) Four section complete with accessories are issued together in a special frame.

b. *Demolition Accessories (Less Electrical)*

(1) *Crimpers*. It is a tool used to attach igniters and detonators to fuzes.

(2) *Igniter Safety Fuze*. These igniters are made of brass, and work on the principle of a spring-operated striker controlled by the withdrawal of a pin. The firing caps are easily damaged by damps, and the igniters should be kept sealed in their tins.

(3) *Safety Fuze*. It is used to initiate Number 27 detonators and at the same time to provide delay to enable the man lighting the fuze to retire to a safe distance before the charge detonates. It is contained in black or red cover and burns at the rate of 60 centimetres per min + 6 seconds). It absorbs water and becomes useless if kept exposed to air for long time or if handled roughly. The burning rate must be checked before use.

(4) *Fuze Instantaneous*. It is a thick orange coloured cord. It is neither a safety fuze nor a detonating cord and is only used in booby traps. Burns at the rate of 27.5 metres per second.

(5) *Detonator Number 27*. These are used with safety fuze to initiate primer or detonating cord. Each detonator is of a thin metal tube, 4.5 centimetres long, .65 centimetres in diameter, closed at one end and half filled with very high explosive. NEVER CRIMP THE DETONATOR NEAR THE CLOSED END OR POKE INSIDE THE TUBE WITH STICK ET CETERA.

(6) *Primers*

(a) *1 Ounce CE Primer*. This is a tapered cylinder 3.2 centimetres long and its diameter tapers from 3.5 centimetres. It can fit into gun-cotton slab and it can take a detonator Number 27.

(b) *1 Ounce GC Primer*. It is made of gun cotton and is similar to CE primer. It should be used in preference to this primer.

(7) *Detonating Cord*. It is white flexible cord about .6 centimetres of diameter. It fits into the Number 27 detonator. It does not burn but detonates at a rate of 6000 to 8000 metres per second (about 322 kilometres per minutes).

c. *Operational Demolition*

(1) *Purposes*. The main purposes of demolitions are:-

(a) *In a Defensive role*. To delay the advance of an enemy by destroying road, rail, or water communication; building stores materials, and public services such as power, water supply, and drainage.

(b) *In an Offensive Role*. To weaken the resistance of enemy by destroying his communications, installations, or materials, as in a raid or by sabotage.

(c) *In an Assault Role*. To speed up the advance of our own troops by destroying enemy obstacles or fortifications.

(d) Demolitions considered from the point of view of defence fall into two classes:-

i. Preliminary demolitions.

ii. Reserved demolitions.

(2) *Preliminary Demolitions*. These are used to destroy objectives which must be denied to the enemy and which are not required by our own troops in carrying out the tactical plan. An example of this type of demolition is the cratering of a road, which does not form part of a main

traffic route. As their name implies, they are taken in hand before tactical demolitions and when engineer resources can more easily be made available, and they are timed to be completed before the enemy has any chance of interfering with them. Their preparation is relatively simple, as it is rarely necessary to protect the charges from the long-term effects of the weather or traffic.

(3) *Reserved*

(a) A typical example of this type of demolition is the destruction of a bridge immediately after our own troops have withdrawn across it. Demolitions of this type require the most careful control. They are normally planned at division or above level, and because of the complications involved, their number is kept to the minimum. Reserved demolitions are intimately linked with the tactical plan, and special arrangements must be made for giving the order to fire them.

(b) When the order is given to prepare a reserved demolition, the exact time of firing may not be known. It is therefore prepared as a LONG STANDING DEMOLITION: Special precautions must be taken to protect the charges and equipment, from damage, which may be caused by weather, traffic, or enemy action.

(4) *State of Readiness.* The prepared reserved demolitions could be in one of the following states of readiness: -

(a) *State of Readiness 1 (Safe).* It is safe against pre-mature detonation. All the charges are prepared, fixed and firing circuits completed but not connected. The detonators are kept out of the charges.

(b) *State of Readiness 2 (Armed).* In this state all firing circuits are completed and connected and the detonators are placed in the charges and the demolition is ready to be fired. In this state the risk of premature detonation is acceptable.

(5) *Authorised Commander*. It is that commander who is authorized to fire a particular reserve demolition. He will change the state of readiness and give orders for firing the demolition.

(6) *Duties of Demolition Guard Commander*. He is responsible for the protection of target prepared for demolition against enemy interference or attempt to capture it before firing. The duties in this regard are:-

(a) Commands demolition guard and firing.

(b) Protect the target from being captured and enemy interference.

(c) Control of traffic on the target.

(d) Communicate orders to firing party commander regarding change in state of readiness or to fire the demolition.

(e) Keep the authorised commander informed about the tactical situation and the state of readiness.

(f) Intimate the results of fired demolition to the authorised commander.

(7) *Duties of Firing Party*. Firing party from engineers is responsible for maintaining and firing the demolitions. Their duties in this regard are:-

(a) Maintain the state of readiness ordered.

(b) Fire the demolition and ensure its success.

(c) Provide report after firing the demolition.

(8) *Protection.* Following must be borne in mind:-

(a) *Ground Attack*. The demolition guard should be able to defend against infantry or armour attack. The guard should have anti-tank and automatic weapons and if possible wire obstacles and mines to cater for various contingencies.

(b) *Heliborne Attack*. Cater for additional trenches for automatic weapons so that the enemy aircraft/helicopters can be engaged after landing.

(c) *Swimmer and Floating Mines*. Make arrangements to detect such threats well in time and cover gaps and water line with automatic weapons during day and night.

(d) *All Round Defence*. The defensive position must cater for all round defence and good observation and surveillance at all times during day and night.

(9) *Command and Control*. There must be foolproof communications with firing party and liaison officer (if on site). In addition there is a need for establishing control point for the following reasons:-

(a) Liaison Officer, demolition guard commander and firing party commander stay closer to each other.

(b) 2IC of the guard remains with forward sections and has radio link with control point.

(c) Control of traffic through control point.

(10) *Firing*. Following method be adopted:-

(a) Troops move 200 to 300 metres away from the target (alternate position) just before the demolition is to be fired. Fire and move to be carried out if enemy is in contact.

(b) Firing party should initiate firing both by fuze and electric circuits. Electric initiation can be from the control point.

(c) During all this, firing party must be supported by demolition guard till successful firing is completed.

7. Water Crossing Equipment

a. *Standard Service Equipment*

(1) Assault Boat M2

(a) Dimensions and Weight

i. Length 3.2 metres.

ii. Width 1.8 metres.

iii. Depth 64 centimetres.

iv. Weight 181.5 kilograms.

(b) *Carrying Capacity*

i. Buoyancy when 20.3 centimetres free board is 1.8 tonnes.

ii. Buoyancy when 7.6 centimetres free is 2.4 tonnes.

iii. It can carry 12 fully equipped infantrymen in addition to three crewmembers from engineers. (In a current up to 1.2 metres/ sec).

(c) *Use.* It can be used for following purposes:-

i. To carry infantry across the water obstacles in the initial stages of assault across it.

ii. Two assault boats can be joined to provide a pontoon, which can be used for carrying loads or can be used in raft construction.

iii. It can be used to construct three, five and seven pontoons raft.

iv. Can be used in Class 9 bridge construction.

(d) *Propulsion*

i. By padditionales provided with the boats.

ii. By OBM (Out Board Motor) of 25 horse power (HP).

(e) *Transportation*

i. 7 x M2 boats in each 2 ½ ton truck.

ii. 10 men can carry it for short distance of about 200 metres.

(2) *Infantry Support Raft*

(a) Details of equipment for one Raft Set.

Serial	*Nomenclature*	*Quantity*
i.	Rope sasal 15.25 metres	6

Serial	*Nomenclature*	*Quantity*
ii.	Boat assault M2 with pin without padditionale	8
iii.	Bracket OBM	1
iv.	Clamps, kerbs, side-rails 45.7 centimetres jaw	16
v.	Marker luminous 4.5 centimetres	65
vi.	Kerb side-rails 10.2 x 15.2 centimetres wide 3.7 metres long	6
vii.	Pin assault boat M2	6
viii.	Padditionales boat single blade 1.5 metres long (9 per case)	72
ix	Tread way plywood complete with two pins and two ramp side connectors	11
x.	Connector side ramp	21

(b) Load Carrying Capacity

	Velocity of Current	*Class*
i.	1.5 metres per second	8
ii.	1.8 to 2.1 metres per second	6-7

(c) *Site Requirements and Preparation*

i. Current up to 1.5 metres per second.

ii. Firm banks, which allow loading on the raft without using landing stages.

iii. Strong natural hold fast.

iv. Bank almost level with deck of raft or slightly lower.

v. Minimum depth of water to be .6 metres.

b. *Improvised Means*

(1) *General*

(a) Improvisation is the art of fixing together the materials readily available and utilizing them with imagination to suit a particular requirement. The scope of this improvisation in the field of water crossing expedients is vast. The large number of rivers and their tributaries necessitate the effective use of these expedients for military operations. Some of the expedients which can be easily improvised are as under:-

i. Bamboo expedient.

ii. Logs expedient.

iii. Basket Ball/Foot Ball Bladder.

iv. Tubes Inner (Jeep or Motor Cycle).

v. Two 19 litres Kerosene Tins.

vi. Six 19 litres Kerosene Tins.

vii. Water Post.

viii. Improvised Method of Ferrying Casualties.

ix. 2.5 tonne (2 ½ ton) Superstructure Float.

x. Ground Sheet Float.

(2) *Bamboo Expedient*. Bamboos are available in plenty and can be cut to the required length and tied by lashing in 2-3 layers to improvise a raft. This will be comparatively lighter and can be operated easily by using bamboo poles. The lashings have to be checked and replaced after a week or two.

(3) *Log Expedient*. This is similar to the bamboo raft. Most of the rivers have thick growth of different types of trees on the banks. Trees of similar size can be cut to the required length for the rafts. These logs can be tied together to make a raft, which will be able to carry stores and personnel. The raft made out of comparatively dry logs would be lighter in operation. Bamboo poles have to be used for operation. The

lashings have to be checked and replaced after a week or two.

(4) *Tube Inner (Jeep or Motor Cycle).* It consists of an inner tube of a jeep or motorcycle, which weighs only about 1 kilogram and can easily support two men fully equipped. With a strong net covering it can also be used to carry heavy weapons, ammunition boxes and other stores.

(5) *Two (19 Litres Kerosene) Tines.* Two (19 litres kerosene) tins lashed together with two bamboo sticks are required for these expedients. This will assist a fully equipped man to swim across a water obstacle with confidence and ease.

(6) *Six (19 Litres Kerosene) Tins.* Six (19 litres kerosene) tins lashed with four long bamboo sticks and eight short bamboo sticks are required for this expedient. This will support two men.

(7) *Water Pots.* Two water pots are used as wings to support one man. The cord is passed under the arms. This expedient can conveniently support one equipped man.

(8) *Improvised Methods of Ferrying Casualties*. Eight ground sheets, four bamboo poles, material in bales i.e., bhoosa, gunny paper or cotton waste and eight yards rope are required for this expedient. The material is securely bound within two ground sheets; the dimensions of each bale being approximately 90 centimetres by 30 centimetres. By ensuring that the second ground sheet of each bale overlaps the fastening of the first, the bale will be rendered waterproof. The bamboo poles are lashed together to the bales.

(9) *2.5 Tonne (2½ Ton) Superstructure Float*. One superstructure, two tarpaulins and 4 x 200 litres drums are required for this expedient. Two tarpaulins are fixed on the superstructure and the drums secured with ropes and lashings. This expedient will facilitate one rifle section with arms and equipment to go across.

(10) *Ground Sheet Float*

(a) A ground sheet, line bedding and dry grass is required. Wrap the grass in the ground sheet which should be tied around with line bedding and the

ends of this expedient can be secured with lashings. This will assist one man to swim across.

(b) It is important to remember that some of the above expedient will not necessarily support the whole weight of the man. Weak swimmers and non-swimmers must be trained to allow the water to support their weight, only using these expedients as an additional assistance. Provided a man keeps his shoulders below the surface and breathes normally, he will be buoyant.

(c) *Reconnaissance Boat Ferry*. The reconnaissance boat is standard engineer equipment. It can be carried by one man, whether deflated or inflated. Two ropes of the required length are tied to it, one at each end and rolled on signal cable laying drums. One small iron picket is driven into the ground on each bank as anchor when the ferry is not in operation. The boat is rowed across carrying one rope rolled on a cable-laying drum in it to establish the ferry. Two men on each bank are required to operate the ferry.

(11) *Floating Platform*

(a) *Description*. The wheeled floating platform comprises a plywood sheet 1.8 x 1.2 metres mounted on 4 tricycle wheels of 25.5 centimetres diameter with 2 straight MS Bar axles of 2.5 centimetres diameter connecting the wheels. Four rubber floats are attached to it underneath for giving it sufficient buoyancy to carry across about 363 kilograms weight.

(b) Various components are as under:-

i. Pulling rope for establishing ferry.

ii. Carry rope handles.

iii. Wheels.

iv. Long float 3.7 metres long in Khaki drill covers.

v. 3 Ton vehicle tyre tube in khaki drill covers.

vi. Hooks for carrying rope handles.

vii. Axle.

viii. Hooks for pulling ropes.

ix. Eyeholes welded on the axle to pass the ropes through for securing the tubes with the platform.

(12) *Floating Platform 2*

(a) *Description*. It can be pulled by one man or carried by 2 men from Assembly Area/FUP to the water obstacle. Total weight is 59 kilograms.

(b) While the platform is being loaded for the first time after being lowered into the water, man with a rope with one end tied to the platform and other to his belt swims across to establish the ferry as per reconnaissance boat ferry.

SECTION 108 – **SIGNALS**

Characteristics and Principles of Signal Communication

1. Signal communication provides the battalion, company and platoon commanders necessary means through which they can:-

a. Exercise command and control over their subordinates.

b. Administer and maintain their troops.

c. Receive orders and instructions from superior headquarters.

d. Disseminate intelligence.

e. Cooperate with other arms and services.

2. **Essential Characteristics**. To meet the arduous demands of war, signal communication must fulfil certain basic requirements, which are hinged to the principles of war and of administration. Though, these requirements are generally constant but their priorities and emphasis on them vary according to the situation. These requirements or characteristics are discussed in the following paragraphs, which must be clearly understood by all personnel concerned with provision of signal communication.

3. **Flexibility**. It is the ability to react to the changing situations and sudden tactical developments successfully and without undue loss of time. Communication must be flexible to keep pace with the tactical mobility of the fighting troops and cater for changes necessitated either due to developments in own operational plans or by enemy action. To achieve flexibility, the following are useful guidelines:-

a. Keep abreast of the latest tactical developments, appreciate intention of the higher commander and plan ahead.

b. Learn the characteristics of all means of communication and utilize the most flexible one, warranted by the situation.

c. Never depend on a single system and always have an alternative means of communication available.

d. Maintain a reserve of signal resources at all times. Once the existing resources are used up, create fresh reserves by withdrawing resources not really committed or needed elsewhere.

e. Streamline the battle procedure and drills so as to reduce signal planning time.

f. Attain a very high degree of training in establishing and maintenance of communication.

g. Look after your signal equipment and have an efficient maintenance system.

4. **Reliability**. Failure of communication in battle can be dangerous or even disastrous. To avert such situations, alternative means must always be provided. Sound planning, high standard of training, good maintenance of equipment and avoidance of over work reduce breakdown risks and promote reliability.

5. **Security**. All means of communication are subject to enemy interception in varying degrees. Whichever system or means of communication is employed, it must provide the requisite amount of communication security. Measures and practices designed to protect friendly transmission from interception and traffic analysis by the enemy should be followed rigidly.

6. Precautions must also be taken to ensure physical security of isolated radio detachments, messengers and wire parties working in forward areas. Tapping or serving of telephone wires and jamming of radio nets by the enemy also need consideration. However, these hazards should not prevent the signal planner from taking calculated risks.

7. **Speed**. Provision of communication must keep pace with the speed of operation. At least one and preferably two or more means of communication must be available at all times. Remember that no system of communication, however, elaborate it might be, is of any use if it is not available in time.

8. **Simplicity**. A signal plan must always be simple and easy to understand. Complicated plans, apart from being difficult to interpret, are hard to execute and create maintenance problems. Simplicity is therefore the keynote of signal planning.

9. **Economy**. Signal resources being limited and hard to get, should be utilized most judiciously. Their indiscreet use will not only jeopardize the principles of reliability and flexibility but also hinder creation of reserves. Greatest economy must therefore be exercised in the employment and committal of signal resources. Some signal reserves must always be at hand to cater for sudden change in plans or to replace unexpected heavy casualties.

10. **Principles of Communication**. Like all other sciences the science of communications also has certain principles. Negation of these principles can be catastrophic and their observance will help in the establishment of rapid, dependable and safe communication. The principles given below apply to every infantry battalion, under all conditions except when otherwise ordered by the brigade headquarters:-

a. *Command Responsibility*. Provision of communication is a command responsibility. Every battalion, company and platoon commander is therefore responsible for the organization and smooth functioning of his communication systems at all times.

b. *Higher to Lower*. The higher headquarters is responsible for establishment and maintenance of communication with its subordinate elements.

c. *Supporting to Supported*. Supporting elements are responsible for establishing communication with the supported element. An exception to the rule is that when infantry is supporting armour, the latter is responsible for establishing radio communication with the former. Supporting elements should also maintain communication with their own higher headquarters.

d. *Left to Right.* Lateral communication is normally provided from left to right.

e. *Under Command*. Elements placed under command a headquarters, establish communication with that headquarters and disconnect communication with their own higher headquarters.

f. *On Move.* Communication must be maintained at all times including when on the move.

g. *Communication Restoration*. Disrupted or broken down communication must be immediately restored and traffic routed over the alternative system.

Means Available

11. Irrespective of their role, all infantry battalions are equipped with the following means of signal communication. Detailed characteristics of these means are discussed in the succeeding paragraphs:-

a. Radio.

b. Wire.

c. Runner.

12. To meet a given situation, the general rule is that the medium, which provides maximum reliability, flexibility, security and speed with minimum of effort, should be employed as the primary means. However, emphasis on these factors will vary according to the situation and restrictions imposed by the higher headquarters. Hence there can be no fixed rule, which may be applied in every case.

13. When planning communication, irrespective of all other considerations, never place reliance on any one single means, the rule being, that at least one alternative means must always be provided. Different means should be so employed as to supplement each other and be available for instant use when needed.

14. **Radio**. Radio is an important means of communication in the army. It employs transmission of information through space using electromagnetic waves, which need no physical path. Radio communication has certain inherent characteristics, which are discussed in the following paragraphs:-

a. *Advantages.* The main advantages of radio communication are:-

(1) *Flexibility*. Radio communication can be established and readjusted quickly. Stations can join or leave radio network without difficulty or dislocation of communication.

(2) *Speed*. Communication can be established rapidly in comparison with other means. Once established, communication is available for instant use.

(3) *Mobility*. It does not require construction of any transmission medium. Thus radio stations are not tied down to a particular place and can change location if desired. Also, infantry radio sets being light and portable, can be easily man packed.

(4) *Adaptability*. Communication is possible across country and obstacles like marshes, minefields and rugged or undulating ground et cetera. Radio communication can be maintained even when on the move whether marching or using transport.

(5) *Economy*. Radio communication does not require man-made transmission medium. Moreover, simultaneous transmission of information is possible to a number of stations located widely apart.

(6) *Physical Security*. Radio stations are generally located at headquarters, where men and material are comparatively safer. Further, the electromagnetic path between various radio stations is invulnerable to enemy sabotage and bombardment et cetera.

(7) *Personal Control*. Voice radio nets and links provide personal contact between users and enable commanders to project their personalities, which is an essential element of command.

b. *Limitations*. The following limitations affect the use of radio:-

(1) *Insecurity*. All radio transmissions are susceptible to enemy interception hence it cannot be used with complete freedom. Authorized codes et cetera do provide a degree of protection against enemy traffic analysis but none against direction finding and interception. The system is therefore basically insecure.

(2) *Interference*. Transmissions are liable to atmospheric, mutual and technical interference. Deliberate interference (jamming) by the enemy further reduces its reliability.

(3) *Range*. The power of the radio set, type of antenna, geography of the terrain and skill of the operator affect the range of radio communication.

c. In spite of the above inherent limitations, radio can still be used to advantage by:-

(1) Rigid compliance with the principles of communication security.

(2) Strict observation of radio net discipline.

(3) Skill and unwavering determination of the user.

d. *Employment*. Within infantry battalions, radio sets are used for voice communication only. The battalion and company/platoon commander each require a command radio net and a rear link set working back to the next higher headquarters. As a general rule, radio nets should be kept as light as possible because overcrowded nets tend to become cumbersome and may cause traffic delays.

15. **Wire**. Wire communication at infantry battalion level is a simple affair. A pair of field wire is laid over the ground between various headquarters and is connected to a telephone set or switchboard ends. The system is called wire telephony and is ideal for static or semi mobile conditions. Advantages and limitations of the system are given below:-

a. *Advantages*. Telephony is the principal mean of infantry communication when not on move. Its advantages are:-

(1) *Security*. So long as the enemy is denied physical access to the field wires, telephony is a secure mean of communication.

(2) *Reliability*. Telephony is virtually free from atmospheric interference and immune to enemy jamming. Well-planned and properly laid out wire routes provide dependable communication. However, the chance of enemy's physical interference with the wire routes in forward areas cannot be ruled out completely.

(3) *Stability*. Telephony provides considerably stable and high quality speech circuits.

(4) *Simplicity*. Telephony is the simplest means of communication, which can be installed and used without any special skill.

(5) *Adaptability*. It is suited to most types of terrain and conditions. Laying and maintenance of wire routes can be done on foot or by vehicles.

(6) *Personal Contact*. Telephony is medium through which commanders can converse with each other as if talking face to face. Views can be exchanged and important points discussed without procedural preliminaries.

(7) *Capacity*. Telephony is inherently capable of handling greater volume of traffic as compared to radio.

b. *Limitations*. Telephony suffers from the following limitations: -

(1) *Slow*. Installation of wire communication is slow and time consuming, especially when routes are to be laid on rough country or on foot.

(2) *Inflexible*. Wire routes once laid cannot be easily reoriented. Also, the user is tied down to the place where the wire is terminated. Unlike a radio set, telephone cannot be moved from place to place.

(3) *Uneconomical*. Laying and maintenance of wire routes requires substantial effort, both physical and material. It also involves carriage of heavy and bulky stores.

(4) *Vulnerable*. Wire routes are vulnerable to physical damage by enemy action (sabotage and bombardment et cetera) as well as by own heavy transport and tanks et cetera.

c. *Employment*. Wire communication system is used for telephony down to platoon headquarters and other important elements like screens and Observation Posts et cetera.

16. **Runners**. Runners, as a medium of communication are available to infantry commanders at all levels. However, if they are not intelligent and well trained, the system cannot function satisfactorily. It has few advantages and numerous limitations.

a. *Advantages*. The advantages are:-

(1) *Flexibility*. Runners can be sent to any place within the battalion area, at any time of the day or night, but the distance for foot runners should not be too long.

(2) *Capacity*. Runners can carry long and bulky despatches, maps, air photos et cetera that cannot be transmitted by any other means of communication.

(3) *Security*. Baring the chances of capture by the enemy, this is a secure method.

b. *Limitations*. The main limitations of this system are:-

(1) *Slow*. Runners take time to move from one place to another. Longer the distance more is the time required.

(2) *Distortion*. If the runner is not intelligent and well trained, even a simple verbal message may be distorted by him. Verbal messages must therefore be simple and brief.

(3) *Vulnerability*. Runners are liable to be captured by the enemy or wounded enroute.

(4) *Straying*. Bad weather and terrain conditions may cause a runner to stray. This may result in delay or not reaching the destination at all.

c. *Employment*. As a general rule, foot runners should not be employed on long routes, they must be thoroughly briefed on the routes to be followed and the action to be taken on being captured by the enemy.

d. *Information*. Whenever any runner leaves a headquarters, the distant end must be informed of this and the expected time of arrival at the destination.

e. *Scheduled Service*. Like the signal despatch service, runners despatch services to the companies/platoons may be scheduled by the signal officer, if needed.

Voice Procedure

17. A standard voice procedure is necessary for efficiency and security in wireless communication. A thorough knowledge of this voice procedure is therefore an essential/basic requirement for all users of wireless sets.

18. **Principles**. Remember the catchword “BASS” and you will not forget the following basic principles of voice procedure:-

a. B – Brevity.

b. A – Accuracy.

c. S – Security.

d. S – Speed.

19. Successful application of these principles depends on the correct use of the standard procedure.

20. **Call Sign System**

a. *Link*. Wireless channel between control and a substation.

b. *Net*. More than two stations working together on the same frequency.

c. *Call Sign*. Identification of a link. Consists of a figure or two figures between 1 and 99, and in certain cases a letter is added e.g., 4A. 96 B.

d. *Fixed Call Sign*. Used within units and for certain stations in army formation nets. In engineer units fixed call signs are used within companies.

21. **Prowords**

a. All after or all before ……..repeat all after or all before……

b. Correction – Cancel last phrase sent and substitute ……..

c. From……to……Repeat from, …..to …..

d. Grid references – Used before any GRIDDLE reference or grid reference in clear.

e. I say again – I will repeat last group or phrase.

f. Out – Finished with you.

g. Out to you–Finished with you, will now call some other station.

h. Over–Finished. Reply expected.

j. Read back–Repeat message back to me.

k. Roger so far–Acknowledge what has been sent so far.

l. Say again – repeat whole message.

m. Wait–A few seconds pause follows. Other stations must NOT interrupt.

n. Wait out–Finished for time being, will call you again.

o. Wilco–Message received, understood and will be compiled with.

p. Word after or Word before ……. Repeat word after or word before.

q. Words twice–1 (or you) say every phrase or word twice.

r. Wrong–What you just said is wrong. Correct version is …….

22. **Form of Transmission**

a. Hullo – Will be used in the initial call of every transmission thereafter if will be dropped.

b. Call sign to draw attention of station required.

c. Subject matter – The substance of message.

d. Call sign is repeated here only, when working conditions are bad.

e. Ending Signal – Over "or Out" always ends a transmission.

23. **Preliminary**

a. Think before speaking. Is the message really necessary?

b. Make a note of what you are going to say.

c. Bring codes, maps and paper et cetera, which you may have to refer to during conversation.

24. **Security**

a. Assume that the enemy is listening.

b. Be brief.

c. Use correct procedure.

d. Don't use unauthorised codes, clear names of peoples, formations or units, Jargon is NOT permitted.

e. Don't reply in other languages for security.

f. Never say "Sir".

g. Use veiled speech judiciously.

25. Wireless appointment titles. These titles are not SECRET and only conceal the level of the headquarters:-

	Appointment	*Title*
a.	Commander at any level	Imam
b.	Deputy Commander or 2IC	Wazir
c.	Senior Coordinating Staff Officer or Colonel Staff	Markhor.
d.	General Staff Officer.	Markhor Minor
e.	General Intelligence Staff Officer/Unit Intelligence Officer	Chota
f.	Administrative Staff Officer	Tattoo
g.	Armour Representative	Bakhtar
h.	Artillery Representative	Gharbeen
j.	Counter Bombardment Representative	Atishbar
k.	Engineer Representative	Farhad
l.	Signals Representative	Kabutar
m.	Infantry Representative	Cheeta
n.	Supply and Transport Representative	Salwa
o.	Medical Representative	Taweez
p.	Ordinance Representative	Ameen
q.	Electrical and Mechanical Engineers Representative	Lohar
r.	Remount Venture and Forms Corps Representative	Ghora
s.	Military Police Representative	Zabit

26. **How to Speak**

a. Divide your transmission into short, complete understandable phrases.

b. Speak at steady medium speed. Allow time for it to be written down.

c. Speak louder than for normal conversation but do not shout.

d. Don't stress important words or slur over less important ones.

e. Keep mouth close to the microphone.

f. Avoid tendency to let voice drop at the end of phrases and sentences.

27. **Spellings**

a. The word of pronounceable abbreviation will be pronounced before being spelt e.g., HOURS "I spell", Hotel Oscar Uniform Romeo Sierra.

b. "I spell" is NOT used before spelling out:-

(1) Address groups.

(2) Call signs.

(3) NIS.

28. **Figures**

Exact multiples of a hundred or thousand are spoken as in the following examples. In expressing time multiples of a hundred will always be given:-

0700 hours – Figures zero seven hundred hours.

1000 rounds – Figures one thousand rounds.

1000 hours – Figures one zero hundred hours.

40000 rounds – Figures four zero thousand rounds.

29. **How to Call**

a. *Single Call*

(1) Control – One substation and vice versa.

(a) Control–Hullo Seven …. the message ……Over. 7 – Seven. Roger, Out.

(b) 29 – Hullo two nine ……. the message ……. Over. Control – Two nine. Roger. Out.

(2) *Substation – Substation.*

7 – Hullo seven for four Zero – the message …… Over.

40 – Four Zero – wait. (short pause). Roger. Out.

b. *Multiple Call*

(1) Control Substations.

Control – Hullo Seven. Hullo Four Zero – the message …….over. 7 – Seven – wilco. Out.

40 – Four Zero..-wilco. Out.

(2) *Substation – Substations.*

7 – Hullo Seven for Two nine. Four zero and six

Six – the message……..over.

29 – Two nine. Roger. Out.

40 – Four Zero. Roger. Out.

66 – Six six. Roger. Out.

(3) *Substation – Control and Substation.*

40 – Hullo four zero for Four Six and Seven – the message over.

Control ….Four Zero. Wait. Out.

7 – Seven. Roger. Out.

Control – Hullo Four Zero. Reference – Roger. Over.

40 – Four Zero. Roger. Out.

c. *Collective Call*

Control – Hullo – Charlie Charlie Five ……. message ….over.

7 – Seven.	Roger. Out.
29 – Two nine.	Roger. Out.
40 – Four zero.	Roger. Out.

d. *The Net or All Stations Call*

(1) *Control – All substations.*

Control – Hullo. all stations – the message --------- over.

7 – Seven. Yes. Out.

29 – Two nine. Yes. Out.

40 – Four zero. Yes. Out.

66 – Six six. Yes. Out.

(2) *Substation – All Stations.*

40 – Hullo four zero for all station. The message …… over.

Control – Four zero, Roger. Out.

7 – Seven. Roger. Out.

29 – Two nine. Roger. Out.

66 - Six six. Roger. Out.

Griddle Code

30. Griddle is a grid reference code, which is used within a unit up to battalion level. It is normally compiled by the intelligence section. The basis of the code is 26 squares of 1000 metres each. A grid intersection point, South West corner of proposed square, is selected by the unit so that the proposed square covers the area over which the unit is likely to operate. The size of the square is necessarily limited to 26,000 metres owing to the employment of two randomly arranged alphabets.

31. **Preparation of the Key**. Letter off the grid lines from the grid intersection point, i.e., allot one letter from each alphabet to each grid line West to East and South to North. An example is given in figure 37 below. The grid intersection used is 8535.

Eastings: **CYKLMFWXHTADEPJOVIQUBRNZSG**

Northings: **JSTNDRQIAWFPKBMUVYEHZLGXCO**

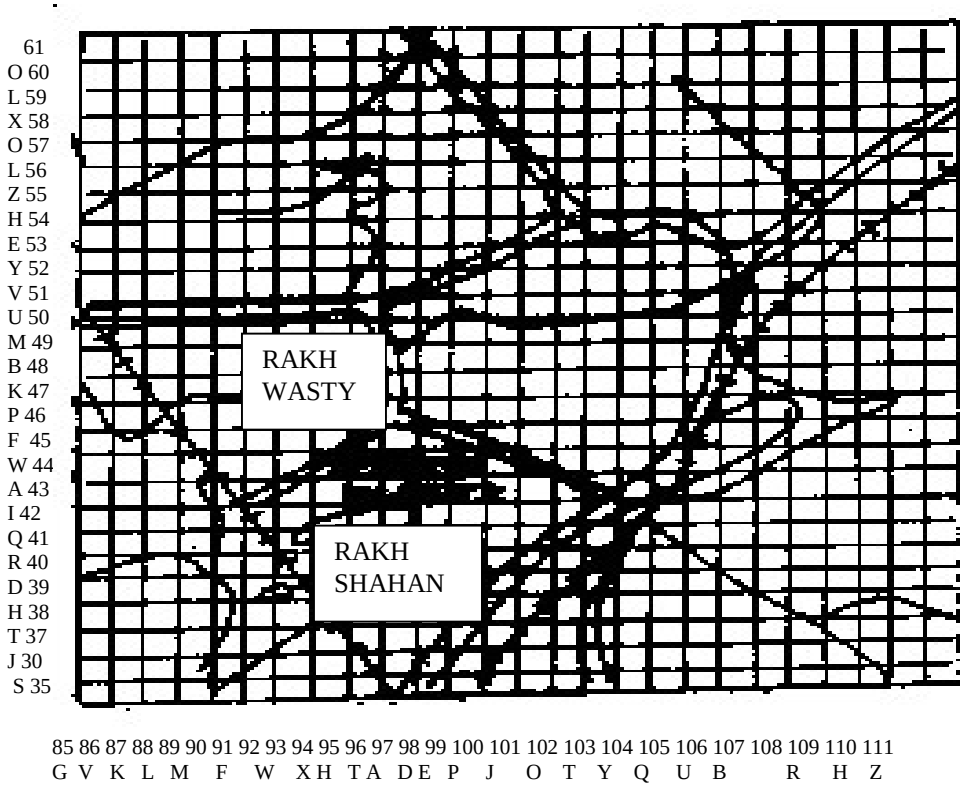


Figure 37 - Preparation of the Key

32. **The Use of Code, Four Figures Grid References**. A four figures grid reference is indicated by one letter for Northings.

Example : Kufri-Square 0343-EX encoded as “QA”

33. **Six Figure Grid Reference**. A six figures grid reference is indicated by two letters for Eastings and two for Northings.

Example : **ROAD JUNCTION** 936508

a. *Eastings*. 93 would be encoded as H. To encode the last figure of the Eastings. Find a figure 6 in the Eastings. There are three of them 96,96 06. The letter equivalent is then substituted for the

figure 6 could therefore be encoded as Y or D or R 936 Eastings could therefore be encoded as HY or HD or HR.

b. *Northings*. 50 would be encoded as for Eastings, the letter equivalent being taken from Northings 9 could therefore be encoded as N or B or X with a resultant Northings code of UN or UB or UX.

c. The complete reference could therefore be encoded in nine different ways is: -

HYUN HYUB HYUX

HDUN HDUB HDUX

HRUN HRUB HRUX

34. **Eight Figures Grid Reference**. An eight figures grid reference is indicated by three letters for Eastings and three for Northings:-

Example: **ROAD JUNCTION** 99875275

a. *Eastings*. Encode the first three figures of Eastings as in example above. Code-JL or JP JZ the fourth figure of Eastings is indicated as for the third figure, in this example by finding figure in the Eastings and substituting the letter equivalents. 7 could therefore be encoded as K or E or N giving nine possible ways of encoding Eastings e.g.,

JLK JPK JZK

JLE JPE JZE

JLN JPN JZN

b. *Northings*. These are indicated as for Eastings, the letter equivalent being taken from Northings 5275 could therefore be encoded as:-

YTS YKS YGS

YTF YKF YGF

YTZ YKZ YGZ

c. The complete eight figure grid reference could, therefore be encoded as JLKYTS or JPEYKF and a further in 79 different ways.

35. **Twelve Figures Grid Reference**. The procedure used for eight figures grid reference applies to twelve figures grid reference (mainly used by survey units of artillery). The difference is that it will now be extended by four letters.

36. A twelve figures grid reference could, therefore, be encoded in some hundreds of say.

Example	**BRIDGE** 999925384703
Eastings	Northings
999925	384703

Encoded **JMSXA HNKRE** as one example.

37. The height when sent with eight or twelve figure grid reference is transmitted in clear.

38. The security of the code lies in the grid intersection point and the jumbled alphabets, the issue of which is a unit responsibility. These parts of the code will be SECRET, and documents containing this information will be so marked and thereafter safeguarded in accordance with the instructions contained in "Classification and Treatment of Protected Documents". Care should be exercised in the jumbling of the alphabets so as to avoid any stereotyped arrangements.

39. During active operations, keys will be changed every 24 hours, otherwise key may be changed weekly. When compromise of a key or grid intersection pointing is suspected, new keys and a new grid intersection point will be issued. Units are advised to prepare an emergency key list which will be immediately issued, should compromise of a key list in use is suspected.

40. In voice transmissions the proword"Grid Reference Killo X-ray Oscar Alfa".

41. Qualification of an encoded grid reference by a geographical feature must be avoided wherever possible e.g., avoid saying "bridge Killo Alfa".

42. When a message contains a mixture of Slidex for words and phrases and Griddle for grid reference, it is essential that the recipient is aware of the switch from Slidex to Griddle and vice versa.

43. When the slidex portion of message is followed by a Griddle Reference, the Griddle encoded reference will be preceded by the proword"Grid Reference". If the Griddle Reference is followed by further slidex diagram the letter will be preceded by the proword "Slidex" subsequent slidex diagram (separated by plain

language), do not required to be preceded by the proword “Slidex”, provided there are no Griddle Reference in between.

44. During operations, certain units or subunits may require to operate over an area greater than 26000 by 26000 metres. In these cases, several grid intersections points “1”. “2” et cetera using the same can be used with any grid intersection point key. It is necessary to indicate which grid intersection is referred to, the number of the grid intersection point will be given before the encoded grid reference.

Example: (Using grid intersection point 2):-

“A moving to grid reference figures two Foxtrot November Sierra Alfa”.

45. Similarly, in an emergency, such as pursuit, or withdrawal, new grid intersection point be given to an “O” group or sent by safe means.

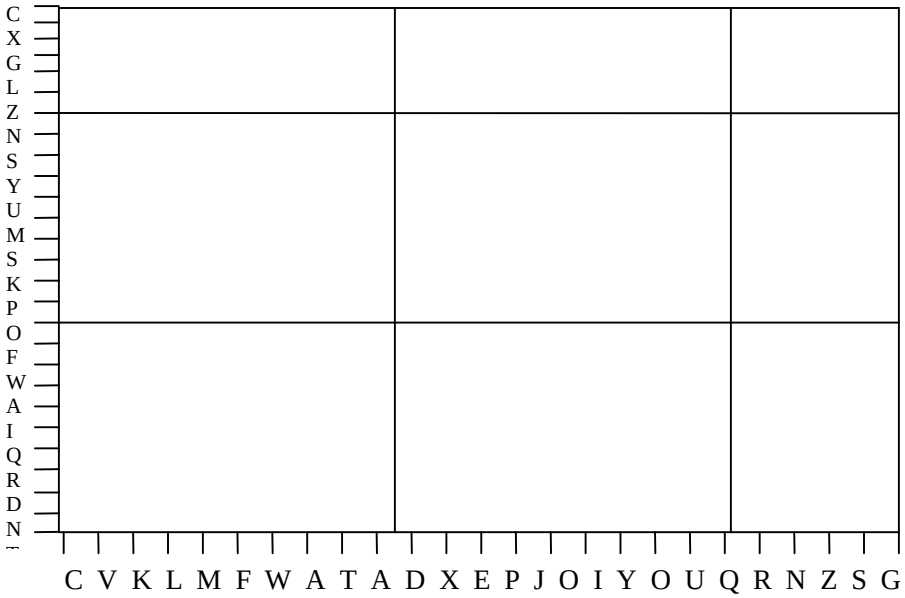


Figure 38 – Grid Intersection

46. Should talc not be available and it is considered undesirable to superimpose the 26,000 by 26,000 metres square on the map, once the grid intersection point has been given, the code can be laid out as below:-

Eastings

85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 00 01 02 03 04 05 06 07 08

C Y K L M F W X H T A D E P J O I V Q U B R N Z S G

Northings

35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60

S J T N D R I A W F P K B M U V Y E H Z L G X G O

47. In the scale 1/100,000, there are maps which have grid lines spaced at 10 kilometres intervals. The instructions in paragraph 35 above will NOT apply when this type of map is used.

48. The following instructions will be followed where this type of map is issued.

"Select a grid intersection point as in paragraph 34. Mark off East and North from the grid intersection point, one centimetre spacing up to a maximum of 26 centimetres". These marking will then be lettered off from the intersection point as in paragraph 35.

CHAPTER 20
FIRE SUPPORT PLANNING
SECTION 109 – **RESPONSIBILITY AND PROCEDURE**

Introduction

1. Tactics is the combination of manoeuvre and fire power. A tactical plan is thus a combination of manoeuvre plan and fire support plan. The size, extent and time of manoeuvre is directly related to the quantum of fire support available to support that manoeuvre. Therefore, manoeuvre plan and fire support plan are complementary and must be considered concurrently to match the conflicting requirements of manoeuvre and fire support in such a way that supported troops accomplish their mission by suffering least casualties.

2. Fire power comprised fire of all weapons of armour, artillery infantry and air made available to a commander. Therefore, the plan made to utilize the fire of all weapons is called 'Fire Support Plan'.

3. Artillery fire is seldom enough to fully support the manoeuvre and its effectiveness depends upon concentration. Therefore, only important fire tasks are allotted to artillery which cannot be undertaken by others with equal effectiveness. The plan made to contain only those tasks from the fire support plan which are allotted to artillery, is called 'Artillery Fire Plan'.

Responsibility

4. The making of fire support plan is the joint responsibility of the supported commanders and their artillery commanders. In general, it is for the supported commanders to say where fire is required (targets), in which form it is needed (types of fire support), what it is to achieve (effects of fire) and when it is to be applied (timings). It is then the duty of their artillery commanders to advise them on all these points and to assist them in the preparation of fire support plans. However, for those types, of fire support which are not closely connected with the movements or dispositions of the forward troops, e.g., CB, HF and DFs in depth, artillery commanders are then responsible to incorporate these in the fire support plan, and such information is considered necessary to be passed to the subordinate formations or units.

5. Based on the fire support plan, preparation of artillery fire plan is solely the responsibility of artillery commanders.

Essential Information

6. Form the above it follows that the supported commanders must associate their artillery commander in their planning from its very inception. Capability of available artillery will be an important factor to be considered by the supported commanders in their appreciation of the situation and making of outline plan. Their artillery commanders will advise them on the consideration of this factor.

7. Before actual consideration of manoeuvre and fire support can begin, the supported commanders must tell their mission, give latest information about the enemy and own troops, spell out their concept of operation to their artillery commanders and give their tentative outline plan to contain:-

a. *For Attack*

(1) Phases and troops.

(2) Objectives or bounds and distances to them.

(3) Forming up places, start lines, frontage and rate of advance.

(4) Action on objectives and limits of exploitation.

(5) Timings including H hour.

(6) Threats of counter-attack.

b. *For Defence*

(1) Defensive layout including location and tasks of reserves.

(2) Approaches and probable axes, composition and strength of enemy attacks.

(3) Probable location, composition and strength of enemy reserves, headquarters and supporting weapons.

(4) Localities to be counter-attacked when lost.

8. During fire support planning at every level, the following specific information must be given to the artillery commander responsible for the operation or task:-

a. Targets to be included in the fire support plan.

b. Effect required on each target.

c. Safety of own troops at each stage.

d. Restrictions. If any.

e. Timings.

9. **Targets to be Included in the Fire Support Plan**. Targets or target areas to be engaged during the fire support plan must be indicated by the supported commander on ground, map or air photograph and identified by the artillery commander.

10. **Effects Required**. Normally, with the fire power available, artillery commander will not be able to guarantee more than neutralization of the targets. The extent of neutralization to be expected will depend on the number of weapons, rate of fire and amount of ammunition employed. The artillery commander must, therefore, be given the priority targets and effect of fire required on each. This will lead to the selection of:-

a. The best and most appropriate type of fire (air support, tanks, infantry mortars or artillery) to engage each target in the fire support plan.

b. The best and most appropriate artillery weapon and ammunition to be used for each artillery target.

c. Number of firing units and scale of ammunition required to achieve the desired effect on each artillery targets.

11. **Safety**. It is the supported commander’s responsibility to say how close his own troops should be to artillery fire, based on the advice of his artillery

commander. Normal safe distance to which own troops can approach the point of burst in reasonably safety during operations are given here as a guide:-

	Weapon	*Safe Distance In Metres*	
		Overhead Fire	*Flanking Fire*
a.	105 mm Howitzer	200	200
b.	122 mm Howitzer	300	300
c.	155 mmGun/155 mm Howitzer	400	400
d.	130 mm Gun	600	600
e.	155 mm Gun	600	400
f.	8 inch Howitzer	500	550
g.	120 mm Mortar	550	550

12. **Restrictions**. The supported commander may lay down restrictions on registration and use of testers which are likely to prejudice surprise, or on the use of smoke in the fire plan which may hinder own operations.

13. **Timings**. Before deciding upon H hour, and other timings, the supported commander must consider the state of readiness of artillery and the time required for registration or other technical preparations. Similarly, the time required for redeployment of guns and registration of further targets et cetera must be kept in mind for fixing timings for subsequent phases.

Procedure

14. The basic pattern of making manoeuvre and fire support plan is that:-

a. Outline manoeuvre and fire support plans (including sub-allotment resources) are made successively at each level going down the chain of command.

b. Detailed manoeuvre and fire support plans are prepared, coordinated and approved successively at each level going up the chain of command up to where the outline plans were initiated.

15. Generally, the following procedure is followed in the making of a fire support plan by the supported commander along with their artillery commanders at different levels:-

a. Higher commander advised by his artillery commander makes his outline plans by phases to include:-

(1) Tasks of subordinate formations or units.

(2) Fire support resources (weapons, ammunition, air effort) allotted to the subordinate formations or units for the tasks given to them or allotted to the phase of the operation.

(3) Outline pattern of fire support connected with the movement of troops in each phase (types of fire support and their sequence).

(4) Policy regarding registration, CB, HF et cetera.

b. Based on the outline plan, the subordinate commanders along with their artillery commanders, carry-out reconnaissance and prepare their manoeuvre and fire support plans with particular reference to the following in the order given hereunder:-

(1) Availability and use of infantry, armour and air in the fire support plan (decide which targets are to be engaged by them and how).

(2) Rough division of artillery ammunition between phases and reserve for each phase.

(3) Types of artillery fire support for each phase in the precedence suggested hereunder and the requirement of ammunition for them:-

(a) *Priority One*. The assault stage. In attack, covering fire and in defence, DFs;

(b) *Priority Two*. Reorganization stage. In attack DF and in defence covering fire for counter-attacks.

(c) *Priority Three*. Preparatory stage. In attack smoke screens to cover move from FUP to start line and in defence DF or DFs in depth.

(d) *Priority Four*. In both attack and defence CB and HF and use of artillery for deception if not being undertaken by higher commander.

(4) The artillery commander keeps a running total of the ammunition and if it runs short, then the supported commander decides to eliminate the lower priority tasks or reduce the effect of fire on them or may even modify the manoeuvre plan to make better use of fire support.

(5) The finalized fire support plan is entered in the task table by the artillery commander and approved by the supported commander.

(6) Task tables are forwarded to higher headquarters where these are successively coordinated, modified where necessary and approved at each level.

(7) Approved task tables are then issued to all concerned by the higher headquarters.

16. Detailed procedures for preparing fire support plans for various operations of war are covered in the succeeding sections.

SECTION 110 – **FIRE SUPPORT PLANNING FOR ATTACK**

Introduction

1. The attack may be a quick attack against weak defences to take full advantage of the unpreparedness of the enemy or deliberate attack against well organized defences warranting detailed planning and thorough coordination. The procedure of planning (manoeuvre and fire support) of each is discussed in the succeeding paragraphs.

Fire support plan for Quick Attacks

2. Quick attack is an extension of advance to exploit the initial success achieved soon after contact in order to break and disrupt enemy position without wasting any time. It may vary from a quick attack by subunit of just one target arranged by FOO to a quick attack by a formation of many targets to be supported by divisional artillery.

3. **Fire Support Plan for Subunit Attack**. This will comprise just one target being attacked by infantry company/platoon or tank squadron/troops. The fire support plan will be made by the subunit commander advised by the accompanying FOO. Since speed is of utmost importance and quantum of fire support is very little, the planning is confined to the decisions with regards to use of HE or smoke screen and the timings as under:-

a. Target to be engaged by one battery with HE when ready or on an agreed time, for certain duration and rate of fire (one gun may fire smoke).or

b. Target to be blinded with a quick smoke screen by one battery on a timed programme, with an alternative HE task on the target to be fired for a certain duration and rate of fire if smoke screen proves ineffective.

4. **Mechanics of Making Fire Support Plan for Quick Attack**

a. *The Supported Commander*

(1) Gives out his manoeuvre (or attack) plan by phased, to include the following:-

(a) Troops.

(b) Objectives and distances to them.

(c) Forming up place and start line.

(d) Rate of advance.

(e) H hour.

(2) Decides which of his own support weapons (MGs infantry mortars, BSs, recoilless rifles, tanks) he can make available for the fire support plan.

(3) In consultation with artillery commander decides on a rough division of the available ammunition between phases, and for each phase decides what proportion of the ammunition will be kept as a reserve for on-call and impromptu targets and DFs.

b. The fire support plan is made by phases, each phase being dealt with as follows in the sequence suggested hereunder:-

(1) *Covering Fire to Support Assault*

(a) The supported commander states the enemy localities which are likely to interfere, when he wants them neutralized relative to the manoeuvre plan, by whom (artillery, infantry, armour or air) and how (HE or smoke).

(b) The artillery commander allots target numbers and recommends the punishment required by each artillery target, i.e., number and type of guns, number and type of rounds and rate of fire.

(c) The armour adviser, if present, is consulted with regard to targets to be engaged by tanks.

(d) The commanders of the infantry support weapons, if present are, consulted about the targets to be engaged by infantry support weapons.

(e) If air support is available, the artillery commander recommends on its employment.

(f) The Supported Commander approves or modifies these recommendations.

(g) The artillery commander keeps a running total of the ammunition expenditure. If resources are limited the supported commander decides on the priorities.

(2) *DF to Support Reorganization*. Having decided on the covering fire to support the assault, the supported commander, in consultation with the artillery commander, decides on the following:-

(a) On call targets.

(b) Targets on the flanks, or screening of the flanks by smoke.

(c) Pre-arranged DFs.

(3) *Other Fire*. When the covering fire and DF for all the phases has been planned, the supported commanders, will consider the following depending upon the availability of ammunition for them:-

(a) Smoke screens to cover move from forming up place (FUP) to the start line. This may be so important that ammunition will have to be found for it.

(b) Preparatory bombardment.

(c) Counter bombardment (CB) against guns and mortors (if not being arranged by the higher commander).

(d) Depending on the time available, detailed timings for the engagement of targets are either worked out jointly by the supported commander and the

artillery commander, or by the latter alone, based on the manoeuvre plan and the supported commander's statement of when he wants each target neutralized relative to that plan.

c. The details of the fire support plan, including the tasks allotted to all supporting weapons (air, armour, artillery, infantry weapons), are entered on the task table (PAFZ-6047) by the artillery commander. He completes the task table and gets it approved by the supported commander. Fire support plan is then issued to all concerned.

d. The artillery commander makes out artillery fire plan for the tasks allotted to artillery in the fire support plan and issues it to the artillery firing units.

Fire Support Plans for Deliberate Attacks

5. Fire support plans for deliberate attacks against well prepared defensive positions may be made at unit (infantry battalion/armoured regiment), brigade, division and higher level. Since thorough preparation is of utmost importance to the success of operation, reconnaissance is carried out in detail, targets are accurately registered and support plans are thoroughly prepared and coordinated at each level by the respective commanders advised by their artillery commanders. Based on fire support plans, artillery fire plans are prepared and issued by the higher artillery headquarters to all concerned.

6. **Sequence of Making Fire Support Plan for Deliberate Attacks**. The following sequence of events in making fire support plan and artillery fire plan for a deliberate attack is suggested:-

a. The higher commander gives out his outline plan, which should include the tasks given to subordinate units and the fire support resources allotted for carrying out these tasks. It may also include the outline methods of fire support. The outline method of fire support and allotment of artillery resources will be made by the commander in consultation with his artillery commander. The artillery commander on behalf of the supported commander may lay down any restrictions on registration and ammunition expenditure.

b. Subordinate commanders who are to make the fire support carry-out their reconnaissance accompanied by their artillery and armour

commanders and make their manoeuvre and fire support plans as under:-

(1) The supported commander gives out his plan of attack in phases to include the following for each phase:-

(a) Troops

(b) Objectives, and distances to them.

(c) Start line.

(d) Timings.

(e) Rate of advance

(f) Reorganization.

(2) The supported commander decides which of his own support weapons (MGs, infantry mortors, BSs, recoilless rifles, tanks) he can make available for the fire support plan and which targets are to be engaged by artillery and air.

(3) The supported commander, in consultation with the artillery commander, decides on a rough division of artillery ammunition available to him between phases and for each phase decides what proportion of the ammunition will be kept as reserve for covering fire on-call, impromptu targets, and DF.

(4) *Assault Stage*. The assault stage being the most important stage is dealt with first for each phase in the sequence, suggested hereunder:-

(a) *Artillery*. The supported commander states the enemy localities which are likely to interfere and when he wants them neutralized or destroyed by the artillery relative to the manoeuvre plan. The artillery commander supports the assault by covering fire and allots target number and recommends the punishment required by each target, i.e., number and type of guns, number and type of rounds and rate of fire.

(b) *Armour*. The armour adviser if available is consulted with regard to targets to be engaged by tanks.

(c) *Infantry Weapons*. The supported commander, advised by the artillery commander and the commanders of the infantry support weapons if present, decides which targets are to be engaged by these weapons and how.

(d) *Air Force*. If air support is available the supported commander, advised by the artillery commanders decides on its employment.

(5) *Reorganization Stage*. Having decided on the covering fire to fit the assault stage, the supported commander, in consultation with the artillery commander, decides on the following for the reorganization stage of each phase:-

(a) On-call targets.

(b) Pre-arranged DFs on call.

(c) Targets to be registered or surveyed and allotment of fire units.

(d) Use of smoke, if any.

(6) *Preparatory Stage*. When the fire support for the assault and reorganization stages of various phases has been planned, the supported commander in consultation with the artillery commander, will consider the following items subject to availability of ammunition:-

(a) Smoke screen to cover move from FUP to the start line. This may be so important that ammunition will have to be found for it.

(b) Preparatory bombardment.

(c) CB against guns and motors (if not being arranged by the higher commander).

(d) Use of the artillery for the deceptions.

(7) Depending on the time available, the detailed timings for the engagement of targets to be engaged and timed concentration are either worked out jointly by the supported commander and the artillery commander, or by the latter alone, based on the manoeuvre plan and the supported commanders requirement when he wants each target centralized relative to that plan.

(8) Throughout this process, the artillery commander keeps a running total of the ammunition expenditure. If resources are limited, the supported commander decides on the priorities.

(9) The details of the fire support plan, including the tasks allotted to all supporting weapons, (air, armour, artillery, infantry weapons) are entered on the task table (PAFB-6047) by the artillery commander. He completes the task table, has it approved by the supported commander, and gives him a copy. The final approval of the completed task table by the supported commander is most essential. It will often be given only after the next lower echelon of supported commanders and their artillery commanders have had a chance of examining and discussing the fire support planned suggesting any modifications they thinks necessary.

c. The subordinate commanders then submit their fire support plans to the higher headquarters through artillery channels where these are coordinated by the higher artillery commanders with the subordinate artillery commanders usually in a coordinating conference. Any points for decision are referred to the higher commander.

d. Fire support plan is then approved, with modification if necessary, by the higher commander in coordinating conference attended by the subordinate commanders along with their artillery commanders.

Determination of H Hour

7. The time of H hour will be ordered by the supported commander after consideration of number of factors. These will include the time required by the artillery commander to produce the degree of guarantee of effective fire which the supported commander considers necessary with particular reference to:-

a. The number of targets to be registered by firing and the observers available for registration.

b. The number of targets to be surveyed, map spotted and predicted.

c. The time needed for the passage of fire plan orders, including circulation of the results of the registrations.

d. The technical work to be done in command posts.

e. The time needed for preparation of ammunition.

f. Any requirement to provide fire before H hour.

g. Time required for the guns to come into action.

Distribution of Ammunition

8. Depending upon the tactical situation, the ammunition allotted to a phase can be distributed to various types of fire supports on the following percentage (as a guide only):-

a.	Covering fire	-	60%
b.	Preparatory Bombardment and CB	-	15%
c.	Reserve including DF	-	25%

SECTION 111 – **DEFENSIVE FIRE PLANNING**

General

1. The aim of DF is to disorganize the enemy's preparations for attack and to break up his assault when it is delivered.

2. DF tasks may be conveniently subdivided into:-

a. Close DF, including DF (SOS).

b. DF in depth.

Close DFs

3. Close DF tasks are sited with the aim of breaking up an enemy attack when it is forming up or during the assault. They are chosen by the unit commander advised by his direct support battery commander and are based on his appreciation of emery's likely approaches and pattern of attack. Likely close DFs are:-

a. For each approach:-

(1) FUPs.

(2) Line of approach/assault from FUPs to own forward defended localities.

(3) Any tactical feature on the enemy's line of approach.

b. Gaps between localities.

c. Grounds, dead to own infantry/armour weapons.

d. Tactical features, occupation of which will jeopardize the defences.

e. Areas covering obstacles and demolitions.

f. Own localities.

g. Behind own localities.

4. As the battalion's weapons may be destroyed or neutralized by enemy fire, some close DF tasks must be sited to cover the dangerous approaches in close proximity to our own troops. Keeping in mind the normal inaccuracies in firings, close DFs can be sited at 400 metres for field, 500 metres for medium and 550 metres for mortar and heavy from own FDLs. If DF tasks are sited closer than these distances, there is a danger that a few rounds might land in own positions and supported commanders must approve such curtailments. But when there are no inaccuracies and when own troops are well protected, the safe distances for various guns/mortars as given for attack are equally applicable to defence.

5. There will often be a requirement for DF to be brought down on enemy, held up on protective wire and minefields, which are usually about 200 metres in front of infantry localities, provided the infantry is well dug in. This distance provides a reasonable minimum safety from the closest DF tasks for light/field weapons.

6. There may be occasions when own artillery fire will have to be brought to bear on our own positions when being over run by the enemy. Therefore, while making defensive fire plan, close DFs should be planned on or even behind own positions which are located across the most threatened enemy approach, so that if targets appear in these areas, artillery fire could be brought down without any loss of time. However, to distinguish, such DFs from the others and to avoid mistake and doubts at the gun end, such DF target numbers should be followed by letters "OL" (meaning own localities).

7. **DF (SOS).** Of the close DFs chosen by the unit commander, one or more DFs covering the most likely approach may be designated as DF (SOS) task depending upon the number of firing units allotted to that unit. Guns of the firing units allotted DF (SOS) are usually laid on that task or even kept loaded and laid, when not otherwise engaged, as decided upon between the unit commander and the battery commander. DF (SOS) may be changed periodically to conform to the

changing threat and is a matter for arrangement between the unit commander and his battery commander.

DF in Depth

8. DF task in depth are sited with the aim of disorganizing the enemy's preparations for attack in the assembly areas and of striking at his reserves and headquarters once the attack is launched. They are usually chosen by the brigade commander advised by the direct support regiment commander but more may be added by the divisional commander or commander artillery, if necessary.

Grouping of DFs

9. When DF tasks are tactically inter-related e.g., when it is appreciated that the enemy will appear at two or more places simultaneously, they should be grouped for simultaneous engagement. Maximum number of targets in a group is governed by the amount of artillery likely to be available. For example, a group of DFs planned to break up an assault on a battalion front should be capable of being engaged by the artillery whose support is guaranteed to that battalion. In this case additional in-support units from the divisional artillery should be superimposed so that even if their fire is withdrawn, the effect of DF is not reduced. Where groups of DF target in depth have been planned at brigade level, the number should not be greater than which can be effectively engaged by the artillery, guaranteed to the brigade.

Number of DFs

10. It is not intended to limit the number of DF task which can be arranged but too many will defeat the aim of planning as more than about 30 to 35 cannot be handled efficiently by a regiment command post. As a guide 8 to 12 DFs are selected for a battalion deployed in defence. On this basis and assuming that about one third of all DFs will be selected in depth, the total on a brigade front are not likely to exceed a manageable number.

Use of Standard Linear Concentration

11. The use of the standard linear concentration on a DF task has several disadvantages and its use should be discouraged except when its special characteristics make it more effective than a simple concentration. It's limitation are as follows:-

a. The density of fire in this, even when engaged by a whole regiment.

b. It is comparatively slow to move, due to technical work involved at the command post, than a simple concentration.

c. Unless the ground in the target area is flat, it tends to lose its shape when move.

Use of Smoke

12. Armour is very sensitive to smoke in the battlefield, therefore, a few WP smoke shells fired along with HE on DFs outside the range of our anti-tank weapons will panic the attacking armour and disable it to carry-out accurate fire from long-ranges. Similarly, enemy air cannot carry-out precision bombardment during preparatory stage of enemy attack, if own localities are blanketed with smoke. In order to deceive the enemy air, preparation to fire a coloured smoke on or near the enemy locality as soon as a coloured smoke shell is fired by the enemy to direct his aircraft will help in confusing the enemy aircraft that may attack his own troops. However, use of smoke against enemy armour or air must be fully considered to avoid the adverse effect on own troops.

Use of Illumination

13. The attacker often makes use of night to conceal his operation and to achieve surprise. Moreover, even if his approach or presence is discovered through various security means, the accuracy and effectiveness of DF cannot be ascertained without NVDs or illumination.

14. Sudden illumination of the battle area surprises the enemy who was depending upon the cover of darkness for the conduct of his operation. Moreover, it facilitates own observation of the attacker and in determining his location, strength and composition and thereby enabling the observer to bring effective DF. Battlefield illumination should, therefore, be pre-planned to cover the likely approaches. Also illuminating rounds should be fired with DF so that its effectiveness can be ascertained and the fire is adjusted, if necessary.

Procedure of DF Planning

15. **Action by Unit Commander and Battery Commander**. Initially the unit commander is responsible for choosing the close DFs in support of his unit and in this, he is advised by his direct support battery commander on the following aspects:-

a. Artillery units available for DF and DF (SOS).

b. Relative suitability of targets for engagement by artillery, infantry and armour.

c. Type of artillery weapons and ammunition to be used for each artillery targets.

d. The safety factor for artillery weapons with regard to proximity to own troops.

16. The battery commander produces DF Task Table for the agreed DF tasks in support of the unit, which must be approved by the unit commander. One copy of the battery commander's task table goes to the direct support regiment commander for coordination and one copy per command post of artillery guaranteed to the unit where it will be used for answering calls for DF till it is replaced by the coordinated DF task table or the final DF plan of the regiment/divisional artillery. Except in case of extreme urgency, a battery commander's DF target list should never be passed by wireless because, if intercepted. It will give away the DF plan to the enemy and thus indicate the pattern of unit's defensive layout.

Procedure for Calling for DF

17. A DF task may be called by:-

a. Normal target procedure.

b. The target number when the response has been preplanned.

c. A nickname when targets have been grouped and the units to engage have been detailed in the task table.

18. **Normal Target Procedure**. When DF Task Table has not been issued but only a DF Target List has been issued, then, in addition to the target number, units to engage and the scale, if not laid down in the formation standing orders, must also be ordered, e.g., "DELTA, FOXTROT 3612. Oscar Tango 40 degree 3 rounds gun fire, fire over".

19. **Response Pre-Planned**. When the DF Table has been issued and the units to engage the DF tasks along with the ammunition to be fired has been preplanned, then it is necessary to only order the target number and fire e.g.:-

"DELTA FOXTROT 5643 fire over"

20. **Grouped Targets**. When the task table for grouped targets has been issued, then the fire can be called for by just ordering the nickname of the group,

on which the fire units concerned will fire the laid down number of rounds or their respective targets, e.g.:-

"TOP HAT fire, over"

SECTION 112 – **FIRE PLANNING**

General

1. In all operations, the company commander should directly carry-out the planning of employment of his firing units/weapons and how to help them with his indirect fire. While planning, the commanders of additional detachments will assist the company commander.

2. The commander while presenting his concept of operation will begin planning of fire (scheme of manoeuvre or position of units is included in it) and he will assist in the requirements of fire. This assistance is included which has been designed by the battalion commander.

3. **Planning of Offensive Fire**. In attack, the commander deliberates on the mission, enemy, terrain, weather, troops and the provided time and then decides the scheme of manoeuvre. In doing so, he determines how he will manoeuvre his platoons and weapons. He will select the positions along with the passage from where surveillance elements can provide cover to the movement of company, granting that during the attack, if an element of fire is used and the commander selects a position that will provide support to the manoeuvring elements.

4. Once the scheme of manoeuvre is determined with the forward observation officer, then the platoon commander for his assistance, carry-out planning of indirect fire. On the base of guidance for the commander, platoon commander develops a list of targets. Targets, personnel, equipment (included vehicles) and the features of fortifications and terrain which are engaged with the fire.

5. Forward observation officer, up to the objectives carry-out planning of the targets of known and unknown probable places of the enemy. At objective, in the rear and flanks, he executes the planning of target. The determination of probable locations of the enemy is made with the study of a map and the information of enemy.

6. If houses are few and if they exist, then the planning of fire is carried out on the prominent features (known points) of the terrain, so that from a known point, the targets that appear could be engaged. On a map and ground, the prominent features of terrain should be such so that they could be identified easily. The number of such targets should be less.

7. There are four types of targets for which the forward observation officer can make the planning. They are planned targets, targets on demand, foremost target and targets appearing on the spot:-

a. The planning is carried out to hit a planned target on fixed time. Before or after a fixed time or incident, the time in minutes is given.

b. In relation to a fixed time, the planning to hit a target on demand is carried out.

c. Foremost target is a target on demand for which the unit commander feels its manoeuvre un-secure and is critical for his unit and as far as possible it should be fired upon.

d. The target appearing on the spot is a target for which no planning has been carried out. Such targets appear unexpectedly because these targets appear for a very short time and they must be hit at earliest.

8. Apart from engagement of the targets, FOO carries out the special type of planning of supporting fire, which includes the prepared fire and screen fire.

9. A prepared indirect fire has a fierce volume, and is prepared according to the plan for supporting an attack. This fire commences on a fixed time in relation to the time of an offensive and may be continued, after the fixed time. The fire on a fixed time or when the need arises can be commenced. A prepared mortar is not fired because it expends an immense quantity of ammunition.

10. The smoke is involved in the screen, which is used to conceal the friendly units from the enemy. It also includes specific offensive use of smoke screens i.e., to prepare a screen for crossing a river, to consolidate the position of a unit on an objective or for any type of movement. In a region of smoke, the rounds of high explosive are fired. Friendly troops do not use it, thus rendering the enemy unable to use it to conceal his movement.

11. When the list of targets is compiled and is approved by the company commander then it is furnished to the battery commander through safest means. Battery commander scrutinizes the list and consolidates the lists of targets of the

companies. He prepares a replica of targets and allots target number to every target. A target number comprises two letters of alphabet and four digits, e.g., Delta Foxtrot 5643.

CHAPTER - 21

THE ENEMY

SECTION 113 – CHARACTERISTICS AND EMPLOYMENT OF INFANTRY FORMATIONS/UNITS

General

1. Infantry constitutes the bulk of Indian land forces. Indians are continuing their efforts to improve the performance of this important arm through modernization. They are also seeking to acquire strategic third dimension capability through the use of helicopters and air transported forces. The tactical doctrine is being developed taking into consideration, all the modern developments and requirements of threat estimated in the coming years.

Types of Formations

2. *Infantry Division*. Indian Infantry Division is organized on similar lines as we have in our army.

a. *Mountain Division*. These will be reorganized by the year 2010 and will be known as Reorganized Army Mountain Infantry Divisions.

b. *Reorganized Army Plain Infantry Divisions (RAPIDs).* These are of two types, RAPID (Defence) and RAPID (Strike).

c. *Independent Parachute Brigade*. India has Parachute Brigades based on fixed wing aircraft.

d. *Amphibious Brigade*. Indians have recently raised an amphibious brigade.

Role, Capabilities and Tasks of an Infantry Division

3. **General**. The Infantry is organized into divisions, brigades and battalions. Battalions are further divided into companies, platoons and sections. Appropriate organizations have been evolved and the formations/units are suitably equipped and trained to enable them to operate without impediments in various types of terrain (mountainous/desert et cetera) and specific roles (airborne

operations et cetera). A division is the basic combat formation and forms the bulk of Foxland Army. A major general commands it. It normally consists of three infantry brigades, one artillery brigade, and an armour regiment and balanced elements of all supporting arms/services. The number of brigades, however, may vary and a division may have more than three brigades depending upon the role assigned to a particular division. The organization is flexible enough to allow absorption of all formations and units placed under command/in support for specific tasks.

4. **Capabilities**

a. *Advance*. Can advance 25 kilometres in a day and has total capability of 50 kilometres in five days against light opposition. This may imply quick attack up to battalion level or a brigade level encounter battles:-

b. *Attack*. Can launch an attack with two brigades at a time. However, this type of attack will require additional support from artillery and engineers et cetera. It may be given following missions:-

(1) Break-in and capture of enemy defences. This may involve opposed crossing of natural/artificial obstacles.

(2) Holding of a firm base from which further offensive operations may be carried out.

c. *Defensive Operations*

(1) Can hold compact division defensive sector 20-30 kilometres wide and deep.

(2) When defence oriented anti-tank obstacles exist, it can hold an extended defensive system from 60-80 kilometres in width.

(3) Can create a force for counter penetration; counter-attack from within its own resources.

d. *Withdrawal*. Can undertake withdrawal up to 32 kilometres in a day when transport is available. In case transport is not available, it will be up to a max of 16 kilometres in a day.

e. *Organization*. Given at Annex ‘BB’.

Role, Capabilities and Tasks of Mountain Division

5. **General**. The Mountain Division is capable of fighting independently or as part of a Corps at an altitude up to 6000 metres. Although, it is primarily designed for operations in the mountains, it can operate equally effectively in the jungles and plains with some organizational changes such as provision of additional anti-tank weapons and transport.

6. **Capabilities**

a. *Advance*

(1) Can advance at an average rate of 8 -10 kilometres per day on more than one axis.

(2) Can be employed for a period of seven days i.e. a turn round of 3 days from the road head.

b. *Attack*. Can launch a phased deliberate attack with one brigade at a time and with one more brigade sent out as an outflanking move on a mule track to assist the attack on the main axis.

c. *Defence*. Can hold a sector up to 30 kilometres in width and depth, independently or as part of a corps.

d. Organization. Given at Annex 'CC'.

e. Difference in Infantry and Mountain Division. Given at Annex 'DD'.

RAPID(S) and RAPID (D)

7. **General**. Foxland has two types of infantry divisions that take-up defence i.e. plain infantry/mountain divisions and RAPID (D). These could either be deployed in the plains of Punjab or in the desert against which an attack may have to be undertaken. These have been organized to provide infantry with manoeuvre elements, which have swift cross-country capability to exploit opportunities. RAPID(S) are organized to operate in conjunction with the armoured and mechanised formations. RAPID(S) is to be grouped with offensive formations and RAPID (D) to be used for defensive balance.

Mission/Tasks

8. **Defensive Operations**. RAPIDs have only two infantry brigades for holding ground. Gaps have to be accepted within and between defensive sectors. The disadvantages are expected to be offset by a qualitative improvement in

surveillance, indirect fire support plans, covering troops and armour heavy mobile reserves.

9. **Forward Zone**. Provide the flexibility to design a more meaningful battle in the forward zone by employment of superior surveillance to ensure early warning and accurate identification of axes of advance and attack.

10. **Defence Based on Obstacles**. Initial occupation of defence is based on a careful tactical and technical appreciation to decide the priority of enemy's likely bridgehead sites. Effective covering force operations to identify axes of attack and impose delay allowing the defender to adjust his dispositions on the main obstacle by occupying contingency positions. The limit of penetration is likely to remain along the line of the main obstacle. Availability of mechanised infantry forces will permit occupation of contingency positions in gaps.

11. **Defence Based on Key Terrain behind Forward Obstacles**. Strong covering and para military forces may be deployed on the forward obstacles. Defence in depth will be based on important tactical ground or obstacle. They will deny key terrain avenues of approach and will provide firm bases for offensive manoeuvre of mechanised forces.

12. **Defence – Nodal Point Concept**. While there will be a decrease in the overall number of nodal and strong points that can be occupied in a divisional defensive sector, the important variations are:-

a. It will be possible to fight the battle of the gaps more effectively. Groups comprising mechanised infantry, Reconnaissance& Support battalion elements and armour will be used.

b. Air may be available for dedicated allocation to selected nodal points.

13. **Offensive Operations**

a. *Attack across a Linear Obstacle*. RAPIDs are capable of undertaking leaning on operations without major allocation of resources from outside. However, additional forces will be made available for the creation of a divisional infantry reserves, where necessary.

b. *Follow Up Support Operations*. RAPIDs are capable of more efficient follow up support operations in the wake of advance by mechanised forces. It can provide a safe corridor and armour heavy

mobile pivots for conduct of reactionary tasks using infantry pivots for aggressive manoeuvre.

c. *Attack against a Nodal Point*. The armoured brigade will lead the advance to the objective and drive in minor opposition, isolate the objective, provide a firm base and assist infantry brigades in its capture.

d. *Limited Offensive Operations*. Armoured brigades of RAPIDs will enhance significantly the offensive capability of the Corps in limited offensive operations in support of strategic counter-offensive. During consolidation, it will improve reaction to and defeat enemy counter-attacks.

14. **Organization**. Given at Annex 'EE'.

Role, Mission and Employment of Parachute Brigade

15. **General**. Parachute Brigade is trained and equipped both for airborne and air assault roles. It comprises of a brigade headquarters, three parachute battalions and supporting arms.

16. **Role/Mission**

a. *Tasks When Cooperating with Main Land Forces*

(1) Facilitate the task of main force by creating a rear threat through seizing and holding dominating ground or defiles, vital to the advance of the main force. The link-up in such a situation is normally planned to take place within 72 hours.

(2) Attack enemy's headquarters and communication centres.

(3) Divert and block the approach(s) of enemy reserves.

(4) Neutralize enemy defences in an area selected for a sea-borne landing.

(5) Delay and cut-off enemy's retreat.

b. *Independent Tasks*

(1) Seize and hold an isolated area, against limited opposition, for use as a land, air or sea base for attack on the enemy and his communication zone.

(2) Raid enemy territory to capture material or to destroy installations.

17. **Employment**. Brigade will have to comply with orders from Army or Central Command Headquarters and act accordingly.

18. **Organization**. Given at Annex 'FF'.

19. **Types of Infantry Battalions**

a. *Standard Infantry Battalion*. Designed to operate in the plains as part of Infantry Division.

b. *Infantry Battalion Modified 'M'*. Forms part of mountain division.

c. *Infantry Battalion Modified 'D'*. These are designed to operate in desert terrain.

d. *Infantry Battalion Modified 'Q'*. Battalions organized to operate in Andaman and Nicobar Islands.

e. *Mechanised Infantry Battalion*. It is part of armoured and mechanised formations.

f. *Parachute Infantry Battalion*. Forms part of parachute brigade.

g. *Para-Commando Battalions*. These are designed to carry-out commando tasks.

h. *Scout Battalion*. Designed to operate in mountainous terrain along northern border.

j. *Reconnaissance & Support Battalions*. These are organized on a separate establishment.

k. *Anti-Tank Guided Missile (ATGM) Battalion*. The ATGM battalion is organized, trained and equipped to augment the anti-tank defence in a corps defensive zone.

20. **Standard Infantry Battalion**. An infantry battalion is a unit designed to close with and destroy or capture the enemy and to hold ground with the support of other arms and services. Various capabilities of Foxland infantry battalion according to operation/terrain are as following:-

a. *Defence in Plains*

(1) Is capable of fighting defensive battle ahead of main defences in the form of screens to repulse at least one attack before withdrawing.

(2) Besides fighting the battle of protective detachments, is capable of fighting aggressively in the security zone in the form of leaving stay behind parties, anti-tank nests, launching of spoiling attacks, and conducting effective patrolling.

(3) Is capable of fighting effectively on the independent/ isolated bases of operations – with the support of 2nd line troops till physically attacked and thrown back.

(4) Under normal terrain conditions, is capable of defending an area covering frontages up to 3000 metres in width and 2000 metres in depth, against deliberately planned and co-ordinated attacks within 10 days hours time.

(5) In an obstacle oriented defence, an infantry battalion is capable of defending an area by covering frontages up to 6000 metres in width and 3000 metres in depth, against deliberately planned and coordinated attacks within 5 to 6 days hours time.

(6) Is capable of launching spontaneous local counter-attacks with the help of committed reserves of company/company minus strength from the depth or flanks with or without the support of armour.

(7) In a Ditch Cum Bund (DCB) oriented defence, is capable of sealing a breach in the bund by readjusting weapons and section positions.

(8) Is capable of wearing down the attacker by causing maximum attrition on him through a coordinated weight of defensive fire.

(9) Is not capable of launching deliberate counter-attacks for having no uncommitted reserves at battalion level.

b. *Attack in Plains*

(1) Is capable of launching a set piece phased attack from a secured firm base with armour in close support/fire support role and capture an area up to 400 metres wide and up to 1000 – 1500 metres in depth.

(2) Is capable of launching a set piece phased attack across an obstacle oriented defence from a secured firm base with

armour in fire support role and capture an area up to 400 metres wide and 1000 metres in depth.

(3) Can launch a quick day attack with armour as predominant arm on a company location (1000 x 1500 metres) in 2-3 hours time.

(4) Is capable of exploiting beyond the objective, with or without the support of armour, up to 500 metres in depth.

(5) Is capable of achieving great concentration of coordinated fire support both of direct and indirect firing weapons including use of air. Is also capable of deceiving through a well-planned deceptive fire.

(6) Is capable of attacking in waves.

c. *Miscellaneous Information*

(1) Speed is 4.5 kilometres per hour in plains by day and half of this by night cross-country.

(2) In mountains, additional one hour for every 300 metres of ascent and 450 metres of descent is added to the above mentioned timings.

(3) It can cover a distance of 30-40 kilometres in 24 hours.

(4) Rifleman can carry 22.5 kilograms in battle order and 32 kilograms in Field Service Marching Order (FSMO).

(5) *Echelons of an Infantry Battalion*

(a) *F Echelon Transport*. This echelon consists of transport carrying essential stores, equipment and ammunition that will be required for the immediate fighting. The headquarters company commander is usually incharge of this echelon.

(b) *A Echelon Transport*. It consists of vehicles carrying ammunition, equipment and stores that will be required during the battle to supplement or replenish the F echelon. This echelon is normally controlled by the Mechanical Transport Officer (MTO).

(c) *B Echelon Transport*. It consists of transport carrying administrative personnel, stores and equipment that will not be required until there is a pause

in the fighting. B Echelon will normally be under brigade or division control.

d. *Organization*. Given at annex 'GG'.

Modified Infantry Battalions

21. **Infantry Battalion Modified 'M'**. Capabilities of mountain infantry battalions are as following:-

a. *Defence in Mountains*

(1) Is capable of defending a frontage of 6000-8000 metres occupying defensive localities on dominating heights.

(2) Is capable of fighting successive battles in depth by occupying series of counter penetration positions.

(3) Is capable of reinforcing a threatened locality by shifting troops from less threatened localities/areas.

(4) Is capable of fighting against infiltration by adopting both active and passive measures.

(5) Is capable of fighting an aggressive defensive battle in the given area of responsibility through active patrolling, conducting raids, ambushes and implementing an effective surveillance/obstacle plan.

b. *Attack in Mountains*

(1) Can launch a set piece attack – may be multidirectional, from a secured base to capture a company/company plus locality sited in depth.

(2) Capable of launching set piece phased attack from a secured base along a ridge/spur line in echelons to capture a company/company plus locality sited in depth.

(3) Capable of launching attack by shallow infiltration in the immediate vicinity of FDLs to capture a company/company plus locality.

(4) Capable of launching attack through deep infiltration on objectives in depth, with or without helicopters support and hold ground up to 48 – 72 hours till link up is achieved or the battalion is asked to ex-filtrate.

(5) Capable of fighting series of battles in depth till exhausted.

(6) Capable of remaining self contained on man pack/mule pack basis up to 48 – 72 hours retaining the potential of fighting normal battles.

(7) Capable of replenishing quickly from dumping well forward by employing follow up/reserve echelons for the purpose.

(8) Capable of establishing efficient and reliable wireless communication down to section level by establishing relay station (s) to off set the effects of hill screening.

(9) Is capable of supporting the attack through a coordinated fire support of direct and indirect weapons to the nearest distances of objective taking maximum risks.

22. **Infantry Battalion Modified 'D'**. Units placed under command infantry division operate in desert with following increase/decrease in equipment/ manpower from standadrd Infantry Battalion:-

a. *Increase*. 1 x Jeeps 4x4 GS

b. *Decrease*. 1 x Driver MT and 3 xMotorcycles.

23. **Infantry Battalion Modified 'Q'**. Battalions have following additional authorization of manpower/equipment from standard Infantry Battalion:-

a.	Drivers		-	8
b.	*Vehicles*			
	(1)	Bicycle	-	4
	(2)	Motorcycle	-	1
	(3)	Truck 2.5 Ton	-	2
	(4)	Trk 5 Ton	-	6

24. **Mechanised Infantry Battalion**. The concept is based on the need to provide protection, mobility, radio communication and firepower to the infantry to operate effectively in mobile operations. Mechanized infantry battalion is an integral part of armoured brigade. Companies are affiliated to the regiments. Organization of the mechanised infantry battalion is given at Annex 'HH'. Its tasks are:-

a. Close protection of tanks in close country or built-up areas.

b. Holding ground, over run by tanks and forming a firm base from which the tanks can operate further.

c. Seizing, denying or holding ground.

d. Special protective tasks, which consist of defence of tank harbours, escort of maintenance convoys and the protection of tanks moving at night.

e. Reconnaissance.

f. Collection of Prisoners of War.

25. **Parachute Infantry Battalion**. The parachute battalion is a unit specially organized, equipped and trained for airborne tasks. The tactical employment of a parachute battalion is the same as of a normal infantry battalion after it has been reorganized following an airborne assault landing. Parachute infantry is specialized and should not be confused with infantry, which is air-transported. Organization of parachute infantry battalion is given at Annex 'JJ'. Characteristics and tasks of parachute infantry battalion are as under:-

a. *Characteristics*. Parachute infantry, by means of its transportation to the battlefield have the following additional major characteristics:-

(1) Strategic mobility and surprise.

(2) Morale effects.

(3) Vulnerability on landing.

(4) Lack of tactical mobility and firepower.

(5) Sensitivity to weather.

(6) Need for air superiority and suitable dropping zones.

(7) Need for detailed preparation and early land link up or sustained maintenance by air.

b. *Tasks*. Suitable tasks for parachute infantry fall into the following main groups:-

(1) Operations in close cooperation with the main ground forces.

(2) Independent operations of a strategic nature.

(3) Guerrilla activity in small bodies within enemy occupied territory.

(4) Special internal security duties.

26. **Para-Commando Battalion**. It is organized into a battalion headquarters, an administrative team and three commando groups. Each commando group consists of three commando teams and a Medium Machine-Gun (MMG) section. Organization is given at Annex 'KK'.

27. **Scout Battalion**. The various scout battalions are organized, equipped and trained to operate in the mountainous terrain along the northern borders. Outline organizations of scout battalion (Garh) & scout battalion (Komaun) are given at Annexes 'LL' and 'MM' respectively.

28. **Reconnaissance and Support (R&S) Battalion**. Keeping in view the weaknesses of the erstwhile motorized battalions and futuristic requirement of the army, R&S battalions have been raised to provide adequate mobility, firepower and surveillance capability. Organization is given at Annex 'NN'.

29. **Anti-Tank Guided Missile (ATGM) Battalion**. The ATGM (MILAN) battalion is organized, trained and equipped to augment the anti-tank defence in a corps defensive zone. The battalion is organized into a headquarters company and five missile companies. Organization is given at Annex 'NN'.

30. **Border Security Forces (BSF)**

a. *Role and Employment*

(1) *In Peace*

(a) Promote a sense of security amongst the people living in the border areas.

(b) Prevent trans-border crimes, unauthorized entry into, or exit from Foxland territory.

(c) Prevent smuggling and other illegal activities.

(d) Collect intelligence in the course of patrolling, trans-border observations and interrogation of suspects.

(2) *During Hostilities*

(a) Sectors where enemy's major threat is not expected may be held by BSF units.

(b) BSF units may be employed for extension of flanks of main defence line.

(c) Undertake limited aggressive actions against enemy's para-military or irregular forces.

(d) Protective duties like protection of airfields, bridges, and administrative installations against commando/parachute raids.

(e) Convoy protection duties, escort of prisoners, protection of PW transit camps.

(f) Control of refugees and administration of captured enemy territory.

(g) Organization and acquisition of intelligence of areas adjoining the international border.

(h) Provision of guides for army within a BSF unit's area of responsibility.

b. *Organization*. Given at Annex 'OO'.

SECTION 114 – **TACT1CAL DOCTRINES OF FOXLAND ARMY**

DEFENSIVE DOCTRINE

General

1. The aim of defence in Foxland Army is to prevent penetration into areas or ground essential for the conduct of war and to blunt the enemy's superiority until an offensive can be resumed. Ground of tactical importance is always kept secure, and physically occupied, if it has been chosen as the vital ground also. Defensive battle is fought forward of this ground and counter-attacked to recapture, in case of its loss. Counter-attack is launched for those areas whose loss will render the vital ground vulnerable to attack.

2. Basic considerations that are taken into account in defence by Foxland Army are:-

a. Vital ground.

b. Enemy threat and approaches.

c. Assessment of own tasks.

d. Use of obstacles.

e. Deception, concealment and camouflage.

f. Good Observation Posts.

g. Availability of time and resources.

h. Logistic cover.

j. Reserves.

Forms of Defence

3. **Positional Defence**. Entails holding or control of specific area for a specified period.

4. **Mobile Defence**. The defenders aim at the destruction of the enemy by canalising the attacking force into a suitable area and then launching a counter-attack with strong mobile reserves to destroy him.

Basic Requirements of Defence

5. **Enemy**. Information about the enemy and air threat is collected from all possible sources, collated and correct deductions are made.

6. **Ground**. Vital ground, loss of which will provide enemy observation into defences and provide him with a firm base, will be held by troops physically. Vital ground/ground of tactical importance will very often be nominated by next higher commander, i.e. brigade commander nominates ground of tactical importance in battalion defended area.

7. **Frontages**. Normal frontages with 2 x LMGs per section are:-

a.	Division	-	80 – 120 kilometres
b.	Brigade	-	20 – 40 kilometres
c.	Battalion	-	7 – 10 kilometres
d.	Company	-	1100 – 1400 metres
e.	Platoon	-	300 – 350 metres
f.	Section	-	70 – 90 metres

8. **Mutual Support**. Fire of one subunit should cover the front of another.

9. **Depth**. Should aim at:-

a. Absorbing the weight of attack and breaking it up by a process of attrition.

b. Deny enemy ground observation and reconnaissance of defensive positions.

c. Enabling defended localities sited in depth to act as firm bases, from which counter-attacks can be launched.

d. Providing a cushion to defences, covering the ground selected as vital.

e. Gaining time to discover enemy intentions.

10. **All Round Defence**. Implies the following:-

a. **Organization**. Bulk of the weapons are deployed to cover the most likely lines of approaches, though some fire can be brought to bear against an attacker from any direction.

b. **Sited for All Round Defence**. Weapons are so located that an attack from any direction will meet with an approximately equal volume of fire.

11. **Coordinated Fire Plan**. Coordinated fire plan involving artillery, armour, anti-tank weapons, MGs, mortars and air is made. Coordination takes place at every level, commencing from the lowest to upwards.

12. **Air Defence**. Forms part of coordinated fire plan. Air defence guns are employed to protect Vital Points (VPs). Camouflage and concealment, dispersion, control of move, preparation of dummy positions are some of the passive measures, which are adopted at unit level.

13. **Coordinated Anti-Tank and Obstacle Plan**. It is based on mines, demolitions and obstacles of all kinds, which together with tanks and anti-tank weapons are sited in conjunction with each other.

14. **Reserves**. It will rarely be possible at brigade level and seldom at battalion level to keep infantry reserves uncommitted to a ground holding role. A reserve of armour and gun ammunition for counter-attack or counter penetration is held at division and if necessary, at brigade level. Reserves are maintained at following levels:-

a. *Battalion*. Maximum of a committed company from the depth locality could be mustered to react within 30 to 45 minutes of the attack.

b. *Brigade*. Maximum of a tank squadron and about two companies may be kept as uncommitted reserves to react within 2 to 3 hours from the decision of their employment.

c. *Division*. Possible composition could be a tank regiment less a squadron and an infantry battalion, kept uncommitted to react within 4 to 6 hours of the decision for their employment.

15. **Domination of Ground between Opposing Forces**. It will vary from the employment of covering troops or screens, establishment of sniper posts by day and active patrolling by night. Surveillance devices and night vision aids will help the defender in getting early warning.

16. **Security**. Security can be achieved by adopting measures to prevent surprise, preserve own freedom of action and deny information to the enemy. Necessary steps must, therefore, be taken to avoid tactical surprise. Considerable emphasis is laid on patrolling, gaining intelligence, camouflage and concealment, communication security against enemy interception and direction finding agencies. Today's field surveillance means can detect move and concentrations up to a considerable depth, beyond which observation is possible through satellites and Remotely Piloted Vehicles (RPVs). Innovative means of surprise and deception have to be evolved. The commander must ensure that the enemy is kept under constant surveillance during all phases of the defensive battle. Optimum utilization of available surveillance resources is essential for which comprehensive plan should be made.

17. **Reliable Signal Communication**. All signal communications must be duplicated.

18. **Echelons of Defence**. Foxland has the following echelons of defence:-

a. *Covering Troops*. Where space is available, a force generally composed of all arms works ahead of the defensive positions to observe, harass and delay the enemy's advance. Composition will depend upon the following:-

(1) Task.

(2) Availability of troops.

(3) Amount of delay to be imposed.

(4) Type of terrain.

(5) Strength and composition of enemy.

(6) Should be mobile, balanced and have good signal communication.

b. *Screens*. These are detachments generally composed of all arms occupying an area of ground for attack and to deny close observation of main defences. Composition will depend upon the following:-

(1) Degree of resistance to be offered.

(2) Nature of the enemy threat.

(3) Availability of troops, strength should not adversely effect the main defensive positions.

(4) Ground and distance in the front of the defensive positions.

(5) Task of the screens are:-

(a) Hamper the enemy patrolling.

(b) Report the strength and direction of enemy's attack.

(c) Deny enemy close observation and reconnaissance.

(d) Assist in withdrawal of covering troops.

(e) Force early deployment of enemy.

(f) Deceive the enemy regarding the location of the defensive positions.

c. *Main Defensive Positions*. Foxland Army does cater for depth in defence through employment of protective elements and integral depth in main defences. Priority is given to those areas which are visualized to be more threatened. Less threatened areas have relatively less protective elements, more gaps in the main defences and little depth. All main defences have defensive minefields that are effectively covered by coordinated fire of all weapons. It will be difficult to employ mechanised or manual method of breaching in the face of the enemy especially when defence is based on a water obstacle, which has a fortified high far bank.

d. *Reserves*. Refer to para 14 above.

19. **Foxland Conduct of Defence**. It has three stages:-

a. *The Prep Stage*. At this stage the defence is being organized and the attacker may be making preparations for the attack including reconnaissance to find out the extent of the defences. Foxland

preparatory stage is a period of intense activity and involves the following:-

(1) Move and occupation of defences including deployment of covering troops and reserves.

(2) Development of defences.

(3) Interference with enemy's preparation for attack.

(4) Collection of intelligence.

(5) Security of own forces.

(6) *Resistance Stage*. At this stage, the attacker is attempting to breakthrough the defences. It consists of :-

(7) Defensive Fire (DF).

(8) Spoiling attacks.

(9) Breaking up the assault.

(10) Local counter-attack.

(11) Counter penetration.

b. *Counter-Attack Stage*. At this stage the attacker has pen into the defences and deliberate counter-attack is launched. Salient features of this stage are:-

(1) Deliberate counter-attack is launched if ground of tactical importance is lost or penetration reaches a point, where it threatens the overall security of defences. It is not necessary that a commander must with hold his reserves till the above situation has arisen. Deliberate counter-attack may be launched to forestall any attempt for further penetration, recapture ground of tactical importance and fully restore the situation.

(2) Circumstances, when such counter-attack is advisable, includes the following situations:-

(a) When the enemy has been subjected to severe attrition and has spent his effort before reaching the limit of penetration.

(b) The enemy build-up on any captured locality may not be fast enough and therefore opportunities be availed to inflict severe losses on him.

(3) On rare occasions, when a deliberate counter-attack is envisaged at brigade level, only those troops are employed which are not engaged by the enemy. The deliberate counter-attack is coordinated and fully supported by tanks, artillery and air effort. The conduct of deliberate counter-attack differs in no way from the normal attack. It has however, the advantage of attacking over known ground against an enemy, who has had little time to prepare defences. The local counter-attack, although part of the resistance stage is given some attention here for purposes of information. It is launched up to brigade level to destroy and evict the enemy and regain control of the lost area. Troops in depth are employed to block penetration by readjustment, reinforcement of existing positions or by occupation of alternative positions. At platoon or company level, it will seldom be possible to lift troops without weakening of defensive localities. The same holds good at battalion level, for battalion under attack. At brigade level, it may be possible to lift some troops from battalion not under attack.

20. **Extended Defences**. Force is deployed on wider front with wider gaps then is tactically desirable. A battalion however, is not dispersed so much that its companies become isolated and are incapable of mutual support. The gaps are blocked by natural or artificial obstacles, patrols and artillery fire. Alternative defended area in depth is prepared, to which troops can be moved rapidly and a mobile reserve is based on armour and mechanised infantry for counter-attack.

21. **Defence on Ditch Cum Bund (DCB)**. Following are the types of defence oriented obstacles:-

a. Single Canal.

b. Twin Canal.

c. Single/double DCB, dry or wet.

d. Rivers.

22. **Foxland Pattern of Deployment on DCB**

a. *Ahead of DCB.* Various protective and surveillance elements are:-

(1) *Border Out Posts (BOPs*). Provided by Border Security Force (BSF), reinforced by regular troops when required.

(2) *Screens*. These play an important role as in lands of resistance and are adequately strengthened with MGs and rocket launchers supported by maximum possible artillery. Selected screen positions may be given ATGMs.

(3) *Protective Patrols*. Deployed to cover large gaps between screens and provide vital information regarding routes of ingress of enemy.

(4) *Reconnaissance Patrols*. Maintain surveillance and pass back detailed information from build-up to ingress stage. Classified as:-

(a) Trans-border patrols.

(b) Early warning patrols.

(c) Link patrols.

(5) *Stay Behind Parties*. These are taken from the protective detachments and tasked to maintain contact and provide continuous flow of information.

(6) *Mobile Rocket Launcher and MG Teams*. Deployed forward with screens, protective patrols and reinforced BOPs to cover gaps.

(7) *Minefield Nests*. Located within defensive minefields to destroy critical equipment.

(8) *Armour*. Terrain and resources permitting, some armour may be deployed in conjunction with screens and protective patrols.

(9) *Tank Hunting Teams*. Armed with portable weapons, these are employed along likely tank approaches.

b. *On the DCB Battalion Defended Sector*

(1) Three basic patterns are:-

(a) *No Gap Pattern*. Two companies forward on the obstacles and two in depth. Each Battalion covers an area of approximately 5 kilometres. There are no gaps between companies and Battalions, and require larger number of infantry Battalions to hold the obstacles.

(b) *Rectangular Pattern*. Two companies forward on the obstacles and two in depth. There are gaps between platoons, companies and Battalions. Each Battalion covers an area of approximately 5.5 kilometres. Defences in the gaps are prepared and manned, when necessary, by troops from depth companies.

(c) *Triangular Pattern*. Three companies forward on the obstacles and one in depth. This deployment facilitates covering large frontage i.e. Battalion covering a front of 7-10 Kilometres. The gaps are less but there is lesser depth in the defence.

(2) Fifty percent weapons are deployed on the obstacles. Rough guide line is:-

(a)	Forward of the obstacle	-	33%
(b)	On the obstacle	-	50%
(c)	Depth of the obstacle	-	17%

(3) Mortars are split to cover the entire battalion front.

(4) No separate un-committed reserves are kept. The Battalion can at best muster maximum of a company size force from the depth to react within 30 to 45 minutes of the H hour.

23. **Defence on River Line**. Main defensive positions are sited some distance from the river. The ground occupied is such that the enemy must capture it, if he is to consolidate the bridgehead. Depth and reserves in the main position is never sacrificed for the sake of holding dominating ground on the far bank. Enemy is not given free run to the water line. Basic design of battle is to cause attrition, while the water line is being approached and destroy the force, which may effect crossing by counter-attack, before the enemy is able to build-up sufficient force for break out battle.

24. **Defence on the Canal Line**. The main characteristics of defence on the canal line are similar to a river line defence, except that the efficacy of the canal as an obstacles varies. The banks are narrower 10'-40' and capable of taking limited B vehicles traffic. The canal banks are 4'-12' higher than the surrounding. Defensive positions are sited on the bank itself and result in a linear defence. As in any extended defence, a number of alternative positions in depth are prepared for occupation at short notice. At Battalion, level it will be normal to

prep 3 to 4 alternative company defensive localities in depth behind the home bank. Likely crossing places are held in strength.

Employment of Other Arms

25. **Armour**. It is used in the following roles:-

a. Reconnaissance.

b. Covering Troops.

c. Screens.

d. Anti -Tank.

e. Counter pen.

f. Counter-attack.

g. Special missions.

26. **Artillery**. The main tasks are:-

a. To harass and delay enemy approach.

b. To disorganize the assembly and forming up of enemy by DF tasks in depth.

c. Engage assaulting troops by closed DF/DF(SOS).

d. Counter Bombardment to neutralize enemy move/guns.

e. Assist in counter penetration.

f. Support counter-attack.

27. **Air Defence Artillery**. Employed for the protection of the following:-

a. Artillery gun areas.

b. Defiles, both natural and artificial.

c. Headquarters.

d. Divisional Administrative Area (DAA).

e. Concentration of troops, vehicles and armour.

28. **Engineers**. Tasks of engineers in defence include the following:-

a. Demolitions and cratering.

b. Laying and recording of tactical, defensive and nuisance minefields.

c. Construction and improvement of anti-tank obstacles.

d. Fortification of buildings and strengthening of command posts, signal exchanges and shelter for medical installations.

e. Laying and recording booby traps.

f. Maintenance and improvement of routes including those required for counter-attack force.

g. Construction and maintenance of bridges.

h. Construction of special devices for deception, concealment and camouflage.

j. Provision of defence stores is of engineers responsibility.

k. Issue of extra tools and assist with plant for digging pits for heavy weapons.

l. Water supply.

m. Construction and maintenance of air strips.

n. Occupation of defensive area in depth after completion of engineers tasks.

29. **Air**. Role of Foxland Airforce in defence are:-

a. Offensive air support during all stages of the defence.

b. Provide air transport support during all stages of the defence.

Offensive Doctrine

30. **Foxland Philosophy of Offensive**. Foxland intends adopting "Blitzkrieg" type of tactics in their offensive manoeuvres. They advocate boldness in execution and encourage/accept taking calculated risks. The accent is on deep and bold manoeuvres by their mechanised and armoured formations with intimate air support. Air assault units and parachute borne formations are likely to be used extensively in rear areas in conjunction with ground operations. Introduction of strike RAPID divisions, mechanised divisions and modernisation of other components of their armed forces are the steps in that direction. The new concepts of offensive were put to test in a major exercise conducted in the recent past and were found to be viable. Scarlet thread of Foxland Blitzkrieg version is

"simultaneity of attacks against various depth levels, harmonised to break the time and space bonds of enemy's sequentially reacting reserves".

31. **Aim of Attack**. The aim of attack is to:-

a. Destroy an enemy force.

b. Seize and hold ground which is either required for own operations or to deny its use to the enemy.

c. Contain an enemy force or divert its attention from other areas of operations.

32. **Types of Attack**. Foxland categorize the attack operations in two:-

a. Quick Attack.

b. Deliberate Attack.

33. **Basic Considerations of Attack**

a. *Combat Superiority*. It is desirable to concentrate a superior force at the decisive point and time for successful operations. Where this is not possible, calculated risks may be taken.

b. *Surprise*. The initiative is with the attacker, as such he should aim at achieving absolute surprise, thereby forcing the enemy to fight the battle under circumstances unfavourable to him.

c. *Firm Base*. The attack must start from a firm base which includes a secure start line. This ensures the following:-

(1) The enemy cannot wrest the initiative at the very outset by a spoiling attack and dislocate the attacker even before it has been launched.

(2) The attacking troops can fall back on the firm base, should the attack fail.

(3) A commander will not have to react to enemy counter-attacks and thus loose the initiative.

d. *Depth and Frontages*. There is an interrelation between depth and frontage in a major attack. Only by reducing the frontages, sufficient weight be achieved to ensure the depth of penetration required. As a general rule, the greater the frontage, the lesser will be the depth thus a balance between the two is necessary. Attack should be organized in depth and on a narrow front. This will:-

(1) Ensure concentration of force.

(2) Enable deep penetration.

(3) Ensure maintenance of momentum.

(4) Allow quick mopping up.

e. *Maintenance of Momentum*. The impetus of attack must not be allowed to die down. Momentum is maintained by:-

(1) Attack in depth.

(2) Maintaining an adequate reserve (attack force organize into assault and reserves echelons).

(3) Correct positioning and timely employment of reserves.

(4) Rapid regrouping and bringing up of fresh troops.

f. *Reserves*. Adequate reserves must be held to deal with any unexpected situation. Reserves once committed, must be replaced or recreated. Troops holding the firm base could also be used as reserves depending on situation.

g. *Reorganization*. Attacking troops must reorganized on capture of each objective. This process starts with the lowest subunit and is successively carried out up the chain of command.

h. *Fire Support*. Adequate fire support, including close air support, if available, must be provided till the capture of final objective and subsequent exploitation. Defensive fire must be available as soon as the objective has been captured.

j. *Favourable Air Situation*. It is essential for the success of any attack, especially in plains and desert.

k. *Axis of Maintenance*. An axis of maintenance must be cleared and kept open for the maintenance of the force. This need not be a road. The axis of attack and axis of maintenance need not necessary be the same.

34. **Stages of Attack**

a. *Preparatory*. Activities for domination of enemy, logistic arrangements, collection of information, planning and issue of orders, surprise and deception measures, concentration of forces, deployment of artillery, deployment of early warning means, development of communication arrangements continue.

b. *Break-in*. Punching of a hole through the crust of enemy defences and breaching of obstacles. First essential will be to secure a bridgehead and then holding of this bridgehead.

c. *Dog Fight*. Enlarging the bridgehead and facing enemy counter-attacks. Consists of sharp attacks with limited objectives.

d. *The Breakout*. It is a continuation of dog fight. Begins when enemy available reserves on battlefield are all committed and his resistance begins to crumble.

e. *Pursuit*. Following the withdrawing enemy if included in the plan.

f. *Sub Stages*. The break-in, the dog fight and the breakout periods are further divided into three sub stages:-

(1) The preparatory stage.

(2) The assault stage.

(3) The reorganization stage.

35. **Infantry Battalion in Attack**

a. *The Preparatory Stage*. It is during this stage that all the preparations for an attack are completed. The sequence of events at a battalion level is the same as for own attack.

b. *Warning Order*. On being given his task, the battalion commander will issue a warning order. Contents of warning order are the same as for own warning order.

c. *Collection of Information*. The battalion commander will have received a fair amount of information about the ground and the enemy disposition at the preliminary orders of the brigade commander. This information is, however, not sufficient for the detailed planning of the attack. The Battalion commander must, therefore, consider what additional information he requires and the best means of obtaining it. The battalion intelligence officer obtains this from the brigade headquarters and the troops in contact. Any information not available with then will be obtained by patrols. Before the attack is launched information about the enemy must be as complete as possible. Not until the battalion commander has the necessary information, will he be able to make his final plan for the attack.

d. *Reconnaissance and Appreciation*. Having issued his warning order and obtained such info as is possible in the time available, the battalion commander will carry-out preliminary reconnaissance with his 'R' group and makes his appreciation.

e. *Objectives*. Objectives must be recognizable on the ground. While the battalion objectives will be laid down by the brigade commander, the battalion commander will have to decide on the company objectives. He will take the following into consideration:-

(1) Strength and dispositions of the enemy.

(2) Ground and obstacles which have to be overcome.

(3) The frontage and depth of the objectives.

(4) Troops and supporting fire required to capture and hold the objectives.

f. *Phases*. If a number of company objectives have to be captured, the assault stage of an attack may have to be divided into phases depending on the ground and resources available particularly fire support. Each phase will work on a separate time table and is normally given its own Start Line (SL). The Battalion commander must allow time for brief re-organization between phases.

g. *Grouping*. Having decided the phases, the Battalion commander will group the companies into assault and reserves echelons for each phase, giving such additional resources to each company as considered necessary. The troops of the reserves echelon may be earmarked as the assault each for a subsequent phase.

h. *Axis of Adv*. The axis of advance indicates general direction of assault. When selecting this, the battalion commander must consider the likely assembly area, FUP and SL. The axis, if possible should afford cover from enemy observation, be short, and suitable for move up of reorganization stores.

j. *Deployment Areas*

(1) *Assembly Area*. 2 to 3 kilometres from troops in contact and out of the range of enemy's surveillance devices. Stay is normally from 2-6 hours and may extend up to 24 hours.

(2) *FUP*. Out of enemy direct firing weapons range. Stay should be minimum, normally 15 to 30 minutes.

k. *Rate of Advance*. In a deliberate attack against well prep defences, the advance may not be faster than 100 metres in 4 minutes. On the other hand, in a day attack over favourable ground and against hastily prepared defence, the rate of advance may be as fast as 100 metres in 2 minutes.

l. *Frontages (Night Attack)*

(1) *Frontages*

(a)	Battalion	-	400 metres
(b)	Company	-	200 metres (Could be as less as 60 metres)
(c)	Platoon	-	75 metres

(2) *Depths*

(a)	Battalion	-	400 metres
(b)	Company	-	125 metres
(c)	Platoon	-	50 metres
(d)	Rear Companies	-	225 metres

m. *Flanks Protection*. In the assault initially, flank protection is provided usually by units through which the battalion is attacking. Flank protection for the assault when it has reached greater depth is planned by the battalion commander by the use of tanks, artillery, mortars, detachments of recoilless guns, medium MGs and protective patrols.

n. *Reorganization*. The reorganization on the objective must be planned carefully. Defensive fire tasks for artillery and mortars and the probable location of medium MGs, tanks and recoilless guns should be laid down before hand and adjusted on the capture of the objective. Troops must dig in fast and deep to break up any counter-attack and to ensure protection against enemy fire from ground and air. Failure to do so, may result in serious consequences and loss of captured ground or heavy casualties. The battalion commander will consider what reorganization stores will be required and will make arrangements for their move forward.

o. *Exploitation*. At battalion level, the extent of exploitation that can be planned ahead is limited and the limit of exploitation for the battalion will normally be laid down by the brigade commander.

The battalion commander must however, consider what troops will be necessary for carrying out this task. While planning for an attack, if it is known that the enemy is occupying certain defended localities, the capture of these defended localities must be planned as part of the attack either in one phase or in two or more phases. Although it may appear that the capture of one defended locality decreases the defence potential of another defended locality considerably, the commander must plan for the capture of the second defended locality as well. From his appreciation, he may come to the conclusion that the capture of the second defended locality would require lesser effort and would therefore plan accordingly. To state, however, that after the capture of the first defended locality, he will exploit to the second defended locality, would be incorrect.

36. **Rifle Company in Attack**

a. The sequence of planning at company level will be similar to that at battalion level. Having received his preliminary orders and issued his own warning order, the company commander will carry-out his reconnaissance. Prepares his outline plan and issue his preliminary orders to his 'O' group. He will issue his final orders after he has carried out a further reconnaissance and has received detailed orders from the battalion commander. His orders must contain the following:-

(1) All relevant information including the battalion commander's outline plan.

(2) Company commander's intention.

(3) Formation for the attack.

(4) Platoon objectives.

(5) Inter platoon and companies boundaries.

(6) Assembly area, FUP and SL.

(7) Axis of advance.

(8) Rate of advance.

(9) Method of crossing the obstacles.

(10) Action on capture of objective.

(11) Fire plan including employment of armour, MMG and RL detachments.

(12) H hour.

(13) Success signal.

(14) Administrative details.

b. After crossing the SL, company will advance to its objective at a steady pace. Running should be avoided except during the last stage of the assault or when crossing ground is exposed to enemy fire. The assault troops should not open fire due to the danger of cross fire within the company. Fire may be opened only when it is essential to destroy a known enemy position which is holding up the progress of the assault.

c. The reserve platoon should generally be given the task of capturing the enemy defences in depth. This will ensure that the momentum of the attack is maintained.

d. Normally the attack will be planned on certain timings for which a timed artillery programme will be made. It would be advisable to maintain this timed programme so that the company is on the objective as soon as the covering fire lifts. For movement up to the objective, the company will generally depend on the covering fire of other arms as well as its own supporting weapons. On reaching the objective, the company will have to fight with its own weapons.

37. **Rifle Platoon in Attack**

a. The platoon must go into action absolutely certain of its objective. Once on the move, it must not be stopped by anything less than really effective fire. It must be understood that to close with the enemy and to destroy him, is the primary duty of all assaulting troops.

b. The aim of the platoon should be to advance as close to the enemy's position as possible under the covering fire of supporting weapons. This may, however, not be possible due to the intensity of the enemy's fire. In such a case the platoon will have to use fire and move tactics to continue the adv.

c. Should the platoon be held up in its advance, it will be so disposed that it is able to bring down the maximum fire on the enemy in

order to assist other platoons to continue their advance. Having done this, the platoon commander must still endeavour to get his men forward whenever such action is possible.

d. When the platoon has reached within assaulting distance, it will charge the objective destroying the enemy with grenades and the bayonets. The whole area of the objective will then be searched and cleared. Small enemy detachments which may come to life again after the first shock of the assault must be destroyed. This close fighting may lead to disorganization, the platoon commander must, therefore, regain control over his platoon as quickly as possible.

38. **Obstacles Crossing**

a. *Mine Breaching*

(1) Whenever attack formations come across minefield which is not discernible, they must press on boldly. Three tanks casualties per armed regiment is quite acceptable in this regard.

(2) Minefield breaching will only be undertaken when it becomes difficult to bypass it or to find a gap through it.

(3) Assault across the minefield on the home side of the canal/ ditch will be undertaken by the elements of strike formations and not by the holding formations elements which have cleared the security zone. First wave of infantry will pass through the lane created by firing CLMC i.e. charge lane mine clearance.

(4) Trawl PT 55 will be used for making vehicle's safe lanes.

b. *Water Obstacles*

(1) Infantry will either swim across or use boats for crossing.

(2) Amphibious BMPs will be used by mechanised infantry to cross the wet gaps whenever situation permits.

39. **Attack through Minefield**. Foxland methods regarding attack through a minefield are listed below:-

a. *Options*. A commander may choose one of the three options available:-

(1) An armed attack using trawls/ploughs to clear the lanes through minefield, supported by infantry and artillery.

(2) Infantry assault through the minefield either with or without safe lanes, to form a bridgehead across the minefield. In case of low intensity minefield, mechanised forces may launch an assault from the line of march.

(3) It may be possible to clear gaps by silent engineers parties under cover of smoke or darkness.

b. *Number and Types of Lanes Required*. The supporting weapons and vehicles will normally require a lane of 5 metres width. This width may have to be increased up to 10-12 metres later, for two way traffic. Normally, an assault brigade attack with two up will require three lanes of 5 metres width each as under:-

(1) One per battalion.

(2) One as brigade reserve.

c. *Lanes*. The lanes should be sufficiently apart so that enemy artillery fire against one gap does not interfere with work of the other. This implies that gaps will be 400 to 500 metres apart. If infantry lanes are required, minimum of one per platoon will be necessary.

40. **Clearance of Security Zone**. Foxland is likely to employ a mechanised brigade and an infantry brigade of RAPID (D) to clear the security zone. This infantry brigade is likely to come from the holding corps reserves. Armoured brigade (holding corps reserves) may also be employed in lieu of mechanised brigade in case the protective detachments employed by the enemy are too strong.

41. **Concept of Fighting the Battle of Security Zone**. Holding formations will be employed to clear the security zone. This task would probably be given to mechanised division in desert sector. Mechanised brigade of RAPID (D) and corps reserves of holding corps will play a vital role in this regard. Mechanised formations will organize itself into following forces:-

a. *Direct Pressure Force*. Its main task is to deny enemy units any chance of rest, re-grouping or re-supply by exerting relentless pressure.

b. *Encircling Force*. Two combat groups will generally be employed as the Encircling Force. The organization of the mechanised division permits double envelopment.

c. *Interception Force*. Interception Force is generally a composite group composed of tanks and infantry elements, which intercept and destroy the withdrawing enemy.

42. **The Concept of Breakout/Main Defensive Positions**. Main defensive position of enemy will be based on some kind of obstacles system i.e. canal, ditch or minefield. They claim that under existing operational environments, they will be able to establish the bridgehead over the obstacles, build-up enough armoured/ mechanised strength inside the bridgehead the same night and breakout at first light. This operation will be undertaken by their strike formations. Holding formations would have already secured area up to and including home side of the water obstacles/minefield on night D-1/D Day. Holding formations would have already established contact all along the front and would have found weak spots/gaps in the defences. At corps level enemy will attempt a min of two bridgeheads, each by a division size force. As per new organization, one could be an infantry division and the other RAPID (S). An infantry brigade of the corps reserves may be released to any one of these divisions to assist them in their task of establishment/expanding the bridgehead. RAPID (S) may be given some infantry from holding formations if need arises.

43. **Weaknesses of Foxland Offensive**

a. *While Clearing Security Zone*

(1) *Lack of Artillery*. In certain quick attacks during the break-in phase, artillery support may not be adequate. Mechanised/armoured columns will attempt to outflank and trap the defender without giving him any reaction time. This is possible only by undertaking deep outflanking cross country manoeuvre, wherein towed artillery of the mechanised formations and independent armoured brigades will not be able to accompany and support these manoeuvres. These formations would thus have inadequate artillery support and will have to rely mainly on air and attack helicopters support if situation/resources permit.

(2) *Imbalance in Defence*. If any flank of the holding formations is threatened by the enemy, it will not have much reaction capability as it would have already employed its reserves to clear the security zone.

b. *Break in and Dog Fight*

(1) Separation of armour from infantry due to obstacles.

(2) Lack of infantry in case of RAPID(S) while making bridge-head.

(3) Difficulty to conceal concentration of major forces, hence loss of surprise at the very outset of hostilities.

(4) Due to paucity of infantry, bridgehead will be weak and vulnerable to enemy's counter-attacks for a prolonged period of time.

(5) Bridging equipment and trawls will remain highly vulnerable to enemy tank hunting parties.

c. *Breakout Phase*

(1) Inadequate artillery support during the initial stages of breakout.

(2) Inadequate infantry for clearing areas in depth.

(3) Axis being opened after 48 hours, during which time intimate logistic support will not be available.

(4) Giving battles to enemy at two or more levels at a time will have a telling effect on the resources. There will be no reserves to stabilize the situation, if need be.

(5) Vulnerability to early warning devices of the enemy, which can compromise the entire coordination between various formations.

(6) Logistic based on wheels may not be able to keep pace with manoeuvre elements.

(7) Lack of Self Propelled (SP) artillery to support fast moving mechanised operations.

SECTION 115 – RIFLE **COMPANY ACTING AS THE BATTALION RESERVE**

General

1. The company may be held as the battalion's reserve during an attack. The battalion commander commits the reserve to best influence the action to maintain the momentum of the attack. To exploit the success of the other attacking companies, the reserve should attack the enemy from a new direction. Because of the various missions which the reserve may be assigned, its commander must

keep abreast of the tactical situation, knows the mission and tactical plans of the other companies, and be familiar with the terrain and enemy situation in the objective area. The reserve must react quickly and effectively when committed.

2. The battalion commander may want the reserve company commander with him during the fight in order to keep the company commander abreast of the situation. This also makes it easier when the battalion commander needs to pass detailed instructions to him. If the reserve company commander remains with his company, he must monitor the battalion radio net.

Mission of the Reserve

3. The reserve may be assigned one or more of the following mission:-

   a. Protect the flanks or rear of the battalion.

   b. Maintain contact with adjacent units.

   c. Clear a position which has been overrun or bypassed by the attacking companies.

   d. Support the attacking companies by fire.

   e. Assume the mission of an attacking company.

   f. Attack from a new direction.

   g. Protect or assist during the consolidation on an objective.

   h. Guard and evacuate prisoners.